

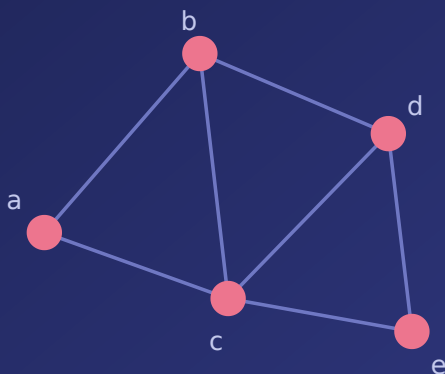
C S C 2 4 2

# Discrete Structures

*Study Manual*

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Victor Ayodeji



| p | q | $p \rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

$A \cup B$        $A \cap B$        $A \subseteq B$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

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*Every line is a link. The page numbers are read back from the rendered book, never typed by hand.*

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## BEFORE YOU START

# How to use this manual

*Read this page once. It will save you hours, and it explains one thing about this course that nothing else will tell you.*

## The thing you need to know first

This course has an official course text of 161 pages. **It does not contain the exam.** That is not a complaint, it is a measurement. Every past paper leans heavily on graph theory, trees, the properties of relations, ordered sets and Boolean algebra. The official text covers none of them. Searched end to end, it contains the word "vertex" zero times, "reflexive" zero

times, "adjacency" zero times, "Handshaking" zero times, "planar" zero times. Its five uses of the word "graph" are all about plotting a function, and its twelve uses of "tree" all belong to one worked example about trees growing in a street.

Meanwhile, on the 2025/2026 paper, roughly two marks in every five come from exactly that missing material. A student who revises only the course text walks into the hall having never seen two fifths of the paper.

So this manual teaches **both**: everything the course text covers, and everything the papers demand that it does not. Wherever a topic falls in the second group, it is flagged like this, once, at the top of the section:

#### NOT IN THE COURSE TEXT

The official 161-page text does not cover this topic at all, and the exam tests it every year. Everything you need is taught here from zero. If you go back to the course text to revise this, you will not find it. That is the text's gap, not yours.

## How every past question is presented

Past questions are the whole point of a manual like this, and they are easy to present dishonestly. If a book tidies a question into its own neat prose, it has done the hardest half of the work for you, namely working out what is being asked, and left you only the arithmetic. Worse, it hides the examiner's habits of phrasing, which are exactly what you are training to recognise, and it quietly repairs the paper's own mistakes, so you never learn that papers contain mistakes.

This manual therefore never restates a past question. Every one arrives in three parts, in this order.

#### AS PRINTED

#### EXAMPLE ONLY

The examiner's exact words appear in a box like this one, carrying the paper's own numbering, spelling, symbols and mark text, down to the last comma. **(2 marks)**

#### NOTE ON THE PAPER

If a question contains a typo, a contradiction or an ambiguity, it stays in the quote and is flagged here, underneath. It is never silently corrected, because when you meet it in the hall it will not have been corrected for you either. Handling a broken question is part of the skill.

#### BREAK IT DOWN

|                  |                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | what the question hands you, listed plainly.                                                                                                                                                                          |
| <b>ASKED</b>     | what it actually wants. The instruction verb decides this, and it is where marks are most often thrown away: "state" is not "prove", "differentiate" is not "define", and "determine which is true" is not "convert". |
| <b>MARKS SAY</b> | what the mark count tells you about the depth expected, and roughly how long to spend.                                                                                                                                |
| <b>TEACHES</b>   | the section of this manual that covers it.                                                                                                                                                                            |
| <b>TRAP</b>      | the specific way people lose marks on this one.                                                                                                                                                                       |

Only then comes the worked answer. Questions this manual wrote itself, for practice, are labelled as practice and never wear an "As printed" bar, because they were never printed by anybody.

## The boxes, and what each one is for

Navigate by shape and colour. Each box does exactly one job.

### TEACH

One idea, in plain language, built from what you already know. Never two ideas in one box.

### ● MUST MEMORISE

### THE RESERVED COLOUR

Crimson appears nowhere else in this book. When you see it, the content is going into your memory, not just past your eyes. Each one carries a hook to make that easier.

◆ **Memory hook:** crimson means learn this by heart.

### WORKED EXAMPLE

### PQ 25/26 Q1C

A full solution with the reasoning shown on every line. The chip on the right names where the question came from, so you always know whether you are looking at a real exam question and which paper it is from. Numeric ones open by listing what you were given and what you must find, before any formula appears.

### ⚠ TRAP

The mistake made every year. Read these twice.

### RECALL AND CHECK

A question with its answer hidden below a dashed rule. Cover the answer, say yours out loud, then look. Reading the answer without trying first teaches you almost nothing, however productive it feels.

### FILL IN THE GAP

Cloze practice. Write the missing words on paper, not in your head.

### C++

A program, with its real output. Every listing in this manual was compiled and run, and the output shown is what the compiler actually printed, not what it ought to have printed.

## How to work through it

Work with a pen and blank paper. When a worked example ends, cover it and redo it from the question. When you reach a past paper, sit it under exam conditions before you read a single answer: three hours, four questions, no notes. A paper you have read the answers to teaches you far less than one you have struggled through and then marked.

The parts are ordered so that nothing is used before it is taught. If you meet a symbol you do not recognise, it was introduced earlier, and the Formula and Notation Reference at the back lists every one.

## What the exam looks like

The format changed part way through the papers we hold, so be careful with the older ones.

| SESSION          | CODE PRINTED   | INSTRUCTION                | TIME           | UNITS      | MARKS PER QUESTION |
|------------------|----------------|----------------------------|----------------|------------|--------------------|
| 2020/2021        | CSC311         | Attempt any THREE          | 2 hours        | 2          | 23.5               |
| 2021/2022        | CSC416         | Attempt any THREE          | 2 hours        | 2          | 23.5               |
| 2023/2024        | CSC 242        | not on our copy            | not stated     | not stated | 23.5               |
| 2024/2025        | CSC416         | Attempt any FOUR           | 3 hours        | 3          | 17.5               |
| <b>2025/2026</b> | <b>CSC 242</b> | <b>Answer any Four (4)</b> | <b>3 hours</b> | <b>3</b>   | <b>17.5</b>        |

It is one course that has been renamed twice, so all five papers are valid practice. The current shape is the last row: **six questions, answer any four, 17.5 marks each, 70 marks in three hours**. That is about 44 minutes a question. This manual solves all five papers in full, then gives you three more in the current shape to sit.

### ⚠ TRAP

### BEFORE YOU START REVISING

You choose four questions out of six, so it is tempting to plan on dodging a topic. Look at what that costs on the 2025/2026 paper: graphs appear in Q1, Q4, Q5 and Q6, and counting appears in Q3, Q4 and Q6. There is no set of four questions you can answer while avoiding graph theory, and no honest way to avoid counting either.

The choice protects you against a bad day on one topic. It does not let you leave one out.

## FOUNDATIONS

# The Language of Discrete Maths

*Every symbol this course uses, introduced once, plainly, before anything is built on it. If you have ever stared at a line of mathematics and not known how to say it out loud, start here.*

|           |           |             |           |               |                   |             |          |           |
|-----------|-----------|-------------|-----------|---------------|-------------------|-------------|----------|-----------|
| $\in$     | $\notin$  | $\subseteq$ | $\subset$ | $\cup$        | $\cap$            | $\emptyset$ | $\Sigma$ | $\forall$ |
| is in     | is not in | subset      | proper    | union         | meet              | empty       | sum      | all       |
| $\exists$ | $\neg$    | $\wedge$    | $\vee$    | $\rightarrow$ | $\leftrightarrow$ | $\equiv$    | $\times$ | $\Sigma$  |

# Numbers, Symbols and Notation

*Discrete mathematics has a small vocabulary. Learn it once here and nothing later in this book will look like a foreign language.*

*By the end of this section you will know what "discrete" means, be able to name and write the standard number sets, read and write the membership and subset symbols out loud, and use factorial and summation notation without hesitating.*

## What "discrete" means, and why the course exists

### TEACH

### THE ONE WORD IN THE TITLE

**Discrete** means separated, made of individual pieces you can count, with nothing in between. Its opposite is **continuous**, which flows without gaps.

The whole numbers 1, 2, 3 are discrete: between 2 and 3 there is no other whole number. Temperature is continuous: between 20 degrees and 21 there are infinitely many values, and you can always find another.

A computer is a discrete machine. It stores whole bits, executes whole instructions, and holds finitely many values. That is the whole reason this course exists: **the mathematics of computing is the mathematics of things you can count and separate**, not the mathematics of smooth curves.

### TEACH

### WHERE EACH PART GOES IN COMPUTING

Keep this map in your head. It also answers a question the examiner likes to ask directly (2025/2026 Q2a asks for five justifications for including discrete structures in a computer science curriculum).

| TOPIC               | WHAT IT IS FOR IN COMPUTING                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Logic               | Circuit design, conditions in programs, database queries, proving a program correct, artificial intelligence |
| Sets                | Data types, database tables, collections, search spaces                                                      |
| Relations           | Relational databases, dependency between tasks, equivalence of states                                        |
| Functions           | Every subprogram, hashing, encryption, compression                                                           |
| Counting            | Algorithm complexity, password strength, probability of collisions                                           |
| Graphs              | Networks, routing, the web, social networks, scheduling, compilers                                           |
| Trees               | File systems, search structures, parsing, decision procedures                                                |
| Proof and induction | Showing an algorithm always works, not just on the cases you tried                                           |
| Recurrence          | The running time of a recursive algorithm                                                                    |



(a) Provide five Justification the inclusion of Discrete structure to Computer Science students curriculum. **(5 marks)**

**NOTE ON THE PAPER**

A word is missing and one is capitalised mid-sentence. The sentence should read "Provide five justifications **FOR** the inclusion of discrete structures **IN** the computer science students' curriculum". The meaning is not in doubt: give five reasons the subject belongs on the degree. Answer it as the five-reason question it is.

**BREAK IT DOWN**

|                  |                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Nothing. This is an opinion question with a right answer.                                                                                                              |
| <b>ASKED</b>     | <b>Provide five</b> justifications. The number is explicit, so give exactly five, numbered. Four costs a mark; six wastes time.                                        |
| <b>MARKS SAY</b> | 5 marks, one per justification. Each wants a claim <i>and</i> a concrete example of where it bites. A bare list of topic names is not a justification.                 |
| <b>TEACHES</b>   | This section, F.1, using the table above.                                                                                                                              |
| <b>TRAP</b>      | Listing five <b>topics</b> ("logic, sets, graphs, trees, counting") instead of five <b>reasons</b> . The question asks why the subject is included, not what is in it. |

Five justifications, each a claim with a concrete example attached.

1. **Computers are discrete machines, so discrete mathematics is their native mathematics.** A machine stores whole bits and executes whole instructions; it has no smooth quantities. Continuous mathematics describes a falling stone, but nothing inside a processor. *Example:* an integer overflow is a fact about a finite set of representable values, and only discrete reasoning predicts it.
2. **It supplies the language in which algorithms are specified and their cost measured.** Counting techniques give the running time of a procedure before it is ever run. *Example:* knowing that comparing every pair of  $n$  records takes  $n(n - 1)/2$  comparisons tells you the method will not scale, without writing a line of code.
3. **It is the basis of correctness, not merely of testing.** Logic and proof by induction establish that a program works for *every* input, where testing only covers the cases you thought of. *Example:* proving a loop invariant holds shows a sorting routine is right for all inputs, including the ones nobody tried.
4. **Its structures are the data structures of the subject.** Sets, relations, functions, graphs and trees are not analogies for computing objects; they are those objects. *Example:* a relational database is built directly on relations, and a file system is a tree.
5. **It underpins the applied areas the degree leads to.** Networking, cryptography, compilers and artificial intelligence all rest on it. *Example:* routing traffic is a shortest-path problem on a graph, and public-key cryptography rests on number theory over the integers.

**How to write it under time pressure:** number them 1 to 5, give each a bold opening claim and one sentence of example. Five short numbered paragraphs read as five justifications; one long paragraph reads as one.

## The number sets

A **set** is any collection of distinct objects, gathered together and treated as one thing. Part Two studies sets properly; for now you only need the five standard ones, because every question in this course lives in one of them.

They are written with a special doubled letter so nobody confuses the set of all integers with a variable called  $Z$ .

| SYMBOL       | NAME                 | WHAT IS IN IT                                                                  |
|--------------|----------------------|--------------------------------------------------------------------------------|
| $\mathbb{N}$ | the natural numbers  | 0, 1, 2, 3, ... the counting numbers                                           |
| $\mathbb{Z}$ | the integers         | ... -2, -1, 0, 1, 2, ... whole numbers, positive and negative                  |
| $\mathbb{Q}$ | the rational numbers | anything writable as a fraction of two integers, such as $\frac{3}{4}$ or $-7$ |
| $\mathbb{R}$ | the real numbers     | every point on the number line, including $\pi$ and the square root of 2       |
| $\mathbb{C}$ | the complex numbers  | reals extended with the square root of $-1$ (not examined here)                |

◆ **Memory hook:** Natural, **Z**ahlen (German for numbers, which is where  $\mathbb{Z}$  comes from), **Q**uotient, **R**eal, **C**omplex. Each doubled stroke says "this is a whole universe of numbers, not a variable".

## ⚠ TRAP

DOES  $\mathbb{N}$  CONTAIN 0?

Textbooks disagree. Some start the natural numbers at 0, some at 1. In computing, 0 is almost always included, and this manual includes it.

What matters in an exam: **say which convention you are using**. One short sentence ("taking  $\mathbb{N}$  to include 0") protects you completely, and a marker cannot penalise a stated convention. What loses marks is leaving it ambiguous and then relying on it.

## Membership: the two symbols you will use most

## TEACH

## ∈ AND ∉

To say that an object is in a set we write  $\in$ . The statement  $3 \in \mathbb{Z}$  is read "**3 is an element of the integers**", or more casually "3 is in  $\mathbb{Z}$ ". The symbol is a stylised Greek epsilon, for the Greek word meaning "is".

To say an object is not in a set, strike it through:  $\notin$ . So  $\frac{1}{2} \notin \mathbb{Z}$  reads "**one half is not an element of the integers**", which is true, because one half is not a whole number.

The left side is always an **object** and the right side is always a **set**. That direction never changes, and it is the single most common slip in first attempts.

● MUST MEMORISE

READ THESE OUT LOUD

| WRITTEN                 | SAID                                         | TRUE?                             |
|-------------------------|----------------------------------------------|-----------------------------------|
| $5 \in \mathbb{N}$      | 5 is an element of the natural numbers       | yes                               |
| $-5 \in \mathbb{N}$     | minus 5 is an element of the natural numbers | no, $\mathbb{N}$ has no negatives |
| $-5 \in \mathbb{Z}$     | minus 5 is an element of the integers        | yes                               |
| $\pi \notin \mathbb{Q}$ | pi is not an element of the rationals        | yes, pi is not a fraction         |
| $\pi \in \mathbb{R}$    | pi is an element of the reals                | yes                               |

◆ **Memory hook:**  $\in$  opens towards the set, like a mouth swallowing the element.

## Two ways to describe a set

The 2025/2026 paper asks for exactly this, in two marks, so learn both names.

● MUST MEMORISE

ROSTER AND SET-BUILDER

**1. The roster method**, also called listing or the tabular method. Write the members out between braces, separated by commas.

$$A = \{2, 4, 6, 8\}$$

**2. Set-builder notation**, also called the rule method. State the property that decides membership.

$$A = \{x \mid x \text{ is an even number and } 0 < x < 10\}$$

The vertical bar is read "**such that**". Some books use a colon instead, which means the same thing. The whole line is read: "A is the set of all x such that x is even and x is between 0 and 10".

◆ **Memory hook:** the roster method names the members, the way a class register does. Set-builder gives the entry rule, the way a door policy does.

WORKED EXAMPLE

METHOD

Write each set both ways.

**(a) The odd numbers under 10.**

Roster:  $\{1, 3, 5, 7, 9\}$ .

Set-builder:  $\{x \mid x \in \mathbb{N}, x \text{ is odd}, x < 10\}$ .

**(b) The days of the week starting with T.**

Roster:  $\{\text{Tuesday, Thursday}\}$ .

Set-builder:  $\{d \mid d \text{ is a day of the week and } d \text{ begins with T}\}$ .

**(c) The integers whose square is 4.**

Set-builder:  $\{x \mid x \in \mathbb{Z} \text{ and } x^2 = 4\}$ .

Roster:  $\{-2, 2\}$ . Notice the roster form does the work of solving the equation for you, which is exactly why an examiner asks for it.

**Now redo it:** write "the prime numbers below 12" both ways, then check:  $\{2, 3, 5, 7, 11\}$ .

**⚠ TRAP****SETS DO NOT REPEAT AND DO NOT CARE ABOUT ORDER**

$\{1, 2, 2, 3\}$  is the same set as  $\{1, 2, 3\}$ . Writing an element twice adds nothing, because a set records only *whether* a thing is in.

$\{3, 1, 2\}$  is also the same set as  $\{1, 2, 3\}$ . Order carries no information either.

This matters later. When order *does* matter you are no longer dealing with a set but with an **ordered pair** or a sequence, and the brackets change from braces to parentheses:  $(1, 2)$  is not  $(2, 1)$ . Part Three is built entirely on that distinction.

## Factorial

**● MUST MEMORISE****N FACTORIAL**

For a whole number  $n$ ,  $n!$  (said "n factorial") means multiply every whole number from  $n$  down to 1.

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

$n$  a whole number, 0 or more.  $n!$  the product of every whole number from  $n$  down to 1. By definition  $0! = 1$ .

The first few, which you should know on sight:

| N  | 0 | 1 | 2 | 3 | 4  | 5   | 6   | 7    |
|----|---|---|---|---|----|-----|-----|------|
| N! | 1 | 1 | 2 | 6 | 24 | 120 | 720 | 5040 |

◆ **Memory hook:**  $0! = 1$  looks wrong but is forced: there is exactly **one** way to arrange nothing, namely do nothing. Every counting formula in Part Five breaks if you set it to 0.

**TEACH****WHY FACTORIALS GROW SO FAST**

$10!$  is already 3 628 800, and  $20!$  is bigger than a billion billion. This is not a curiosity: it is the reason some problems are hopeless to solve by trying every possibility. The travelling-salesman question in Part Six turns on exactly this.

Useful cancelling trick, used constantly in Part Five:

$$10! / 8! = (10 \times 9 \times 8!) / 8! = 10 \times 9 = 90$$

Never expand both factorials and divide. Cancel first. It turns an impossible arithmetic problem into a two-second one.

## Summation

### ● MUST MEMORISE

### SIGMA NOTATION

The Greek capital sigma,  $\Sigma$ , is shorthand for "add up". It is read "the sum of".

$$\sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$$

**i** the index, a counter. **i=1** below the sigma, where the counter starts. **n** above the sigma, where it stops. **a<sub>i</sub>** the thing being added, written in terms of i.

So  $\sum_{i=1}^4 i^2$  means  $1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$ .

◆ **Memory hook:** sigma is the Greek S, for Sum. Read it left to right as a sentence: "sum, from i equals 1, up to n, of a-sub-i".

### RECALL AND CHECK

1. Say out loud what  $7 \notin \mathbb{Q}$  would mean, and say whether it is true.
2. What is  $6! / 4!$ ?
3. Write  $\{x \mid x \in \mathbb{Z} \text{ and } -2 \leq x \leq 2\}$  by the roster method.
4. Evaluate  $\sum_{i=1}^3 (2i + 1)$ .

#### ANSWERS

1. "7 is not an element of the rationals". It is false:  $7 = 7/1$ , so 7 is rational.
2.  $6!/4! = 6 \times 5 = 30$ . Cancel, do not expand.
3.  $\{-2, -1, 0, 1, 2\}$ .
4.  $(2+1) + (4+1) + (6+1) = 3 + 5 + 7 = 15$ .

### FILL IN THE GAP

1. The symbol  $\in$  is read \_\_\_\_\_.
2.  $\mathbb{Z}$  is the set of \_\_\_\_\_.
3. The two mathematical ways to describe a set are the \_\_\_\_\_ method and \_\_\_\_\_ notation.
4.  $0!$  equals \_\_\_\_\_.
5. In set-builder notation the vertical bar is read \_\_\_\_\_.

#### ANSWERS

1. is an element of. 2. the integers. 3. roster (listing), set-builder (rule). 4. 1. 5. such that.

### FOUNDATIONS • F.2

## Reading and Writing Mathematics

*Mathematical English is a dialect. A handful of phrases mean something narrower than they do in ordinary speech, and marks are lost every year by reading them casually.*

*By the end of this section you will know exactly what "if", "only if", "if and only if", "necessary" and "sufficient" translate to in symbols, what the instruction verbs on an exam paper demand, and how to lay out an answer so a marker can give you every mark you earned.*

## The five phrases that carry the marks

### TEACH

### TWO ARROWS, BEFORE WE USE THEM

Two symbols appear all through this section, so they are introduced here and studied properly in Part One.

$\rightarrow$  is the **implication**, also called the conditional.  $p \rightarrow q$  is read "if p then q", and it is a single statement claiming that whenever p holds, q holds too.

$\leftrightarrow$  is the **biconditional**.  $p \leftrightarrow q$  is read "p if and only if q", and it claims the two always hold together: each one implies the other. It points both ways because it works both ways.

Here p and q stand for whole statements. Section 1.1 gives them their exact truth tables; for now these readings are enough.

### TEACH

### NECESSARY AND SUFFICIENT

Take a statement of the form "if p then q", written  $p \rightarrow q$ . English dresses it up in several ways, and they are not interchangeable.

**p is sufficient for q.** Having p is enough to guarantee q. Being a dog is sufficient for being an animal.

**q is necessary for p.** Without q you cannot have p. Being an animal is necessary for being a dog.

So one implication,  $p \rightarrow q$ , says two things at once: p is enough, and q is required. Which side is which is the whole question, and it is the thing the examiner tests.

| ENGLISH                             | SYMBOLS               |
|-------------------------------------|-----------------------|
| if p, then q                        | $p \rightarrow q$     |
| p implies q                         | $p \rightarrow q$     |
| q if p                              | $p \rightarrow q$     |
| p only if q                         | $p \rightarrow q$     |
| p is sufficient for q               | $p \rightarrow q$     |
| q is necessary for p                | $p \rightarrow q$     |
| q whenever p                        | $p \rightarrow q$     |
| q unless not p                      | $p \rightarrow q$     |
| p if and only if q                  | $p \leftrightarrow q$ |
| p is necessary and sufficient for q | $p \leftrightarrow q$ |

◆ **Memory hook:** the two dangerous ones are "**p if q**", which is  $q \rightarrow p$ , and "**p only if q**", which is  $p \rightarrow q$ . "If" alone introduces the *hypothesis*; "only if" introduces the *conclusion*. They point opposite ways.

## ⚠ TRAP

## "WHENEVER" AND "ONLY IF" REVERSE PEOPLE EVERY YEAR

The 2024/2025 paper uses both, deliberately.

**"I go to the beach whenever it is a sunny summer day."** The "whenever" clause is the *hypothesis*. Let p be "it is a sunny summer day" and q "I go to the beach". This is  $p \rightarrow q$ , not  $q \rightarrow p$ . The sunny day triggers the beach trip, not the other way round.

**"When I stay up late, it is necessary that I sleep until noon."** Here "necessary" marks the *conclusion*. Let p be "I stay up late" and q "I sleep until noon". This is  $p \rightarrow q$ .

Method that never fails: rewrite the sentence as a plain "if ... then ..." **before** you touch symbols. Do that in the margin, then translate. Two extra seconds, and the whole question stops being about English.



Translate each into symbols, with  $p$  = "it is raining",  $q$  = "the match is cancelled".

**(a) "The match is cancelled if it is raining."**

Rewrite: if it is raining, then the match is cancelled. So  $p \rightarrow q$ .

**(b) "The match is cancelled only if it is raining."**

"Only if" marks the conclusion, so rain is required for cancellation. Rewrite: if the match is cancelled, then it is raining. So  $q \rightarrow p$ . Note that this is the reverse of (a), from a sentence that differs by one word.

**(c) "Rain is necessary and sufficient for cancellation."**

Both directions at once:  $p \leftrightarrow q$ .

**(d) "The match is cancelled whenever it rains."**

"Whenever" marks the hypothesis:  $p \rightarrow q$ , the same as (a).

**Now redo it:** translate "passing the exam is necessary for graduating". Answer below the fold of your hand: let  $g$  be graduating and  $e$  be passing, then  $g \rightarrow e$ .

## What the instruction verbs demand

### TEACH

### READ THE VERB BEFORE YOU START WRITING

Marks are allocated to the verb, not to the topic. A perfect definition earns nothing when the question said "prove". This table is drawn from the verbs the five papers actually use.

| VERB                       | WHAT EARNS THE MARKS                                                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Define                     | One precise sentence. No example needed unless asked.                                                                    |
| State                      | The claim itself, in full, with its conditions. Not a worked instance of it.                                             |
| Explain                    | The idea in your own words, plus why it is so. Usually an example is expected.                                           |
| Differentiate between      | A stated point of contrast, both sides, on the same axis. A list of two definitions side by side does not differentiate. |
| Show that / Prove          | An argument that settles every case, not a check of one or two.                                                          |
| Determine whether          | A verdict plus the reason. A bare yes or no scores almost nothing.                                                       |
| Construct / Draw           | The object itself, labelled.                                                                                             |
| Compute / Calculate / Find | The number, with the working that produced it.                                                                           |
| With the aid of examples   | The example is compulsory and carries marks of its own.                                                                  |

The 2025/2026 paper opens with ten marks of "differentiate between the following", five pairs at two marks each. The commonest way to lose half of those ten marks is to write two definitions and stop.

A contrast needs a shared axis. Compare this pair of answers on the two ways of describing a set, from F.1:

**Scores poorly:** "The roster method lists the elements. Set-builder notation uses a rule."

**Scores fully:** "Both name exactly the same collection, and either can describe any set that the other can. They differ in what they make easy. The roster method writes the members out, so it is direct but only practical for a small finite set:  $\{2, 4, 6, 8\}$ . Set-builder states the property that decides membership, so it handles infinite sets and sets defined by a condition:  $\{x \mid x \text{ is even and } 0 < x < 10\}$ . The roster shows you *what* is in; set-builder shows you *why*."

Same facts. The second one answers the verb, because every sentence contrasts the two on a stated axis: what they name, what they make easy, what they show.

## How to lay out an answer

### ● MUST MEMORISE

### THE FOUR-LINE ANSWER

For anything with a number in it, write these four things, in this order, every time.

1. **Given.** The values the question handed you, listed.
2. **Find.** What is being asked, in your own words.
3. **Formula or rule.** Stated before it is used, with what each letter means.
4. **Substitute, work, and answer.** With units or a unit of counting.

This is worth real marks even when your arithmetic goes wrong, because a marker can see the method and award it. An unexplained number cannot be given method marks, however correct it is.

♦ **Memory hook:** **G**iven, **F**ind, **F**ormula, **S**ubstitute. Every worked example in this book opens that way, so you will have copied the habit a hundred times before the exam.

### TEACH

### THREE WORDS THIS COURSE USES PRECISELY

**Definition.** A naming. It cannot be true or false, only used or misused. " $n!$  means the product of every whole number from  $n$  down to 1" is a definition.

**Theorem.** A claim that has been proved. It is true, and something else depends on it. "There is no largest prime number" is a theorem: it is not a naming, it asserts something, and it has been established beyond doubt.

**Proof.** An argument that settles a claim for *every* case it covers. Checking three examples is not a proof, it is evidence. Part Nine builds proofs properly.

An exam that says "prove" and receives three examples gets the marks for three examples, which is usually zero.

## RECALL AND CHECK

1. Translate "q is necessary for p" into symbols.
2. Translate "p only if q" into symbols.
3. A question says "determine whether 51 is prime". You write "No". How many of the 2 marks should you expect?
4. What is the difference between a definition and a theorem?

---

### ANSWERS

1.  $p \rightarrow q$ . The necessary thing is the conclusion.
2.  $p \rightarrow q$ . "Only if" also marks the conclusion.
3. About one, at best. "Determine whether" wants the verdict *and* the reason: exhibit what settles it, for instance "no, because  $51 = 3 \times 17$ ".
4. A definition names something and cannot be true or false. A theorem makes a claim that has been proved true.

## FILL IN THE GAP

1. "p is sufficient for q" in symbols is \_\_\_\_\_.
2. "p if and only if q" in symbols is \_\_\_\_\_.
3. An argument that settles a claim for every case is called a \_\_\_\_\_.
4. The four-line answer runs Given, \_\_\_\_\_, Formula, \_\_\_\_\_.

---

### ANSWERS

1.  $p \rightarrow q$ . 2.  $p \leftrightarrow q$ . 3. proof. 4. Find, Substitute.

## PART ONE

# Logic: The Grammar of Reasoning

*Before you can prove anything, you need a language precise enough that a statement cannot wriggle. Logic is that language, and it is the one part of this course the examiner returns to on every single paper.*

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ | $p \rightarrow q$ | $p \leftrightarrow q$ |
|-----|-----|--------------|------------|-------------------|-----------------------|
| T   | T   | T            | T          | T                 | T                     |
| T   | F   | F            | T          | F                 | F                     |
| F   | T   | F            | T          | T                 | F                     |
| F   | F   | F            | F          | T                 | T                     |

# Propositions and Truth Tables

Everything in this part is built from one idea: a sentence that is definitely true or definitely false, and nothing in between.

By the end of this section you will be able to say whether a sentence is a proposition, write an English sentence in symbols, build a truth table for any compound proposition however tangled, and name a proposition as a tautology, a contradiction or a contingency.

## What a proposition is

### TEACH

### THE STARTING POINT

A **proposition** is a declarative sentence that is either true or false, but not both.

Three phrases in that definition are doing work. **Declarative** means it declares something, so it is not a question or a command. **Either true or false** means it has a definite truth value, even if nobody present knows which. **Not both** means it cannot be true in one breath and false in the next.

| SENTENCE                | PROPOSITION? | WHY                                                                                                                           |
|-------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| Lagos is in Nigeria.    | Yes, true    | It declares something, and that something is the case.                                                                        |
| $2 + 2 = 5$             | Yes, false   | Being false does not stop it being a proposition. It only needs a truth value.                                                |
| What time is the exam?  | No           | A question. It declares nothing, so there is nothing to be true.                                                              |
| Close the door.         | No           | A command, for the same reason.                                                                                               |
| $x + 1 = 3$             | Not yet      | Its truth depends on $x$ . Until $x$ is fixed it has no truth value. Section 1.4 turns sentences like this into propositions. |
| This sentence is false. | No           | If it is true it is false, and if it is false it is true. It breaks the "not both" rule.                                      |

### ● MUST MEMORISE

### DEFINITION

A **proposition** is a declarative sentence that is either true or false, but not both.

Propositions are written with letters, usually  $p$ ,  $q$ ,  $r$ ,  $s$ . A letter standing for a proposition is a **propositional variable**. The two truth values are written **T** for true and **F** for false.

◆ **Memory hook:** if you can sensibly answer "is that true?", it is a proposition. You cannot answer that about a question or an order.

## The connectives

### TEACH

### BUILDING BIGGER STATEMENTS

Single propositions are not very interesting. We join them with **connectives** to make **compound propositions**, and the whole of this part is the study of how the truth of the pieces fixes the truth of the whole.

Each connective is defined by a **truth table**: a table listing every possible combination of truth values for the pieces, with the resulting value for the whole. A truth table is not an explanation of a connective, it *is* the connective. There is nothing more to know about "and" in logic than the four rows of its table.

### MUST MEMORISE

### THE CONNECTIVES

| NAME          | SYMBOL                | READ AS             | TRUE EXACTLY WHEN              |
|---------------|-----------------------|---------------------|--------------------------------|
| Negation      | $\neg p$              | not p               | p is false                     |
| Conjunction   | $p \wedge q$          | p and q             | both are true                  |
| Disjunction   | $p \vee q$            | p or q              | at least one is true           |
| Exclusive or  | $p \oplus q$          | p or q but not both | exactly one is true            |
| Implication   | $p \rightarrow q$     | if p then q         | p is false, or q is true       |
| Biconditional | $p \leftrightarrow q$ | p if and only if q  | both have the same truth value |

◆ **Memory hook:**  $\wedge$  points up like the A in "and";  $\vee$  is the other way up, for "or".  $\rightarrow$  points from the assumption to the conclusion.  $\leftrightarrow$  points both ways because it works both ways.

## The tables themselves

Here are all six side by side. Learn to reproduce this from memory, because almost every logic question on every paper starts by using it.

| P | Q | $\neg P$ | $P \wedge Q$ | $P \vee Q$ | $P \oplus Q$ | $P \rightarrow Q$ | $P \leftrightarrow Q$ |
|---|---|----------|--------------|------------|--------------|-------------------|-----------------------|
| T | T | F        | T            | T          | F            | T                 | T                     |
| T | F | F        | F            | T          | T            | F                 | F                     |
| F | T | T        | F            | T          | T            | T                 | F                     |
| F | F | T        | F            | F          | F            | T                 | T                     |

Look at row 3 of  $p \rightarrow q$ :  $p$  is false,  $q$  is true, and the implication is **true**. Now row 4: both false, and it is **true** again.

Students reject this because it feels wrong. "If I pass, I will celebrate" seems to say nothing about what happens when I fail. Exactly: it says nothing, and a promise that says nothing about a case has not been broken in that case. An implication is a promise, and a promise is false only when it is **broken**, which needs  $p$  true and  $q$  false.

**$p \rightarrow q$  is false in exactly one case:  $p$  true,  $q$  false.** Learn that as a single fact and the other three rows follow.

## TEACH

## WHY "OR" NEEDS CARE

English uses "or" in two ways. "Tea or coffee?" usually means one or the other but not both. "Applicants must have a degree or five years of experience" means at least one, and having both is fine.

Logic splits these. **Disjunction** ( $\vee$ ) is the inclusive one: true when at least one part is true, including when both are. **Exclusive or** ( $\oplus$ ) is the other: true when exactly one is true. Compare the rows where  $p$  and  $q$  are both true:  $\vee$  gives T,  $\oplus$  gives F. That single row is the whole difference.

## Building a table for a compound proposition

## TEACH

## THE METHOD, ONCE, PROPERLY

Any compound proposition, however tangled, is handled the same way.

**Step 1. Count the letters.** With  $n$  distinct propositional variables the table has  $2^n$  rows: 2 letters give 4 rows, 3 give 8, 4 give 16.

**Step 2. Fill the letter columns mechanically.** Under the first letter write T for the top half of the rows and F for the bottom half; under the next, alternate in blocks half that size; keep halving. Done this way you cannot miss a combination or write one twice.

**Step 3. Work outward from the innermost bracket.** Give every intermediate expression its own column. Do not jump to the final answer in one step: you cannot check a step you did not write down, and the marks are for the working.

**Step 4. Read the final column.** All T is a tautology, all F a contradiction, a mixture a contingency.

A compound proposition is a

**tautology** if it is true for every assignment of truth values, that is, the final column is all T;

**contradiction** if it is false for every assignment, all F;

**contingency** if it is neither, true for some assignments and false for others.

◆ **Memory hook:** a tautology tells you nothing because it is always true, a contradiction is always false, and a contingency depends, which is what "contingent" means in ordinary English.

## Now the real questions

The 2020/2021 paper asks for the definitions and the tables together. Read what it actually says, including its own spelling.

AS PRINTED

2020/2021 · Q5(A)

5. (a) Define and draw the truth table for each of the following logic proposition terminologies:

- (i) Negation    (ii) Conjunction    (iii) Exclusive or    (iv) Implication  
(v) Tautology    (vi) Contingency

(6marks)

### NOTE ON THE PAPER

The paper prints "Contingency". The word is **CONTINGENCY**. Write the correct spelling in your answer; nobody has ever lost a mark for spelling the examiner's typo correctly. Notice too that (v) and (vi) are not connectives at all, so "draw the truth table for a tautology" does not quite make sense as written. What is wanted is an example whose table is all T, and one whose table is mixed.

### BREAK IT DOWN

**GIVEN** Six terms: four connectives and two names for a shape of final column.

**ASKED** Two things for each. **Define** it in words, and **draw** its truth table. Half the marks are in the words, so a page of tables with no definitions scores about half.

**MARKS SAY** 6 marks over six items is one mark each: one sentence and one small table apiece. Do not write an essay on implication.

**TEACHES** This section, 1.1.

**TRAP** Giving the table for  $\vee$  when asked for exclusive or. They differ in exactly one row, and that row is the answer.



**(i) Negation.** The negation of  $p$ , written  $\neg p$ , is the proposition "it is not the case that  $p$ ". It reverses the truth value.

| P | $\neg P$ |
|---|----------|
| T | F        |
| F | T        |

**(ii) Conjunction.** The conjunction of  $p$  and  $q$ , written  $p \wedge q$ , is " $p$  and  $q$ ". It is true only when both are true.

| P | Q | $P \wedge Q$ |
|---|---|--------------|
| T | T | T            |
| T | F | F            |
| F | T | F            |
| F | F | F            |

**(iii) Exclusive or.** The exclusive or of  $p$  and  $q$ , written  $p \oplus q$ , is true when exactly one of them is true and false when they agree.

| P | Q | $P \oplus Q$ |
|---|---|--------------|
| T | T | F            |
| T | F | T            |
| F | T | T            |
| F | F | F            |

**(iv) Implication.** The implication  $p \rightarrow q$ , read "if  $p$  then  $q$ ", is false only when  $p$  is true and  $q$  is false. Here  $p$  is the **hypothesis** and  $q$  the **conclusion**.

| P | Q | $P \rightarrow Q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

**(v) Tautology.** A compound proposition that is true for every assignment of truth values to its variables. The standard example is  $p \vee \neg p$ , "either it is raining or it is not".

| P | $\neg P$ | $P \vee \neg P$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |

**(vi) Contingency.** A compound proposition that is true for some assignments and false for others, so it is neither a tautology nor a contradiction. Any ordinary compound will

do, for instance  $p \wedge q$ , whose column above reads T, F, F, F.

For completeness, the third member of that family: a **contradiction** is false for every assignment, for example  $p \wedge \neg p$ .

| P | $\neg P$ | $P \wedge \neg P$ |
|---|----------|-------------------|
| T | F        | F                 |
| F | T        | F                 |

**Now redo it:** cover this answer and write out all six definitions and tables on blank paper. Time yourself; you should be inside six minutes.

## A harder table, from the recent papers

The same skill on propositions with more structure. This one is worth quoting carefully, because part of it appears on two different papers.

AS PRINTED

2024/2025 · Q1(B)

b. Construct a truth table for each of these compound propositions.

- i.  $p \leftrightarrow \neg p$       ii.  $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$   
 iii.  $(p \oplus q) \vee (p \oplus \neg q)$       iv.  $((p \rightarrow q) \rightarrow r) \rightarrow s$

4 marks

AS PRINTED

2021/2022 · Q5(D)

(d) Construct a truth table for each of the following compound propositions

- (i)  $(p \leftrightarrow q) \ominus (p \leftrightarrow \neg q)$   
 (ii)  $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

(5 marks)

(5 marks)

### NOTE ON THE 2021/2022 VERSION

Part (i) prints the connective as  $\ominus$ , a circled *minus*. There is no standard logical connective with that symbol. In context it can only mean **EXCLUSIVE OR**, normally written  $\oplus$ , and the question is unanswerable on any other reading. Quote the glyph as printed, say in one line what it has to mean, and answer on that reading.

Note also that this paper prints "(5 marks)" with a space here, while every other mark on it is closed up as "5marks". Part (ii) is word for word the same proposition as 2024/2025 Q1 b(ii) above.

### NOTE ON THE 2024/2025 VERSION

Its part (ii) is word for word the 2021/2022 part (ii), so one table answers both papers and that one table is worth 5 marks on one paper and a share of 4 on the other. Note also that part (iv) has **FOUR** letters, so its table needs 16 rows: four times the work of the others for the same share of the marks. In the hall, do i, ii and iii first.

**BREAK IT DOWN**

- GIVEN** Four compound propositions of increasing size: one letter, two, two, then four.
- ASKED** **Construct** a truth table. Not "simplify", not "say whether it is a tautology". Build the table; any conclusion you then draw from it is a bonus that shows you understood it.
- MARKS SAY** 4 marks for four tables is one mark each, and one of them is 16 rows. So the marks are for method, not endurance: set the columns out cleanly and the mark is yours.
- TEACHES** This section, 1.1, plus 1.3 for the shortcut on (ii).
- TRAP** Rushing (iv) and losing a row. Fill the letter columns by halving, as in Step 2, and you cannot go wrong.

**WORKED EXAMPLE****PQ 24/25 Q1(B) I AND II**

**(i)  $p \leftrightarrow \neg p$ .** One letter, so 2 rows. The biconditional is true when both sides agree, and  $p$  never agrees with  $\neg p$ .

| P | $\neg P$ | $P \leftrightarrow \neg P$ |
|---|----------|----------------------------|
| T | F        | F                          |
| F | T        | F                          |

Every row is F, so  **$p \leftrightarrow \neg p$  is a contradiction**. Say so: naming it shows you understood the table you just drew.

**(ii)  $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$ .** Two letters, so 4 rows. Build the pieces first:  $\neg p$ , then  $q \rightarrow \neg p$ , then  $p \leftrightarrow q$ , then join them.

| P | Q | $\neg P$ | $Q \rightarrow \neg P$ | $P \leftrightarrow Q$ | WHOLE |
|---|---|----------|------------------------|-----------------------|-------|
| T | T | F        | F                      | T                     | F     |
| T | F | F        | T                      | F                     | F     |
| F | T | T        | T                      | F                     | F     |
| F | F | T        | T                      | T                     | T     |

Taking the rows one at a time, remembering that  $\rightarrow$  is false only for T then F, and  $\leftrightarrow$  is true only when its two sides match:

Row 1:  $q$  is T and  $\neg p$  is F, so  $q \rightarrow \neg p$  is F.  $p \leftrightarrow q$  with both T is T. F against T gives F.

Row 2:  $q$  is F, so  $q \rightarrow \neg p$  is T whatever follows it.  $p \leftrightarrow q$  with T and F is F. T against F gives F.

Row 3:  $q$  is T and  $\neg p$  is T, so T.  $p \leftrightarrow q$  with F and T is F. T against F gives F.

Row 4:  $q$  is F so the implication is T.  $p \leftrightarrow q$  with both F is T. T against T gives **T**.

Three F rows and one T row, so this is a **contingency**. It is true exactly when  $p$  and  $q$  are both false.

**Now redo it:** parts (iii) and (iv) are yours. For (iii) expect a surprise in the final column, and say what it means. For (iv) remember 16 rows, and fill the  $p$  column as 8 T then 8 F.

## RECALL AND CHECK

1. In which single case is  $p \rightarrow q$  false?
2. How many rows does a truth table with 5 letters have?
3. In which row do  $p \vee q$  and  $p \oplus q$  disagree?
4. Is " $x > 3$ " a proposition?

### ANSWERS

1. When  $p$  is true and  $q$  is false, and in no other.
2.  $2^5 = 32$ .
3. When both are true:  $\vee$  gives T,  $\oplus$  gives F.
4. No. Its truth depends on  $x$ , so it has no truth value until  $x$  is fixed. It is a predicate, and Section 1.4 shows how to turn it into a proposition.

## FILL IN THE GAP

1. A declarative sentence that is either true or false but not both is called a \_\_\_\_\_.
2. The connective  $\wedge$  is called \_\_\_\_\_ and is true only when \_\_\_\_\_ parts are true.
3. A compound proposition whose final column is entirely T is a \_\_\_\_\_, entirely F a \_\_\_\_\_, and mixed a \_\_\_\_\_.
4. With  $n$  distinct variables a truth table has \_\_\_\_\_ rows.

### ANSWERS

1. proposition. 2. conjunction, both. 3. tautology, contradiction, contingency. 4.  $2^n$ .

## PART ONE · 1.2

# Implication, Converse, Inverse, Contrapositive

*One implication has three relatives. Exactly one of them says the same thing as the original, and every paper in our set asks you which.*

*By the end of this section you will be able to form the converse, inverse and contrapositive of any conditional statement, in English or in symbols, and prove which one preserves truth value.*

## The three relatives

### TEACH

### BUILT FROM ONE IMPLICATION

Start with a conditional statement  $p \rightarrow q$ . There are exactly three ways to rearrange it, and each has a name.

The **converse** swaps the two parts:  $q \rightarrow p$ .

The **inverse** negates both parts, keeping the order:  $\neg p \rightarrow \neg q$ .

The **contrapositive** does both, swap and negate:  $\neg q \rightarrow \neg p$ .

Watch what happens to a concrete case. Let  $p$  be "it is a dog" and  $q$  "it is an animal", so  $p \rightarrow q$  reads "if it is a dog then it is an animal", which is true.

| NAME           | FORM                        | THE DOG EXAMPLE                          | TRUE?                                 |
|----------------|-----------------------------|------------------------------------------|---------------------------------------|
| Original       | $p \rightarrow q$           | If it is a dog, it is an animal.         | yes                                   |
| Converse       | $q \rightarrow p$           | If it is an animal, it is a dog.         | <b>no</b> , a cat is a counterexample |
| Inverse        | $\neg p \rightarrow \neg q$ | If it is not a dog, it is not an animal. | <b>no</b> , the same cat              |
| Contrapositive | $\neg q \rightarrow \neg p$ | If it is not an animal, it is not a dog. | yes                                   |

### ● MUST MEMORISE

### THE ONE THAT MATTERS

For any conditional statement  $p \rightarrow q$ :

**converse** =  $q \rightarrow p$

**inverse** =  $\neg p \rightarrow \neg q$

**contrapositive** =  $\neg q \rightarrow \neg p$

**Only the contrapositive is logically equivalent to the original.** The converse and the inverse are equivalent to each other, and neither follows from the original.

◆ **Memory hook:** Converse **C**hanges order. Inverse **I**nverts (negates) both. Contrapositive does both, and doing both is the only safe operation. In the dog example: knowing it is not an animal really does tell you it is not a dog.

### TEACH

### PROVING IT RATHER THAN ASSERTING IT

The 2024/2025 paper does not ask you to state this, it asks you to **prove** it, for six marks. A truth table is the proof.

| P | Q | $\neg P$ | $\neg Q$ | $P \rightarrow Q$ | $Q \rightarrow P$ | $\neg P \rightarrow \neg Q$ | $\neg Q \rightarrow \neg P$ |
|---|---|----------|----------|-------------------|-------------------|-----------------------------|-----------------------------|
| T | T | F        | F        | T                 | T                 | T                           | T                           |
| T | F | F        | T        | F                 | T                 | T                           | F                           |
| F | T | T        | F        | T                 | F                 | F                           | T                           |
| F | F | T        | T        | T                 | T                 | T                           | T                           |

Read the columns, not the rows. The column for  $p \rightarrow q$  is T, F, T, T. The column for the contrapositive  $\neg q \rightarrow \neg p$  is T, F, T, T: **identical**, so they are equivalent.

The converse column is T, T, F, T and the inverse column is T, T, F, T: identical to each other, and different from the original in rows 2 and 3. So the converse and inverse are equivalent to each other and neither is equivalent to  $p \rightarrow q$ . That single table proves every claim in this section at once.

## The real question

AS PRINTED

2024/2025 · Q1(C)

c. Consider the conditional statement:

"The home team wins whenever it is raining."

Let  $p$  represent: "It is raining."

Let  $q$  represent: "The home team wins."

- i. Using this, find the following: the contrapositive, the converse, and the inverse of the statement. **3 marks**
- ii. Prove that only the contrapositive of a conditional statement  $p \rightarrow q$  has the same truth value as the original conditional statement. Use the statement from c(i) in your explanation. **6 marks**

### NOTE ON THE PAPER

The paper hands you the letters, which is a gift:  $p$  is the *rain*, not the win. Read that assignment twice before writing, because the sentence puts the win first and the rain second, while the letters run the other way.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | The sentence "The home team wins whenever it is raining.", with $p$ = it is raining and $q$ = the home team wins.                                                                                             |
| <b>ASKED</b>     | Part i: <b>find</b> three rearrangements, in English. Part ii: <b>prove</b> a general claim, then illustrate it with this sentence. "Prove" means a truth table or a full argument, not an assertion.         |
| <b>MARKS SAY</b> | 3 marks for three statements is one each, so short English sentences suffice. 6 marks for the proof is the bulk of the question: that is a full table with the columns compared and a conclusion written out. |
| <b>TEACHES</b>   | This section, 1.2, and 1.1 for the table method.                                                                                                                                                              |
| <b>TRAP</b>      | "Whenever" marks the hypothesis, so the statement is $p \rightarrow q$ (rain implies win), not $q \rightarrow p$ . Getting that backwards inverts all three answers in part i and costs the lot.              |

**GIVEN**  $p$  = "It is raining",  $q$  = "The home team wins", and the sentence "The home team wins whenever it is raining."

**FIND** The contrapositive, converse and inverse, then a proof that only the contrapositive preserves truth value.

**RULE** converse  $q \rightarrow p$ , inverse  $\neg p \rightarrow \neg q$ , contrapositive  $\neg q \rightarrow \neg p$ .

**Step 1. Get the direction right.** "The home team wins whenever it is raining" rewrites as "if it is raining, then the home team wins". So the statement is  $p \rightarrow q$ .

**Step 2 (part i). Apply each definition.**

| NAME           | SYMBOLS                     | IN ENGLISH                                             |
|----------------|-----------------------------|--------------------------------------------------------|
| Contrapositive | $\neg q \rightarrow \neg p$ | If the home team does not win, then it is not raining. |
| Converse       | $q \rightarrow p$           | If the home team wins, then it is raining.             |
| Inverse        | $\neg p \rightarrow \neg q$ | If it is not raining, then the home team does not win. |

**Step 3 (part ii). Prove it with a truth table.** Build every column and compare them against  $p \rightarrow q$ .

| P | Q | $\neg P$ | $\neg Q$ | $P \rightarrow Q$ | $\neg Q \rightarrow \neg P$ | $Q \rightarrow P$ | $\neg P \rightarrow \neg Q$ |
|---|---|----------|----------|-------------------|-----------------------------|-------------------|-----------------------------|
| T | T | F        | F        | T                 | T                           | T                 | T                           |
| T | F | F        | T        | F                 | F                           | T                 | T                           |
| F | T | T        | F        | T                 | T                           | F                 | F                           |
| F | F | T        | T        | T                 | T                           | T                 | T                           |

The  $p \rightarrow q$  column reads T, F, T, T. The contrapositive column reads T, F, T, T. They agree in every row, so  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ . The converse reads T, T, F, T and disagrees with the original in rows 2 and 3, and the inverse matches the converse rather than the original.

**Step 4. Say it in words, using their sentence.** Suppose it rains and the home team loses. That single situation makes the original false, because the promise was broken. It also makes the contrapositive false: the team did not win, yet it was raining. The two statements fail together and hold together, which is what equivalence means.

Now the converse. Suppose the home team wins on a dry day. The original is untouched, since it only ever promised something about rainy days, so it stays true. But the converse, "if the home team wins then it is raining", is false. One statement true, the other false, so they are not equivalent. The inverse fails on that same dry, winning day.

**Conclusion.** Of the three rearrangements only the contrapositive always carries the same truth value as the original.

**Now redo it:** cover this and do the whole question for "I go to the beach whenever it is a sunny summer day", which is 2024/2025 Q1(a)(ii).

## The rest of that question

AS PRINTED

2024/2025 · Q1(A)

a. State the converse, contrapositive, and inverse of each of these conditional statements.

- i. If it snows tonight, then I will stay at home.
- ii. I go to the beach whenever it is a sunny summer day.
- iii. When I stay up late, it is necessary that I sleep until noon.

4.5 marks

### BREAK IT DOWN

**GIVEN** Three English sentences, each hiding an implication in a different disguise: a plain "if ... then", a "whenever", and a "necessary that".

**ASKED** **State** three rearrangements of each, so nine sentences in all. English is fine; symbols are not required but help you avoid errors.

**MARKS SAY** 4.5 marks over nine statements is half a mark each. Write them as short sentences and move on; there are no marks for commentary here.

**TEACHES** This section for the three forms, F.2 for decoding "whenever" and "necessary".

**TRAP** All three sentences put the pieces in a different order from  $p \rightarrow q$ . Identify  $p$  and  $q$  explicitly, in the margin, before writing a single answer.

### WORKED EXAMPLE

PQ 24/25 Q1(A)

#### (i) "If it snows tonight, then I will stay at home."

Already in standard form.  $p$  = it snows tonight,  $q$  = I will stay at home.

**Converse:** If I stay at home tonight, then it will snow.

**Inverse:** If it does not snow tonight, then I will not stay at home.

**Contrapositive:** If I do not stay at home tonight, then it will not snow.

#### (ii) "I go to the beach whenever it is a sunny summer day."

"Whenever" marks the hypothesis.  $p$  = it is a sunny summer day,  $q$  = I go to the beach.

Rewritten: if it is a sunny summer day, then I go to the beach.

**Converse:** If I go to the beach, then it is a sunny summer day.

**Inverse:** If it is not a sunny summer day, then I do not go to the beach.

**Contrapositive:** If I do not go to the beach, then it is not a sunny summer day.

#### (iii) "When I stay up late, it is necessary that I sleep until noon."

"When" marks the hypothesis and "it is necessary that" marks the conclusion.  $p$  = I stay up late,  $q$  = I sleep until noon. Rewritten: if I stay up late, then I sleep until noon.

**Converse:** If I sleep until noon, then I stayed up late.

**Inverse:** If I do not stay up late, then I do not sleep until noon.

**Contrapositive:** If I do not sleep until noon, then I did not stay up late.

**Now redo it:** nine sentences, blank paper, four minutes. If any of yours reverses the roles of  $p$  and  $q$ , go back to F.2 and re-read the phrase table.



The negation of "I will stay at home" is "I will not stay at home". It is **not** "I will go out", however natural that sounds. Staying at home and going out might not exhaust the possibilities, and logic only knows what you wrote down.

Similarly, the negation of " $x > 3$ " is " $x \leq 3$ ", not " $x < 3$ ". Dropping the equality case is the commonest numeric slip on this topic.

#### RECALL AND CHECK

1. Which one of the three relatives is equivalent to the original?
2. Write the contrapositive of "if  $n$  is even then  $n^2$  is even".
3. The converse is equivalent to which other statement?
4. What is the negation of " $x \geq 5$ "?

#### ANSWERS

1. The contrapositive.
2. If  $n^2$  is not even, then  $n$  is not even.
3. The inverse. Both are T, T, F, T.
4.  $x < 5$ .

#### FILL IN THE GAP

1. The converse of  $p \rightarrow q$  is \_\_\_\_\_.
2. The inverse of  $p \rightarrow q$  is \_\_\_\_\_.
3. The contrapositive of  $p \rightarrow q$  is \_\_\_\_\_.
4. Only the \_\_\_\_\_ has the same truth value as the original.

#### ANSWERS

1.  $q \rightarrow p$ . 2.  $\neg p \rightarrow \neg q$ . 3.  $\neg q \rightarrow \neg p$ . 4. contrapositive.

#### PART ONE · 1.3

## Logical Equivalence and the Laws

*Two statements that always agree can be swapped freely. That is what lets you simplify a condition, redesign a circuit, or rewrite a database query without changing what it does.*

*By the end of this section you will be able to test two propositions for equivalence with a truth table, state De Morgan's laws properly as laws, and use the standard laws to transform one expression into another.*

## What equivalence means

### ● MUST MEMORISE

### DEFINITION

Two compound propositions are **logically equivalent** when they have the same truth value for every possible assignment of truth values to their variables. We write  $P \equiv Q$ . Equivalently:  $P \equiv Q$  exactly when the biconditional  $P \leftrightarrow Q$  is a **tautology**.

◆ **Memory hook:**  $\equiv$  is the equals sign with an extra bar, meaning "equal in every row, not just this one". It is a claim *about* two propositions, so it is not itself a connective.

### TEACH

### HOW TO SHOW TWO THINGS ARE EQUIVALENT

There are two accepted methods, and the question usually tells you which it wants.

**Method 1: truth table.** Build a column for each side and check they match row for row. Always works, never elegant, and cannot fail through a wrong step. Use it when the question says "show that" or "construct a truth table".

**Method 2: a chain of known laws.** Rewrite one side step by step until it becomes the other, naming the law at every step. Shorter when it works, and it is what "using logical equivalences" means when a paper says so, as 2024/2025 Q3(c) does.

The naming is not decoration. In a law-chain proof the marks are for the justifications, so a chain of correct lines with no names attached scores poorly.

## De Morgan's laws

### DE MORGAN'S LAWS

**The claim.** Negating a conjunction turns it into the disjunction of the negations, and negating a disjunction turns it into the conjunction of the negations. In words: *not (both) is the same as (not one) or (not the other)*, and *not (either) is the same as (not one) and (not the other)*.

**In symbols.**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

**Conditions.** None beyond  $p$  and  $q$  being propositions. The laws hold universally, which is what makes them safe to apply anywhere.

**The limiting case, and why the wrong version survives.** The common error is to write  $\neg(p \wedge q)$  as  $\neg p \wedge \neg q$ , keeping the connective instead of flipping it. Those two agree in exactly the rows where  $p$  and  $q$  have the **same** truth value, and differ in the other two. So a student who checks only the TT and FF rows sees the wrong law confirmed. Check a row where  $p$  and  $q$  disagree, and the error appears at once.

**The same laws for sets.** Replacing "and" by intersection, "or" by union and "not" by complement gives the set forms, which Part Two uses:

the complement of  $(A \cap B) = (\text{complement of } A) \cup (\text{complement of } B)$

the complement of  $(A \cup B) = (\text{complement of } A) \cap (\text{complement of } B)$

**Consequence.** A negation can always be pushed inwards until it sits on single variables. That is why every Boolean expression can be simplified mechanically, and why a compiler can re-write your `if` conditions.

Say it about breakfast. "It is not the case that I had both tea and toast" means "I did not have tea, or I did not have toast" (at least one of them was missing). It does **not** mean I had neither.

"It is not the case that I had either tea or toast" is the stronger one: it means I had no tea **and** no toast.

So negation flips the connective as it moves inside. Forgetting to flip it is the mistake De Morgan exists to prevent.

| P | Q | $P \wedge Q$ | $\neg(P \wedge Q)$ | $\neg P$ | $\neg Q$ | $\neg P \vee \neg Q$ |
|---|---|--------------|--------------------|----------|----------|----------------------|
| T | T | T            | F                  | F        | F        | F                    |
| T | F | F            | T                  | F        | T        | T                    |
| F | T | F            | T                  | T        | F        | T                    |
| F | F | F            | T                  | T        | T        | T                    |

Columns four and seven are both F, T, T, T. The law is proved.

## The full list of laws

### ● MUST MEMORISE

### THE LAWS OF LOGIC

Here **T** means a tautology (always true) and **F** a contradiction (always false).

| NAME            | LAW                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------|
| Identity        | $p \wedge T \equiv p$ $p \vee F \equiv p$                                                 |
| Domination      | $p \vee T \equiv T$ $p \wedge F \equiv F$                                                 |
| Idempotent      | $p \vee p \equiv p$ $p \wedge p \equiv p$                                                 |
| Double negation | $\neg(\neg p) \equiv p$                                                                   |
| Commutative     | $p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$                                 |
| Associative     | $(p \vee q) \vee r \equiv p \vee (q \vee r)$                                              |
| Distributive    | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$                                 |
| De Morgan       | $\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$ |
| Absorption      | $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$                             |
| Negation        | $p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$                                       |
| Implication     | $p \rightarrow q \equiv \neg p \vee q$                                                    |
| Contrapositive  | $p \rightarrow q \equiv \neg q \rightarrow \neg p$                                        |
| Biconditional   | $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$                   |

♦ **Memory hook:** the one that unlocks most problems is **implication**:  $p \rightarrow q \equiv \neg p \vee q$ . An arrow is hard to manipulate; an "or" is easy. Turn every arrow into an or as your first move, and most chains fall out.

## A real question

### AS PRINTED

2020/2021 · Q4(C)

(c) Show that the following are logically equivalent

(i)  $(p \rightarrow q) \wedge (p \rightarrow r)$  and  $p \rightarrow (q \wedge r)$ .

**(3marks)**

(ii)  $p \leftrightarrow q$  and  $(p \wedge q) \vee (\neg p \wedge \neg q)$

**(3marks)**

**BREAK IT DOWN**

- GIVEN** Two pairs of compound propositions, the first on three letters and the second on two.
- ASKED** **Show that** each pair is equivalent. A truth table is a complete answer, and is safer than a law chain under time pressure.
- MARKS SAY** 3 marks each. Part (i) has three letters so its table is 8 rows; part (ii) has two letters and 4 rows. Both are quick if the columns are set out before you start filling.
- TEACHES** This section for equivalence, 1.1 for building the table.
- TRAP** Part (i) needs 8 rows, and the temptation is to do 4. Count the distinct letters, not the brackets.

**WORKED EXAMPLE****PQ 20/21 Q4(C)(I)****GIVEN**  $(p \rightarrow q) \wedge (p \rightarrow r)$  and  $p \rightarrow (q \wedge r)$ .**FIND** Whether the two columns agree in every row.**RULE** Three letters, so  $2^3 = 8$  rows.

| P | Q | R | $P \rightarrow Q$ | $P \rightarrow R$ | $(P \rightarrow Q) \wedge (P \rightarrow R)$ | $Q \wedge R$ | $P \rightarrow (Q \wedge R)$ |
|---|---|---|-------------------|-------------------|----------------------------------------------|--------------|------------------------------|
| T | T | T | T                 | T                 | T                                            | T            | T                            |
| T | T | F | T                 | F                 | F                                            | F            | F                            |
| T | F | T | F                 | T                 | F                                            | F            | F                            |
| T | F | F | F                 | F                 | F                                            | F            | F                            |
| F | T | T | T                 | T                 | T                                            | T            | T                            |
| F | T | F | T                 | T                 | T                                            | F            | T                            |
| F | F | T | T                 | T                 | T                                            | F            | T                            |
| F | F | F | T                 | T                 | T                                            | F            | T                            |

Column 6 reads T, F, F, F, T, T, T, T. Column 8 reads T, F, F, F, T, T, T, T. Identical in all eight rows, so  $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ .

Sanity check in words: promising  $q$  and promising  $r$  separately, on the same condition  $p$ , is the same as promising both together on that condition. It should feel obvious, and the table is why it is certain.

**GIVEN**  $p \leftrightarrow q$  and  $(p \wedge q) \vee (\neg p \wedge \neg q)$ .

**FIND** Whether the two agree in every row.

**RULE** Two letters, 4 rows.

| P | Q | $P \leftrightarrow Q$ | $P \wedge Q$ | $\neg P$ | $\neg Q$ | $\neg P \wedge \neg Q$ | $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ |
|---|---|-----------------------|--------------|----------|----------|------------------------|--------------------------------------------|
| T | T | T                     | T            | F        | F        | F                      | T                                          |
| T | F | F                     | F            | F        | T        | F                      | F                                          |
| F | T | F                     | F            | T        | F        | F                      | F                                          |
| F | F | T                     | F            | T        | T        | T                      | T                                          |

Column 3 is T, F, F, T and column 8 is T, F, F, T. Equivalent.

In words: "p and q agree" is the same as "either both are true, or both are false". That is exactly what the biconditional means, so the right-hand side is really the definition written out.

**Now redo it:** on blank paper, show that  $p \rightarrow q \equiv \neg p \vee q$  with a truth table. It is the law you will lean on most.

⚠ TRAP

EQUIVALENT IS NOT "SIMILAR"

Equivalence needs agreement in **every** row. Two propositions that match in three rows out of four are not "nearly equivalent", they are simply different, and the one row where they differ is the counterexample you should quote.

When a question asks you to show two things are **not** equivalent, do not build the whole table and say "they differ". Name the row: "take p true and q false; then the first is F and the second is T". One row, fully specified, is a complete disproof.

RECALL AND CHECK

1. State both of De Morgan's laws.
2. Rewrite  $p \rightarrow q$  without using the arrow.
3. Simplify  $p \vee (p \wedge q)$ .
4. How many rows are needed to test an equivalence on four letters?

ANSWERS

1.  $\neg(p \wedge q) \equiv \neg p \vee \neg q$  and  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ .
2.  $\neg p \vee q$ .
3. p, by the absorption law.
4.  $2^4 = 16$ .

#### FILL IN THE GAP

1.  $P \equiv Q$  means the two have the same truth value for \_\_\_\_\_ assignment.
2.  $\neg(p \vee q) \equiv$  \_\_\_\_\_.
3. The implication law says  $p \rightarrow q \equiv$  \_\_\_\_\_.
4.  $p \wedge \neg p \equiv$  \_\_\_\_\_.

#### ANSWERS

1. every. 2.  $\neg p \wedge \neg q$ . 3.  $\neg p \vee q$ . 4. F (a contradiction).

#### PART ONE · 1.4

## Predicate Logic and Quantifiers

*Propositional logic cannot say "every student has a laptop" without inventing a letter for each student. Predicate logic can, and that is the whole reason it exists.*

*By the end of this section you will be able to say exactly how predicate logic differs from propositional logic, use both quantifiers correctly, translate an English sentence with "some", "all", "no" or "every" into symbols, negate a quantified statement, and decide the truth of a quantified statement over the real numbers.*

### The limit of propositional logic

#### TEACH

#### WHY WE NEED SOMETHING MORE

In Part One so far,  $p$  was an entire sentence with a fixed truth value. That is fine for "it is raining", but it cannot see inside a sentence.

Consider "every student in this class has a laptop". With propositional letters you would need one letter per student, and you would still have no way to say "every". Worse, " $x + 1 = 3$ " cannot be a proposition at all, because its truth depends on  $x$ .

**Predicate logic** fixes both problems by splitting a sentence into a **subject** and a **property**, and then adding words for "all" and "some".

A **predicate** is a statement containing one or more variables, which becomes a proposition once the variables are given values. We write  $P(x)$ , read "P of x".

If  $P(x)$  is " $x > 3$ ", then  $P(5)$  is " $5 > 3$ ", which is true, and  $P(1)$  is " $1 > 3$ ", which is false.  $P(x)$  on its own is neither.

| AXIS                         | PROPOSITIONAL LOGIC             | PREDICATE LOGIC                                        |
|------------------------------|---------------------------------|--------------------------------------------------------|
| Smallest unit                | a whole sentence, $p$           | a predicate with variables, $P(x)$                     |
| Can see inside a sentence    | no                              | yes, subject and property are separate                 |
| Quantifiers                  | none                            | $\forall$ and $\exists$                                |
| Can express "all" and "some" | no                              | yes                                                    |
| Example                      | $p = \text{"Ada has a laptop"}$ | $L(x) = \text{"x has a laptop"}$ , so $\forall x L(x)$ |

◆ **Memory hook:** propositional logic treats a sentence as a sealed box. Predicate logic opens the box and finds a subject and a property inside.



## The two quantifiers

### ● MUST MEMORISE

### UNIVERSAL AND EXISTENTIAL

The **universal quantifier**  $\forall$ , read "for all" or "for every".  $\forall x P(x)$  is true exactly when  $P(x)$  is true for *every*  $x$  in the domain.

The **existential quantifier**  $\exists$ , read "there exists" or "for some".  $\exists x P(x)$  is true exactly when  $P(x)$  is true for *at least one*  $x$  in the domain.

The **domain** (or universe of discourse) is the collection the variable ranges over. It must always be stated, because the same statement can flip truth value when the domain changes.

|                | $\forall$ UNIVERSAL                   | $\exists$ EXISTENTIAL                |
|----------------|---------------------------------------|--------------------------------------|
| Read as        | for all, for every, for each          | there exists, for some, at least one |
| True when      | every element satisfies $P$           | at least one element satisfies $P$   |
| False when     | one counterexample exists             | every element fails $P$              |
| To prove it    | a general argument covering all cases | produce one witness                  |
| To disprove it | produce one counterexample            | a general argument that none works   |
| Pairs with     | $\rightarrow$ usually                 | $\wedge$ usually                     |

◆ **Memory hook:**  $\forall$  is an upside-down **A** for **All**.  $\exists$  is a back-to-front **E** for **Exists**. The asymmetry in the last two rows is worth learning on its own: *one example kills a "for all" and one example makes a "there exists"*.

### ⚠ TRAP

### $\forall$ TAKES AN ARROW, $\exists$ TAKES AN AND

This is the single most common error in translation, and it appears on every paper that asks for one.

"Every student in the class has a laptop" is  $\forall x (S(x) \rightarrow L(x))$ , with an arrow. Writing  $\forall x (S(x) \wedge L(x))$  instead would claim that *everything in the universe* is a student and has a laptop, which is far stronger and almost certainly false.

"Some student has a laptop" is  $\exists x (S(x) \wedge L(x))$ , with an and. Writing  $\exists x (S(x) \rightarrow L(x))$  instead is nearly useless: it is satisfied by any non-student at all, since a false hypothesis makes the implication true.

If the domain is already restricted to students, the qualifier disappears and you write  $\forall x L(x)$  or  $\exists x L(x)$ . Say which domain you are using.

## Negating a quantified statement

### DE MORGAN'S LAWS FOR QUANTIFIERS

**The claim.** A negation moved across a quantifier flips it. Saying "not everything has the property" is the same as saying "something lacks it", and saying "nothing has the property" is the same as saying "everything lacks it".

**In symbols.**

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

**Conditions.** None. They hold for any predicate over any domain.

**The limiting case.** Over a **finite** domain  $\{a_1, \dots, a_n\}$ ,  $\forall$  is a chain of conjunctions and  $\exists$  a chain of disjunctions, so these two laws collapse into the propositional De Morgan laws of section 1.3 applied  $n$  times. They are not a second pair of laws but the same pair, stretched over a domain. Over an **empty** domain they still hold, and the check is worth doing once:  $\forall x P(x)$  is true there (nothing fails) and  $\exists x P(x)$  is false (nothing succeeds), so both sides of the first law come out false.

**Consequence.** Any negation can be pushed inwards until it rests on a bare predicate, flipping every quantifier it passes. This is how "no student has a cat, a dog and a ferret" gets written two different but equivalent ways, which 2024/2025 Q2(a)(iv) exploits.

### TEACH

### IN PLAIN ENGLISH

"Not all swans are white" means "some swan is not white". You have not claimed all swans are non-white, only that at least one is.

"No swan is green" means "every swan is not green". Here the negation does apply to all of them.

Notice that "no" is a negated **existential**: "no  $x$  has property  $P$ " is  $\neg \exists x P(x)$ , which becomes  $\forall x \neg P(x)$ . Whenever an exam sentence starts with "No", write it as a negated "there exists" first, then convert if you want.

## The translation question

### AS PRINTED

2024/2025 · Q2(A)

a. Let  $C(x)$  be the statement "x has a cat," let  $D(x)$  be the statement "x has a dog," and let  $F(x)$  be the statement "x has a ferret." Express each of these statements in terms of  $C(x)$ ,  $D(x)$ ,  $F(x)$ , quantifiers, and logical connectives. Let the domain consist of all students in your class.

- i. A student in your class has a cat, a dog, and a ferret.
- ii. All students in your class have a cat, a dog, or a ferret.
- iii. Some student in your class has a cat and a ferret, but not a dog.
- iv. No student in your class has a cat, a dog, and a ferret.
- v. For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

7.5 marks

### NOTE ON THE PAPER

The domain is given as "all students in your class", which is a real gift: it means you do **NOT** need a student predicate  $S(x)$ , and the  $\forall/\rightarrow$  pairing collapses. Students who invent  $S(x)$  anyway are not wrong, but they do more work for the same marks. Note also the paper's spacing, " $F(x)$ " with a space, which is just typesetting.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Three predicates $C, D, F$ , and a domain restricted to students in the class.                                                                                                                            |
| <b>ASKED</b>     | <b>Express</b> five English sentences in symbols. Nothing is to be proved or evaluated; the marks are purely for correct translation.                                                                     |
| <b>MARKS SAY</b> | 7.5 over five items is 1.5 each, so each answer is one line of symbols. There is no credit for prose here.                                                                                                |
| <b>TEACHES</b>   | This section, 1.4.                                                                                                                                                                                        |
| <b>TRAP</b>      | Item (v) is not three separate statements joined by "and" about one student. It says each animal has <i>some</i> owner, and the owners may be three different people. That distinction is the whole mark. |

Throughout, the domain is the students in the class, so a bare  $\forall x$  or  $\exists x$  already means "every student" or "some student".

**(i) "A student in your class has a cat, a dog, and a ferret."**

"A student" here means at least one, so it is existential, and the three pets belong to the same student, so they join with "and".

$$\exists x (C(x) \wedge D(x) \wedge F(x))$$

**(ii) "All students in your class have a cat, a dog, or a ferret."**

Universal, and "or" is inclusive: having two of them is fine.

$$\forall x (C(x) \vee D(x) \vee F(x))$$

**(iii) "Some student in your class has a cat and a ferret, but not a dog."**

Existential. "But" is simply "and" with a change of tone, and "not a dog" negates D.

$$\exists x (C(x) \wedge F(x) \wedge \neg D(x))$$

**(iv) "No student in your class has a cat, a dog, and a ferret."**

"No" is a negated existential, so start there and then convert.

$$\neg \exists x (C(x) \wedge D(x) \wedge F(x))$$

which by the quantifier law equals  $\forall x \neg(C(x) \wedge D(x) \wedge F(x))$ , and by De Morgan equals  $\forall x (\neg C(x) \vee \neg D(x) \vee \neg F(x))$ .

All three are correct and any one earns the mark. Quoting the first and then showing the conversion demonstrates more, and costs one line.

Notice this is exactly the negation of (i), which is worth saying in your answer.

**(v) "For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet."**

Three separate existence claims, one per animal, and they need not be the same student. So it is a conjunction of three existentials, each with its own variable.

$$\exists x C(x) \wedge \exists y D(y) \wedge \exists z F(z)$$

Using three different letters is not fussiness. Writing  $\exists x (C(x) \wedge D(x) \wedge F(x))$  instead would say one student owns all three, which is statement (i), a different and much stronger claim. That contrast is exactly what the examiner set (i) and (v) to test.

**Now redo it:** cover the answers and translate all five again. Then translate a sixth of your own: "exactly one student has a ferret", which is harder than it looks.

## Deciding the truth of a quantified statement

a. Determine the truth value of each of these statements if the domain consists of all real numbers.

- i.  $\exists x(x^3 = -1)$     ii.  $\exists x(x^4 < x^2)$     iii.  $\forall x((-x)^2 = x^2)$     iv.  $\forall x(2x > x)$

**4 marks**

**BREAK IT DOWN**

|                  |                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Four quantified statements, domain the real numbers.                                                                                                                     |
| <b>ASKED</b>     | <b>Determine the truth value.</b> That means a verdict plus the reason: a witness for a true existential, a counterexample for a false universal, an argument otherwise. |
| <b>MARKS SAY</b> | 1 mark each, so one line each. But a bare "true" is worth little; name the number.                                                                                       |
| <b>TEACHES</b>   | This section, and the asymmetry table above.                                                                                                                             |
| <b>TRAP</b>      | Part (iv) looks obviously true and is false. The domain is <i>all</i> reals, which includes zero and the negatives.                                                      |

**WORKED EXAMPLE****PQ 24/25 Q4(A)**

**(i)  $\exists x(x^3 = -1)$ .** An existential, so one witness settles it. Take  $x = -1$ : then  $x^3 = -1$ . **TRUE.**

**(ii)  $\exists x(x^4 < x^2)$ .** Another existential. Try a fraction:  $x = \frac{1}{2}$  gives  $x^2 = 0.25$  and  $x^4 = 0.0625$ , and  $0.0625 < 0.25$ . **TRUE.**

Worth understanding rather than guessing:  $x^4 < x^2$  means  $x^2(x^2 - 1) < 0$ , which needs  $x^2 < 1$  and  $x \neq 0$ . So it holds exactly for the non-zero numbers strictly between  $-1$  and  $1$ .

**(iii)  $\forall x((-x)^2 = x^2)$ .** A universal, so it needs an argument covering everything, not an example.  $(-x)^2 = (-1)^2 \times x^2 = 1 \times x^2 = x^2$ , for every real  $x$ . **TRUE.**

**(iv)  $\forall x(2x > x)$ .** A universal, so one counterexample kills it. Take  $x = 0$ : then  $2x = 0$  and  $x = 0$ , and  $0 > 0$  is false. **FALSE.**

Take  $x = -3$  for a second counterexample:  $2x = -6$ , and  $-6 > -3$  is false. In fact  $2x > x$  only for the positive reals, and the domain here is all of them.

**Now redo it:** decide  $\forall x(x^2 \geq x)$  over the reals, then over the integers. The answers differ, which is the point of always stating the domain.

**Translating a sentence with structure****AS PRINTED****2024/2025 · Q2(C)**

- c. Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, and quantifiers with a domain consisting of all people, and logical connectives. **4 marks**

**BREAK IT DOWN**

|                  |                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | One English sentence and a domain of all people.                                                                                                         |
| <b>ASKED</b>     | <b>Express</b> it in symbols. You must invent the predicates yourself, which is part of the marks: define them before you use them.                      |
| <b>MARKS SAY</b> | 4 marks for one sentence is generous, so they are paying for the setup: naming predicates, choosing quantifiers, nesting correctly.                      |
| <b>TEACHES</b>   | This section. The nesting of two quantifiers is the new element.                                                                                         |
| <b>TRAP</b>      | "Someone's mother" hides a second quantifier. It means there exists a person of whom she is the mother, so the answer nests $\exists$ inside $\forall$ . |

**Step 1. Name the predicates.** Over the domain of all people, let

$F(x)$  = "x is female"

$P(x)$  = "x is a parent"

$M(x, y)$  = "x is the mother of y"

$M$  takes two arguments because motherhood is a relation between two people, not a property of one. Part Three studies such relations properly.

**Step 2. Read the shape of the sentence.** It says: for every person, if (female and parent) then (mother of someone).

**Step 3. Write it.**

$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$

**Step 4. Check each piece.** The outer quantifier is universal because the claim is about every person, and it takes an arrow, as universals do. The hypothesis is a conjunction, because both conditions must hold. The conclusion is existential, because being someone's mother only requires one such someone. The  $y$  is bound inside, so it cannot be confused with the  $x$ .

An equally correct variant pulls the existential out to the front:  $\forall x \exists y ((F(x) \wedge P(x)) \rightarrow M(x, y))$ . Both are accepted.

**Now redo it:** express "every student has an adviser" and then "some adviser advises every student". They differ only in the order of the two quantifiers, and they mean very different things.

### ⚠ TRAP

### ORDER OF NESTED QUANTIFIERS CHANGES THE MEANING

$\forall x \exists y L(x, y)$  says: for each  $x$  there is *some*  $y$ , and it may be a different  $y$  each time. "Everyone has a mother."

$\exists y \forall x L(x, y)$  says: there is *one*  $y$  that works for every  $x$ . "Someone is the mother of everyone."

Same symbols, opposite claims, and the first is true while the second is absurd. Two quantifiers of the same kind can be swapped freely; two of different kinds never can.

### RECALL AND CHECK

1. What does  $\forall$  pair with, an arrow or an and? And  $\exists$ ?
2. Write "no student passed" using  $\exists$ , then using  $\forall$ .
3. How do you disprove  $\forall x P(x)$ ?
4. Is  $\forall x \exists y L(x, y)$  the same as  $\exists y \forall x L(x, y)$ ?

#### ANSWERS

1.  $\forall$  pairs with  $\rightarrow$ ,  $\exists$  pairs with  $\wedge$ .
2.  $\neg \exists x P(x)$ , equivalently  $\forall x \neg P(x)$ , where  $P(x)$  is "x passed".
3. Produce a single counterexample: one  $x$  with  $P(x)$  false.
4. No. The first lets  $y$  depend on  $x$ ; the second demands one  $y$  for all  $x$ .

## FILL IN THE GAP

1.  $\forall$  is the \_\_\_\_\_ quantifier and is read \_\_\_\_\_.
2.  $\exists$  is the \_\_\_\_\_ quantifier and is read \_\_\_\_\_.
3.  $\neg\forall x P(x)$  is equivalent to \_\_\_\_\_.
4. A statement with a variable that becomes a proposition once the variable is given a value is called a \_\_\_\_\_.

## ANSWERS

1. universal, "for all". 2. existential, "there exists". 3.  $\exists x \neg P(x)$ . 4. predicate.

## PART ONE • 1.5

# Rules of Inference and Valid Arguments

*An argument moves from things you accept to something new. The rules below are the moves that are always safe, and the two famous ones that are not.*

*By the end of this section you will be able to say what makes an argument valid, name and apply the standard rules of inference, spot the two classic fallacies, and construct acceptable logic statements leading to a conclusion, which the 2025/2026 paper asks for directly.*

## What "valid" means

### ● MUST MEMORISE

### VALIDITY IS ABOUT SHAPE, NOT TRUTH

An **argument** is a list of propositions called **premises**, followed by one called the **conclusion**.

An argument is **valid** when the conclusion is true in every case where all the premises are true. Equivalently, the conjunction of the premises implies the conclusion, and that implication is a tautology.

Validity says nothing about whether the premises are actually true. A valid argument from false premises can reach a false conclusion. What validity guarantees is that the *step* is safe.

◆ **Memory hook:** valid means the wiring is sound. Sound means the wiring is good *and* the input is true. An exam asks about the wiring.

### TEACH

### A VALID ARGUMENT WITH A FALSE CONCLUSION

Premise 1: If it is a bird, it can fly.

Premise 2: A penguin is a bird.

Conclusion: A penguin can fly.

The conclusion is false, yet the argument is **valid**: the shape is the safest one there is. The fault lies in premise 1, which is false. That is the distinction the word "valid" exists to make.

## The rules of inference

### ● MUST MEMORISE

### THE STANDARD RULES

Each rule reads: given the premises above the line, you may write the conclusion below it.

| NAME                   | PREMISES                                  | CONCLUSION        | IN WORDS                                         |
|------------------------|-------------------------------------------|-------------------|--------------------------------------------------|
| Modus ponens           | $p \rightarrow q$ , and $p$               | $q$               | The rule fired, so the result follows.           |
| Modus tollens          | $p \rightarrow q$ , and $\neg q$          | $\neg p$          | The result did not happen, so the cause did not. |
| Hypothetical syllogism | $p \rightarrow q$ , and $q \rightarrow r$ | $p \rightarrow r$ | Chain two implications.                          |
| Disjunctive syllogism  | $p \vee q$ , and $\neg p$                 | $q$               | One of two, and not the first, so the second.    |
| Addition               | $p$                                       | $p \vee q$        | If $p$ is true, " $p$ or anything" is true.      |
| Simplification         | $p \wedge q$                              | $p$               | If both, then certainly one.                     |
| Conjunction            | $p$ , and $q$                             | $p \wedge q$      | Two separate facts combine.                      |
| Resolution             | $p \vee q$ , and $\neg p \vee r$          | $q \vee r$        | The rule automated theorem provers run on.       |

◆ **Memory hook:** *ponens* is Latin for affirming and *tollens* for denying. Modus ponens affirms the hypothesis; modus tollens denies the conclusion. Those two carry most of the marks.

### TEACH

### MODUS TOLLENS IS THE CONTRAPOSITIVE AT WORK

Modus tollens says: from  $p \rightarrow q$  and  $\neg q$ , conclude  $\neg p$ .

That is nothing more than section 1.2 in action. Since  $p \rightarrow q$  is equivalent to its contrapositive  $\neg q \rightarrow \neg p$ , and you have  $\neg q$ , modus ponens on the contrapositive gives  $\neg p$ .

So the two rules are the same rule, seen from either end of an implication. Saying that in an answer shows the marker you understand the structure rather than having memorised a table.



## The two fallacies

### ⚠ TRAP

### THE TWO INVALID MOVES THAT LOOK VALID

**Affirming the consequent.** From  $p \rightarrow q$  and  $q$ , concluding  $p$ . This is invalid, and it is the converse mistake from 1.2 wearing a different hat.

If it is raining, the ground is wet. The ground is wet. Therefore it is raining.

No: a burst pipe would do it. The wet ground is consistent with rain but does not establish it.

**Denying the antecedent.** From  $p \rightarrow q$  and  $\neg p$ , concluding  $\neg q$ . Also invalid, and it is the inverse mistake.

If it is raining, the ground is wet. It is not raining. Therefore the ground is not wet.

No: the pipe again.

Both fallacies come from treating  $p \rightarrow q$  as if it ran both ways. Only the contrapositive runs the other way, which is precisely why 1.2 matters here.

## The question the newest paper asks

### AS PRINTED

2025/2026 · Q4(C)

(c) Develop three good examples of acceptable logic statements leading to a conclusion.  
(3 marks)

### NOTE ON THE PAPER

The wording is loose. "Acceptable logic statements leading to a conclusion" means **VALID ARGUMENTS**: premises followed by a conclusion that genuinely follows. It does not mean three true sentences. Answer it as "give three valid arguments, naming the rule of inference each uses", which is what earns the marks, and say in one line that this is how you are reading it.

### BREAK IT DOWN

**GIVEN** Nothing but the instruction. You supply the content.

**ASKED** **Develop** three examples, each being premises plus a conclusion that follows. Since you choose them, choose ones you can name.

**MARKS SAY** 3 marks, one per example. Each needs premises, a conclusion, and the name of the rule; a bare pair of sentences is half an answer.

**TEACHES** This section, 1.5.

**TRAP** Offering an argument that is not actually valid, most often by affirming the consequent. Stick to modus ponens, modus tollens and hypothetical syllogism, which are impossible to get wrong.

Reading the question as asking for three valid arguments, each named.

**Example 1, modus ponens.**

Premise 1: If a program compiles without errors, then it produces an executable file.

Premise 2: This program compiles without errors.

Conclusion: Therefore it produces an executable file.

Form:  $p \rightarrow q$ ,  $p$ , therefore  $q$ . Valid, because the only row where  $p \rightarrow q$  and  $p$  are both true also has  $q$  true.

**Example 2, modus tollens.**

Premise 1: If a set has 5 elements, then it has 32 subsets.

Premise 2: This set does not have 32 subsets.

Conclusion: Therefore this set does not have 5 elements.

Form:  $p \rightarrow q$ ,  $\neg q$ , therefore  $\neg p$ . Valid. (Section 2.1 established the premise: a set of  $n$  elements has  $2^n$  subsets, and  $2^5 = 32$ .)

**Example 3, hypothetical syllogism.**

Premise 1: If a student passes the practical, then the student qualifies for the project.

Premise 2: If a student qualifies for the project, then the student graduates.

Conclusion: Therefore if a student passes the practical, that student graduates.

Form:  $p \rightarrow q$ ,  $q \rightarrow r$ , therefore  $p \rightarrow r$ . Valid, by chaining.

**Finish with one line that earns the last mark:** in each case the conclusion is true in every situation where the premises are both true, which is the definition of a valid argument.

**Now redo it:** write three more of your own, one of them using disjunctive syllogism, and check each against the rules table.

## Logic in computing

(b) Discuss the place of Logic in Computer Science highlighting practical areas of relevance?

**(2 marks)**

**NOTE ON THE PAPER**

Printed with a question mark on a command, which is a habit of this examiner and carries no meaning. Two marks means two or three sentences with named areas, not an essay.

Logic is the formal language in which computer science states precise conditions and reasons about them, and it appears at every level of the subject.

**Hardware.** Every digital circuit is a Boolean expression made physical: AND, OR and NOT gates implement the connectives directly, so simplifying an expression means using fewer components. Part Eight is this.

**Programming.** Every condition in an if or while is a proposition, and the compiler applies De Morgan's laws and the implication law to optimise them.

**Databases.** A query is a predicate; the engine uses quantifiers and logical equivalence to rewrite it into a faster but equivalent form.

**Verification.** Proving that a program meets its specification, and automated theorem proving, both run on the rules of inference, with resolution as the workhorse.

**Artificial intelligence.** Knowledge representation and rule-based reasoning are predicate logic with an inference engine attached.

For two marks, three of these named with a one-line reason each is a full answer.

#### RECALL AND CHECK

1. State modus ponens and modus tollens.
2. Name the fallacy: "if  $p$  then  $q$ ;  $q$ ; therefore  $p$ ".
3. Can a valid argument have a false conclusion?
4. From " $p \vee q$ " and " $\neg p$ ", what follows and by which rule?

#### ANSWERS

1. Modus ponens: from  $p \rightarrow q$  and  $p$ , conclude  $q$ . Modus tollens: from  $p \rightarrow q$  and  $\neg q$ , conclude  $\neg p$ .
2. Affirming the consequent. It is invalid.
3. Yes, if a premise is false. Validity constrains the step, not the inputs.
4.  $q$ , by disjunctive syllogism.

#### FILL IN THE GAP

1. An argument is \_\_\_\_\_ when the conclusion is true in every case where all premises are true.
2. From  $p \rightarrow q$  and  $\neg q$  we conclude \_\_\_\_\_, by \_\_\_\_\_.
3. Concluding  $\neg q$  from  $p \rightarrow q$  and  $\neg p$  is the fallacy of \_\_\_\_\_.
4. Chaining  $p \rightarrow q$  with  $q \rightarrow r$  to get  $p \rightarrow r$  is called \_\_\_\_\_.

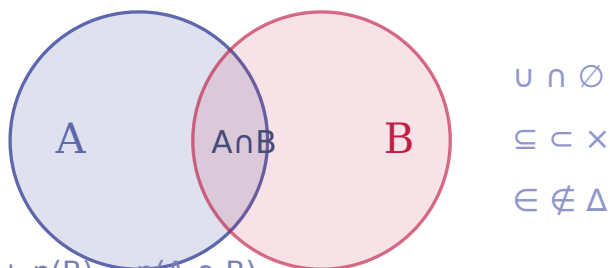
#### ANSWERS

1. valid. 2.  $\neg p$ , modus tollens. 3. denying the antecedent. 4. hypothetical syllogism.

## PART TWO

# Sets and Their Operations

*A set is the simplest idea in this course and the one everything else is built from. Relations are sets, functions are sets, graphs are sets, and a database table is a set with a nice interface.*



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

## Sets, Membership and the Nine Types

The 2025/2026 paper asks you to discuss nine types of set with an example of each. Here they are, with the vocabulary that surrounds them.

By the end of this section you will be able to define a set, state its cardinality, name and give an example of each of the nine standard types, and use the subset symbols correctly.

### The definition and the vocabulary

#### ● MUST MEMORISE

#### SET, ELEMENT, CARDINALITY

A **set** is a well-defined collection of distinct objects. The objects are its **elements** or **members**.

**Well-defined** is doing real work: for any object it must be unambiguous whether it belongs. "The set of tall students" is not a set until "tall" is pinned down. "The set of students over 1.8 m" is.

The number of elements in a set  $A$  is its **cardinality**, written  $n(A)$  or  $|A|$ . So if  $A = \{2, 4, 6\}$  then  $n(A) = 3$ .

◆ **Memory hook:** distinct and well-defined. Repeats do not count twice, and "maybe" is not an answer to whether something is in.

## The nine types of set

### ● MUST MEMORISE

### THE NINE TYPES, WITH EXAMPLES

| TYPE                      | DEFINITION                                                                 | EXAMPLE                                                     |
|---------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Empty</b> (null, void) | Contains no elements at all. Written $\emptyset$ or $\{ \}$ .              | $\{x \mid x \in \mathbb{R}, x^2 = -1\} = \emptyset$         |
| <b>Singleton</b> (unit)   | Contains exactly one element.                                              | $\{5\}$ , or the set of even prime numbers $\{2\}$          |
| <b>Finite</b>             | Its elements can be counted and the count ends.                            | $\{a, e, i, o, u\}$ , $n = 5$                               |
| <b>Infinite</b>           | Counting its elements never ends.                                          | $\mathbb{N}$ , the natural numbers                          |
| <b>Equal</b>              | Two sets with exactly the same elements. Order and repeats are irrelevant. | $\{1, 2, 3\} = \{3, 2, 1, 1\}$                              |
| <b>Equivalent</b>         | Two sets with the same cardinality, whatever their elements.               | $\{a, b, c\}$ and $\{1, 2, 3\}$                             |
| <b>Subset</b>             | Every element of A is also in B. Written $A \subseteq B$ .                 | $\{1, 2\} \subseteq \{1, 2, 3\}$                            |
| <b>Universal</b>          | The set containing everything under discussion. Written U.                 | $U = \{1, 2, \dots, 10\}$ in a question about single digits |
| <b>Disjoint</b>           | Two sets with no element in common, so $A \cap B = \emptyset$ .            | $\{1, 3, 5\}$ and $\{2, 4, 6\}$                             |

Two more you should be able to name if pressed, since some books count them among the nine: a **proper subset** ( $A \subset B$ , a subset that is not the whole of B) and the **power set** (the set of all subsets, covered in 2.3).

◆ **Memory hook:** group them in threes. By *size*: empty, singleton, finite/infinite. By *comparison*: equal, equivalent, subset. By *role*: universal, disjoint, power set.

### ⚠ TRAP

### EQUAL VERSUS EQUIVALENT

These two words are close in English and precise in mathematics, and the examiner knows it.

**Equal** sets have the same *elements*:  $\{1, 2, 3\} = \{3, 2, 1\}$ .

**Equivalent** sets have the same *number* of elements:  $\{a, b, c\}$  and  $\{1, 2, 3\}$  are equivalent but very much not equal.

Every pair of equal sets is equivalent. The reverse fails constantly.

⚠ TRAP

$\emptyset$  IS NOT  $\{\emptyset\}$

$\emptyset$  is the empty set: it has nothing in it, so  $n(\emptyset) = 0$ .

$\{\emptyset\}$  is a set containing one thing, and that thing happens to be the empty set. So  $n(\{\emptyset\}) = 1$ . It is a singleton, not an empty set.

Think of  $\emptyset$  as an empty box and  $\{\emptyset\}$  as a box with an empty box inside it. The second is not empty.

Also: never write  $\{\emptyset\}$  meaning "the empty set", and never write " $\emptyset = \{ \}$ " with anything between the braces. Both cost marks in the power-set questions of 2.3.

## Subsets

● MUST MEMORISE

$\subseteq$  AND  $\subset$

$A \subseteq B$  ("A is a subset of B") means every element of A is also an element of B. A may equal B.

$A \subset B$  ("A is a proper subset of B") means  $A \subseteq B$  and  $A \neq B$ , so B has at least one element A lacks.

Standard facts, all worth knowing on sight:

$\emptyset \subseteq A$  for every set A. The empty set is a subset of everything, because it has no element that could fail to be in A.

$A \subseteq A$  for every set A.

If  $A \subseteq B$  and  $B \subseteq C$  then  $A \subseteq C$ , so subethood carries through a chain.

A set with n elements has  $2^n$  subsets.

◆ **Memory hook:**  $\subseteq$  is to sets what  $\leq$  is to numbers, and  $\subset$  is what  $<$  is. The extra line means "or equal".

⚠ TRAP

$\in$  AND  $\subseteq$  ARE NOT INTERCHANGEABLE

Let  $A = \{1, 2, 3\}$ .

$1 \in A$  is true. The number 1 is an element.

$\{1\} \subseteq A$  is true. The *set*  $\{1\}$  is a subset.

$1 \subseteq A$  is nonsense. The number 1 is not a set, so it cannot be a subset.

$\{1\} \in A$  is false. The elements of A are numbers, not sets, so the set  $\{1\}$  is not among them.

The rule:  **$\in$  relates an element to a set,  $\subseteq$  relates a set to a set.** Check which side of the symbol each object belongs on before you write it.

## Listing the members of a described set

AS PRINTED

2021/2022 · Q1(A)

1. (a) List the members of the following sets. (5marks)
- (i)  $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
  - (ii)  $\{x \mid x \text{ is a positive prime number less than } 20\}$
  - (iii)  $\{x \mid x \text{ is a perfect square and } x < 100\}$
  - (iv)  $\{x \mid x \text{ is an integer such that } x^2 = 2\}$
  - (v)  $\{x \mid x \text{ is an integer such that } 2x = x\}$

### NOTE ON THE PAPER

The marks are printed *before* the five items, on the instruction line, which is why the quote above carries them there.

**ITEM (III) IS GENUINELY AMBIGUOUS AND YOU SHOULD SAY SO.** It reads " $\{x \mid x \text{ is a perfect square and } x < 100\}$ " without saying that  $x$  is a positive integer. Read strictly, 0 qualifies, and so do non-integer squares such as 0.25, which would make the set infinite. The intended answer is the perfect squares of whole numbers below 100. State the reading you use and give the set; a stated assumption protects the mark, a silent one does not.

### BREAK IT DOWN

- GIVEN** Five sets in set-builder notation.
- ASKED** **List the members**, that is, convert each from set-builder to roster form.
- MARKS SAY** 5 marks, one per item. Each is one line. Do not explain; just list, with braces.
- TEACHES** This section, 2.1, and section F.1 for the notation.
- TRAP** Two items have **fewer** members than they look. Item (iv) is empty, and item (v) has exactly one member. Writing "no solution" instead of the empty set loses the mark: the answer to "list the members" is always a set.



(i)  $\{x \mid x \text{ is a real number such that } x^2 = 1\}$

$x^2 = 1$  gives  $x = 1$  or  $x = -1$ , and both are real.

**$\{1, -1\}$**

(ii)  $\{x \mid x \text{ is a positive prime number less than } 20\}$

A prime has exactly two positive divisors, itself and 1, so 1 is not prime.

**$\{2, 3, 5, 7, 11, 13, 17, 19\}$** , eight members.

(iii)  $\{x \mid x \text{ is a perfect square and } x < 100\}$

*Reading used:*  $x$  is a whole number that is the square of a whole number. Then  $0^2 = 0$  up to  $9^2 = 81$  qualify, and  $10^2 = 100$  does not, because the condition is strictly less than 100.

**$\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$** , ten members.

If your reading excludes 0, say so and give the same set without it, which is nine members. Either is defensible; leaving the reading unstated is not.

(iv)  $\{x \mid x \text{ is an integer such that } x^2 = 2\}$

$x^2 = 2$  gives  $x = \sqrt{2}$  or  $x = -\sqrt{2}$ , and neither is an integer. No integer qualifies, so the set is empty.

$\emptyset$ , also written  $\{ \}$ .

**Do not write  $\{ \emptyset \}$ .** That is the set *containing* the empty set, which has one member and is not empty at all.

(v)  $\{x \mid x \text{ is an integer such that } 2x = x\}$

$2x = x$  gives  $x = 0$ , which is an integer, so exactly one value qualifies.

**$\{0\}$** , a singleton set.

**Note the contrast between (iv) and (v)**, which is almost certainly why the examiner set them next to each other: one is the empty set with no members, the other is a singleton whose single member happens to be 0.  $\emptyset$  and  $\{0\}$  are different sets, and confusing them is the standard slip here.

**Now redo it:** list  $\{x \mid x \text{ is an integer such that } x^2 < 5\}$ . The answer is  $\{-2, -1, 0, 1, 2\}$ , five members.

## The real question

b) i) Discuss nine (9) types of sets, giving an example of each.

**(3 marks)**

ii) List two mathematical ways to describe a set.

**(2 marks)**

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Nothing. Both parts are pure recall.                                                                                                                                                                                                   |
| <b>ASKED</b>     | Part i: <b>discuss</b> nine types <b>giving an example of each</b> . The example is compulsory and carries marks. Part ii: <b>list</b> two ways to describe a set.                                                                     |
| <b>MARKS SAY</b> | 3 marks for nine types is a third of a mark each, so a table with one line and one example per row is exactly right. Do not write paragraphs. Part ii is 2 marks for two named methods, so name them <i>and</i> show one line of each. |
| <b>TEACHES</b>   | This section for the types, F.1 for the two descriptions.                                                                                                                                                                              |
| <b>TRAP</b>      | Giving nine types but forgetting the examples throws away half the marks, because "giving an example of each" is part of the instruction.                                                                                              |

### WORKED EXAMPLE

PQ 25/26 Q1(B)

**(i) Nine types of set.** Answer as a table; it is faster to write and easier to mark.

| TYPE            | MEANING                                    | EXAMPLE                           |
|-----------------|--------------------------------------------|-----------------------------------|
| Empty set       | has no elements                            | $\emptyset = \{ \}$               |
| Singleton set   | has exactly one element                    | $\{7\}$                           |
| Finite set      | has a countable, ending number of elements | $\{a, e, i, o, u\}$               |
| Infinite set    | counting never ends                        | $\mathbb{N} = \{0, 1, 2, \dots\}$ |
| Equal sets      | same elements                              | $\{1, 2, 3\}$ and $\{3, 1, 2\}$   |
| Equivalent sets | same number of elements                    | $\{a, b\}$ and $\{9, 4\}$         |
| Subset          | every element of one lies in the other     | $\{1, 2\} \subseteq \{1, 2, 3\}$  |
| Universal set   | contains everything under discussion       | $U = \{1, 2, \dots, 10\}$         |
| Disjoint sets   | no element in common                       | $\{1, 3\}$ and $\{2, 4\}$         |

**(ii) Two mathematical ways to describe a set.**

**1. The roster (listing or tabular) method:** write the elements between braces.

Example:  $A = \{2, 4, 6, 8\}$ .

**2. Set-builder (rule) notation:** state the property membership requires. Example:  $A = \{x \mid x \text{ is even and } 0 < x < 10\}$ , read "the set of all  $x$  such that  $x$  is even and  $x$  lies between 0 and 10".

**Now redo it:** cover the table and write all nine types with fresh examples of your own. Three minutes.

## RECALL AND CHECK

1. What is  $n(\{\emptyset\})$ ?
2. Give two sets that are equivalent but not equal.
3. How many subsets does  $\{a, b, c, d\}$  have?
4. True or false:  $\{2\} \in \{1, 2, 3\}$ .

### ANSWERS

1. It is a singleton whose only element is the empty set.
2.  $\{a, b, c\}$  and  $\{1, 2, 3\}$ , for instance. Same size, different elements.
3.  $2^4 = 16$ .
4. False.  $\{2\}$  is a subset, not an element:  $\{2\} \subseteq \{1, 2, 3\}$  is the true statement.

## FILL IN THE GAP

1. The number of elements in a set is its \_\_\_\_\_.
2. Two sets with no element in common are \_\_\_\_\_.
3. A set with  $n$  elements has \_\_\_\_\_ subsets.
4. Sets with the same elements are \_\_\_\_\_; sets with the same number of elements are \_\_\_\_\_.

### ANSWERS

1. cardinality. 2. disjoint. 3.  $2^n$ . 4. equal, equivalent.

## PART TWO · 2.2

# Set Operations and Venn Diagrams

*Five operations, each with a picture. Draw the picture and the answer is usually obvious; skip the picture and the same question becomes a guessing game.*

*By the end of this section you will be able to compute union, intersection, difference, symmetric difference and complement, draw and read a Venn diagram, and state De Morgan's laws for sets.*

## The five operations

### ● MUST MEMORISE

### THE FIVE OPERATIONS

| OPERATION            | SYMBOL            | DEFINITION                                   | IN WORDS                           |
|----------------------|-------------------|----------------------------------------------|------------------------------------|
| Union                | $A \cup B$        | $\{x \mid x \in A \text{ or } x \in B\}$     | everything in either, counted once |
| Intersection         | $A \cap B$        | $\{x \mid x \in A \text{ and } x \in B\}$    | only what is in both               |
| Difference           | $A - B$           | $\{x \mid x \in A \text{ and } x \notin B\}$ | in A but not in B                  |
| Symmetric difference | $A \triangle B$   | $(A - B) \cup (B - A)$                       | in one or the other, not in both   |
| Complement           | $A'$ or $\bar{A}$ | $\{x \mid x \in U \text{ and } x \notin A\}$ | everything outside A               |

◆ **Memory hook:**  $\cup$  is a **u**p-cup that holds everything, so it is **u**nion.  $\cap$  points down and only what is caught in both bowls stays. The complement needs a universal set to be meaningful at all, so U must always be stated.

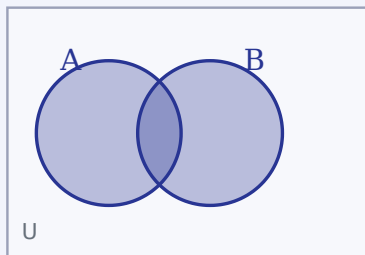
## The pictures

### TEACH

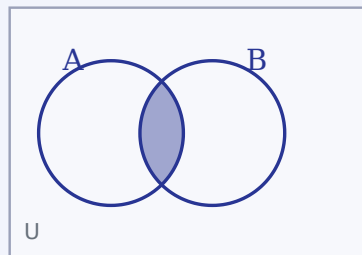
### VENN DIAGRAMS

A **Venn diagram** draws each set as a circle inside a rectangle, where the rectangle is the universal set  $U$ . Overlapping circles create regions, and every element of  $U$  lives in exactly one region.

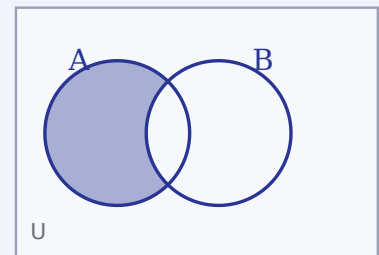
Shade the region the operation describes, and the answer becomes something you can read off rather than compute.



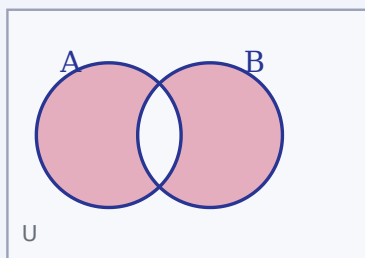
$A \cup B$  union



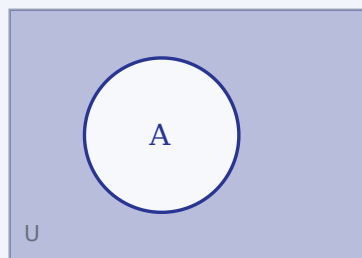
$A \cap B$  intersection



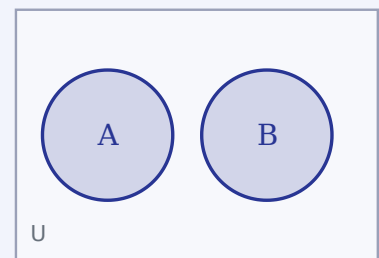
$A - B$  difference



$A \triangle B$  symmetric difference



$A'$  complement



disjoint:  $A \cap B = \emptyset$

## WORKED EXAMPLE

## METHOD

**GIVEN**  $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ ,  $A = \{1, 2, 3, 4\}$ ,  $B = \{3, 4, 5, 6\}$ .

**FIND** All five operations.

$A \cup B$  = everything in either =  $\{1, 2, 3, 4, 5, 6\}$ . Note 3 and 4 appear once, not twice.

$A \cap B$  = what is in both =  $\{3, 4\}$ .

$A - B$  = in A, not in B =  $\{1, 2\}$ .

$B - A$  = in B, not in A =  $\{5, 6\}$ . Note this differs from  $A - B$ , so the order of the two sets matters.

$A \triangle B = (A - B) \cup (B - A) = \{1, 2\} \cup \{5, 6\} = \{1, 2, 5, 6\}$ . It is the union minus the intersection.

$A'$  = everything in U outside A =  $\{5, 6, 7, 8\}$ .

**Check.**  $n(A) = 4$ ,  $n(B) = 4$ ,  $n(A \cap B) = 2$ , and  $n(A \cup B) = 4 + 4 - 2 = 6$ , which matches the six elements listed. Always run that check; it catches a dropped element instantly.

## The real question

### AS PRINTED

2025/2026 · Q1(C)

(c)  $A = \{11, 12, 13, 14\}$

$B = \{13, 14, 15, 16\}$

From the two sets above, calculate:

- |      |                         |          |
|------|-------------------------|----------|
| i)   | Intersection set        | (½ mark) |
| ii)  | Difference ( $A - B$ )  | (1 mark) |
| iii) | Difference ( $B - A$ ). | (1 mark) |

### BREAK IT DOWN

**GIVEN**  $A = \{11, 12, 13, 14\}$  and  $B = \{13, 14, 15, 16\}$ , which overlap in two elements.

**ASKED** **Calculate** three sets. Answers are sets, so write them in braces.

**MARKS SAY** 2.5 marks total and one line each. This is a gift question: take it, check it, move on.

**TEACHES** This section, 2.2.

**TRAP** Parts ii and iii are asked separately *because* they differ. Writing the same answer twice, or writing the symmetric difference for both, throws away two of the 2.5 marks.

**GIVEN**  $A = \{11, 12, 13, 14\}$ ,  $B = \{13, 14, 15, 16\}$ .

**FIND**  $A \cap B$ ,  $A - B$ ,  $B - A$ .

**RULE** Intersection keeps what is in both;  $A - B$  keeps what is in A and not in B.

**(i) Intersection.** Go through A one element at a time and ask whether it is in B. 11: no. 12: no. 13: yes. 14: yes.

$$A \cap B = \{13, 14\}$$

**(ii)  $A - B$ .** Elements of A that are *not* in B, which is the two that just failed the test.

$$A - B = \{11, 12\}$$

**(iii)  $B - A$ .** Now start from B. 13: in A, drop it. 14: in A, drop it. 15: not in A, keep. 16: not in A, keep.

$$B - A = \{15, 16\}$$

**Check.** The three answers should partition the union:  $\{11, 12\} \cup \{13, 14\} \cup \{15, 16\} = \{11, 12, 13, 14, 15, 16\}$ , which is  $A \cup B$ , with 6 elements. And  $4 + 4 - 2 = 6$ . Consistent.

**Now redo it:** add  $A \triangle B$  and  $A'$  taking  $U = \{10, 11, \dots, 18\}$ . Answers:  $\{11, 12, 15, 16\}$  and  $\{10, 15, 16, 17, 18\}$ .

## De Morgan's laws for sets

### DE MORGAN'S LAWS (SET FORM)

**The claim.** The complement of an intersection is the union of the complements, and the complement of a union is the intersection of the complements. Complementing flips the operation.

**In symbols.**

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

**Conditions.** A and B must be subsets of a stated universal set U, since complement is meaningless without one.

**The limiting case.** Take  $B = U$ . Then  $B'$  is the empty set, and the first law reads  $(A \cap U)' = A' \cup \emptyset$ , which is  $A' = A'$ : true, and empty of content. The law only says something when both sets are proper parts of U, and it says most when they overlap. The same collapse happens at  $A = B$ , where it reduces to  $A' = A'$  again. A worked answer that lands on one of these has almost certainly lost a term somewhere.

**Why it is the same law as in logic.** Membership in an intersection is a conjunction ("x is in A and x is in B") and membership in a union is a disjunction. Complement is negation. So the set law is the logic law of section 1.3, read through the definition of membership. That is exactly the proof the 2024/2025 paper asks for.

c. Use set builder notation and logical equivalences to establish the first De Morgan law  $A \cap \bar{B} = \bar{A} \cup \bar{B}$

**3 marks**

### NOTE ON THE PAPER

Two typesetting faults in one line. The overbar sits above **B** alone when the law needs it above the whole of  $A \cap B$ , and the relation symbol images as a short dash where an equals sign belongs. The law being asked for is unambiguous from its name: the complement of an intersection equals the union of the complements. Answer that, and note the misprint in one line so the marker knows you read it rather than copied it.

### BREAK IT DOWN

- GIVEN** The name of the law and a garbled rendering of it.
- ASKED** **Establish** it, using **set builder notation and logical equivalences**. The method is prescribed: translate membership into logic, apply the logic law, translate back. A Venn diagram alone will not answer this question.
- MARKS SAY** 3 marks for about five lines of derivation, one mark per genuine step.
- TEACHES** This section, plus 1.3 for the logical form.
- TRAP** Drawing two Venn diagrams and asserting they match. That illustrates the law, it does not establish it, and the question named the method it wants.

### WORKED EXAMPLE

PQ 24/25 Q3(C)

**To prove:**  $(A \cap B)' = A' \cup B'$ .

Two sets are equal when they have exactly the same elements, so take an arbitrary element  $x$  and show that belonging to the left is equivalent to belonging to the right, step by step.

| LINE | STATEMENT                                       | REASON                           |
|------|-------------------------------------------------|----------------------------------|
| 1    | $(A \cap B)' = \{x \mid x \notin (A \cap B)\}$  | definition of complement         |
| 2    | $= \{x \mid \neg(x \in A \cap B)\}$             | meaning of $\notin$              |
| 3    | $= \{x \mid \neg(x \in A \wedge x \in B)\}$     | definition of intersection       |
| 4    | $= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$ | <b>De Morgan's law for logic</b> |
| 5    | $= \{x \mid x \notin A \vee x \notin B\}$       | meaning of $\notin$              |
| 6    | $= \{x \mid x \in A' \vee x \in B'\}$           | definition of complement         |
| 7    | $= A' \cup B'$                                  | definition of union              |

Line 4 is the only real step; everything else is unpacking a definition. That is worth saying out loud in your answer, because it shows you know where the content is.

**The other law, for completeness:** the same seven lines with  $\cap$  and  $\cup$  exchanged, and  $\wedge$  and  $\vee$  exchanged, give  $(A \cup B)' = A' \cap B'$ .

**Now redo it:** write the seven lines for the second law from memory, naming the reason on each.



For any two finite sets,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$n(A)$  the number of elements in  $A$ .  $n(A \cup B)$  the number in the union.  $n(A \cap B)$  the number in both, subtracted because adding  $n(A)$  and  $n(B)$  counts those elements twice.

This is the two-set case of the inclusion and exclusion principle, which Part Five develops for three sets and more. It is worth 7 marks on the 2025/2026 paper by itself.

◆ **Memory hook:** add both, then pay back the overlap you counted twice.

### RECALL AND CHECK

1. With  $A = \{1,2,3\}$ ,  $B = \{3,4\}$ , find  $A \triangle B$ .
2. Is  $A - B$  the same as  $B - A$ ?
3. State  $(A \cup B)'$  in terms of  $A'$  and  $B'$ .
4.  $n(A) = 20$ ,  $n(B) = 15$ ,  $n(A \cap B) = 6$ . Find  $n(A \cup B)$ .

#### ANSWERS

1.  $\{1, 2, 4\}$ . The 3 is in both, so it is excluded.
2. No.  $A - B = \{1,2\}$  but  $B - A = \{4\}$ .
3.  $A' \cap B'$ .
4.  $20 + 15 - 6 = 29$ .

### FILL IN THE GAP

1.  $A \cap B$  is the set of elements in \_\_\_\_\_  $A$  and  $B$ .
2.  $A \triangle B$  equals  $(A - B) \cup (B - A)$ .
3.  $(A \cap B)' = \text{_____}$ .
4.  $n(A \cup B) = n(A) + n(B) - \text{_____}$ .

#### ANSWERS

1. both. 2.  $\cup$  (union). 3.  $A' \cup B'$ . 4.  $- n(A \cap B)$ .

## PART TWO · 2.3

# Bit Strings, Power Sets and Cartesian Products

Three ways of building something new out of a set. Each one is asked about directly, and the bit-string representation is how a computer actually stores a set.

By the end of this section you will be able to write any subset of a universal set as a bit string and compute unions, intersections and complements by bitwise operations, list a power set without missing a subset, and form a Cartesian product and count its size.

## Bit strings

### NOT IN THE COURSE TEXT

The official 161-page text never mentions bit strings, yet the 2023/2024 and 2024/2025 papers together spend more than 15 marks on them. Everything you need is here.

Fix a universal set  $U$  and write its elements in a definite order, usually increasing. A subset  $A$  of  $U$  can then be described by a string of 0s and 1s with one position per element of  $U$ :

- put **1** in position  $i$  if the  $i$ -th element of  $U$  is in  $A$ ,
- put **0** in position  $i$  if it is not.

That string is the **bit string** (or characteristic vector) of  $A$ . With  $U = \{1, 2, \dots, 10\}$  there are ten positions, so every subset is ten bits, and a computer can hold it in a single machine word.

This is not a classroom curiosity. It is how set types are implemented, and it is why testing membership is one machine instruction rather than a search.

### ● MUST MEMORISE

### SETS AS BIT STRINGS

With  $U$  ordered, the subset  $A$  has bit string  $b_1b_2 \dots b_n$  where  $b_i = 1$  exactly when the  $i$ -th element of  $U$  is in  $A$ .

#### WHERE

$U$  is the universal set, **listed in a fixed order** that is agreed before anything is encoded. Change the order and every bit string changes.  $n$  is the number of elements in  $U$ , so every bit string for that  $U$  is exactly  $n$  bits long.  $b_i$  is the bit in position  $i$ , reading left to right from  $i = 1$ .

The operations then become **bitwise**:

| SET OPERATION   | BIT OPERATION | RULE PER POSITION         |
|-----------------|---------------|---------------------------|
| $A \cup B$      | bitwise OR    | 1 if either bit is 1      |
| $A \cap B$      | bitwise AND   | 1 only if both bits are 1 |
| $A'$            | bitwise NOT   | flip every bit            |
| $A \triangle B$ | bitwise XOR   | 1 if the bits differ      |

◆ **Memory hook:** union is or, intersection is and, complement is not. The set operations and the logic connectives are the same operations wearing different clothes, which is exactly what section 1.3 and 2.2 kept saying.

# WORKED EXAMPLE

# METHOD

**GIVEN**  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  in increasing order,  $A = \{1, 3, 5, 7, 9\}$ .

**FIND** The bit string of A.

**RULE** 1 where the position's element is in A, 0 otherwise.

Walk the positions in order. Position 1 holds the element 1, which is in A, so 1. Position 2 holds 2, not in A, so 0. Continuing:

| POSITION | 1   | 2  | 3   | 4  | 5   | 6  | 7   | 8  | 9   | 10 |
|----------|-----|----|-----|----|-----|----|-----|----|-----|----|
| ELEMENT  | 1   | 2  | 3   | 4  | 5   | 6  | 7   | 8  | 9   | 10 |
| IN A?    | yes | no | yes | no | yes | no | yes | no | yes | no |
| BIT      | 1   | 0  | 1   | 0  | 1   | 0  | 1   | 0  | 1   | 0  |

So the bit string is **1010101010**, which the papers group in fours from the left as **10 1010 1010**. The grouping is only for legibility, like spaces in a phone number.

# AS PRINTED

2023/2024 · Q3(A)

3. (a) Suppose that a universal set,  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ . Express the following sets with bit strings where the  $i^{\text{th}}$  bit in the string is 1 if  $i$  is in the set and 0 otherwise.

(i)  $\{3, 4, 5\}$  **(1 marks)**

(ii)  $\{1, 3, 6, 10\}$  **(1 marks)**

(iii)  $\{2, 3, 4, 7, 8, 9\}$  **(1 marks)**

# NOTE ON THE PAPER

"(1 marks)" is printed three times: singular value, plural noun. Harmless. The question helpfully defines the convention itself, so use exactly the ordering it gives, with position  $i$  holding the element  $i$ .

# BREAK IT DOWN

**GIVEN**  $U = \{1, \dots, 10\}$ , and the convention that bit  $i$  is 1 when  $i$  is in the set.

**ASKED** **Express** three sets as bit strings. Ten bits each, no working needed.

**MARKS SAY** 1 mark each, so speed matters more than presentation. Write the ten bits and move on.

**TEACHES** This section, 2.3.

**TRAP** Producing nine or eleven bits. Count them before moving on; the length must equal  $n(U)$  every time.

### WORKED EXAMPLE

PQ 23/24 Q3(A)

(i) {3, 4, 5}. Ones in positions 3, 4, 5, zeros elsewhere.

**0011100000** (grouped: 00 1110 0000)

(ii) {1, 3, 6, 10}. Ones in positions 1, 3, 6, 10.

**1010010001** (grouped: 10 1001 0001)

(iii) {2, 3, 4, 7, 8, 9}. Ones in positions 2, 3, 4, 7, 8, 9.

**0111001110** (grouped: 01 1100 1110)

**Check every time:** count the 1s and compare with the size of the set. Three, four and six ones respectively, matching  $n = 3, 4, 6$ . Then count the total length: ten in each. Two checks, three seconds, and they catch every slip this question can produce.

### AS PRINTED

2024/2025 · Q3(D)

d. The bit string for the set {1, 3, 5, 7, 9} with universal set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} is 10 1010 1010.

i. What is the bit string for the complement of this set?

**2.5 marks**

### NOTE ON THE PAPER

The universal set closes with two round brackets instead of a brace, and there is no space before the 10. Both are typing slips and change nothing:  $U$  is {1, 2, ..., 10}.

### WORKED EXAMPLE

PQ 24/25 Q3(D)(I)

**GIVEN** The bit string 1010101010 for {1, 3, 5, 7, 9} over  $U = \{1, \dots, 10\}$ .

**FIND** The bit string of the complement.

**RULE** Complement is bitwise NOT: flip every bit.

Flipping 1010101010 gives **0101010101**, grouped as **01 0101 0101**.

**Reading it back as a set**, the 1s now sit in positions 2, 4, 6, 8, 10, so the complement is {2, 4, 6, 8, 10}, the even numbers up to 10. That is exactly right, since the original held the odd ones.

Always convert back and sanity-check the meaning. It takes five seconds and turns a mechanical answer into one you know is correct.

### AS PRINTED

2024/2025 · Q3(D)(II)

ii. The bit strings for the sets {1, 2, 3, 4, 5} and {1, 3, 5, 7, 9} are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

**2.5 marks**

### BREAK IT DOWN

|                  |                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Two ten-bit strings, already computed for you.                                                                                                                                         |
| <b>ASKED</b>     | Find the union and intersection, and the method is prescribed: <b>use bit strings</b> . So do the bitwise work and show it.                                                            |
| <b>MARKS SAY</b> | 2.5 marks for two operations. Show the two strings lined up and the result underneath; that layout is the working.                                                                     |
| <b>TEACHES</b>   | This section, 2.3.                                                                                                                                                                     |
| <b>TRAP</b>      | Converting to sets, doing the union there, and converting back. That gets the right answer by the wrong method and risks losing the method marks, since the question named the method. |

### WORKED EXAMPLE

PQ 24/25 Q3(D)(II)

Line the strings up in columns and apply the rule position by position.

| POSITION    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-------------|---|---|---|---|---|---|---|---|---|----|
| {1,2,3,4,5} | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0  |
| {1,3,5,7,9} | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0  |
| OR (UNION)  | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0  |
| AND (MEET)  | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0  |

**Union** = 1111101010, grouped **11 1110 1010**. Reading the 1s back: positions 1, 2, 3, 4, 5, 7, 9, so the union is **{1, 2, 3, 4, 5, 7, 9}**.

**Intersection** = 1010100000, grouped **10 1010 0000**. The 1s sit at 1, 3, 5, so the intersection is **{1, 3, 5}**.

**Check.**  $n(\text{union}) = 7$ ,  $n(\text{first}) = 5$ ,  $n(\text{second}) = 5$ ,  $n(\text{intersection}) = 3$ , and  $5 + 5 - 3 = 7$ . The counting formula from 2.2 agrees.

**Now redo it:** compute the symmetric difference of the same two sets by XOR. You should get 01 0100 1010, that is {2, 4, 7, 9}.

## The power set

### ● MUST MEMORISE

### POWER SET

The **power set** of  $A$ , written  **$P(A)$** , is the set of *all* subsets of  $A$ , including the empty set and  $A$  itself.

If  $n(A) = n$  then  **$n(P(A)) = 2^n$** .

**$n$**  the number of elements in  $A$ .  **$2^n$**  the number of subsets, because each element is independently either in or out, giving two choices repeated  $n$  times.

◆ **Memory hook:** the power set is one binary decision per element, in or out. That is why its size is  $2^n$ , and why bit strings and power sets are the same idea: **every subset is a bit string**, so the number of subsets is the number of  $n$ -bit strings.

**GIVEN**  $A = \{1, 2, 3\}$ .

**FIND**  $P(A)$ .

**RULE**  $2^3 = 8$  subsets. List by size so none is missed.

**Listing by size** is the safe method, because you can count each row against the binomial coefficients of Part Five.

| SIZE | SUBSETS                     | HOW MANY |
|------|-----------------------------|----------|
| 0    | $\emptyset$                 | 1        |
| 1    | $\{1\}, \{2\}, \{3\}$       | 3        |
| 2    | $\{1,2\}, \{1,3\}, \{2,3\}$ | 3        |
| 3    | $\{1,2,3\}$                 | 1        |

Total  $1 + 3 + 3 + 1 = 8 = 2^3$ . Correct.

**Listing by bit pattern** is the method a program uses, and it explains the order the 2024/2025 paper prints. Count from 000 to 111 in binary, reading each pattern as "which elements are in":

| COUNT | BITS (FOR 3,2,1) | SUBSET        |
|-------|------------------|---------------|
| 0     | 000              | $\{ \}$       |
| 1     | 001              | $\{1\}$       |
| 2     | 010              | $\{2\}$       |
| 3     | 011              | $\{1, 2\}$    |
| 4     | 100              | $\{3\}$       |
| 5     | 101              | $\{1, 3\}$    |
| 6     | 110              | $\{2, 3\}$    |
| 7     | 111              | $\{1, 2, 3\}$ |

That is exactly the order printed in 2024/2025 Q3(b), and knowing why makes the program in Part Eleven write itself.

### ⚠ TRAP

### DO NOT FORGET THE TWO FREE SUBSETS

Every power set contains  $\emptyset$  and the whole set  $A$ . They are the two most commonly forgotten, and forgetting either makes the count wrong, which is the first thing a marker checks.

Count first, then list. If you expect  $2^n$  and you have written  $2^n - 1$ , you know exactly what is missing before you hand it in.

## The Cartesian product

### ● MUST MEMORISE

### CARTESIAN PRODUCT

The **Cartesian product** of A and B, written  $A \times B$ , is the set of all **ordered pairs** whose first entry comes from A and second from B:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

$$n(A \times B) = n(A) \times n(B).$$

**(a, b)** an ordered pair: round brackets, and the order matters, so  $(1, 2) \neq (2, 1)$ .  $n(A) \times n(B)$  the count, by the product rule of Part Five:  $n(A)$  choices for the first entry, and for each of them  $n(B)$  choices for the second.

◆ **Memory hook:** braces for a set, where order does not matter; round brackets for a pair, where it does.  $A \times B$  is every way of picking one from each, so it is a grid, and a grid's area is the product of its sides.

### AS PRINTED

2025/2026 · Q2(B)

(b) A university has the set of students:

$S = \{\text{John, Mary, Peter}\}$  and the set of courses:

$C = \{\text{CSC701, CSC702}\}$

Find the Cartesian product  $S \times C$

(2 marks)

### BREAK IT DOWN

**GIVEN** S with 3 students, C with 2 courses.

**ASKED** **Find**  $S \times C$ . Note the order: students first, courses second, because it says  $S \times C$  and not  $C \times S$ .

**MARKS SAY** 2 marks for six pairs. List them all; a formula with no list does not answer "find".

**TEACHES** This section, 2.3.

**TRAP** Writing the pairs with braces instead of round brackets, or writing them in the order (course, student). Both are marked wrong because the pair is ordered and the question fixed the order.

**GIVEN**  $S = \{\text{John, Mary, Peter}\}$ ,  $C = \{\text{CSC701, CSC702}\}$ .

**FIND**  $S \times C$ .

**RULE** Every ordered pair with the student first. Expect  $n(S) \times n(C) = 3 \times 2 = 6$  pairs.

Take each student in turn and pair with every course. Working systematically is what stops a pair going missing.

$S \times C = \{(\text{John, CSC701}), (\text{John, CSC702}), (\text{Mary, CSC701}), (\text{Mary, CSC702}), (\text{Peter, CSC701}), (\text{Peter, CSC702})\}$

Six pairs, matching  $3 \times 2$ . It is a grid: three rows of students, two columns of courses.

**What it means here:**  $S \times C$  is every possible student and course combination, that is, every enrolment that *could* exist. The enrolments that actually exist form a subset of it, and a subset of a Cartesian product is precisely what Part Three calls a **relation**. That single sentence is the bridge from sets to relations, and it is worth writing in your answer.

**Now redo it:** write  $C \times S$  and confirm it is a different set with the same six students and courses. That difference is why the product is not commutative.

a. Explain why  $(A \times B) \times (C \times D)$  and  $A \times (B \times C) \times D$  are not the same.

**3.5 marks**

#### BREAK IT DOWN

**GIVEN** Two ways of bracketing a product of four sets.

**ASKED** **Explain why** they differ. This is an argument about the *shape* of the elements, not a calculation.

**MARKS SAY** 3.5 marks for a short explanation, so they want the structure spelled out, ideally with a small concrete example.

**TEACHES** This section, 2.3.

**TRAP** Answering "because the Cartesian product is not associative" and stopping. That names the fact without explaining it. The marks are in showing what a typical element of each side looks like.



**The point: an ordered pair has exactly two slots, and what sits in each slot is decided by the brackets.**

**Left side,  $(A \times B) \times (C \times D)$ .** This is a product of two things: the set  $A \times B$ , and the set  $C \times D$ . So a typical element is an ordered pair whose first entry is itself a pair from  $A \times B$  and whose second entry is a pair from  $C \times D$ :

**$((a, b), (c, d))$**

Two slots at the top level, each holding a pair.

**Right side,  $A \times (B \times C) \times D$ .** This is a product of three things:  $A$ , then  $B \times C$ , then  $D$ . A typical element is an ordered *triple*:

**$(a, (b, c), d)$**

Three slots at the top level, with the middle one holding a pair.

**Why that settles it.** An element of the left is a pair; an element of the right is a triple. A pair is never equal to a triple, whatever is inside them, so the two sets share no elements at all and cannot be equal. The Cartesian product is **not associative**.

**A concrete example seals the marks.** Take  $A = B = C = D = \{1\}$ . Then

$$(A \times B) \times (C \times D) = \{((1,1), (1,1))\}$$

$$A \times (B \times C) \times D = \{(1, (1,1), 1)\}$$

Both sets have exactly one element, so they are *equivalent*, but those elements are differently shaped, so the sets are not *equal*. Section 2.1's distinction between equal and equivalent is doing real work here, and saying so is worth a mark.

**Worth adding:** because the counts multiply the same way either side,  $n$  of both is  $n(A)n(B)n(C)n(D)$ . Equal size, different structure. That is why people expect them to be the same and why they are not.

## RECALL AND CHECK

1. Write the bit string of  $\{2, 5\}$  over  $U = \{1, 2, 3, 4, 5\}$ .
2. How many elements are in  $P(A)$  when  $n(A) = 6$ ?
3.  $n(A) = 4$  and  $n(B) = 7$ . Find  $n(A \times B)$ .
4. Which bitwise operation gives the intersection?

### ANSWERS

1. 01001.
2.  $2^6 = 64$ .
3.  $4 \times 7 = 28$ .
4. AND.

### FILL IN THE GAP

1. In a bit string, position  $i$  holds a 1 when the  $i$ -th element of \_\_\_\_\_ is in the set.
2. The power set of  $A$  is the set of all \_\_\_\_\_ of  $A$ , and it has \_\_\_\_\_ elements.
3.  $A \times B$  is the set of all \_\_\_\_\_  $(a, b)$ .
4. Union corresponds to the bitwise \_\_\_\_\_ operation.

---

### ANSWERS

1. the universal set  $U$ . 2. subsets,  $2^n$ . 3. ordered pairs. 4. OR.

## PART THREE

# Relations, Matrices and Ordered Sets

*A relation is any set of ordered pairs. From that one idea come relational databases, dependency graphs, equivalence of states, and the ordering that makes a scheduler work.*

$$R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$$

|   |   |   |   |  |            |                       |               |
|---|---|---|---|--|------------|-----------------------|---------------|
| 1 | 1 | 0 | 0 |  | reflexive  | <input type="radio"/> | antisymmetric |
| 1 | 1 | 0 | 0 |  | symmetric  | <input type="radio"/> | equivalence   |
| 0 | 0 | 0 | 1 |  | transitive | <input type="radio"/> | partial order |
| 0 | 0 | 0 | 0 |  |            |                       |               |

# Relations and Their Representations

A relation records which things are connected to which. There are four ways to write one down, and the exam uses all four.

By the end of this section you will be able to define a relation on a set, list the pairs of a relation given by a rule, and move freely between the set of pairs, the matrix, the arrow diagram and the directed graph.

## NOT IN THE COURSE TEXT

The official 161-page text never defines a relation as a set of ordered pairs, never mentions reflexive, symmetric, antisymmetric or transitive, and never mentions partial or total orders. Every paper we hold examines them: 2024/2025 gives 7.5 marks to it, 2023/2024 gives 10, 2025/2026 gives 6. Everything you need is taught here from zero.

## What a relation is

### TEACH

### BUILT STRAIGHT OUT OF PART TWO

Section 2.3 ended with the Cartesian product  $A \times B$ , the set of *every* ordered pair with a first entry from  $A$  and a second from  $B$ . That is the set of all connections that *could* exist.

A **relation** is simply a choice of which of them actually do. It is any subset of  $A \times B$ .

If  $A = \{\text{John, Mary}\}$  and  $B = \{\text{CSC701, CSC702}\}$ , the product has four pairs. The relation "is enrolled in" might be  $\{(\text{John, CSC701}), (\text{Mary, CSC702})\}$ : a subset, recording the two enrolments that exist.

That is the whole definition. Everything in this part is a consequence.

### ● MUST MEMORISE

### DEFINITION

A **relation** from  $A$  to  $B$  is a subset of  $A \times B$ .

A **relation on  $A$**  is a subset of  $A \times A$ , that is, a relation from a set to itself. Almost every exam question is of this kind.

We write  **$a R b$**  to mean  $(a, b) \in R$ , read "a is related to b".

A relation on a set of  $n$  elements is a subset of  $A \times A$ , which has  $n^2$  pairs, so there are  $2^{n^2}$  different relations on it.

◆ **Memory hook:** the Cartesian product lists every possible arrow; a relation is the set of arrows you actually drew.

### ⚠ TRAP

### THE ORDER IN THE PAIR IS THE DIRECTION OF THE ARROW

$(a, b) \in R$  means  $a$  is related to  $b$ , **not**  $b$  to  $a$ . In the relation "is the parent of",  $(\text{Ada, Ben})$  says Ada is Ben's parent, and  $(\text{Ben, Ada})$  would say the reverse.

Every property in section 3.2 is a statement about which direction the pairs run, so getting the convention backwards inverts every answer. Fix it now: **first entry acts on second entry**.

## The four representations

### TEACH

### ONE RELATION, FOUR PICTURES

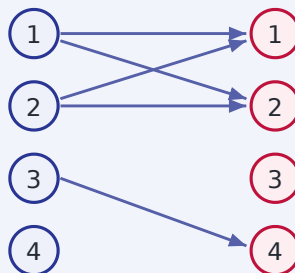
Take  $A = \{1, 2, 3, 4\}$  and  $R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$ . Here is the same relation written four ways. Learn to convert between them instantly, because different exam questions hand you different forms.

#### 1. As a set of ordered pairs

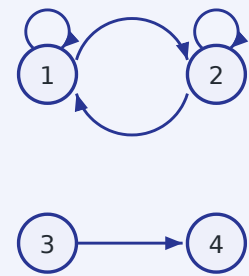
$$R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$$

#### 2. As a matrix (row = first entry, column = second)

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 |
| 2 | 1 | 1 | 0 | 0 |
| 3 | 0 | 0 | 0 | 1 |
| 4 | 0 | 0 | 0 | 0 |



#### 4. As a directed graph



loops at 1 and 2 = the pair

### ● MUST MEMORISE

### THE RELATION MATRIX

For a relation  $R$  on  $A = \{a_1, \dots, a_n\}$ , the **relation matrix** (also called the logical matrix)  $M$  has

$$M_{ij} = 1 \text{ if } (a_i, a_j) \in R, \text{ and } 0 \text{ otherwise.}$$

**i** the row, indexing the **first** entry of the pair. **j** the column, indexing the **second**.  $M_{ij}$  the entry in row  $i$ , column  $j$ .

Every property of a relation becomes a visible pattern in this matrix, which is what makes 3.2 easy rather than fiddly.

◆ **Memory hook: row acts on column.** Row  $i$ , column  $j$  means "does  $a_i$  relate to  $a_j$ ". Same order as the pair.

## Listing a relation given by a rule

### AS PRINTED

2023/2024 · Q3(D)(I)

(i) Which ordered pairs are in the relation  $R = \{(a, b) \mid a \text{ divides } b\}$ ?

(2 marks)

#### NOTE ON THE PAPER

The set is given earlier in the same question as  $A = \{1, 2, 3, 4\}$ , so the relation is on that set. "a divides b" means b is a whole multiple of a, so 2 divides 4 but 4 does not divide 2, so the pairs run one way only. That is the point of choosing divisibility, as section 3.2 will show.

**BREAK IT DOWN****GIVEN**  $A = \{1, 2, 3, 4\}$  and the rule "a divides b".**ASKED** Which ordered pairs are in the relation. So list them, in braces, as a set of pairs.**MARKS SAY** 2 marks for a list. Work through it systematically rather than by inspiration.**TEACHES** This section, 3.1.**TRAP** Forgetting that every number divides itself, so (1,1), (2,2), (3,3), (4,4) are all in. And forgetting that 1 divides everything.**WORKED EXAMPLE****PQ 23/24 Q3(D)(I)****GIVEN**  $A = \{1, 2, 3, 4\}$ ,  $R = \{(a, b) \mid a \text{ divides } b\}$ .**FIND** Every pair in R.**RULE** a divides b when  $b \div a$  leaves no remainder. Test all  $4 \times 4 = 16$  pairs.

Take each a in turn and ask which b it divides.

| A | DIVIDES WHICH OF 1, 2, 3, 4?            | PAIRS                      |
|---|-----------------------------------------|----------------------------|
| 1 | all of them, since 1 divides everything | (1,1), (1,2), (1,3), (1,4) |
| 2 | 2 and 4                                 | (2,2), (2,4)               |
| 3 | 3 only                                  | (3,3)                      |
| 4 | 4 only                                  | (4,4)                      |

 $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$ 

Eight pairs. As a matrix, with rows and columns in the order 1, 2, 3, 4:

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 2 | 0 | 1 | 0 | 1 |
| 3 | 0 | 0 | 1 | 0 |
| 4 | 0 | 0 | 0 | 1 |

Two patterns in that matrix are worth noticing now, because section 3.2 gives each of them a name and tests you on it. The whole diagonal is 1, because every number divides itself. And every 1 sits on or above the diagonal, so no two different numbers divide each other.

**Now redo it:** list the same relation on  $\{1, 2, 3, 4, 5, 6\}$ . You should get 14 pairs.

(c) Consider these relations on the set of integers:

$$R1 = \{(a, b) \mid a \leq b\} \quad R2 = \{(a, b) \mid a > b\}$$

$$R3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R4 = \{(a, b) \mid a = b\} \quad R5 = \{(a, b) \mid a = b + 1\}$$

$$R6 = \{(a, b) \mid a + b \leq 3\}$$

(i) List 5 ordered pairs in each of the relation.

(5marks)

(ii) Which of the relations are: (a) Reflexive (b) Symmetric (c) Transitive (6½marks)

#### NOTE ON THE PAPER

The domain here is **THE INTEGERS**, not a small finite set, so these relations are infinite. You cannot list them, only sample them, which is exactly why part (i) asks for five pairs each rather than all of them. Part (ii) is answered in section 3.2, once the properties are defined.

#### WORKED EXAMPLE

PQ 20/21 Q5(C)(I)

Five pairs from each. Choose a spread, including negatives and a repeated value, because that also prepares part (ii).

| RELATION | RULE                       | FIVE PAIRS                              |
|----------|----------------------------|-----------------------------------------|
| R1       | $a \leq b$                 | (1,1), (1,2), (0,5), (-3,2), (-7,-7)    |
| R2       | $a > b$                    | (2,1), (5,0), (0,-4), (-2,-9), (10,3)   |
| R3       | $a = b \text{ or } a = -b$ | (3,3), (3,-3), (-5,5), (0,0), (7,7)     |
| R4       | $a = b$                    | (0,0), (1,1), (-2,-2), (9,9), (100,100) |
| R5       | $a = b + 1$                | (1,0), (2,1), (0,-1), (-4,-5), (8,7)    |
| R6       | $a + b \leq 3$             | (1,1), (0,3), (-5,2), (2,1), (-1,-1)    |

**Check each one before writing it.** For R5, (1,0) works because  $1 = 0 + 1$ . For R6, (0,3) works because  $0 + 3 = 3$ , and the rule says " $\leq 3$ ", so equality is allowed. That boundary case is the one people get wrong.

**Now redo it:** give five more pairs for each, all different from these.

## RECALL AND CHECK

1. Complete the definition: a relation on  $A$  is a subset of \_\_\_\_.
2. In the matrix of  $R$ , what does a 1 in row 3, column 2 mean?
3. How many relations are there on a set with 3 elements?
4. Is  $(2,4)$  in the relation "a divides b"? Is  $(4,2)$ ?

### ANSWERS

1.  $A \times A$ .
2. That  $(a_3, a_2) \in R$ , so the third element relates to the second.
3.  $2^{3^2} = 2^9 = 512$ .
4.  $(2,4)$  yes, since  $4 = 2 \times 2$ .  $(4,2)$  no, since 2 is not a whole multiple of 4.

## FILL IN THE GAP

1. A relation from  $A$  to  $B$  is a \_\_\_\_\_ of  $A \times B$ .
2. In a relation matrix, the row indexes the \_\_\_\_\_ entry of the pair.
3. A loop in the directed graph of  $R$  corresponds to a pair of the form \_\_\_\_\_.
4. There are \_\_\_\_\_ relations on a set of  $n$  elements.

### ANSWERS

1. subset. 2. first. 3.  $(a, a)$ . 4.  $2^{n^2}$ .

## PART THREE · 3.2

# The Five Properties of a Relation

*Reflexive, symmetric, antisymmetric, transitive and irreflexive. Three of the five papers we hold ask you to classify a list of relations against these, so this is the highest-value page in Part Three.*

*By the end of this section you will be able to state each property precisely, test any relation against all five quickly and reliably, spot each one in a matrix at a glance, and produce the counterexample that proves a property fails.*



## The five definitions

### ● MUST MEMORISE

### THE FIVE PROPERTIES

Throughout,  $R$  is a relation on a set  $A$ .

| PROPERTY             | DEFINITION                                                       | IN PLAIN WORDS                                   |
|----------------------|------------------------------------------------------------------|--------------------------------------------------|
| <b>Reflexive</b>     | $(a, a) \in R$ for <b>every</b> $a \in A$                        | everything relates to itself, with no exceptions |
| <b>Irreflexive</b>   | $(a, a) \notin R$ for every $a \in A$                            | nothing relates to itself                        |
| <b>Symmetric</b>     | whenever $(a, b) \in R$ , also $(b, a) \in R$                    | every arrow has a matching return arrow          |
| <b>Antisymmetric</b> | whenever $(a, b) \in R$ and $(b, a) \in R$ , then $a = b$        | no two <i>different</i> elements point both ways |
| <b>Transitive</b>    | whenever $(a, b) \in R$ and $(b, c) \in R$ , also $(a, c) \in R$ | any two-step path has a one-step shortcut        |

◆ **Memory hook:** reflexive looks in a mirror, symmetric goes both ways, antisymmetric forbids both ways unless it is the same thing twice, transitive takes shortcuts.

### ⚠ TRAP

### ANTISYMMETRIC IS NOT THE OPPOSITE OF SYMMETRIC

This is the mistake that costs the most marks in this whole part.

A relation can be **both** symmetric and antisymmetric. Take  $R = \{(1,1), (2,2)\}$  on  $\{1, 2, 3\}$ . Every arrow has its return arrow, since each is its own return, so it is symmetric. And no two *different* elements point both ways, so it is antisymmetric too.

A relation can also be **neither**. Take  $R = \{(1,2), (2,1), (2,3)\}$ . It fails symmetry, because  $(2,3)$  has no  $(3,2)$ . It fails antisymmetry, because  $(1,2)$  and  $(2,1)$  are both in with  $1 \neq 2$ .

The four combinations all occur. So test the two properties **separately** and never infer one from the other.

### ⚠ TRAP

### REFLEXIVE NEEDS EVERY ELEMENT

$R = \{(1,1), (2,2), (3,3)\}$  on  $A = \{1, 2, 3, 4\}$  is **not** reflexive, because  $(4,4)$  is missing. Three out of four is a fail.

This is exactly the trap in 2025/2026 Q2(c), and it is why "determine whether  $R$  is reflexive" is worth two marks rather than one: the mark is for naming the missing pair, not for the verdict.

Also note that **not reflexive** and **irreflexive** are different.  $R = \{(1,1), (2,3)\}$  on  $\{1,2,3\}$  is not reflexive (2 and 3 lack loops) and not irreflexive either (1 has one). Most relations are neither.

## Seeing the properties in the matrix

### TEACH

### THE FASTEST RELIABLE METHOD

Write the matrix and every property becomes something you can see. This is far quicker than checking pairs one at a time, and much harder to get wrong.

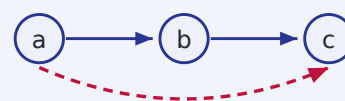
| PROPERTY      | WHAT TO LOOK AT                                    | PASSES WHEN                                                 |
|---------------|----------------------------------------------------|-------------------------------------------------------------|
| Reflexive     | the main diagonal                                  | every diagonal entry is 1                                   |
| Irreflexive   | the main diagonal                                  | every diagonal entry is 0                                   |
| Symmetric     | the matrix versus its mirror image in the diagonal | $M_{ij} = M_{ji}$ everywhere                                |
| Antisymmetric | each off-diagonal pair of positions                | never both $M_{ij}$ and $M_{ji}$ equal to 1 when $i \neq j$ |
| Transitive    | every two-step path                                | $M_{ij} = M_{jk} = 1$ forces $M_{ik} = 1$                   |

**Reflexive: diagonal all 1**    **Symmetric: mirror across diagonal**    **Transitive: shortcut must exist**

$\boxed{1} \ 0 \ 1$   
 $0 \ 1 \ \boxed{0}$   
 $1 \ 1 \ 1 \ \boxed{\phantom{0}}$

$\begin{matrix} 1 & 1 & 0 \\ \boxed{1} & 0 & 1 \\ 0 & 1 & 1 \end{matrix}$

$a \rightarrow b$  and  $b \rightarrow c$   
 forces  $a \rightarrow c$



the dashed arrow must be present

### ● MUST MEMORISE

### HOW TO DISPROVE EACH PROPERTY

Every one of these properties is a "for all" statement, so a **single counterexample** disproves it, and quoting that counterexample is what earns the mark. Never write "R is not transitive" alone.

| TO SHOW R IS NOT ... | PRODUCE                                                  |
|----------------------|----------------------------------------------------------|
| reflexive            | one element $a$ with $(a, a) \notin R$                   |
| symmetric            | one pair $(a, b) \in R$ with $(b, a) \notin R$           |
| antisymmetric        | a pair $(a, b)$ and $(b, a)$ both in $R$ with $a \neq b$ |
| transitive           | $(a, b)$ and $(b, c)$ in $R$ with $(a, c) \notin R$      |

◆ **Memory hook:** one bad case kills a "for all". Find it, write it, move on.

## The repeated question

AS PRINTED

2024/2025 · Q4(B)

b. Consider the following relations on  $\{1, 2, 3, 4\}$ :

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$ ,

$R2 = \{(1, 1), (1, 2), (2, 1)\}$ ,

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$ ,

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$ ,

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$ ,

$R6 = \{(3, 4)\}$ .

i. Which of these relations are reflexive?

ii. Which of these relations are symmetric?

iii. Which of these relations are antisymmetric?

iv. Which of these relations are transitive?

v. Which of these relations are reflexive?

7.5 marks

### NOTE ON THE PAPER

**PARTS I AND V ARE THE SAME QUESTION.** Both read "Which of these relations are reflexive?" Since ii, iii and iv cover symmetric, antisymmetric and transitive, part v was almost certainly meant to be **IRREFLEXIVE** or **ASYMMETRIC**. Answer what is printed: give the reflexive answer again, and add one line noting the apparent duplication and supplying the irreflexive answer as well. That covers you whichever was intended, and costs three lines. Never silently answer a question the paper did not ask.

### BREAK IT DOWN

**GIVEN** Six relations on  $\{1, 2, 3, 4\}$ , listed as sets of pairs.

**ASKED** **Which of these relations are** [property]. A list of names is the answer. Reasons are not demanded, but one short reason per failure is cheap insurance and shows method.

**MARKS SAY** 7.5 marks over five parts is 1.5 each, so about six verdicts per part. That is a lot of checking for the time, which is why the matrix method matters.

**TEACHES** This section, 3.2.

**TRAP** Confusing symmetric with antisymmetric, and forgetting that a relation can be both or neither. Also:  $R6 = \{(3,4)\}$  is a single pair, and single-pair relations are vacuously symmetric-free, antisymmetric and transitive. Vacuous cases are where marks go missing.

**Method.** Build the matrix of each relation first. Then read the diagonal for reflexive, compare with the mirror for symmetric and antisymmetric, and check two-step paths for transitive.

**The six matrices** (rows and columns in the order 1, 2, 3, 4):

| R1 | 1 | 2 | 3 | 4 | R2 | 1 | 2 | 3 | 4 | R3 | 1 | 2 | 3 | 4 |
|----|---|---|---|---|----|---|---|---|---|----|---|---|---|---|
| 1  | 1 | 1 | 0 | 0 | 1  | 1 | 1 | 0 | 0 | 1  | 1 | 1 | 0 | 1 |
| 2  | 1 | 1 | 0 | 0 | 2  | 1 | 0 | 0 | 0 | 2  | 1 | 1 | 0 | 0 |
| 3  | 0 | 0 | 0 | 1 | 3  | 0 | 0 | 0 | 0 | 3  | 0 | 0 | 1 | 0 |
| 4  | 1 | 0 | 0 | 1 | 4  | 0 | 0 | 0 | 0 | 4  | 1 | 0 | 0 | 1 |

| R4 | 1 | 2 | 3 | 4 | R5 | 1 | 2 | 3 | 4 | R6 | 1 | 2 | 3 | 4 |
|----|---|---|---|---|----|---|---|---|---|----|---|---|---|---|
| 1  | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 1 | 1 | 1  | 0 | 0 | 0 | 0 |
| 2  | 1 | 0 | 0 | 0 | 2  | 0 | 1 | 1 | 1 | 2  | 0 | 0 | 0 | 0 |
| 3  | 1 | 1 | 0 | 0 | 3  | 0 | 0 | 1 | 1 | 3  | 0 | 0 | 0 | 1 |
| 4  | 1 | 1 | 1 | 0 | 4  | 0 | 0 | 0 | 1 | 4  | 0 | 0 | 0 | 0 |

(i) Which are reflexive?

Read each diagonal and require all four 1s.

| R  | DIAGONAL   | REFLEXIVE?                      |
|----|------------|---------------------------------|
| R1 | 1, 1, 0, 1 | no, (3,3) missing               |
| R2 | 1, 0, 0, 0 | no, (2,2), (3,3), (4,4) missing |
| R3 | 1, 1, 1, 1 | <b>yes</b>                      |
| R4 | 0, 0, 0, 0 | no, and in fact irreflexive     |
| R5 | 1, 1, 1, 1 | <b>yes</b>                      |
| R6 | 0, 0, 0, 0 | no, and irreflexive             |

**Answer: R3 and R5.**

(ii) Which are symmetric?

Check  $M_{ij} = M_{ji}$  for every off-diagonal position.

| R  | TEST                                                 | SYMMETRIC? |
|----|------------------------------------------------------|------------|
| R1 | (3,4) is in but (4,3) is not                         | no         |
| R2 | (1,2) and (2,1) both in; no other off-diagonal pairs | yes        |
| R3 | (1,4) and (4,1) both in, (1,2) and (2,1) both in     | yes        |
| R4 | (2,1) is in but (1,2) is not                         | no         |
| R5 | (1,2) is in but (2,1) is not                         | no         |
| R6 | (3,4) is in but (4,3) is not                         | no         |

**Answer: R2 and R3.**

(iii) Which are antisymmetric?

Look for any off-diagonal position where both  $M_{ij}$  and  $M_{ji}$  are 1. One such pair kills it.

| R  | TEST                                | ANTISYMMETRIC? |
|----|-------------------------------------|----------------|
| R1 | (1,2) and (2,1) both in, $1 \neq 2$ | no             |
| R2 | (1,2) and (2,1) both in             | no             |
| R3 | (1,2) and (2,1) both in             | no             |
| R4 | no returning pair anywhere          | yes            |
| R5 | no returning pair anywhere          | yes            |
| R6 | only one pair, so nothing to return | yes, vacuously |

**Answer: R4, R5 and R6.**

(iv) Which are transitive?

For each, hunt for a two-step path with no shortcut.

| R  | TEST                                                                                                                          | TRANSITIVE?    |
|----|-------------------------------------------------------------------------------------------------------------------------------|----------------|
| R1 | (4,1) and (1,2) are in, but (4,2) is not                                                                                      | no             |
| R2 | (2,1) and (1,2) are in, but (2,2) is not                                                                                      | no             |
| R3 | (4,1) and (1,2) are in, but (4,2) is not                                                                                      | no             |
| R4 | (4,3) and (3,1) are in, and (4,1) is in; (4,3) and (3,2) in, and (4,2) in; (3,2) and (2,1) in, and (3,1) in. All paths close. | yes            |
| R5 | every path closes; it is the relation $a \leq b$                                                                              | yes            |
| R6 | only (3,4); no two-step path exists                                                                                           | yes, vacuously |

**Answer: R4, R5 and R6.**

(v) Which are reflexive? (as printed, repeating part i)

**Answer: R3 and R5,** as in part (i).

**Note for the marker:** parts (i) and (v) are identically worded on the paper. If (v) was intended as **irreflexive**, the answer is **R4 and R6**, since those are the two whose diag-

onals are entirely 0.

**Summary table**, which is what you should transcribe into the answer book:

|                      | R1 | R2         | R3         | R4         | R5         | R6         |
|----------------------|----|------------|------------|------------|------------|------------|
| <b>REFLEXIVE</b>     | no | no         | <b>yes</b> | no         | <b>yes</b> | no         |
| <b>SYMMETRIC</b>     | no | <b>yes</b> | <b>yes</b> | no         | no         | no         |
| <b>ANTISYMMETRIC</b> | no | no         | no         | <b>yes</b> | <b>yes</b> | <b>yes</b> |
| <b>TRANSITIVE</b>    | no | no         | no         | <b>yes</b> | <b>yes</b> | <b>yes</b> |
| <b>IRREFLEXIVE</b>   | no | no         | no         | <b>yes</b> | no         | <b>yes</b> |

**Now redo it:** cover everything and rebuild the summary table from the six lists of pairs. Twelve minutes is a realistic target, which tells you how much of the 44 minutes this question would consume.

### WORKED EXAMPLE

PQ 20/21 Q5(C)(II)

The same skill on the infinite relations from 3.1, where you cannot write a matrix and must argue instead.

| R                          | REFLEXIVE?                         | SYMMETRIC?                                     | TRANSITIVE?                                            |
|----------------------------|------------------------------------|------------------------------------------------|--------------------------------------------------------|
| R1: $a \leq b$             | <b>yes</b> , $a \leq a$ always     | no, $1 \leq 2$ but not $2 \leq 1$              | <b>yes</b> , $a \leq b$ and $b \leq c$ give $a \leq c$ |
| R2: $a > b$                | no, $a > a$ is never true          | no, $2 > 1$ but not $1 > 2$                    | <b>yes</b> , $a > b$ and $b > c$ give $a > c$          |
| R3: $a = b$<br>or $a = -b$ | <b>yes</b> , $a = a$ always        | <b>yes</b> , the rule is unchanged by swapping | <b>yes</b> , see below                                 |
| R4: $a = b$                | <b>yes</b>                         | <b>yes</b>                                     | <b>yes</b>                                             |
| R5: $a = b + 1$            | no, $a = a + 1$ is impossible      | no, $2 = 1 + 1$ but $1 \neq 2 + 1$             | no, $(3, 2)$ and $(2, 1)$ in but $3 \neq 1 + 1$        |
| R6: $a + b \leq 3$         | no, take $a = 2$ : $2 + 2 = 4 > 3$ | <b>yes</b> , $a + b = b + a$                   | no, $(3, 0)$ and $(0, 3)$ in but $3 + 3 = 6 > 3$       |

**Why R3 is transitive.**  $a R3 b$  means  $|a| = |b|$ , since  $a = b$  or  $a = -b$  is exactly the statement that they have the same absolute value. So if  $|a| = |b|$  and  $|b| = |c|$  then  $|a| = |c|$ . Rewriting the rule in terms of absolute value turns a fiddly case check into one line, and R3 is then visibly reflexive, symmetric and transitive all at once, which is the combination section 3.3 takes up.

**Now redo it:** decide antisymmetry for all six. You should find R1, R2, R4 and R5 antisymmetric, and R3, R6 not.

## RECALL AND CHECK

1. Give a relation that is both symmetric and antisymmetric.
2. What single pair would you name to show  $\{(1,2),(2,3)\}$  is not transitive?
3. Is the empty relation on  $\{1,2\}$  transitive?
4. Where do you look in a matrix to test reflexivity?

### ANSWERS

1.  $\{(1,1)\}$  on  $\{1\}$ , or any relation consisting only of loops.
2.  $(1,3)$  is missing, although  $(1,2)$  and  $(2,3)$  are present.
3. Yes, vacuously: there is no two-step path to fail. It is also symmetric and antisymmetric, and not reflexive.
4. The main diagonal; it must be all 1s.

## FILL IN THE GAP

1. R is reflexive when  $(a, a) \in R$  for \_\_\_\_\_ a in A.
2. R is \_\_\_\_\_ when  $(a,b)$  and  $(b,a)$  in R forces  $a = b$ .
3. To disprove transitivity, produce  $(a,b)$  and  $(b,c)$  in R with \_\_\_\_\_ not in R.
4. In the matrix, symmetry means  $M_{ij} =$  \_\_\_\_\_.

### ANSWERS

1. every. 2. antisymmetric. 3.  $(a, c)$ . 4.  $M_{ji}$ .

## PART THREE • 3.3

# Equivalence Relations and Partitions

Three properties together make the most useful relation in mathematics: one that behaves like equality without being equality.

By the end of this section you will be able to define an equivalence relation, test one in the order that finds a failure fastest, and explain the partition it creates.

## The definition

### ● MUST MEMORISE

### EQUIVALENCE RELATION

A relation R on a set A is an **equivalence relation** when it is all three of:

**reflexive:**  $(a, a) \in R$  for every a

**symmetric:**  $(a, b) \in R$  forces  $(b, a) \in R$

**transitive:**  $(a, b)$  and  $(b, c)$  in R force  $(a, c) \in R$

All three, with no exceptions. Failing any one means it is not an equivalence relation, however close it looks.

◆ **Memory hook:** Reflexive, Symmetric, Transitive: **RST**. Say it out loud before every answer, and check them in that order.

An equivalence relation is "sameness in some respect". Ordinary equality is the strictest example, but most useful ones are looser.

"Has the same remainder when divided by 3" is an equivalence relation on the integers. So is "was born in the same year", "has the same absolute value", and "computes the same output". None of these is equality, yet each lets you treat different things as interchangeable for a purpose.

That is why it matters in computing: two states of a machine that behave identically can be merged, two expressions that always evaluate the same can be swapped, and a cache can return one answer for many requests. Every one of those is an equivalence relation being exploited.

## Testing in the fastest order

Reflexivity is the cheapest to test and the most commonly broken. It needs one glance at the diagonal, and a single missing loop settles the whole question.

So the efficient order is:

1. **Reflexive?** Scan the diagonal. If one loop is missing, stop: the answer is no, and you have your counterexample.
2. **Symmetric?** Compare each pair with its reverse. One unmatched pair settles it.
3. **Transitive?** The expensive one, so do it last, and only if the first two passed.

Under exam time this ordering routinely saves several minutes on a four-mark question.

## The real questions

(c) A social networking platform records users who can view their own profiles. The relation is defined as:

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$$

(2 marks)

Determine whether  $R$  is reflexive. Justify your answer.

### NOTE ON THE PAPER

The mark text is printed *above* the instruction rather than after it, which is only a layout quirk. Note that the underlying set is not stated explicitly; it is implied to be  $\{1, 2, 3, 4\}$  by the pairs present. Say that you are taking  $A = \{1, 2, 3, 4\}$  in your answer, because the verdict depends on it: if the platform had a fifth user,  $R$  would not be reflexive.



**BREAK IT DOWN**

- GIVEN** Six pairs, four of them loops, on an implied set  $\{1, 2, 3, 4\}$ .
- ASKED** **Determine whether**  $R$  is reflexive, and **justify**. The justification is explicitly demanded, so a bare "yes" is half an answer at best.
- MARKS SAY** 2 marks: one for the verdict, one for the reason. Name the loops.
- TEACHES** 3.2 for the property, this section for the context.
- TRAP** Being distracted by  $(1,2)$  and  $(2,3)$ . Reflexivity is only about loops; the other pairs are irrelevant to this question. Also, state the set: an unstated domain is where this question hides its difficulty.

**WORKED EXAMPLE****PQ 25/26 Q2(C)**

- GIVEN**  $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$ , on the set  $A = \{1, 2, 3, 4\}$  implied by the elements appearing.
- FIND** Whether  $R$  is reflexive, with justification.
- RULE**  $R$  is reflexive when  $(a, a) \in R$  for every  $a$  in  $A$ .

**Step 1. Fix the set.** The elements mentioned are 1, 2, 3 and 4, so  $A = \{1, 2, 3, 4\}$ .

**Step 2. Check each element for its loop.**

| A | IS (A, A) IN R?      |
|---|----------------------|
| 1 | yes, (1,1) is listed |
| 2 | yes, (2,2) is listed |
| 3 | yes, (3,3) is listed |
| 4 | yes, (4,4) is listed |

**Step 3. Verdict and justification.**

**Yes,  $R$  is reflexive.** Every element of  $A = \{1, 2, 3, 4\}$  is related to itself, because all four pairs  $(1,1)$ ,  $(2,2)$ ,  $(3,3)$  and  $(4,4)$  are present in  $R$ . The remaining pairs  $(1,2)$  and  $(2,3)$  do not affect reflexivity, which depends only on the loops.

**Worth one extra line, and it is free:**  $R$  is *not* symmetric, since  $(1,2)$  is in  $R$  but  $(2,1)$  is not, and it is not transitive, since  $(1,2)$  and  $(2,3)$  are in  $R$  but  $(1,3)$  is not. So  $R$  is reflexive but not an equivalence relation. In context that says every user can view their own profile, but viewing is not mutual and does not chain.

**Now redo it:** what single pair would you add to make  $R$  symmetric? And what would you then need for transitivity?

**AS PRINTED****2025/2026 · Q2(D)**

- (d) Let  $A = \{1,2,3,4\}$  and let  $R = \{(1,1),(2,2),(3,3),(4,4),(1,3),(3,1),(2,4),(4,2)\}$ . Determine whether  $R$  is an equivalence relation. Provide detailed explanations. **(4 marks)**

#### BREAK IT DOWN

|                  |                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | $A = \{1, 2, 3, 4\}$ explicitly, and eight pairs: four loops and two mutual pairs.                                                                                         |
| <b>ASKED</b>     | <b>Determine whether</b> $R$ is an equivalence relation, with <b>detailed explanations</b> . So all three properties must be checked individually and the reasoning shown. |
| <b>MARKS SAY</b> | 4 marks: roughly one per property plus one for the conclusion. Structure the answer under three headings and the marks fall out.                                           |
| <b>TEACHES</b>   | This section, 3.3.                                                                                                                                                         |
| <b>TRAP</b>      | Transitivity looks fiddly with eight pairs. Work it systematically: list every two-step path and check its shortcut. Do not eyeball it and assert.                         |

**GIVEN**  $A = \{1,2,3,4\}$ ,  $R = \{(1,1),(2,2),(3,3),(4,4),(1,3),(3,1),(2,4),(4,2)\}$ .

**FIND** Whether  $R$  is reflexive, symmetric and transitive, hence whether it is an equivalence relation.

**RULE** All three properties must hold. Test in the order reflexive, symmetric, transitive.

The matrix makes all three visible at once:

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 |
| 2 | 0 | 1 | 0 | 1 |
| 3 | 1 | 0 | 1 | 0 |
| 4 | 0 | 1 | 0 | 1 |

### 1. Reflexive

The diagonal reads 1, 1, 1, 1. All four pairs (1,1), (2,2), (3,3), (4,4) are present, so every element relates to itself. **R is reflexive.**

### 2. Symmetric

Take each non-loop pair and look for its reverse.

| PAIR IN R | REVERSE NEEDED | PRESENT? |
|-----------|----------------|----------|
| (1,3)     | (3,1)          | yes      |
| (3,1)     | (1,3)          | yes      |
| (2,4)     | (4,2)          | yes      |
| (4,2)     | (2,4)          | yes      |

The loops are their own reverses, so they hold automatically. The matrix is its own mirror image across the diagonal. **R is symmetric.**

### 3. Transitive

Every two-step path must have its one-step shortcut. Ignoring paths where one step is a loop, since those give back a pair already present, the genuine two-step paths are:

| FIRST STEP | SECOND STEP | SHORTCUT NEEDED | PRESENT? |
|------------|-------------|-----------------|----------|
| (1,3)      | (3,1)       | (1,1)           | yes      |
| (3,1)      | (1,3)       | (3,3)           | yes      |
| (2,4)      | (4,2)       | (2,2)           | yes      |
| (4,2)      | (2,4)       | (4,4)           | yes      |

There is no path from the {1,3} side to the {2,4} side, because no pair joins them, so no further check is needed. **R is transitive.**

Conclusion

**R is reflexive, symmetric and transitive, so R IS an equivalence relation on A.**

**The detail that earns the last mark.** An equivalence relation cuts its set into groups, and this one is easy to see. Element 1 is related to 1 and 3; element 2 is related to 2 and 4. So A splits into

**{1, 3} and {2, 4}**

Those are the **equivalence classes**. Every element lies in exactly one, none is empty, and together they make up all of A. Here the relation is "has the same parity", since 1 and 3 are odd while 2 and 4 are even, which is why it behaves so neatly.

**Now redo it:** remove (2,4) and (4,2) from R and re-test. It stays an equivalence relation, but the classes become {1,3}, {2}, {4}.

## Equivalence classes and partitions

### ● MUST MEMORISE

### CLASSES AND PARTITIONS

Given an equivalence relation R on A, the **equivalence class** of a, written [a], is the set of everything related to a:

$$[a] = \{x \in A \mid (a, x) \in R\}$$

A **partition** of A is a collection of non-empty subsets that are pairwise disjoint and whose union is all of A. Every element is in exactly one piece.

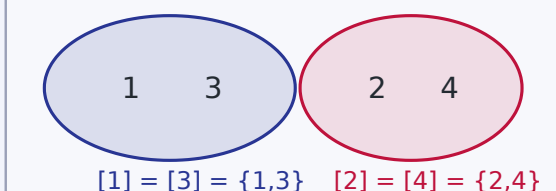
**The fundamental fact:** the equivalence classes of an equivalence relation always form a partition of A, and conversely every partition defines an equivalence relation ("lies in the same piece").

◆ **Memory hook:** an equivalence relation and a partition are the same information written two ways. Sorting things into boxes and declaring things equivalent are one act.

### TEACH

### THE PICTURE

A, partitioned by R



Every element sits in exactly one class.

No class is empty.

The classes together make up all of A.

*That is precisely a partition.*

## RECALL AND CHECK

1. Name the three properties an equivalence relation must have.
2. Which property should you test first, and why?
3. Is "is less than or equal to" on the integers an equivalence relation?
4. What are the equivalence classes of "same remainder on division by 3" over the integers?

### ANSWERS

1. Reflexive, symmetric, transitive.
2. Reflexive: it is the cheapest to check and the most often broken, and one missing loop ends the question.
3. No. It is reflexive and transitive but not symmetric, since  $1 \leq 2$  while  $2 \leq 1$  is false. It is a partial order instead, which is section 3.4.
4. Three classes: the multiples of 3, those leaving remainder 1, and those leaving remainder 2.

## FILL IN THE GAP

1. An equivalence relation is reflexive, \_\_\_\_\_ and \_\_\_\_\_.
2. The set of everything related to a is called the \_\_\_\_\_ of a.
3. The equivalence classes of an equivalence relation form a \_\_\_\_\_ of the set.
4. In a partition, every element lies in \_\_\_\_\_ piece.

### ANSWERS

1. symmetric, transitive. 2. equivalence class. 3. partition. 4. exactly one.

## PART THREE · 3.4

# Partial and Total Orders

*Swap symmetry for antisymmetry and you stop describing sameness and start describing ranking. That is the difference between an equivalence relation and an order.*

*By the end of this section you will be able to define a partially ordered set and a totally ordered set, tell them apart by the one question that separates them, and prove which a given table is.*

### NOT IN THE COURSE TEXT

The official text never uses the words "partially ordered", "totally ordered" or "poset". The 2025/2026 paper gives 4.5 marks to exactly this, in Q2(e).

## The definitions

### ● MUST MEMORISE

### PARTIAL ORDER

A relation  $R$  on  $A$  is a **partial order** when it is

**reflexive**: every element relates to itself

**antisymmetric**:  $(a,b)$  and  $(b,a)$  both in  $R$  forces  $a = b$

**transitive**:  $(a,b)$  and  $(b,c)$  in  $R$  force  $(a,c)$  in  $R$

The set together with the order, written  $(A, R)$ , is a **partially ordered set**, or **poset**.

◆ **Memory hook: RAT**: Reflexive, Antisymmetric, Transitive. Compare with **RST** for an equivalence relation. The only difference is the middle letter, and that one letter is the difference between "the same as" and "no worse than".

### ● MUST MEMORISE

### TOTAL ORDER

A partial order is a **total order** (also called a linear order or a chain) when it also satisfies **comparability**:

for every  $a$  and  $b$  in  $A$ , either  $(a, b) \in R$  or  $(b, a) \in R$ .

That is, **no two elements are incomparable**. Every pair can be put in order.

A set with a total order is a **totally ordered set**, or a **chain**.

◆ **Memory hook**: every total order is a partial order. The extra demand is that nothing sits off to the side: you can line all the elements up in a single row, which is why it is also called linear.

Once you know a relation is a partial order, ask one thing:

**"Is there a pair I cannot compare?"**

If yes, it is partial only. If no, it is total.

**Examples of partial but not total:**

Divisibility on  $\{1, 2, 3, 4\}$ : 2 and 3 are incomparable, since neither divides the other.  
Subset inclusion on the subsets of  $\{1,2\}$ :  $\{1\}$  and  $\{2\}$  are incomparable, since neither contains the other.

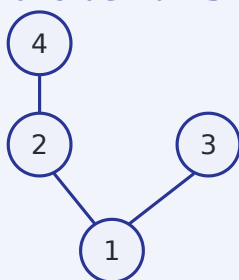
"Is a prerequisite of" on university courses: two unrelated electives are incomparable.

**Examples of total:**

" $\leq$ " on the integers, on the reals, or on any set of numbers: any two numbers can be compared.

Alphabetical order on words: any two words have a first one.

**Partial order: divisibility on  $\{1,2,3,4\}$**



2 and 3 are incomparable:  
no line joins them, so it is  
partial and not total.

**Total order:  $\leq$  on  $\{1,2,3,4\}$**



A single chain. Every pair  
is comparable, so it is total.

### ⚠ TRAP

### ANTISYMMETRIC DOES NOT MEAN "NOT SYMMETRIC"

Section 3.2 made this point and it returns here with marks attached. Antisymmetry forbids two **different** elements pointing at each other. It says nothing about loops, which are always allowed and indeed required by reflexivity.

So a partial order is full of pairs  $(a, a)$ , and that is not a violation of antisymmetry. The definition already says "then  $a = b$ ", which those pairs satisfy.

## The real question

AS PRINTED

2025/2026 · Q2(E)

(e) Differentiate between a partially ordered set and a totally-ordered set. Hence or otherwise determine with proof the status of the table below in terms of partially-ordered or total ordered set.

| Student | CGPA |
|---------|------|
| A       | 4.80 |
| B       | 4.50 |
| C       | 4.10 |

(4½ marks)

### NOTE ON THE PAPER

The relation is never stated. The table gives three students and three distinct CGPAs, and you are expected to supply the obvious order, "has a CGPA at least as high as". **SAY WHICH RELATION YOU ARE USING IN YOUR FIRST LINE.** Without it the question has no determinate answer, and stating it is part of what the 4.5 marks are for. Note also the inconsistent hyphenation ("totally-ordered" then "total ordered"), which changes nothing.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Three students with three distinct CGPAs.                                                                                                                                                                                                   |
| <b>ASKED</b>     | Two tasks. First <b>differentiate</b> the two kinds of ordered set, which needs a stated point of contrast, not two definitions side by side. Then <b>determine with proof</b> which the table is, so all the order axioms must be checked. |
| <b>MARKS SAY</b> | 4.5 marks: roughly 2 for the difference and 2.5 for the classification with proof. The word "proof" means check every property explicitly.                                                                                                  |
| <b>TEACHES</b>   | This section, 3.4, plus 3.2 for the properties.                                                                                                                                                                                             |
| <b>TRAP</b>      | Answering "it is a total order" with no proof. The marks are in the four checks: reflexive, antisymmetric, transitive, comparable. The last one is what settles total versus partial.                                                       |



Part 1: the difference

Both are sets carrying a relation that is reflexive, antisymmetric and transitive, so both rank their elements consistently and neither allows two different elements to outrank each other.

**They differ on comparability.** In a **partially ordered set** there may be pairs of elements that the relation does not connect in either direction; such elements are *incomparable*, and the ranking has gaps. In a **totally ordered set** every pair is comparable, so all the elements line up in one chain with no gaps.

Consequently every total order is a partial order, but not the reverse. Divisibility on  $\{1, 2, 3, 4\}$  is partial only, because 2 and 3 are incomparable; the relation  $\leq$  on numbers is total, because any two numbers can be compared.

Part 2: classifying the table

- GIVEN**  $S = \{A, B, C\}$  with  $\text{CGPA}(A) = 4.80$ ,  $\text{CGPA}(B) = 4.50$ ,  $\text{CGPA}(C) = 4.10$ .  
**FIND** Whether  $(S, R)$  is partially or totally ordered, with proof.  
**RELATION** Taking the natural relation  **$x R y$  when  $\text{CGPA}(x) \geq \text{CGPA}(y)$** , that is,  $x$  ranks at least as high as  $y$ . This must be stated because the paper does not state it.

Writing the pairs out, since  $\text{CGPA}(A) > \text{CGPA}(B) > \text{CGPA}(C)$ :

$$R = \{(A,A), (B,B), (C,C), (A,B), (A,C), (B,C)\}$$

**1. Reflexive.** Each student's CGPA is at least as high as itself, since  $4.80 \geq 4.80$ ,  $4.50 \geq 4.50$  and  $4.10 \geq 4.10$ . So  $(A,A)$ ,  $(B,B)$  and  $(C,C)$  are all in  $R$ . **Reflexive.**

**2. Antisymmetric.** Suppose  $x R y$  and  $y R x$ . Then  $\text{CGPA}(x) \geq \text{CGPA}(y)$  and  $\text{CGPA}(y) \geq \text{CGPA}(x)$ , so  $\text{CGPA}(x) = \text{CGPA}(y)$ . The three CGPAs listed are distinct, so this forces  $x = y$ . **Antisymmetric.**

**3. Transitive.** Suppose  $x R y$  and  $y R z$ . Then  $\text{CGPA}(x) \geq \text{CGPA}(y) \geq \text{CGPA}(z)$ , so  $\text{CGPA}(x) \geq \text{CGPA}(z)$  and  $x R z$ . Checking the only non-trivial case directly:  $(A,B)$  and  $(B,C)$  are in  $R$ , and  $(A,C)$  is in  $R$ . **Transitive.**

So  $(S, R)$  is a **partially ordered set**. Now the deciding test.

**4. Comparability.** Take the three pairs of distinct students:

| PAIR    | COMPARISON                    | COMPARABLE? |
|---------|-------------------------------|-------------|
| A and B | $4.80 \geq 4.50$ , so $A R B$ | yes         |
| A and C | $4.80 \geq 4.10$ , so $A R C$ | yes         |
| B and C | $4.50 \geq 4.10$ , so $B R C$ | yes         |

Every pair is comparable, because any two real numbers can be compared.

**Conclusion.**  $(S, R)$  is reflexive, antisymmetric and transitive, so it is a partially ordered set; and every pair of students is comparable, so it is in fact a **totally ordered set**. The students form a single chain

$$A > B > C \quad (4.80, 4.50, 4.10)$$

which is exactly what a class ranking is.

**The remark that shows real understanding:** totality here depends on the three CGPAs being *distinct* and on CGPA being a single number. If two students shared a CGPA, antisymmetry would fail and  $R$  would not be a partial order at all. And if students were ranked on two criteria at once, say CGPA *and* attendance, two students could each lead on one measure, making them incomparable and the order merely partial. That is precisely how a partial order arises in practice.

**Now redo it:** add student D with CGPA 4.50 and re-classify. Antisymmetry now fails, so it is no longer a partial order at all.

## TEACH

## WHERE ORDERS APPEAR IN COMPUTING

Worth a line if a question asks for relevance.

**Task scheduling.** "Must finish before" is a partial order on jobs; independent jobs are incomparable and can run in parallel. Sorting a partial order into a legal sequence is a topological sort.

**Version control.** "Is an ancestor of" is a partial order on commits; branches are incomparable commits.

**Type systems.** "Is a subtype of" is a partial order.

**Sorting.** Every sorting algorithm assumes a *total* order on the keys. That is why a comparison function must never report two different items as mutually less-than: it would break antisymmetry and the sort would misbehave.

## RECALL AND CHECK

1. Name the three properties of a partial order.
2. What one extra condition makes it a total order?
3. Give two incomparable elements under divisibility on  $\{1, \dots, 6\}$ .
4. Which property differs between an equivalence relation and a partial order?

### ANSWERS

1. Reflexive, antisymmetric, transitive.
2. Comparability: for every  $a$  and  $b$ , either  $a R b$  or  $b R a$ .
3. 4 and 6, for instance: neither divides the other. Also 2 and 5, or 3 and 4.
4. The middle one. An equivalence relation is symmetric; a partial order is antisymmetric.

## FILL IN THE GAP

1. A partial order is reflexive, \_\_\_\_\_ and transitive.
2. A set with a partial order is called a \_\_\_\_\_.
3. A partial order in which every pair is comparable is a \_\_\_\_\_ order.
4. Two elements neither of which relates to the other are \_\_\_\_\_.

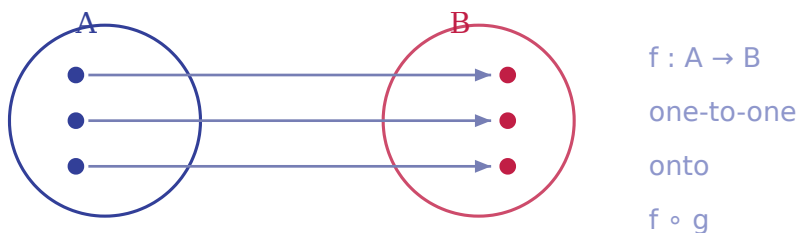
### ANSWERS

1. antisymmetric. 2. partially ordered set (poset). 3. total (linear). 4. incomparable.

## PART FOUR

# Functions and Correspondence

*A function is a relation that never gives two answers to the same question. Every subprogram you write is one, and the five kinds of correspondence in this part appear on paper after paper.*



# Functions, Domain and Range

A function assigns exactly one output to each input. That single word "exactly" is the whole definition, and it is what separates a function from an ordinary relation.

By the end of this section you will be able to say whether a given rule or diagram is a function, name the domain, codomain and range, and explain the difference between the last two.

## From relations to functions

### ● MUST MEMORISE

### DEFINITION

A **function**  $f$  from  $A$  to  $B$ , written  $f : A \rightarrow B$ , is a relation from  $A$  to  $B$  in which **every** element of  $A$  is paired with **exactly one** element of  $B$ .

If  $f$  pairs  $a$  with  $b$  we write  $f(a) = b$ , and call  $b$  the **image** of  $a$ , and  $a$  a **pre-image** of  $b$ .

Two conditions are hiding in "every ... exactly one":

**Totality:** nothing in  $A$  is left unassigned.

**Single-valuedness:** nothing in  $A$  gets two different outputs.

◆ **Memory hook:** a function is a machine with one slot and one answer. Put the same thing in twice and you must get the same thing out; put anything valid in and something must come out.

### ● MUST MEMORISE

### DOMAIN, CODOMAIN, RANGE

For  $f : A \rightarrow B$ ,

**domain** =  $A$ , the set of all allowed inputs.

**codomain** =  $B$ , the set the outputs are declared to live in.

**range** (or image) = the set of values actually produced,  $\{f(a) \mid a \in A\}$ .

The range is always a subset of the codomain, and it may be smaller.

◆ **Memory hook:** the codomain is what you *promised*, the range is what you *delivered*. When they are equal, the function is onto, which is section 4.2.

### TEACH

### WHY RANGE AND CODOMAIN DIFFER

Take  $f : \mathbb{R} \rightarrow \mathbb{R}$  with  $f(x) = x^2$ .

The domain is all reals and the codomain is all reals, because that is how the function was declared. But no square is negative, so the range is only the non-negative reals. The function promised to return a real number and delivered only half of them.

In programming terms the codomain is the declared return type and the range is the set of values the routine can actually return. A routine declared to return an integer might in fact only ever return 0 or 1.

**⚠ TRAP****TWO WAYS A RELATION FAILS TO BE A FUNCTION**

**An input with no output.** On  $A = \{1, 2, 3\}$ , the relation  $\{(1,a), (2,b)\}$  is not a function: 3 has no image. Every element of the domain must be used.

**An input with two outputs.**  $\{(1,a), (1,b), (2,c), (3,d)\}$  is not a function either: 1 is sent to two places.

An output used twice is **fine**:  $\{(1,a), (2,a), (3,a)\}$  is a perfectly good function. The restriction is on inputs, never on outputs. Getting that backwards is the commonest error in the diagram question of 4.2.

**WORKED EXAMPLE****METHOD**

Which of these relations on  $A = \{1, 2, 3\}$  to  $B = \{a, b, c\}$  are functions?

| RELATION                         | FUNCTION? | WHY                                   |
|----------------------------------|-----------|---------------------------------------|
| $\{(1,a), (2,b), (3,c)\}$        | yes       | each input once, each with one output |
| $\{(1,a), (2,a), (3,a)\}$        | yes       | repeated outputs are allowed          |
| $\{(1,a), (2,b)\}$               | no        | 3 has no image, so it is not total    |
| $\{(1,a), (1,b), (2,c), (3,c)\}$ | no        | 1 has two images                      |

For the second one, the domain is  $\{1,2,3\}$ , the codomain is  $\{a,b,c\}$ , and the range is just  $\{a\}$ .

**Now redo it:** is  $\{(1,c), (2,c), (3,b)\}$  a function? What is its range? (Yes; range  $\{b, c\}$ .)

**TEACH****THE TWO-LINE NOTATION**

For a function on a small finite set, the tidiest way to write it is a two-line table with each input directly above its output:

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

meaning  $f(1) = 2$ ,  $f(2) = 3$ ,  $f(3) = 1$ . It is only notation, and it is not a matrix in the sense of section 3.1: you cannot multiply two of these or row-reduce them. The name is borrowed, the operations are not.

## RECALL AND CHECK

1. What two conditions must a relation satisfy to be a function?
2. For  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x^2$ , state the codomain and the range.
3. May two different inputs share an output?
4. May one input have two outputs?

### ANSWERS

1. Every element of the domain is used (totality), and none is sent to more than one place (single-valuedness).
2. Codomain  $\mathbb{R}$ ; range the non-negative reals.
3. Yes, always allowed.
4. No. That is exactly what a function forbids.

## FILL IN THE GAP

1. In  $f : A \rightarrow B$ , the set  $A$  is the \_\_\_\_\_ and  $B$  is the \_\_\_\_\_.
2. The set of values actually produced is the \_\_\_\_\_.
3. If  $f(a) = b$  then  $b$  is the \_\_\_\_\_ of  $a$ .
4. The range is always a \_\_\_\_\_ of the codomain.

### ANSWERS

1. domain, codomain. 2. range (image). 3. image. 4. subset.

## PART FOUR • 4.2

# The Five Correspondence Types

Ten marks on two separate papers, for one diagram you can learn in ten minutes. This is the best value in the whole course.

By the end of this section you will be able to define one-to-one and onto, draw all five correspondence types from memory, and say which is which from a diagram.

## The two properties

### ● MUST MEMORISE

### ONE-TO-ONE AND ONTO

Let  $f : A \rightarrow B$ .

**One-to-one** (also **injective**): different inputs always give different outputs. Formally,  $f(a_1) = f(a_2)$  forces  $a_1 = a_2$ . *No output is hit twice.*

**Onto** (also **surjective**): every element of  $B$  is hit by something. Formally, for every  $b \in B$  there is an  $a \in A$  with  $f(a) = b$ . *Equivalently, the range equals the codomain.*

**Bijective**: both one-to-one and onto. A perfect pairing, with every element of  $A$  matched to exactly one of  $B$  and nothing left over on either side.

◆ **Memory hook**: one-to-one is about **collisions** in  $B$  (there are none). Onto is about **coverage** of  $B$  (it is complete). One looks for doubles, the other looks for gaps.

Compare the sizes of A and B and some cases become impossible, which is a fast sanity check on any answer.

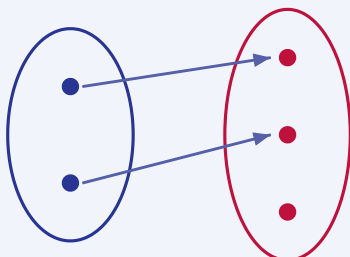
| SIZES         | ONE-TO-ONE POSSIBLE?                | ONTO POSSIBLE?                        | BIJECTION POSSIBLE? |
|---------------|-------------------------------------|---------------------------------------|---------------------|
| $n(A) < n(B)$ | yes                                 | <b>no</b> , too few inputs to cover B | no                  |
| $n(A) = n(B)$ | yes                                 | yes                                   | yes                 |
| $n(A) > n(B)$ | <b>no</b> , two inputs must collide | yes                                   | no                  |

The "no" in the last row is the pigeonhole principle of section 5.3 arriving early: with more inputs than outputs, some output is used twice.

## The five diagrams

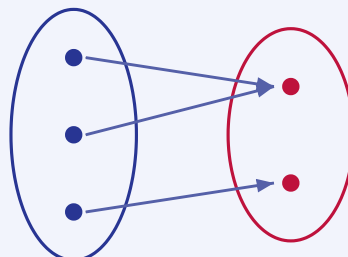
The exam asks for a diagram, so the diagram is the answer. Each one below uses a small A and B so the property is visible at a glance.

(i) One-to-one, not onto



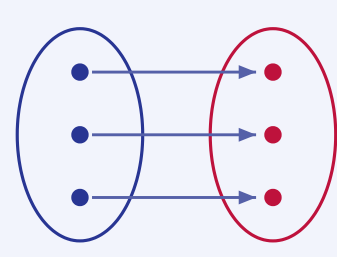
no collisions, but one target is never reached

(ii) Onto, not one-to-one



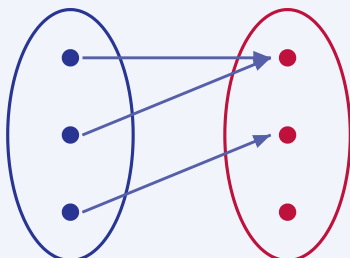
every target reached, but the top one is hit twice

(iii) One-to-one and onto

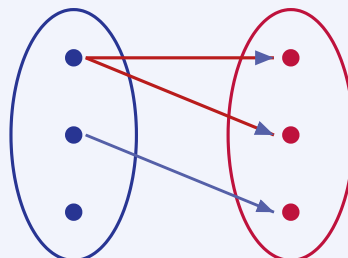


a perfect pairing: no collisions, no gaps

(iv) Neither one-to-one nor onto (v) Not a function



the top target is hit twice AND the bottom one is missed



the top input has TWO arrows, and the bottom input has none

### Reading a diagram

Look at the RIGHT circle for both i and ii. Two arrows landing on one dot means not one-to-one.

A dot with no arrow landing on it means not onto.

Then check the LEFT circle: two arrows leaving one dot, or a dot with none, and it is not a function at all.

## The real question

AS PRINTED

2021/2022 · Q2(C)

(c) With the aid of a diagram, explain the following types of function correspondences types:

(i) One-to-One, not Onto      (ii) Onto, not One-to-One      (iii) One-to-One and Onto

(iv) Neither One-to-One nor Onto      (v) Not a Function      **(10 marks)**

AS PRINTED

2020/2021 · Q4(B)

(b) With the aid of a diagram, explain the following types of function correspondences types:

(i) One-to-one, not onto      (ii) Onto, not one-to-one      (iii) One-to-one and onto

(iv) Neither one-to-one nor onto      (v) Not a function      **(10 marks)**

### NOTE ON THE PAPER

"the following types of function correspondences types" says "types" twice, on both papers. Harmless. The two versions are the same question down to the five items and the 10 marks, and differ only in capitalisation: 2021/2022 writes "One-to-One, not Onto", 2020/2021 writes "One-to-one, not onto". One prepared answer serves both.

### BREAK IT DOWN

**GIVEN** Five named correspondence types.

**ASKED** **With the aid of a diagram, explain** each. Two things are demanded per item: a drawing **and** a sentence. A page of diagrams with no words scores about half, and so does a page of words with no diagrams.

**MARKS SAY** 10 marks over five items is 2 each, so one diagram plus one or two sentences per item. This is the highest marks-per-minute question in the whole set: it is pure recall and needs no calculation.

**TEACHES** This section, 4.2, and 4.1 for what a function is.

**TRAP** Item (v) is not a correspondence type at all, it is the failure case. Its diagram must break the **function** rule, on the left circle, and not merely be an odd pattern of arrows on the right.



Open with the two definitions, because every item below refers to them and stating them once earns the understanding marks.

**One-to-one (injective):** different inputs give different outputs, so no element of  $B$  receives two arrows.

**Onto (surjective):** every element of  $B$  receives at least one arrow, so the range equals the codomain.

**(i) One-to-one, not onto.** Every element of  $B$  receives at most one arrow, but at least one receives none. Take  $A = \{1, 2\}$ ,  $B = \{a, b, c\}$  with  $f(1) = a$ ,  $f(2) = b$ . Nothing maps to  $c$ . This needs  $n(A) < n(B)$ .

**(ii) Onto, not one-to-one.** Every element of  $B$  is reached, but at least one is reached twice. Take  $A = \{1, 2, 3\}$ ,  $B = \{a, b\}$  with  $f(1) = a$ ,  $f(2) = a$ ,  $f(3) = b$ . Both targets are covered, and  $a$  is hit twice. This needs  $n(A) > n(B)$ .

**(iii) One-to-one and onto (bijective).** Every element of  $B$  receives exactly one arrow. Take  $A = \{1, 2, 3\}$ ,  $B = \{a, b, c\}$  with  $f(1) = a$ ,  $f(2) = b$ ,  $f(3) = c$ . This is a perfect pairing and requires  $n(A) = n(B)$ . Only a bijection has an inverse, which is section 4.3.

**(iv) Neither one-to-one nor onto.** Some element of  $B$  is hit twice and some other is missed. Take  $A = \{1, 2, 3\}$ ,  $B = \{a, b, c\}$  with  $f(1) = a$ ,  $f(2) = a$ ,  $f(3) = b$ . Here  $a$  is hit twice and  $c$  is never reached.

**(v) Not a function.** The mapping breaks the definition in 4.1, so the fault is on the *input* side. Either some element of  $A$  has two arrows leaving it, or some element of  $A$  has none. Take  $A = \{1, 2, 3\}$ ,  $B = \{a, b, c\}$  with 1 sent to both  $a$  and  $b$ , and 3 sent nowhere. Both faults are present, and either alone is enough.

Draw the five pictures exactly as above, label the circles  $A$  and  $B$ , and write the one-line reason under each. Keep the reason on the right circle for (i) to (iv) and on the left circle for (v).

**Now redo it:** draw all five from memory on blank paper in four minutes, then check the arrow counts against the diagrams above.

The exam sometimes gives a formula instead of a picture. The test is the same, done algebraically.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1.$$

One-to-one? Suppose  $f(a) = f(b)$ . Then  $2a + 1 = 2b + 1$ , so  $2a = 2b$  and  $a = b$ . **Yes.**

Onto? Given any  $y$ , solve  $y = 2x + 1$  for  $x = (y - 1)/2$ , which is a real number. **Yes.** So  $f$  is a bijection.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2.$$

One-to-one?  $f(2) = 4 = f(-2)$  with  $2 \neq -2$ . **No.**

Onto? There is no real  $x$  with  $x^2 = -1$ . **No.** So it is type (iv), neither.

Notice how each "no" is settled by a single explicit counterexample, and each "yes" by a general argument. That asymmetry is section 1.4's rule about quantifiers, showing up again.

## RECALL AND CHECK

1. Define one-to-one and onto in one sentence each.
2. If  $n(A) = 5$  and  $n(B) = 3$ , can  $f : A \rightarrow B$  be one-to-one?
3. What must be true of the range for  $f$  to be onto?
4. In a diagram, what tells you it is not a function?

### ANSWERS

1. One-to-one: different inputs always give different outputs. Onto: every element of the codomain is an output of something.
2. No. With more inputs than outputs two inputs must share a target.
3. The range equals the codomain.
4. An element of  $A$  with two arrows leaving it, or with none.

## FILL IN THE GAP

1. Another word for one-to-one is \_\_\_\_\_.
2. Another word for onto is \_\_\_\_\_.
3. A function that is both is called \_\_\_\_\_.
4. A function is onto exactly when its \_\_\_\_\_ equals its codomain.

### ANSWERS

1. injective. 2. surjective. 3. bijective (a bijection). 4. range.

## PART FOUR • 4.3

# Composition and Inverse Functions

*Feeding one function's output into another is composition, and the exam asks for the same worked pair on two different papers.*

*By the end of this section you will be able to compose two functions in either order, explain why order matters, state when an inverse exists, and answer the conditions question on commuting linear functions.*

## Composition

### ● MUST MEMORISE

### COMPOSITION

For  $f : B \rightarrow C$  and  $g : A \rightarrow B$ , the **composition**  $f \circ g$  is the function from  $A$  to  $C$  given by

$$(f \circ g)(x) = f(g(x))$$

**g** runs first, on  $x$ . **f** runs second, on whatever  $g$  produced.  $\circ$  is read "composed with", or "after".

◆ **Memory hook: read it right to left.** In  $f \circ g$  the letter nearest the  $x$  is the one that goes first. Say it as "f after g" and the order stops being confusing.

**⚠ TRAP****THE ORDER IS NOT DECORATION**

$f \circ g$  and  $g \circ f$  are usually **different functions**, and the exam asks for both precisely to see whether you noticed.

Everyday version: putting on socks then shoes is not the same as shoes then socks. Composition, like getting dressed, is not commutative.

So compute each one separately, from its own definition. Never compute one and assume the other.

## The question that appears twice

**AS PRINTED****2023/2024 · Q3(C)**

(c) Find  $f \circ g$  and  $g \circ f$ , where  $f(x) = x^2 + 1$  and  $g(x) = x + 2$ , are functions from  $\mathbb{R}$  to  $\mathbb{R}$ . State if  $f \circ g = g \circ f$ .

**(3½ marks)****AS PRINTED****2024/2025 · Q2 B(III)**

iii. Find  $f \circ g$  and  $g \circ f$ , where  $f(x) = x^2 + 1$  and  $g(x) = x + 2$ , are functions from  $\mathbb{R}$  to  $\mathbb{R}$

**2 marks****NOTE ON THE PAPER**

The same question on two papers, one year apart, with the same functions. The only differences: 2023/2024 adds "State if  $f \circ g = g \circ f$ " and pays 3½ marks, while 2024/2025 pays 2 and does not ask for the comparison. Both print "are functions from  $\mathbb{R}$  to  $\mathbb{R}$ ", which reads oddly because the subject of the sentence is  $f$  and  $g$ . Answer both with the same working, and add the comparison line either way: it costs one sentence and 2023/2024 pays for it.

**BREAK IT DOWN**

**GIVEN**  $f(x) = x^2 + 1$  and  $g(x) = x + 2$ , both from the reals to the reals.

**ASKED** **Find** both compositions, then **state** whether they are equal. Two computations and one verdict.

**MARKS SAY** 3.5 marks: roughly 1.5 per composition and 0.5 for the comparison. Show the substitution step; that is where the method mark sits.

**TEACHES** This section, 4.3.

**TRAP** Expanding  $(x + 2)^2$  as  $x^2 + 4$ . It is  $x^2 + 4x + 4$ . That single slip loses both the answer and the comparison.

**GIVEN**  $f(x) = x^2 + 1$ ,  $g(x) = x + 2$ , both  $\mathbb{R}$  to  $\mathbb{R}$ .

**FIND**  $f \circ g$ ,  $g \circ f$ , and whether they are equal.

**RULE**  $(f \circ g)(x) = f(g(x))$ :  $g$  first, then  $f$ .

$f \circ g$

Apply  $g$  first, then feed the result into  $f$ .

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x + 2) \quad [g \text{ has run}] \\ &= (x + 2)^2 + 1 \quad [\text{put } x + 2 \text{ wherever } f \text{ has } x] \\ &= x^2 + 4x + 4 + 1 \\ &= \mathbf{x^2 + 4x + 5}\end{aligned}$$

$g \circ f$

Now the other order:  $f$  first, then  $g$ .

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x^2 + 1) \quad [f \text{ has run}] \\ &= (x^2 + 1) + 2 \quad [g \text{ adds 2 to whatever it is given}] \\ &= \mathbf{x^2 + 3}\end{aligned}$$

Are they equal?

No.  $x^2 + 4x + 5$  and  $x^2 + 3$  are different functions.

Saying so is enough for the mark, but naming a witness is better and costs one line. At  $x = 0$ :

$$(f \circ g)(0) = 0 + 0 + 5 = 5, \quad (g \circ f)(0) = 0 + 3 = 3.$$

Since  $5 \neq 3$ , the two compositions differ, so  $\mathbf{f \circ g \neq g \circ f}$ .

They agree only where  $x^2 + 4x + 5 = x^2 + 3$ , that is  $4x = -2$ , so at the single point  $x = -\frac{1}{2}$ . Agreeing at one point is not being the same function.

**Now redo it:** compose the same  $f$  with  $h(x) = 3x$ . You should get  $(f \circ h)(x) = 9x^2 + 1$  and  $(h \circ f)(x) = 3x^2 + 3$ .

## When do two functions commute?

bi. Let  $f(x) = ax + b$  and  $g(x) = cx + d$ , where  $a, b, c$  and  $d$  are constants. Determine necessary and sufficient conditions on the constants  $a, b, c$  and  $d$  so that  $f \circ g = g \circ f$ .

**2 marks**

### NOTE ON THE PAPER

The label is printed "bi.", welding part b to its first roman numeral. The phrase "necessary and sufficient" is section F.2's vocabulary: it means find the condition that holds **EXACTLY** when the two compositions agree, so the argument must run both ways, not just one.

**GIVEN**  $f(x) = ax + b$ ,  $g(x) = cx + d$ , with  $a, b, c, d$  constant.

**FIND** The condition on  $a, b, c, d$  that holds exactly when  $f \circ g = g \circ f$ .

**RULE** Compose both ways, then compare coefficients.

**Step 1. Compose both ways.**

$$(f \circ g)(x) = f(cx + d) = a(cx + d) + b = acx + ad + b$$

$$(g \circ f)(x) = g(ax + b) = c(ax + b) + d = acx + cb + d$$

**Step 2. Compare.** Both have the same coefficient of  $x$ , namely  $ac$ , so the  $x$  terms always agree whatever the constants are. The two functions are equal exactly when the constant terms agree:

$$ad + b = cb + d$$

**Step 3. Say why this is both necessary and sufficient.** If  $ad + b = cb + d$  then the two expressions are identical, so the compositions agree everywhere: the condition is **sufficient**. Conversely, if the compositions agree everywhere then they agree at  $x = 0$ , where they take the values  $ad + b$  and  $cb + d$ , so those must be equal: the condition is **necessary**. Hence it is exactly the right condition.

**A tidier form**, worth a line: rearranging gives  $ad - d = cb - b$ , that is  $d(a - 1) = b(c - 1)$ . In that shape it is easy to read off the familiar special cases: if  $a = c = 1$  both sides are 0, so any two translations commute; and if  $b = d = 0$  both sides are 0, so any two functions through the origin commute.

**Now redo it:** check the earlier example. There  $f(x) = x^2 + 1$  is not linear, so this condition does not apply; but for  $f(x) = 2x + 3$  and  $g(x) = 4x + 1$ , is  $d(a-1) = b(c-1)$ ? Here  $1 \times 1 = 1$  and  $3 \times 3 = 9$ , so no, and the two do not commute.

ii. If  $f$  and  $f \circ g$  are one-to-one, does it follow that  $g$  is one-to-one? Justify your answer.

**2 marks**

**Answer: yes,  $g$  must be one-to-one.**

**Proof.** Suppose  $g(a) = g(b)$  for some  $a$  and  $b$  in the domain of  $g$ . Applying  $f$  to both sides gives  $f(g(a)) = f(g(b))$ , that is  $(f \circ g)(a) = (f \circ g)(b)$ . But  $f \circ g$  is one-to-one, so  $a = b$ . Since  $g(a) = g(b)$  forced  $a = b$ ,  $g$  is one-to-one.

**Notice what was not needed.** The hypothesis that  $f$  is one-to-one was never used. Only the injectivity of the composition mattered. Saying that shows the marker you followed the logic rather than pattern-matching, and it is true:  $f \circ g$  one-to-one alone forces  $g$  one-to-one.

**The converse fails**, which is worth one more line. If  $g$  is one-to-one,  $f \circ g$  need not be: take  $g(x) = x$  on  $\{1, 2\}$  and  $f$  sending both 1 and 2 to the same value. Then  $g$  is one-to-one and  $f \circ g$  is not.

## Inverse functions

### ● MUST MEMORISE

### INVERSE

A function  $f : A \rightarrow B$  has an **inverse**  $f^{-1} : B \rightarrow A$  exactly when  $f$  is a **bijection**, that is, one-to-one *and* onto.

The inverse undoes  $f$ :  $f^{-1}(f(a)) = a$  for every  $a$  in  $A$ , and  $f(f^{-1}(b)) = b$  for every  $b$  in  $B$ .

**Why both conditions are needed.** If  $f$  is not one-to-one, two inputs share an output and the inverse would not know which to return, so it would not be single-valued. If  $f$  is not onto, some  $b$  has no pre-image and the inverse would be undefined there, so it would not be total. Those are exactly the two conditions of section 4.1.

◆ **Memory hook:** only a perfect pairing can be run backwards. Collisions break single-valuedness, gaps break totality.

### WORKED EXAMPLE

### METHOD

**Find the inverse of  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 2x + 1$ .**

Section 4.2 showed this  $f$  is a bijection, so an inverse exists.

Write  $y = 2x + 1$  and solve for  $x$ :

$$y = 2x + 1 \Rightarrow y - 1 = 2x \Rightarrow x = (y - 1)/2$$

So  $f^{-1}(y) = (y - 1)/2$ , usually rewritten with  $x$  as the variable:  $f^{-1}(x) = (x - 1)/2$ .

**Check it, always.**  $f^{-1}(f(x)) = ((2x + 1) - 1)/2 = 2x/2 = x$ . Correct.

By contrast  $f(x) = x^2$  on the reals has no inverse: it is neither one-to-one nor onto, and "the square root" cannot decide between 2 and  $-2$ .

### RECALL AND CHECK

1. In  $f \circ g$ , which function acts first?
2. With  $f(x) = x + 1$  and  $g(x) = 3x$ , find both compositions.
3. What must be true of  $f$  for  $f^{-1}$  to exist?
4. Is composition commutative?

#### ANSWERS

1.  $g$ , the one written nearest  $x$ .
2.  $(f \circ g)(x) = 3x + 1$  and  $(g \circ f)(x) = 3(x + 1) = 3x + 3$ .
3.  $f$  must be a bijection: one-to-one and onto.
4. No.  $f \circ g$  and  $g \circ f$  generally differ.

### FILL IN THE GAP

1.  $(f \circ g)(x)$  means \_\_\_\_\_.
2. Composition is read from \_\_\_\_\_ to \_\_\_\_\_.
3. An inverse exists exactly when the function is a \_\_\_\_\_.
4. For  $f(x) = x^2 + 1$  and  $g(x) = x + 2$ ,  $(f \circ g)(x) =$  \_\_\_\_\_.

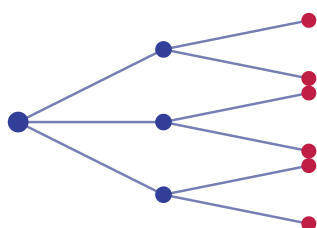
### ANSWERS

1.  $f(g(x))$ . 2. right, left. 3. bijection. 4.  $x^2 + 4x + 5$ .

## PART FIVE

# Counting, Permutations and Combinations

*Counting sounds like the easiest thing in this book and is the one that loses the most marks. Every question here is answered by deciding two things: are the choices stages or alternatives, and does order matter.*



$$3 \times 2 = 6$$

$$P(n, r)$$

$$C(n, r)$$

$$n(A \cup B)$$

stages multiply

order matters

order does not

overlap subtracts



## The Product and Sum Rules

Two rules answer most counting questions on these papers, and choosing between them is the whole skill. Stages multiply. Alternatives add.

By the end of this section you will be able to break a counting task into parts, decide whether those parts are compulsory or alternative, apply the product rule, the sum rule and the subtraction rule, and answer the auditorium chairs, ATM PIN, three-digit-number and password questions in full.

### What a counting question is asking

#### TEACH

#### COUNTING WITHOUT LISTING

A counting question asks **how many** objects of some kind exist. The honest way to answer is to list them all and count the list. That works for six letters and fails for a million PIN codes, so we need rules that give the size of the list without writing it out.

Every such rule comes from the same move: **break the task into parts**, count each part, then combine. The only question is how to combine, and there are exactly two answers.

| IF THE PARTS ARE                                                                       | THEN     | BECAUSE                                                            |
|----------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------|
| <b>compulsory:</b> you do part 1 <i>and</i> part 2 <i>and</i> part 3                   | multiply | every way of doing part 1 pairs with every way of doing part 2     |
| <b>alternative:</b> you do part 1 <i>or</i> part 2 <i>or</i> part 3, never two at once | add      | the possibilities are separate piles and nothing sits in two piles |

Those two words, **compulsory** and **alternative**, are the ones the 2025/2026 paper uses when it asks you to name the parts of a task. They are worth learning in exactly that language.

### The product rule

#### ● MUST MEMORISE

#### PRODUCT RULE

If a task is carried out in a sequence of stages, and  
 stage 1 can be done in  $n_1$  ways,  
 stage 2 in  $n_2$  ways *whatever* stage 1 did,  
 ... and stage  $k$  in  $n_k$  ways,

then the whole task can be carried out in

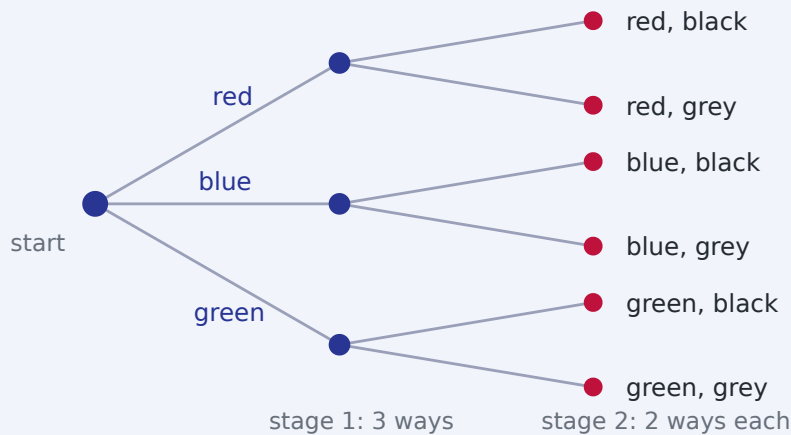
$n_1 \times n_2 \times \dots \times n_k$  ways.

#### WHERE

$k$  is the number of stages, one for each decision you must make.  $n_i$  is the number of choices at stage  $i$ .  $\times$  is ordinary multiplication.

♦ **Memory hook: and means times.** A shirt *and* a pair of trousers: 3 shirts and 2 trousers give 6 outfits, not 5. Say the task out loud; if the joining word is "and", multiply.

Draw the choices as a tree. Each stage adds a level, and each complete path from the root to a tip is one outcome. Three first choices, each splitting into two, gives  $3 \times 2 = 6$  tips.



*The product rule is a tree. Three shirts, then two pairs of trousers for each: six paths, so six outfits.*

The condition that stage 2 has  $n_2$  ways **whatever stage 1 did** is what makes every branch split the same number of times. If the second choice depended on the first, the branches would have different widths and the count would not be a single product.

## The sum rule

### ● MUST MEMORISE

### SUM RULE

If a task can be done in one of  $k$  alternative ways, and

alternative 1 can be done in  $n_1$  ways, alternative 2 in  $n_2$  ways, ... alternative  $k$  in  $n_k$  ways,

and **no outcome counts under two alternatives at once**, then the task can be done in

$n_1 + n_2 + \dots + n_k$  ways.

#### WHERE

**k** is the number of alternatives, one for each separate case. **no overlap** is a genuine condition, not decoration. If two alternatives can both produce the same outcome, adding counts that outcome twice.

◆ **Memory hook: or means plus.** Piles you can choose between are added; stages you must pass through are multiplied.

⚠ TRAP

THE WORD "OR" IS NOT PROOF OF A SUM

English "or" is a bad guide on its own. In the password question below, each character is "an uppercase letter *or* a digit", and that or is counted by **adding**:  $26 + 10 = 36$  symbols. But choosing eight characters is eight compulsory stages, so those 36s are **multiplied**. One sentence, both rules, in that order.

The reliable test is not the word. It is this question: **am I describing one decision with several options, or several decisions I must all make?** Options add. Decisions multiply.

## The subtraction rule

● MUST MEMORISE

SUBTRACTION RULE

To count the objects with some property, when the ones *without* it are easier to count:

$$(\text{number with the property}) = (\text{total}) - (\text{number without it})$$

WHERE

**total** is the count with the condition ignored entirely, which is nearly always the easy number. **number without it** is the count of the cases the condition rules out, and it must be counted over the same total, not over some smaller pool. **the two groups** must cover everything between them and share nothing, which is what makes one subtraction exact.

This is the counting form of the complement from section 2.2: the objects with the property and the objects without it fill the whole collection between them and share nothing.

◆ **Memory hook: "at least one" means "not none"**. Whenever a question says at least one, count everything, count the cases with none, and subtract. Counting "at least one" directly means splitting into exactly one, exactly two, exactly three and so on, which is longer and easier to get wrong.

## The auditorium chairs

AS PRINTED

2021/2022 · Q2(A)

2. (a) The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently? **(6marks)**

NOTE ON THE PAPER

"labeled" is the American spelling; the paper uses it throughout. The mark text is closed up as "(6marks)", which is how this whole paper prints marks. Neither affects the answer. What does affect the answer is "not exceeding 100": that means 100 is allowed, so the integers run 1 to 100 inclusive and there are 100 of them, not 99.

### BREAK IT DOWN

|                  |                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | A label is an uppercase English letter followed by a positive integer that is at most 100.                                                                                   |
| <b>ASKED</b>     | <b>The largest number of chairs that can be labelled differently</b> , which is the number of distinct labels available.                                                     |
| <b>PARTS</b>     | Two, and both are compulsory: every label has a letter <b>and</b> a number. So this is the product rule.                                                                     |
| <b>MARKS SAY</b> | 6 marks for one multiplication is generous, so the marks are for the reasoning: name the two stages, count each, justify the counts, then multiply.                          |
| <b>TEACHES</b>   | This section, 5.1.                                                                                                                                                           |
| <b>TRAP</b>      | Two of them. "Uppercase English letter" is 26, not 52. "Positive integer not exceeding 100" is 100, not 99 and not 101: positive excludes 0, and not exceeding includes 100. |

### WORKED EXAMPLE

PQ 21/22 Q2(A)

**GIVEN** label = uppercase letter, then a positive integer at most 100.

**FIND** How many different labels exist.

**RULE** Product rule: the two parts are compulsory.

**Stage 1: choose the letter.** The uppercase English letters are A to Z, so there are **26** choices.

**Stage 2: choose the integer.** Positive integers not exceeding 100 are 1, 2, 3, ..., 100. Counting them, there are **100** choices.

Stage 2 offers the same 100 numbers whichever letter was picked, so the product rule applies directly:

$$\text{number of labels} = 26 \times 100 = \mathbf{2600}$$

**Answer: 2600 chairs can be labelled differently.**

Say the last sentence in the paper's own terms: since each chair needs its own label, 2600 is the largest number of chairs that can be labelled differently.

**Now redo it:** if the label were two uppercase letters followed by a positive integer not exceeding 100, how many? Three compulsory stages now:  $26 \times 26 \times 100 = 67600$ .

## The ATM PIN codes

### AS PRINTED

2025/2026 · Q6(B)

(b) ATM PIN code in Nigeria can contain the digits 0,1,2,...,9 and digits may be repeated. In the light of the above, how many 4 to 6 digit pin codes are there? **(5 marks)**

### NOTE ON THE PAPER

Two sentences have been run together with no full stop: "digits may be repeated In the light of the above". Read the capital "In" as the start of the second sentence. The phrase "4 to 6 digit" means codes of length 4, length 5 or length 6, and those three lengths are the alternatives that make this a sum-rule question sitting on top of a product-rule one.

**BREAK IT DOWN**

|                  |                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Each position holds one of the ten digits 0 to 9, and repeats are allowed.                                                                                                         |
| <b>ASKED</b>     | <b>How many</b> codes of length 4, 5 or 6 there are, in total.                                                                                                                     |
| <b>PARTS</b>     | Two levels. Within one length the positions are compulsory parts, so multiply. Across the three lengths the cases are alternative parts, so add.                                   |
| <b>MARKS SAY</b> | 5 marks for three powers and one addition. Show each length separately, then total. A bare final number risks losing the method marks.                                             |
| <b>TEACHES</b>   | This section, 5.1.                                                                                                                                                                 |
| <b>TRAP</b>      | Answering only for 6 digits, or multiplying the three lengths together. A code is 4 <b>or</b> 5 <b>or</b> 6 digits long, never two lengths at once, so the three counts are added. |

**WORKED EXAMPLE****PQ 25/26 Q6(B)**

- GIVEN** Digits 0 to 9, so 10 choices per position. Repetition allowed. Lengths 4, 5 and 6.
- FIND** The total number of such codes.
- RULE** Product rule within a length, sum rule across lengths.

**Step 1. Count the codes of each length.** Because repeats are allowed, every position still has all 10 digits available, no matter what the earlier positions took. So each position is a stage with 10 ways.

| LENGTH   | PRODUCT RULE                              | COUNT   |
|----------|-------------------------------------------|---------|
| 4 digits | $10 \times 10 \times 10 \times 10 = 10^4$ | 10000   |
| 5 digits | $10^5$                                    | 100000  |
| 6 digits | $10^6$                                    | 1000000 |

**Step 2. Combine the three lengths.** A code has exactly one length, so no code is counted twice and the sum rule applies:

$$\begin{aligned}
 \text{total} &= 10^4 + 10^5 + 10^6 \\
 &= 10000 + 100000 + 1000000 \\
 &= \mathbf{1110000}
 \end{aligned}$$

**Answer: there are 1110000 PIN codes of 4 to 6 digits.** Written with separators for readability, 1 110 000.

**Why repetition matters.** If digits could not repeat, each position would remove one digit from the pool and the length-4 count would be  $10 \times 9 \times 8 \times 7$  instead of  $10^4$ . Section 5.2 counts that case. The words "digits may be repeated" are the paper telling you which of the two applies.

**Now redo it:** how many codes of 4 to 6 digits start with a nonzero digit? Each length loses its first-position zero, giving  $9 \times 10^3 + 9 \times 10^4 + 9 \times 10^5 = 9000 + 90000 + 900000 = 999000$ .

## Three-digit numbers, and naming the rule

AS PRINTED

2025/2026 · Q6(C)

- (c) i. How many three digit natural numbers can be formed using digits 0, 1,  $\dots$ , 9? (1 mark)  
ii. Identify the number of parts in the task and the type of the parts (compulsory or alternative). (4 marks)  
iii. Which rule applies here? (1 mark)

### NOTE ON THE PAPER

Parts ii and iii say "the task" and "here" without saying which task. They sit directly under (c) i, and (c) i is the nearest task, so answer them about the three-digit numbers. Part (b) immediately above is also a counting task with a different answer, so the safe examination move is to answer for (c) i and then add one line covering (b) as well. That costs a sentence and cannot be marked wrong. Both answers are given below.

### BREAK IT DOWN

**GIVEN** The ten digits 0 to 9, and a three-digit natural number to build from them.

**ASKED** (i) **How many** such numbers. (ii) **Identify** the number of parts and their type. (iii) **Which rule** applies. Only the first is a calculation; the other two ask you to describe your own method.

**MARKS SAY** 1, 4 and 1. The 4 marks are the largest on this part and they are for the description, not the arithmetic. Name each part, say how many ways it has, and say why it is compulsory.

**TEACHES** This section, 5.1.

**TRAP** Answering  $10^3 = 1000$ . A three-digit number cannot begin with 0, because 012 is the two-digit number 12. The hundreds position has 9 choices, not 10.

**GIVEN** Digits 0 to 9, forming a three-digit natural number.

**FIND** The count, the parts, and the rule.

**RULE** Product rule, with one restricted position.

i. How many three-digit natural numbers

A three-digit natural number runs from 100 to 999. Build it position by position:

| POSITION | ALLOWED DIGITS                                           | WAYS |
|----------|----------------------------------------------------------|------|
| hundreds | 1 to 9, since a leading 0 would make it a shorter number | 9    |
| tens     | 0 to 9                                                   | 10   |
| units    | 0 to 9                                                   | 10   |

$$\text{count} = 9 \times 10 \times 10 = \mathbf{900}$$

**Answer: 900.** Check it another way, which is worth doing on a 1-mark part: the numbers 100 to 999 inclusive number  $999 - 100 + 1 = 900$ . The two methods agree.

ii. The number of parts and their type

**Three parts, all compulsory.** Set out in full, since 4 marks are on offer:

1. **Part 1, choose the hundreds digit:** 9 ways. Compulsory, because every three-digit number has a hundreds digit.
2. **Part 2, choose the tens digit:** 10 ways. Compulsory.
3. **Part 3, choose the units digit:** 10 ways. Compulsory.

They are compulsory rather than alternative because the number is not finished until all three have been done. You do not pick *one* of the three parts; you do part 1 **and** part 2 **and** part 3.

**For part (b) above,** the structure is different and worth one line: that task has **three alternative parts**, one per allowed length (4, 5 and 6 digits), and each of those parts is itself built from compulsory sub-parts, one per digit position.

iii. Which rule applies

**The product rule**, because the parts of (c) i are compulsory: the counts 9, 10 and 10 are multiplied.

And again for (b): **the sum rule** across the three lengths, sitting on top of the product rule within each length.

**Now redo it:** how many three-digit natural numbers have no repeated digit? Hundreds 9 ways, tens 9 (zero is back in, but the hundreds digit is gone), units 8. So  $9 \times 9 \times 8 = 648$ .

## The password question

(b) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? **(5 marks)**

#### NOTE ON THE PAPER

This is the hardest counting question in the whole set, and it is worth 5 marks out of 22.5. It combines all three rules of this section: add across the three lengths, multiply across the character positions, and subtract to handle "at least one digit". Nothing about it is printed wrongly. The difficulty is real, not a defect.

#### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Password length 6, 7 or 8. Each character is an uppercase letter or a digit. At least one character must be a digit.                                                                                                                |
| <b>ASKED</b>     | <b>How many possible passwords</b> there are in total.                                                                                                                                                                              |
| <b>PARTS</b>     | Three alternative lengths, so add. Within a length, one compulsory part per character position, so multiply. The "at least one" condition is handled by subtracting.                                                                |
| <b>MARKS SAY</b> | 5 marks. Roughly: 1 for the 36 symbols, 1 for seeing that "at least one digit" means total minus letters-only, 2 for the three lengths, 1 for the total. Show every line.                                                           |
| <b>TEACHES</b>   | This section, 5.1: product, sum and subtraction rules together.                                                                                                                                                                     |
| <b>TRAP</b>      | Trying to count "at least one digit" directly by cases (exactly one digit, exactly two, and so on). It can be done, but it is seven cases per length and almost nobody finishes it correctly under exam pressure. Subtract instead. |



**GIVEN** Length 6, 7 or 8; each character from 26 uppercase letters or 10 digits; at least one digit required.

**FIND** The number of possible passwords.

**RULE** Sum over lengths; product over positions; subtraction for "at least one".

**Step 1. Count the symbols available per character.** A character is an uppercase letter **or** a digit, and no symbol is both, so the sum rule gives

$$26 + 10 = \mathbf{36} \text{ symbols.}$$

**Step 2. Turn "at least one digit" into a subtraction.** Let  $P_n$  be the number of valid passwords of length  $n$ . Then

$$\begin{aligned} P_n &= (\text{all strings of length } n \text{ over the 36 symbols}) - (\text{those using no digit at all}) \\ &= 36^n - 26^n \end{aligned}$$

A string with no digit is made only of the 26 letters, which is where  $26^n$  comes from. Every string either contains a digit or does not, and never both, so the subtraction is exact.

**Step 3. Evaluate each length.**

$$P_6 = 36^6 - 26^6 = 2176782336 - 308915776 = \mathbf{1867866560}$$

$$P_7 = 36^7 - 26^7 = 78364164096 - 8031810176 = \mathbf{70332353920}$$

$$P_8 = 36^8 - 26^8 = 2821109907456 - 208827064576 = \mathbf{2612282842880}$$

**Step 4. Add the three lengths.** A password has exactly one length, so the sum rule applies:

$$P = P_6 + P_7 + P_8 = \mathbf{2684483063360}$$

**Answer: 2684483063360 possible passwords**, or 2 684 483 063 360 written with separators.

**If you run out of time**, the formula  $P = (36^6 - 26^6) + (36^7 - 26^7) + (36^8 - 26^8)$  with the three rules named earns most of the 5 marks on its own. Write the formula first, then evaluate. Never the other way round.

**Now redo it:** same question but passwords are exactly six characters and must contain at least one letter. Answer:  $36^6 - 10^6 = 2176782336 - 1000000 = 2175782336$ .

## RECALL AND CHECK

1. When do you multiply the counts of the parts, and when do you add them?
2. What condition must alternatives satisfy before the sum rule is valid?
3. How many three-digit natural numbers are there, and why is it not 1000?
4. Turn "at least one vowel" into a subtraction.
5. Labels are one uppercase letter then an integer from 1 to 100. How many?

### ANSWERS

1. Multiply when the parts are compulsory, that is, when all of them must be done. Add when they are alternatives and exactly one of them happens.
2. No outcome may count under two alternatives, otherwise it is counted twice.
3. 900. The hundreds digit cannot be 0, so it has 9 choices, not 10:  $9 \times 10 \times 10$ .
4. (total number of strings) – (number of strings with no vowel).
5.  $26 \times 100 = 2600$ .

## FILL IN THE GAP

1. Parts that must all be done are \_\_\_\_\_, and their counts are \_\_\_\_\_.
2. Parts you choose between are \_\_\_\_\_, and their counts are \_\_\_\_\_.
3. The number of PIN codes of 4 to 6 digits, repeats allowed, is \_\_\_\_\_.
4. "At least one" is counted as total minus \_\_\_\_\_.

### ANSWERS

1. compulsory, multiplied. 2. alternative, added. 3. 1110000. 4. none.

## PART FIVE • 5.2

# Permutations and Combinations

*One question decides which formula you need: does the order of the chosen objects change the answer? If it does, you are permuting. If it does not, you are combining.*

*By the end of this section you will be able to compute factorials, use the permutation and combination formulas, count arrangements of a word with repeated letters, handle "at least one" selections, and answer the SCHOOL, the six boys and four girls, and the prayer partners questions in full.*

## Factorial notation

### ● MUST MEMORISE

### FACTORIAL

Section F.1 introduced this notation. It is restated here because every formula in this part is built from it, and you should not have to turn back mid-question.

For a positive integer  $n$ ,  **$n$  factorial**, written  $n!$ , is the product of every positive integer from  $n$  down to 1:

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

and by definition

$$0! = 1$$

#### WHERE

$n$  is a whole number that is not negative.  $!$  is read "factorial", so  $5!$  is read "five factorial". It is not an exclamation mark and never punctuation.

◆ **Memory hook: count down and multiply.** And  $0! = 1$  is not a trick: there is exactly one way to arrange nothing, namely do nothing, so the count is 1, not 0.

### TEACH

### THE VALUES YOU SHOULD KNOW BY SIGHT

| N  | 0 | 1 | 2 | 3 | 4  | 5   | 6   | 7    | 8     | 10      |
|----|---|---|---|---|----|-----|-----|------|-------|---------|
| N! | 1 | 1 | 2 | 6 | 24 | 120 | 720 | 5040 | 40320 | 3628800 |

Factorials grow violently. That is the reason counting questions ask for an expression as often as a number:  $20!$  has nineteen digits and no examiner wants it written out.

**The cancelling trick**, which you will use constantly. Because  $n! = n \times (n - 1)!$ , a ratio of factorials collapses:

$$10! / 7! = (10 \times 9 \times 8 \times 7!) / 7! = 10 \times 9 \times 8 = 720$$

Never compute  $10!$  and  $7!$  separately and divide. Cancel first, always.

## Permutations: order matters

### ● MUST MEMORISE

### PERMUTATION

A **permutation** is an arrangement of objects **in a definite order**. An **r-permutation** of  $n$  distinct objects is an ordered selection of  $r$  of them, and the number of these is

$$P(n, r) = n! / (n - r)!$$

#### WHERE

$n$  is how many distinct objects are available.  $r$  is how many are being chosen and arranged, with  $0 \leq r \leq n$ .  $P(n, r)$  is also written  $nPr$  or  ${}^nP_r$ , which is the label printed on most calculators.

**The special case  $r = n$ :** arranging all  $n$  objects gives  $P(n, n) = n! / 0! = n!$ . Every permutation question about "all of them" is just a factorial.

◆ **Memory hook: P for position.** A permutation cares where each object lands, so AB and BA are two different answers.

### TEACH

### WHERE THE FORMULA COMES FROM

You already have this from the product rule. To fill  $r$  ordered positions from  $n$  objects with no repeats:

position 1:  $n$  ways

position 2:  $n - 1$  ways, since one object is used up

position 3:  $n - 2$  ways

...

position  $r$ :  $n - r + 1$  ways

Multiplying gives  $n \times (n - 1) \times \dots \times (n - r + 1)$ , which is  $r$  factors counting down from  $n$ . Writing that as a ratio of factorials:

$$n \times (n - 1) \times \dots \times (n - r + 1) = n! / (n - r)!$$

because dividing by  $(n - r)!$  cancels exactly the tail of the product that we stopped short of. So  $P(n, r)$  is not a new idea; it is the product rule packaged.

**Worked:**  $P(6, 4) = 6! / 2! = 720 / 2 = 360$ , and by the counting-down form  $6 \times 5 \times 4 \times 3 = 360$ . The two agree, as they must.

## Combinations: order does not matter

### ● MUST MEMORISE

### COMBINATION

A **combination** is a selection of objects **without regard to order**. The number of ways of choosing  $r$  objects from  $n$  distinct objects is

$$C(n, r) = n! / (r! (n - r)!)$$

WHERE

$n$  is how many objects are available.  $r$  is how many are chosen, with  $0 \leq r \leq n$ .  $C(n, r)$  is also written  $nCr$  or  ${}^nC_r$ .

**Why the extra  $r!$**  Each unordered selection of  $r$  objects can be arranged in  $r!$  different orders, and  $P(n, r)$  counts every one of those orders separately. Dividing by  $r!$  collapses them back into one:

$$C(n, r) = P(n, r) / r!$$

◆ **Memory hook: C for committee.** A committee of three is the same committee however you read out the names, so order is thrown away and you divide by  $r!$ .

### PERMUTATION: order matters

Choose 2 of A, B, C and line them up

A B          B A

A C          C A

B C          C B

$$P(3, 2) = 3! / 1! = 6$$

six ordered pairs

A B and B A are different

### COMBINATION: order does not

Choose 2 of A, B, C as a set

—— {A, B}

—— {A, C}

—— {B, C}

$$C(3, 2) = 3! / (2! 1!) = 3$$

three sets

each set collapses 2! orders into one

*The same three objects, the same  $r = 2$ , and two different answers. Six divided by  $2!$  is three.*

Ask the question in this exact form: **if I swap two of the chosen objects, is that a different answer?**

| TASK                                     | SWAP CHANGES IT?              | USE         |
|------------------------------------------|-------------------------------|-------------|
| Arranging letters into a word            | yes, TAB is not BAT           | permutation |
| Filling the posts of chair and secretary | yes, the two people swap jobs | permutation |
| Picking 4 students for a team            | no, the same four people      | combination |
| Choosing a subset of a set               | no, a set has no order        | combination |

The wording that gives it away: **arrange, order, line up, rank, first and second** point to a permutation; **select, choose, pick, committee, team, subset, group** point to a combination.

One warning about that list. It is a guide, not a law. Read the task, not the verb: a question that says "select" and then gives the chosen people different jobs is a permutation despite the word.

## Arranging a word with repeated letters

### ● MUST MEMORISE

### ARRANGEMENTS WITH REPEATS

The number of distinct arrangements of  $n$  objects, of which one kind repeats  $n_1$  times, another  $n_2$  times, and so on, is

$$n! / (n_1! \times n_2! \times \dots \times n_k!)$$

#### WHERE

$n$  is the total number of objects, letters counted as often as they appear.  $n_i$  is how many times the  $i$ -th repeated object occurs. Objects appearing once contribute  $1! = 1$  and can be left out.

**Why divide.** Pretend the repeats are distinguishable for a moment, say  $O_1$  and  $O_2$ . Then all  $n$  objects are distinct and there are  $n!$  arrangements. But swapping  $O_1$  and  $O_2$  produces a word that looks identical, so every genuine word has been counted  $2!$  times. Divide by  $2!$  to correct it.

◆ **Memory hook: divide out what you cannot tell apart.** One factorial on the bottom for each repeated letter, using how many times it repeats.

### AS PRINTED

2025/2026 · Q4(D)

(d) Find out the number of ways that the letters of the word "SCHOOL" can be arranged?  
(2½ marks)

### NOTE ON THE PAPER

A command punctuated as a question, which this paper does in several places. Nothing else is wrong with it, and the quotation marks around the word are the paper's own. The word is the whole difficulty: SCHOOL has six letters and the letter O appears twice.

### BREAK IT DOWN

- GIVEN** The six letters of the word SCHOOL: S, C, H, O, O, L.
- ASKED** **The number of ways** they can be arranged. Arranged means ordered, so this is a permutation.
- MARKS SAY** 2.5 marks. That is enough for the formula, the substitution and the answer, and not enough to list 360 words. Do not list them.
- TEACHES** This section, 5.2.
- TRAP** Answering  $6! = 720$ . The two O's are identical, so half of those 720 orderings spell the same word twice over. The answer is 360.

### WORKED EXAMPLE

PQ 25/26 Q4(D)

- GIVEN** The letters S, C, H, O, O, L. Total  $n = 6$ , with the letter O repeated twice.
- FIND** The number of distinct arrangements.
- FORMULA**  $n!$  divided by the factorial of each repeat count.

#### Step 1. Count the letters and the repeats.

| LETTER          | S | C | H | O | L |
|-----------------|---|---|---|---|---|
| TIMES IT OCCURS | 1 | 1 | 1 | 2 | 1 |

Total letters  $n = 1 + 1 + 1 + 2 + 1 = 6$ . One letter repeats, namely O, and it repeats twice.

#### Step 2. Apply the formula.

$$\begin{aligned}\text{arrangements} &= 6! / 2! \\ &= 720 / 2 \\ &= 360\end{aligned}$$

**Answer: the letters of SCHOOL can be arranged in 360 ways.**

**State the reasoning in one line for the marker:** if the six letters were all different there would be  $6! = 720$  arrangements, but swapping the two O's changes nothing visible, so every distinct word has been counted  $2!$  times and 720 must be halved.

**Now redo it:** how many arrangements has the word LEVEL? Five letters with L twice and E twice, so  $5! / (2! 2!) = 120 / 4 = 30$ .

## Selecting with a condition

AS PRINTED

2025/2026 · Q3(B)

(b) In a group of students, there are 6 boys and 4 girls. Out of 10 students, 4 students have to be selected. Find out how many different ways the students can be selected such that at least one boy should be selected. **(5 marks)**

### NOTE ON THE PAPER

"Out of 10 students" restates what the first sentence already said, since  $6 + 4 = 10$ . That is a helpful redundancy rather than a defect: it confirms the ten students are the whole pool. The phrase to act on is "at least one boy", which is the subtraction rule of section 5.1 in a combination setting.

**BREAK IT DOWN**

|                  |                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | 6 boys and 4 girls, 10 students in all. Four are to be selected.                                                                                                |
| <b>ASKED</b>     | <b>How many different ways</b> the selection can be made with <b>at least one boy</b> among the four.                                                           |
| <b>ORDER</b>     | A group of four is selected, with no jobs or ranking attached, so order does not matter. This is a combination.                                                 |
| <b>MARKS SAY</b> | 5 marks. Enough for the total, the excluded case, the subtraction and a check. Showing the case-by-case method as a verification is worth doing if time allows. |
| <b>TEACHES</b>   | This section, 5.2, using the subtraction rule from 5.1.                                                                                                         |
| <b>TRAP</b>      | Reading "at least one boy" as "exactly one boy" and answering $C(6, 1) \times C(4, 3) = 24$ . At least one means one, two, three or four.                       |

**WORKED EXAMPLE****PQ 25/26 Q3(B)**

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| <b>GIVEN</b>   | 6 boys, 4 girls, 10 students; choose 4.                                        |
| <b>FIND</b>    | The number of selections containing at least one boy.                          |
| <b>FORMULA</b> | $C(n, r) = n! / (r! (n - r)!)$ , with the subtraction rule for "at least one". |

**Step 1. Count all selections of four, ignoring the condition.**

$$\begin{aligned}
 C(10, 4) &= 10! / (4! \times 6!) \\
 &= (10 \times 9 \times 8 \times 7) / (4 \times 3 \times 2 \times 1) \\
 &= 5040 / 24 \\
 &= \mathbf{210}
 \end{aligned}$$

**Step 2. Count the selections that break the condition.** A selection fails only when it contains **no boy**, that is, when all four chosen are girls. There are exactly 4 girls, so all four must be taken:

$$C(4, 4) = 4! / (4! \times 0!) = 1$$

Note  $0! = 1$  here; that definition is doing real work.

**Step 3. Subtract.**

$$\text{at least one boy} = 210 - 1 = \mathbf{209}$$

**Answer: 209 different selections contain at least one boy.**

**Verification by cases**, which is the long way and a good check:

| BOYS | GIRLS | WAYS                     | VALUE              |
|------|-------|--------------------------|--------------------|
| 1    | 3     | $C(6, 1) \times C(4, 3)$ | $6 \times 4 = 24$  |
| 2    | 2     | $C(6, 2) \times C(4, 2)$ | $15 \times 6 = 90$ |
| 3    | 1     | $C(6, 3) \times C(4, 1)$ | $20 \times 4 = 80$ |
| 4    | 0     | $C(6, 4) \times C(4, 0)$ | $15 \times 1 = 15$ |

$$24 + 90 + 80 + 15 = \mathbf{209}.$$
 The two methods agree.

The case table also shows why subtracting is the better exam move: four combinations and three additions, against one subtraction.

**Now redo it:** how many selections of four contain at least one girl? Total 210, minus the all-boy selections  $C(6, 4) = 15$ , gives 195.



## The prayer partners

AS PRINTED

2025/2026 · Q3(C)

(c) Suppose 10 male and 10 female students needs to pray with one another in a one male-one female order:

- (i) How many Prayer sessions will there be? **(1 mark)**
- (ii) How many prayer units are generated here? **(1½ marks)**
- (iii) Calculate the choice of prayer partners. **(3 marks)**

### NOTE ON THE PAPER

Two things to flag. First, "students needs" does not agree in number, and "Prayer" is capitalised in (i) but not in (ii). Those are harmless. Second, and this one matters: the paper never says what a **SESSION** is or what a **UNIT** is, so parts (i) and (ii) can be read more than one way. The answer below states the reading it uses before computing anything, gives the numbers that reading produces, and then gives the other reading's numbers too. Part (iii) is unambiguous and carries the most marks.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | 10 male and 10 female students. They pray in pairs, each pair being one male and one female.                                                                                                             |
| <b>ASKED</b>     | (i) the number of <b>sessions</b> , (ii) the number of <b>units</b> , (iii) <b>calculate</b> the choice of prayer partners.                                                                              |
| <b>MARKS SAY</b> | 1, 1.5 and 3. The weight is on (iii), which is a full permutation, so budget your time there. Parts (i) and (ii) are one line each.                                                                      |
| <b>TEACHES</b>   | This section, 5.2, with the product rule from 5.1.                                                                                                                                                       |
| <b>TRAP</b>      | Answering (iii) with $10 \times 10 = 100$ . That counts the possible pairs, which is part (ii). The <b>choice of partners</b> means assigning every male a different female at once, and that is $10!$ . |

**GIVEN** 10 males, 10 females, praying one male with one female.

**FIND** Sessions, units, and the number of ways to choose partners.

**RULE** Product rule for the pairs, permutation for the assignment.

**Step 0. State the reading.** Write this on the script before answering, because the paper does not supply it:

*A prayer unit is one male paired with one female. A session is one round in which all twenty students are paired at the same time, giving ten units per session. Every male is to pray with every female.*

i. How many prayer sessions

In one session each of the 10 males prays with one female, so a session uses up 10 of the possible male-female pairings. Every male must eventually meet all 10 females, so each male needs 10 turns:

$$\text{number of sessions} = \mathbf{10}$$

ii. How many prayer units

A unit is one male with one female. Choosing the male is 10 ways and choosing the female is 10 ways, and both choices must be made, so the product rule gives

$$\text{units} = 10 \times 10 = \mathbf{100}$$

Consistency check: 10 sessions with 10 units in each is 100 units, which matches. The two answers agree, which is exactly why stating the reading first was worth a line.

iii. Calculate the choice of prayer partners

Now the real work. In a single session, every male is given a different female. Line the males up in a fixed order and hand out the females:

male 1: 10 choices of partner

male 2: 9 choices, since one female is taken

male 3: 8 choices

...

male 10: 1 choice

All ten choices must be made, so the product rule applies:

$$\text{choices} = 10 \times 9 \times 8 \times \dots \times 2 \times 1 = 10! = \mathbf{3628800}$$

In the language of this section, the ten females are being arranged among the ten males, so the count is  $P(10, 10) = 10! / 0! = 10!$ .

**Answer: 3628800 ways**, or 3 628 800 with separators.

**The other reading, for safety.** If instead a "session" means a single male-female meeting, then there are 100 sessions and 10 units in each round. Whichever reading a marker has in mind, the two numbers on the table are 10 and 100, and part (iii) is  $10! = 3628800$  either way. Writing your reading down protects the marks.

**Now redo it:** with 5 males and 5 females, the units are  $5 \times 5 = 25$  and the choice of partners is  $5! = 120$ .

## RECALL AND CHECK

1. State the formulas for  $P(n, r)$  and  $C(n, r)$ .
2. What is  $0!$ , and why?
3. Which is larger,  $P(8, 3)$  or  $C(8, 3)$ , and by what factor?
4. How many arrangements has the word SCHOOL, and why is it not 720?
5. Four students are picked from ten for a team. Permutation or combination?

### ANSWERS

1.  $P(n, r) = n! / (n - r)!$  and  $C(n, r) = n! / (r! (n - r)!)$ .
2.  $0! = 1$ , because there is exactly one way to arrange nothing.
3.  $P(8, 3)$  is larger, by a factor of  $3! = 6$ :  $P(8, 3) = 336$  and  $C(8, 3) = 56$ .
4. 360. The two O's are identical, so  $6!$  counts every word twice and must be divided by  $2!$ .
5. A combination: the team is the same team however the names are ordered.

## FILL IN THE GAP

1. An arrangement in a definite order is a \_\_\_\_\_.
2. A selection with no regard to order is a \_\_\_\_\_.
3.  $C(n, r)$  is  $P(n, r)$  divided by \_\_\_\_\_.
4. Choosing 4 from 10 can be done in \_\_\_\_\_ ways.
5. Ten males paired with ten females can be partnered in \_\_\_\_\_ ways.

### ANSWERS

1. permutation. 2. combination. 3.  $r!$ . 4. 210. 5. 3628800.

## PART FIVE • 5.3

# Inclusion, Exclusion and the Pigeonhole Principle

*When two groups overlap, adding their sizes counts the overlap twice. Subtracting it once is the whole of inclusion and exclusion, and it is worth seven marks on the newest paper.*

*By the end of this section you will be able to state and apply the inclusion and exclusion principle for two and three sets, read the overlap off a Venn diagram, state the pigeonhole principle in both its forms, and answer the Ibibio and Efik question and the class-size question in full.*

## Why adding is wrong

### TEACH

### THE DOUBLE COUNT

The sum rule of section 5.1 came with a condition: **no outcome may count under two alternatives**. Sets of people almost always break that condition, because a person can be in both.

Suppose 30 students take Maths and 25 take Physics. Adding gives 55. But if 10 take both, those 10 were counted once in the 30 and again in the 25, so 55 overstates the class by exactly 10. The true total is 45.

That is the entire idea. Add the parts, then take the overlap back out once, because it went in twice.

### ● MUST MEMORISE

### INCLUSION AND EXCLUSION, TWO SETS

Section 2.2 introduced this as the counting formula, where it belonged with the set operations. Here it is again, this time with the rearrangements the exam actually asks for.

For any two finite sets A and B,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

#### WHERE

$n(A)$  is the cardinality of A, the number of things in it, as defined in section 2.1.  $A \cup B$  is the union: everything in A or in B or in both.  $A \cap B$  is the intersection: the things in both, which is the overlap that was counted twice.

**The rearrangements you will actually use.** Exam questions give you three of the four quantities and ask for the fourth, so learn all three forms, not just the first:

$$n(A \cap B) = n(A) + n(B) - n(A \cup B) \quad [\text{find the overlap}]$$

$$n(A \text{ only}) = n(A) - n(A \cap B) \quad [\text{strip the overlap out of A}]$$

$$n(B \text{ only}) = n(B) - n(A \cap B)$$

◆ **Memory hook: in, in, out.** Put A in, put B in, take the middle out. And a completed answer always satisfies  $A \text{ only} + \text{both} + B \text{ only} = n(A \cup B)$ .

### THE PRINCIPLE OF INCLUSION AND EXCLUSION

**The claim.** The size of a union is found by adding the sizes of the sets and then removing every overlap that the addition counted more than once. For two sets that means subtracting the single overlap; for more sets the corrections alternate in sign.

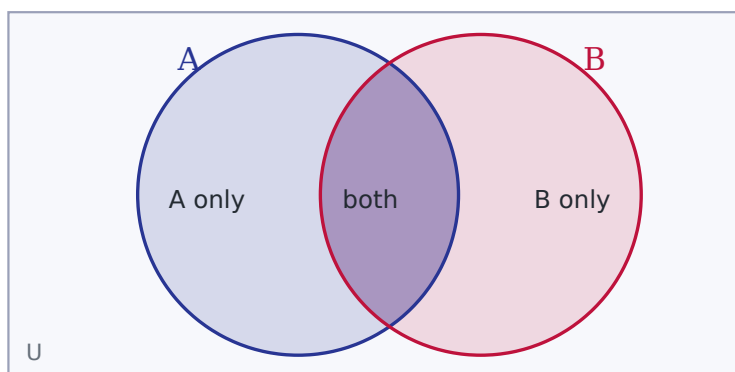
**In symbols.**

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

**Conditions.** The sets must be **finite**, or the counts do not exist to be added. Nothing else is required: the sets may overlap in any way at all, which is precisely the point of the principle.

**The limiting case.** When the sets are disjoint, every overlap term is 0 and the principle collapses to the sum rule of section 5.1:  $n(A \cup B) = n(A) + n(B)$ . So the sum rule is not a separate law but the special case of this one where nothing is shared.



$n(A)$  counts  
A only + both

$n(B)$  counts  
B only + both

so "both" is in  
the sum twice

Three regions, not two. Adding  $n(A)$  and  $n(B)$  covers the middle region twice, so it is subtracted once.

### ● MUST MEMORISE

### INCLUSION AND EXCLUSION, THREE SETS

For three finite sets A, B and C,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

WHERE

$n(A)$ ,  $n(B)$ ,  $n(C)$  are the sizes of the three sets on their own, each counted with no regard to the others.  $n(A \cap B)$  and the other two pair terms are the sizes of each two-way overlap, subtracted because each was counted in two of the singles.  $n(A \cap B \cap C)$  is the size of the three-way overlap, added back because the singles put it in three times and the pairs took it out three times.

**Read the pattern rather than memorising the letters:** add the singles, subtract the pairs, add the triple. Signs alternate, and the number of sets in each term goes up by one.

The final term is added back because the elements in all three sets were added three times by the singles and then subtracted three times by the pairs, leaving them counted zero times. One addition restores them.

◆ **Memory hook: plus singles, minus pairs, plus triple.** Odd-sized groups are added, even-sized groups are subtracted.

## The Ibibio and Efik question

AS PRINTED

2025/2026 · Q3(A)

(a) In a group of 100 students, 72 students can speak Ibibio and 43 students can speak Efik. Based on these data, answer the following questions:

- (i) Find the number of students who can speak both Ibibio and Efik. **(3 marks)**
- (ii) Find the number of students who can speak Ibibio only. **(2 marks)**
- (iii) Find the number of students who can speak Efik only. **(2 marks)**

#### NOTE ON THE PAPER

**THIS QUESTION CONTRADICTS ITSELF, AND IT IS THE MOST IMPORTANT DEFECT ON THE PAPER.** The opening sentence names the same language twice: "72 students can speak Ibibio and 43 students can speak Ibibio". Parts (i) to (iii) then ask about Ibibio **AND EFİK**, so the

second figure must have been meant as Efik. Read literally, the data mention Efik nowhere and 7 marks are unanswerable.

**WHAT TO DO IN THE HALL.** Do not stall and do not silently rewrite the question. Write one line: *"The second figure is taken as 43 students who can speak Efik, since parts (i) to (iii) ask about Efik and the question is otherwise unanswerable."* Then answer on that reading. A stated assumption that is obviously the intended one is normal mathematical practice and protects every mark; a silent one does not.

#### BREAK IT DOWN

|                          |                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>             | 100 students in the group. 72 speak Ibibio. 43 speak Efik, on the stated reading.                                                                                                      |
| <b>ASKED</b>             | (i) <b>both</b> languages, (ii) <b>Ibibio only</b> , (iii) <b>Efik only</b> . Three different regions of the same diagram.                                                             |
| <b>HIDDEN ASSUMPTION</b> | That every one of the 100 speaks at least one of the two, so $n(A \cup B) = 100$ . The question does not say so, but without it the three parts have no unique answer. State this too. |
| <b>MARKS SAY</b>         | 3, 2 and 2. Part (i) is worth more because the other two follow from it in one subtraction each. Get (i) right and the rest are free.                                                  |
| <b>TEACHES</b>           | This section, 5.3.                                                                                                                                                                     |
| <b>TRAP</b>              | Answering (ii) with 72. That is everyone who speaks Ibibio, including those who also speak Efik. "Only" means the overlap must come out.                                               |

**GIVEN**  $n(U) = 100$ ,  $n(I) = 72$  speak Ibibio,  $n(E) = 43$  speak Efik.

**FIND**  $n(I \cap E)$ , then  $n(I \text{ only})$  and  $n(E \text{ only})$ .

**FORMULA**  $n(I \cup E) = n(I) + n(E) - n(I \cap E)$ .

**Step 0. State the two readings you are working under.**

1. The second figure of 43 refers to Efik, not Ibibio.
2. Every student speaks at least one of the two languages, so  $n(I \cup E) = 100$ .

Assumption 2 is forced by the arithmetic as well as by common sense:  $72 + 43 = 115$ , which is 15 more than the group, so at least 15 students must be in both. If some students spoke neither, the overlap would be even larger, and the question gives no way to know how many.

i. Students who speak both

Rearrange the two-set formula to make the overlap the subject:

$$\begin{aligned} n(I \cap E) &= n(I) + n(E) - n(I \cup E) \\ &= 72 + 43 - 100 \\ &= 115 - 100 \\ &= \mathbf{15} \end{aligned}$$

**Answer: 15 students speak both Ibibio and Efik.**

ii. Students who speak Ibibio only

The 72 Ibibio speakers include the 15 who also speak Efik. Strip them out:

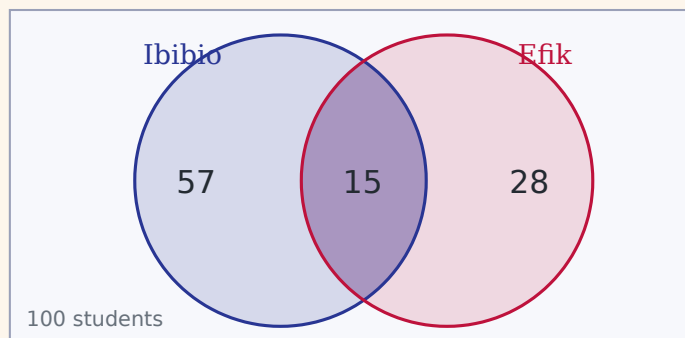
$$n(I \text{ only}) = n(I) - n(I \cap E) = 72 - 15 = \mathbf{57}$$

iii. Students who speak Efik only

$$n(E \text{ only}) = n(E) - n(I \cap E) = 43 - 15 = \mathbf{28}$$

**Check, and write the check down.** The three regions must fill the group exactly:

$$57 + 15 + 28 = 100 \quad \checkmark$$



$$57 + 15 = 72$$

$$28 + 15 = 43$$

$$\begin{aligned} 57 + 15 + 28 \\ = 100 \end{aligned}$$

*The finished diagram. Every given number can be read back off it, which is what makes it a check and not a decoration.*

Drawing this diagram is worth doing even when the question does not ask for it. It takes twenty seconds and it catches the "only" trap automatically.

**Now redo it:** in a group of 80, 50 speak Yoruba and 45 speak Hausa, everyone speaking at least one. Both:  $50 + 45 - 80 = 15$ . Yoruba only: 35. Hausa only: 30. Check:  $35 + 15 + 30 = 80$ .

## The class-size question

AS PRINTED

2021/2022 · Q5(B)

(b) In a discrete structures' class, every student is a major in either Computer Science or Maths or both. There of 25 students who major in Computer, while 13 in Maths. If 8 students major in both Computing Science & Maths, how many students are in the entire class? **(4marks)**

### NOTE ON THE PAPER

"There of 25 students" is a typing slip for "There are 25 students", and the subject is called "Computer Science" in one clause, "Computer" in the next and "Computing Science" in the third. All three name the same set. The sentence that saves the question is the first one: **EVERY STUDENT MAJORS IN COMPUTER SCIENCE OR MATHS OR BOTH**, which states outright the assumption that the Ibibio question left you to supply. So here  $n(A \cup B)$  is the class, and it is what you are asked for.

### BREAK IT DOWN

- GIVEN**  $n(C) = 25$  major in Computer Science,  $n(M) = 13$  major in Maths,  $n(C \cap M) = 8$  major in both, and nobody majors in neither.
- ASKED** **How many students are in the entire class**, which is  $n(C \cup M)$ .
- MARKS SAY** 4 marks for one formula. Write the principle, then substitute, then evaluate, then answer in a sentence. A bare 30 earns less.
- TEACHES** This section, 5.3.
- TRAP** Answering  $25 + 13 = 38$ , forgetting the 8 who were counted in both figures.

### WORKED EXAMPLE

PQ 21/22 Q5(B)

- GIVEN**  $n(C) = 25$ ,  $n(M) = 13$ ,  $n(C \cap M) = 8$ , every student in at least one.
- FIND**  $n(C \cup M)$ , the size of the class.
- FORMULA**  $n(C \cup M) = n(C) + n(M) - n(C \cap M)$ .

Because every student majors in one subject or the other or both, the class *is* the union, so the union is what to compute:

$$\begin{aligned}n(C \cup M) &= 25 + 13 - 8 \\&= 38 - 8 \\&= 30\end{aligned}$$

**Answer: there are 30 students in the entire class.**

**Break it into regions as a check:** Computer Science only is  $25 - 8 = 17$ , Maths only is  $13 - 8 = 5$ , and both is 8. Then  $17 + 5 + 8 = 30$ , which agrees.

**One ambiguity worth a line on the script.** The sentence "There of 25 students who major in Computer" could be misread as saying the class has 25 students in total. It cannot mean that: 8 of them major in both, and 13 major in Maths, so Maths-only alone would need 5 more students than a 25-strong class has room for once the 25 Computer Science majors are placed. The reading above is the only consistent one.

**Now redo it:** if instead 30 major in Computer Science, 20 in Maths and 12 in both, the class has  $30 + 20 - 12 = 38$  students.



## The pigeonhole principle

### THE PIGEONHOLE PRINCIPLE

**The claim.** If more objects are placed into boxes than there are boxes, then some box must hold more than one object. Nothing about the objects, the boxes or the way they are placed can prevent it.

**In symbols.**

If  $k + 1$  or more objects go into  $k$  boxes, some box holds at least 2.

If  $N$  objects go into  $k$  boxes, some box holds at least  $\lceil N / k \rceil$ .

**Conditions.** The number of boxes must be finite, and every object must go into some box. It says nothing about **which** box is crowded, and it cannot: that is a feature, since it lets you prove something exists without being able to find it.

**The limiting case.** When  $N = k$  the principle promises only  $\lceil k / k \rceil = 1$ , which is no information at all, and correctly so:  $k$  objects can sit one to a box with nothing shared. The principle bites from  $N = k + 1$  onward, and that single extra object is the whole of it.

### ● MUST MEMORISE

### PIGEONHOLE PRINCIPLE

**If  $k + 1$  or more objects are placed into  $k$  boxes, then at least one box contains two or more objects.**

WHERE

$k$  is the number of boxes, sometimes called pigeonholes. **objects** are whatever is being sorted into them: people, numbers, letters, socks.

**The generalised form**, which is the one that answers real questions: if  $N$  objects are placed into  $k$  boxes, then at least one box contains at least

$\lceil N / k \rceil$  objects,

where  $\lceil x \rceil$  is the **ceiling** of  $x$ : the smallest whole number that is not less than  $x$ . So  $\lceil 7 / 3 \rceil = \lceil 2.33 \rceil = 3$ .

◆ **Memory hook: more pigeons than holes means sharing.** The principle never tells you *which* box is crowded, only that one of them is. That is exactly why it is useful: it proves something exists without finding it.

Every pigeonhole argument has the same three steps, and the marks are in step 2.

1. **Name the objects.** What is being placed?
2. **Name the boxes.** This is the creative step. The boxes are usually the possible *values* of something, not physical containers.
3. **Compare the counts** and apply the principle.

**Example.** Among any 13 people, at least two were born in the same month. Objects: the 13 people. Boxes: the 12 months. Since  $13 > 12$ , two people share a box, that is, a birth month.

**Example, generalised form.** Among 100 people, at least  $\lceil 100 / 12 \rceil = \lceil 8.33 \rceil = 9$  share a birth month.

**Example with an unobvious box.** In any group of 8 people, two have the same number of friends within the group. Objects: 8 people. Boxes: the possible friend counts, 0 to 7, which is 8 boxes, so the counting alone is not enough. The extra step: nobody can have 0 friends and somebody else have 7 at the same time, because a person with 7 friends is friends with everyone, including the supposed loner. So only 7 of the 8 boxes can be occupied, and 8 people into 7 boxes forces a repeat.

Where this stands on the papers: none of the five papers we hold has set a pigeonhole question outright. It is in the official course text, it takes two minutes to learn, and the 2023/2024 paper does ask a closely related impossibility question about five cities that is settled by a counting argument in Part Six. Learn the statement and the three steps; that is the whole of the return on it.

## RECALL AND CHECK

1. State the inclusion and exclusion principle for two sets.
2. 40 students take Maths, 30 take Physics, 12 take both and everyone takes at least one. How many students are there?
3. In the same group, how many take Maths only?
4. Write the three-set formula from the pattern, without looking.
5. State the pigeonhole principle, and say what  $\lceil 17 / 5 \rceil$  is.

### ANSWERS

1.  $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ .
2.  $40 + 30 - 12 = 58$ .
3.  $40 - 12 = 28$ .
4. Add the three singles, subtract the three pairs, add the triple.
5. If  $k + 1$  or more objects go into  $k$  boxes, some box holds two or more.  $\lceil 17 / 5 \rceil = \lceil 3.4 \rceil = 4$ .

### FILL IN THE GAP

1.  $n(A \cup B) = n(A) + n(B) -$  \_\_\_\_\_.
2. The number who speak Ibibio only is 72 minus \_\_\_\_\_, giving \_\_\_\_\_.
3. In the three-set formula the pairs are \_\_\_\_\_ and the triple is \_\_\_\_\_.
4. Placing 10 objects into 3 boxes forces some box to hold at least \_\_\_\_\_.

### ANSWERS

1.  $n(A \cap B)$ . 2. 15, 57. 3. subtracted, added. 4.  $\lceil 10 / 3 \rceil = 4$ .

### PART FIVE · 5.4

## The Binomial Theorem

*Expanding a bracket raised to a power is a counting problem in disguise, and the numbers that come out are the combinations you met in section 5.2.*

*By the end of this section you will be able to read the binomial coefficient notation, state and apply the binomial theorem, build and use Pascal's triangle, pick out a single term without expanding the whole thing, and explain why the coefficients of a row add to a power of two.*

### TEACH

### WHERE THIS STANDS ON THE PAPERS

Say this plainly before spending time here. **None of the five papers we hold sets a binomial theorem question.** It is a full module of the official course text, so it is examinable and cheap to learn, but it should be the last thing you revise, not the first.

It earns its place for a second reason: it is the clearest proof that combinations are not just a formula for committees. The coefficients of an expansion *are* counts of selections, and seeing that makes section 5.2 harder to forget.

## The binomial coefficient

### ● MUST MEMORISE

### BINOMIAL COEFFICIENT

The **binomial coefficient** is the combination count of section 5.2 under a second name and a second notation. For whole numbers  $n$  and  $r$  with  $0 \leq r \leq n$ ,

**$C(n, r) = n! / (r! (n - r)!)$** , also written **(  $n$  over  $r$  )** and read " **$n$  choose  $r$** ".

### WHERE

**$n$**  is the power the bracket is raised to.  **$r$**  selects which term of the expansion you are looking at, counting from 0. " **$n$  choose  $r$** " is the spoken form. When a question says "choose", it means this.

**Two identities worth knowing on sight.**

$C(n, r) = C(n, n - r)$  [choosing  $r$  to keep is choosing  $n - r$  to leave]

$C(n, 0) = C(n, n) = 1$  [one way to take none, one way to take all]

◆ **Memory hook: the coefficient counts the ways.** The number in front of a term is the number of ways of picking that many copies of one letter from the brackets.

## The theorem

### ● MUST MEMORISE

### BINOMIAL THEOREM

For any numbers  $x$  and  $y$  and any whole number  $n$  that is not negative,

$$(x + y)^n = C(n, 0) x^n + C(n, 1) x^{n-1} y + C(n, 2) x^{n-2} y^2 + \dots + C(n, n) y^n$$

The general term, the one to quote when a question asks for a single term rather than the whole expansion, is

$$\text{term } r = C(n, r) x^{n-r} y^r, \text{ for } r = 0, 1, 2, \dots, n.$$

#### WHERE

$r$  counts from 0, so the expansion has  $n + 1$  terms, not  $n$ . **the exponents** always add to  $n$ : as the power of  $x$  falls, the power of  $y$  rises by the same step.

◆ **Memory hook: the powers see-saw and the exponents always sum to  $n$ .** If a line of your working breaks that rule, it is wrong, and you can see it without checking anything else.

### TEACH

### WHY THE COEFFICIENTS ARE COMBINATIONS

Write  $(x + y)^3$  as three brackets side by side and do not collect anything yet:

$$(x + y)(x + y)(x + y)$$

Multiplying out means going through the three brackets and taking either  $x$  or  $y$  from each. Every choice gives one product of three letters, and every such product is a term.

How many of those products come out as  $x^2y$ ? Exactly as many as there are ways of picking **which one** of the three brackets supplied the  $y$ , and that is  $C(3, 1) = 3$ . So the  $x^2y$  term is  $3x^2y$ , and it is a combination count, not a coincidence.

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

This also explains the second identity above. Choosing which brackets give  $y$  is the same job as choosing which give  $x$ , so  $C(3, 1) = C(3, 2)$  and the coefficients read the same forwards and backwards.

## Pascal's triangle

### TEACH

### GETTING THE COEFFICIENTS WITHOUT FACTORIALS

Write  $C(n, r)$  in rows,  $n$  going down and  $r$  across. Each row starts and ends with 1, and every other entry is the sum of the two directly above it. That triangle is the fastest way to get the coefficients for small  $n$ , which is all any exam will ask for.

|        |   |   |   |   |    |   |         |
|--------|---|---|---|---|----|---|---------|
| sum 1  |   |   | 1 |   |    |   | $n = 0$ |
| sum 2  |   |   | 1 |   | 1  |   | $n = 1$ |
| sum 4  |   |   | 1 |   | 2  |   | $n = 2$ |
| sum 8  |   |   | 1 |   | 3  |   | $n = 3$ |
| sum 16 |   | 1 |   | 4 |    | 6 | $n = 4$ |
| sum 32 | 1 |   | 5 |   | 10 |   | $n = 5$ |

$4 + 6 = 10$

Row  $n$  gives the coefficients of  $(x + y)$  to the power  $n$ . Each entry is the sum of the two above it, and each row adds to a power of two.

**The rule behind the triangle**, called Pascal's identity, is

$$C(n, r) = C(n - 1, r - 1) + C(n - 1, r)$$

and it says something sensible about choosing. Fix one particular object. Any selection of  $r$  either takes that object, leaving  $r - 1$  to pick from the other  $n - 1$ , or it does not, leaving all  $r$  to pick from the other  $n - 1$ . Those two cases are alternatives with no overlap, so the sum rule of section 5.1 adds them.

### ● MUST MEMORISE

### THE ROW SUM

The coefficients of row  $n$  add to  $2^n$ :

$$C(n, 0) + C(n, 1) + \dots + C(n, n) = 2^n$$

#### WHERE

$n$  is the row number, counting from 0 at the apex of the triangle.  $C(n, r)$  is the entry  $r$  places along that row, also counting from 0.  $2^n$  is 2 multiplied by itself  $n$  times, so row 4 adds to 16 and row 5 to 32.

**Why, in one line.** Put  $x = 1$  and  $y = 1$  in the binomial theorem. The left side becomes  $(1 + 1)^n = 2^n$  and every power on the right becomes 1, leaving just the coefficients.

**And what it means.**  $C(n, r)$  counts the subsets of size  $r$  of a set with  $n$  elements, so adding over every  $r$  counts *all* the subsets. That is  $2^n$ , which is exactly the size of the power set from section 2.3. Two sections of this book, arrived at from opposite directions, meeting on the same number.

◆ **Memory hook: every row doubles.** 1, 2, 4, 8, 16, 32. If your coefficients do not add to the next power of two, one of them is wrong.

## Using it

### WORKED EXAMPLE

### FULL EXPANSION

Expand  $(x + 2)^4$ .

**GIVEN** The bracket  $x + 2$  raised to the power 4, so  $n = 4$  and the second term is  $y = 2$ .

**FIND** The expansion in full.

**FORMULA** term  $r = C(4, r) x^{4-r} 2^r$ .

**Step 1. Read the coefficients off row 4:** 1, 4, 6, 4, 1.

**Step 2. Build each term.** The power of  $x$  falls from 4 to 0 and the power of 2 rises from 0 to 4.

| R | C(4, R) | $x^{4-R}$ | $2^R$ | TERM    |
|---|---------|-----------|-------|---------|
| 0 | 1       | $x^4$     | 1     | $x^4$   |
| 1 | 4       | $x^3$     | 2     | $8x^3$  |
| 2 | 6       | $x^2$     | 4     | $24x^2$ |
| 3 | 4       | $x$       | 8     | $32x$   |
| 4 | 1       | 1         | 16    | 16      |

$$(x + 2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$$

**Check it**, and this check costs one line. Put  $x = 1$  on both sides. The left is  $(1 + 2)^4 = 3^4 = 81$ , and the right is  $1 + 8 + 24 + 32 + 16 = 81$ . They agree, so no coefficient was dropped.

**Now redo it:** expand  $(x + 1)^5$ . Row 5 is 1, 5, 10, 10, 5, 1 and every power of 1 is 1, so it is  $x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1$ . Check at  $x = 1$ : the sum is  $32 = 2^5$ .

**Find the coefficient of  $x^3$  in  $(2x + 1)^5$ .**

Expanding all six terms would work and would waste four minutes. Use the general term instead and solve for  $r$ .

**GIVEN**  $n = 5$ , first term  $2x$ , second term  $1$ .

**FIND** The number multiplying  $x^3$ .

**FORMULA** term  $r = C(5, r) (2x)^{5-r} 1^r$ .

**Step 1. Find which  $r$  gives  $x^3$ .** The power of  $x$  in term  $r$  is  $5 - r$ , so

$$5 - r = 3, \text{ so } r = 2$$

**Step 2. Evaluate that one term.**

$$\begin{aligned} \text{term } 2 &= C(5, 2) (2x)^3 1^2 \\ &= 10 \times 8x^3 \times 1 \\ &= 80x^3 \end{aligned}$$

**Answer: the coefficient of  $x^3$  is 80.**

**The mistake to avoid** is taking the coefficient to be  $C(5, 2) = 10$  and forgetting that the 2 in  $2x$  is also raised to the power 3. Whenever the bracket has a number attached to a letter, that number carries its own power.

**Now redo it:** the coefficient of  $x^2$  in  $(x + 3)^4$ . Here  $4 - r = 2$  so  $r = 2$ , and the term is  $C(4, 2) x^2 3^2 = 6 \times 9 x^2 = 54x^2$ .

## RECALL AND CHECK

1. State the binomial theorem.
2. How many terms has the expansion of  $(x + y)^7$ ?
3. Write row 4 of Pascal's triangle and say what its entries add to.
4. Expand  $(x + 2)^4$ .
5. Why is  $C(n, r) = C(n, n - r)$ ?

### ANSWERS

1.  $(x + y)^n$  is the sum over  $r$  of  $C(n, r) x^{n-r} y^r$ , for  $r$  from 0 to  $n$ .
2. Eight. The terms run  $r = 0$  to  $r = 7$ , so there are  $n + 1$  of them.
3. 1, 4, 6, 4, 1, adding to  $16 = 2^4$ .
4.  $x^4 + 8x^3 + 24x^2 + 32x + 16$ .
5. Choosing which  $r$  objects to take is the same act as choosing which  $n - r$  to leave behind, so the two counts are equal.

### FILL IN THE GAP

1. In every term of  $(x + y)^n$  the two exponents add to \_\_\_\_\_.
2. The expansion of  $(x + y)^n$  has \_\_\_\_\_ terms.
3. Row  $n$  of Pascal's triangle adds to \_\_\_\_\_.
4. The coefficient of  $x^3$  in  $(2x + 1)^5$  is \_\_\_\_\_.

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### ANSWERS

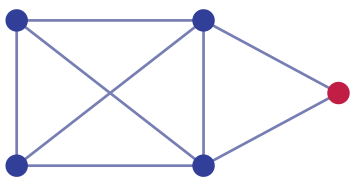
1.  $n$ . 2.  $n + 1$ . 3.  $2^n$ . 4. 80.



## PART SIX

# Graph Theory: Vertices and Edges

*The heaviest part of the exam and the emptiest part of the course text. Every paper we hold gives graphs at least a third of its marks, and the official text does not define a vertex. Everything here is taught from zero.*



$\deg(v)$  degree of a vertex

$\sum \deg = 2e$  handshaking

$v - e + f = 2$  planarity

$K_5, K_{3,3}$  the non-planar pair

# Graphs, Vertices, Edges and Degree

A graph is dots and lines. Almost every mark in this part comes from naming the dots and lines precisely, so the vocabulary is the subject, not the preamble.

By the end of this section you will be able to define a graph, a vertex and an edge, count the order and the size of a graph, compute the degree of a vertex, say what incidence and adjacency mean, name loops, parallel edges, pendant and isolated vertices, and answer the incidence and short-answers questions in full.

## NOT IN THE COURSE TEXT

The official 161-page text contains no graph theory at all. It never uses the word vertex, never defines an edge, and never states the Handshaking theorem. Meanwhile 2025/2026 gives graphs roughly 30 of its marks, 2024/2025 gives a whole 17.5-mark question to one graph, and 2020/2021, 2021/2022 and 2023/2024 each build two questions on them. This part is your only source, so nothing here assumes you have seen a graph before.

## What a graph is

### ● MUST MEMORISE

### GRAPH, VERTEX, EDGE

A **graph**  $G = (V, E)$  consists of a set  $V$  of **vertices** and a set  $E$  of **edges**.

A **vertex** (plural **vertices**) is a point, drawn as a dot or a small circle, standing for an object: a city, a computer, a person, a state of a program. A vertex is also called a **node**.

An **edge** is a line that joins two vertices, standing for a relationship between the objects: a road, a cable, a friendship, a transition.

#### WHERE

$V$  is the vertex set, and it must not be empty: a graph has at least one vertex.  $E$  is the edge set, and it may be empty. A graph with no edges is still a graph.  $G = (V, E)$  is the standard way of naming a graph by its two parts, and it is worth writing out in an answer.

◆ **Memory hook: dots and lines, objects and relationships.** The picture is not the graph. Two very different-looking drawings can be the same graph, because only which dots are joined matters, never where they sit on the page.

### ● MUST MEMORISE

### ORDER AND SIZE

The **order** of a graph is the number of vertices, written  $|V|$  or  $n$ .

The **size** of a graph is the number of edges, written  $|E|$  or  $e$ .

#### WHERE

$|V|$  and  $|E|$  are cardinalities, the same bars used for set size in section 2.1. **order** counts dots. **size** counts lines. The two words are not interchangeable, and 2025/2026 Q4(a) pays 2 marks for knowing which is which.

◆ **Memory hook: order counts the objects, size counts the connections.** Everyday English suggests "size" should mean how big the graph is; in graph theory it means how many edges it has. Learn it against your instinct.

## Degree, and the words that go with it

### ● MUST MEMORISE

### DEGREE AND VALENCY

The **degree of a vertex**  $v$ , written  **$\deg(v)$** , is the number of edges meeting  $v$ , with a loop counted twice.

**Valency** is another word for exactly the same thing. Some papers use one word, some the other, and 2025/2026 asks you to tell them apart. They are the same quantity under two names.

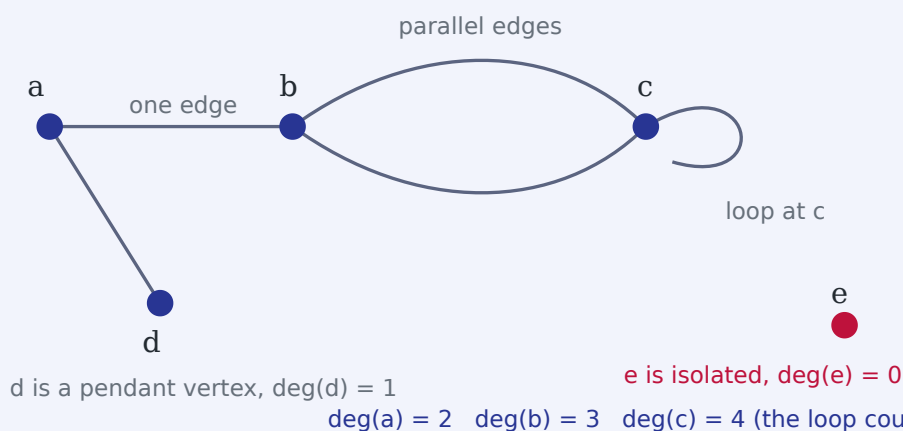
#### WHERE

**$\deg(v)$**  is a whole number, 0 or more. **a loop** at  $v$  counts **2**, because both of its ends meet  $v$ . **parallel edges** each count separately: two edges between the same pair contribute 2 to each endpoint.

◆ **Memory hook: count the line-ends touching the dot, not the neighbours.** A vertex joined to one neighbour by three parallel edges has degree 3, not 1.

| TERM                   | MEANING                                                                               | TEST FOR IT                                   |
|------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>incident</b>        | A vertex is incident to an edge when the edge has that vertex as one of its two ends. | Does the line touch the dot?                  |
| <b>adjacent</b>        | Two vertices are adjacent when an edge joins them.                                    | Is there a line between the two dots?         |
| <b>loop</b>            | An edge whose two ends are the same vertex.                                           | Does the line start and finish at one dot?    |
| <b>parallel edges</b>  | Two or more edges joining the same pair of vertices. Also called multiple edges.      | More than one line between the same two dots? |
| <b>pendant vertex</b>  | A vertex of degree 1. Also called a leaf or an end vertex.                            | Exactly one line touches it?                  |
| <b>isolated vertex</b> | A vertex of degree 0.                                                                 | No line touches it at all?                    |

**Incidence and adjacency are not the same relation**, and the exam asks about incidence specifically. **Incidence relates a vertex to an edge; adjacency relates a vertex to another vertex.** Saying "the vertices are incident" is wrong, and so is "the vertex is adjacent to the edge".



*Every term at once. Note that c has degree 4: two parallel edges to b, plus a loop worth two.*

## Degree and valency, told apart

a) With the aid of two examples each, differentiate between the following:

- |                                                      |           |
|------------------------------------------------------|-----------|
| i) Degree and Valency of a node in a graph           | (2 marks) |
| ii) Predicate Logic and Propositional logic          | (2 marks) |
| iii) Universal Quantifier and Existential Quantifier | (2 marks) |
| iv) Adjacency and Path Matrice                       | (2 marks) |
| v) A logical matrix and relation matrix              | (2marks)  |

#### NOTE ON THE PAPER

Three defects, all harmless once seen. Item ii) prints "Prepositional logic"; the word is **PROPOSITIONAL**, since a preposition is a part of speech and has nothing to do with logic. Item iv) prints "Path Matrice", a slip for "Path Matrix" or "Path Matrices". Item v) prints "(2marks)" closed up, unlike the four above it. Quote them as printed and answer the intended questions.

This section answers i). Item ii) is section 1.4, item iii) is section 1.4, and items iv) and v) are section 6.5. The 10 marks are spread across three parts of this book, which is worth knowing before you start writing.

#### BREAK IT DOWN

**GIVEN** Five pairs of terms, 2 marks each.

**ASKED** **Differentiate, with the aid of two examples each.** The examples are not optional decoration: they are named in the instruction, so an answer without them cannot score full marks.

**MARKS SAY** 2 marks per pair. Read as roughly 1 for the distinction and 1 for the two examples. Keep each answer to four or five lines and move on; there are ten items on this question.

**TEACHES** Item i) here in 6.1. See 1.4 for ii) and iii), and 6.5 for iv) and v).

**TRAP** Item i) is a trick in the sense that there is **no** numerical difference between degree and valency. Saying so confidently, and explaining where each word comes from, is the answer. Inventing a difference loses the marks.

**Degree and valency of a node in a graph.**

**The distinction, stated plainly.** They are two names for one quantity: the number of edges meeting the node, a loop counting twice. There is no graph and no node for which the degree and the valency differ. The difference is one of usage, not of value:

|                           | DEGREE                                                    | VALENCY                                                            |
|---------------------------|-----------------------------------------------------------|--------------------------------------------------------------------|
| Where the word comes from | graph theory and mathematics                              | chemistry, where an atom's valency is the number of bonds it forms |
| How it is written         | $\deg(v)$ , or $d(v)$                                     | usually written out in words                                       |
| Where you will meet it    | the Handshaking theorem, degree sequences, most textbooks | older texts, chemical graph theory, and this course's own papers   |

**Two examples, as the question demands.**

1. In the figure above, vertex  $b$  is met by one edge to  $a$  and two parallel edges to  $c$ . So  **$\deg(b) = 3$** , and equally the **valency of  $b$  is 3**.
2. Vertex  $c$  carries two parallel edges to  $b$  and one loop. The loop contributes 2, so  **$\deg(c) = 2 + 2 = 4$** , and the **valency of  $c$  is 4**. Both words give the same 4, and both would give the same 4 for any vertex you choose.

**The sentence that earns the mark:** "Degree and valency are synonyms for the number of edge-ends at a vertex; the terms differ only in origin, degree being the graph-theoretic word and valency the term borrowed from chemistry."

**Now redo it:** what are the degree and the valency of the pendant vertex  $d$ ? Both are 1, and both would be 0 for the isolated vertex  $e$ .

**Short notes: size, valency, and the rest**

(a) With relevant examples, write short notes on the following

- |                                            |           |
|--------------------------------------------|-----------|
| (i) Size of a graph                        | (2 marks) |
| (ii) The valency of a vertex in a graph    | (2 marks) |
| (iii) In-degree and Out-degree of a vertex | (2 marks) |
| (iv) Graph isomorphism                     | (2 marks) |
| (v) Eulerian graph                         | (2 marks) |

**NOTE ON THE PAPER**

Ten marks in five two-mark notes, and they run right across this part: (i) and (ii) belong here, (iii) is section 6.4, and (iv) and (v) are section 6.6. Each answer wants a definition plus a relevant example, and two marks buys about four lines. Items (i) and (ii) are answered below; the other three are answered where they are taught, and cross-referenced back here.

## (i) Size of a graph

**Definition.** The size of a graph  $G = (V, E)$  is the number of edges it has, written  $|E|$ . It is not the number of vertices; that is the **order**,  $|V|$ .

**Example 1.** The graph in the figure above has edges  $ab$ ,  $bc$ ,  $bc$  again, the loop at  $c$ , and  $ad$ . That is five edges, so its size is 5. Its order is 5 as well, since it has vertices  $a$ ,  $b$ ,  $c$ ,  $d$  and  $e$ , and the coincidence is worth pointing out precisely because it shows the two counts are different questions.

**Example 2.** A graph on 4 vertices with every pair joined has  $C(4, 2) = 6$  edges, so its size is 6 and its order is 4.

## (ii) The valency of a vertex in a graph

**Definition.** The valency of a vertex is the number of edges incident to it, a loop counting twice. It is the same quantity as the degree,  $\deg(v)$ .

**Example 1.** In a triangle, every vertex is joined to the other two, so every vertex has valency 2.

**Example 2.** In a star with one centre joined to four outer vertices, the centre has valency 4 and each outer vertex has valency 1, so each outer vertex is a pendant vertex.

## Incidence

## AS PRINTED

2020/2021 · Q1(C)

(c) When is a Vertex said to be Incident to an Edge? Identify all the edges each of the vertices is incident to in the diagram below. **(5marks)**

## AS PRINTED

2021/2022 · Q5(C)

(c) When is a Vertex said to be Incident to an Edge? Show an example **(3½marks)**

## NOTE ON THE PAPER, AND ON THE FIGURE

The same question on two papers, one wanting a worked figure for 5 marks and the other an example for 3.5. The definition is identical; only the second half changes.

**THE 2020/2021 FIGURE CANNOT BE FULLY READ.** It shows four vertices numbered 1 to 4 and six edges labelled  $e_1$  to  $e_6$ . Vertices 1, 2 and 3 and the edges  $e_1$  to  $e_5$  are clear, including that  $e_1$  and  $e_2$  are parallel, as are  $e_4$  and  $e_5$ . The sixth edge,  $e_6$ , is drawn as a long curve on the right passing around vertex 4, and at the resolution we hold it cannot be settled whether it is a loop at vertex 2 or a second path from 2 to 3. Vertex 4 is joined to nothing either way, which is plainly the point the question is testing. The answer below states the reading it uses, tabulates on that reading, and shows what changes under the other. **READ THE ARROWLESS CURVE ON YOUR OWN COPY BEFORE COMMITTING.**

### BREAK IT DOWN

|                  |                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | A graph with vertices 1, 2, 3, 4 and edges $e_1$ to $e_6$ .                                                                                                                                              |
| <b>ASKED</b>     | Two things. <b>When</b> a vertex is said to be incident to an edge, which is a definition; and <b>identify all the edges each vertex is incident to</b> , which is a table with one row per vertex.      |
| <b>MARKS SAY</b> | 5 marks: about 1.5 for the definition and 3.5 for the four rows. Do not answer only the first half; it is worth under a third of the marks.                                                              |
| <b>TEACHES</b>   | This section, 6.1.                                                                                                                                                                                       |
| <b>TRAP</b>      | Skipping vertex 4 because nothing touches it. Its row is the one the examiner is watching for: it is incident to <b>no</b> edges, so it is an isolated vertex, and saying so is worth a mark on its own. |

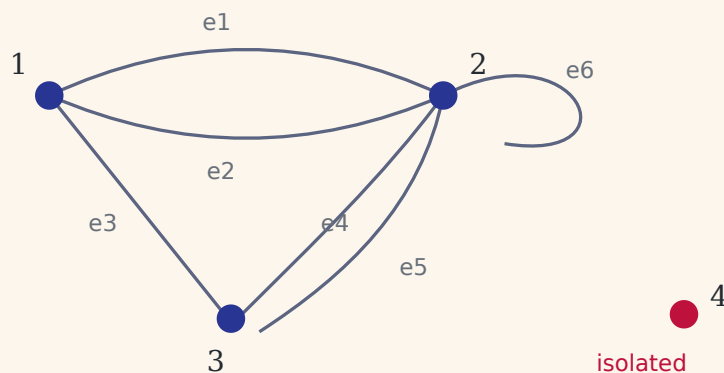


- GIVEN** Vertices 1, 2, 3, 4. Edges e1 and e2 both joining 1 and 2; e3 joining 1 and 3; e4 and e5 both joining 2 and 3; e6 a loop at 2 on the reading used here.
- FIND** The definition of incidence, then every edge at every vertex.
- RULE** A vertex is incident to an edge exactly when it is one of that edge's two endpoints.

**Part 1: the definition.** A vertex  $v$  is said to be **incident** to an edge  $e$  when  $v$  is one of the two endpoints of  $e$ ; equivalently, when  $e$  touches  $v$  in the drawing. The relation runs both ways in speech: we say that  $e$  is incident to  $v$ , and that  $v$  is incident to  $e$ . For a loop, the single vertex is incident to the edge, and the edge contributes 2 to that vertex's degree.

**Part 2: the table.** State the reading first, then tabulate.

*Reading used: e6 is a loop at vertex 2.*



*The figure as read, with e6 taken as a loop at vertex 2 and vertex 4 joined to nothing.*

| VERTEX | EDGES IT IS INCIDENT TO                     | DEGREE |
|--------|---------------------------------------------|--------|
| 1      | e1, e2, e3                                  | 3      |
| 2      | e1, e2, e4, e5, e6 (the loop, twice)        | 6      |
| 3      | e3, e4, e5                                  | 3      |
| 4      | none; vertex 4 is an <b>isolated vertex</b> | 0      |

**Check the total.** The degrees add to  $3 + 6 + 3 + 0 = 12$ , which is twice the six edges. That is the Handshaking theorem of section 6.3, and it is the fastest way to catch a miscounted table.

**Under the other reading**, where e6 joins 2 to 3 rather than looping, vertex 2 is incident to e1, e2, e4, e5, e6 with degree 5 and vertex 3 to e3, e4, e5, e6 with degree 4. The degrees then add to  $3 + 5 + 4 + 0 = 12$ , still twice six, so the check passes either way. Only the two middle rows change.

**For the 2021/2022 version**, which asks only to "Show an example", one edge and its two vertices is enough: draw an edge  $e$  joining  $u$  and  $v$ , and write " $u$  is incident to  $e$  and  $v$  is incident to  $e$ ;  $u$  and  $v$  are adjacent". Add a loop as a second example to show you know it counts twice.

**Now redo it:** in a triangle with vertices  $x, y, z$  and edges  $f_1 = xy, f_2 = yz, f_3 = zx$ , which edges is  $y$  incident to?  $f_1$  and  $f_2$ , so  $\deg(y) = 2$ .

## Where graphs are used

AS PRINTED

2020/2021 · Q2(B)

(b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. **(6marks)**

AS PRINTED

2020/2021 · Q3(C)

(c) Highlight, with a relevant example each, four application areas in computing where graph models are applicable. **(8marks)**

AS PRINTED

2021/2022 · Q3(B)

(b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. **(6marks)**

AS PRINTED

2021/2022 · Q5(A)

(a) Highlight, with a relevant example each, four computing domains where graph models are applicable. **(6marks)**

### NOTE ON THE PAPER

One question in three costumes, worth 6, 8 and 6 marks, and 2021/2022 Q3(b) repeats the first of them word for word. Two things change between them and both change your answer. **REAL-LIFE** versus **COMPUTING**: the first wants roads and social networks, the second wants compilers and routing tables. And **TWO EXAMPLES EACH** versus **A RELEVANT EXAMPLE EACH**: 2020/2021 Q2(b) asks for eight examples in total, the others for four. Count them before you start writing.

**Answer every version from one table**, taking the top half for the real-life question and the bottom half for the computing one. In each case say what the vertices are and what the edges are; that is the modelling step, and it is what separates a full answer from a list of words.

Four real-life scenarios, with two examples each

| SCENARIO                         | VERTICES                           | EDGES                            | TWO EXAMPLES                                                                                                              |
|----------------------------------|------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Transport networks</b>        | towns, junctions, stations         | roads, rail lines, flight routes | a road map used to find the shortest drive; an airline route map used to plan connections                                 |
| <b>Social networks</b>           | people                             | friendships, follows, calls      | a friend graph used to suggest people you may know; a call graph used to spot a fraud ring                                |
| <b>Utility distribution</b>      | substations, reservoirs, junctions | power lines, pipes               | an electricity grid checked for a route that survives one cable failing; a water network sized to reach every district    |
| <b>Scheduling and allocation</b> | tasks, exams, rooms                | a conflict between two of them   | an examination timetable where joined exams may not share a slot; radio frequencies assigned so neighbouring masts differ |

Four computing application areas, with an example each

| AREA                                      | VERTICES                  | EDGES                                            | EXAMPLE                                                                     |
|-------------------------------------------|---------------------------|--------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Computer networks</b>                  | routers and hosts         | links between them                               | a routing table computed as shortest paths across the link graph            |
| <b>Compilers</b>                          | basic blocks of a program | a jump or fall-through from one block to another | the control-flow graph a compiler walks to decide which code is unreachable |
| <b>Databases and the web</b>              | pages, records, entities  | hyperlinks, foreign keys                         | a search engine ranking pages by how the link graph points at them          |
| <b>Artificial intelligence and search</b> | states of a problem       | a legal move from one state to the next          | a route planner or a game solver searching the state graph for a goal       |

**One closing line worth writing**, because it shows why the topic exists rather than just listing where it appears: in every one of these, the objects and the connections are all that matter, and the geometry of the drawing is irrelevant. That is precisely what a graph abstracts.

## RECALL AND CHECK

1. Define a graph, a vertex and an edge.
2. What is the difference between the order and the size of a graph?
3. How much does a loop add to a vertex's degree, and why?
4. When is a vertex incident to an edge?
5. What is a pendant vertex, and what is an isolated vertex?
6. Is there any graph in which a vertex's degree and its valency differ?

### ANSWERS

1. A graph  $G = (V, E)$  is a set  $V$  of vertices, which are points standing for objects, together with a set  $E$  of edges, each joining two vertices and standing for a relationship.
2. Order is the number of vertices,  $|V|$ . Size is the number of edges,  $|E|$ .
3. Two, because both of its ends meet that vertex.
4. When the vertex is one of the edge's two endpoints.
5. A pendant vertex has degree 1; an isolated vertex has degree 0.
6. No. They are two names for the same number.

## FILL IN THE GAP

1. A graph is written  $G = (\text{_____}, \text{_____})$ .
2. The number of edges is the \_\_\_\_\_ of the graph.
3. A vertex of degree 1 is called a \_\_\_\_\_ vertex.
4. Two vertices joined by an edge are \_\_\_\_\_; a vertex at the end of an edge is \_\_\_\_\_ to it.
5. Valency is another word for \_\_\_\_\_.

### ANSWERS

1.  $V, E$ . 2. size. 3. pendant. 4. adjacent, incident. 5. degree.

## PART SIX · 6.2

# The Types of Undirected Graph

*This is the single most repeated question in the whole set. It has appeared on three of the five papers we hold, worth 12, 12 and 9 marks, and it is always phrased the same way.*

*By the end of this section you will be able to name and draw every standard type of undirected graph, count the edges of a complete graph, decide which type a given drawing is, and answer the types question and the communication-links questions in full.*

Three papers, one question:

| PAPER           | AS ASKED                                       | MARKS |
|-----------------|------------------------------------------------|-------|
| 2020/2021 Q1(a) | with the aid of relevant examples              | 12    |
| 2021/2022 Q1(c) | with diagrammatical illustrations              | 12    |
| 2023/2024 Q4(a) | in details, and with diagrammatic illustration | 9     |

That is 33 marks across three papers for one prepared answer. Every version demands **drawings**, so an answer in words alone throws away most of the marks however correct it is. Learn the list, and practise sketching all of it in under ten minutes.

## The types

### ● MUST MEMORISE

### THE FAMILIES BY WHAT EDGES ARE ALLOWED

The first three types are told apart by one question: **what kinds of edge does the graph permit?**

| TYPE                | LOOPS? | PARALLEL EDGES? | IN ONE LINE                                                                 |
|---------------------|--------|-----------------|-----------------------------------------------------------------------------|
| <b>Simple graph</b> | no     | no              | at most one edge between any two vertices, and none from a vertex to itself |
| <b>Multigraph</b>   | no     | yes             | a simple graph that also allows repeated edges between the same pair        |
| <b>Pseudograph</b>  | yes    | yes             | anything goes: the most general undirected graph                            |

#### WHERE

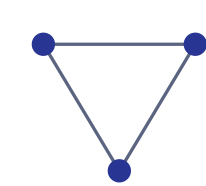
**loop** is an edge from a vertex to itself, defined in section 6.1. **parallel edges** are two or more edges joining the same pair, also from 6.1. **the nesting** matters: every simple graph is a multigraph, and every multigraph is a pseudograph. The families contain one another, they do not exclude one another.

◆ **Memory hook: simple, multi, pseudo.** Simple forbids both, multi allows repeats, pseudo allows repeats and self-loops. Say them in that order and the permissions only ever grow.

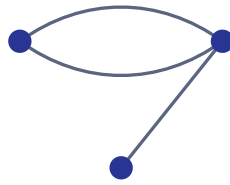
The remaining types are told apart by **shape** rather than by permission.

| TYPE                      | DEFINITION                                                                                                             | SYMBOL          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Null (empty) graph</b> | Has vertices but no edges at all. Every vertex is isolated.                                                            | $N_n$           |
| <b>Trivial graph</b>      | Has exactly one vertex and no edges. The smallest graph there is.                                                      | $K_1$           |
| <b>Complete graph</b>     | Simple, and <i>every</i> pair of distinct vertices is joined by an edge.                                               | $K_n$           |
| <b>Regular graph</b>      | Every vertex has the same degree. If that degree is $k$ it is <b><math>k</math>-regular</b> .                          |                 |
| <b>Cycle graph</b>        | $n$ vertices joined in a single closed ring; 2-regular.                                                                | $C_n$           |
| <b>Path graph</b>         | $n$ vertices in a line, not closed up.                                                                                 | $P_n$           |
| <b>Wheel graph</b>        | A cycle $C_n$ plus one extra vertex joined to every vertex of the cycle.                                               | $W_n$           |
| <b>Bipartite graph</b>    | The vertices split into two sets so that every edge runs between the sets and none inside either.                      |                 |
| <b>Complete bipartite</b> | Bipartite, and every vertex of one set is joined to every vertex of the other.                                         | $K_{m,n}$       |
| <b>Connected graph</b>    | There is a route between every pair of vertices. Otherwise <b>disconnected</b> , and it falls into <b>components</b> . |                 |
| <b>Weighted graph</b>     | Every edge carries a number: a distance, a cost, a capacity.                                                           |                 |
| <b>Subgraph</b>           | A graph made from some of another graph's vertices and some of its edges.                                              | $H \subseteq G$ |

◆ **Memory hook: K for complete, C for cycle, P for path, W for wheel, N for null.** The subscript is always the number of vertices, except in  $K_{m,n}$  where it is the two side sizes.



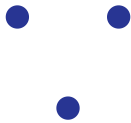
Simple



Multigraph



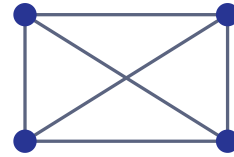
Pseudograph



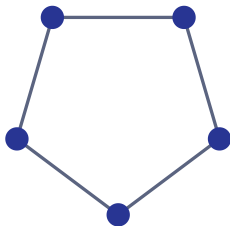
Null  $N_3$



Trivial  $K_1$



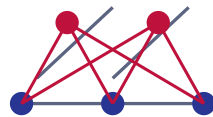
Complete  $K_4$



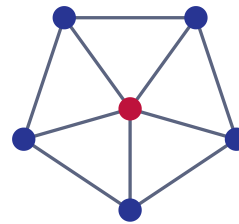
Cycle  $C_5$



Path  $P_4$



Complete bipartite  $K_{2,3}$



Wheel  $W_5$

Nine of the types at a glance. Crimson marks the vertex or the side that makes each shape what it is: the hub of the wheel, the two sides of the bipartite graph.

## ● MUST MEMORISE

## EDGES OF A COMPLETE GRAPH

The complete graph on  $n$  vertices,  $K_n$ , has

$$C(n, 2) = n(n - 1) / 2 \text{ edges,}$$

and every vertex has degree  $n - 1$ .

### WHERE

$n$  is the number of vertices.  $C(n, 2)$  is the combination count of section 5.2: an edge is an unordered pair of distinct vertices, and order does not matter because an edge from  $a$  to  $b$  is the same edge as one from  $b$  to  $a$ .  $n - 1$  is the degree because each vertex is joined to every other vertex, that is, to all but itself.

### The values you will need:

| $N$            | 2 | 3 | 4 | 5  | 6  | 7  |
|----------------|---|---|---|----|----|----|
| EDGES OF $K_N$ | 1 | 3 | 6 | 10 | 15 | 21 |

◆ **Memory hook: every pair, counted once.** The "counted once" is the divide by 2, and forgetting it is the standard error:  $n(n - 1)$  counts each edge from both ends.

## The question that appears three times

AS PRINTED

2020/2021 · Q1(A)

1. (a) Explain, with the aid of relevant examples, the types of undirected graph.

(12marks)

AS PRINTED

2021/2022 · Q1(C)

(c) Explain the types of undirected graph, with diagrammatical illustrations.

(12marks)

AS PRINTED

2023/2024 · Q4(A)

(a) Explain in details (and with diagrammatic illustration) the types of undirected graph.

(9 marks)

### NOTE ON THE PAPER

The same question three times, differing only in wording and marks. The 2023/2024 version has no question number printed above it: the scan runs straight from Q3(d) into "(a) Explain in details ...", which is plainly Question 4 but is nowhere labelled as such. The phrase "Explain in details" is not standard English but its meaning is not in doubt. All three demand pictures.

### BREAK IT DOWN

**GIVEN** Nothing. This is a recall question with no data.

**ASKED** **Explain** the types, **with examples** or **with diagrams**. Two things per type: what it is, and a drawing of one.

**MARKS SAY** 12, 12 and 9. At roughly 1 to 1.5 marks per type, aim for **eight to ten types** on the 12-mark versions and six or seven on the 9-mark version. More types briefly beats three types at length.

**TEACHES** This section, 6.2.

**TRAP** Two. Writing prose with no drawings, when the instruction names them. And drifting into **directed** graphs: the question says undirected, so digraphs and the two directional degrees earn nothing here. They are section 6.4 and a different question.



**How to lay the answer out.** One short paragraph per type, each with a small sketch beside it. Open with a single framing sentence, then work down the list. The framing sentence is worth writing because it shows the list is organised rather than remembered:

*"An undirected graph is one whose edges have no direction. The types divide first by which edges are permitted, and then by the shape the edges make."*

Group 1: told apart by what edges are allowed

1. **Simple graph.** No loops and no parallel edges, so at most one edge between any two vertices. *Example:* a triangle on a, b, c.
2. **Multigraph.** Parallel edges are allowed, loops are not. *Example:* two towns joined by two separate roads.
3. **Pseudograph.** Both loops and parallel edges are allowed; the most general undirected graph. *Example:* a network where a router also has a cable looping back to it-self for testing.

Group 2: told apart by shape

1. **Null or empty graph  $N_n$ .** Vertices but no edges; every vertex is isolated. *Example:* three computers with no cables between them.
2. **Trivial graph  $K_1$ .** One vertex, no edges. The smallest graph there is, and a special case of the null graph.
3. **Complete graph  $K_n$ .** Simple, with every pair of distinct vertices joined. It has  $n(n - 1)/2$  edges and every vertex has degree  $n - 1$ . *Example:*  $K_4$ , four vertices and six edges.
4. **Regular graph.** Every vertex has the same degree;  $k$ -regular if that degree is  $k$ . *Example:* a cycle is 2-regular, and  $K_n$  is  $(n - 1)$ -regular.
5. **Cycle graph  $C_n$ .** The vertices form one closed ring. *Example:*  $C_5$ , a pentagon.
6. **Path graph  $P_n$ .** The vertices in a line, the ring left open. Its two end vertices are pendant.
7. **Wheel graph  $W_n$ .** A cycle plus a hub joined to every vertex on it. *Example:* a bicycle wheel, rim and spokes.
8. **Bipartite graph.** The vertices split into two sets with every edge running between the sets and none inside either. *Example:* students on one side, courses on the other, an edge for each enrolment. If every cross pair is joined it is **complete bipartite**,  $K_{m,n}$ .
9. **Connected and disconnected graphs.** Connected means a route exists between every pair of vertices. A disconnected graph falls into separate **components**. *Example:* the figure in section 6.1 is disconnected, because vertex 4 is joined to nothing.
10. **Weighted graph.** Each edge carries a number such as a distance or a cost. *Example:* the five-city map of section 6.8, whose edges are labelled with mileages.
11. **Subgraph.** A graph formed from some of the vertices and some of the edges of another. *Example:* a triangle inside  $K_4$ .

**Close with one sentence on the nesting**, which few candidates write and which shows real understanding: *every simple graph is a multigraph, and every multigraph is*

a pseudograph, so the three families contain one another rather than exclude one another. The shape-based types cut across all three.

**Now redo it:** which type is a single square with its two diagonals drawn in? Four vertices, every pair joined, no loops or repeats: it is the complete graph  $K_4$ , and it is also 3-regular.

## Counting the possible links

AS PRINTED

2020/2021 · Q1(B)

(b) Assuming you are required to set up a simple communication link between CSIS, CST Lab, CLR, CUCRID and Senate Building, how many of such simple communication links are possible? Support your answer with a diagrammatical illustration. **(6½marks)**

AS PRINTED

2021/2022 · Q2(B)

(b) Assuming CU has contracted you to help set up simple communication links between the SENATE Building, CUCRID, CLR & CSIS NOC. How many of such simple communication links are possible? Support your answer with a diagrammatical illustration. **(7½marks)**

### NOTE ON THE PAPER

These two look identical and are not. **COUNT THE BUILDINGS.** The 2020/2021 version names **FIVE** (CSIS, CST Lab, CLR, CUCRID, Senate Building) and the 2021/2022 version names **FOUR** (SENATE Building, CUCRID, CLR, CSIS NOC). Five gives 10 links and four gives 6. Copying the answer across from one paper to the other is the fastest way to lose 6.5 marks, and it is exactly what a candidate who has memorised "the answer is 10" will do.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | A list of named buildings, and the word <b>simple</b> , which is doing real work: no building links to itself, and no pair is linked twice.                                                                                                       |
| <b>ASKED</b>     | <b>How many such links are possible</b> , plus a <b>diagrammatical illustration</b> .                                                                                                                                                             |
| <b>MARKS SAY</b> | 6.5 and 7.5, which is far more than one combination is worth. The marks are in the modelling and the drawing: say that buildings are vertices and links are edges, name the complete graph, give the formula, then draw it with every edge shown. |
| <b>TEACHES</b>   | This section, 6.2, using $C(n, 2)$ from section 5.2.                                                                                                                                                                                              |
| <b>TRAP</b>      | Answering $n(n - 1)$ instead of $n(n - 1)/2$ . A link between CLR and CUCRID is the same link as one between CUCRID and CLR, so each pair must be counted once.                                                                                   |

- GIVEN** Five buildings in the 2020/2021 version, four in the 2021/2022 version. Links are simple: no self-links, no repeats.
- FIND** The number of possible links, and a diagram.
- FORMULA** Links =  $C(n, 2) = n(n - 1)/2$ , the edge count of the complete graph  $K_n$ .

**Step 1. Model it.** Let each building be a **vertex** and each possible communication link an **edge**. "How many links are possible" asks for the case where every pair is joined, which is the **complete graph**  $K_n$ .

**Step 2. Count for five buildings (2020/2021).**

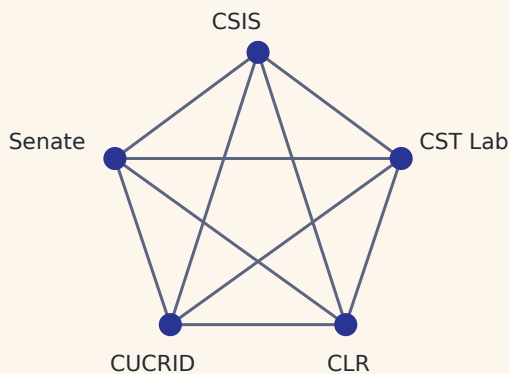
$$\text{links} = C(5, 2) = 5 \times 4 / 2 = 20 / 2 = 10$$

List them once to show the count is right: CSIS to each of the other four is 4 links; CST Lab to the three not yet used is 3; CLR to the two remaining is 2; CUCRID to Senate is 1. Then  $4 + 3 + 2 + 1 = 10$ , and no pair is repeated because each building only reaches forward down the list.

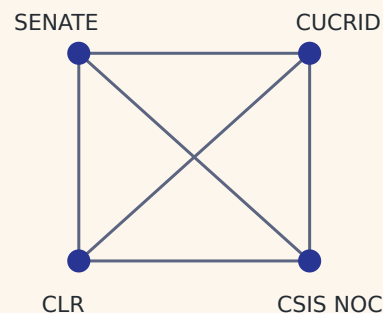
**Step 3. Count for four buildings (2021/2022).**

$$\text{links} = C(4, 2) = 4 \times 3 / 2 = 12 / 2 = 6$$

By the same listing,  $3 + 2 + 1 = 6$ .



$$K_5: 5 \times 4 / 2 = 10 \text{ links}$$



$$K_4: 4 \times 3 / 2 = 6 \text{ links}$$

*Both answers drawn. Five buildings give ten links; four give six. Draw every edge, and label the vertices with the names the question used.*

**The sentence that ties it together:** "Since every pair of buildings may be linked and a link has no direction, the possible links are the edges of the complete graph on the buildings, so there are  $n(n - 1)/2$  of them."

**Now redo it:** with six buildings, how many links?  $C(6, 2) = 6 \times 5 / 2 = 15$ .

## RECALL AND CHECK

1. What separates a simple graph, a multigraph and a pseudograph?
2. How many edges has  $K_7$ , and what is the degree of each vertex?
3. What is a regular graph? Give two examples.
4. Define a bipartite graph.
5. Name the type: five vertices in a closed ring, nothing else.
6. A graph on 8 vertices has no edges. Name it.

### ANSWERS

1. What edges are allowed: simple forbids loops and parallel edges; multigraph allows parallel edges but no loops; pseudograph allows both.
2.  $7 \times 6 / 2 = 21$  edges, each vertex of degree 6.
3. Every vertex has the same degree. A cycle is 2-regular;  $K_4$  is 3-regular.
4. Its vertices split into two sets with every edge running between the sets and none inside either.
5. The cycle graph  $C_5$ .
6. The null or empty graph  $N_8$ .

## FILL IN THE GAP

1. A graph that allows loops and parallel edges is a \_\_\_\_\_.
2.  $K_n$  has \_\_\_\_\_ edges and every vertex has degree \_\_\_\_\_.
3. Five buildings can be joined by \_\_\_\_\_ simple links; four buildings by \_\_\_\_\_.
4. A graph with vertices but no edges is the \_\_\_\_\_ graph.

### ANSWERS

1. pseudograph. 2.  $n(n - 1)/2$ ,  $n - 1$ . 3. 10, 6. 4. null or empty.

## PART SIX · 6.3

# The Handshaking Theorem

*One line of arithmetic that every graph must obey. It is asked for by name on three papers, and it silently decides a fourth: the whole of 2024/2025 Question Five turns on it.*

*By the end of this section you will be able to state and prove the Handshaking theorem, use it to find a missing degree count, use its corollary to prove that a described graph cannot exist, and answer the 17.5-mark Question Five of 2024/2025 in full.*

## THE HANDSHAKING THEOREM

**The claim.** In any undirected graph, the sum of the degrees of all the vertices equals twice the number of edges. Counting degrees counts every edge exactly twice, once from each end.

**In symbols.**

$\sum \deg(v) = 2|E|$ , the sum running over every vertex  $v$  in  $V$ .

**The proof, which is one sentence.** Every edge has two endpoints, so each edge contributes exactly 1 to the degree of each of its two ends, that is, 2 to the total. Adding over all  $|E|$  edges gives  $2|E|$ . A loop is no exception: both of its ends are at the same vertex, which is precisely why a loop is defined to add 2 to that vertex's degree.

**Conditions.** The graph must be **finite** and **undirected**. Loops and parallel edges are allowed, provided a loop counts 2. For a directed graph the corresponding statement is different and is given in section 6.4.

**The corollary that does the real work.** The number of vertices of **odd** degree is always **even**. Since the total  $2|E|$  is even, and the even-degree vertices contribute an even amount, the odd-degree vertices must also contribute an even amount between them, which is only possible if there is an even number of them.

**The limiting case.** With no edges at all the theorem reads  $0 = 0$ , true and empty. It is at its most useful in the opposite direction: given a degree sum that is **odd**, no graph with those degrees exists at all, and that impossibility is what 2023/2024 Q5(b) is testing.

### ● MUST MEMORISE

### HANDSHAKING, AND WHAT TO DO WITH IT

$$\sum \deg(v) = 2|E|$$

#### WHERE

$\Sigma$  is summation, introduced in section F.1: add the quantity that follows over everything listed under it.  $\deg(v)$  is the degree of vertex  $v$  from section 6.1, a loop counting twice.  $|E|$  is the number of edges, the size of the graph.  $2|E|$  is even **always**, whatever the graph, and that single fact is the source of every impossibility proof in this section.

**The three ways it is used**, and the exam uses all three:

1. **Find the edges from the degrees.**  $|E| = (\sum \deg(v)) / 2$ .
2. **Find a missing degree.** If all degrees but one are known and  $|E|$  is known, the last one is forced.
3. **Prove something cannot exist.** If the degrees a question describes add to an odd number, no such graph exists, and saying so *is* the answer.

◆ **Memory hook: every handshake is counted twice, once by each hand.** That is where the name comes from: at a party, the total number of hands shaken is twice the number of handshakes.

## Stating it, as two papers ask

### AS PRINTED

2021/2022 · Q4(A)

4. (a) Briefly explain the Handshaking Theorem?

(5½marks)

5. (a) Define the Handshaking Theorem.

(3 marks)

#### NOTE ON THE PAPER

2021/2022 punctuates a command as a question ("Briefly explain ...?"), which it also does in Q1(a) and Q2(c). The two versions want the same thing at different lengths: 3 marks buys the statement and the reason, 5.5 buys the statement, the proof, the corollary and an example. Write to the marks.

#### WORKED EXAMPLE

PQ 21/22 Q4(A) · PQ 23/24 Q5(A)

For 3 marks (2023/2024)

**Statement.** In any undirected graph, the sum of the degrees of all the vertices is equal to twice the number of edges:  $\sum \deg(v) = 2|E|$ .

**Reason.** Each edge has two endpoints and so contributes 1 to the degree of each, that is, 2 to the total. Summing over all edges gives  $2|E|$ .

**Example.** A triangle has 3 vertices each of degree 2 and 3 edges:  $2 + 2 + 2 = 6 = 2 \times 3$ .

For 5.5 marks (2021/2022), add these

**The corollary.** The number of vertices of odd degree is even.

**Why the corollary follows.** Split the sum into the even-degree vertices and the odd-degree ones. The even-degree part is even. The whole sum is  $2|E|$ , also even. So the odd-degree part is even too; but a sum of odd numbers is even only when there is an even count of them.

**A use.** No graph can have exactly three vertices of degree 5 and nothing else, because  $5 + 5 + 5 = 15$  is odd.

**Where the name comes from,** worth one line: if each vertex is a person and each edge a handshake, the degree of a person is the number of hands they shook, and the total counts every handshake from both sides.

**Now redo it:** a graph has 6 vertices each of degree 3. How many edges?  $\sum \deg = 18$ , so  $|E| = 18 / 2 = 9$ .

## Proving a graph cannot exist

(b) An Airplane pilot has been scheduled for a trip to five (5) different cities. He can travel directly between two cities through the same route.

(i) Can he visit exactly three (3) cities from each of the cities on the flight plan?

Compute your answer.

(3 marks)

(ii) Support your answer with a diagrammatical illustration.

(2 marks)

### NOTE ON THE PAPER

The second sentence is doing the work of "every pair of cities is joined by at most one route, and a route may be flown in either direction". That is the paper's way of saying the graph is **SIMPLE** and **UNDIRECTED**. Read part (i) as: can a simple undirected graph on 5 vertices have every vertex of degree exactly 3? The word "Compute" is the clue that an arithmetic argument, not a drawing, is what earns the 3 marks.

### BREAK IT DOWN

|                  |                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | 5 cities. Routes are undirected and there is at most one between any pair.                                                                               |
| <b>ASKED</b>     | (i) <b>Can</b> every city connect to exactly 3 others, and <b>compute</b> the answer. (ii) <b>Support</b> it with a diagram.                             |
| <b>MARKS SAY</b> | 3 and 2. The 3 marks are for the degree-sum computation. Answering only "no" earns almost nothing; the arithmetic is the answer.                         |
| <b>TEACHES</b>   | This section, 6.3.                                                                                                                                       |
| <b>TRAP</b>      | Trying to draw it, failing, and concluding it is impossible. Failing to draw something proves nothing about whether it exists. The degree sum proves it. |

- GIVEN**  $n = 5$  vertices (cities), each required to have degree exactly 3, graph simple and undirected.
- FIND** Whether such a graph exists.
- FORMULA**  $\sum \deg(v) = 2|E|$ , so the degree sum must be even.

**Step 1. Compute the degree sum.** If each of the 5 cities connects to exactly 3 others, then every vertex has degree 3, so

$$\sum \deg(v) = 5 \times 3 = 15$$

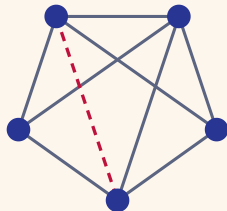
**Step 2. Apply the theorem.** By the Handshaking theorem this sum must equal  $2|E|$ , so

$$2|E| = 15, \text{ giving } |E| = 7.5$$

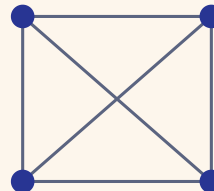
**Step 3. Conclude.** The number of edges must be a whole number, and 7.5 is not. Equivalently, 15 is odd and  $2|E|$  is always even. **No such flight plan exists: the pilot cannot visit exactly three cities from each of five cities.**

**The corollary says the same thing faster,** and is worth adding: every one of the 5 vertices would have odd degree, giving 5 odd-degree vertices, but the number of odd-degree vertices is always even.

**Part (ii): the illustration.** A drawing cannot show something impossible, so show **why** it fails and **what does** work instead.



5 cities, trying for degree 3 each:  
 whichever edge is added last,  
 one city is left at degree 2 or 4



4 cities, degree 3 each: this works.  
 $\text{Sum} = 4 \times 3 = 12 = 2 \times 6$  edges.  
 An odd degree needs an even number of cities

*Left: the attempt on five cities always strands one vertex at the wrong degree. Right: on four cities it works, which shows the obstruction is the parity, not the degree 3.*

**Now redo it:** can six cities each connect to exactly three others?  $\sum \deg = 18$ , even, so  $|E| = 9$  and there is no obstruction. It can be done, for instance as two triangles joined by a perfect matching.



## The 2024/2025 Question Five

AS PRINTED

2024/2025 · QUESTION FIVE

A graph designed by a 400-level Graphics student consists of 24 vertices and 30 edges. The graph has five vertices of degree 4, seven pendant vertices, and seven vertices of degree 2. All other vertices have a degree of either 3 or 4. After completing the design, the student seeks your assistance as a computer science student to help solve the following questions for further review:

- a. How many vertices of degree 4 are there in the given graph? **4 marks**
- b. What is the total number of vertices of degree 4 in the corresponding complete graph? **1 mark**
- c. Provide a proof to determine whether the Handshaking Theorem holds for this scenario. **3 marks**
- d. Write a C++ code snippet to compute the remaining number of vertices that have either degree 3 or 4. **3 marks**
- e. Demonstrate how the Handshaking Theorem can be applied in Data Communication. **2 marks**
- f. Prove that if  $G$  is a graph with  $n$  vertices and  $n$  edges, and no vertex has a degree of 0 or 1, then the degree of every vertex in  $G$  is 2. **3 marks**
- g. Explain what it means for two graphs  $G$  and  $H$  to be isomorphic. **1.5 marks**

### NOTE ON THE PAPER

**THIS QUESTION LOOKS SELF-CONTRADICTIONARY AND IS NOT.** It tells you there are "five vertices of degree 4" and then asks in part (a) how many vertices of degree 4 there are. The answer is **NOT** five, and a candidate who writes five has fallen into the best trap on any of these papers.

The sentence that resolves it is "All other vertices have a degree of either 3 or 4". More degree-4 vertices are hiding in that remainder, and part (a) is asking for the **TOTAL**. Read the preamble as a partial inventory, not a complete one.

Part (d) is a C++ question and is answered in section 11.2, where the code can be compiled and run. Part (g) is answered in section 6.6, where isomorphism is taught. Everything else is answered below.

### BREAK IT DOWN

- GIVEN**  $|V| = 24$ ,  $|E| = 30$ . Five vertices of degree 4, seven pendant vertices (degree 1 by definition, section 6.1), seven of degree 2. All remaining vertices have degree 3 or 4.
- ASKED** Seven parts. The whole 17.5-mark question rests on one degree sum, so compute that first and almost everything follows.
- MARKS SAY** 4 for part (a) alone, which tells you part (a) is not a one-line read-off. If your answer to a 4-mark question is a number copied from the question, re-read the question.
- TEACHES** This section, 6.3, with (d) in 11.2 and (g) in 6.6.
- TRAP** Answering (a) with 5. Also forgetting that **pendant means degree 1**: the question never says "degree 1" anywhere, and if you do not know the word you cannot start.

- GIVEN**  $|V| = 24$ ,  $|E| = 30$ ; 5 vertices of degree 4, 7 pendant (degree 1), 7 of degree 2, the rest of degree 3 or 4.
- FIND** (a) the total number of degree-4 vertices; (b) the same count in the corresponding complete graph; (c) a proof that Handshaking holds; (e) an application; (f) a proof about  $n$  vertices and  $n$  edges.
- FORMULA**  $\sum \deg(v) = 2|E|$ .

a. How many vertices of degree 4 are there

**Step 1. The total degree.** By the Handshaking theorem,

$$\sum \deg(v) = 2|E| = 2 \times 30 = \mathbf{60}$$

**Step 2. Account for the vertices the question names.** A pendant vertex has degree 1.

| GROUP                | HOW MANY  | DEGREE EACH | DEGREE TOTAL |
|----------------------|-----------|-------------|--------------|
| stated degree 4      | 5         | 4           | 20           |
| pendant              | 7         | 1           | 7            |
| degree 2             | 7         | 2           | 14           |
| <b>accounted for</b> | <b>19</b> |             | <b>41</b>    |

**Step 3. What is left.**

$$\text{vertices remaining} = 24 - 19 = 5$$

$$\text{degree remaining} = 60 - 41 = \mathbf{19}$$

**Step 4. Split the remainder.** Those 5 vertices have degree 3 or 4. Let  $x$  have degree 4 and  $y$  have degree 3. Then

$$x + y = 5 \quad \text{and} \quad 4x + 3y = 19$$

Substituting  $y = 5 - x$  into the second equation:

$$4x + 3(5 - x) = 19$$

$$4x + 15 - 3x = 19$$

$$x + 15 = 19$$

$$x = \mathbf{4}, \text{ so } y = \mathbf{1}$$

So the remainder is four vertices of degree 4 and one of degree 3, and this is the only fit.

**Step 5. Total.**

$$\text{vertices of degree 4} = 5 \text{ (stated)} + 4 \text{ (found)} = \mathbf{9}$$

**Answer: the graph has 9 vertices of degree 4.**

**Check the whole inventory**, and write the check down:  $9 \times 4 + 1 \times 3 + 7 \times 2 + 7 \times 1 = 36 + 3 + 14 + 7 = 60$ , which is  $2 \times 30$ . The vertices count too:  $9 + 1 + 7 + 7 = 24$ . Both agree.

b. Degree-4 vertices in the corresponding complete graph

**State the reading, because "corresponding" is not defined.** The complete graph corresponding to this one is the complete graph on the same 24 vertices,  $K_{24}$ .

In  $K_n$  every vertex is joined to every other, so every vertex has degree  $n - 1$ . Here  
degree of every vertex of  $K_{24} = 24 - 1 = 23$

Since  $23 \neq 4$ , **no vertex of  $K_{24}$  has degree 4, so the answer is 0.**

**Add the alternative reading in one line**, since it costs nothing and the phrase is genuinely ambiguous: the only complete graph in which every vertex has degree 4 is  $K_5$ , which has 5 such vertices. If "corresponding" was meant to point at the complete graph of degree 4, the answer is 5. State whichever you use.

c. Proof that the Handshaking theorem holds here

"Provide a proof to determine whether" means: check it, and show the check.

**Left side, the degree sum.** Using the full inventory from part (a):

$$\begin{aligned}\sum \deg(v) &= (9 \times 4) + (1 \times 3) + (7 \times 2) + (7 \times 1) \\ &= 36 + 3 + 14 + 7 \\ &= \mathbf{60}\end{aligned}$$

**Right side, twice the edges.**

$$2|E| = 2 \times 30 = \mathbf{60}$$

**Conclusion.** The two sides agree, so  $\sum \deg(v) = 2|E|$  and the Handshaking theorem holds for this graph. Add the reason it had to: the theorem holds for *every* finite undirected graph, so agreement is not a coincidence but a confirmation that the degree inventory in part (a) is consistent. Had the two sides differed, the fault would have been in the inventory, not in the theorem.

e. The Handshaking theorem in Data Communication

Model a network as a graph: each **node** (router, switch, host) is a vertex, and each physical **link** between two nodes is an edge. The degree of a node is the number of its ports that are in use.

**Application 1: counting cables from port counts.** The theorem says the total number of used ports across the network is twice the number of links. So a network of 40 nodes with 6 used ports each needs  $(40 \times 6) / 2 = 120$  cables, and a bill of materials that says anything else is wrong.

**Application 2: validating a proposed topology.** If a design specifies an odd total of used ports, that design cannot be built: no graph has an odd degree sum. This catches a specification error before any cable is bought, which is the practical point.

**Application 3: the odd-degree corollary in redundancy planning.** In a topology where every node must have an odd number of links, the number of nodes must be even. That constrains what a resilient ring or mesh design can look like.

f. If  $G$  has  $n$  vertices and  $n$  edges and no degree is 0 or 1

**To prove:** every vertex of  $G$  has degree exactly 2.

**Proof.** By the Handshaking theorem,

$$\sum \deg(v) = 2|E| = 2n$$

No vertex has degree 0 or 1, so every vertex has degree at least 2. There are  $n$  vertices, so

$$\sum \deg(v) \geq 2 + 2 + \dots + 2 \text{ (} n \text{ times)} = 2n$$

But the sum *equals*  $2n$ . So the inequality is an equality.

Now suppose some vertex  $u$  had degree 3 or more. Then the other  $n - 1$  vertices still contribute at least 2 each, so

$$\sum \deg(v) \geq 3 + 2(n - 1) = 2n + 1 > 2n,$$

contradicting the theorem. Hence no vertex has degree 3 or more, and since none has degree 0 or 1, **every vertex has degree exactly 2**.  $\square$

**What such a graph looks like**, worth a closing line: a graph in which every vertex has degree 2 is a disjoint union of cycles. With  $n$  vertices and  $n$  edges and no isolated or pendant vertices,  $G$  is a cycle, or several cycles side by side.

**Now redo it:** a graph has 10 vertices and 12 edges, with four pendant vertices and four of degree 2. What degrees do the remaining two vertices share? Degree sum 24, accounted  $4 \times 1 + 4 \times 2 = 12$  over 8 vertices, leaving 2 vertices sharing 12, so both have degree 6.

#### RECALL AND CHECK

1. State the Handshaking theorem and prove it in one sentence.
2. A graph has 7 vertices each of degree 4. How many edges?
3. Can a graph have exactly five vertices of degree 3 and no others? Why?
4. What is the degree of every vertex of  $K_{24}$ ?
5. Why is the number of odd-degree vertices always even?

#### ANSWERS

1.  $\sum \deg(v) = 2|E|$ , because each edge contributes 1 to the degree of each of its two endpoints.
2.  $\sum \deg = 28$ , so  $|E| = 14$ .
3. No. The degree sum would be 15, which is odd, and  $2|E|$  is always even.
4. 23, since every vertex is joined to all the other 23.
5. The total degree is even and the even-degree vertices contribute an even amount, so the odd ones must too, which needs an even number of them.

#### FILL IN THE GAP

1.  $\sum \deg(v) =$  \_\_\_\_\_.
2. A pendant vertex has degree \_\_\_\_\_.
3. In 2024/2025 Question Five the degree sum is \_\_\_\_\_ and the number of degree-4 vertices is \_\_\_\_\_.
4. The number of odd-degree vertices in any graph is always \_\_\_\_\_.

#### ANSWERS

1.  $2|E|$ . 2. 1. 3. 60, 9. 4. even.

#### PART SIX · 6.4

## Directed Graphs, In-degree and Out-degree

Put an arrowhead on every edge and the degree splits in two: arrows coming in, and arrows going out. Two papers set the same tabulation question on a hand-drawn figure.

By the end of this section you will be able to define a directed graph, compute in-degree and out-degree, state the directed form of the Handshaking theorem, tabulate a digraph vertex by vertex,

and answer both tabulation questions and the short note on in-degree and out-degree.

## Directed graphs

### ● MUST MEMORISE

### DIGRAPH, ARC, IN-DEGREE, OUT-DEGREE

A **directed graph**, or **digraph**, is a graph in which every edge has a direction. A directed edge is called an **arc**, and it is written as an **ordered** pair  $(u, v)$ , drawn as an arrow from  $u$  to  $v$ .

For a vertex  $v$ :

the **in-degree**  $\deg^-(v)$  is the number of arcs pointing **into**  $v$ ;

the **out-degree**  $\deg^+(v)$  is the number of arcs pointing **out of**  $v$ .

#### WHERE

$(u, v)$  is an ordered pair, so  $(u, v)$  and  $(v, u)$  are **different arcs**. This is the whole difference from section 6.1, where an edge joining  $u$  and  $v$  was one thing however you named it.

$\deg^-(v)$  counts arrowheads landing on  $v$ .  $\deg^+(v)$  counts arrow tails leaving  $v$ . a **loop** at  $v$  adds 1 to each, since it both leaves  $v$  and arrives at  $v$ .

◆ **Memory hook: minus is in, plus is out.** The superscript minus looks like an arrow flattened as it arrives; the plus is the vertex sending traffic outward. And a mnemonic that never fails: the **in-degree** counts the arrow**heads**.

### ● MUST MEMORISE

### HANDSHAKING FOR DIGRAPHS

In any directed graph,

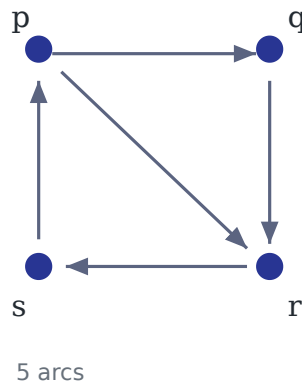
$$\sum \deg^-(v) = \sum \deg^+(v) = |E|$$

#### WHERE

$|E|$  is the number of arcs. **the two sums** are each equal to  $|E|$ , **not** to  $2|E|$ : every arc has exactly one tail and exactly one head, so it is counted once in each sum rather than twice in one.

**How this differs from section 6.3**, which is the point most often lost: in an undirected graph the degrees sum to  $2|E|$ ; in a digraph the in-degrees sum to  $|E|$  and the out-degrees sum to  $|E|$ , and adding the two sums together recovers the  $2|E|$ .

◆ **Memory hook: one tail and one head per arc.** Both column totals of your table must equal the arc count. That is a free check, and it catches a misread arrowhead immediately.



| vertex | in | out |
|--------|----|-----|
| p      | 1  | 2   |
| q      | 1  | 1   |
| r      | 2  | 1   |
| s      | 1  | 1   |
| total  | 5  | 5   |

both totals equal the arc count

A digraph and its table. The arcs are  $p$  to  $q$ ,  $q$  to  $r$ ,  $r$  to  $s$ ,  $s$  to  $p$ , and  $p$  to  $r$ . Both column totals come to 5, which is the check.

## The short note

### WORKED EXAMPLE

PQ 25/26 Q4(A)(III)

**The question** is quoted in full in section 6.1: "(iii) In-degree and Out-degree of a vertex", 2 marks, wanting short notes with relevant examples.

**Definition.** In a directed graph, the in-degree of a vertex  $v$ , written  $\deg^-(v)$ , is the number of arcs whose arrowhead lands on  $v$ . The out-degree,  $\deg^+(v)$ , is the number of arcs whose tail leaves  $v$ . A loop at  $v$  adds 1 to each.

**Example 1.** In the figure above, vertex  $p$  has one arc arriving (from  $s$ ) and two leaving (to  $q$  and to  $r$ ), so  $\deg^-(p) = 1$  and  $\deg^+(p) = 2$ .

**Example 2.** On a web link graph, a page's in-degree is the number of pages linking *to* it and its out-degree the number it links *out* to. Search engines rank on the first and not the second, which is a one-line way of showing you understand the difference matters.

**The line worth adding:**  $\sum \deg^-(v) = \sum \deg^+(v) = |E|$ , because every arc has exactly one head and one tail.

## Tabulating a digraph

AS PRINTED

2021/2022 · Q1(B)

(b) In the Graph diagram below, present in tabular form the following:

(6½marks)

- (i) The number of Vertices
- (ii) The number of edges
- (iii) The total number of In-degree & Out-degree from all the vertices.

(c) In the Graph diagram below, present in tabular form the following:

- (i) Identify each vertex and state the total number of vertices. **(1 marks)**
- (ii) Identify each edge per vertex and state total number of edges. **(1 marks)**
- (iii) Identify each vertex's in-degree and state the total number of the in-degrees. **(2 marks)**
- (iv) Identify each vertex's out-degree and state the total number of the out-degrees. **(2 marks)**

#### NOTE ON THE PAPER, AND ON BOTH FIGURES

The same question twice, expanded from three parts to four. 2023/2024 prints "(1 marks)" twice, a singular value with a plural noun.

**NEITHER FIGURE CAN BE FULLY READ FROM THE SCANS WE HOLD, AND THIS MANUAL WILL NOT PRETEND OTHERWISE.** Both are small directed graphs on four labelled vertices with a fifth vertex sitting alone and joined to nothing, drawn with several curved arcs between the same pairs and with arrowheads a couple of pixels across.

For 2021/2022 Q1(b) the arc count and the arc directions are **NOT** recoverable at the resolution available, so no reading of it is given here at all. For 2023/2024 Q5(c) the multi-arc bundles are clear but the two crossing diagonals in the middle are not; that reading is given below, flagged, with the arithmetic that lets you re-do it in seconds once you have read the arrowheads off your own copy.

**THE MARKS ARE FOR THE METHOD, NOT FOR THE FIGURE.** Both questions pay for a table laid out correctly with totals that reconcile. Learn the layout, and fill it from whatever your own copy shows.

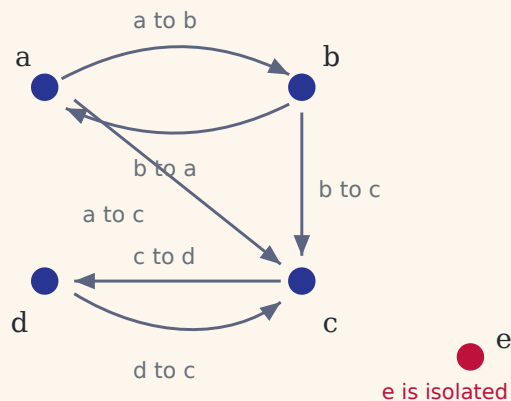
#### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | A directed graph drawn on the paper, with labelled vertices, several arcs, and at least one vertex joined to nothing.                                                                                                                                            |
| <b>ASKED</b>     | <b>Present in tabular form:</b> the vertices and their count, the edges and their count, then each vertex's in-degree and out-degree with totals. The instruction "in tabular form" is explicit, so a paragraph loses marks even if every number in it is right. |
| <b>MARKS SAY</b> | 6.5 and 6 respectively, split roughly evenly across the parts. In-degree and out-degree carry 2 marks each on 2023/2024, which is where the care belongs.                                                                                                        |
| <b>TEACHES</b>   | This section, 6.4.                                                                                                                                                                                                                                               |
| <b>TRAP</b>      | Three. Forgetting the isolated vertex, which must still appear in the table with 0 and 0. Counting a two-arc bundle as one edge. And giving a single "degree" column instead of two.                                                                             |

**The method, in four steps.** It works on any figure, including the one on your own paper.

1. **List the vertices first**, reading the labels off the drawing and including any vertex with no arcs. Write the count.
2. **Walk the arcs one bundle at a time**, writing each as an ordered pair (tail, head). Where several curves join the same pair, count the curves, not the pairs. Write the count.
3. **Build a row per vertex** with three columns: arcs out, arcs in, and the two degrees. Fill it from the list in step 2, not from the drawing, so each arc is used exactly once.
4. **Total both degree columns.** Each total must equal the arc count from step 2. If they do not, an arc was missed or double counted.

**Worked on a clearly drawn example**, so the method is visible without an unreadable figure in the way. *This figure is mine, not the paper's.*



*Six arcs on five vertices, one of them isolated. Every arrowhead is unambiguous, which is the point of drawing it again.*

**Step 1. Vertices:** a, b, c, d, e. **Total vertices = 5.** Note e, joined to nothing.

**Step 2. Arcs** as ordered pairs: (a, b), (b, a), (b, c), (c, d), (d, c), (a, c). **Total arcs = 6.**

**Step 3 and 4. The table.**

| VERTEX       | ARCS OUT       | OUT-DEGREE | ARCS IN                | IN-DEGREE |
|--------------|----------------|------------|------------------------|-----------|
| a            | (a, b), (a, c) | 2          | (b, a)                 | 1         |
| b            | (b, a), (b, c) | 2          | (a, b)                 | 1         |
| c            | (c, d)         | 1          | (b, c), (a, c), (d, c) | 3         |
| d            | (d, c)         | 1          | (c, d)                 | 1         |
| e            | none           | 0          | none                   | 0         |
| <b>Total</b> |                | <b>6</b>   |                        | <b>6</b>  |

**The check.** Both totals are 6, which is the arc count from step 2. That is  $\sum \deg^+(v) = \sum \deg^-(v) = |E|$ , and it is the only verification available in the exam hall. Do it every time.



**Now redo it:** add an arc  $(e, a)$  to the figure. Then  $|E| = 7$ ,  $\deg^+(e) = 1$ ,  $\deg^-(a) = 2$ , and both totals become 7.

- GIVEN** A digraph on vertices a, b, c, d and an isolated vertex e. Bundles read from the scan: one arc b to a; two arcs a to b; two arcs d to a; two arcs b to c; three arcs d to c. The two crossing diagonals are read as two arcs a to c and one arc b to d.
- FIND** The vertices and their count, the arcs and their count, each in-degree with its total, each out-degree with its total.
- RULE**  $\sum \deg^-(v) = \sum \deg^+(v) = |E|$ .

**Flag first, then answer.** Write this line before the table: "*The two crossing diagonals are taken as two arcs from a to c and one from b to d; the remaining bundles are unambiguous.*" An examiner marking a hand-drawn figure will accept a stated reading; a wrong reading presented as certain earns nothing.

**(i) Vertices.** a, b, c, d, e. **Total = 5.** Vertex e carries no arcs at all and must still be listed.

**(ii) Arcs, per vertex of origin.**

| BUNDLE             | ARCS               |
|--------------------|--------------------|
| b to a             | 1                  |
| a to b             | 2                  |
| d to a             | 2                  |
| b to c             | 2                  |
| d to c             | 3                  |
| a to c             | 2 (stated reading) |
| b to d             | 1 (stated reading) |
| <b>Total edges</b> | <b>13</b>          |

**(iii) and (iv) In-degree and out-degree.**

| VERTEX       | OUT-DEGREE | FROM                   | IN-DEGREE | FROM                         |
|--------------|------------|------------------------|-----------|------------------------------|
| a            | 4          | 2 to b, 2 to c         | 3         | 1 from b, 2 from d           |
| b            | 4          | 1 to a, 2 to c, 1 to d | 2         | 2 from a                     |
| c            | 0          | none                   | 7         | 2 from b, 3 from d, 2 from a |
| d            | 5          | 2 to a, 3 to c         | 1         | 1 from b                     |
| e            | 0          | none                   | 0         | none                         |
| <b>Total</b> | <b>13</b>  |                        | <b>13</b> |                              |

**The check passes:** both totals are 13, which equals the arc count. Note also that c is a **sink**, with arcs arriving and none leaving, and that e is isolated in both directions.

**If your copy reads the diagonals the other way,** say one arc each way, then  $|E| = 13$  still holds only if the count of diagonal arcs is unchanged; with two arcs a to c replaced by one a to c and one c to a, the arc total stays 13,  $\deg^+(a)$  falls to 3,  $\deg^+(c)$  rises to 1

and  $\deg^-(c)$  falls to 6, while  $\deg^-(a)$  rises to 4. Redo the two affected rows and check the totals again. Nothing else changes.

**For the 2021/2022 version**, whose figure is not readable at all from the scan, use the four-step method above on your own copy. Its part (iii) asks only for "the total number of In-degree & Out-degree from all the vertices", which is the pair of column totals, and by the digraph Handshaking rule both of them equal the number of arcs. If you can count the arcs you can answer part (iii) even if you misread an arrowhead.

**⚠ TRAP**

**NEVER PRESENT AN UNREADABLE FIGURE AS READ**

Both of these questions are drawn by hand and scanned. If an arrowhead is genuinely ambiguous on your copy, the professional move is the same one this manual has used: **state the reading you are working under, then answer under it**. One sentence protects the whole table.

What loses marks is silence. A table with no stated reading and a misread arc looks like a candidate who cannot count, rather than one working from a poor photocopy.

**RECALL AND CHECK**

1. What makes a digraph different from an undirected graph?
2. Define in-degree and out-degree.
3. What do the in-degrees sum to, and what do the degrees of an undirected graph sum to?
4. How much does a loop add to each of the two degrees in a digraph?
5. A digraph has 13 arcs. What must each of your two column totals be?

ANSWERS

1. Every edge has a direction, so an arc is an ordered pair and  $(u, v)$  differs from  $(v, u)$ .
2. In-degree counts arcs arriving at the vertex; out-degree counts arcs leaving it.
3. In-degrees sum to  $|E|$ . Undirected degrees sum to  $2|E|$ .
4. One to each, since it both leaves and arrives at that vertex.
5. 13 each. If either differs, an arc has been missed or counted twice.

**FILL IN THE GAP**

1. A directed edge is called an \_\_\_\_\_ and is written as an \_\_\_\_\_ pair.
2.  $\deg^-(v)$  is the \_\_\_\_\_-degree.
3.  $\sum \deg^+(v) = \underline{\hspace{2cm}}$ .
4. A vertex with arcs arriving and none leaving is a \_\_\_\_\_.

ANSWERS

1. arc, ordered. 2. in. 3.  $|E|$ . 4. sink.

## Adjacency, Path and Relation Matrices

A graph can be stored as a grid of ones and zeros. Once it is, questions about routes become questions about multiplying the grid by itself, and the newest paper gives 6.5 marks to how a computer holds it.

By the end of this section you will be able to write the adjacency matrix of a graph or digraph, read degrees off it, build the path matrix from powers of the adjacency matrix, tell a logical matrix from a relation matrix, and answer the four matrix questions on the 2025/2026 paper.

### The adjacency matrix

#### ● MUST MEMORISE

#### ADJACENCY MATRIX

Number the vertices 1 to  $n$  in a fixed order. The **adjacency matrix**  $A$  of the graph is the  $n$  by  $n$  grid whose entry in row  $i$ , column  $j$  is

$a_{ij}$  = the number of edges from vertex  $i$  to vertex  $j$ , which for a simple graph is **1** if they are joined and **0** if they are not.

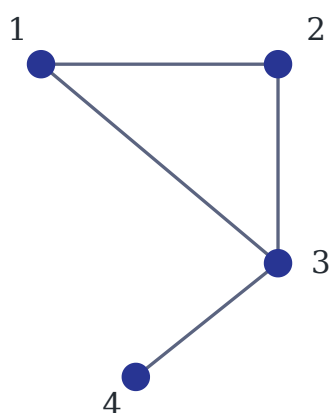
#### WHERE

$n$  is the order of the graph, so  $A$  is always square and always  $n$  by  $n$ . **the vertex order** must be fixed and stated first. Re-number the vertices and you get a different-looking matrix for the same graph.  $a_{ii}$ , on the leading diagonal, records loops at vertex  $i$ .

**Three properties to state whenever you are asked to explain one.**

1. For an **undirected** graph,  $A$  is **symmetric**:  $a_{ij} = a_{ji}$ , because an edge joining  $i$  and  $j$  joins  $j$  and  $i$ . For a **digraph** it usually is not, since an arc has a direction.
2. **The row sum is the degree**. Adding across row  $i$  gives  $\deg(i)$  for an undirected simple graph, and  $\deg^+(i)$  for a digraph; the column sum gives  $\deg^-(i)$ .
3. A **simple** graph has zeros all along the leading diagonal, because a simple graph has no loops.

◆ **Memory hook: row to column, tail to head.** Read  $a_{ij}$  as "from  $i$  to  $j$ ". For an undirected graph the direction does not matter and the matrix comes out symmetric, which is itself the proof that the graph is undirected.



|        |   |   |   |   |       |
|--------|---|---|---|---|-------|
| to:    | 1 | 2 | 3 | 4 |       |
| from 1 | 0 | 1 | 1 | 0 | deg 2 |
| from 2 | 1 | 0 | 1 | 0 | deg 2 |
| from 3 | 1 | 1 | 0 | 1 | deg 3 |
| from 4 | 0 | 0 | 1 | 0 | deg 1 |

symmetric, zero diagonal, row sums are the degrees  
degree sum  $8 = 2 \times 4$  edges

A graph and its adjacency matrix. Everything you can say about the graph can be read off the grid, which is the point of building it.

## The path matrix

### ● MUST MEMORISE

### WALKS, AND THE PATH MATRIX

**The key fact.** If  $A$  is the adjacency matrix, then the entry in row  $i$ , column  $j$  of  $A^k$  is the number of **walks of length exactly  $k$**  from vertex  $i$  to vertex  $j$ .

The **path matrix** (also called the reachability matrix)  $P$  records only whether a route exists at all:

$p_{ij} = 1$  if there is a path from  $i$  to  $j$ , and  $0$  if there is none.

#### WHERE

a **walk of length  $k$**  is a sequence of  $k$  edges followed end to end; vertices and edges may repeat.  $A^k$  is  $A$  multiplied by itself  $k$  times, using ordinary arithmetic.  $P$  is built from  $A + A^2 + \dots + A^{n-1}$  by replacing every non-zero entry with 1. Nothing longer than  $n - 1$  edges is needed: a path that visits no vertex twice cannot use more.

**The difference the exam asks about, in one line:** the **adjacency** matrix answers "can I get there in **one** step?"; the **path** matrix answers "can I get there **at all**?"

◆ **Memory hook: adjacency is one hop, path is any number of hops.** Powers of  $A$  count the routes of each exact length; flattening the sum to ones and zeros forgets how many and remembers only whether.

**GIVEN** The graph in the figure above: vertices 1, 2, 3, 4 with edges 1-2, 1-3, 2-3 and 3-4.

**FIND** A, then  $A^2$ , then the path matrix P.

**RULE** The  $(i, j)$  entry of  $A^k$  counts walks of length  $k$ ; P flattens the sum of the powers to ones and zeros.

**Step 1. Write A**, stating the vertex order 1, 2, 3, 4.

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 |
| 2 | 1 | 0 | 1 | 0 |
| 3 | 1 | 1 | 0 | 1 |
| 4 | 0 | 0 | 1 | 0 |

**Step 2. Square it.** Entry  $(i, j)$  of  $A^2$  is row  $i$  of A dotted with column  $j$  of A. For instance entry  $(1, 1)$ :

$$(0 \times 0) + (1 \times 1) + (1 \times 1) + (0 \times 0) = 2$$

and there are indeed two walks of length 2 from vertex 1 back to itself, namely 1 to 2 to 1 and 1 to 3 to 1.

| $A^2$ | 1 | 2 | 3 | 4 |
|-------|---|---|---|---|
| 1     | 2 | 1 | 1 | 1 |
| 2     | 1 | 2 | 1 | 1 |
| 3     | 1 | 1 | 3 | 0 |
| 4     | 1 | 1 | 0 | 1 |

**Read one entry aloud to prove you understand it.** Entry  $(3, 3)$  is 3, so there are three walks of length 2 from vertex 3 to itself: out and back along each of its three edges, to 1, to 2 and to 4. Entry  $(3, 4)$  is 0, because there is no walk of exactly two edges from 3 to 4, only the direct one of length 1.

**Step 3. Build the path matrix.** Add A,  $A^2$  and  $A^3$ , then replace every non-zero by 1. Every entry comes out non-zero, so

| P | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 1 | 1 |
| 3 | 1 | 1 | 1 | 1 |
| 4 | 1 | 1 | 1 | 1 |

**What P tells you at a glance:** every vertex reaches every other, so **the graph is connected**. A path matrix with a block of zeros in one corner is the signature of a disconnected graph, and that is the usual reason for computing it.

**Now redo it:** delete the edge 3-4. Vertex 4 then reaches nothing, so row 4 and column 4 of P become all zero apart from the diagonal, and the graph is disconnected.

## Adjacency against path matrix

### WORKED EXAMPLE

PQ 25/26 Q1 A) IV)

**The question** is quoted in full in section 6.1: "iv) Adjacency and Path Matrice", 2 marks, wanting the difference **with two examples each**. The paper's "Matrice" is a slip for "Matrix" or "Matrices".

|                            | ADJACENCY MATRIX                           | PATH MATRIX                                       |
|----------------------------|--------------------------------------------|---------------------------------------------------|
| Records                    | direct connection, one edge                | reachability, any number of edges                 |
| Entry $a_{ij}$ or $p_{ij}$ | 1 if i and j are joined by an edge         | 1 if some path runs from i to j                   |
| Obtained from              | the drawing directly                       | $A, A^2, \dots$ summed and flattened to 1s and 0s |
| Answers                    | "is there a road between these two towns?" | "can I drive from this town to that one?"         |

**Two examples for the adjacency matrix.** (1) In the graph above,  $a_{12} = 1$  because 1 and 2 are joined;  $a_{14} = 0$  because they are not. (2) In  $K_3$ , every off-diagonal entry is 1 and every diagonal entry is 0.

**Two examples for the path matrix.** (1) In the same graph  $p_{14} = 1$ , because 1 to 3 to 4 is a path, even though  $a_{14} = 0$ . That single pair is the clearest possible illustration of the difference and is worth writing down. (2) In a graph of two separate triangles, P has ones inside each triangle's block and zeros between the blocks, showing at once that the graph is disconnected.

## Logical matrix against relation matrix

### ● MUST MEMORISE

### LOGICAL AND RELATION MATRICES

A **logical matrix** (also called a Boolean or binary matrix) is any matrix whose entries are all 0 or 1, where the entries are read as **false** and **true** rather than as numbers.

A **relation matrix** is the particular logical matrix that records a relation  $R$  from a set  $A$  to a set  $B$ , as defined in section 3.1:

$m_{ij} = 1$  exactly when  $(a_i, b_j)$  is in  $R$ , and **0** otherwise.

#### WHERE

$a_i$  is the  $i$ -th element of  $A$  in a fixed listed order, giving the rows.  $b_j$  is the  $j$ -th element of  $B$ , giving the columns. For a relation on one set,  $A$  and  $B$  are the same and the matrix is square.  $m_{ij}$  is 1 or 0, and it is a **truth value**, not a count.

**The relationship between the three terms**, which is what a "differentiate between" question is really asking for:

- ▶ Every relation matrix **is** a logical matrix. The reverse does not hold: a logical matrix is only a relation matrix once you say which sets index its rows and columns.
- ▶ An adjacency matrix of a **simple** graph is also a logical matrix, and it is exactly the relation matrix of the relation "is joined to". A multigraph's adjacency matrix is **not** logical, because its entries count parallel edges and may exceed 1.

◆ **Memory hook: logical says what the entries are, relation says what they mean.** The first is a description of the contents; the second is a description of the job.



**The question** is quoted in full in section 6.1: "v) A logical matrix and relation matrix", printed "(2marks)" with no space.

**The distinction.** A logical matrix is any matrix all of whose entries are 0 or 1, read as false and true. A relation matrix is a logical matrix whose rows are labelled by the elements of one set and whose columns are labelled by the elements of another, with a 1 marking exactly the pairs that belong to the relation. So every relation matrix is a logical matrix given a job to do; not every logical matrix has been given that job.

**Two examples of a logical matrix.** (1) The 2 by 2 grid with rows (1, 0) and (0, 1) is logical, and until its rows and columns are labelled it records nothing in particular. (2) The truth table of a two-variable connective, written as a grid of Ts and Fs, is a logical matrix in disguise.

**Two examples of a relation matrix.** (1) On  $A = \{1, 2, 3\}$  the relation "is less than" has 1 exactly above the diagonal:

|   | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0 | 1 | 1 |
| 2 | 0 | 0 | 1 |
| 3 | 0 | 0 | 0 |

(2) The adjacency matrix of the simple graph drawn above is the relation matrix of "is joined to" on  $\{1, 2, 3, 4\}$ .

**The line that earns the second mark:** section 3.2 showed that the properties of a relation can be read straight off its matrix, the diagonal for reflexive and the mirror image for symmetric. That is only possible because the rows and columns are labelled, which is precisely what makes it a relation matrix rather than a bare logical one.

## Explaining the adjacency matrix, and its memory cost

(a) i. Clearly explain Adjacency matrix

(2½ marks)

ii. Discuss memory management in Adjacency matrix

(4 marks)

### NOTE ON THE PAPER

Part (ii) is the most unusual question on any of the five papers: it is a **SYSTEMS** question wearing a discrete maths hat, and at 4 marks it is worth more than part (i). "Memory management" here means how the matrix is stored, what it costs, and when a different structure is the better choice. A candidate who only defines the matrix again cannot score it.

#### BREAK IT DOWN

|                  |                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Nothing. Both parts are explanation.                                                                                                               |
| <b>ASKED</b>     | (i) <b>Clearly explain</b> the adjacency matrix. (ii) <b>Discuss</b> memory management in it. "Discuss" invites several points, not one.           |
| <b>MARKS SAY</b> | 2.5 and 4. Part (ii) wants roughly four distinct points, one per mark: the cost, why it is that cost, when it is wasteful, and what to do instead. |
| <b>TEACHES</b>   | This section, 6.5.                                                                                                                                 |
| <b>TRAP</b>      | Answering (ii) by repeating (i). They are different questions and the marker is looking for storage, not definition.                               |

i. Clearly explain the adjacency matrix

**Definition.** Fix an order on the  $n$  vertices of a graph. The adjacency matrix  $A$  is the  $n$  by  $n$  matrix whose entry  $a_{ij}$  is the number of edges from vertex  $i$  to vertex  $j$ , which is 1 or 0 for a simple graph.

**What it is for.** It turns a drawing into data. A picture cannot be given to a program; a grid of numbers can, and every graph algorithm in computing starts from a representation of this kind.

**Its properties,** which is where the half marks live: it is square; it is symmetric exactly when the graph is undirected; its diagonal is zero exactly when there are no loops; its row sums are the degrees; and the entries of  $A^k$  count the walks of length  $k$ .

**An example.** Give the four-vertex matrix drawn above, with its vertex order stated, and point at one entry:  $a_{34} = 1$  because vertices 3 and 4 are joined, while  $a_{14} = 0$  because they are not.

ii. Memory management in the adjacency matrix

Four points, one per mark.

1. **The cost is  $n^2$ , whatever the graph.** An adjacency matrix stores an entry for every ordered pair of vertices, so a graph of  $n$  vertices needs  $n^2$  cells regardless of how many edges it actually has. A graph on 1000 vertices needs 1000000 cells even if it has only 5 edges. The storage depends on the **order**, never on the **size**.
2. **It is wasteful on sparse graphs, and most real graphs are sparse.** A graph is **sparse** when its edge count is far below the  $n(n - 1)/2$  maximum, and **dense** when it is near it. Road networks, social graphs and computer networks are all sparse: a city has millions of junctions and each meets three or four roads. For these the matrix is almost entirely zeros, and storing zeros costs exactly as much as storing ones.
3. **Two savings are available without changing structure.** First, an undirected graph's matrix is symmetric, so only the upper triangle need be held, roughly halving the space to  $n(n - 1)/2$  cells. Second, a simple graph's entries are single bits, so 64 of them pack into one machine word instead of one integer each; that is a factor of 32 or 64 on the same data.
4. **The alternative is an adjacency list.** Store, for each vertex, only the list of its neighbours. That costs about  $n + 2|E|$  for an undirected graph rather than  $n^2$ , which is dramatically less when  $|E|$  is small. The trade-off is the point to state: the matrix answers "are  $i$  and  $j$  adjacent?" in constant time by one lookup, while the list must scan a neighbour list; but the list walks all the neighbours of a vertex in time proportional to their number, while the matrix must scan a whole row of  $n$  cells to find them.

**The closing sentence that makes it a discussion rather than a list:** choose the adjacency matrix for small or dense graphs and for algorithms that ask "is this pair joined?" repeatedly; choose the adjacency list for large sparse graphs and for algorithms that walk neighbours, which is most traversal and shortest-path work.

|                               | ADJACENCY MATRIX      | ADJACENCY LIST                 |
|-------------------------------|-----------------------|--------------------------------|
| Space                         | $n^2$                 | about $n + 2 E $               |
| "Are $i$ and $j$ joined?"     | one lookup            | scan $i$ 's list               |
| "List the neighbours of $i$ " | scan $n$ cells        | read the list, $\deg(i)$ items |
| Best for                      | small or dense graphs | large sparse graphs            |

**Now redo it:** a graph has 500 vertices and 800 edges. The matrix needs  $500^2 = 250000$  cells; the list needs about  $500 + 1600 = 2100$ . The list is the right choice by a factor of over a hundred.

#### RECALL AND CHECK

1. What does  $a_{ij}$  record, and what must be fixed before you can write  $A$  at all?
2. Why is the adjacency matrix of an undirected graph symmetric?
3. What do the entries of  $A^2$  count?
4. Give the one-line difference between an adjacency matrix and a path matrix.
5. How much space does an adjacency matrix need, and how much does an adjacency list need?

#### ANSWERS

1. The number of edges from vertex  $i$  to vertex  $j$ . The order of the vertices must be fixed and stated.
2. An edge joining  $i$  and  $j$  also joins  $j$  and  $i$ , so  $a_{ij} = a_{ji}$ .
3. The number of walks of length exactly 2 between each pair.
4. Adjacency records one-step connection; the path matrix records whether any route exists at all.
5.  $n^2$  cells for the matrix, about  $n + 2|E|$  for the list.

#### FILL IN THE GAP

1. The row sums of an adjacency matrix are the \_\_\_\_\_.
2. The entries of  $A^k$  count the \_\_\_\_\_ of length \_\_\_\_\_.
3. Every relation matrix is a \_\_\_\_\_ matrix, but not conversely.
4. An adjacency matrix costs \_\_\_\_\_ cells whatever the number of edges.

#### ANSWERS

1. degrees. 2. walks,  $k$ . 3. logical. 4.  $n^2$ .

#### PART SIX • 6.6

## Isomorphism, Euler and Hamilton

*Three questions about the same drawing: is this the same graph as that one, can I walk every edge once, and can I visit every vertex once. Only the middle one has an easy test, and knowing that is half the marks.*

By the end of this section you will be able to define graph isomorphism and use invariants to disprove it, define Euler paths and circuits and apply Euler's test, define Hamilton paths and circuits and say why no simple test exists, and answer both short notes and the isomorphism part of 2024/2025 Question Five.

## Isomorphism

### ● MUST MEMORISE

### GRAPH ISOMORPHISM

Two graphs  $G$  and  $H$  are **isomorphic**, written  $G \cong H$ , when there is a **bijection**  $f$  from the vertices of  $G$  to the vertices of  $H$  such that

**$u$  and  $v$  are adjacent in  $G$  exactly when  $f(u)$  and  $f(v)$  are adjacent in  $H$ .**

#### WHERE

$f$  is a bijection, that is, one-to-one and onto, as defined in section 4.2. So  $G$  and  $H$  must have the same number of vertices before anything else can be checked. "**exactly when**" is two conditions in one:  $f$  must carry edges to edges **and** non-edges to non-edges.  $\cong$  is read "is isomorphic to".

**In plain words:** the two graphs are the same graph with the vertices renamed and the picture redrawn. Nothing about position, length or crossing matters; only which vertices are joined.

◆ **Memory hook: same structure, different labels.** An isomorphism is a relabelling that preserves every join and every non-join.

### ● MUST MEMORISE

### INVARIANTS: HOW TO DISPROVE ISOMORPHISM

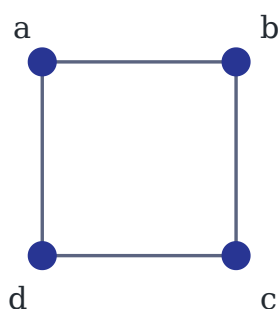
An **invariant** is a property that isomorphic graphs must share. If two graphs differ on any invariant, they are **not** isomorphic, and naming the invariant is a complete proof.

- ▶ the number of vertices;
- ▶ the number of edges;
- ▶ the **degree sequence**, that is, the list of all degrees in order;
- ▶ the number of components, and whether the graph is connected;
- ▶ the length of the shortest cycle, and whether any cycle exists at all;
- ▶ whether the graph is bipartite.

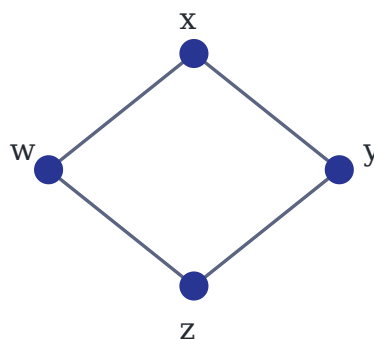
Section 6.7 adds one more to this list once it has the vocabulary for it.

**The asymmetry that the exam rewards.** Invariants can only ever **disprove**. Two graphs agreeing on every invariant in the list above may still fail to be isomorphic, so to **prove** isomorphism you must exhibit the bijection itself and check it. Say that sentence in any answer about isomorphism; it is the difference between knowing the definition and understanding it.

◆ **Memory hook: invariants say no, only a mapping says yes.** Count vertices, count edges, sort the degrees. If any of the three differ, stop and write the answer.



G: a square



H: a diamond

$$G \cong H$$

a to w

b to x

c to y

d to z

same graph, redrawn

Two drawings, one graph. Both are  $C_4$ : four vertices, four edges, every degree 2, and the named bijection carries every edge to an edge.

### WORKED EXAMPLE

PQ 24/25 Q5(G) · PQ 25/26 Q4(A)(IV)

**The two questions** are quoted in full in sections 6.3 and 6.1: 2024/2025 Q5(g) asks you to "Explain what it means for two graphs  $G$  and  $H$  to be isomorphic" for 1.5 marks, and 2025/2026 Q4(a)(iv) asks for a short note on "Graph isomorphism" with a relevant example for 2 marks. One answer serves both; give the example either way.

**The explanation.** Two graphs  $G$  and  $H$  are isomorphic when there is a one-to-one and onto function  $f$  from the vertices of  $G$  to the vertices of  $H$  such that two vertices  $u$  and  $v$  are adjacent in  $G$  if and only if  $f(u)$  and  $f(v)$  are adjacent in  $H$ . Informally,  $H$  is  $G$  with its vertices renamed and its picture redrawn: the two are the same graph in every respect except the labels and the geometry.

**What must match, and what need not.** The number of vertices, the number of edges and the degree of every corresponding vertex must match. The positions of the vertices on the page, the lengths of the edges and whether the edges cross need not: none of those is part of what a graph is.

**The example.** The square  $a, b, c, d$  and the diamond  $w, x, y, z$  above are isomorphic under  $a$  to  $w$ ,  $b$  to  $x$ ,  $c$  to  $y$ ,  $d$  to  $z$ . Check it edge by edge:  $ab$  goes to  $wx$ ,  $bc$  to  $xy$ ,  $cd$  to  $yz$ ,  $da$  to  $zw$ , and all four are edges of  $H$ . The two drawings look different and are the same graph.

**The extra line for the full 1.5 marks:** to disprove isomorphism it is enough to find one invariant on which the graphs differ, such as the degree sequence; but to prove it, the bijection must actually be exhibited, because agreeing on every standard invariant is not sufficient.

**Now redo it:** are a 4-cycle and a path on 4 vertices isomorphic? No. Both have 4 vertices, but the cycle has 4 edges and degree sequence 2, 2, 2, 2, while the path has 3 edges and degree sequence 1, 2, 2, 1. Either invariant settles it.

## Euler paths and circuits

### ● MUST MEMORISE

### EULER PATH AND EULER CIRCUIT

An **Euler path** is a walk that uses **every edge of the graph exactly once**. An **Euler circuit** is an Euler path that finishes where it started.

A graph that has an Euler circuit is called an **Eulerian graph**.

#### WHERE

**every edge exactly once** is the condition. Vertices may be revisited as often as you like; it is the **edges** that are rationed. **circuit** means the walk returns to its starting vertex; a path need not.

◆ **Memory hook: Euler is about edges, Hamilton is about vertices.** Both names begin with a letter that helps: **E**uler for **e**edges. Get that one association right and neither definition can be confused again.

### EULER'S THEOREM ON CIRCUITS AND PATHS

**The claim.** Whether a graph can be traced without lifting the pen or repeating an edge depends only on how many of its vertices have odd degree, and on nothing else.

#### In symbols.

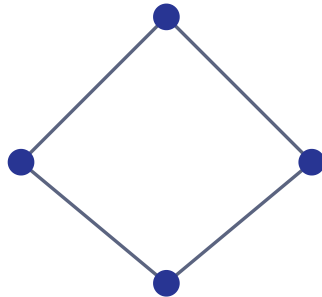
A connected graph has an **Euler circuit** exactly when **every** vertex has even degree. A connected graph has an **Euler path but no circuit** exactly when it has **exactly two** vertices of odd degree, and any such path must start at one of them and finish at the other.

**Conditions.** The graph must be **connected**, apart from isolated vertices, which no walk needs to reach. Loops and parallel edges are allowed. Note that "exactly one odd vertex" never arises: by the Handshaking corollary of section 6.3, the number of odd-degree vertices is always even.

**Why the degree condition is the right one.** Every time a walk passes through a vertex it uses one edge to arrive and one to leave, so it consumes edges there in pairs. A vertex of odd degree therefore cannot be passed through cleanly: the walk must either start or finish there. With no odd vertices the walk can close up into a circuit; with two it runs between them; with four or more it cannot be a single walk at all.

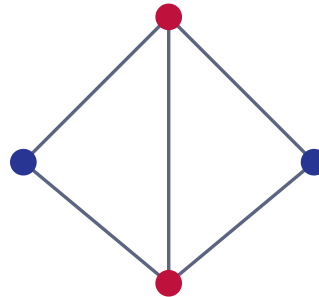
**The limiting case.** A graph with no edges satisfies the condition vacuously and has an empty Euler circuit, which is true and of no use. The theorem earns its keep in the negative direction: the seven bridges of Königsberg form a graph with **four** odd vertices, so no walk crosses every bridge once, and that impossibility is the result the whole subject grew from.

all degrees 2, all even



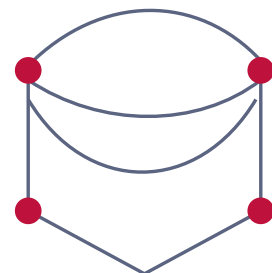
Euler circuit exists

two odd vertices, in crimson



Euler path only, between them

four odd vertices



neither path nor circuit

The whole theorem in three pictures. Count the odd-degree vertices: none, two, or more than two.

### WORKED EXAMPLE

PQ 25/26 Q4(A)(V)

**The question** is quoted in full in section 6.1: "(v) Eulerian graph", 2 marks, wanting a short note with a relevant example.

**Definition.** A connected graph is **Eulerian** when it contains an Euler circuit, that is, a closed walk using every edge of the graph exactly once. By Euler's theorem a connected graph is Eulerian exactly when every one of its vertices has even degree.

**Example 1, Eulerian.** The 4-cycle drawn above: every vertex has degree 2, which is even, so the circuit a to b to c to d to a uses all four edges once and returns to the start.

**Example 2, not Eulerian.** The seven bridges of Königsberg, the problem Euler solved in 1736 and the origin of the whole subject. Four land masses joined by seven bridges give a graph whose four vertices all have odd degree (3, 3, 3 and 5), so no walk crosses every bridge exactly once. The townspeople had been trying; Euler proved it could not be done.

**The distinction worth adding for the second mark:** a graph with exactly two odd vertices has an Euler **path** but no circuit, and is therefore **not** Eulerian. Only the all-even case earns the name.



## Hamilton paths and circuits

### ● MUST MEMORISE

### HAMILTON PATH AND CIRCUIT

A **Hamilton path** is a walk that visits **every vertex of the graph exactly once**. A **Hamilton circuit** is a Hamilton path that returns to its starting vertex.

#### WHERE

**every vertex exactly once** is the condition, so here it is the **vertices** that are rationed and edges may be left unused entirely. **the starting vertex** of a circuit is visited twice in the sense that the walk returns to it, which is not counted as a repeat visit.

**There is no easy test, and saying so is part of the answer.** Euler circuits are settled by counting degrees. For Hamilton circuits **no such simple criterion is known**, and deciding whether one exists is one of the classic computationally hard problems. Only sufficient conditions exist, such as Dirac's: a simple graph on  $n \geq 3$  vertices in which every vertex has degree at least  $n/2$  has a Hamilton circuit. That is a one-way test, so a graph failing it may still have one.

◆ **Memory hook: Euler counts degrees, Hamilton counts on luck.** If an exam question asks you to find a Hamilton circuit, you are expected to find it by inspection; if it asks whether one exists in general, the answer involves saying there is no efficient test.

### ⚠ TRAP

### EULER AND HAMILTON ARE EASY TO SWAP

|                  | EULER                                | HAMILTON                               |
|------------------|--------------------------------------|----------------------------------------|
| Uses each        | <b>edge</b> exactly once             | <b>vertex</b> exactly once             |
| May repeat       | vertices                             | nothing, and may skip edges            |
| Test             | count odd-degree vertices            | none known; only sufficient conditions |
| Everyday version | a postal round covering every street | a delivery visiting every house        |

The travelling salesman question of section 6.8 is a Hamilton circuit question with weights attached: visit every city once, return home, and do it as cheaply as possible. That is why it has no quick method either.

**GIVEN** The four-vertex graph of section 6.5: vertices 1, 2, 3, 4 with edges 1-2, 1-3, 2-3 and 3-4.

**FIND** Whether it has an Euler circuit, an Euler path, and a Hamilton path or circuit.

**RULE** Euler by counting odd degrees; Hamilton by inspection.

**Step 1. The degrees,** read off the adjacency matrix row sums in section 6.5:  $\deg(1) = 2$ ,  $\deg(2) = 2$ ,  $\deg(3) = 3$ ,  $\deg(4) = 1$ .

**Step 2. Count the odd ones.** Vertices 3 and 4 have odd degree, and the other two are even. That is **exactly two** odd vertices.

**Step 3. Apply Euler's theorem.** Exactly two odd vertices means **no Euler circuit, but an Euler path exists**, and it must run between the two odd vertices, that is, between 3 and 4. One such path:

4 to 3 to 1 to 2 to 3

Check it: the edges used are 34, 31, 12, 23. That is all four edges, each once. Vertex 3 is visited twice, which an Euler path is allowed to do.

**Step 4. Hamilton, by inspection.** A Hamilton path must touch each of 1, 2, 3, 4 exactly once. Try 4 to 3 to 1 to 2: every vertex once, so **a Hamilton path exists**.

A Hamilton **circuit** would have to return from 2 to 4, and there is no edge 2-4. In fact vertex 4 has degree 1, and a circuit must both enter and leave every vertex, needing degree at least 2. So **no Hamilton circuit exists**, and the pendant vertex is the whole reason.

**That last observation generalises** and is worth remembering: a graph with any vertex of degree 1 can never have a Hamilton circuit.

**Now redo it:** add the edge 2-4. Degrees become 2, 3, 3, 2, so still exactly two odd vertices and still an Euler path but no circuit. Now 1 to 2 to 4 to 3 to 1 is a Hamilton circuit, so the answers to the two questions moved independently.

## RECALL AND CHECK

1. Define graph isomorphism.
2. Name three invariants, and say what invariants can and cannot prove.
3. State Euler's condition for a circuit and for a path.
4. What is an Eulerian graph?
5. What is the difference between an Euler circuit and a Hamilton circuit?
6. Why can a graph with a pendant vertex have no Hamilton circuit?

### ANSWERS

1. A bijection between the vertex sets under which two vertices are adjacent in one graph exactly when their images are adjacent in the other.
2. Vertex count, edge count, degree sequence (also connectedness, shortest cycle, bipartiteness). They can disprove isomorphism but never prove it; only exhibiting the bijection proves it.
3. A connected graph has an Euler circuit exactly when every degree is even, and an Euler path but no circuit exactly when exactly two degrees are odd.
4. A connected graph that has an Euler circuit, equivalently one whose every vertex has even degree.
5. Euler uses every edge once and may repeat vertices; Hamilton visits every vertex once and may skip edges.
6. A circuit must enter and leave every vertex it visits, which needs degree at least 2.

## FILL IN THE GAP

1. Isomorphic graphs must have the same \_\_\_\_\_ sequence.
2. An Euler circuit exists exactly when every vertex has \_\_\_\_\_ degree.
3. Euler is about \_\_\_\_\_; Hamilton is about \_\_\_\_\_.
4. The Königsberg graph has \_\_\_\_\_ odd vertices, so no such walk exists.

### ANSWERS

1. degree. 2. even. 3. edges, vertices. 4. four.

## PART SIX · 6.7

# Planarity, Euler's Formula and Kuratowski

*Can this graph be drawn on paper with no edges crossing? Two graphs cannot, every other obstruction is built from those two, and the newest paper gives 10 marks to proving it.*

*By the end of this section you will be able to define a planar graph, state and use Euler's formula for planar graphs, derive and apply the two edge bounds, state Kuratowski's theorem, say why  $K_5$  and  $K_{3,3}$  matter, and prove both non-planar in full.*

## Planar graphs

### ● MUST MEMORISE

### PLANAR GRAPH AND FACES

A graph is **planar** when it can be drawn in the plane with **no two edges crossing** except at their shared endpoints. Such a drawing is called a **plane drawing** of the graph.

A plane drawing cuts the plane into **faces**, or regions. The unbounded region outside the whole drawing is a face too, and forgetting it is the standard error.

#### WHERE

**planar** is a property of the **graph**: it asks whether *some* crossing-free drawing exists.

**plane** is a property of a particular **drawing**, one that happens to have no crossings. **a**

**face** is a region enclosed by edges, counting the infinite outer region as one.

**Planarity is an invariant**, in the sense of section 6.6: isomorphic graphs are either both planar or both non-planar, since a relabelling cannot create or remove a crossing. So it joins the list of properties that can disprove isomorphism.

**Why a drawing with crossings proves nothing.**  $K_4$  is usually drawn as a square with both diagonals, which cross; but it can be redrawn as a triangle with one vertex inside, which does not. So a crossing you happen to have drawn never shows a graph is non-planar. Only a counting argument does, which is exactly what this section supplies.

◆ **Memory hook: planar means it can be drawn flat, not that you drew it flat.** And never forget the outside face.

### EULER'S FORMULA FOR PLANAR GRAPHS

**The claim.** Every plane drawing of the same connected planar graph cuts the plane into the same number of regions, and that number is fixed by the vertex and edge counts alone. Vertices minus edges plus faces is always two, whatever the graph and however it is drawn.

**In symbols.**

$$v - e + f = 2$$

**Conditions.** The graph must be **connected** and **planar**, and  $f$  must be counted from a **plane drawing**, that is, one with no crossings, including the unbounded outer face. For a graph with  $k$  components the formula becomes  $v - e + f = 1 + k$ .

**The limiting case.** For a tree, which section 7.1 defines as a connected graph with no cycles, there are no enclosed regions at all, so  $f = 1$ , and the formula gives  $v - e + 1 = 2$ , that is,  $e = v - 1$ . The edge count of a tree is not a separate fact to memorise but this formula at its smallest possible  $f$ .

For a **simple connected planar** graph with  $v \geq 3$  vertices,

$$e \leq 3v - 6$$

and if in addition the graph has **no triangles**, for example if it is bipartite,

$$e \leq 2v - 4$$

WHERE

$v$  is the number of vertices and  $e$  the number of edges.  $3v - 6$  is the general bound, coming from the fact that every face of a simple graph is bounded by at least 3 edges.  $2v - 4$  is the triangle-free bound, coming from the fact that without triangles every face needs at least 4 edges.

**Where they come from**, which is worth knowing because the derivation is short and it is what makes the bounds usable rather than magic. Count the pairs (face, edge on its boundary) in two ways. Every edge borders exactly two faces, so that count is  $2e$ . Every face is bounded by at least 3 edges in a simple graph, so that count is at least  $3f$ . Hence

$$2e \geq 3f, \text{ so } f \leq 2e/3.$$

Substituting into Euler's formula  $f = 2 - v + e$  gives

$$2 - v + e \leq 2e/3$$

$$6 - 3v + 3e \leq 2e$$

$$e \leq 3v - 6$$

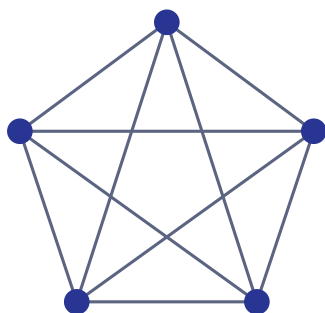
With no triangles every face needs at least 4 edges, so  $2e \geq 4f$  and the same substitution gives  $e \leq 2v - 4$ .

**How to use them.** These are **one-way** tests. If a graph *breaks* the bound it is definitely not planar. If it satisfies the bound it may or may not be planar, and the bound has told you nothing.

◆ **Memory hook: too many edges for too few vertices means crossings are forced.**

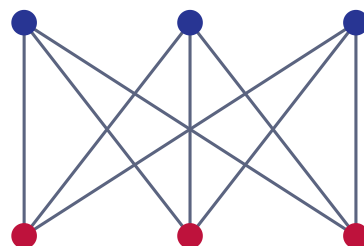
Three  $v$  minus six in general; two  $v$  minus four when there are no triangles to help.

## The two graphs that matter



$K_5$   $v = 5, e = 10$

every crossing drawn here is unavoidable



$K_{3,3}$   $v = 6, e = 9$ , bipartite

The two obstructions.  $K_5$  joins five vertices to each other;  $K_{3,3}$  joins each of three vertices to each of three others.

### KURATOWSKI'S THEOREM

**The claim.** There are only two ways a graph can fail to be planar. A graph is planar if and only if it contains no subgraph that is a subdivision of  $K_5$  or of  $K_{3,3}$ . Every non-planar graph, however large, hides one of those two inside it.

**In symbols.**

$G$  is planar  $\Leftrightarrow G$  contains no subdivision of  $K_5$  and no subdivision of  $K_{3,3}$ .

**Conditions.** A **subdivision** of a graph is what you get by adding extra vertices in the middle of its edges, which never changes whether the graph can be drawn flat. The theorem holds for all finite simple graphs. Note the "if and only if": unlike the edge bounds, this is a complete characterisation, not a one-way test.

**The limiting case, and why it is remarkable.** Any graph on 4 or fewer vertices is planar, because  $K_5$  needs 5 vertices and  $K_{3,3}$  needs 6, so neither can be hidden inside anything smaller.  $K_4$  is therefore the largest complete graph that can be drawn flat, and  $K_5$  is the first that cannot. The force of the theorem is that the list stops at two: an infinite family of obstructions could have existed and does not.

## The 10-mark planarity question

AS PRINTED

2025/2026 · Q5(B)

- (b) i. Briefly state and explain Kuratowski's Theorem. (2 marks)
- ii. Why are  $K_5$  and  $K_{3,3}$  important in graph theory? (2 marks)
- iii. Show that  $K_5$  is non-planar using Euler's formula, Recall that Euler's formula for a  $K_5$  is  $e \leq 3v - 6$  and for a  $K_{3,3}$  is  $e \leq 2v - 4$ . Also show that  $K_{3,3}$  is non-planar. (6 marks)

### NOTE ON THE PAPER

Part iii has a comma where a full stop belongs ("using Euler's formula, Recall that"), and the subscripts are printed inline as " $K_5$ " and " $K_{3,3}$ " rather than as  $K_5$  and  $K_{3,3}$ .

### ONE THING IN IT IS MATHEMATICALLY LOOSE AND YOU SHOULD HANDLE IT CAREFULLY.

The paper calls  $e \leq 3v - 6$  and  $e \leq 2v - 4$  "Euler's formula". They are not. Euler's formula is  $v - e + f = 2$ ; these two are **COROLLARIES** of it, derived by combining it with a count of the edges bounding each face. The safe answer states Euler's formula properly, notes that the two inequalities follow from it, and then uses them exactly as the question directs. That earns the marks the question offers and shows you know the difference, without arguing with the examiner.

Note also that the bound  $e \leq 2v - 4$  is not a general planar bound: it needs the graph to have no triangles.  $K_{3,3}$  is bipartite so it has none, which is exactly why the paper offers that bound for it.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | The two inequalities, handed to you in the question. Part iii is a "show that", so the tools are supplied and the marks are for using them correctly.                                                                          |
| <b>ASKED</b>     | (i) <b>State and explain</b> Kuratowski. (ii) <b>Why</b> the two graphs matter. (iii) <b>Show</b> both are non-planar, which means a computation for each with a stated conclusion.                                            |
| <b>MARKS SAY</b> | 2, 2 and 6. Part iii carries three marks per graph: roughly one for the counts, one for the substitution, one for the conclusion. Do both graphs; a candidate who does only $K_5$ loses half of the largest part.              |
| <b>TEACHES</b>   | This section, 6.7, with the complete and complete bipartite graphs from 6.2.                                                                                                                                                   |
| <b>TRAP</b>      | Miscounting the edges. $K_5$ has $C(5, 2) = 10$ edges, not 5; $K_{3,3}$ has $3 \times 3 = 9$ , not 6. Get either count wrong and the inequality comes out satisfied, and you will confidently prove a non-planar graph planar. |

- GIVEN**  $K_5$ , the complete graph on 5 vertices, and  $K_{3,3}$ , the complete bipartite graph with 3 vertices on each side. The bounds  $e \leq 3v - 6$  and  $e \leq 2v - 4$ .
- FIND** A statement of Kuratowski's theorem, the reason these two graphs matter, and a proof that each is non-planar.
- FORMULA** Euler's formula  $v - e + f = 2$ , and the two bounds derived from it.

i. State and explain Kuratowski's theorem

**Statement.** A graph is planar if and only if it contains no subgraph that is a subdivision of  $K_5$  or of  $K_{3,3}$ .

**Explanation.** A subdivision of a graph is obtained by inserting extra vertices along its edges, which cannot affect whether the graph can be drawn without crossings, since the inserted vertices simply sit on the curves already drawn. The theorem therefore says that these two graphs are the **only** obstructions to planarity: any graph that cannot be drawn flat contains a copy of one of them, possibly with extra vertices along its edges, and any graph containing neither can be drawn flat.

**The line that shows you understand it:** this is an "if and only if", so unlike the edge bounds it decides the question in both directions. The bounds can only ever say "not planar"; Kuratowski also says "planar".

ii. Why  $K_5$  and  $K_{3,3}$  are important

Three reasons, one per half mark and a spare:

1. **They are the two smallest non-planar graphs**, and by Kuratowski's theorem they are the **only** obstructions to planarity. Every non-planar graph in existence contains a subdivision of one of them, so understanding two small graphs settles a question about infinitely many large ones.
2. **They mark the exact boundary.**  $K_4$  is planar and  $K_5$  is not, so  $K_4$  is the largest complete graph that can be drawn flat.  $K_{2,3}$  is planar and  $K_{3,3}$  is not. The two graphs are where planarity stops.
3. **They matter in practice.**  $K_{3,3}$  is the classic "three utilities to three houses" puzzle, which has no solution for exactly this reason. Planarity decides whether a circuit can be laid out on a single-layer board without wires crossing, so these two graphs are the test for when a second layer becomes unavoidable.

iii. Show that  $K_5$  is non-planar

**State the tool first.** Euler's formula for a connected planar graph is  $v - e + f = 2$ . Combining it with the fact that every face of a simple graph is bounded by at least three edges gives the bound the question supplies: for any simple connected planar graph with  $v \geq 3$ ,

$$e \leq 3v - 6$$

**Step 1. Count  $v$  and  $e$  for  $K_5$ .** It has 5 vertices, and being complete it joins every pair, so by section 6.2

$$v = 5, \quad e = C(5, 2) = 5 \times 4 / 2 = 10$$

**Step 2. Evaluate the bound.**

$$3v - 6 = 3 \times 5 - 6 = 15 - 6 = 9$$



### Step 3. Compare and conclude.

$$e = 10 > 9 = 3v - 6$$

Every simple connected planar graph on at least 3 vertices satisfies  $e \leq 3v - 6$ .  $K_5$  does not. **Therefore  $K_5$  is non-planar.**  $\square$

iii, continued. Show that  $K_{3,3}$  is non-planar

**Step 1. Count  $v$  and  $e$  for  $K_{3,3}$ .** Three vertices on each side gives  $3 + 3 = 6$  vertices, and every vertex of one side joins every vertex of the other, so by the product rule of section 5.1

$$v = 6, \quad e = 3 \times 3 = 9$$

**Step 2. Check that the tighter bound applies.** This is the step most answers skip and it is worth a mark.  $K_{3,3}$  is bipartite, and a bipartite graph has no odd cycles, so in particular it has **no triangles**. Every face of a plane drawing of it would therefore be bounded by at least four edges, which is what gives the bound

$$e \leq 2v - 4$$

**Step 3. Evaluate and compare.**

$$2v - 4 = 2 \times 6 - 4 = 12 - 4 = 8$$

$$e = 9 > 8 = 2v - 4$$

**Therefore  $K_{3,3}$  is non-planar.**  $\square$

**Why the general bound would have failed here,** worth one line because it shows why the question supplied two different inequalities: for  $K_{3,3}$ ,  $3v - 6 = 12$ , and  $e = 9$  is comfortably under 12. The general bound is satisfied, so it proves nothing. Only the triangle-free bound is sharp enough, and it applies only because the graph is bipartite.

| GRAPH     | V | E  | $3V - 6$    | $2V - 4$                      | VERDICT                          |
|-----------|---|----|-------------|-------------------------------|----------------------------------|
| $K_4$     | 4 | 6  | 6           | not applicable, has triangles | bound satisfied; $K_4$ is planar |
| $K_5$     | 5 | 10 | 9           | not applicable, has triangles | $10 > 9$ , non-planar            |
| $K_{3,3}$ | 6 | 9  | 12, no help | 8                             | $9 > 8$ , non-planar             |

**Now redo it:** is  $K_6$  planar?  $v = 6$  and  $e = C(6, 2) = 15$ , while  $3v - 6 = 12$ . Since  $15 > 12$  it is non-planar, which also follows from Kuratowski because  $K_6$  contains  $K_5$ .

- GIVEN**  $K_4$  drawn as a triangle with one vertex inside, joined to all three corners. No edges cross.
- FIND** The number of faces, and a check of Euler's formula.
- FORMULA**  $v - e + f = 2$ .

**Count.**  $v = 4$  and  $e = 6$ .

**Solve for f.**

$$4 - 6 + f = 2, \text{ so } f = 4$$

**Verify by looking at the drawing.** The inner vertex splits the triangle into three small regions, and there is the unbounded region outside:  $3 + 1 = 4$ . The formula and the picture agree, and the outer face is the one that is easy to miss.

**The bound as a sanity check:**  $3v - 6 = 6$  and  $e = 6$ , so  $K_4$  sits exactly on the bound. A graph on the bound is called **maximal planar**: it is planar, and adding any further edge would make it non-planar.

## RECALL AND CHECK

1. Define a planar graph, and say why a drawing with crossings proves nothing.
2. State Euler's formula and say what  $f$  counts.
3. State the two edge bounds and the condition on the second one.
4. State Kuratowski's theorem.
5. Show  $K_5$  is non-planar in two lines.
6. Why does the bound  $3v - 6$  fail to settle  $K_{3,3}$ ?

## ANSWERS

1. A graph is planar when some drawing of it has no crossing edges. A particular drawing with crossings may just be a bad drawing, as with  $K_4$ .
2.  $v - e + f = 2$ , where  $f$  counts the faces of a plane drawing, including the unbounded outer face.
3.  $e \leq 3v - 6$  for a simple connected planar graph with  $v \geq 3$ , and  $e \leq 2v - 4$  when it also has no triangles.
4. A graph is planar if and only if it contains no subdivision of  $K_5$  or  $K_{3,3}$ .
5.  $v = 5$  and  $e = 10$ , but  $3v - 6 = 9$ , and  $10 > 9$ .
6. Because  $3v - 6 = 12$  and  $e = 9$  satisfies it. The bound is not sharp enough; the triangle-free bound of 8 is.

## FILL IN THE GAP

1. Euler's formula is \_\_\_\_\_, where  $f$  includes the \_\_\_\_\_ face.
2.  $K_5$  has \_\_\_\_\_ edges and the bound allows \_\_\_\_\_.
3.  $K_{3,3}$  has \_\_\_\_\_ edges and the triangle-free bound allows \_\_\_\_\_.
4. Kuratowski's theorem says the only obstructions to planarity are \_\_\_\_\_ and \_\_\_\_\_.

## ANSWERS

1.  $v - e + f = 2$ , outer. 2. 10, 9. 3. 9, 8. 4.  $K_5$ ,  $K_{3,3}$ .

## PART SIX · 6.8

# Shortest Routes and the Travelling Salesman

*The same five-city map has appeared on three papers, worth 10 marks every time, always with the same distances and always asking the same thing. It is the most reliable 10 marks in this book.*

*By the end of this section you will be able to state the travelling salesman problem, count how many tours a graph has, apply the nearest-neighbour method and say honestly what it does and does not guarantee, and produce the optimal Detroit tour with a proof that it is optimal.*

## ● MUST MEMORISE

## WEIGHTED GRAPHS AND THE TRAVELLING SALESMAN PROBLEM

A **weighted graph** is a graph in which every edge carries a number  $w(u, v)$ , its **weight**: a distance, a cost, a time. The **weight of a route** is the sum of the weights of the edges it uses.

The **travelling salesman problem** asks: starting from one vertex, visit **every other vertex exactly once** and return to the start, with the **smallest possible total weight**.

## WHERE

$w(u, v)$  is the weight on the edge joining  $u$  and  $v$ , read straight off the labels on the drawing. a **tour** is a **Hamilton circuit**, as defined in section 6.6: every vertex once, back to the start. **smallest total weight** is what makes it an optimisation problem rather than an existence one. On a complete graph a tour always exists; the work is finding the cheapest.

◆ **Memory hook: every city once, home at the end, cheapest possible.** Drop any one of those three and it is a different problem with a different answer.

On a complete graph with  $n$  vertices, the number of **distinct** tours starting and ending at a fixed vertex is

$$(n - 1)! / 2$$

WHERE

$(n - 1)!$  counts the orders in which the other  $n - 1$  cities can be visited, by the permutation rule of section 5.2. **the division by 2** is because a tour and the same tour driven backwards use the same edges and cost the same, so they are one route, not two.

**For the five-city map:**  $(5 - 1)! / 2 = 24 / 2 = 12$  distinct tours. That is few enough to list them all, which is why this question is answerable exactly in an exam.

**Why the problem is famous, in one line worth writing:** the count grows factorially, so 10 cities give 181440 tours and 20 cities give more than  $10^{16}$ . No method is known that finds the exact optimum quickly for large  $n$ , which is why heuristics like nearest neighbour exist at all.

◆ **Memory hook: fix the start, permute the rest, halve for the reverse.**

## The question that appears three times

AS PRINTED

2020/2021 · Q2(A)

2. (a) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? **(10marks)**

AS PRINTED

2021/2022 · Q4(C)

(c) Given the graph below, a travelling missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? **(10marks)**

AS PRINTED

2023/2024 · Q4(B)

(b) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance?

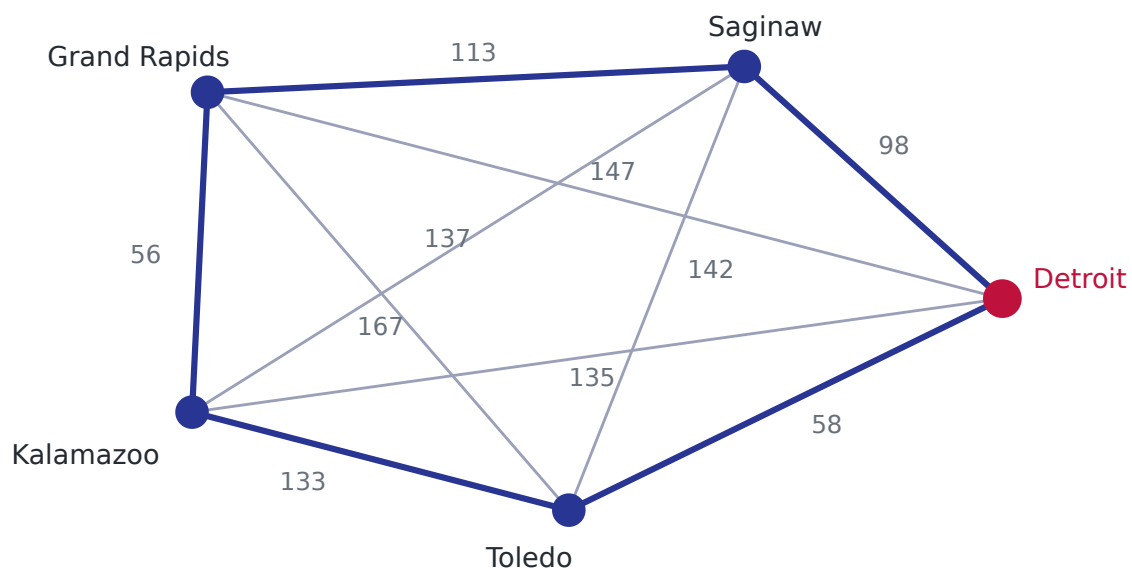
#### NOTE ON THE PAPER, AND ON THE FIGURE

Three papers, one question, 30 marks in total, and the same figure each time. The wording differs only in "A Christian missionary" against "a travelling missionary" and in whether the marks are printed closed up.

**WHY THE 2023/2024 QUOTE ABOVE CARRIES NO MARK TEXT.** On that paper the diagram sits between the question and its marks, and "(10 marks)" is printed on its own line *below* the figure. Splicing it onto the end of the sentence would join words the paper does not join, so the quote stops where the paper's sentence stops and the marks are reported here instead. They are 10, the same as the other two versions.

**THE FIGURE IS THE SAME ON ALL THREE AND IT IS READABLE.** It is a complete graph on five cities, so all  $C(5, 2) = 10$  edges are drawn and labelled. The labels were read at 500 to 700 dpi and are listed in the worked answer below. Unlike the two digraphs of section 6.4, nothing here is in doubt.

**WHAT "IN WHICH ORDER" MEANS FOR YOUR ANSWER.** The question asks for an order, so the answer is a sequence of city names, not just a number. Give the route, give its total, and give the argument that no route is shorter. All three are needed for 10 marks.



The optimal tour, in indigo: Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit. Total

*The five-city map as printed on all three papers, with the optimal tour drawn over it. All ten edges exist; only five are used.*

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Five cities, all pairs joined, with the ten distances labelled on the figure. The tour must start and finish at Detroit.                                                                                                                                                           |
| <b>ASKED</b>     | <b>In which order</b> the cities should be visited to travel the <b>minimum</b> distance. An order, a total, and a reason it is minimal.                                                                                                                                           |
| <b>MARKS SAY</b> | 10, the largest single part on any of these papers. Budget: 1 for tabulating the distances, 1 for the tour count, 4 for the enumeration or the method, 2 for the route, 2 for showing it is optimal.                                                                               |
| <b>TEACHES</b>   | This section, 6.8, with Hamilton circuits from 6.6 and permutations from 5.2.                                                                                                                                                                                                      |
| <b>TRAP</b>      | Giving the nearest-neighbour route and calling it the minimum without checking. Nearest neighbour is a heuristic: it is fast, it is often good, and it is <b>not guaranteed</b> to be optimal. With only 12 tours you can check them all, so there is no excuse for guessing here. |

**GIVEN** Five cities forming a complete weighted graph, start and finish at Detroit. The ten distances are tabulated in step 1.

**FIND** The order of visits giving the minimum total distance, and that minimum.

**RULE** A tour is a Hamilton circuit; there are  $(n - 1)!/2 = 12$  of them; the minimum is found by evaluating all 12.

**Step 1. Tabulate the distances,** reading them off the figure. Do this first, always: it takes thirty seconds and every later line depends on it.

|              | GRAND RAPIDS | SAGINAW | KALAMAZOO | TOLEDO | DETROIT |
|--------------|--------------|---------|-----------|--------|---------|
| GRAND RAPIDS | 0            | 113     | 56        | 167    | 147     |
| SAGINAW      | 113          | 0       | 137       | 142    | 98      |
| KALAMAZOO    | 56           | 137     | 0         | 133    | 135     |
| TOLEDO       | 167          | 142     | 133       | 0      | 58      |
| DETROIT      | 147          | 98      | 135       | 58     | 0       |

**Step 2. Count the tours.** Five cities with Detroit fixed as the start gives

$$(5 - 1)! / 2 = 4! / 2 = 24 / 2 = \mathbf{12} \text{ distinct tours.}$$

Twelve is small enough to evaluate exhaustively, and doing so is what turns "here is a short route" into "here is **the** shortest route".

**Step 3. Evaluate all twelve.** Abbreviate Grand Rapids as GR, Saginaw as S, Kalamazoo as K, Toledo as T and Detroit as D. Each row starts and ends at D.

| TOUR         | DISTANCES ADDED               | TOTAL      |
|--------------|-------------------------------|------------|
| D S GR K T D | $98 + 113 + 56 + 133 + 58$    | <b>458</b> |
| D K GR S T D | $135 + 56 + 113 + 142 + 58$   | 504        |
| D S K GR T D | $98 + 137 + 56 + 167 + 58$    | 516        |
| D GR K S T D | $147 + 56 + 137 + 142 + 58$   | 540        |
| D GR K T S D | $147 + 56 + 133 + 142 + 98$   | 576        |
| D GR S K T D | $147 + 113 + 137 + 133 + 58$  | 588        |
| D K GR T S D | $135 + 56 + 167 + 142 + 98$   | 598        |
| D K S GR T D | $135 + 137 + 113 + 167 + 58$  | 610        |
| D K T GR S D | $135 + 133 + 167 + 113 + 98$  | 646        |
| D GR S T K D | $147 + 113 + 142 + 133 + 135$ | 670        |
| D GR T K S D | $147 + 167 + 133 + 137 + 98$  | 682        |
| D GR T S K D | $147 + 167 + 142 + 137 + 135$ | 728        |

**Step 4. Read off the answer.**

**Order: Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit.**

**Minimum distance: 458.**

**Why it is optimal, which is the part worth two marks:** all twelve possible tours have been evaluated and none is shorter than 458. This is not an estimate. The worst tour costs 728, so the choice of order matters by over 270, which is a good closing observation.

**Note the route may be written either way round.** Detroit, Toledo, Kalamazoo, Grand Rapids, Saginaw, Detroit is the same tour driven backwards and costs the same 458. Either is a correct answer; that is exactly what the division by 2 in step 2 was accounting for.

**The nearest-neighbour method, and its honest status.** If time is short, the heuristic is: from the current city, always go to the nearest unvisited one, and return home at the end.

| AT           | UNVISITED, WITH DISTANCES | NEAREST          |
|--------------|---------------------------|------------------|
| Detroit      | GR 147, S 98, K 135, T 58 | Toledo, 58       |
| Toledo       | GR 167, S 142, K 133      | Kalamazoo, 133   |
| Kalamazoo    | GR 56, S 137              | Grand Rapids, 56 |
| Grand Rapids | S 113                     | Saginaw, 113     |
| Saginaw      | return home               | Detroit, 98      |

$$\text{total} = 58 + 133 + 56 + 113 + 98 = \mathbf{458}$$

**On this map the heuristic happens to find the optimum**, because the route it builds is the optimal tour driven backwards. Say **"happens to"**. Nearest neighbour carries no guarantee: it commits to a cheap first step that can force an expensive last one, and on other maps it does exactly that. If you use it, use it to *find* a candidate and then justify the candidate by comparison, as above. A 10-mark answer that says "nearest neighbour gives 458, therefore 458 is the minimum" has asserted the thing it was asked to establish.

**Now redo it:** how many tours would six cities have?  $(6 - 1)!/2 = 120/2 = 60$ , five times as many for one extra city. That growth is the whole difficulty of the problem.

#### ⚠ TRAP

#### A HEURISTIC IS NOT A PROOF

The three papers all say "the minimum distance". A minimum is a claim about **every** alternative, so it can only be established by ruling the alternatives out. With 12 tours, list them. With 60 you would explain the method and evaluate a well-chosen few, stating plainly that the result is a good route rather than a proven optimum.

Writing that distinction down costs one sentence and is the difference between an answer that is correct and one that is merely right.



## RECALL AND CHECK

1. State the travelling salesman problem.
2. How many distinct tours has a complete graph on  $n$  vertices, and why the division by 2?
3. How many tours has the five-city map?
4. Give the optimal Detroit route and its total.
5. What does the nearest-neighbour method guarantee?

---

### ANSWERS

1. Starting from one vertex of a weighted graph, visit every other vertex exactly once and return to the start, at minimum total weight.
2.  $(n - 1)!/2$ . Fixing the start leaves  $(n - 1)!$  orders for the rest, and each tour is counted twice because driving it backwards costs the same.
3.  $4!/2 = 12$ .
4. Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit, totalling 458.
5. Nothing. It is a heuristic that often gives a good route and is not guaranteed to give the shortest.

## FILL IN THE GAP

1. A tour is a \_\_\_\_\_ circuit with weights attached.
2. A complete graph on  $n$  vertices has \_\_\_\_\_ distinct tours.
3. The five-city map has \_\_\_\_\_ tours and its minimum is \_\_\_\_\_.
4. Nearest neighbour is a \_\_\_\_\_, so it is not a \_\_\_\_\_.

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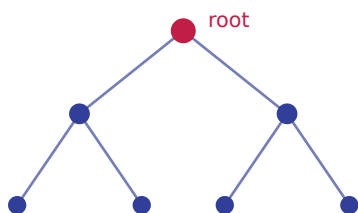
### ANSWERS

1. Hamilton. 2.  $(n - 1)!/2$ . 3. 12, 458. 4. heuristic, proof.

## PART SEVEN

# Trees and Their Traversals

*A tree is a graph with no way back. That single restriction makes it the structure computing actually uses: file systems, expressions, search, and the chain-letter question three of these papers keep asking.*



$$e = v - 1$$

always

$$l = (m - 1)i + 1$$

full m-ary

at most  $m$  to the  $h$

leaves at height  $h$

pre, in, post

three traversals

## Trees and Rooted-Tree Terminology

Two papers ask for the key terminology of a rooted tree, worth 7.5 marks each, and a third asks where trees are used. The vocabulary is the question.

By the end of this section you will be able to define a tree and a forest, prove that a tree has one fewer edge than it has vertices, name every part of a rooted tree with a labelled diagram, and answer the terminology and application questions in full.

### NOT IN THE COURSE TEXT

The official text has no trees in it. Its twelve occurrences of the word "tree" are all in one Venn diagram about trees growing on a street. It never uses root, leaf or traversal in the technical sense. Yet 2024/2025 gives a whole 17.5-mark question to trees, and 2020/2021, 2021/2022 and 2023/2024 give them two parts each. Everything here is taught from zero.

## What a tree is

### ● MUST MEMORISE

### TREE AND FOREST

A **tree** is a connected undirected graph with **no cycles**.

A **forest** is an undirected graph with no cycles that need not be connected: each of its components is a tree.

#### WHERE

**connected** means there is a route between every pair of vertices, as defined in section 6.2.

**no cycles** means no route leaves a vertex and returns to it without repeating an edge.

This is the restriction that makes a tree a tree. **acyclic** is the one-word term for "no cycles", and a forest is exactly an acyclic graph.

**Three equivalent definitions**, any of which may be quoted. A graph on  $v$  vertices is a tree exactly when

1. it is connected and has no cycles; or
2. there is **exactly one** path between every pair of vertices; or
3. it is connected and has exactly  $v - 1$  edges.

◆ **Memory hook: connected, and no way back.** Adding any edge to a tree creates a cycle; removing any edge disconnects it. A tree is as connected as a graph can be without being redundant.

A tree with  $v$  vertices has exactly

$$e = v - 1 \text{ edges.}$$

WHERE

$v$  is the number of vertices and  $e$  the number of edges. **the**  $- 1$  is the root, or any one vertex you like: every other vertex is reached by exactly one edge.

**It is not a separate fact.** Section 6.7 gave Euler's formula  $v - e + f = 2$  for a connected planar graph. A tree has no cycles, so it encloses no region and its only face is the outer one:  $f = 1$ . Then

$$v - e + 1 = 2, \text{ so } e = v - 1.$$

By the Handshaking theorem of section 6.3 the degrees therefore sum to  $2(v - 1)$ , which is a useful check on any tree you draw.

◆ **Memory hook: one edge per vertex, except the first.** A tree of 17 vertices has 16 edges, always.

## Rooted trees

A **rooted tree** is a tree in which one vertex has been singled out as the **root**, and every edge is then understood as pointing away from the root.

WHERE

**the root** is chosen, not discovered: any vertex of a tree may be made the root, and different choices give different rooted trees from the same underlying tree. **away from the root** is what gives every other vertex a unique parent, and it is the whole reason the family vocabulary below makes sense.

◆ **Memory hook: pick a vertex and hang the tree from it.** Everything else in this section is the vocabulary of what happens once you have.

Let  $T$  be a rooted tree and  $v$  one of its vertices.

| TERM                                   | MEANING                                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Root</b>                            | The single distinguished vertex. It has no parent, and it is the only vertex that does not.                                     |
| <b>Parent</b>                          | The vertex immediately above $v$ on the unique path from $v$ to the root. Every vertex but the root has exactly one.            |
| <b>Child</b>                           | Any vertex whose parent is $v$ . A vertex may have any number of children, including none.                                      |
| <b>Siblings</b>                        | Vertices that share a parent.                                                                                                   |
| <b>Ancestors</b>                       | Every vertex on the path from $v$ up to the root, not counting $v$ itself.                                                      |
| <b>Descendants</b>                     | Every vertex having $v$ as an ancestor, that is, everything hanging below $v$ .                                                 |
| <b>Internal vertex</b>                 | A vertex with at least one child. Also called a branch vertex.                                                                  |
| <b>Leaf</b>                            | A vertex with no children. Also called a terminal or external vertex.                                                           |
| <b>Subtree at <math>v</math></b>       | $v$ together with all its descendants and the edges between them; itself a rooted tree with root $v$ .                          |
| <b>Level (depth) of <math>v</math></b> | The number of edges on the path from the root to $v$ . The root is at level 0.                                                  |
| <b>Height of <math>T</math></b>        | The largest level of any vertex, that is, the length of the longest root-to-leaf path.                                          |
| <b>Ordered rooted tree</b>             | A rooted tree in which the children of each vertex are given a left-to-right order. The traversals of section 7.3 require this. |

◆ **Memory hook: it is a family tree drawn upside down.** Parents above, children below, siblings alongside, ancestors all the way up, descendants all the way down. Every term means what it means in a family, except that the root is drawn at the top.

## The terminology question

AS PRINTED

2020/2021 · Q2(C)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. **(7½marks)**

AS PRINTED

2021/2022 · Q3(C)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. **(7½marks)**

### NOTE ON THE PAPER

Word for word the same question on two papers, the same marks, the same capitalisation of "Rooted Tree". One prepared answer covers 15 marks across the set. "Highlight" here means list and define; "diagrammatical illustration" is not optional, and an answer without a drawing cannot reach full marks.

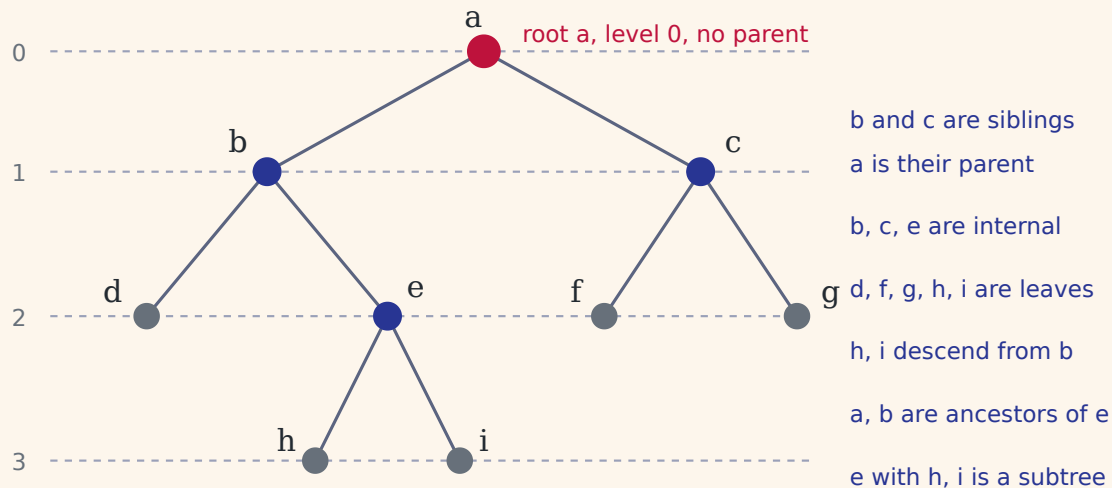
### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Nothing. Pure recall.                                                                                                                                                                                                                   |
| <b>ASKED</b>     | <b>Highlight the various key terminologies</b> , and <b>support with a diagram</b> .<br>Two deliverables: a labelled list and one labelled tree.                                                                                        |
| <b>MARKS SAY</b> | 7.5. At roughly half a mark per term, aim for <b>ten to twelve terms</b> plus the diagram. Define each in one line; do not write a paragraph on any of them.                                                                            |
| <b>TEACHES</b>   | This section, 7.1.                                                                                                                                                                                                                      |
| <b>TRAP</b>      | Drawing a tree and labelling only the root and the leaves. The marks are spread across the terms, so the diagram should carry as many labels as it can hold: an internal vertex, a pair of siblings, a subtree, a level and the height. |

**Open with the definition**, because every term below depends on it: a rooted tree is a connected acyclic graph in which one vertex has been designated the root, with all edges understood as directed away from it.

**Then the terms.** Use the table above, giving each in one line. Twelve terms at half a mark each covers the paper.

**Then the diagram, fully labelled.** This is the half of the answer that is easiest to earn and easiest to skip.



Levels down the left. The height is 3: the deepest level any vertex reaches.

9 vertices and 8 edges, as  $e = v - 1$  requires.

*One diagram carrying every term the question asks for. Draw the level lines: they earn the marks for level and height at no extra cost.*

**Close with the two counting facts**, which show the terms are not just names: this tree has 9 vertices and 8 edges, satisfying  $e = v - 1$ , and its degrees sum to 16, which is twice the 8 edges as the Handshaking theorem requires.

**Now redo it:** in the diagram above, name the parent of h, the siblings of f, and the level of i.  
Answers: e; g; level 3.

## Where trees are used

Highlight, with a relevant example each, three application domains of Tree models in computing. **(4½ marks)**

### NOTE ON THE PAPER

**THIS PART HAS NO LETTER.** After Q4(a) and Q4(b) the paper runs straight into this sentence with its marks but no "(c)", just as Question 4 itself carries no number. It is plainly the third part of Question 4, and it is quoted here exactly as printed, with no letter supplied.

Note the count: **THREE** domains with **ONE** example each, not the four with two examples that the graph version of this question asks for in section 6.1. Read the numbers before writing.

Three domains, each with what the vertices are, what the edges are, and one concrete example. Naming the modelling is what separates this from a list of words.

1. **File systems and hierarchical storage.** Vertices are directories and files; edges join a directory to what it contains. The root vertex is the root directory, every file has exactly one parent directory, and there are no cycles, which is precisely why the route from the root to any file is unique. *Example:* resolving a path name is a walk from the root down to a leaf.
2. **Compilers and expression evaluation.** Vertices are operators and operands; the children of an operator are the things it acts on. *Example:* a compiler parses source code into a syntax tree and then walks it to emit machine code, which is the expression-tree work of section 7.3.
3. **Searching and decision procedures.** Vertices are stored keys or states; edges are the comparisons or moves leading from one to the next. *Example:* a binary search tree finds a record in a number of comparisons proportional to the tree's height rather than to the number of records, which is why the height bound of section 7.2 matters.

**A fourth, if the question asks for more: data compression,** where a Huffman coding tree gives short codes to frequent symbols by placing them near the root.

## RECALL AND CHECK

1. Define a tree, and give two other equivalent definitions.
2. How many edges has a tree with 17 vertices?
3. What is the difference between a tree and a rooted tree?
4. Define leaf, internal vertex, level and height.
5. Why does a tree have exactly one face?

## ANSWERS

1. A connected acyclic undirected graph. Equivalently: exactly one path between every pair of vertices; or connected with exactly  $v - 1$  edges.
2. 16, since  $e = v - 1$ .
3. A rooted tree has one vertex singled out as the root, which gives every other vertex a unique parent.
4. A leaf has no children; an internal vertex has at least one; the level of a vertex is its distance from the root; the height is the largest level.
5. It has no cycles, so it encloses no region; only the unbounded outer face remains, and Euler's formula then gives  $e = v - 1$ .



## FILL IN THE GAP

1. A tree is a \_\_\_\_\_ graph with no \_\_\_\_\_.
2. A tree on  $v$  vertices has \_\_\_\_\_ edges.
3. The only vertex with no parent is the \_\_\_\_\_.
4. A vertex with no children is a \_\_\_\_\_.
5. The root sits at level \_\_\_\_\_.

## ANSWERS

1. connected, cycles. 2.  $v - 1$ . 3. root. 4. leaf. 5. 0.

## PART SEVEN • 7.2

# m-ary and Balanced Trees

Two formulas answer almost every tree counting question on these papers, and the chain-letter problem they solve has appeared on three of the five in three different costumes.

By the end of this section you will be able to define  $m$ -ary, full  $m$ -ary, binary and balanced trees, use the two counting formulas to find the vertices, leaves or internal vertices of a full  $m$ -ary tree, prove the height bound on leaves, and answer the balanced-tree, chain-letter, quinary and playoff questions in full.

## The families

### ● MUST MEMORISE

### M-ARY, FULL AND BINARY

A rooted tree is  **$m$ -ary** when every internal vertex has **at most**  $m$  children.

It is a **full  $m$ -ary tree** when every internal vertex has **exactly**  $m$  children.

An  $m$ -ary tree with  $m = 2$  is a **binary tree**; with  $m = 3$  a **ternary tree**; with  $m = 5$  a **quinary tree**, which is the word the 2024/2025 paper uses.

#### WHERE

$m$  is a positive whole number, the branching factor. **at most  $m$**  is the  $m$ -ary condition; **exactly  $m$**  is the **full** condition, and the difference between them decides whether the counting formulas below apply at all. **a leaf** has no children, so the condition is only ever about internal vertices.

◆ **Memory hook:  $m$ -ary caps the branching, full fixes it.** Every full  $m$ -ary tree is  $m$ -ary; the reverse is not true, and the formulas need *full*.

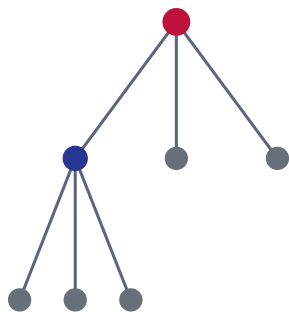
An  $m$ -ary tree of height  $h$  is **balanced** when every leaf lies at level  $h$  or at level  $h - 1$ .

## WHERE

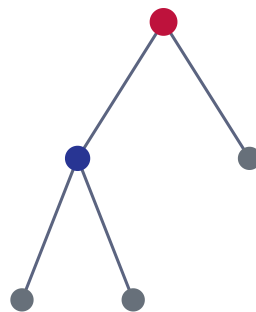
$h$  is the height, the deepest level any vertex reaches, from section 7.1.  **$h$  or  $h - 1$**  is the whole condition: the leaves sit on the bottom two levels and nowhere higher. No leaf may dangle near the top while another branch runs deep.

**Why balance is worth having.** A balanced tree of  $n$  vertices has height about  $\log_m n$ , which is as small as it can be. Since searching costs one comparison per level, a balanced tree is what makes a search structure fast: an unbalanced tree can degenerate into a single long chain, at which point searching it is no better than reading a list.

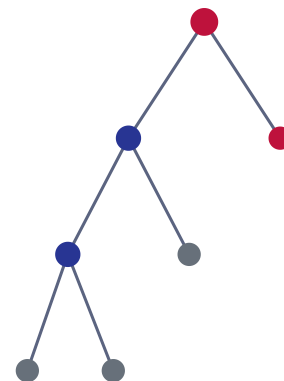
◆ **Memory hook: all the leaves on the bottom two shelves.** If any leaf is higher than that, the tree is not balanced.



Full ternary, balanced  
every internal vertex has exactly 3  
leaves at levels 1 and 2  
leaves on levels 1 and 2 only



Full binary, balanced



Full binary, NOT balanced  
a leaf at level 1, height 3

*All three are full: every internal vertex has exactly  $m$  children. Only the first two are balanced. The crimson leaf on the right sits far above the bottom two levels, which is what breaks it.*

## The two counting formulas

### ● MUST MEMORISE

### COUNTING A FULL M-ARY TREE

In a **full** m-ary tree with **i** internal vertices:

$$n = mi + 1 \quad (\text{total vertices})$$

$$l = n - i = (m - 1)i + 1 \quad (\text{leaves})$$

#### WHERE

**m** is the branching factor: every internal vertex has exactly m children. **i** is the number of **internal** vertices, the ones that do have children. **n** is the total number of vertices and **l** the number of leaves, so  $n = i + l$  always.

**Where  $n = mi + 1$  comes from.** Every vertex except the root is the child of exactly one internal vertex. Each of the i internal vertices supplies exactly m children, so there are mi non-root vertices, and adding the root gives  $mi + 1$ .

**Then the leaves follow** with no new idea:  $l = n - i = (mi + 1) - i = (m - 1)i + 1$ .

**Rearranged for the exam,** because questions usually give you the leaves and want the rest:

$$i = (l - 1) / (m - 1) \quad \text{and then} \quad n = mi + 1$$

◆ **Memory hook: every vertex but the root is somebody's child.** That one sentence gives  $n = mi + 1$ , and everything else in this box is algebra on it.

### ● MUST MEMORISE

### THE HEIGHT BOUND

An m-ary tree of height h has **at most  $m^h$  leaves**.

#### WHERE

**h** is the height, from section 7.1.  **$m^h$**  is m multiplied by itself h times. **at most** is exact: equality holds precisely when the tree is full **and** every leaf is at level h.

**Why.** Level 0 holds the root alone. Each vertex has at most m children, so level 1 holds at most m vertices, level 2 at most  $m \times m = m^2$ , and level k at most  $m^k$ . Every leaf lies at some level no deeper than h, and the deepest level can hold at most  $m^h$  of them.

◆ **Memory hook: each level multiplies by m.** The product rule of section 5.1 applied h times: h independent choices of which child to descend to, m ways each.

## The balanced m-ary tree question

### AS PRINTED

2020/2021 · Q3(A)

3. (a) Explain, with examples, what a Balanced M-ary Tree means?

(7½marks)

### AS PRINTED

2021/2022 · Q4(B)

(b) Explain, with examples, what a Balanced M-ary Tree means.

(8½marks)

### NOTE ON THE PAPER

The same question twice, worth 7.5 and 8.5 marks, differing only in the final punctuation: 2020/2021 ends a command with a question mark, 2021/2022 with a full stop. Sixteen marks across the set for one prepared answer. "With examples" is in the instruction, so drawings are required, and the plural means more than one.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Nothing. Recall with illustration.                                                                                                                                                                                                                                            |
| <b>ASKED</b>     | <b>Explain</b> what a balanced $m$ -ary tree means, <b>with examples</b> . Two ideas are bundled in the phrase, and both need defining: <i>m</i> -ary and <i>balanced</i> .                                                                                                   |
| <b>MARKS SAY</b> | 7.5 and 8.5, which is a lot for one definition. The marks are for unpacking it: define $m$ -ary, define full, define height and level, define balanced, then give a balanced example <b>and</b> an unbalanced one, and say why balance matters.                               |
| <b>TEACHES</b>   | This section, 7.2, with the vocabulary of 7.1.                                                                                                                                                                                                                                |
| <b>TRAP</b>      | Defining "balanced" and stopping. The term has two words in it, and an answer that never says what $m$ -ary means has answered half the question. The second trap is giving only a balanced example: the <b>counter</b> example is what shows you know where the boundary is. |

**Build the answer in four layers**, each earning its own marks.

1. m-ary

A rooted tree is m-ary when every internal vertex has at most m children. It is **full** m-ary when every internal vertex has exactly m children. For  $m = 2$  it is called binary, for  $m = 3$  ternary, for  $m = 5$  quinary.

2. Level and height

The level of a vertex is the number of edges from the root to it, the root being at level 0. The height of the tree is the largest level any vertex reaches.

3. Balanced

An m-ary tree of height  $h$  is **balanced** when every leaf is at level  $h$  or at level  $h - 1$ : all the leaves lie on the bottom two levels. Putting the two words together, a **balanced m-ary tree** is a rooted tree in which no internal vertex has more than m children and every leaf sits on one of the two deepest levels.

4. The examples, drawn

Refer to the three trees drawn above. The first is a full ternary tree of height 2 whose leaves lie at levels 1 and 2, so it is balanced. The second is a full binary tree of height 2 with leaves at levels 1 and 2, also balanced. The third is a full binary tree of height 3 with one leaf at level 1, which is two levels above the bottom, so it is **not** balanced even though it is full.

**That third example is the one that earns marks**, because it shows that full and balanced are independent: a tree can be one without the other.

5. Why it matters, for the last mark

A balanced m-ary tree with  $n$  vertices has height roughly  $\log_m n$ , the smallest a tree of that size can have. Search cost is proportional to height, so balance is what keeps a search structure fast; an unbalanced tree can stretch into a chain and lose the advantage entirely.

**Now redo it:** is a full binary tree with leaves at levels 2, 3 and 3 balanced? Its height is 3, so leaves are allowed at levels 3 and 2. Yes, it is balanced.

## The chain letter, in three costumes

(b) Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter? **(8marks)**

3. (a) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

(i) How many people have seen the card, including the first person? **(5marks)**

(ii) How many people sent out the card? **(5marks)**

(d) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four (4) other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

(i) How many people have seen the card, including the first person? **(6 marks)**

(ii) How many people sent out the card? **(3½ marks)**

#### NOTE ON THE PAPER

**ONE PROBLEM, THREE PAPERS, 22.5 MARKS.** 2020/2021 tells it as a chain letter and asks both questions in one run-on sentence for 8 marks; the two later papers tell it as convocation invitation cards and split it into (i) and (ii) for 10 and 9.5 marks. The numbers are identical every time: four recipients each, 100 people who did not pass it on.

Both card versions end with "If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out." That is a subordinate clause with no main clause, so the sentence never finishes. The numbered parts supply the question. Nothing is missing from the data.

The sentence doing the mathematical work is "**NO ONE RECEIVES MORE THAN ONE CARD**". That is what makes the picture a **TREE**: one parent per person, and no cycles.

#### BREAK IT DOWN

|                  |                                                                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Each sender sends to exactly 4 others. Nobody receives twice. 100 people received it and did not send it on.                                                                                                            |
| <b>ASKED</b>     | (i) how many have <b>seen</b> it, including the first person; (ii) how many <b>sent it out</b> .                                                                                                                        |
| <b>MODEL</b>     | A <b>full 4-ary tree</b> . Vertices are people. The root is the first person. A person's children are the four they sent to. Senders are the <b>internal</b> vertices; the 100 who did not send are the <b>leaves</b> . |
| <b>MARKS SAY</b> | 8, 10 and 9.5 across the papers, far more than two formulas are worth. The marks are for the modelling: say what the vertices are, why it is a tree, why it is full, and which quantity is which.                       |
| <b>TEACHES</b>   | This section, 7.2.                                                                                                                                                                                                      |
| <b>TRAP</b>      | Answering (i) with 100, or with 400. And getting the two questions the wrong way round: "seen it" is <b>all</b> the vertices $n$ , "sent it out" is the <b>internal</b> vertices $i$ .                                  |

**GIVEN**  $m = 4$  recipients per sender;  $l = 100$  people who read it but did not send it on; nobody receives it twice.

**FIND**  $n$ , the number who have seen it, and  $i$ , the number who sent it out.

**FORMULA** For a full  $m$ -ary tree,  $l = (m - 1)i + 1$  and  $n = mi + 1$ .

**Step 1. Model it, and say so.** Draw each person as a vertex and each sending as an edge from sender to recipient. Because **no one receives more than one letter**, every person other than the first has exactly one parent, so the picture is a **rooted tree** with the first person at the root. Because everyone who sends, sends to **exactly four**, every internal vertex has exactly four children, so it is a **full 4-ary tree**.

senders = internal vertices =  $i$

people who read but did not send = leaves =  $l = 100$

people who have seen it = all vertices =  $n$

**Step 2. Find the internal vertices from the leaves.**

$$l = (m - 1)i + 1$$

$$100 = (4 - 1)i + 1$$

$$100 = 3i + 1$$

$$99 = 3i$$

$$i = 33$$

**Step 3. Find the total.**

$$n = mi + 1 = 4 \times 33 + 1 = 132 + 1 = 133$$

**Answers.**

**(i) 133 people have seen the letter, including the first person.**

**(ii) 33 people sent it out.**

**Check, and write the check down.** The vertices split into senders and non-senders with nothing left over:

$$i + l = 33 + 100 = 133 = n \quad \checkmark$$

And count the edges: 33 senders each sending 4 gives 132 letters sent, which is exactly the 132 people other than the first. That is  $e = v - 1$  from section 7.1, so the tree arithmetic closes twice over.

**For the 2020/2021 version**, which asks both questions in one sentence and pays 8 marks, give the same working and end with both numbers named: 133 have seen it, 33 sent it out.

**Now redo it:** same problem with each person sending to three others. Then  $100 = 2i + 1$  gives  $i = 49.5$ , which is not a whole number, so **no such chain exists**: with  $m = 3$  the leaf count is always odd. Saying that is the answer, and it is the same style of impossibility argument as section 6.3.

## The quinary tree

AS PRINTED

2024/2025 · Q6(B)

- b. Consider the expression:  $(3 - 4 \cdot 2)[\text{UNRESOLVED}] - (5 + (5 - 2)[\text{UNRESOLVED}])$
- i. Draw the rooted tree representing this computation **2 marks**
  - iv. Write this expression in prefix notation **2 marks**
  - v. Write this expression in postfix notation. **2 marks**
  - vi. A quinary tree has the height of 3. Find the total number of leaves present on it. **2.5 marks**
  - vii. Prove that a quinary tree is an m-ary tree of height 3 has at most  $m^h$  leaves. **3 marks**

### NOTE ON THE PAPER, AND ON THE TWO MARKS IN GREY

**THE NUMBERING SKIPS.** The parts run i, then iv, v, vi, vii. There is no ii and no iii. The marks still total 17.5 for the question, so nothing is missing from the paper; only the labels are wrong.

**[UNRESOLVED] IS NOT PRINTED ON THE PAPER.** It is this manual's marker, shown in the quote as plain text, for a character that could not be read. The expression carries a single superscript after each of the two closing brackets, roughly two pixels tall in the source photograph. Rendered at 900 dpi they are still unreadable, and the PDF's own text layer dropped them too, returning "(3-4.2)- (5+ (5-2))". **THEY HAVE NOT BEEN GUESSED.** Section 7.3 solves the plain reading and says what changes if they turn out to be exponents.

**PART VII IS NOT GRAMMATICAL:** "Prove that a quinary tree is an m-ary tree of height 3 has at most  $m^h$  leaves" has two verbs where one belongs. The claim being asked for is the general one: an m-ary tree of height h has at most  $m^h$  leaves. Note also that the paper writes the exponent as a small raised h while the height in the sentence is fixed at 3, which is part of the same confusion. Prove the general statement and then apply it at  $m = 5$ ,  $h = 3$ ; that answers both readings.

Parts i, iv and v are answered in section 7.3, where expression trees are taught. Parts vi and vii are answered below.

### BREAK IT DOWN

- GIVEN** A quinary tree, that is  $m = 5$ , of height  $h = 3$ .
- ASKED** (vi) **Find** the total number of leaves. (vii) **Prove** the bound. One computation and one proof.
- MARKS SAY** 2.5 and 3. Part vii is a proof, so induction or a level-by-level argument is expected, not an example.
- TEACHES** This section, 7.2.
- TRAP** Part vi says "the total number of leaves" as though it were determined, but a quinary tree of height 3 could have as few as 3 leaves. What is determined is the **maximum**, and that is what part vii is about. Say so, then give  $5^3 = 125$ .



**GIVEN**  $m = 5$  (quinary),  $h = 3$ .

**FIND** The number of leaves, and a proof of the general bound.

**FORMULA** An  $m$ -ary tree of height  $h$  has at most  $m^h$  leaves.

vi. The number of leaves

**Say what is determined.** A quinary tree of height 3 need not have any particular number of leaves; the quantity fixed by  $m$  and  $h$  is the **maximum**, reached when the tree is full and every leaf sits at level 3. So:

$$\text{leaves} = m^h = 5^3 = 5 \times 5 \times 5 = \mathbf{125}$$

**Answer: at most 125 leaves, and exactly 125 for the full quinary tree of height 3.**

**Check it against the other formula**, which is worth a line. The full quinary tree of height 3 has internal vertices at levels 0, 1 and 2, that is

$$i = 1 + 5 + 25 = 31$$

and then  $l = (m - 1)i + 1 = 4 \times 31 + 1 = \mathbf{125}$ . The two formulas agree, as they must, and  $n = mi + 1 = 5 \times 31 + 1 = 156$  total vertices.

vii. Prove the bound

**To prove:** an  $m$ -ary tree of height  $h$  has at most  $m^h$  leaves.

**The method, in two sentences, before it is used.** This is a proof by **mathematical induction**: show the claim holds at the smallest value of  $h$ , then show that whenever it holds at one height it must also hold at the next, and those two facts together carry it to every height. Section 9.2 sets the method out in full, with the reasons it works and several more examples; everything needed for this proof is in the sentence you have just read.

**Proof, by induction on the height  $h$ .**

**Base case,  $h = 0$ .** A tree of height 0 is a single vertex, which is the root and also a leaf. It has 1 leaf, and  $m^0 = 1$ . So the claim holds.

**Inductive step.** Suppose the claim holds for every  $m$ -ary tree of height  $h - 1$ , for some  $h \geq 1$ . Let  $T$  be an  $m$ -ary tree of height  $h$ . Remove the root. What remains is at most  $m$  subtrees, one for each child of the root, since  $T$  is  $m$ -ary. Each subtree has height at most  $h - 1$ , so by the inductive hypothesis each has at most  $m^{h-1}$  leaves.

Every leaf of  $T$  lies in one of those subtrees, so

$$\text{leaves of } T \leq m \times m^{h-1} = m^h$$

By induction the claim holds for every  $h \geq 0$ .  $\square$

**When is the bound attained?** Exactly when the root has all  $m$  children, every subtree attains its own bound, and no leaf sits above level  $h$ ; that is, when the tree is **full** and all leaves are at level  $h$ . Adding that sentence shows the bound is sharp rather than merely true.

**Applying it at  $m = 5$ ,  $h = 3$**  gives  $5^3 = 125$ , which is part vi.

**Now redo it:** what is the greatest number of leaves a binary tree of height 4 can have?  $m^h = 2^4 = 16$ .

## The playoff tree

AS PRINTED

2020/2021 · Q4(A)

4. (a) Playoff between two teams consists of at most five games. The first team that wins three games wins the playoff.

(i) Construct a tree diagram for five games.

(4marks)

(ii) In how many different ways can the playoff occur?

(3½marks)

### NOTE ON THE PAPER

"Playoff between two teams" is missing its article; it should read "A playoff". More importantly, part (i) says "Construct a tree diagram for five games", but the playoff can finish in three or four games as well: the tree must show **EVERY WAY THE SERIES CAN END**, and its branches are of different lengths. Drawing exactly five levels everywhere would be wrong, because once a team has three wins no further game is played.

### BREAK IT DOWN

**GIVEN** Two teams. At most five games. The first to three wins takes the playoff.

**ASKED** (i) **Construct** the tree diagram. (ii) **In how many different ways** the playoff can occur.

**MODEL** A **binary** tree. Each vertex is a state of the series; the two children are "team A wins the next game" and "team B wins it". A branch stops as soon as either team reaches three wins, so the leaves are exactly the possible outcomes.

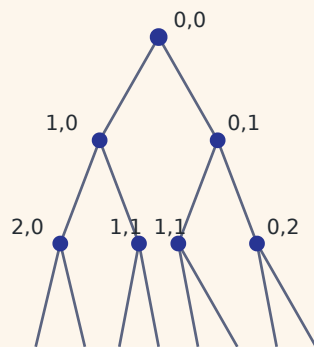
**MARKS SAY** 4 for the diagram, 3.5 for the count. The count is the number of **leaves**, so drawing the tree carefully answers both parts at once.

**TEACHES** This section, 7.2, with the combinations of 5.2.

**TRAP** Drawing all  $2^5 = 32$  sequences of five games. Most of them are impossible, because the series stops the moment a team reaches three wins. The answer is 20, not 32.

- GIVEN** Two teams, first to 3 wins, at most 5 games.
- FIND** The tree of possible series, and the number of ways the playoff can occur.
- RULE** Each game branches two ways; a branch stops at 3 wins. The outcomes are the leaves.

**Part (i): the tree.** Label each vertex with the score so far as (wins for A, wins for B). The root is (0, 0). From any vertex, the left child adds a win for A and the right child a win for B. Stop as soon as either count reaches 3.



| Series ending in | A wins | B wins |
|------------------|--------|--------|
| 3 games          | 1      | 1      |
| 4 games          | 3      | 3      |
| 5 games          | 6      | 6      |
| total            | 10     | 10     |

20 leaves in all

the first three levels: each game branches two ways  
 a branch stops the moment a team reaches 3 wins  
 so branches have different lengths

A winning in 4 games: the last game must be A's third win, so choose which 2 of the first 3 A also won:  $C(3, 2) = 3$

*The top of the tree, and the leaf count worked out by length. Every leaf is one possible series, so counting leaves answers part (ii).*

**Part (ii): how many ways.** Count the leaves by how long the series runs. Take team A winning first.

- ▶ **A wins in 3 games.** Only AAA, so 1 way.
- ▶ **A wins in 4 games.** The fourth game must be A's third win, so A won exactly 2 of the first 3. Choosing which 2:  $C(3, 2) = 3$ .
- ▶ **A wins in 5 games.** The fifth game is A's third win, so A won exactly 2 of the first 4:  $C(4, 2) = 6$ .

A wins in  $1 + 3 + 6 = 10$  ways.

By symmetry B also wins in 10 ways, and the two cases cannot both happen, so the sum rule of section 5.1 gives

$$\text{total} = 10 + 10 = 20$$

**Answer: the playoff can occur in 20 different ways.**

**Why the last game must be the decider,** which is the step most answers skip: if A had already won three games earlier, the series would have stopped there and the later games would never be played. So in any series of exactly  $k$  games, the  $k$ -th is the winner's third win.

**Now redo it:** best of three, first to two wins. A wins in 2 games: 1 way; in 3 games:  $C(2, 1) = 2$  ways. So A wins in 3 ways, B in 3, total 6.

#### RECALL AND CHECK

1. What is the difference between an m-ary and a full m-ary tree?
2. When is an m-ary tree balanced?
3. State the two counting formulas for a full m-ary tree.
4. A full 4-ary tree has 100 leaves. How many vertices in total, and how many are internal?
5. What is the greatest number of leaves a quinary tree of height 3 can have?
6. In the playoff question, why is the answer not  $2^5 = 32$ ?

#### ANSWERS

1. m-ary means every internal vertex has at most m children; full m-ary means exactly m.
2. When every leaf is at level h or h - 1, where h is the height.
3.  $n = mi + 1$  and  $l = (m - 1)i + 1$ .
4.  $i = (100 - 1)/3 = 33$  internal, and  $n = 4 \times 33 + 1 = 133$  in total.
5.  $5^3 = 125$ .
6. The series stops as soon as a team reaches three wins, so most five-game sequences never occur. The answer is 20.

#### FILL IN THE GAP

1. In a full m-ary tree with i internal vertices,  $n = \underline{\hspace{2cm}}$  and  $l = \underline{\hspace{2cm}}$ .
2. An m-ary tree of height h has at most  $\underline{\hspace{2cm}}$  leaves.
3. The chain letter has  $\underline{\hspace{2cm}}$  senders and  $\underline{\hspace{2cm}}$  people who have seen it.
4. A tree is balanced when every leaf is at level  $\underline{\hspace{2cm}}$  or  $\underline{\hspace{2cm}}$ .

#### ANSWERS

1.  $mi + 1$ ,  $(m - 1)i + 1$ . 2.  $m^h$ . 3. 33, 133. 4. h, h - 1.

#### PART SEVEN • 7.3

## Traversals and Expression Trees

Three ways to walk a tree, and each one writes an arithmetic expression in a different notation. The 2024/2025 paper gives 6 marks to the walks and 6 more to the notations.

By the end of this section you will be able to traverse an ordered rooted tree in preorder, inorder and postorder, build the expression tree of an arithmetic expression, read prefix and postfix notation off that tree, and answer both traversal questions on the 2024/2025 paper in full.

## The three traversals

### ● MUST MEMORISE

### PREORDER, INORDER, POSTORDER

Let  $T$  be an ordered rooted tree with root  $r$  and subtrees  $T_1, T_2, \dots, T_k$  hanging from  $r$ , taken left to right. Each traversal is defined by where the root goes.

**Preorder:** visit  **$r$  first**, then traverse  $T_1, \dots, T_k$  in preorder.

**Inorder:** traverse  $T_1$  in inorder, then visit  **$r$** , then traverse  $T_2, \dots, T_k$  in inorder.

**Postorder:** traverse  $T_1, \dots, T_k$  in postorder, then visit  **$r$  last**.

#### WHERE

**ordered** means the children of every vertex have a fixed left-to-right order. Without it none of the three is defined, which is why section 7.1 named ordered rooted trees separately.  $T_1$  is the **leftmost** subtree, and inorder treats it differently from all the others. **a single vertex** is its own traversal in all three orders: the recursion stops there.

**The names say it.** "Pre", "in" and "post" describe where the **root** is visited relative to its subtrees: before them, in among them, or after them. Nothing else differs.

◆ **Memory hook: the prefix tells you when the root is visited.** Pre-order: root first. Post-order: root last. In-order: root in the middle, after the leftmost subtree only.

### TEACH

### HOW TO DO IT WITHOUT LOSING YOUR PLACE

The reliable method by hand is the **tracing** method. Draw a curve that starts at the root, hugs the outside of the whole tree anticlockwise, and returns to the root. Every vertex is passed **three** times: on its left, underneath it, and on its right.

- ▶ **Preorder:** list each vertex the **first** time the curve passes it, on the left.
- ▶ **Inorder:** list each vertex the **second** time, from below.
- ▶ **Postorder:** list each vertex the **third** time, on the right.

One curve gives all three lists, which matters when a question asks for all three of the same tree, as 2024/2025 Q6(a) does.

**The check that catches nearly every slip:** each list must contain **every vertex exactly once**, so its length equals the number of vertices. In preorder the **root comes first**; in postorder the **root comes last**. If either of those fails, stop and redo it.

## The traversal question

### AS PRINTED

### 2024/2025 · Q6(A)

- List the vertices in the order that you would visit them if you traversed the tree in preorder. **2 marks**
- List the vertices in the order that you would visit them if you traversed the tree in inorder. **2 marks**
- List the vertices in the order that you would visit them if you traversed the tree in postorder. **2 marks**

#### NOTE ON THE PAPER, AND ON THE FIGURE

The part opens with the line "a. Consider the following tree" and then prints the figure, with the three numbered parts below it. The quote above starts at part i for that reason: splicing the lead line onto part i would join text the paper separates with a whole diagram.

The tree is a drawing, so its structure is read off the picture. It has 17 vertices, and it is **NOT** binary: the root has 2 children but two other vertices have 3 each, which matters because the in-order rule treats the leftmost subtree differently from the rest. The structure read from the scan is:

a is the root with children b and c; b has children d, e, f; e has children j and k; k has children n and o; c has children g, h, i; h has children l and m; m has children p and q. The leaves are d, f, j, n, o, g, l, p, q and i.

One label is worth flagging: the vertex between d and f is blurred and could be misread as "c", but c is already the root's other child and the run of labels reads d, e, f, g, so it is **E**. Everything else on the figure is unambiguous.

#### BREAK IT DOWN

**GIVEN** An ordered rooted tree of 17 vertices, drawn on the paper.

**ASKED** **List the vertices** in each of the three orders. Three lists, nothing else. No explanation is asked for and none is needed.

**MARKS SAY** 2 each. That is one mark for the method and one for getting all seventeen right, so accuracy is the whole game. Draw the tracing curve once and read all three off it.

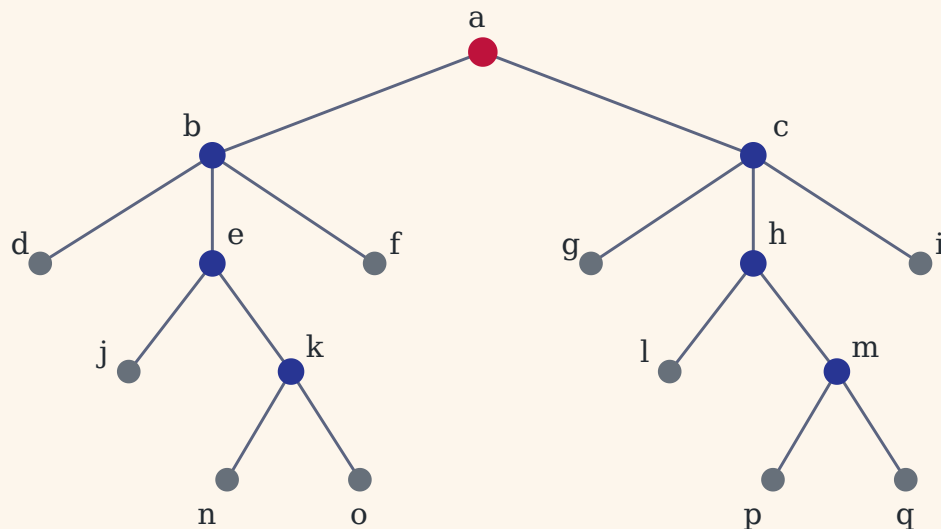
**TEACHES** This section, 7.3.

**TRAP** Applying the **binary** inorder rule, "left subtree, root, right subtree", to vertices that have three children. The general rule is: leftmost subtree, then the root, then **all** the remaining subtrees in order. For b, with children d, e, f, that gives d, then b, then e's subtree, then f.

**GIVEN** The 17-vertex ordered rooted tree drawn below, read off the paper's figure.

**FIND** The preorder, inorder and postorder lists.

**RULE** Preorder visits the root first, inorder after the leftmost subtree, postorder last.



17 vertices, 16 edges. Crimson is the root; indigo the internal vertices; grey the ten leaves.

*The tree as read from the paper. Note that  $b$  and  $c$  each have three children, so this is not a binary tree.*

#### i. Preorder

Root first, then each subtree in turn. Take  $a$ , then everything under  $b$ , then everything under  $c$ .

under  $b$ :  $b$ , then  $d$ , then  $e$ 's subtree, then  $f$

$e$ 's subtree:  $e$ , then  $j$ , then  $k$ 's subtree, which is  $k, n, o$

so under  $b$ :  $b, d, e, j, k, n, o, f$

under  $c$ :  $c, g$ , then  $h$ 's subtree, then  $i$

$h$ 's subtree:  $h, l$ , then  $m$ 's subtree, which is  $m, p, q$

so under  $c$ :  $c, g, h, l, m, p, q, i$

**$a, b, d, e, j, k, n, o, f, c, g, h, l, m, p, q, i$**

#### ii. Inorder

Leftmost subtree, then the root, then the rest. At  $a$ : everything under  $b$ , then  $a$ , then everything under  $c$ .

at  $b$ , children  $d, e, f$ :  $\text{inorder}(d), b, \text{inorder}(e), \text{inorder}(f)$

at  $e$ , children  $j, k$ :  $j, e, \text{inorder}(k)$ ; at  $k$ , children  $n, o$ :  $n, k, o$

so  $\text{inorder}(e) = j, e, n, k, o$ , and under  $b$ :  $d, b, j, e, n, k, o, f$

at  $c$ , children  $g, h, i$ :  $g, c, \text{inorder}(h), i$

at  $h$ , children  $l, m$ :  $l, h, \text{inorder}(m)$ ; at  $m$ :  $p, m, q$

so  $\text{inorder}(h) = l, h, p, m, q$ , and under  $c$ :  $g, c, l, h, p, m, q, i$

**$d, b, j, e, n, k, o, f, a, g, c, l, h, p, m, q, i$**

#### iii. Postorder

All the subtrees first, root last.

under b: d, then postorder(e), then f, then b  
 postorder(e) = j, then postorder(k) = n, o, k, then e, giving j, n, o, k, e  
 so under b: d, j, n, o, k, e, f, b  
 under c: g, then postorder(h), then i, then c  
 postorder(h) = l, then p, q, m, then h, giving l, p, q, m, h  
 so under c: g, l, p, q, m, h, i, c

**d, j, n, o, k, e, f, b, g, l, p, q, m, h, i, c, a**

**Check all three.** Each list has 17 entries and each contains every vertex exactly once. Preorder starts with a; postorder ends with a; inorder has a in the middle, right after the whole of b's subtree. All four conditions hold.

**Now redo it:** traverse just the subtree at k, whose children are n and o. Preorder k, n, o. Inorder n, k, o. Postorder n, o, k.

## Expression trees

### ● MUST MEMORISE

### EXPRESSION TREE, PREFIX AND POSTFIX

An **expression tree** represents an arithmetic expression: each **leaf** is a number, and each **internal vertex** is an operator whose children are the things it acts on, in order.

Traversing that tree gives the three standard notations:

| TRAVERSAL | NOTATION                | ALSO CALLED      | OPERATOR SITS        |
|-----------|-------------------------|------------------|----------------------|
| preorder  | <b>prefix notation</b>  | Polish notation  | before its operands  |
| inorder   | infix notation          | ordinary algebra | between its operands |
| postorder | <b>postfix notation</b> | reverse Polish   | after its operands   |

#### WHERE

**the tree** records the grouping, so the brackets of the original expression are built into its shape. **prefix** and **postfix** need **no brackets at all** and are never ambiguous, which is exactly why compilers and calculators use them. **infix** is the notation people write, and it is the only one of the three that needs brackets and precedence rules.

**Building the tree:** find the operator that is applied **last**, which is the one *least* tightly bound. That becomes the root. Its two sides become its subtrees, and the same rule is applied to each of them.

◆ **Memory hook: the root is the last operation performed.** Work out what you would compute last, put it at the top, and recurse.



## The expression question

### THE QUESTION

2024/2025 Q6(b) is quoted in full in section 7.2, where its parts vi and vii are answered. Its parts i, iv and v are answered here: draw the rooted tree of the expression, then write it in prefix and in postfix notation, 2 marks each.

**THE EXPRESSION CARRIES TWO CHARACTERS THAT COULD NOT BE READ.** As recorded in 7.2, a single superscript follows each of the two closing brackets and neither is legible at any resolution available, including in the PDF's own text layer. They are not guessed. The answer below solves the **PLAIN READING**, in which those characters are absent, and then says exactly what changes if they turn out to be exponents.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | The expression $(3 - 4 \cdot 2) - (5 + (5 - 2))$ on the plain reading, where $\cdot$ is multiplication.                                                                                                                                                                         |
| <b>ASKED</b>     | (i) <b>Draw</b> the rooted tree. (iv) <b>Write</b> it in prefix notation. (v) <b>Write</b> it in postfix notation.                                                                                                                                                              |
| <b>MARKS SAY</b> | 2 each. Parts iv and v are read straight off the tree from part i, so the tree is worth getting right first: one drawing earns all 6 marks.                                                                                                                                     |
| <b>TEACHES</b>   | This section, 7.3.                                                                                                                                                                                                                                                              |
| <b>TRAP</b>      | Putting the wrong operator at the root. Inside the first bracket, multiplication binds tighter than subtraction, so the <b>subtraction</b> is done last and is the root of that subtree, not the multiplication. Getting the root wrong makes all three answers wrong together. |

**GIVEN**  $(3 - 4 \cdot 2) - (5 + (5 - 2))$ , the plain reading of the printed expression.

**FIND** The expression tree, then the prefix and postfix forms.

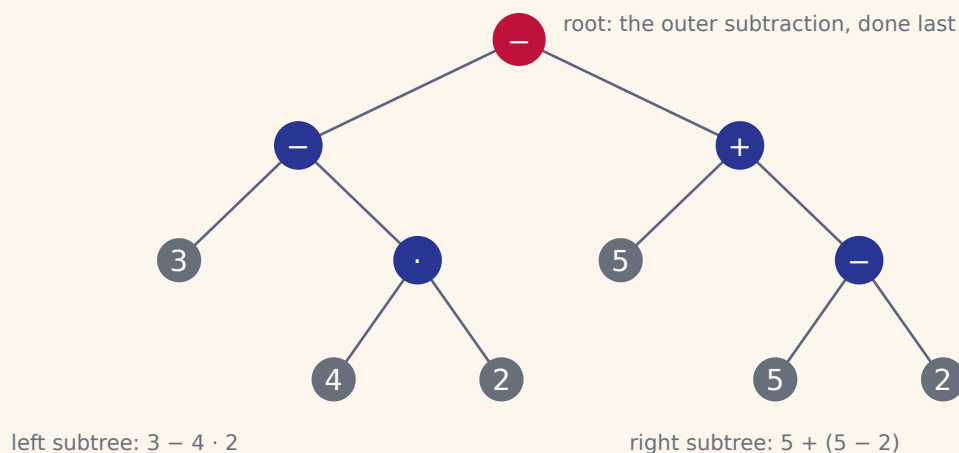
**RULE** The last operation performed is the root. Preorder gives prefix, postorder gives postfix.

i. The rooted tree

**Find the root.** The whole expression is (something)  $-$  (something), and that outer subtraction is what you would do last. So the root is  $-$ .

**Left subtree,  $3 - 4 \cdot 2$ .** Multiplication binds tighter than subtraction, so  $4 \cdot 2$  is computed first and the subtraction last. Its root is  $-$ , with children 3 and the subtree  $4 \cdot 2$ .

**Right subtree,  $5 + (5 - 2)$ .** The bracket is computed first, so the addition is last. Its root is  $+$ , with children 5 and the subtree  $5 - 2$ .



*The expression tree. Operators at the internal vertices, numbers at the leaves, and no brackets anywhere: the shape holds the grouping.*

iv. Prefix notation

Read the tree in **preorder**: root, then left subtree, then right.

root  $-$ , then the left subtree  $(- 3 \cdot 4 2)$ , then the right subtree  $(+ 5 - 5 2)$

**$- - 3 \cdot 4 2 + 5 - 5 2$**

v. Postfix notation

Read the tree in **postorder**: left subtree, right subtree, then root.

left subtree  $(3 4 2 \cdot -)$ , then the right subtree  $(5 5 2 - +)$ , then the root  $-$

**$3 4 2 \cdot - 5 5 2 - + -$**

**Check by evaluating the postfix form**, which costs four lines and proves the tree is right. Read left to right, pushing numbers and applying each operator to the two most recent values:

| READ    | STACK AFTERWARDS |
|---------|------------------|
| 3, 4, 2 | 3, 4, 2          |
| .       | 3, 8 (4 × 2)     |
| −       | −5 (3 − 8)       |
| 5, 5, 2 | −5, 5, 5, 2      |
| −       | −5, 5, 3 (5 − 2) |
| +       | −5, 8 (5 + 3)    |
| −       | −13 (−5 − 8)     |

Directly:  $(3 - 8) - (5 + 3) = -5 - 8 = -13$ . The two agree, so the tree, the postfix form and the arithmetic are all consistent.

**If the two unreadable characters are exponents.** Say this in one line on the script rather than leaving it unaddressed. Each would become an extra internal vertex, an exponentiation operator, sitting above the bracket it follows: the left subtree would become  $(3 - 4 \cdot 2)$  raised to that power, and the inner  $(5 - 2)$  likewise. The tree gains two vertices, the root and the overall shape are unchanged, and both notations gain the two operators in the corresponding places. Nothing about the method changes.

**Now redo it:** give the prefix and postfix forms of  $(2 + 3) \cdot 4$ . The root is  $\cdot$ , so prefix is  $\cdot + 2 3 4$  and postfix is  $2 3 + 4 \cdot$ .

#### ⚠ TRAP

#### PREFIX IS NOT "THE EXPRESSION BACKWARDS"

Postfix is not the reverse of prefix, and neither is produced by moving the operators about by eye. Both are read off the **tree**, and the tree is where the grouping lives. Compare:

| INFIX             | PREFIX          | POSTFIX         |
|-------------------|-----------------|-----------------|
| $3 - 4 \cdot 2$   | $- 3 \cdot 4 2$ | $3 4 2 \cdot -$ |
| $(3 - 4) \cdot 2$ | $\cdot - 3 4 2$ | $3 4 - 2 \cdot$ |

Those two infix expressions differ only by a pair of brackets and they have different values,  $-5$  and  $-2$ . Their prefix and postfix forms use the same five symbols in different orders, with **no brackets at all**. That is the property that makes these notations worth having, and it only works if the tree was built correctly.

## RECALL AND CHECK

1. Define preorder, inorder and postorder in one line each.
2. Which traversal gives prefix notation, and which gives postfix?
3. In the 17-vertex tree, which vertex comes first in preorder and which comes last in postorder?
4. How do you find the root of an expression tree?
5. Why do prefix and postfix need no brackets?
6. Give the postfix form of  $(2 + 3) \cdot 4$ .

---

### ANSWERS

1. Preorder visits the root before its subtrees; postorder after them; inorder after the leftmost subtree and before the rest.
2. Preorder gives prefix; postorder gives postfix.
3. a in both cases: the root is first in preorder and last in postorder.
4. It is the operator applied last, the most loosely bound one.
5. The order of the symbols already records the grouping, so no punctuation is needed to disambiguate it.
6.  $2\ 3\ +\ 4\ \cdot$

## FILL IN THE GAP

1. Preorder visits the root \_\_\_\_\_; postorder visits it \_\_\_\_\_.
2. Prefix notation is also called \_\_\_\_\_ notation.
3. The root of an expression tree is the operator applied \_\_\_\_\_.
4. The postfix form of  $(3 - 4 \cdot 2) - (5 + (5 - 2))$  is \_\_\_\_\_.

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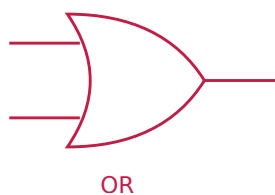
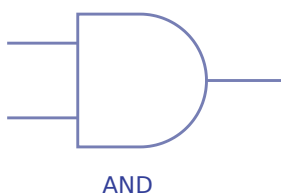
### ANSWERS

1. first, last. 2. Polish. 3. last. 4.  $3\ 4\ 2\ \cdot\ -\ 5\ 5\ 2\ -\ +\ -$

## PART EIGHT

# Boolean Algebra and Logic Circuits

*The same laws as Part One and Part Two, written a third way. Once you see that AND, intersection and conjunction are one operation under three names, this part is a translation exercise.*



$$A \cdot B$$

$$F + S$$

$$(A \cdot B)'$$

## Boolean Algebra and Its Laws

Two values, three operations, and a set of laws you have already met twice. The newest paper spends 7.5 marks on turning two sentences about a door and an alarm into Boolean expressions.

By the end of this section you will be able to write Boolean expressions for AND, OR and NOT, build a Boolean table, name the laws of Boolean algebra and their duals, translate between Boolean algebra, logic and sets, draw the corresponding gates, and answer the door and alarm questions in full.

### NOT IN THE COURSE TEXT

The official text has no Boolean algebra in it: no Boolean expression, no logic gate, no truth table written in ones and zeros. The 2025/2026 paper gives it 7.5 marks. Everything here is taught from zero, but almost none of it is new: it is section 1.3 in different clothing, and the translation table below is the whole trick.

## The two values and three operations

### ● MUST MEMORISE

### BOOLEAN ALGEBRA

A **Boolean variable** takes one of exactly two values, written **1** and **0**, read as **true** and **false**, or as **on** and **off**. The three operations are

| OPERATION       | WRITTEN               | READ           | VALUE IS 1 EXACTLY WHEN   |
|-----------------|-----------------------|----------------|---------------------------|
| Boolean product | $A \cdot B$ , or $AB$ | A <b>AND</b> B | both A and B are 1        |
| Boolean sum     | $A + B$               | A <b>OR</b> B  | at least one of A, B is 1 |
| Complement      | $A'$ , or $\bar{A}$   | <b>NOT</b> A   | A is 0                    |

#### WHERE

· is **not** ordinary multiplication and + is **not** ordinary addition, although · behaves exactly like multiplication on 0 and 1. The one place they part company is  $1 + 1 = 1$ , not 2: there is no value above 1 for it to be.  $A'$  is also written with an overbar. Both appear on these papers.

**The complete arithmetic**, which is worth writing out once:

$$\begin{aligned} 0 \cdot 0 &= 0, & 0 \cdot 1 &= 0, & 1 \cdot 0 &= 0, & 1 \cdot 1 &= 1 \\ 0 + 0 &= 0, & 0 + 1 &= 1, & 1 + 0 &= 1, & 1 + 1 &= 1 \\ 0' &= 1, & 1' &= 0 \end{aligned}$$

◆ **Memory hook: product needs both, sum needs one.** And the only surprise in the whole table is  $1 + 1 = 1$ , which is just "true or true is true".

Every law in this part has already been proved twice in this book, once for propositions in section 1.3 and once for sets in section 2.2. They are the same laws.

| BOOLEAN ALGEBRA | LOGIC (PART ONE)   | SETS (PART TWO)             |
|-----------------|--------------------|-----------------------------|
| 1               | T, a tautology     | U, the universal set        |
| 0               | F, a contradiction | $\emptyset$ , the empty set |
| $A \cdot B$     | $p \wedge q$       | $A \cap B$                  |
| $A + B$         | $p \vee q$         | $A \cup B$                  |
| $A'$            | $\neg p$           | $A'$ , the complement       |
| $A = B$         | $p \equiv q$       | $A = B$                     |

◆ **Memory hook: AND, intersection and conjunction are one operation with three names.** If you can do section 1.3, you can do this part: replace  $\wedge$  by  $\cdot$ ,  $\vee$  by  $+$ , and  $\neg$  by a dash.

For all Boolean variables A, B, C:

| NAME              | IN TERMS OF $\cdot$                         | IN TERMS OF $+$                         |
|-------------------|---------------------------------------------|-----------------------------------------|
| Identity          | $A \cdot 1 = A$                             | $A + 0 = A$                             |
| Domination        | $A \cdot 0 = 0$                             | $A + 1 = 1$                             |
| Idempotent        | $A \cdot A = A$                             | $A + A = A$                             |
| Complement        | $A \cdot A' = 0$                            | $A + A' = 1$                            |
| Double complement | $(A')' = A$                                 |                                         |
| Commutative       | $A \cdot B = B \cdot A$                     | $A + B = B + A$                         |
| Associative       | $(A \cdot B) \cdot C = A \cdot (B \cdot C)$ | $(A + B) + C = A + (B + C)$             |
| Distributive      | $A \cdot (B + C) = A \cdot B + A \cdot C$   | $A + B \cdot C = (A + B) \cdot (A + C)$ |
| De Morgan         | $(A \cdot B)' = A' + B'$                    | $(A + B)' = A' \cdot B'$                |
| Absorption        | $A \cdot (A + B) = A$                       | $A + A \cdot B = A$                     |

**The duality principle.** Every law above comes in a pair, and each half is obtained from the other by swapping  $\cdot$  with  $+$  and **0** with **1**. That is not a coincidence: it is a theorem, and it means you only ever have to memorise half the table.

**Two that are not in Part One's list** and are worth noting. **Distributing  $+$  over  $\cdot$** , that is  $A + B \cdot C = (A + B) \cdot (A + C)$ , has no counterpart in ordinary arithmetic, where  $a + bc$  is certainly not  $(a + b)(a + c)$ . Boolean algebra is not number algebra, and this is where the difference bites hardest. **Absorption** is the law that does most of the work in simplifying circuits.

◆ **Memory hook: swap the dots for plusses and the zeros for ones, and the law still holds.** Learn one column and derive the other.

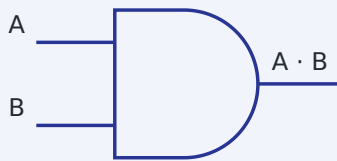


## Gates

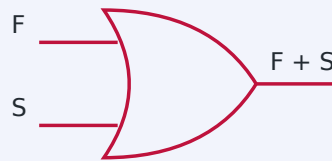
### TEACH

### THE THREE BASIC GATES

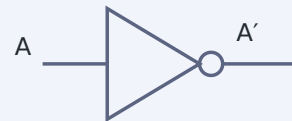
A **logic gate** is a physical component computing one Boolean operation. Circuits are drawn from them, and a Boolean expression and a circuit diagram are two views of the same thing.



AND gate  
output 1 only if both inputs are 1



OR gate  
output 1 if either input is 1



NOT gate  
output is the opposite of t

*The three gates the papers use. The small circle on the NOT gate is the standard mark for inversion.*

Two more names appear often enough to be worth knowing: a **NAND** gate computes  $(A \cdot B)'$  and a **NOR** gate computes  $(A + B)'$ , that is, an AND or an OR followed by an inversion.

## The door and the alarm

### AS PRINTED

2025/2026 · Q5(A)

- (a) i. A secure laboratory door opens only when a valid ID card (A) and a correct finger-print (B) are presented. Express this condition using Boolean Algebra and determine when the door opens. Draw the Boolean table. **(4½ marks)**
- ii. An alarm should sound if either a fire sensor (F) or a smoke sensor (S) is activated. Formulate the Boolean expression and determine the output when only the smoke sensor detects danger. **(3 marks)**

### NOTE ON THE PAPER

Nothing is printed wrongly here, but two words are doing precise work and are easy to read past.

Part i says the door opens "**ONLY WHEN**" both are presented. Section F.2 covered that phrase: "only when" states a **NECESSARY** condition. Read strictly, it says the door does not open without both, which does not by itself promise that presenting both *does* open it. The intended reading, and the one to answer, is that both together are necessary and sufficient, so  $\text{Door} = A \cdot B$ . Say which reading you are using in one line; it costs nothing and shows you noticed.

Part ii says "**EITHER ... OR**", which in this context means the **INCLUSIVE** or: the alarm should certainly sound if *both* sensors fire. So it is  $A + B$ , not exclusive or. A candidate who reads "either ... or" as exclusive gets an alarm that stays silent in a real fire, which is worth saying in the answer.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | Part i: two conditions A and B, both required. Part ii: two sensors F and S, either sufficient.                                                                                                                                                                           |
| <b>ASKED</b>     | Part i: <b>express</b> the condition, <b>determine</b> when the door opens, <b>draw</b> the Boolean table. Three deliverables for 4.5 marks. Part ii: <b>formulate</b> the expression and <b>determine the output</b> for one specific input.                             |
| <b>MARKS SAY</b> | 4.5 and 3. In part i the table is explicitly asked for and is worth roughly 2 of the 4.5, so an answer without it cannot pass. In part ii "determine the output when only the smoke sensor detects danger" is a single substitution: name $F = 0$ , $S = 1$ and evaluate. |
| <b>TEACHES</b>   | This section, 8.1.                                                                                                                                                                                                                                                        |
| <b>TRAP</b>      | Answering part ii with the expression only. It asks for a specific output as well, and that is where the marks split.                                                                                                                                                     |

**GIVEN** Part i: A = valid ID card presented, B = correct fingerprint presented, each 1 if so and 0 if not. Part ii: F = fire sensor activated, S = smoke sensor activated.

**FIND** The Boolean expression for each, the Boolean table for the door, and the alarm output when only S is 1.

**RULE** "and" is the Boolean product; "either or" is the inclusive Boolean sum.

i. The laboratory door

**Assign the variables.** Let A = 1 when a valid ID card is presented and 0 otherwise; B = 1 when a correct fingerprint is presented and 0 otherwise. Let D = 1 when the door opens.

**The expression.** The door opens only when **both** are presented, so the two conditions are combined with AND:

$$D = A \cdot B$$

**Determine when the door opens.** By the arithmetic of the Boolean product,  $D = 1$  exactly when  $A = 1$  and  $B = 1$ . In words: **the door opens only in the single case where a valid ID card and a correct fingerprint are both presented, and stays shut in the other three cases.**

**The Boolean table.** Two variables give  $2^2 = 4$  rows, by the counting of section 5.1.

| A | B | $D = A \cdot B$ | DOOR  |
|---|---|-----------------|-------|
| 0 | 0 | 0               | shut  |
| 0 | 1 | 0               | shut  |
| 1 | 0 | 0               | shut  |
| 1 | 1 | 1               | opens |

**The circuit,** if a diagram is wanted: a single AND gate with inputs A and B and output D, as drawn above.

**A line worth adding.** Rows 2 and 3 are the security property doing its job: a correct fingerprint with no card, or a card with the wrong fingerprint, both leave the door shut. That is what "only when" bought.

ii. The alarm

**Assign the variables.** Let F = 1 when the fire sensor is activated, S = 1 when the smoke sensor is activated, and L = 1 when the alarm sounds.

**The expression.** The alarm sounds if either sensor is activated, and certainly if both are, so this is the **inclusive** or, which is the Boolean sum:

$$L = F + S$$

**Determine the output when only the smoke sensor detects danger.** That is  $F = 0$  and  $S = 1$ . Substituting:

$$L = F + S = 0 + 1 = 1$$

**So the alarm sounds.** Which is exactly what it should do: a smoke sensor firing on its own is the ordinary case of a fire being detected early.

**The full table**, which costs three more lines and secures the mark:

| F | S | $L = F + S$ | ALARM                              |
|---|---|-------------|------------------------------------|
| 0 | 0 | 0           | silent                             |
| 0 | 1 | 1           | <b>sounds</b> (the case asked for) |
| 1 | 0 | 1           | sounds                             |
| 1 | 1 | 1           | sounds                             |

**Why it must be inclusive or.** If exclusive or had been used,  $L = F \oplus S$ , then the last row would give  $L = 0$ : the alarm would fall silent precisely when both sensors detect danger. Saying that sentence shows you chose the operation rather than guessed it, and it is the difference between the two marks and the three.

**Now redo it:** the door should also open for a security override switch  $V$ , regardless of card and fingerprint. Then  $D = A \cdot B + V$ , and with  $A = 0$ ,  $B = 0$ ,  $V = 1$  the output is  $0 \cdot 0 + 1 = 0 + 1 = 1$ , so the door opens.

## Simplifying with the laws

### WORKED EXAMPLE

### METHOD · SIMPLIFICATION

**Simplify  $A \cdot B + A \cdot B'$ .** Name every law you use; that is where the marks are in any simplification question.

$$\begin{aligned}
 &A \cdot B + A \cdot B' \\
 &= A \cdot (B + B') \quad [\text{distributive}] \\
 &= A \cdot 1 \quad [\text{complement: } B + B' = 1] \\
 &= A \quad [\text{identity}]
 \end{aligned}$$

**What it means:** a circuit that fires when  $A$  is on and  $B$  is on, or when  $A$  is on and  $B$  is off, is just a circuit that fires when  $A$  is on. The  $B$  input can be removed entirely, which is a real saving in components.

**A second one, using De Morgan.** Simplify  $(A + B)' + A' \cdot B$ .

$$\begin{aligned}
 &(A + B)' + A' \cdot B \\
 &= A' \cdot B' + A' \cdot B \quad [\text{De Morgan}] \\
 &= A' \cdot (B' + B) \quad [\text{distributive}] \\
 &= A' \cdot 1 \quad [\text{complement}] \\
 &= A' \quad [\text{identity}]
 \end{aligned}$$

**Check it on a table**, which is always available and always convincing:  $A = 0$  gives  $(0 + B)' + 1 \cdot B = B' + B = 1$ , and  $A' = 1$ .  $A = 1$  gives  $(1 + B)' + 0 \cdot B = 0 + 0 = 0$ , and  $A' = 0$ . The two agree in both cases.

## RECALL AND CHECK

1. What are the two Boolean values and the three operations?
2. What is  $1 + 1$ , and why is that not a mistake?
3. State De Morgan's laws in Boolean notation.
4. What is the duality principle?
5. Write the expression for the laboratory door and give its table in one line.
6. Why must the alarm use inclusive rather than exclusive or?

### ANSWERS

1. 1 and 0; the Boolean product  $\cdot$  (AND), the Boolean sum  $+$  (OR), and the complement  $'$  (NOT).
2. 1. The sum is 1 when at least one operand is 1, and there is no value above 1.
3.  $(A \cdot B)' = A' + B'$  and  $(A + B)' = A' \cdot B'$ .
4. Swapping  $\cdot$  with  $+$  and 0 with 1 turns any law into another valid law.
5.  $D = A \cdot B$ , which is 0, 0, 0, 1 as A and B run 00, 01, 10, 11.
6. Because with exclusive or the alarm would fall silent when both sensors detect danger.

## FILL IN THE GAP

1. The Boolean product  $A \cdot B$  is 1 only when \_\_\_\_\_.
2.  $1 + 1 =$  \_\_\_\_\_.
3. The door expression is \_\_\_\_\_ and the alarm expression is \_\_\_\_\_.
4. Duality swaps \_\_\_\_\_ with \_\_\_\_\_, and 0 with 1.
5.  $A \cdot B + A \cdot B'$  simplifies to \_\_\_\_\_.

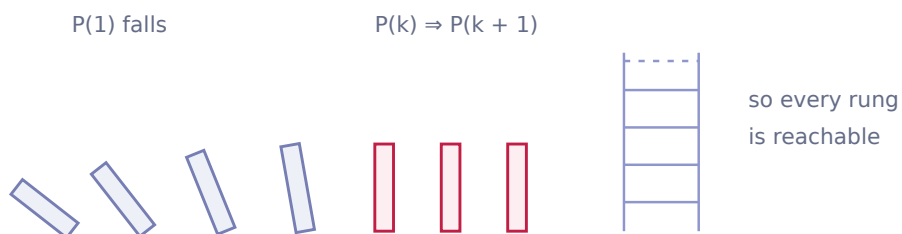
### ANSWERS

1. both are 1. 2. 1. 3.  $D = A \cdot B$ ,  $L = F + S$ . 4.  $\cdot$ ,  $+$ . 5. A.

## PART NINE

# Proof, Induction and Recursion

*Everywhere else in this book you were asked what something is. Here you are asked to show that it must be so. Six strategies cover every proof this course sets, and one of them, induction, is the ladder that climbs forever from a single rung.*



## Proof Techniques

*A proof is not a feeling that something is true. It is a chain of steps from things already agreed to the thing claimed, where every link can be checked by someone who does not trust you. This section gives you the six chains this course uses, and the vocabulary the examiner uses to ask for them.*

*By the end of this section you will be able to say what a theorem, lemma, corollary, proposition, conjecture and axiom each are; write a trivial, vacuous, direct, contrapositive, contradiction or cases proof and choose between them; use the definitions of even, odd, rational and divides fluently; and lay a proof out so a marker can follow it line by line.*

### TEACH

### WHAT IS BEING ASKED FOR

When a question says **prove**, **show that**, **establish** or **demonstrate**, it wants an argument that holds for **every** case covered by the claim, not one example. One example proves nothing except an existence claim.

The mirror image matters just as much. To **disprove** a claim about all cases you need exactly **one** counterexample, and no more. Section 3.2 already used this: a single pair (1, 2) with (2, 1) missing kills symmetry outright.

| THE CLAIM SAYS               | TO PROVE IT                        | TO DISPROVE IT                     |
|------------------------------|------------------------------------|------------------------------------|
| For all $x$ , $P(x)$         | an argument about an arbitrary $x$ | one counterexample                 |
| There exists $x$ with $P(x)$ | one example (a witness)            | an argument about an arbitrary $x$ |

That table is just the quantifier negation of section 1.4 read as advice: the negation of "for all" is "there exists ... not", so disproving a universal claim is proving an existential one, and finding a witness is the whole job.

## The vocabulary of provable things

### ● MUST MEMORISE

### THEOREM, LEMMA, COROLLARY AND THE REST

These words are not interchangeable, and a question that asks you to "differentiate" between them wants the distinction, not a definition of each in isolation.

| WORD               | WHAT IT IS                                          | THE DISTINGUISHING FEATURE                  |
|--------------------|-----------------------------------------------------|---------------------------------------------|
| <b>Axiom</b>       | a statement taken as true without proof             | the starting point; nothing is behind it    |
| <b>Theorem</b>     | a statement shown to be true                        | important in its own right                  |
| <b>Proposition</b> | a statement shown to be true                        | true, proved, but of lesser importance      |
| <b>Lemma</b>       | a statement proved in order to prove something else | a helper; it exists to serve a later proof  |
| <b>Corollary</b>   | a statement that follows at once from a theorem     | comes after, and costs almost no extra work |
| <b>Conjecture</b>  | a statement believed true but not yet proved        | no proof exists yet; it may still be false  |

◆ **Memory hook: a lemma comes before, a corollary comes after, and a conjecture has not arrived at all.** The theorem is the thing in the middle that everyone remembers.

**Worth one line in an answer.** A conjecture that is proved becomes a theorem; a conjecture that is disproved becomes nothing at all. The word records the state of our knowledge, not a property of the statement.



## The definitions every proof in this course leans on

### ● MUST MEMORISE

### EVEN, ODD, RATIONAL, DIVIDES

Almost every proof set on this course is unlocked by writing one of these four definitions down. If you are stuck, the move is nearly always "replace the word by its definition".

| WORD                                   | DEFINITION                                                |
|----------------------------------------|-----------------------------------------------------------|
| m is <b>even</b>                       | $m = 2k$ for some integer $k$                             |
| n is <b>odd</b>                        | $n = 2k + 1$ for some integer $k$                         |
| n is <b>rational</b>                   | $n = p / q$ for some integers $p$ and $q$ with $q \neq 0$ |
| a <b>divides</b> b, written $a \mid b$ | $b = ka$ for some integer $k$                             |

#### WHERE

**k, p, q** are **integers**, and the phrase "for some integer  $k$ " is doing real work: it says such a  $k$  exists, and the proof's next move is always to name it and use it.  **$a \mid b$**  is read "a divides b", and it points the opposite way to a fraction:  $a \mid b$  means  $b$  is the multiple,  $a$  is the divisor. The bar is not a division sign and the statement is not a number, it is either true or false.  $\neq$  means "is not equal to";  $q \neq 0$  because division by zero has no value.

**An integer is either even or odd, never both and never neither.** That sentence is what makes a proof by cases on parity exhaustive, and it is worth stating explicitly when you use it.

◆ **Memory hook: write the definition, then do algebra.** Nine proofs in ten on this course are that and nothing more.

## The six strategies

### ● MUST MEMORISE

### SIX WAYS TO PROVE $P \rightarrow Q$

Every strategy below proves the same shape of statement, the conditional  $p \rightarrow q$  of section 1.2. They differ in what you assume at the start.

| STRATEGY              | YOU ASSUME                       | YOU DERIVE                     | USE IT WHEN                                          |
|-----------------------|----------------------------------|--------------------------------|------------------------------------------------------|
| <b>Trivial</b>        | nothing                          | $q$ is true on its own         | the conclusion is already known true                 |
| <b>Vacuous</b>        | nothing                          | $p$ is false                   | the hypothesis cannot hold                           |
| <b>Direct</b>         | $p$                              | $q$                            | the default; always try this first                   |
| <b>Contrapositive</b> | $\neg q$                         | $\neg p$                       | the negations are easier to work with                |
| <b>Contradiction</b>  | $p$ and $\neg q$                 | $r \wedge \neg r$              | nothing to build from; the claim is a denial         |
| <b>Cases</b>          | $p$ as $p_1 \vee \dots \vee p_n$ | $q$ from each $p_i$ separately | the hypothesis splits naturally, such as odd or even |

#### WHERE

$p$  is the **hypothesis**, also called the premise or antecedent: what you are given.  $q$  is the **conclusion**, also called the consequent: what you must reach.  $\neg q$  is the negation of  $q$ , from section 1.1.  $r \wedge \neg r$  is a **contradiction**: some statement  $r$  and its own negation, both derived. Section 1.3 showed that is false in every row of its table, which is exactly why deriving one destroys the assumption that produced it.

◆ **Memory hook: direct goes forwards, contrapositive goes backwards, contradiction goes nowhere and shows the alternative is impossible.**

Neither is a trick. Both rest on a logical equivalence proved back in section 1.3, and quoting that equivalence by name is worth a mark.

**Contrapositive.**  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ . The two have identical truth tables, so proving either one proves the other. Section 1.2 built that table and 2024/2025 Q1(c)(ii) asked you to prove it; the proof lives there, and this section is where you spend it.

**Contradiction.** If assuming  $p \wedge \neg q$  leads to something false, then  $p \wedge \neg q$  is itself false, so its negation  $\neg p \vee q$  is true, and that is exactly  $p \rightarrow q$  written the other way (section 1.3, the implication law).

**Contrapositive is a special case of contradiction**, which is why the two are so often confused. If you assume  $\neg q$  and reach  $\neg p$  while  $p$  was also assumed, you have  $p \wedge \neg p$ , a contradiction. The practical difference is in the writing: a contrapositive proof is short and clean, a contradiction proof carries an extra assumption all the way through.

**Prefer the contrapositive when it is available.**

**Direct**                       $p \longrightarrow q$                       assume the given, reach the goal

**Contrapositive**         $\neg q \longrightarrow \neg p$                       flip both, then go forwards

**Contradiction**             $p \wedge \neg q \longrightarrow r \wedge \neg r$                       an impossibility  
so the assumption was wrong

*The three arrows worth drawing in the margin of a proof question before you start writing.*

## Direct proofs, worked

### WORKED EXAMPLE

### METHOD · THREE DIRECT PROOFS

#### 1. The sum of two odd integers is even.

**Proof.** Let  $m$  and  $n$  be arbitrary odd integers. By the definition of odd there are integers  $a$  and  $b$  with

$$m = 2a + 1 \quad \text{and} \quad n = 2b + 1$$

Then

$$\begin{aligned} m + n &= (2a + 1) + (2b + 1) \\ &= 2a + 2b + 2 \\ &= 2(a + b + 1) \end{aligned}$$

Since  $a + b + 1$  is an integer,  $m + n$  is twice an integer, so  $m + n$  is even.  $\square$

#### 2. If $n$ is even then $n^2$ is even.

**Proof.** Let  $n$  be an arbitrary integer and suppose  $n$  is even. Then  $n = 2k$  for some integer  $k$ , so

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

and  $2k^2$  is an integer, so  $n^2$  is even.  $\square$

#### 3. Divisibility is transitive: if $a \mid b$ and $b \mid c$ then $a \mid c$ .

**Proof.** Let  $a, b, c$  be integers and assume  $a \mid b$  and  $b \mid c$ . By the definition of divides there are integers  $k$  and  $j$  with

$$b = ka \quad \text{and} \quad c = jb$$

Substituting the first into the second,  $c = j(ka) = (jk)a$ . Since  $jk$  is an integer,  $a \mid c$ .  $\square$

**Notice the shape all three share.** Assume, replace each word by its definition, do ordinary algebra, then read the definition backwards to name what you have reached. That is the entire method, and the last sentence, "since ... is an integer", is the one candidates leave out.

### △ TRAP

### TWO INTEGERS, TWO LETTERS

Writing  $m = 2a + 1$  and  $n = 2a + 1$  with the **same** letter says  $m$  and  $n$  are the same number. The proof then only covers the case  $m = n$  and is worth almost nothing. Two independent objects get two independent letters, every time.

The same slip in reverse: proving something for "an arbitrary even integer  $n = 2k$ " and then later writing  $k = 3$  to check. Checking is fine, but say you are checking. A proof that fixes its arbitrary letter to a number has stopped being a proof.

## Contrapositive, worked

### WORKED EXAMPLE

### METHOD · CONTRAPOSITIVE

#### 1. If $mn$ is even, then $m$ is even or $n$ is even.

**Why not direct?** Assuming  $mn = 2k$  tells you almost nothing about  $m$  and  $n$  separately. But the negation of the conclusion is clean: " $m$  is not even and  $n$  is not even" means both are odd, and odd numbers have a formula. That is the signal to flip.

**The contrapositive.** If  $m$  is odd and  $n$  is odd, then  $mn$  is odd. Note that negating " $m$  is even or  $n$  is even" gives " $m$  is odd **and**  $n$  is odd", by De Morgan's law of section 1.3. Getting that "and" is where the mark is.

**Proof.** Suppose  $m$  and  $n$  are both odd, so  $m = 2a + 1$  and  $n = 2b + 1$  for some integers  $a$  and  $b$ . Then

$$\begin{aligned} mn &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1 \end{aligned}$$

and  $2ab + a + b$  is an integer, so  $mn$  is odd. The contrapositive holds, therefore the original statement holds.  $\square$

#### 2. If $x^3 - 7x^2 + x - 7 = 0$ , then $x \neq 10$ .

**The contrapositive.** If  $x = 10$ , then  $x^3 - 7x^2 + x - 7 \neq 0$ .

**Proof.** Substitute  $x = 10$ :

$$10^3 - 7 \times 10^2 + 10 - 7 = 1000 - 700 + 10 - 7 = \mathbf{303}$$

and  $303 \neq 0$ .  $\square$

**Worth an extra line.** The expression factorises as  $x^2(x - 7) + (x - 7) = (x^2 + 1)(x - 7)$ , and  $x^2 + 1$  is never zero for real  $x$ , so the only real solution is  $x = 7$ . Saying so shows you understood the question rather than only answered it.

#### 3. If $5a + 25b = 156$ , then $a$ or $b$ is not an integer.

**The contrapositive.** If  $a$  and  $b$  are both integers, then  $5a + 25b \neq 156$ .

**Proof.** Suppose  $a$  and  $b$  are integers. Then

$$5a + 25b = 5(a + 5b)$$

and  $a + 5b$  is an integer, so  $5a + 25b$  is a multiple of 5. But  $156 = 5 \times 31 + 1$  leaves remainder 1, so 156 is not a multiple of 5. Hence  $5a + 25b \neq 156$ .  $\square$

**Now redo it:** show that if  $6a + 9b = 100$  then  $a$  or  $b$  is not an integer. Same move:  $6a + 9b = 3(2a + 3b)$  is a multiple of 3, and  $100 = 3 \times 33 + 1$  is not.

## Contradiction, worked

### WORKED EXAMPLE

### METHOD · CONTRADICTION

#### 1. There is no largest integer.

**Why contradiction?** The claim is a denial: it says a thing does not exist. There is nothing to build forwards from, so assume the thing does exist and watch it break.

**Proof.** Suppose, for contradiction, that there is a largest integer, and call it  $N$ . Then  $N + 1$  is also an integer, and  $N + 1 > N$ , so  $N$  is not the largest after all. That contradicts the assumption, which must therefore be false. There is no largest integer.  $\square$

**2.  $\sqrt{2}$  is irrational.** This is the classic, and it is worth knowing by heart because every part of it is a technique you have already met.

**Proof.** Suppose, for contradiction, that  $\sqrt{2}$  is rational. Then by the definition of rational there are integers  $p$  and  $q$  with  $q \neq 0$  and

$$\sqrt{2} = p / q$$

and we may take the fraction in **lowest terms**, so that  $p$  and  $q$  have no common factor other than 1. Squaring and multiplying by  $q^2$ :

$$2 = p^2 / q^2, \text{ so } p^2 = 2q^2$$

So  $p^2$  is even. By the contrapositive of worked example 2 above, if  $p$  were odd then  $p^2$  would be odd, so  **$p$  is even**, say  $p = 2k$ . Substituting:

$$(2k)^2 = 2q^2, \text{ so } 4k^2 = 2q^2, \text{ so } q^2 = 2k^2$$

By the same argument  **$q$  is even**. But then  $p$  and  $q$  are both even and share the factor 2, contradicting "lowest terms". The assumption was false, so  $\sqrt{2}$  is irrational.  $\square$

**Where each technique appeared.** The definition of rational opened it, a contrapositive supplied the step from  $p^2$  even to  $p$  even, and the contradiction closed it. Proofs at this level are almost always built from the smaller ones.

## Cases, worked

### WORKED EXAMPLE

### METHOD · PROOF BY CASES

**For every integer  $n$ , the number  $n^3 - n$  is even.**

**Why cases?** Nothing is given about  $n$  at all, so there is no definition to unfold. But every integer is even or odd, and each of those does come with a formula. Two cases, and they are exhaustive.

**The rule that makes it legal**, which is worth quoting:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q \equiv (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$$

which is another equivalence from section 1.3: proving each case separately is exactly proving the whole.

**Case 1:  $n$  is even**, so  $n = 2k$  for some integer  $k$ .

$$n^3 - n = (2k)^3 - 2k = 8k^3 - 2k = 2(4k^3 - k)$$

and  $4k^3 - k$  is an integer, so  $n^3 - n$  is even.

**Case 2:  $n$  is odd**, so  $n = 2k + 1$  for some integer  $k$ . Expanding the cube by the binomial theorem of section 5.4:

$$(2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1$$

so

$$\begin{aligned} n^3 - n &= 8k^3 + 12k^2 + 6k + 1 - (2k + 1) \\ &= 8k^3 + 12k^2 + 4k \\ &= 2(4k^3 + 6k^2 + 2k) \end{aligned}$$

and  $4k^3 + 6k^2 + 2k$  is an integer, so  $n^3 - n$  is even.

**Both cases are covered and both give the same conclusion**, so  $n^3 - n$  is even for every integer  $n$ .  $\square$

**A one-line check before you trust the algebra:**  $n = 3$  gives  $27 - 3 = 24$ , and  $n = 4$  gives  $64 - 4 = 60$ . Both even. A check is not a proof, but a failed check saves you from writing out a proof of something false.

**The slicker argument**, worth adding after the marks are safe:  $n^3 - n = n(n - 1)(n + 1)$  is a product of three consecutive integers, and among any two consecutive integers one is even. One line, no cases.

The official course text works this same example and drops two errors into the algebra. They are worth knowing because a reader who copies them into an exam loses the marks, and because spotting them is exactly the skill a proof question is testing.

**First**, in the even case it writes  $8k^3 - 2k = 2(4k^2 - k)$ . The bracket should be  $4k^3 - k$ : dividing  $8k^3$  by 2 gives  $4k^3$ , and the cube does not become a square.

**Second**, in the odd case it expands  $(2k + 1)^3$  as  $8k^3 + 6k^2 + 6k + 1$ . The correct expansion is  $8k^3 + 12k^2 + 6k + 1$ ; the squared term is  $3 \times (2k)^2 \times 1 = 12k^2$ , not  $6k^2$ . The same line also prints the cube as a square, writing  $(2k + 1)^2$  where it means  $(2k + 1)^3$ .

**Why the conclusion survives anyway**, which is the interesting part: both wrong brackets still come out as  $2 \times (\text{an integer})$ , so the final claim " $n^3 - n$  is even" is still true. A proof can reach a true conclusion through false steps, and that is precisely why every line has to be checked rather than the answer alone.

## Trivial and vacuous proofs

Both come straight off the conditional's truth table in section 1.2, which is false in one row only, when  $p$  is true and  $q$  is false.

**Trivial proof:  $q$  is already true**, so no row can make  $p \rightarrow q$  false, whatever  $p$  says.

*If it is raining, then  $1 = 1$ .*

Proof:  $1 = 1$  is true, so the conditional is true regardless of the weather.  $\square$

**Vacuous proof:  $p$  is false**, so the one false row is never reached.

*If the door is both open and closed, then  $2 + 2 = 5$ .*

Proof: a door cannot be both open and closed, so the hypothesis is a contradiction and is false. The conditional is therefore true.  $\square$

**These are not jokes, and they matter in computing.** A loop body that never runs satisfies every claim about what it does; a specification that says "for all  $x$  in the empty list, ..." is vacuously true. Section 1.4 made the same point about a universal quantifier over an empty domain.

**In an exam**, a trivial or vacuous proof is normally the answer to "explain why this is true", not to "prove". If you find yourself writing one for a serious-looking question, re-read it: you have probably misidentified  $p$  or  $q$ .



## Choosing a strategy, and one worked mistake

### TEACH

### THE DECISION ORDER

- 1 **Write the claim as  $p \rightarrow q$ .** Name  $p$  and name  $q$  in plain words. Half of all failed proof attempts are a claim never written down.
- 2 **Try direct.** Replace every word in  $p$  by its definition and see how close the algebra takes you to  $q$ .
- 3 **If  $p$  gives you nothing, negate  $q$ .** If  $\neg q$  has a usable formula where  $p$  did not, do the contrapositive. "Or" in the conclusion is a strong signal, because its negation is an "and".
- 4 **If the claim denies existence, use contradiction.** Words like "no", "never", "cannot", "irrational", "infinitely many" all point here.
- 5 **If  $p$  splits, use cases.** Odd or even, positive or negative or zero, and any hypothesis containing "or".
- 6 **Say which one you used, in the first line.** "We prove the contrapositive" costs six words and tells the marker how to read everything that follows.

### ⚠ TRAP

### WHEN THE CONVENTION CHANGES THE ANSWER

Consider the claim "**there is a natural number  $n$  for which  $n^2 + 3n + 2$  is prime**".

Section 2.1 defined a **prime** as a number with exactly two positive divisors, itself and 1. A whole number above 1 that is not prime is **composite**: it can be written as a product of two smaller whole numbers, each at least 2.

**If  $\mathbb{N}$  starts at 1**, the claim is false, and there is a one-line proof:  $n^2 + 3n + 2 = (n + 1)(n + 2)$ , a product of two integers both at least 2, so it is composite for every  $n \geq 1$ . Check:  $n = 1$  gives 6,  $n = 2$  gives 12,  $n = 3$  gives 20.

**If  $\mathbb{N}$  starts at 0**, which is the convention this manual declared in Foundations, the claim is **true**:  $n = 0$  gives  $0 + 0 + 2 = 2$ , and 2 is prime. One witness settles an existence claim, and the whole argument collapses.

The Foundations trap box said "say which convention you are using". This is the case that shows why: the convention is not a matter of taste here, it decides the truth value. **State it in the first line of the answer and you cannot be marked wrong either way.**

**Also note the better proof.** The factorisation  $(n + 1)(n + 2)$  does in one line what a proof by cases on the parity of  $n$  does in fifteen. When a polynomial factorises, factorise it.

## RECALL AND CHECK

1. Distinguish a lemma from a corollary.
2. Write the definitions of even, odd and "a divides b".
3. What do you assume, and what do you derive, in a proof by contrapositive?
4. What do you assume in a proof by contradiction?
5. How many counterexamples disprove a statement about all x?
6. Prove that the product of two odd integers is odd.
7. Why is "if the door is open and closed then  $2 + 2 = 5$ " true?

### ANSWERS

1. A lemma is proved **before** and in order to prove a theorem; a corollary follows **after** a theorem with little extra work.
2. m even:  $m = 2k$  for some integer k. n odd:  $n = 2k + 1$  for some integer k.  $a \mid b$ :  $b = ka$  for some integer k.
3. Assume  $\neg q$ , derive  $\neg p$ .
4. Both p and  $\neg q$ , then derive a statement and its own negation.
5. One.
6.  $(2a + 1)(2b + 1) = 2(2ab + a + b) + 1$ , which is odd.
7. The hypothesis is false, so the conditional is vacuously true.

## FILL IN THE GAP

1. A statement believed true but not yet proved is a \_\_\_\_\_.
2. To prove  $p \rightarrow q$  by contrapositive, assume \_\_\_\_\_ and derive \_\_\_\_\_.
3. An integer n is odd when  $n =$  \_\_\_\_\_ for some integer k.
4. A proof by \_\_\_\_\_ splits the hypothesis into exhaustive alternatives.
5. The negation of "m is even or n is even" is "m is odd \_\_\_\_\_ n is odd".
6.  $\sqrt{2}$  is proved irrational by \_\_\_\_\_.

### ANSWERS

1. conjecture. 2.  $\neg q$ ,  $\neg p$ . 3.  $2k + 1$ . 4. cases. 5. and. 6. contradiction.

## PART NINE • 9.2

# Mathematical Induction

*One proof, infinitely many statements. Induction is how you prove something about every natural number without checking them one at a time, and it is the technique behind the tree bound of section 7.2 that the 2024/2025 paper asked for outright.*

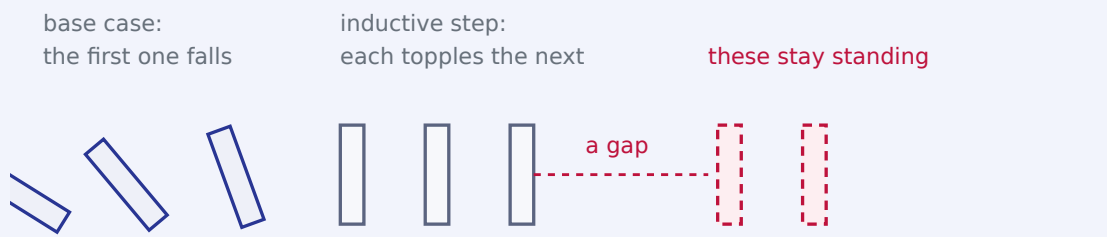
*By the end of this section you will be able to state the principle of mathematical induction, lay out the three steps in the order a marker expects, prove a summation formula, a divisibility claim and an inequality by induction, explain why the base case cannot be skipped, and use strong induction where ordinary induction stalls.*

Suppose there is an infinite ladder and you know two things:

- ▶ you can reach the **first** rung, and
- ▶ from **any** rung you reach, you can reach the next one.

Can you reach every rung? Yes. And notice you were never told how to reach rung 400 directly. The second fact is a machine that manufactures that answer: reach 1, so reach 2, so reach 3, and the argument runs as far as anyone cares to push it.

The same picture with dominoes standing in a line. Knock the first one over, and space them so that each falling one topples its neighbour. Both conditions are needed. A first domino that never falls leaves the line standing; a gap anywhere in the line stops the chain there and everything past the gap survives.



*Both conditions carry weight. Remove either and the chain stops.*

### THE PRINCIPLE OF MATHEMATICAL INDUCTION

**The claim.** Let  $P(n)$  be a statement about the integer  $n$ . If  $P$  is true at a starting value, and if the truth of  $P$  at any value forces its truth at the next value, then  $P$  is true at every value from the start onwards. One check plus one implication buys infinitely many statements.

#### In symbols.

If  $P(a)$  is true, and  $P(k) \rightarrow P(k + 1)$  for every  $k \geq a$ , then  $P(n)$  is true for every  $n \geq a$ .

**Conditions.** The statement must be indexed by an **integer** that increases in steps of one; there is no induction over the real numbers, because there is no "next" real number. The starting value  $a$  is usually 0 or 1 but may be any integer, and the conclusion then only covers  $n \geq a$ . The inductive step must hold for **every**  $k$  from  $a$  onwards, not merely for the small ones you tried.

**The limiting case.** With no base case the principle says nothing at all, however impeccable the inductive step. "If  $n = n + 1$  then  $n + 1 = n + 2$ " is a perfectly valid inductive step for a statement that is false everywhere; nothing ever gets it started. The base case is not a formality, it is half the proof.

Write these three headings on the page before you write any mathematics. Markers award them separately.

| STEP                           | WHAT YOU WRITE                                                                                                      |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>1. Base case</b>            | State $P(a)$ for the smallest $a$ in the claim, and <b>verify it by direct computation</b> . Both sides, evaluated. |
| <b>2. Inductive hypothesis</b> | <b>Assume</b> $P(k)$ holds for some arbitrary $k \geq a$ . Write out what $P(k)$ actually says.                     |
| <b>3. Inductive step</b>       | Using that assumption, <b>prove</b> $P(k + 1)$ . Write out what $P(k + 1)$ says as a target, then reach it.         |

**The whole principle on one line**, which is what the three steps above amount to:

**$P(a)$ , together with  $P(k) \rightarrow P(k + 1)$  for every  $k \geq a$ , gives  $P(n)$  for every  $n \geq a$**

WHERE

**$P(n)$**  is the **statement** being proved, a predicate in the sense of section 1.4: for each  $n$  it is either true or false. It is not a number and you never "calculate  $P(n)$ ".  **$a$**  is the **base value**, the smallest  $n$  the claim covers.  **$k$**  is an **arbitrary** value at or above  $a$ . It is not a specific number, and it is not  $n$ ; using  $n$  for both is the most common way to write a proof that says nothing.  **$P(k) \rightarrow P(k + 1)$**  is the **inductive step**, a conditional, so it is proved by assuming the left side, in the manner of section 9.1.

**Close with the conclusion.** "By the principle of mathematical induction,  $P(n)$  holds for all  $n \geq a$ ." A proof that stops after the algebra has left its last mark on the table.

◆ **Memory hook: check one, assume one, earn the next.** And the assumption is the tool, so every inductive step must use it somewhere; if yours does not, you have proved  $P(k + 1)$  outright and did not need induction at all.

## A summation formula

### WORKED EXAMPLE

### METHOD · 1 + 2 + ... + N

**GIVEN** The claim  $P(n): 1 + 2 + 3 + \dots + n = n(n + 1) / 2$ , for every natural number  $n \geq 1$ .

**FIND** A proof by induction on  $n$ .

**FORMULA** Base case, inductive hypothesis, inductive step.

Step 1. Base case,  $n = 1$

The left side is the single term 1. The right side is

$$1 \times (1 + 1) / 2 = 2 / 2 = 1$$

Both sides equal 1, so  $P(1)$  holds.

Step 2. Inductive hypothesis

Assume  $P(k)$  holds for some  $k \geq 1$ , that is

$$1 + 2 + \dots + k = k(k + 1) / 2$$

Step 3. Inductive step

**The target.**  $P(k + 1)$  says

$$1 + 2 + \dots + k + (k + 1) = (k + 1)(k + 2) / 2$$

**Start from the left side and use the hypothesis.** The first  $k$  terms are exactly what the hypothesis measures, so replace them:

$$\begin{aligned} & 1 + 2 + \dots + k + (k + 1) \\ &= k(k + 1) / 2 + (k + 1) \quad [\text{inductive hypothesis}] \\ &= k(k + 1) / 2 + 2(k + 1) / 2 \quad [\text{common denominator}] \\ &= (k + 1)(k + 2) / 2 \quad [\text{take out the factor } (k + 1)] \end{aligned}$$

which is the target, so  $P(k + 1)$  holds.

**Conclusion.** By the principle of mathematical induction,  $1 + 2 + \dots + n = n(n + 1) / 2$  for every  $n \geq 1$ .  $\square$

**A check that costs one line.** At  $n = 100$  the formula gives  $100 \times 101 / 2 = 5050$ .

Adding one more term,  $5050 + 101 = 5151$ , and the formula at  $n = 101$  gives  $101 \times 102 / 2 = 5151$ . That agreement is the inductive step happening in front of you at one particular  $k$ .

**Now redo it:** prove  $1 + 3 + 5 + \dots + (2n - 1) = n^2$ . Base case  $n = 1$ : left 1, right 1. Step:  $k^2 + (2(k + 1) - 1) = k^2 + 2k + 1 = (k + 1)^2$ . Sanity check at  $n = 4$ :  $1 + 3 + 5 + 7 = 16 = 4^2$ .

## A divisibility claim

### WORKED EXAMPLE

### METHOD · $N^3 + 2N$ IS DIVISIBLE BY 3

- GIVEN**  $P(n)$ :  $n^3 + 2n$  is divisible by 3, for every positive integer  $n$ .  
**FIND** A proof by induction.  
**FORMULA** Divisibility is the definition of section 9.1: 3 divides  $x$  when  $x = 3t$  for some integer  $t$ .

Step 1. Base case,  $n = 1$

$$1^3 + 2 \times 1 = 1 + 2 = 3$$

and  $3 = 3 \times 1$ , so  $P(1)$  holds.

Step 2. Inductive hypothesis

Assume  $P(k)$ :  $k^3 + 2k$  is divisible by 3, so by the definition there is an integer  $t$  with

$$k^3 + 2k = 3t$$

**Naming  $t$  is the step candidates skip**, and without it there is nothing to substitute later.

Step 3. Inductive step

**The target.**  $(k + 1)^3 + 2(k + 1)$  is divisible by 3.

**Expand**, using  $(k + 1)^3 = k^3 + 3k^2 + 3k + 1$ :

$$\begin{aligned}(k + 1)^3 + 2(k + 1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= (k^3 + 2k) + (3k^2 + 3k + 3)\end{aligned}$$

**Now use the hypothesis** on the first bracket, and take 3 out of the second:

$$\begin{aligned}&= 3t + 3(k^2 + k + 1) \\ &= 3(t + k^2 + k + 1)\end{aligned}$$

and  $t + k^2 + k + 1$  is an integer, so the total is divisible by 3.  $P(k + 1)$  holds.

**Conclusion.** By induction,  $n^3 + 2n$  is divisible by 3 for every positive integer  $n$ .  $\square$

**The move to learn here** is the regrouping: split the expanded expression so that the hypothesis appears intact as one bracket, and everything left over is visibly a multiple of 3. Every divisibility induction has that shape.

**Check three values:**  $n = 2$  gives  $8 + 4 = 12$ ,  $n = 3$  gives  $27 + 6 = 33$ ,  $n = 4$  gives  $64 + 8 = 72$ . All multiples of 3.

**Now redo it:** prove  $6^n - 1$  is a multiple of 5 for every natural number  $n$ . Base  $n = 0$ :  $1 - 1 = 0$ , which is  $5 \times 0$ . Hypothesis:  $6^k - 1 = 5j$ , so  $6^k = 5j + 1$ . Then  $6^{k+1} - 1 = 6(5j + 1) - 1 = 30j + 5 = 5(6j + 1)$ . Check:  $n = 1$  gives **5**,  $n = 2$  gives **35**,  $n = 3$  gives **215**.

## An inequality, and where the base case really lives

### WORKED EXAMPLE

### METHOD · $n^2$ BELOW $2$ TO THE $n$

**GIVEN**  $P(n): n^2 < 2^n$ , for every integer  $n \geq 5$ .

**FIND** A proof by induction, and an explanation of why the claim starts at 5.

**FORMULA** Base case at  $n = 5$ , then the step.

**First, why 5.** Test the small values and the reason appears:

| $n$ | $n^2$ | $2^n$ | IS $n^2 < 2^n$ ?           |
|-----|-------|-------|----------------------------|
| 1   | 1     | 2     | yes                        |
| 2   | 4     | 4     | <b>no</b> , they are equal |
| 3   | 9     | 8     | <b>no</b>                  |
| 4   | 16    | 16    | <b>no</b> , they are equal |
| 5   | 25    | 32    | yes                        |
| 6   | 36    | 64    | yes                        |

The claim is **false** at  $n = 2, 3$  and  $4$ , and true from  $5$  onwards. That is why the question says  $n \geq 5$ , and it is a good illustration that a base case is a genuine check, not a ritual: the same inductive step would run happily from a base of  $3$  and would then be carrying a false statement.

Step 1. Base case,  $n = 5$

$5^2 = 25$  and  $2^5 = 32$ , and  $25 < 32$ .  $P(5)$  holds.

Step 2. Inductive hypothesis

Assume  $P(k)$  for some  $k \geq 5$ , that is  $k^2 < 2^k$ .

Step 3. Inductive step

**The target.**  $(k + 1)^2 < 2^{k+1}$ .

**Work on the left side.**  $(k + 1)^2 = k^2 + 2k + 1$ . Since  $k \geq 5$  we have  $2k + 1 \leq k^2$ , because that inequality holds for every  $k \geq 3$ . So

$$(k + 1)^2 = k^2 + 2k + 1 \leq k^2 + k^2 = 2k^2$$

**Now use the hypothesis:**  $k^2 < 2^k$ , so

$$2k^2 < 2 \times 2^k = 2^{k+1}$$

Chaining the two,  $(k + 1)^2 < 2^{k+1}$ , which is  $P(k + 1)$ .

**Conclusion.** By induction,  $n^2 < 2^n$  for every integer  $n \geq 5$ .  $\square$

**The technique in inequality inductions** is always this: bound the awkward extra term by something the hypothesis can absorb. Here  $2k + 1$  was bounded by  $k^2$ , and the side condition  $k \geq 3$  that made that legal was stated rather than assumed.

## The examined one

### WORKED EXAMPLE

PQ 24/25 Q6(B) VII, IN THE INDUCTION TEMPLATE

The 2024/2025 paper asked for a proof that an  $m$ -ary tree of height  $h$  has at most  $m^h$  leaves. That question is quoted as printed and answered in section 7.2, where the tree vocabulary lives. It is repeated here in the three-step layout so you can see the same proof through this section's frame, because the two are the same argument written for two different markers.

**GIVEN** An  $m$ -ary tree of height  $h$ , that is a rooted tree in which every vertex has at most  $m$  children.

**FIND** A proof that it has at most  $m^h$  leaves.

**FORMULA** Induction on the **height**  $h$ , not on the number of vertices.

Step 1. Base case,  $h = 0$

A tree of height 0 is a single vertex, which is both the root and a leaf. It has 1 leaf, and  $m^0 = 1$ , so the bound holds with equality.

Step 2. Inductive hypothesis

Assume every  $m$ -ary tree of height  $k$  has at most  $m^k$  leaves.

Step 3. Inductive step

Let  $T$  be an  $m$ -ary tree of height  $k + 1$ . Remove the root. What remains is at most  $m$  subtrees, one per child of the root, and each has height at most  $k$ . By the hypothesis each has at most  $m^k$  leaves. Every leaf of  $T$  lies in one of them, so

$$\text{leaves of } T \leq m \times m^k = m^{k+1}$$

**Conclusion.** By induction the bound holds at every height.  $\square$

**Note what the induction was on.** Choosing the right quantity to induct on is most of the skill. Inducting on the number of vertices here would be painful, because removing a vertex from an  $m$ -ary tree need not leave an  $m$ -ary tree of any convenient height. Height was the quantity that made the step easy.



## Strong induction

### TEACH

### WHEN ONE STEP BACK IS NOT ENOUGH

Ordinary induction assumes  $P(k)$  and proves  $P(k + 1)$ : it reaches back exactly one rung. Sometimes the argument needs to reach further, or to reach back to a value it cannot predict.

**Strong induction** assumes  $P(a)$ ,  $P(a + 1)$ , ...,  $P(k)$  all at once, and proves  $P(k + 1)$  from any of them it likes. The principle is the same and is itself provable from ordinary induction, so no new permission is needed.

**The standard example:** every integer  $n \geq 2$  is a product of prime numbers.

**Proof.** Base case: 2 is itself prime, so it is a product of one prime. Assume every integer from 2 up to  $k$  is a product of primes, and consider  $k + 1$ . Either  $k + 1$  is prime, in which case it is a product of one prime and we are done; or it is not, in which case  $k + 1 = ab$  for some integers  $a$  and  $b$  with  $2 \leq a \leq k$  and  $2 \leq b \leq k$ . Both  $a$  and  $b$  are covered by the hypothesis, so each is a product of primes, and multiplying those products gives one for  $k + 1$ .  $\square$

**Why ordinary induction fails here.** The factors  $a$  and  $b$  are not  $k$ , and there is no telling in advance where they land. Only an assumption covering **every** smaller value is any use. That is the signal for strong induction: **the step goes back an unpredictable distance**. The same signal appears in section 9.4, where a second-order recurrence needs two previous terms rather than one.

### ⚠ TRAP

### THREE WAYS AN INDUCTION PROOF LOSES MARKS

**1. Assuming what is to be proved.** Writing " $1 + 2 + \dots + (k + 1) = (k + 1)(k + 2) / 2$ " at the top of step 3 and then manipulating both sides until they agree is circular. Start from **one** side, use the hypothesis, and arrive at the other. If you do work on both sides, say plainly that you are showing two expressions are equal, and never treat the target as available.

**2. Never using the hypothesis.** An inductive step that does not cite  $P(k)$  anywhere has proved  $P(k + 1)$  from nothing, which either means induction was unnecessary or, far more often, that a step is missing. **Point at the line where you used it.**

**3. A base case that is asserted rather than checked.** "Clearly true for  $n = 1$ " earns nothing. Evaluate both sides and show they agree. The inequality above is the reason: at  $n = 4$  that claim is false, and only an evaluated base case catches it.

## RECALL AND CHECK

1. Name the three steps of an induction proof.
2. What exactly do you assume in step 2, and about which value?
3. Why can the base case not be skipped?
4. State the formula for  $1 + 2 + \dots + n$  and evaluate it at  $n = 100$ .
5. What does strong induction assume that ordinary induction does not?
6. Give the signal that tells you strong induction is needed.

### ANSWERS

1. Base case, inductive hypothesis, inductive step.
2. That  $P(k)$  is true, for an arbitrary  $k$  at or above the base value.
3. The inductive step alone only passes truth along; with nothing true to start from it proves nothing, and it will happily carry a false statement forever.
4.  $n(n + 1) / 2$ , which is  $100 \times 101 / 2 = 5050$ .
5. Every earlier case,  $P(a)$  through  $P(k)$ , rather than  $P(k)$  alone.
6. The step needs a value an unpredictable distance back, as when a number is split into two unknown factors.

## FILL IN THE GAP

1. Induction proves a statement for every \_\_\_\_\_ from the base value onwards.
2. Step 2 assumes  $P(\text{_____})$  and step 3 proves  $P(\text{_____})$ .
3.  $1 + 3 + 5 + \dots + (2n - 1) = \text{_____}$ .
4.  $n^2 < 2^n$  first holds at  $n = \text{_____}$ .
5. An inductive step that never cites the hypothesis is \_\_\_\_\_.

### ANSWERS

1. integer. 2.  $k, k + 1$ . 3.  $n^2$ . 4. 5. 5. not an induction proof at all: either it was unnecessary, or a step is missing.

## PART NINE • 9.3

# Sequences, Series and Progressions

*A sequence is an ordered list with a rule behind it. Two rules cover almost everything this course does with them: keep adding the same amount, or keep multiplying by the same amount. Getting the indexing right is where the marks actually go.*

*By the end of this section you will be able to define a sequence and a term, tell an arithmetic progression from a geometric one, write the general term and the sum of each, read a rule off a printed sequence, use sigma notation and the four standard sums, and state which indexing convention you are using so that an off-by-one cannot cost you.*

# Sequences

## ● MUST MEMORISE

## SEQUENCE, TERM, FINITE AND INFINITE

A **sequence** is a function from a subset of the integers, usually  $\{0, 1, 2, \dots\}$  or  $\{1, 2, 3, \dots\}$ , to a set  $S$ . The image of the integer  $n$  is written  $a_n$  and is called a **term** of the sequence. The whole sequence is written  $\{a_n\}$ .

### WHERE

$n$  is the **index**, an integer, and it says the position of the term, not its value.  $a_n$  is the **term at position  $n$** : one number, not the whole list. The subscript is not multiplication.  $\{a_n\}$  in braces is the **sequence itself**, the entire ordered list.  $S$  is the set the terms are drawn from, usually the real numbers on this course.

**Calling it a function is the useful part.** Section 4.1 defined a function as a rule assigning exactly one output to each input; a sequence is that rule with the inputs restricted to integers. So each position has exactly one term, which is why "the third term" is a well-formed phrase.

**Finite and infinite.** A **finite** sequence has a last term and its **length** is the number of terms. An **infinite** sequence runs on without end.

| SEQUENCE                  | KIND              | THE RULE BEHIND IT                |
|---------------------------|-------------------|-----------------------------------|
| 2, 4, 6, 8, ..., 20       | finite, length 10 | add 2 to the previous term        |
| 10, 6, 2, -2, ...         | infinite          | subtract 4 from the previous term |
| 1, 1, 2, 3, 5, 8, 13, ... | infinite          | add the previous two terms        |

A **progression** is a sequence whose terms can be described by a formula. The third row above is the **Fibonacci sequence**, in which each term is the sum of the two before it; section 9.4 solves it properly.

◆ **Memory hook: the index is the seat number, the term is who is sitting in it.**

## Arithmetic progressions

### ● MUST MEMORISE

### ARITHMETIC PROGRESSION

An **arithmetic progression** is a sequence in which each term is the previous term plus a fixed amount:

$$a, a + d, a + 2d, a + 3d, \dots$$

The general term and the sum of the first  $n$  terms are

$$a_n = a + (n - 1)d \quad \text{and} \quad S_n = n(a + a_n) / 2 = n(2a + (n - 1)d) / 2$$

#### WHERE

**a** is the **first term**, the one at position 1. **d** is the **common difference**, the fixed amount added each time; it may be negative, which makes the sequence decrease, and it is found as any term minus the one before it. **n** is the **position** of the term you want, counting the first term as position 1. **a<sub>n</sub>** is the **n-th term**. **S<sub>n</sub>** is the **sum** of the first  $n$  terms, also called the arithmetic **series**. **(n - 1)** is the number of **steps taken** to get from the first term to the  $n$ -th, which is one fewer than the number of terms. That subtraction is the single most common error in this topic.

**Why the sum formula looks like that.** Write the sum forwards and backwards under itself and add column by column. Each column gives  $a + a_n$ , and there are  $n$  columns, so twice the sum is  $n(a + a_n)$ . It is the first term plus the last, times how many pairs there are. The  $n = 1$  to 100 case of section 9.2 is exactly this trick.

◆ **Memory hook: arithmetic adds, and the  $n$ -th term has taken  $n$  minus one steps.**

### △ TRAP

### TWO INDEXING CONVENTIONS, ONE SEQUENCE

The definition above starts counting at 1, so  $a_1 = a$  and  $a_n = a + (n - 1)d$ . The official course text writes the same progression as

$$a, a + d, a + 2d, \dots, a + nd$$

which counts from 0, so that its  $a_n = a + nd$  and its first term is  $a_0$ . **Neither is wrong.** They describe the same list of numbers and disagree only about what to call the positions.

**The danger is mixing them inside one answer**, which produces an answer exactly one term out. So do two things:

- ▶ **Say which you are using**, in five words: "counting the first term as  $a_1$ ".
- ▶ **Answer the question in the words it asked.** "Find the 10th term" means the tenth number in the list, whatever you have chosen to call its index. Count the list if you are unsure; ten numbers is not many.

This is the same discipline as the Foundations convention box on whether  $\mathbb{N}$  contains 0, and for the same reason: a stated convention cannot be marked wrong.

**GIVEN** The sequence 7, 12, 17, 22, ...

**FIND** (a) the 10th term, (b) a general formula for the  $n$ -th term, and (c) the sum of the first 10 terms.

**FORMULA**  $a_n = a + (n - 1)d$  and  $S_n = n(a + a_n) / 2$ , counting the first term as  $a_1$ .

**Identify it first.** The differences are  $12 - 7 = 5$ ,  $17 - 12 = 5$ ,  $22 - 17 = 5$ . Constant, so this is an arithmetic progression with  $a = 7$  and  $d = 5$ .

(a) The 10th term

$$a_{10} = a + (10 - 1)d = 7 + 9 \times 5 = 7 + 45 = \mathbf{52}$$

**Check by listing**, which is cheap and catches an off-by-one at once: 7, 12, 17, 22, 27, 32, 37, 42, 47, **52**. Ten numbers, and the tenth is 52.

(b) The general term

$$a_n = 7 + (n - 1) \times 5 = 7 + 5n - 5 = \mathbf{5n + 2}$$

**Test it at both ends:**  $n = 1$  gives 7, and  $n = 10$  gives  $50 + 2 = 52$ . Both agree.

(c) The sum of the first 10 terms

$$S_{10} = 10 \times (7 + 52) / 2 = 10 \times 59 / 2 = \mathbf{295}$$

**Now redo it:** for 2, 4, 6, ..., 20, we have  $a = 2$ ,  $d = 2$  and 10 terms, so the sum is  $10 \times (2 + 20) / 2 = \mathbf{110}$ .

## Geometric progressions

### ● MUST MEMORISE

### GEOMETRIC PROGRESSION

A **geometric progression** is a sequence in which each term is the previous term times a fixed amount:

$$a, ar, ar^2, ar^3, \dots$$

The general term and the sum of the first  $n$  terms are

$$a_n = ar^{n-1} \quad \text{and} \quad S_n = a(r^n - 1) / (r - 1) \quad \text{for } r \neq 1$$

WHERE

**a** is the **first term**. **r** is the **common ratio**, the fixed multiplier, found as any term divided by the one before it. It may be a fraction, which makes the terms shrink, and it may be negative, which makes the signs alternate. **n - 1** is again the number of **multiplications performed** to reach position  $n$  from position 1.  **$r \neq 1$**  is required because at  $r = 1$  the formula divides by zero; that case is just  $n$  copies of  $a$ , so  $S_n = na$ .

**The infinite sum.** If the ratio is small enough that the terms shrink, that is if  $r$  lies strictly between  $-1$  and  $1$ , the infinite series has a finite total:

$$S = a / (1 - r)$$

Outside that range the terms do not shrink and the sum has no finite value.

◆ **Memory hook: arithmetic adds d, geometric times by r.** To tell which one a printed sequence is: take differences, and if they are not constant, take ratios.

**GIVEN** (a) 4, 8, 16, 32, ... (b) 1, 1/3, 1/9, 1/27, ... (c) 1, 3, 5, 7, 9, ...

**FIND** The rule and the general term of each, plus one sum.

**FORMULA** Differences first; if they are not constant, ratios. Then  $a_n = a + (n - 1)d$  or  $a_n = ar^{n-1}$ .

(a) 4, 8, 16, 32

**Differences** are 4, 8, 16: not constant. **Ratios** are  $8 / 4 = 2$ ,  $16 / 8 = 2$ ,  $32 / 16 = 2$ : constant. So it is geometric with  $a = 4$  and  $r = 2$ , and

$$a_n = 4 \times 2^{n-1}$$

**A neater equivalent form**, worth writing beside it:  $4 \times 2^{n-1} = 2^2 \times 2^{n-1} = 2^{n+1}$ .

Check at  $n = 1$ :  $2^2 = 4$ . At  $n = 4$ :  $2^5 = 32$ .

**The 10th term** is  $2^{11} = \mathbf{2048}$ , and the sum of the first 10 terms is

$$S_{10} = 4(2^{10} - 1) / (2 - 1) = 4 \times 1023 = \mathbf{4092}$$

(b) 1, 1/3, 1/9, 1/27

The denominators are powers of 3, so the ratio is 1/3 and the first term is 1:

$$a_n = (1/3)^{n-1}$$

**Here the infinite sum exists**, because 1/3 lies between  $-1$  and  $1$ :

$$S = a / (1 - r) = 1 / (1 - 1/3) = 1 / (2/3) = \mathbf{3/2}$$

So adding infinitely many terms gives 1.5 exactly, not something unbounded. That is worth one sentence of comment in an exam, because it is the point of the formula.

(c) 1, 3, 5, 7, 9

Differences are all 2, so it is arithmetic with  $a = 1$ ,  $d = 2$ , and

$$a_n = 1 + (n - 1) \times 2 = \mathbf{2n - 1}$$

which is the sequence of odd numbers, exactly the one section 9.2 summed to  $n^2$ .

- GIVEN** The sequence 1, 5, 17, 53, 161, 485.  
**FIND** The rule that generates it.  
**FORMULA** None applies directly: the two standard tests, a constant difference and a constant ratio, both fail, so the rule has to be found by comparing each term with the one before it.

**Find the rule behind 1, 5, 17, 53, 161, 485.**

**Differences:** 4, 12, 36, 108, 324. Not constant, so it is not arithmetic. **Ratios:** 5, 3.4, about 3.12, about 3.04, about 3.01. Not constant, so it is not geometric either. But the ratios are creeping towards 3, which is the hint.

Try "times 3 plus something".

| TERM  | $3 \times \text{PREVIOUS}$ | ACTUAL | DIFFERENCE |
|-------|----------------------------|--------|------------|
| $a_2$ | $3 \times 1 = 3$           | 5      | 2          |
| $a_3$ | $3 \times 5 = 15$          | 17     | 2          |
| $a_4$ | $3 \times 17 = 51$         | 53     | 2          |
| $a_5$ | $3 \times 53 = 159$        | 161    | 2          |
| $a_6$ | $3 \times 161 = 483$       | 485    | 2          |

The extra is 2 every time, so the rule is

$$a_n = 3a_{n-1} + 2, \text{ with } a_1 = 1$$

**That is a recurrence relation**, a rule that defines each term from earlier ones rather than from  $n$  directly. Section 9.4 takes this exact sequence and produces a closed formula for  $a_n$  that needs no earlier terms at all.

**The general procedure**, when a sequence resists both standard shapes:

- 1 Take differences.** Constant means arithmetic.
- 2 Take ratios.** Constant means geometric.
- 3 Take second differences.** Constant means the terms follow a quadratic in  $n$ .
- 4 Compare each term with a multiple of its predecessor.** A constant leftover means a recurrence of the form  $a_n = ca_{n-1} + b$ .
- 5 Try the sum of the previous two.** That is Fibonacci.

## Sigma notation and the standard sums

### ● MUST MEMORISE

### FOUR SUMS WORTH KNOWING BY HEART

Summation notation was introduced in Foundations; here are the four closed forms this course expects, each valid for  $n \geq 1$ .

| SUM                         | CLOSED FORM               | CHECK AT $N = 5$                |
|-----------------------------|---------------------------|---------------------------------|
| $1 + 2 + \dots + n$         | $n(n + 1) / 2$            | $5 \times 6 / 2 = 15$           |
| $1^2 + 2^2 + \dots + n^2$   | $n(n + 1)(2n + 1) / 6$    | $5 \times 6 \times 11 / 6 = 55$ |
| $1^3 + 2^3 + \dots + n^3$   | $(n(n + 1) / 2)^2$        | $15^2 = 225$                    |
| $1 + r + r^2 + \dots + r^n$ | $(r^{n+1} - 1) / (r - 1)$ | at $r = 2$ : $2^6 - 1 = 63$     |

#### WHERE

$\sum$  is the **summation sign**: the letter below it is the index, the value below it is where the index starts, the value above it is where it stops, and the expression after it is what to add.  $n$  is the **last index**, so these sums have  $n$  terms when they start at 1 and  $n + 1$  terms when they start at 0. The geometric row starts at  $r^0 = 1$ , which is why its exponent is  $n + 1$ .  $r$  is the **common ratio** as above, and  $r \neq 1$ .

**The third one is the square of the first**, which is a memorable accident worth using: the sum of the first  $n$  cubes equals the square of the sum of the first  $n$  numbers.

**Two rules that let you take a sum apart**, and they are used constantly:

$$\sum (x_k + y_k) = \sum x_k + \sum y_k$$

$$\sum c x_k = c \sum x_k, \text{ for a constant } c$$

♦ **Memory hook: squares over six, cubes are the square of the first sum.** And every one of these four can be proved by induction, which is section 9.2 and a fair exam question.

### WORKED EXAMPLE

### METHOD · EVALUATING A SUM

**GIVEN** The sum of  $k^2$  for  $k$  running from 1 to 10.

**FIND** Its value, and the value of the sum of  $(3k + 2)$  over the same range.

**FORMULA**  $\sum k^2 = n(n + 1)(2n + 1) / 6$  and  $\sum k = n(n + 1) / 2$ , with the two splitting rules above.

**The squares**, at  $n = 10$ :

$$10 \times 11 \times 21 / 6 = 2310 / 6 = \mathbf{385}$$

**The linear one.** Split it first, which is where the rules earn their place:

$$\begin{aligned} \sum (3k + 2) &= 3 \sum k + \sum 2 \\ &= 3 \times (10 \times 11 / 2) + 10 \times 2 \\ &= 3 \times 55 + 20 \\ &= 165 + 20 = \mathbf{185} \end{aligned}$$

**Note the second term.** Summing the constant 2 ten times gives 20, not 2. Forgetting that a constant inside a sum still gets added once per index is the standard slip here.



## RECALL AND CHECK

1. Define a sequence, and say what a term is.
2. What distinguishes an arithmetic progression from a geometric one? Give one example of each.
3. Write the  $n$ -th term of an arithmetic progression and say what every letter means.
4. Write the sum of the first  $n$  terms of a geometric progression, and state its condition.
5. Find the rule behind 4, 8, 16, 32 and give its 10th term.
6. Evaluate the sum of the first 5 cubes.

### ANSWERS

1. A function from a subset of the integers to a set; a term  $a_n$  is the image of the index  $n$ , that is the number in position  $n$ .
2. Arithmetic **adds** a constant difference  $d$ , geometric **multiplies** by a constant ratio  $r$ . For example 7, 12, 17, 22 with  $d = 5$ , and 4, 8, 16, 32 with  $r = 2$ .
3.  $a_n = a + (n - 1)d$ :  $a$  is the first term,  $d$  the common difference,  $n$  the position, and  $n - 1$  the number of steps taken.
4.  $S_n = a(r^n - 1) / (r - 1)$ , valid for  $r \neq 1$ ; at  $r = 1$  the sum is  $na$ .
5. Geometric,  $a = 4$ ,  $r = 2$ , so  $a_n = 4 \times 2^{n-1} = 2^{n+1}$ , and  $a_{10} = 2048$ .
6.  $(5 \times 6 / 2)^2 = 15^2 = 225$ .

## FILL IN THE GAP

1. In an arithmetic progression the fixed amount added is the common \_\_\_\_\_.
2. In a geometric progression the fixed multiplier is the common \_\_\_\_\_.
3. The  $n$ -th term of a geometric progression is \_\_\_\_\_.
4. An infinite geometric series has a finite sum only when  $r$  lies between \_\_\_\_\_ and \_\_\_\_\_.
5. The sum of the first  $n$  squares is \_\_\_\_\_.
6. 1, 5, 17, 53, 161, 485 follows the rule  $a_n =$  \_\_\_\_\_.

### ANSWERS

1. difference. 2. ratio. 3.  $ar^{n-1}$ . 4.  $-1, 1$ . 5.  $n(n + 1)(2n + 1) / 6$ . 6.  $3a_{n-1} + 2$ , with  $a_1 = 1$ .

## PART NINE • 9.4

# Recurrence Relations

A recurrence defines each term from the terms before it. That is how a recursive program describes its own running time, how the chain letter of section 7.2 grows, and how Fibonacci is written down. Solving one means replacing "look back" with a formula that answers straight away.

By the end of this section you will be able to define a recurrence relation and its order, state initial conditions and say why they are needed, generate terms from a relation, solve a first-order rela-

tion by iteration, solve a homogeneous second-order relation with the characteristic equation in both the distinct-root and repeated-root cases, and recognise a non-homogeneous one.

## ● MUST MEMORISE

## RECURRENCE RELATION, ORDER, INITIAL CONDITIONS

A **recurrence relation** for a sequence  $\{a_n\}$  is an equation expressing  $a_n$  in terms of one or more of the previous terms:

$$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k})$$

The **initial conditions** are the terms given outright, before the relation takes effect. A sequence is a **solution** of the relation if its terms satisfy it.

### WHERE

$a_n$  is the term being defined, and  $a_{n-1}$  and its neighbours are the terms already known.  $f$  is any rule at all: it need not be linear and need not involve every previous term.  $k$  is the **order** of the relation, the number of steps back it reaches. First order looks back one term, second order two. **the initial conditions** are  $k$  values supplied separately, and there must be exactly as many of them as the order, because the relation cannot start without them.

**Why the initial conditions are not optional.** The relation  $a_n = 2a_{n-1}$  describes 1, 2, 4, 8, 16, ... and equally describes 5, 10, 20, 40, ... and 0, 0, 0, 0, .... The relation fixes the **shape**; the initial conditions pick out which sequence of that shape you actually have. A question that gives a recurrence and no initial condition has not specified a sequence.

**Solving** a recurrence means finding a **closed form**: a formula for  $a_n$  in terms of  $n$  alone, needing no earlier terms. The point is cost. Computing  $a_{1000}$  from a first-order relation takes a thousand steps; from a closed form it takes one.

◆ **Memory hook: the relation gives the shape, the initial conditions choose the sequence, and solving removes the looking back.**

## TEACH

## READING THE ORDER OFF A RELATION

| RELATION                  | ORDER           | WHY                                                          |
|---------------------------|-----------------|--------------------------------------------------------------|
| $a_n = 3a_{n-1} + 2$      | first           | reaches back one term                                        |
| $F_n = F_{n-1} + F_{n-2}$ | second          | reaches back two, so it needs two initial values             |
| $T_n = T_{n-1} + T_{n-3}$ | third           | the furthest reach is three, even though $T_{n-2}$ is absent |
| $S_n = S_{n/2} + 1$       | no finite order | the distance back grows with $n$ , so no fixed $k$ covers it |

**The order is the furthest reach, not the number of terms mentioned.** The third row skips  $a_{n-2}$  entirely and is still third order, and it still needs three initial values.

**The last row is worth knowing** because it is the shape of a divide-and-conquer algorithm: halve the problem, do one unit of work. The binary search that section 11.2 writes in C++ has exactly that recurrence, and it is why halving is so much faster than stepping.

## Generating terms, and solving by iteration

### WORKED EXAMPLE

### METHOD · THE TOWER OF HANOI

Three pegs and  $n$  discs of different sizes, stacked smallest on top. Move the whole stack to another peg, one disc at a time, never placing a larger disc on a smaller one. How many moves does it take?

**GIVEN**  $H_n$  = the least number of moves for  $n$  discs. Moving  $n$  discs means: move the top  $n - 1$  to the spare peg, move the largest one across, then move the  $n - 1$  back on top.

**FIND** The recurrence, the first few terms, and a closed form.

**FORMULA**  $H_n = 2H_{n-1} + 1$ , with  $H_1 = 1$ .

Generate the terms before trying to be clever:

| N | $H_N = 2H_{N-1} + 1$ | VALUE | COMPARE WITH $2^N$ |
|---|----------------------|-------|--------------------|
| 1 | given                | 1     | $2 - 1$            |
| 2 | $2 \times 1 + 1$     | 3     | $4 - 1$            |
| 3 | $2 \times 3 + 1$     | 7     | $8 - 1$            |
| 4 | $2 \times 7 + 1$     | 15    | $16 - 1$           |
| 5 | $2 \times 15 + 1$    | 31    | $32 - 1$           |

The pattern is unmistakable, so the conjecture is

$$H_n = 2^n - 1$$

**Now prove it**, because a pattern spotted in five rows is a conjecture, not a theorem. Induction, by section 9.2.

**Base case**,  $n = 1$ : the formula gives  $2^1 - 1 = 1$ , which matches  $H_1 = 1$ .

**Hypothesis**:  $H_k = 2^k - 1$ .

**Step**:  $H_{k+1} = 2H_k + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1$ , which is the formula at  $k + 1$ .  $\square$

**The method just used has a name: solving by iteration.** Write out terms until the pattern is visible, conjecture the closed form, then prove it by induction. For a first-order relation it is usually the fastest route, and the induction is what turns a guess into an answer.

- GIVEN**  $a_n = 3a_{n-1} + 2$  with  $a_1 = 1$ , the sequence 1, 5, 17, 53, 161, 485 of section 9.3.  
**FIND** A closed form for  $a_n$ .  
**FORMULA** Iteration, with one substitution that turns the relation into a plain geometric progression.

**The trick worth learning.** The "+ 2" is what stops this being geometric. Add 1 to both sides and watch it disappear:

$$a_n + 1 = 3a_{n-1} + 3 = 3(a_{n-1} + 1)$$

So the sequence  $b_n = a_n + 1$  satisfies  $b_n = 3b_{n-1}$ , which is geometric with ratio 3 and first term  $b_1 = a_1 + 1 = 2$ . By section 9.3,

$$b_n = 2 \times 3^{n-1}$$

and therefore

$$a_n = 2 \times 3^{n-1} - 1$$

**Check every printed term,** which takes four lines and is worth every one:

| N | $2 \times 3^{N-1} - 1$ | VALUE | THE SEQUENCE SAYS |
|---|------------------------|-------|-------------------|
| 1 | $2 \times 1 - 1$       | 1     | 1                 |
| 2 | $2 \times 3 - 1$       | 5     | 5                 |
| 3 | $2 \times 9 - 1$       | 17    | 17                |
| 4 | $2 \times 27 - 1$      | 53    | 53                |
| 5 | $2 \times 81 - 1$      | 161   | 161               |
| 6 | $2 \times 243 - 1$     | 485   | 485               |

**Note the shape of the trick.** Tower of Hanoi was  $H_n = 2H_{n-1} + 1$  and the same substitution works there:  $H_n + 1 = 2(H_{n-1} + 1)$ , giving  $H_n + 1 = 2^n$  and  $H_n = 2^n - 1$  with no induction needed at all. Any relation  $a_n = ca_{n-1} + b$  with  $c \neq 1$  yields to it: add  $b / (c - 1)$  to both sides.

## Linear homogeneous relations with constant coefficients

### ● MUST MEMORISE

### THE CHARACTERISTIC EQUATION METHOD

A relation is **linear** with **constant coefficients** and **homogeneous** of order  $k$  when it can be written

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

Its **characteristic equation** is obtained by replacing  $a_{n-i}$  with  $x^{k-i}$  and moving everything to one side:

$$x^k - c_1 x^{k-1} - \dots - c_k = 0$$

WHERE

**linear** means every a term appears to the first power on its own: no squares, no products of two terms, no  $a_n$  inside a function. **constant coefficients** means each  $c_i$  is a fixed number, not something depending on  $n$ . **homogeneous** means there is no extra term that does not involve the sequence; a stray "+ 2" makes it non-homogeneous. **x** is an ordinary unknown, and the roots of the characteristic equation are called the **characteristic roots**. **k** is the order, so the characteristic equation has degree  $k$  and the general solution carries  $k$  constants, fixed by the  $k$  initial conditions.

**The two cases you will meet at order 2.** Solve  $x^2 - c_1 x - c_2 = 0$  and look at its roots:

| ROOTS                         | GENERAL SOLUTION          |
|-------------------------------|---------------------------|
| two distinct roots $x_1, x_2$ | $a_n = A x_1^n + B x_2^n$ |
| one repeated root $x_1$       | $a_n = (A + Bn) x_1^n$    |

**Then find A and B** by substituting the two initial conditions and solving the resulting pair of simultaneous equations.

**Why powers of a root work at all**, which is worth one line in an answer: trying  $a_n = x^n$  in the relation and dividing through by  $x^{n-k}$  gives exactly the characteristic equation. So the guess "a power of something" is not a rabbit out of a hat; the characteristic equation is precisely the condition that makes it work.

◆ **Memory hook: degree equals order, roots give powers, and a repeated root buys an extra n.**

**GIVEN**  $a_n = 5a_{n-1} - 6a_{n-2}$ , with  $a_0 = 1$  and  $a_1 = 4$ .

**FIND** A closed form for  $a_n$ .

**FORMULA** Characteristic equation  $x^2 - 5x + 6 = 0$ , then  $a_n = A x_1^n + B x_2^n$ .

Step 1. Write the characteristic equation

$$x^2 - 5x + 6 = 0$$

**Watch the signs.** The relation subtracts  $6a_{n-2}$ , so moving it across makes the constant term  $+6$ .

Step 2. Find the roots

$$(x - 3)(x - 2) = 0, \text{ so } x_1 = 3 \text{ and } x_2 = 2$$

Two real, distinct roots, so this is the first case.

Step 3. Write the general solution

$$a_n = A \times 3^n + B \times 2^n$$

Step 4. Use the initial conditions

$$n = 0: A + B = 1$$

$$n = 1: 3A + 2B = 4$$

From the first,  $B = 1 - A$ . Substituting into the second:  $3A + 2(1 - A) = 4$ , so  $A + 2 = 4$  and  **$A = 2$** , hence  **$B = -1$** .

Step 5. The answer

$$a_n = 2 \times 3^n - 2^n$$

**Check it against the relation**, which is the only way to know the signs came out right:

| N | FROM THE RELATION                 | FROM THE FORMULA |
|---|-----------------------------------|------------------|
| 0 | given, 1                          | $2 - 1 = 1$      |
| 1 | given, 4                          | $6 - 2 = 4$      |
| 2 | $5 \times 4 - 6 \times 1 = 14$    | $18 - 4 = 14$    |
| 3 | $5 \times 14 - 6 \times 4 = 46$   | $54 - 8 = 46$    |
| 4 | $5 \times 46 - 6 \times 14 = 146$ | $162 - 16 = 146$ |

**GIVEN**  $a_n = 10a_{n-1} - 25a_{n-2}$ , with  $a_0 = 3$  and  $a_1 = 17$ .

**FIND** A closed form for  $a_n$ .

**FORMULA** Characteristic equation  $x^2 - 10x + 25 = 0$ , then  $a_n = (A + Bn)x_1^n$ .

**The characteristic equation** is  $x^2 - 10x + 25 = 0$ , that is  $(x - 5)^2 = 0$ , so there is a **single** root  $x_1 = 5$ , repeated.

**Why the general solution needs the extra  $n$ .** With only one root there is only one power to build from, so  $A \times 5^n$  carries just one constant and cannot satisfy two initial conditions in general. The second independent solution is  $n \times 5^n$ , so

$$a_n = (A + Bn) \times 5^n$$

**Fit the constants.**

$$n = 0: A = 3$$

$$n = 1: (A + B) \times 5 = 17, \text{ so } A + B = 17 / 5, \text{ so } B = 17/5 - 3 = 2/5$$

**The answer**, written twice because the second form is tidier and markers accept either:

$$a_n = (3 + (2/5)n) \times 5^n = 3 \times 5^n + 2n \times 5^{n-1}$$

**Check.**

| N | FROM THE RELATION                   | FROM THE FORMULA  |
|---|-------------------------------------|-------------------|
| 0 | given, 3                            | $3 \times 1 = 3$  |
| 1 | given, 17                           | $15 + 2 = 17$     |
| 2 | $10 \times 17 - 25 \times 3 = 95$   | $75 + 20 = 95$    |
| 3 | $10 \times 95 - 25 \times 17 = 525$ | $375 + 150 = 525$ |

**Do not be alarmed by a fractional constant.** A and B are whatever the initial conditions force them to be;  $2/5$  is a perfectly good answer, and the check above confirms it produces integers throughout.

The Fibonacci sequence is  $F_n = F_{n-1} + F_{n-2}$ . The official course text takes  $F_0 = 1$  and  $F_1 = 1$ , giving

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

Other books start  $F_0 = 0$ ,  $F_1 = 1$ , which shifts every index by one. As always: **say which you are using**.

**The characteristic equation** is  $x^2 - x - 1 = 0$ . This one does not factorise into whole numbers, so use the **quadratic formula**: the solutions of  $ax^2 + bx + c = 0$  are

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

Here  $a = 1$ ,  $b = -1$  and  $c = -1$ , so  $b^2 - 4ac = 1 + 4 = 5$  and

$$x = (1 \pm \sqrt{5}) / 2$$

The larger of them, about 1.618, is the golden ratio, and it is why Fibonacci numbers grow at roughly 1.618 times per step: the ratio  $13 / 8 = 1.625$  and  $34 / 21$  is about 1.619.

**The point for this course** is not the closed form, which is unpleasant, but the two structural facts: the relation is **second order**, so it needs **two** initial conditions, and irrational characteristic roots can produce a sequence of whole numbers. Both are fair short-answer questions.

A relation is **non-homogeneous** when it carries an extra term that does not involve the sequence:

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + f(n)$$

Its solution has two parts added together:

- ▶ the solution of the **associated homogeneous** relation, the same relation with  $f(n)$  removed, found by the characteristic equation above;
- ▶ a **particular solution**, any one sequence that satisfies the full relation, found by trying a form that matches  $f(n)$ : a constant for a constant  $f$ , a linear expression for a linear  $f$ , a multiple of  $t^n$  for  $f(n) = t^n$ .

**You have already done one.**  $a_n = 3a_{n-1} + 2$  is non-homogeneous with  $f(n) = 2$ . Its associated homogeneous relation  $a_n = 3a_{n-1}$  has solution  $A \times 3^n$ , and a constant particular solution  $a_n = p$  requires  $p = 3p + 2$ , so  $p = -1$ . The general solution is  $A \times 3^n - 1$ , and  $a_1 = 1$  gives  $3A - 1 = 1$ , so  $A = 2/3$  and  $a_n = (2/3) \times 3^n - 1 = 2 \times 3^{n-1} - 1$ , exactly the answer the substitution trick gave.

**Two routes, one answer.** That agreement is the sort of check worth showing: it costs three lines and demonstrates the method rather than the memory.

**One caution on trial solutions.** If the form you try is already a solution of the homogeneous relation, it cannot also fix the extra term, and you must multiply the trial by  $n$ . That is the same phenomenon as the repeated root above, and for the same reason.



| SITUATION                               | RECURRENCE                | SECTION |
|-----------------------------------------|---------------------------|---------|
| People at level $n$ of the chain letter | $s_n = 4s_{n-1}, s_0 = 1$ | 7.2     |
| Leaves of a full $m$ -ary tree by level | $l_n = m l_{n-1}$         | 7.2     |
| Subsets of an $n$ -element set          | $p_n = 2p_{n-1}, p_0 = 1$ | 2.1     |
| Rows in a truth table with $n$ letters  | $r_n = 2r_{n-1}, r_0 = 1$ | 1.1     |
| Comparisons made by binary search       | $T_n = T_{n/2} + 1$       | 11.2    |

**Four of those five are the same relation with a different letter**, and each solves to a power:  $4^n, m^n, 2^n, 2^n$ . Noticing that a counting problem is a first-order geometric recurrence is often faster than the counting argument, and it is a good sentence to write when a question asks you to justify a formula you already know.

## RECALL AND CHECK

1. Define a recurrence relation.
2. What are initial conditions, and why does a relation need them?
3. How many initial conditions does a relation of order  $k$  need?
4. What does it mean to solve a recurrence relation?
5. Write the characteristic equation of  $a_n = 5a_{n-1} - 6a_{n-2}$  and give its roots.
6. What changes in the general solution when the characteristic equation has a repeated root?
7. Distinguish a homogeneous relation from a non-homogeneous one.

## ANSWERS

1. An equation expressing  $a_n$  in terms of one or more earlier terms of the same sequence.
2. The terms given outright before the relation applies. Without them the relation fixes only the shape of the sequence, and infinitely many sequences share that shape.
3. Exactly  $k$ .
4. Finding a closed form: a formula for  $a_n$  in terms of  $n$  alone, requiring no earlier terms.
5.  $x^2 - 5x + 6 = 0$ , with roots 3 and 2.
6. The solution becomes  $(A + Bn)x_1^n$ : the repeated root buys a factor of  $n$  on the second constant.
7. A homogeneous relation involves only terms of the sequence; a non-homogeneous one carries an extra term  $f(n)$  that does not.

### FILL IN THE GAP

1. The number of steps a relation reaches back is its \_\_\_\_\_.
2. The Tower of Hanoi with  $n$  discs takes \_\_\_\_\_ moves.
3. The characteristic equation of  $a_n = 10a_{n-1} - 25a_{n-2}$  has the single root \_\_\_\_\_.
4. Fibonacci is a relation of order \_\_\_\_\_.
5.  $a_n = 3a_{n-1} + 2$  with  $a_1 = 1$  solves to \_\_\_\_\_.

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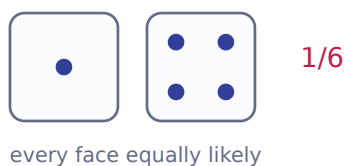
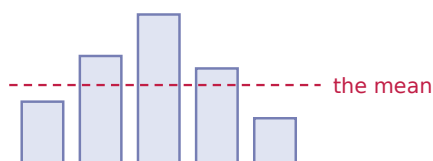
### ANSWERS

1. order. 2.  $2^n - 1$ . 3. 5. 4. two. 5.  $2 \times 3^{n-1} - 1$ .

## PART TEN

# Descriptive Statistics and Discrete Probability

*Two topics named on the syllabus that no past paper has yet asked about. This part is short by design and complete by obligation: it covers them properly, so that a first appearance cannot cost you, and it shows how probability is simply the counting of Part Five divided through by a total.*



# Descriptive Statistics

Given a list of numbers, describe it in one or two numbers. Where is the middle, and how spread out is it? Five measures answer both questions, and each is a formula you can compute by hand in under a minute.

By the end of this section you will be able to compute the mean, median, mode, range, variance and standard deviation of a data set, choose between the mean and the median when the data is skewed, work from a frequency table, and say what each measure does and does not tell you.

## NAMED ON THE SYLLABUS, MISSING FROM THE COURSE TEXT

The course outline lists a module called Descriptive Statistics. Its unit contains no statistics: it is propositional logic from the first line to the last, the same material this manual teaches in Part One. Probed against the whole text, the words median, variance, standard deviation, quartile, arithmetic mean, frequency distribution and histogram occur **ZERO** times between them. No past paper in the five available sessions has asked a statistics question either. So the risk is small but not nil: the topic is named, and a named topic can appear. This section teaches it from zero and keeps it short, which is the honest proportion.

## Measures of the middle

### ● MUST MEMORISE

### MEAN, MEDIAN, MODE

Three answers to "where is the centre", and they can disagree violently on the same data.

| MEASURE       | HOW TO COMPUTE IT                                                                    | WHAT IT IS                   |
|---------------|--------------------------------------------------------------------------------------|------------------------------|
| <b>Mean</b>   | add every value, divide by how many there are                                        | the balance point            |
| <b>Median</b> | sort, then take the middle value; with an even count take the mean of the middle two | the halfway mark by position |
| <b>Mode</b>   | the value occurring most often                                                       | the most popular value       |

In symbols, for values  $x_1, x_2, \dots, x_n$ :

$$\bar{x} = (\sum x_i) / n$$

#### WHERE

$x_i$  is the **i-th data value**, so the subscript is a position in the list, as with a sequence in section 9.3. **n** is the **number of values** in the set, not the largest of them.  $\sum x_i$  is the **total** of all the values, the summation of section 9.3 with  $i$  running from 1 to  $n$ .  $\bar{x}$  is written  $x$  with a bar above it and is read "**x bar**". It is the standard symbol for the mean of a sample.

**The mode may not be unique**, and it may not exist in any useful sense: a set where every value occurs once has no mode, and a set with two equally popular values has two. Say so rather than picking one.

**The median needs the data sorted first.** Taking the middle of an unsorted list is the most common error in this topic, and it produces a number that means nothing.

◆ **Memory hook: mean is the average, median is the middle, mode is the most.** All three start with m, and so do average, middle and most, in the same order.

## Measures of spread

### ● MUST MEMORISE

### RANGE, VARIANCE, STANDARD DEVIATION

The centre alone hides everything. The sets 19, 20, 21 and 0, 20, 40 have the same mean and nothing else in common.

**range = largest value – smallest value**

$$\sigma^2 = (\sum (x_i - \bar{x})^2) / n$$

$$\sigma = \sqrt{(\sigma^2)}$$

#### WHERE

$x_i - \bar{x}$  is a **deviation**: how far one value sits from the mean. Deviations are positive above the mean and negative below it. **the square** is there because the deviations always add to exactly zero, so they must be made positive before summing; squaring does that and also weights the far-out values more heavily.  $\sigma^2$  is the **variance**, the mean of the squared deviations. Its units are the square of the data's units, which is why it is rarely quoted on its own.  $\sigma$  is the **standard deviation**, the square root of the variance, back in the same units as the data. The symbol is a lower-case Greek sigma, and it is not the summation sign, which is the capital. **n** is the number of values.

**Dividing by n or by n – 1.** The formula above divides by n and describes the data set you have. If the data is a **sample** drawn from a larger population and you want to estimate that population's spread, divide by n – 1 instead and write  $s^2$  and s. Both are correct answers to different questions, so **state which you are computing**; this is the same discipline as the indexing convention in section 9.3.

◆ **Memory hook: square the distances, average them, then take the root to come home.** The root is only there to undo the squaring.

**GIVEN** Ten test scores: 21, 15, 33, 12, 21, 18, 24, 15, 20, 21.

**FIND** Mean, median, mode, range, variance and standard deviation.

**FORMULA**  $\bar{x} = (\sum x_i) / n$  and  $\sigma^2 = (\sum (x_i - \bar{x})^2) / n$ , with  $n = 10$ .

**Step 1. Sort the data**, because three of the six measures need it and none of them is harmed by it:

12, 15, 15, 18, 20, 21, 21, 21, 24, 33

**Step 2. The mean.**

$$\sum x_i = 12 + 15 + 15 + 18 + 20 + 21 + 21 + 21 + 24 + 33 = \mathbf{200}$$

$$\bar{x} = 200 / 10 = \mathbf{20}$$

**Step 3. The median.** There are 10 values, an even count, so the middle two are the 5th and 6th: 20 and 21.

$$\text{median} = (20 + 21) / 2 = \mathbf{20.5}$$

**Step 4. The mode.** Counting occurrences: 12 once, 15 twice, 18 once, 20 once, 21 three times, 24 once, 33 once.

$$\text{mode} = \mathbf{21}$$

**Step 5. The range.**

$$\text{range} = 33 - 12 = \mathbf{21}$$

**Step 6. The variance.** Tabulate the deviations, because doing this in your head is how arithmetic errors happen.

| $x_i$        | $x_i - 20$ | $(x_i - 20)^2$ |
|--------------|------------|----------------|
| 12           | -8         | 64             |
| 15           | -5         | 25             |
| 15           | -5         | 25             |
| 18           | -2         | 4              |
| 20           | 0          | 0              |
| 21           | 1          | 1              |
| 21           | 1          | 1              |
| 21           | 1          | 1              |
| 24           | 4          | 16             |
| 33           | 13         | 169            |
| <b>Total</b> | <b>0</b>   | <b>306</b>     |

**Check the middle column adds to zero** before going on. It always does, and if yours does not, the mean is wrong and everything after it would be too.

$$\sigma^2 = 306 / 10 = \mathbf{30.6}$$

**Step 7. The standard deviation.**

$$\sigma = \sqrt{30.6} \approx 5.53$$

**If the sample form is wanted instead**, divide by  $n - 1 = 9$ :  $s^2 = 306 / 9 = 34$  and  $s = \sqrt{34} \approx 5.83$ . State which one you have given.

**Reading the answer back.** Scores centre on about 20 and typically sit within about 5.5 of it. The one score of 33 is more than two standard deviations out, which is what pulls the mean above the median. Saying that sentence is what turns six numbers into a description.

⚠ **TRAP**

**WHEN THE MEAN IS THE WRONG ANSWER**

In the data above the mean is 20 and the median 20.5. They are close because one outlier is mild. Change the 33 to 133 and watch:

new total = 300, so the mean becomes **30**

the median is unchanged at **20.5**, because the middle two values did not move

The mean has left the data: nine of the ten scores are now below it. **The median is resistant to outliers and the mean is not.** That is the single most useful fact in this section, and the reason incomes and house prices are reported as medians.

**What to write in an exam.** If a data set has an extreme value, say which measure you are quoting and why. "The median is quoted here because the value 133 distorts the mean" earns the mark that "average = 30" does not.

**The mode's own trap:** it is the only one of the three that works on data with no numeric order at all. The most common programming language in a class has a mode, but no mean.

**GIVEN** The number of programming errors found in each of 20 submitted assignments, as a frequency table.

**FIND** The mean, median and mode.

**FORMULA**  $\bar{x} = (\sum f_i x_i) / (\sum f_i)$ , where  $f_i$  is the frequency of the value  $x_i$ .

| ERRORS $x_i$ | FREQUENCY $f_i$ | $f_i x_i$ | RUNNING COUNT |
|--------------|-----------------|-----------|---------------|
| 0            | 3               | 0         | 3             |
| 1            | 5               | 5         | 8             |
| 2            | 7               | 14        | 15            |
| 3            | 3               | 9         | 18            |
| 4            | 2               | 8         | 20            |
| <b>Total</b> | <b>20</b>       | <b>36</b> |               |

**The mean.**

$$\bar{x} = 36 / 20 = \mathbf{1.8} \text{ errors per assignment}$$

**The median.** With 20 values the middle two are the 10th and 11th. The running count reaches 8 after the value 1 and 15 after the value 2, so both the 10th and the 11th value are 2.

$$\text{median} = \mathbf{2}$$

**The mode** is the value with the largest frequency, which is **2**, occurring 7 times.

**Note what the frequency column saves you.** Writing out all twenty values and adding them would give the same 36; the table is the same arithmetic with the repetitions collected, exactly as the arrangements formula of section 5.2 collects repeated letters.

**A mean need not be a possible value.** No assignment had 1.8 errors. The mean describes the set, not any member of it, and saying so pre-empts the marker's follow-up question.



## RECALL AND CHECK

1. Define the mean, the median and the mode.
2. What must you do to the data before finding the median?
3. Why are deviations squared before they are averaged?
4. What is the difference between the variance and the standard deviation?
5. Which measure of centre resists an outlier?
6. For 12, 15, 15, 18, 20, 21, 21, 21, 24, 33, give the mean, median, mode and range.

### ANSWERS

1. Mean: the total divided by the count. Median: the middle value once sorted, or the mean of the middle two when the count is even. Mode: the most frequent value.
2. Sort it.
3. Because the deviations sum to zero, so they must be made positive first; squaring does that and weights the extremes more.
4. The standard deviation is the square root of the variance, so it is in the same units as the data.
5. The median.
6. Mean 20, median 20.5, mode 21, range 21.

## FILL IN THE GAP

1. The \_\_\_\_\_ is the total divided by the count.
2. With an even number of values, the median is the mean of the \_\_\_\_\_.
3. The sum of all deviations from the mean is always \_\_\_\_\_.
4. The standard deviation is the \_\_\_\_\_ of the variance.
5. Range = \_\_\_\_\_ - \_\_\_\_\_.

### ANSWERS

1. mean. 2. middle two. 3. zero. 4. square root. 5. largest value, smallest value.

## PART TEN • 10.2

# Discrete Probability

*Probability is Part Five with a division at the end. Count the outcomes you want, count all the outcomes there are, divide. Everything else in this section is a rule for handling two events at once, and every one of those rules is a set identity from Part Two wearing a P.*

*By the end of this section you will be able to define a random experiment, sample space and event; distinguish independent, dependent, mutually exclusive and equally likely events; state the axioms; apply the complement, addition, multiplication and conditional rules; use Bayes' theorem; and recognise which counting rule of Part Five a probability question is really asking for.*

### IN THE COURSE TEXT, NEVER YET EXAMINED

Discrete probability is a module of the course text and is properly written there, unlike the statistics module. No paper in the five available sessions has asked a probability question. It is taught here in full anyway, and two errors printed in the source module are flagged where they occur, because a reader who memorises them would answer confidently and wrongly.

## The vocabulary

### ● MUST MEMORISE

### EXPERIMENT, SAMPLE SPACE, EVENT

| TERM                     | DEFINITION                                                                                                         | EXAMPLE:<br>ROLLING ONE<br>DIE  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Random experiment</b> | a procedure that can be repeated, whose outcome is not determined in advance and is not affected by the repetition | the roll itself                 |
| <b>Outcome</b>           | one possible result of the experiment                                                                              | 4                               |
| <b>Sample space S</b>    | the <b>set</b> of all possible outcomes                                                                            | {1, 2, 3, 4, 5, 6}              |
| <b>Event E</b>           | a <b>subset</b> of the sample space                                                                                | "an even number"<br>= {2, 4, 6} |

**A sample space is a set and an event is a subset of it.** That one sentence imports the whole of Part Two: the union of two events is the event that at least one happens, the intersection is the event that both happen, and the complement is the event that it does not happen. Nothing new has to be learned about combining events, only about counting them.

For an experiment whose outcomes are all **equally likely**,

$$P(E) = n(E) / n(S)$$

WHERE

**P(E)** is the **probability** of the event E, a number between 0 and 1 inclusive. **n(E)** is the **number of favourable outcomes**, the cardinality of the event, in the notation of section 2.1. **n(S)** is the **total number of outcomes**, the size of the sample space. **equally likely** is a condition, not decoration: the formula is wrong the moment some outcomes are more probable than others, and a loaded die is the standard counterexample.

◆ **Memory hook: favourable over total, and only when every outcome carries the same weight.**

| KIND                      | MEANING                                                         | EXAMPLE                                        |
|---------------------------|-----------------------------------------------------------------|------------------------------------------------|
| <b>Independent</b>        | the outcome of one does not affect the probability of the other | two separate coin tosses                       |
| <b>Dependent</b>          | the outcome of one changes the probability of the other         | drawing two cards without replacing the first  |
| <b>Mutually exclusive</b> | they cannot both happen; their intersection is empty            | a single toss landing heads, and landing tails |
| <b>Equally likely</b>     | each has the same probability                                   | the six faces of a fair die                    |

**Mutually exclusive in set language** is  $A \cap B = \emptyset$ , which is precisely the disjoint sets of section 2.1. That is why the word **disjoint** is used for the same idea.

**"Without replacement" is the phrase that signals dependence.** Take a card and put it back, and the second draw faces the same deck; keep it, and the deck has changed. Exam questions turn on that phrase, and it is worth underlining it when you see it.

## ⚠ TRAP

## MUTUALLY EXCLUSIVE IS NOT INDEPENDENT

The official course text writes, under its probability axioms, "if the occurrence of one event is not influenced by another event, they are called mutually exclusive or disjoint". **That sentence defines independence and attaches the wrong name to it.** The two ideas are not merely different; on events that can happen at all they are **opposites**.

|                    | MUTUALLY EXCLUSIVE         | INDEPENDENT                                     |
|--------------------|----------------------------|-------------------------------------------------|
| Means              | they cannot both happen    | one happening tells you nothing about the other |
| In symbols         | $P(A \cap B) = 0$          | $P(A \cap B) = P(A) \times P(B)$                |
| Knowing A happened | tells you B did <b>not</b> | tells you nothing at all                        |

**Why they are opposites.** If A and B are mutually exclusive and both have non-zero probability, then learning that A happened tells you B certainly did not: that is the strongest possible influence, so they cannot be independent. Formally,  $P(A \cap B) = 0$  while  $P(A) \times P(B) > 0$ , so the independence equation fails.

**The same module prints a second error,** in its properties list: "for two non-disjoint events A and B, the probability of the union of two events  $P(A \cup B) = P(A) + P(B)$ ". That formula is the one for **disjoint** events. For non-disjoint events the overlap has been counted twice and must come off, exactly as in section 5.3:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The module's own formula list gives this correct version three paragraphs later, so the two statements contradict each other. Use the one with the subtraction; it is inclusion and exclusion, and it is right.

## The axioms and the rules

### ● MUST MEMORISE

### THE AXIOMS AND THE FIVE RULES

**The axioms**, which every probability obeys:

- ▶  $0 \leq P(E) \leq 1$  for every event  $E$
- ▶  $P(\emptyset) = 0$  for the impossible event, and  $P(S) = 1$  for the certain event
- ▶ if  $E_1, E_2, \dots, E_n$  are pairwise mutually exclusive, then  $P(E_1 \cup \dots \cup E_n) = P(E_1) + \dots + P(E_n)$

**The rules**, which are the axioms put to work:

| RULE               | FORMULA                                   | WHEN                                  |
|--------------------|-------------------------------------------|---------------------------------------|
| Complement         | $P(A') = 1 - P(A)$                        | always                                |
| Addition           | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ | always                                |
| Addition, disjoint | $P(A \cup B) = P(A) + P(B)$               | only when $A \cap B = \emptyset$      |
| Multiplication     | $P(A \cap B) = P(A) \times P(B)$          | only when $A$ and $B$ are independent |
| Conditional        | $P(B   A) = P(A \cap B) / P(A)$           | always, provided $P(A) > 0$           |

#### WHERE

**A'** is the **complement** of  $A$ , the event that  $A$  does not happen, written with the same prime as the set complement of section 2.2. **A  $\cup$  B** is the event that **at least one** of  $A, B$  happens, and **A  $\cap$  B** is the event that **both** happen. **P(B | A)** is the **conditional probability** of  $B$  **given**  $A$ : the probability of  $B$  once you already know  $A$  has happened. The bar is read "given" and is not division. **P(A) > 0** is required because the formula divides by it, and conditioning on something impossible is meaningless.

**The general multiplication rule**, which works whether or not the events are independent and is just the conditional rule rearranged:

$$P(A \cap B) = P(A) \times P(B | A)$$

and independence is exactly the case where  $P(B | A) = P(B)$ , which collapses it to the simple form.

◆ **Memory hook: or adds, and multiplies, not subtracts the overlap, given divides.**

| PROBABILITY                               | SETS (PART TWO)                           | COUNTING (PART FIVE)         |
|-------------------------------------------|-------------------------------------------|------------------------------|
| $P(A') = 1 - P(A)$                        | $A' = U - A$                              | the subtraction rule, 5.1    |
| $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ | $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ | inclusion and exclusion, 5.3 |
| $P(A \cap B) = P(A)P(B)$                  | $n(A \times B) = n(A)n(B)$                | the product rule, 5.1        |

Divide every count in the middle or right column by  $n(S)$  and the left column appears.

**Probability is not a new subject; it is counting normalised so the total is 1.** If a probability question defeats you, answer the counting question instead and divide at the end.

## Worked, from the simplest upward

### WORKED EXAMPLE

### METHOD · COINS, DICE AND CARDS

**GIVEN** A fair coin, a fair die, and a standard deck of 52 cards.

**FIND** Several basic probabilities, each by counting the sample space first.

**FORMULA**  $P(E) = n(E) / n(S)$ , the outcomes being equally likely in each case.

**One coin.**  $S = \{H, T\}$ , so  $n(S) = 2$  and  $P(\text{heads}) = 1 / 2$ .

**One die.**  $S = \{1, 2, 3, 4, 5, 6\}$ , so  $n(S) = 6$ .

$$P(\text{any given face}) = 1/6$$

$$P(\text{even}) = n(\{2, 4, 6\}) / 6 = 3 / 6 = 1/2$$

$$P(\text{odd}) = 3 / 6 = 1/2$$

**One card from 52.**

$$P(\text{ace}) = 4 / 52 = 1/13$$

$$P(\text{diamond}) = 13 / 52 = 1/4$$

$$P(\text{red queen}) = 2 / 52 = 1/26$$

**Three coin tosses, exactly two heads.** The sample space is every sequence of three results, so by the product rule of section 5.1 it has  $2^3 = 8$  outcomes:

HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Exactly two heads occurs in HHT, HTH and THH, which is 3 outcomes, so

$$P(\text{exactly two heads}) = 3/8$$

**The counting shortcut**, for when there are too many outcomes to list: choosing which 2 of the 3 positions hold the heads is  $C(3, 2) = 3$ , by section 5.2. The same 3, obtained without writing out the sample space, which is the move you need when there are ten tosses rather than three.

**The addition rule in action.** One card,  $P(\text{ace or diamond})$ . The events overlap in the ace of diamonds, so

$$P(\text{ace} \cup \text{diamond}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$$

Adding without subtracting would give  $17/52$  and would count the ace of diamonds twice.

- GIVEN** (a) A jar of 7 red, 3 green and 4 blue marbles, one drawn at random. (b) A committee of 4 chosen from 6 boys and 4 girls.
- FIND** (a)  $P(\text{the marble is not blue})$ . (b)  $P(\text{the committee contains at least one boy})$ .
- FORMULA**  $P(A') = 1 - P(A)$ , and  $C(n, r)$  from section 5.2 for the counting.

(a) The marbles

The total is  $7 + 3 + 4 = 14$  marbles.

**Directly:** non-blue means red or green, which is  $7 + 3 = 10$  marbles, so

$$P(\text{not blue}) = 10 / 14 = 5/7$$

**By the complement,** which is the same answer by a shorter route:

$$P(\text{blue}) = 4 / 14 = 2/7, \text{ so } P(\text{not blue}) = 1 - 2/7 = 5/7$$

(b) At least one boy

**"At least one" is the standard signal to use the complement,** because the opposite of "at least one boy" is the single simple case "no boys at all". Section 5.2 answered the counting version of this exact question from the 2025/2026 paper; here it is as a probability.

$$\text{total committees} = C(10, 4) = 210$$

$$\text{committees with no boy} = C(4, 4) = 1$$

$$\text{committees with at least one boy} = 210 - 1 = 209$$

so

$$P(\text{at least one boy}) = 209 / 210$$

**Read the answer back.** That is about 0.995, which is worth saying: with six boys among ten candidates, a committee of four with no boy at all is nearly impossible, and there is in fact exactly one such committee, the four girls. A probability that close to 1 should agree with your intuition, and if it does not, recheck the counting.

- GIVEN** (a) Half of all teenagers own a cycle, and 30 per cent own both a bike and a cycle. (b) Six good and three defective laptops are mixed up and tested one at a time at random, without replacement.
- FIND** (a) The probability that a teenager owns a bike given that the teenager owns a cycle. (b) The probability that the first two tested are both defective.
- FORMULA**  $P(B | A) = P(A \cap B) / P(A)$ , and the general multiplication rule  $P(A \cap B) = P(A) \times P(B | A)$ .

(a) The conditional probability

Let A be "owns a cycle" and B be "owns a bike". We are given  $P(A) = 0.5$  and  $P(A \cap B) = 0.3$ .

$$P(B | A) = P(A \cap B) / P(A) = 0.3 / 0.5 = \mathbf{0.6}$$

**So 60 per cent** of cycle owners also own a bike.

**What conditioning did.** The 30 per cent was measured against **all** teenagers; the answer 60 per cent is measured against cycle owners only. Conditioning shrinks the sample space from S to A, and dividing by  $P(A)$  is exactly that shrinking. Say that sentence and the formula stops being something to memorise.

(b) Two defectives in a row

There are  $6 + 3 = 9$  laptops, of which 3 are defective. Let A be "the first is defective" and B be "the second is defective".

$$P(A) = 3 / 9$$

**Now the pool has changed**, because the first laptop was not replaced: 8 remain and 2 of them are defective. So

$$P(B | A) = 2 / 8$$

and therefore

$$P(A \cap B) = P(A) \times P(B | A) = (3/9) \times (2/8) = 6 / 72 = \mathbf{1/12}$$

**The trap this question sets** is using  $P(A) \times P(B) = (3/9) \times (3/9) = 1/9$ , which treats the draws as independent. They are not, because the first laptop is not put back, and the phrase "tested one by one" is what tells you so.

**Cross-check by counting**, in the manner of section 5.2: the number of ways to choose 2 defectives from 3 is  $C(3, 2) = 3$ , and the number of ways to choose any 2 from 9 is  $C(9, 2) = 36$ , giving  $3 / 36 = 1/12$ . The two methods agree.



## Bayes' theorem

### ● MUST MEMORISE

### BAYES' THEOREM

Conditional probability runs one way; Bayes' theorem turns it round.

$$P(A | B) = P(B | A) \times P(A) / P(B)$$

#### WHERE

$P(A)$  is the probability of A **before** the evidence, sometimes called the prior.  $P(B | A)$  is how likely the evidence B is **when** A holds.  $P(B)$  is how likely the evidence is **overall**, and it is usually the part you must compute rather than the part you are given.  $P(A | B)$  is the probability of A **after** the evidence, which is what the question wants.

**Computing  $P(B)$  is the real work.** If A and A' are the only two possibilities, then B happens either with A or with A', and those two cases are mutually exclusive, so

$$P(B) = P(B | A) \times P(A) + P(B | A') \times P(A')$$

which is the axiom of adding disjoint cases, applied to a split of the sample space.

◆ **Memory hook: Bayes reverses the bar.** You are given "evidence given cause" and you want "cause given evidence".

**GIVEN** 60 per cent of a data centre's machines come from supplier A and 40 per cent from supplier B. In the first year 2 per cent of A's machines fail and 5 per cent of B's do. A machine is found to have failed.

**FIND** The probability that it came from supplier B.

**FORMULA**  $P(B | F) = P(F | B) \times P(B) / P(F)$ , with  $P(F) = P(F | A)P(A) + P(F | B)P(B)$ .

**Name everything first**, which is where these questions are won or lost:

$$P(A) = 0.6, \quad P(B) = 0.4$$

$$P(F | A) = 0.02, \quad P(F | B) = 0.05$$

**Step 1. The overall failure rate.**

$$\begin{aligned} P(F) &= 0.02 \times 0.6 + 0.05 \times 0.4 \\ &= 0.012 + 0.020 \\ &= \mathbf{0.032} \end{aligned}$$

**Step 2. Reverse the bar.**

$$P(B | F) = P(F | B) \times P(B) / P(F) = 0.05 \times 0.4 / 0.032 = 0.020 / 0.032 = \mathbf{0.625}$$

**Answer: 62.5 per cent.**

**Why the answer is interesting.** Supplier B provides only 40 per cent of the machines but accounts for 62.5 per cent of the failures, because its failure rate is two and a half times higher. The evidence has moved the probability from 0.4 to 0.625, and that movement is exactly what Bayes' theorem measures.

**Check with the complement.**  $P(A | F)$  should be  $0.012 / 0.032 = 0.375$ , and  $0.625 + 0.375 = 1$ . Given that the machine failed, it came from one supplier or the other, so the two conditional probabilities must add to 1. That check costs one line and catches an arithmetic slip immediately.

**Now redo it:** if the split were 50 per cent each with the same failure rates,  $P(F) = 0.02 \times 0.5 + 0.05 \times 0.5 = 0.035$  and  $P(B | F) = 0.025 / 0.035$ , which is about 0.714.

A **random variable** assigns a number to each outcome of an experiment. It is a function on the sample space, in the sense of section 4.1, and not a variable in the algebraic sense at all.

| KIND              | TAKES                                  | EXAMPLES                                                                                                       |
|-------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Discrete</b>   | countable values, such as 0, 1, 2, ... | the number of heads in three tosses; the number of cars in a car park; the number of correct answers on a test |
| <b>Continuous</b> | every value in a range                 | a person's height; the time a reaction takes; a temperature                                                    |

**Discrete is the one this course is about**, which is what the word discrete in the course title means: countable, separated values rather than a smooth range. Every structure in this book, from a set to a graph to a tree, is discrete in exactly that sense.

## RECALL AND CHECK

1. Define a sample space and an event, and say how they are related.
2. Write the formula for the probability of an event with equally likely outcomes, and state its condition.
3. Distinguish mutually exclusive from independent events.
4. State the addition rule and say when the subtraction term may be dropped.
5. Write the conditional probability formula and say what the bar means.
6. A card is drawn from 52. Give  $P(\text{ace})$ ,  $P(\text{diamond})$  and  $P(\text{ace or diamond})$ .
7. Three good and six bad chips are tested one by one. What is the probability the first two are both good?

### ANSWERS

1. The sample space  $S$  is the set of all possible outcomes; an event is a subset of  $S$ .
2.  $P(E) = n(E) / n(S)$ , valid only when every outcome is equally likely.
3. Mutually exclusive means they cannot both occur,  $P(A \cap B) = 0$ . Independent means one gives no information about the other,  $P(A \cap B) = P(A)P(B)$ . For events of non-zero probability the two are opposites.
4.  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ ; the last term may be dropped only when the events are disjoint, since it is then zero.
5.  $P(B | A) = P(A \cap B) / P(A)$ ; the bar is read "given" and restricts the sample space to  $A$ .
6.  $1/13$ ,  $1/4$  and  $4/13$ .
7.  $(3/9) \times (2/8) = 1/12$ .

## FILL IN THE GAP

1. An event is a \_\_\_\_\_ of the sample space.
2. Every probability lies between \_\_\_\_\_ and \_\_\_\_\_.
3.  $P(A') =$  \_\_\_\_\_.
4. Two events are independent when  $P(A \cap B) =$  \_\_\_\_\_.
5. The phrase "without replacement" tells you the events are \_\_\_\_\_.
6. "At least one" is the signal to use the \_\_\_\_\_.

### ANSWERS

1. subset. 2. 0, 1. 3.  $1 - P(A)$ . 4.  $P(A) \times P(B)$ . 5. dependent. 6. complement.

## PART ELEVEN

# C++ for Discrete Structures

*Two papers demand written C++ worth six, six and seven marks, and the course text does not contain the word "program". So this part teaches the language from nothing, to exactly the depth those questions need, and every listing in it was compiled and run before it was printed.*

```
int main() {  
    cout << ...  
    return 0;  
}
```



|        |            |
|--------|------------|
| { }    | 0011100000 |
| {1}    | 1010010001 |
| {1, 2} | 0111001110 |

what the examiner asked the program to print

## Enough C++ to Answer the Question

*You are not being examined on C++. You are being examined on whether you can express a discrete structure as a program. That takes far less of the language than a programming course would give you, and this section is the whole of what the four examined questions use.*

*By the end of this section you will be able to write and read a complete C++ program: the skeleton, printing and reading, the types this course needs, arithmetic including integer division and remainder, conditions, both kinds of loop, arrays through `vector`, functions with a return value, and the bit operations that turn a subset into a number.*

### NOT IN THE COURSE TEXT, AND NOT A LITTLE BIT

Probed across all 184 thousand characters of the official text, these occur **ZERO** times: C++, cout, cin, include, main, algorithm, pseudocode, and the word program itself. Meanwhile 2023/2024 asks for a seven-mark program and 2024/2025 asks for two more worth six marks each. That is 19 marks of programming with no source behind it. Everything here is therefore taught from zero, and every listing was compiled with warnings turned up and run before it went on the page.

## The skeleton

### ● MUST MEMORISE

### THE SHAPE EVERY PROGRAM HAS

Write this out from memory once, and you have the first mark of any programming question before you have thought about the problem.

#### THE SMALLEST COMPLETE PROGRAM

```
// A comment. Everything after two slashes is ignored by the compiler.
#include <iostream>           // brings cout and cin into the program
using namespace std;        // lets us write cout, not std::cout

int main() {                 // execution starts here
    cout << "Hello" << endl;
    return 0;                // 0 means "finished normally"
}
```

#### WHERE

**#include <iostream>** pulls in the **input and output** facilities. Without it the compiler has never heard of `cout` and the program does not build. **using namespace std;** saves writing `std::` in front of every library name. Real projects avoid it; exam answers use it, and so does this book. **int main()** is the **entry point**. Every program has exactly one, and the machine begins there. **{ }** braces enclose a **block**: a group of statements treated as one. **cout** is the **output stream**, read "see out", and it sends what follows to the screen. **<<** is the **insertion operator**. Read it as an arrow pointing at `cout`: the value flows into the stream. It can be chained, so one line may print several things. **endl** ends the line and is short for "end line". **;** ends a **statement**. It is not optional and it is the single most common thing candidates leave out. **return 0;** hands 0 back to whatever started the program, meaning success.

◆ **Memory hook: include, namespace, main, return.** Four lines, always the same, before any thinking happens.

## Values and variables

### ● MUST MEMORISE

### THE TYPES THIS COURSE NEEDS

C++ makes you say what kind of value a name holds. Five types cover every examined question.

| TYPE         | HOLDS                                           | EXAMPLE                           |
|--------------|-------------------------------------------------|-----------------------------------|
| int          | a whole number, positive or negative            | int n = 24;                       |
| long<br>long | a whole number with room for much larger values | long long big =<br>2684483063360; |
| bool         | true or false, the propositions of section 1.1  | bool found = false;               |
| char         | one character, written in single quotes         | char bit = '1';                   |
| string       | text, written in double quotes                  | string bits = "1010101010";       |

#### WHERE

**int n = 24;** reads: make a whole-number box called n and put 24 in it. The type comes first, then the name, then the value. **=** is **assignment**, not equality. It puts the right side into the left. The test for equality is **==**, with two signs, and confusing the two is the classic C++ error. **string** needs its own line, `#include <string>`, at the top. **'1'** and **"1"** are different things: single quotes make one character, double quotes make text. A bit string is built from the first and printed as the second.

**Why long long matters here.** An `int` runs out at about two thousand million. The password count of section 5.1 is 2684483063360, which is far past that, and storing it in an `int` silently gives a wrong answer rather than an error. If a count can be large, say `long long`.

| OPERATOR  | MEANING                                                          | EXAMPLE        | RESULT              |
|-----------|------------------------------------------------------------------|----------------|---------------------|
| + - *     | add, subtract, multiply                                          | 3 * 4          | 12                  |
| /         | divide; on two whole numbers it <b>throws the remainder away</b> | 7 / 2          | 3, not 3.5          |
| %         | the <b>remainder</b> after whole division                        | 7 % 2          | 1                   |
| == !=     | is equal to, is not equal to                                     | n % 2 == 0     | true when n is even |
| < <= > >= | the ordinary comparisons                                         | low <= high    | true or false       |
| &&    !   | and, or, not: the connectives of section 1.1                     | a > 0 && b > 0 | true when both are  |

#### WHERE

/ between two `int` values performs **integer division**. `7 / 2` is 3 because both operands are whole numbers and the fractional part is discarded, not rounded. This is not a bug and it is used deliberately below to find the midpoint of a range. % is read "mod" and gives what is left over. `n % 2` is 0 exactly when n is even, which is the definition from section 9.1 written as a test. **&&** and **||** are conjunction and disjunction from Part One, spelled with two symbols each. One `&` or one `|` means something else entirely, covered below.

◆ **Memory hook: slash keeps the whole part, percent keeps the rest.** Together they recover the full answer: 7 is  $2 \times 3 + 1$ .

## Reading input, and making choices

### TEACH

### CIN, IF AND ELSE

`cin` is the input stream, and `>>` is the extraction operator: the arrow points away from `cin` and into the variable you name.

READING A NUMBER AND TESTING IT

```
int n;
cout << "Enter an integer: ";    // always prompt; a bare cursor is not a program
cin >> n;                        // wait for the user, then store it in n

if (n % 2 == 0) {
    cout << n << " is even" << endl;
} else {
    cout << n << " is odd" << endl;
}
```

**The condition goes in round brackets and the branch in braces.** The `else` part is optional. Note that the test is `==`: a single `=` would *assign* 0 to the remainder rather than compare with it, and the compiler will let you.

**Dynamic input is what "dynamically" means in an exam question.** The 2023/2024 paper asks for a program that "dynamically takes these sets as input"; that is a request for `cin`, not for the three sets typed into the source as constants. A program with the answer already written into it has not been asked for.



# Loops

## ● MUST MEMORISE

## FOR AND WHILE

Two loops, and the choice between them is mechanical.

A FOR LOOP: USE IT WHEN YOU KNOW HOW MANY TIMES

```
for (int i = 0; i < n; i++) {  
    cout << i << " ";  
}
```

A WHILE LOOP: USE IT WHEN YOU STOP ON A CONDITION

```
while (low <= high) {  
    // keep going until the range is empty  
}
```

### WHERE

**int i = 0** is the **start**, run once before the first pass. **i < n** is the **condition**, tested before every pass; the loop ends the first time it is false. **i++** is the **step**, run after every pass, and it means "add one to i". **i** is the loop variable, and counting from **0** rather than 1 is the C++ convention. A loop over  $n$  items runs  $i = 0, 1, \dots, n - 1$ , which is  $n$  passes. **while** takes only the condition, so anything that makes it eventually false has to happen inside the body. If nothing does, the program never stops.

◆ **Memory hook: for counts, while waits.** A known number of passes is a for; a search that ends when it finds something is a while.

**The off-by-one to expect.**  $i < n$  gives  $n$  passes;  $i \leq n$  gives  $n + 1$  and reads past the end of an  $n$ -element array. This is the same one-fewer-than-you-think that section 9.3 met in  $a_n = a + (n - 1)d$ .

## Many values at once

### ● MUST MEMORISE

### VECTOR, A LIST THAT KNOWS ITS OWN LENGTH

#### DECLARING, FILLING AND READING A VECTOR

```
#include <vector>                                // needed at the top of the file

int n = 3;
vector<int> s(n);                                // three ints, all 0 to begin with

for (int i = 0; i < n; i++) cin >> s[i];        // fill it from the keyboard
cout << s[0] << " and there are " << s.size() << " of them";
```

#### WHERE

**vector<int>** means "a list of int". The type in the angle brackets is what it holds, so **vector<string>** is a list of text. **s(n)** creates it with *n* slots. The size may be a variable, which is exactly what "dynamically" needs and what a plain fixed-size array cannot do. **s[i]** is the **slot at position i**, counting from **0**. So a three-element vector has slots **s[0]**, **s[1]**, **s[2]**, and **s[3]** does not exist. **s.size()** is how many slots there are. The dot means "of this object", and the brackets are there because it is a function call.

**Position 0 is not a quirk to work around.** It is the reason **bits[m - 1]** appears in the bit-string program: element *m* of the universe {1, ..., 10} lives at slot *m* - 1. Write that subtraction down deliberately rather than discovering it by being one out.

## Functions

### ● MUST MEMORISE

### WRITING A FUNCTION AND CALLING IT

A function is a named piece of program that takes values in and hands one value back. It is the section 4.1 idea of a function, made executable.

DEFINITION, THEN USE

```
// returns the larger of two whole numbers
int larger(int a, int b) {
    if (a > b) return a;
    return b;
}

int main() {
    cout << larger(3, 9);           // prints 9
    return 0;
}
```

WHERE

**int larger(...)** the leading **int** is the **return type**: the kind of value handed back. A function that hands nothing back is declared **void**. (**int a, int b**) are the **parameters**: names for the values passed in, each with its own type. These are the domain of section 4.1. **return a;** hands a value back **and stops the function immediately**. Everything after it in that path is not run. **larger(3, 9)** is the **call**. The whole expression stands for the value returned, so it can be printed, stored or used in arithmetic. **const vector<int>& s**, which appears in the programs below, is a parameter that receives the list **without copying it**, and **const** promises the function will not change it. Say that in one line if a marker asks; nothing else about references is examined.

**A function must appear before the line that calls it**, or the compiler has not met it yet. Defining every helper above **main** avoids the question entirely, and every program in the next section does that.

## Bits, and why a subset is a number

### TEACH

### THE TWO OPERATORS THE POWER SET NEEDS

Section 2.3 encoded a subset as a bit string: one bit per element of the universe, 1 when the element is in. C++ can hold that string of bits as a single whole number and test one bit at a time, which turns "list every subset" into "count from 0 upwards".

| WRITTEN                                | READ                  | DOES                                        |
|----------------------------------------|-----------------------|---------------------------------------------|
| <code>1 &lt;&lt; i</code>              | one shifted left by i | a number with a single 1 bit, at position i |
| <code>mask &amp; (1 &lt;&lt; i)</code> | mask AND that         | non-zero exactly when bit i of mask is 1    |

#### WHERE

**<<** here is the **shift** operator, not the printing one. Which meaning applies is decided by what sits on its left: a stream means print, a number means shift. Both spellings are the same two characters and this is a genuine wart of the language. **&** with one ampersand is **bitwise and**: it lines the two numbers up bit by bit and keeps a 1 only where both have one. Two ampersands, **&&**, is the logical and of Part One; the two are not interchangeable. **mask** is the number whose bits name a subset: bit i is 1 when element i is in. **position i** counts from 0 at the right-hand end, so `1 << 0` is 1, `1 << 1` is 2, `1 << 2` is 4.

| MASK | IN BINARY | THE SUBSET OF {1, 2, 3} |
|------|-----------|-------------------------|
| 0    | 000       | { }                     |
| 1    | 001       | {1}                     |
| 2    | 010       | {2}                     |
| 3    | 011       | {1, 2}                  |
| 4    | 100       | {3}                     |
| 5    | 101       | {1, 3}                  |
| 6    | 110       | {2, 3}                  |
| 7    | 111       | {1, 2, 3}               |

**Read the right-hand column downwards** and compare it with the sample output printed on the 2024/2025 paper. They are in the same order, because counting 0 to 7 is the enumeration the examiner printed. That is not a coincidence to be grateful for; it is the reason the power set of an n-element set has  $2^n$  members, which section 2.1 proved by a different route.

There is no compiler in the hall. Nobody will run what you write, which cuts both ways: a missing semicolon will not stop you, and a program that only looks plausible will not fool a marker who can read.

- ▶ **Write the skeleton first**, before you have solved anything. Four lines you already know are four lines of marks.
- ▶ **Comment the steps the question numbered.** 2024/2025 Q4(c) lists five things the function must do; put those five as comments in the body and the marker can find each one.
- ▶ **Indent.** A marker reading handwriting needs the block structure visible, and it costs nothing.
- ▶ **Prompt before every read.** `cout` then `cin`, every time.
- ▶ **Name things as the question names them.** If it says the function is called `isPerfectSquare`, that is its name. Renaming it is the same defect as paraphrasing a quotation.
- ▶ **If you run out of time, write the logic in ordered comments.** A correct method in comments earns more than half a program that stops mid-loop.

## RECALL AND CHECK

1. Write the four lines every C++ program starts and ends with.
2. What is the difference between `=` and `==`?
3. What do `7 / 2` and `7 % 2` give, and why?
4. When do you use a for loop and when a while loop?
5. What is the first slot of a vector called, and what is the last slot of a vector of size `n`?
6. What does `mask & (1 << i)` test?
7. Why does the bit-string program write `bits[m - 1]` rather than `bits[m]`?

## ANSWERS

1. `#include <iostream>, using namespace std;, int main() {, and return 0; }.`
2. `=` assigns the right side into the left; `==` tests whether the two are equal.
3. 3 and 1. Division between whole numbers discards the fraction, and `%` gives what was discarded.
4. A for loop when the number of passes is known in advance; a while loop when you stop on a condition.
5. `s[0]`, and the last is `s[n - 1]`.
6. Whether bit `i` of `mask` is 1, that is whether element `i` is in the subset that `mask` encodes.
7. Because element `m` of the universe `{1, ..., 10}` sits at slot `m - 1`, since slots count from 0.

## FILL IN THE GAP

1. Every statement ends with a \_\_\_\_\_.
2. `cout <<` sends a value \_\_\_\_\_ and `cin >>` reads a value \_\_\_\_\_.
3. A vector's slots are numbered from \_\_\_\_\_.
4. The remainder operator is written \_\_\_\_\_.
5. A function that hands nothing back has return type \_\_\_\_\_.
6. The word "dynamically" in an exam question is asking for \_\_\_\_\_.

### ANSWERS

1. semicolon. 2. out to the screen, in from the keyboard. 3. 0. 4. `%`. 5. `void`. 6. input read at run time with `cin`, rather than values written into the source.

## PART ELEVEN • 11.2

# The Four Examined Programs

*Four programs have been set across two papers, worth 19 marks between them. Every one is below, quoted as printed, broken down, then answered with a listing that was compiled at full warnings and run. The output panels are what the machine actually printed, not what it ought to have printed.*

*By the end of this section you will be able to write the power set program, the bit-string program, the `isPerfectSquare` function with its main, and the degree-counting snippet, and to explain each line of them.*

## TEACH

## HOW THESE LISTINGS WERE CHECKED

Every listing in this part was saved to a file, compiled with warnings turned up, and run with the input the question specifies. Nothing was transcribed from memory or from a listing that had merely been read through.

That matters more here than anywhere else in the book. A wrong number in a worked example is visible to anyone who checks the arithmetic; a program that does not compile looks exactly like a program that does, and a manual that prints one has taught the reader to write it.

**The output panels below are captured output.** Where a prompt and an answer sit on the same line, that is because the program printed the prompt without a newline and the input was supplied from a file rather than typed. It is what the terminal showed.

## 1. The power set

AS PRINTED

2024/2025 · Q3(B)

b. Write a C++ program to generate all possible subsets (also known as the power set) of a given set of integers.

For example, if the user enters the set {1, 2, 3}, the output should include:

```
{ }  
{1}  
{2}  
{1, 2}  
{3}  
{1, 3}  
{2, 3}  
{1, 2, 3}
```

6 marks

### NOTE ON THE PAPER

Nothing is printed wrongly here, but the sample output is doing more work than it looks. It is not a random selection of subsets: it is **ALL EIGHT OF THEM, IN THE EXACT ORDER A COUNTING LOOP PRODUCES**, as the bit table in section 11.1 shows. Reproducing that order is free if you enumerate by counting and awkward if you enumerate any other way, so take the hint.

Note also the empty set prints as `{ }`, with a space between the braces, while every other line has no space inside them. That is one extra line of code and it is the kind of detail a marker checks first.

### BREAK IT DOWN

**GIVEN** A set of integers, entered by the user. The size is not fixed by the question.

**ASKED** **Write a C++ program** that **generates** all possible subsets. One deliverable: a complete, runnable program, not a fragment.

**MARKS SAY** 6. Expect roughly: the skeleton and input, the enumeration loop, the membership test, the printing format, and the correct count of subsets. The sample output is printed for you, which means the format itself carries marks.

**TEACHES** Sections 11.1 (bits, vector, loops) and 2.1 (a set of  $n$  elements has  $2^n$  subsets).

**TRAP** Writing the set into the source as {1, 2, 3}. The question says "a given set of integers" and "if the user enters", so the program must read it. Also: forgetting the empty set, which is subset number 0 and the one candidates most often lose.

- GIVEN**  $n$  integers read from the keyboard into a vector .
- FIND** A program printing all  $2^n$  subsets in the paper's format and order.
- FORMULA** Count mask from 0 to  $2^n - 1$ ; element  $i$  is in the subset exactly when bit  $i$  of mask is 1, that is when  $\text{mask} \& (1 \ll i)$  is non-zero.

POWERSET.CPP

```
// PQ 2024/2025 Question Three (b): the power set of a set of integers.
#include <iostream>
#include <vector>
using namespace std;

// Print the one subset picked out by the bits of `mask`.
void printSubset(const vector<int>& s, int mask) {
    cout << "{";
    bool first = true;
    for (int i = 0; i < (int)s.size(); i++) {
        if (mask & (1 << i)) {           // is element i in this subset?
            if (!first) cout << ", ";
            cout << s[i];
            first = false;
        }
    }
    if (first) cout << " ";               // the empty set prints as { }
    cout << "}" << endl;
}

int main() {
    int n;
    cout << "How many integers are in the set? ";
    cin >> n;

    vector<int> s(n);
    cout << "Enter " << n << " integers: ";
    for (int i = 0; i < n; i++) cin >> s[i];

    int total = 1 << n;                  // 2^n subsets
    cout << "The power set has " << total << " subsets:" << endl;
    for (int mask = 0; mask < total; mask++) printSubset(s, mask);

    return 0;
}
```

WHAT IT PRINTED, FOR THE PAPER'S OWN EXAMPLE

```
How many integers are in the set? Enter 3 integers: The power set has 8 subsets:
{ }
{1}
{2}
{1, 2}
{3}
{1, 3}
{2, 3}
{1, 2, 3}
```

**Line for line against the question.** The eight lines are the paper's eight lines, in the paper's order, including the space inside the empty braces.



- ▶ **The enumeration.** `total = 1 << n` is  $2^n$ , so the loop runs once per subset. That is section 2.1's count, used rather than restated.
- ▶ **The membership test.** `mask & (1 << i)` asks whether bit *i* is set. It is the bit string of section 2.3 read one position at a time.
- ▶ **The comma.** `first` exists so the separator goes *between* elements and not before the first one. Printing `s[i] << ", "` instead would leave a trailing comma on every line.
- ▶ **The cast.** `(int)s.size()` is there because `size()` returns an unsigned type, and comparing it with a signed `int` draws a warning. Not examinable; say one line about it if asked.

**Now redo it:** with the set {5, 7} the program should print 4 subsets: { }, {5}, {7}, {5, 7}. Trace it by hand with `mask = 0, 1, 2, 3` and check the bits against the table in section 11.1.

## 2. The bit strings

AS PRINTED

2023/2024 · Q3(B)

(b) Write a C++ program that dynamically takes these sets in 3(i-iii) above as input and displays the corresponding bit strings for each set. **(7 marks)**

### NOTE ON THE PAPER

"These sets in 3(i-iii) above" points back at part (a) of the same question, which is quoted in full in section 2.3: the universal set is  $U = \{1, 2, \dots, 10\}$  and the three sets are {3, 4, 5}, {1, 3, 6, 10} and {2, 3, 4, 7, 8, 9}. Part (a) also fixes the convention the program must follow, that "the  $i^{\text{th}}$  bit in the string is 1 if *i* is in the set and 0 otherwise", so the program is not free to choose its own bit order.

The load-bearing word is **DYNAMICALLY**. It rules out a program with the three sets typed into the source. Read together with "as input", it is asking for the sets to arrive through `cin` while the program is running.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | $U = \{1, 2, \dots, 10\}$ , and three sets to be supplied at run time. The convention that bit <i>i</i> is 1 when <i>i</i> is in the set.                                                                                                                             |
| <b>ASKED</b>     | <b>Write a C++ program</b> that <b>dynamically takes</b> the sets <b>as input</b> and <b>displays</b> the corresponding bit strings.                                                                                                                                  |
| <b>MARKS SAY</b> | 7, the largest of the programming questions. Expect: the skeleton, reading a variable number of sets, reading a variable number of elements per set, the conversion itself, the correct bit order, and the display. The word "dynamically" is worth marks on its own. |
| <b>TEACHES</b>   | Sections 11.1 (vector, string, nested loops) and 2.3 (what a bit string is).                                                                                                                                                                                          |
| <b>TRAP</b>      | Two of them. Writing the sets as constants, which fails "dynamically"; and putting element <i>m</i> at position <i>m</i> rather than <i>m</i> − 1, which shifts every bit one place and produces a plausible wrong answer.                                            |

- GIVEN** A universe of 10 elements, and sets read at run time.
- FIND** A program printing each set's bit string.
- FORMULA** Start with ten '0' characters; for each member  $m$ , set position  $m - 1$  to '1'.

BITSTRINGS.CPP

```
// PQ 2023/2024 Question Three (b): read sets dynamically, print bit strings.
#include <iostream>
#include <string>
#include <vector>
using namespace std;

const int UNIVERSE = 10;           // U = {1, 2, ..., 10}

// The bit string of one set: position i carries 1 when i is a member.
string bitString(const vector<int>& members) {
    string bits(UNIVERSE, '0');      // start with every position 0
    for (int i = 0; i < (int)members.size(); i++) {
        int m = members[i];
        if (m >= 1 && m <= UNIVERSE) bits[m - 1] = '1';
    }
    return bits;
}

int main() {
    int howMany;
    cout << "How many sets? ";
    cin >> howMany;

    for (int s = 1; s <= howMany; s++) {
        int size;
        cout << "Set " << s << ", how many elements? ";
        cin >> size;

        vector<int> members(size);
        cout << "Enter " << size << " elements from 1 to " << UNIVERSE << ": ";
        for (int i = 0; i < size; i++) cin >> members[i];
        cout << endl;           // close the line the user typed on

        cout << "Set " << s << " bit string: " << bitString(members) << endl;
    }
    return 0;
}
```

WHAT IT PRINTED, GIVEN THE THREE SETS OF PART (A)

```
How many sets? Set 1, how many elements? Enter 3 elements from 1 to 10:
Set 1 bit string: 0011100000
Set 2, how many elements? Enter 4 elements from 1 to 10:
Set 2 bit string: 1010010001
Set 3, how many elements? Enter 6 elements from 1 to 10:
Set 3 bit string: 0111001110
```

**Check against section 2.3**, where the same three sets were converted by hand: {3, 4, 5} gives 0011100000, {1, 3, 6, 10} gives 1010010001, and {2, 3, 4, 7, 8, 9} gives

0111001110. The program agrees with the hand answer in all thirty positions, which is the point of writing both.

- ▶ **Two loops, nested.** The outer one walks the sets, the inner one walks the elements of one set. Each has its own counter, and reusing `i` for both would destroy the outer count.
- ▶ **The subtraction.** `bits[m - 1]` is where element `m` lives, because string positions count from 0. Getting this wrong produces 0001110000 for the first set, which looks right at a glance and is not.
- ▶ **The guard.** `m >= 1 && m <= UNIVERSE` stops a typed 11 from writing outside the string. A program that crashes on bad input has lost the marks it was about to earn.
- ▶ **Why a function.** `bitString` does one job and is called once per set. It also makes the answer readable, which on a 7-mark question is worth doing.

### 3. isPerfectSquare, by binary search

AS PRINTED

2024/2025 · Q4(C)

c. Write a C++ function called `isPerfectSquare` that uses a binary search algorithm to verify if a given integer `num` is a perfect square. The function should:

- Narrow down the search range `[low,high]` by repeatedly dividing it into two parts (midpoint).
- Calculate the square of the midpoint value.
- Check if the square of the midpoint is equal to `num`. If so, return the midpoint as the square root of `num`.
- If the square of the midpoint is greater than or less than `num`, adjust the search range accordingly.
- If the search range is exhausted and no midpoint satisfies the condition, return -1.

Write a main function to:

- Prompt the user to enter an integer `xx`.
- Call `isPerfectSquare(x)`.
- Display whether `xx` is a perfect square. If `xx` is a perfect square, also display its integer square root.

**6 marks**

#### NOTE ON THE PAPER

**THREE NAMES FOR ONE VALUE.** The function's parameter is called **NUM**, the main function is told to prompt for "an integer **xx**", then to call **ISPERFECTSQUARE(X)** with a single `x`, then to report on **xx** again. Only one value is involved. Answer it with one variable, named once, and say in a line which name you adopted; a marker reading three names in a six-mark question is looking to see whether you noticed.

**THE FUNCTION NAME IS NOT YOURS TO CHANGE.** The question names it **ISPERFECTSQUARE**, so that is its name, spelled and capitalised exactly that way. Renaming it to `perfect_square` or `isSquare` is the same defect as paraphrasing a quotation: the question asked for a specific thing and got something else.

**PART IV SAYS "GREATER THAN OR LESS THAN".** Those are two different adjustments, not one, and the answer must make both moves: too big means search below the midpoint, too small means search above it. A single branch answers half the part.

### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | An integer. A five-step specification of the function, and a three-step specification of main.                                                                                                                                                                                                                                                                                      |
| <b>ASKED</b>     | <b>Write a function called <code>isPerfectSquare</code></b> using <b>binary search</b> , plus a <b>main</b> that prompts, calls and reports. Two deliverables.                                                                                                                                                                                                                      |
| <b>MARKS SAY</b> | 6, split across eight numbered requirements. The cheapest way to secure them is to make the numbering visible in the answer, so that each printed requirement has a line the marker can point at.                                                                                                                                                                                   |
| <b>TEACHES</b>   | Sections 11.1 (functions, while, integer division) and 9.4, where $T(n) = T(n/2) + 1$ is exactly this algorithm's cost.                                                                                                                                                                                                                                                             |
| <b>TRAP</b>      | Answering with <code>sqrt(num)</code> . It is shorter, it works, and it scores nothing, because the question asks for a binary search and that is what the marks are for. Also: returning <code>true</code> or <code>false</code> , when the question says to return the midpoint or <code>-1</code> , so the return type is <code>int</code> despite the name beginning with "is". |

- GIVEN** An integer num, and the search range [low, high] starting at [1, num].
- FIND** The integer square root of num, or -1.
- FORMULA** Halve the range each pass: compare  $\text{mid}^2$  with num and keep whichever half can still contain the answer.

PERFECTSQUARE.CPP

```
// PQ 2024/2025 Question Four (c): isPerfectSquare by binary search.
#include <iostream>
using namespace std;

// Returns the integer square root of num, or -1 if num is not a perfect square.
int isPerfectSquare(int num) {
    if (num < 0) return -1;           // no real square root
    if (num == 0) return 0;          // 0 = 0 * 0

    int low = 1, high = num;         // the search range [low, high]
    while (low <= high) {
        int mid = low + (high - low) / 2; // i. the midpoint
        long long square = (long long)mid * mid; // ii. square it

        if (square == num) return mid; // iii. found it
        if (square < num) low = mid + 1; // iv. adjust upward
        else high = mid - 1;           // iv. adjust downward
    }
    return -1;                        // v. range exhausted
}

int main() {
    int x;
    cout << "Enter an integer: ";      // i. prompt
    cin >> x;

    int root = isPerfectSquare(x);      // ii. call it
    if (root == -1) {                   // iii. report
        cout << x << " is not a perfect square." << endl;
    } else {
        cout << x << " is a perfect square." << endl;
        cout << "Its integer square root is " << root << "." << endl;
    }
    return 0;
}
```

WHAT IT PRINTED, RUN TWICE

```
Enter an integer: 49 is a perfect square.
Its integer square root is 7.
```

```
Enter an integer: 50 is not a perfect square.
```

**The name is the paper's.** `isPerfectSquare`, spelled and capitalised exactly as printed, and one variable `x` for the value the question called num, `xx` and `x` in three different places. The comments carry the paper's own numbering, i to v and i to iii, so each requirement has a line beside it.

- **Why the range starts at [1, num].** The square root of num is never larger than num for whole numbers, so the answer, if it exists, is inside. Starting at [1, num] is

safe and needs no justification;  $[1, \text{num}/2 + 1]$  is tighter and needs one.

- **Why  $\text{low} + (\text{high} - \text{low}) / 2$ .** It is the midpoint, written so that  $\text{low} + \text{high}$  is never formed. That sum can exceed what an `int` holds when `num` is large, and the overflow would produce a negative midpoint. Worth one line in an answer.
- **Why long long for the square.** `mid` can be close to `num`, so `mid × mid` can be far past the `int` limit. Squaring inside an `int` is the bug this program is most likely to have, and it is silent.
- **Why the loop ends.** Every pass either returns or moves `low` up or `high` down by at least one, so the range strictly shrinks and `low ≤ high` must eventually fail. That is the termination argument, and it is the sort of thing section 9.2 taught you to state rather than assume.
- **The cost.** Each pass halves the range, so the number of passes is the number of halvings needed to reduce `num` to 1. That is the recurrence  $T(n) = T(n/2) + 1$  of section 9.4, and it is why binary search beats testing 1, 2, 3, ... in turn.

**Now redo it:** trace the function on `num = 49` by hand. Range  $[1, 49]$ , `mid 25`,  $625 > 49$  so `high` becomes 24; `mid 12`,  $144 > 49$  so `high` becomes 11; `mid 6`,  $36 < 49$  so `low` becomes 7; `mid 9`,  $81 > 49$  so `high` becomes 8; `mid 7`,  $49 = 49$ , return **7**. Five passes to search forty-nine numbers.

## 4. Counting the remaining degree-3 and degree-4 vertices

### TEACH

### WHERE THIS QUESTION IS QUOTED

2024/2025 Question Five is printed as a single scenario with seven parts, so it is quoted whole, once, in section 6.3 where the Handshaking theorem is taught. Part (d) is the programming one:

"Write a C++ code snippet to compute the remaining number of vertices that have either degree 3 or 4." 3 marks

Section 6.3 answers the mathematics: 24 vertices and 30 edges force the degree sum to 60; the five of degree 4, seven pendant and seven of degree 2 account for 19 vertices and 41 of that sum, leaving 5 vertices to share 19, which can only be four of degree 4 and one of degree 3. So the graph has **9** vertices of degree 4, not the five the question appears to state. This section turns that reasoning into the snippet the part asks for.

**Note the word "snippet".** The question does not ask for a complete program, so a fragment would be accepted. The listing below is complete anyway, because a complete program can be run and a fragment cannot, and the wrapper is **ours, not the examiner's**: if you reproduce this in an exam, the part inside `main` is the answer and the `include` line is scaffolding.

#### BREAK IT DOWN

|                  |                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GIVEN</b>     | 24 vertices, 30 edges, five of degree 4, seven pendant, seven of degree 2, and every other vertex of degree 3 or 4.                                                                                                                                                                          |
| <b>ASKED</b>     | <b>Write a C++ code snippet to compute the remaining number</b> of vertices of degree 3 or 4.                                                                                                                                                                                                |
| <b>MARKS SAY</b> | 3. Small, so the marks are for the arithmetic being right and visible, not for program structure. Print the intermediate values.                                                                                                                                                             |
| <b>TEACHES</b>   | Sections 6.3 (Handshaking) and 11.1.                                                                                                                                                                                                                                                         |
| <b>TRAP</b>      | Reading "the remaining number of vertices that have either degree 3 or 4" as just $24 - 19 = 5$ . That is the count of vertices left, and it is worth computing, but the question sits inside a scenario whose whole point is the split of those five. Compute both, and say which is which. |

- GIVEN**  $v = 24$ ,  $e = 30$ , and the counts 5, 7, 7 of degree 4, 1 and 2 respectively.
- FIND** How many vertices remain, and how those split between degree 3 and degree 4.
- FORMULA**  $\sum \deg(v) = 2|E|$  from section 6.3, then the pair of simultaneous equations  
 $\text{four} + \text{three} = \text{remaining}$  and  $4 \times \text{four} + 3 \times \text{three} = \text{leftDegree}$ .

DEGREES.CPP

```
// PQ 2024/2025 Question Five (d): the vertices of degree 3 or 4 still to classify.
#include <iostream>
using namespace std;

int main() {
    int vertices = 24, edges = 30;
    int knownDeg4 = 5, pendant = 7, deg2 = 7;

    int accounted = knownDeg4 + pendant + deg2;           // vertices already named
    int remaining = vertices - accounted;                 // still to classify

    int degreeSum = 2 * edges;                           // Handshaking theorem
    int usedDegree = 4 * knownDeg4 + 1 * pendant + 2 * deg2;
    int leftDegree = degreeSum - usedDegree;              // degree left to share

    // Solve four + three = remaining and 4*four + 3*three = leftDegree.
    int four = leftDegree - 3 * remaining;
    int three = remaining - four;

    cout << "Vertices still to classify: " << remaining << endl;
    cout << "Degree left to share: " << leftDegree << endl;
    cout << "Of those, degree 4: " << four << ", degree 3: " << three << endl;
    cout << "Degree-4 vertices in the whole graph: " << knownDeg4 + four << endl;
    return 0;
}
```

WHAT IT PRINTED

```
Vertices still to classify: 5
Degree left to share:      19
Of those, degree 4: 4, degree 3: 1
Degree-4 vertices in the whole graph: 9
```

**Where the two-line solution comes from.** Let four and three be the counts. Then  $\text{four} + \text{three} = 5$  and  $4 \times \text{four} + 3 \times \text{three} = 19$ . Substituting  $\text{three} = 5 - \text{four}$  into the second gives  $4 \times \text{four} + 15 - 3 \times \text{four} = 19$ , so  $\text{four} = 19 - 15 = 4$  and  $\text{three} = 1$ . The single line `four = leftDegree - 3 * remaining` is that substitution done once and for all, which is why it deserves the comment above it.

**The final line is the answer to part (a),** not part (d), and printing it costs nothing.  $5 + 4 = 9$  vertices of degree 4, which is the trap the whole question is built around.



## RECALL AND CHECK

1. How does the power set program enumerate subsets, and why does that give the paper's order?
2. What does the word "dynamically" require of the bit-string program?
3. Why is element  $m$  stored at position  $m - 1$ ?
4. Why must `isPerfectSquare` return an `int` rather than a `bool`?
5. Give two reasons the binary search uses `long long` and `low + (high - low) / 2`.
6. In the degree question, how many vertices of degree 4 does the graph have?

### ANSWERS

1. By counting a mask from 0 to  $2^n - 1$  and reading its bits. Counting in that order is exactly the order the paper printed.
2. That the sets arrive through `cin` at run time rather than being written into the source.
3. Because string and vector positions count from 0, so the ten elements of  $U$  occupy positions 0 to 9.
4. Because the question says to return the midpoint as the square root, or  $-1$  if there is none, and neither of those is a truth value.
5. `long long` because `mid × mid` can exceed the `int` range and overflow silently; the midpoint form because `low + high` can overflow before the division happens.
6. Nine: the five named in the question plus four of the five remaining vertices.

## FILL IN THE GAP

1. A set of  $n$  elements has \_\_\_\_\_ subsets, so the enumeration loop runs that many times.
2. The empty set is the subset with `mask =` \_\_\_\_\_.
3. The bit string of  $\{3, 4, 5\}$  over  $U = \{1, \dots, 10\}$  is \_\_\_\_\_.
4. Answering the `isPerfectSquare` question with `sqrt(num)` scores \_\_\_\_\_.
5. Binary search on a range of  $n$  values costs about \_\_\_\_\_ passes.

### ANSWERS

1.  $2^n$ . 2. 0. 3. 0011100000. 4. nothing, because the question asks for a binary search and that is where the marks are. 5. the number of halvings needed to reduce  $n$  to 1.

## REFERENCE

# Formulas, Notation and Exam Craft

*Everything the book has taught, on as few pages as it will go. Nothing here is new; every row points back at the section that derived it, so this part is for the last hour before the paper, not for learning from.*

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |

$$n! \quad C(n, r) \quad \sum \deg(v) = 2|E|$$

$$- e = v - 1 \quad v - e + f = 2 \quad P(A | B)$$

$$a + (n - 1)d \quad e \leq 3v - 6$$

## Formula and Notation Reference

Every symbol and every formula in this book, in one place, with the section that taught it. Use it to look things up, and to test yourself: cover the middle column and see whether you can supply it.

By the end of this section you will have one page to revise from for each of the three things the paper tests: reading the notation, applying the formula, and knowing which section a question belongs to.

### Notation

| TEACH        |                       |                                                   | SETS AND NUMBERS |
|--------------|-----------------------|---------------------------------------------------|------------------|
| SYMBOL       | READ AS               | MEANS                                             | SECTION          |
| $\in$        | is an element of      | the object on the left is in the set on the right | F.1              |
| $\notin$     | is not an element of  | the object is not in the set                      | F.1              |
| $\subseteq$  | is a subset of        | every element of the left set is in the right set | 2.1              |
| $\subset$    | is a proper subset of | a subset, and not equal to it                     | 2.1              |
| $\cup$       | union                 | everything in either set, or in both              | 2.2              |
| $\cap$       | intersection          | everything in both sets                           | 2.2              |
| $A - B$      | A minus B             | in A and not in B                                 | 2.2              |
| $\Delta$     | symmetric difference  | in exactly one of the two                         | 2.2              |
| $A'$         | the complement of A   | everything in U that is not in A                  | 2.2              |
| $\emptyset$  | the empty set         | the set with no elements                          | F.1              |
| $n(A)$       | the cardinality of A  | how many elements A has                           | 2.1              |
| $A \times B$ | the Cartesian product | all ordered pairs (a, b) with a in A and b in B   | 2.3              |
| $\mathbb{N}$ | the natural numbers   | 0, 1, 2, 3, ... in this book                      | F.1              |
| $\mathbb{Z}$ | the integers          | ..., -2, -1, 0, 1, 2, ...                         | F.1              |
| $\mathbb{Q}$ | the rationals         | every $p / q$ with $q$ not zero                   | F.1              |
| $\mathbb{R}$ | the reals             | every point on the number line                    | F.1              |



| SYMBOL                | READ AS                          | MEANS                                                   | SECTION |
|-----------------------|----------------------------------|---------------------------------------------------------|---------|
| $\neg p$              | not p                            | the negation: true exactly when p is false              | 1.1     |
| $p \wedge q$          | p and q                          | conjunction: true only when both are                    | 1.1     |
| $p \vee q$            | p or q                           | disjunction, inclusive: true when at least one is       | 1.1     |
| $p \oplus q$          | p exclusive or q                 | true when exactly one is                                | 1.1     |
| $p \rightarrow q$     | if p then q                      | the conditional: false only when p is true and q false  | 1.2     |
| $p \leftrightarrow q$ | p if and only if q               | the biconditional: true when the two agree              | 1.2     |
| $\equiv$              | is logically equivalent to       | identical truth tables                                  | 1.3     |
| $\forall x$           | for all x                        | the universal quantifier                                | 1.4     |
| $\exists x$           | there exists an x                | the existential quantifier                              | 1.4     |
| $\therefore$          | therefore                        | marks the conclusion of an argument                     | 1.5     |
| $\Sigma$              | the sum of                       | add the expression over the stated range                | F.1     |
| $n!$                  | n factorial                      | $n \times (n - 1) \times \dots \times 1$ , and $0! = 1$ | F.1     |
| $a \mid b$            | a divides b                      | $b = ka$ for some integer k                             | 9.1     |
| $\lceil x \rceil$     | the ceiling of x                 | round up to the next whole number                       | 5.3     |
| $\square$             | end of proof                     | the argument is complete                                | 9.1     |
| $\deg(v)$             | the degree of v                  | how many edge ends meet at vertex v                     | 6.1     |
| $\deg^-(v)$           | the in-degree of v               | how many arrows arrive at v                             | 6.4     |
| $\deg^+(v)$           | the out-degree of v              | how many arrows leave v                                 | 6.4     |
| $K_n$                 | the complete graph on n vertices | every pair joined by an edge                            | 6.2     |
| $K_{m,n}$             | the complete bipartite graph     | two groups, every cross pair joined                     | 6.2     |
| $A \cdot B$           | the Boolean product              | AND: 1 only when both are 1                             | 8.1     |
| $A + B$               | the Boolean sum                  | OR: 1 when at least one is 1                            | 8.1     |
| $\bar{x}$             | x bar                            | the mean of a data set                                  | 10.1    |
| $\sigma$              | sigma                            | the standard deviation                                  | 10.1    |
| $P(E)$                | the probability of E             | a number from 0 to 1                                    | 10.2    |

|            |                              |                                                  |      |
|------------|------------------------------|--------------------------------------------------|------|
| $P(B   A)$ | the probability of B given A | B's probability once A is known to have happened | 10.2 |
|------------|------------------------------|--------------------------------------------------|------|

**Two collisions to keep straight.**  $P(A)$  is the power set in Part Two and a probability in Part Ten, and which is meant is always settled by whether A is a set of outcomes. And  $A'$  is a set complement in Part Two and a Boolean complement in Part Eight, which is not a collision at all: section 8.1 showed they are the same operation.

## Formulas, by part

| TEACH                                            |                                                                        |          | COUNTING |
|--------------------------------------------------|------------------------------------------------------------------------|----------|----------|
| FORMULA                                          | WHAT THE LETTERS MEAN                                                  | SECTION  |          |
| $n_1 \times n_2 \times \dots \times n_k$         | the product rule: $n_i$ choices at stage i, every stage performed      | 5.1      |          |
| $n_1 + n_2 + \dots + n_k$                        | the sum rule: $n_i$ choices in group i, exactly one group used         | 5.1      |          |
| (total) – (without the property)                 | the subtraction rule, which is what "at least one" asks for            | 5.1      |          |
| $P(n, r) = n! / (n - r)!$                        | permutations: r chosen from n, <b>order matters</b>                    | 5.2      |          |
| $C(n, r) = n! / (r!(n - r)!)$                    | combinations: r chosen from n, <b>order does not</b>                   | 5.2      |          |
| $n! / (n_1! n_2! \dots n_k!)$                    | arrangements of n items where item i repeats $n_i$ times               | 5.2      |          |
| $n(A \cup B) = n(A) + n(B) - n(A \cap B)$        | inclusion and exclusion, two sets                                      | 2.2, 5.3 |          |
| $[N / k]$                                        | generalised pigeonhole: N objects in k boxes, some box holds this many | 5.3      |          |
| $C(n,0)x^n + C(n,1)x^{n-1}y + \dots + C(n,n)y^n$ | the binomial theorem for $(x + y)^n$                                   | 5.4      |          |
| $2^n$                                            | subsets of an n-element set, and rows of an n-letter truth table       | 2.1, 1.1 |          |

| FORMULA                                    | WHAT THE LETTERS MEAN                                                               | SECTION |
|--------------------------------------------|-------------------------------------------------------------------------------------|---------|
| $G = (V, E)$                               | a graph is a vertex set $V$ and an edge set $E$                                     | 6.1     |
| order = $ V $ , size = $ E $               | how many vertices, and how many edges                                               | 6.1     |
| $\sum \deg(v) = 2 E $                      | <b>Handshaking</b> : the degrees sum to twice the edges, so that sum is always even | 6.3     |
| $C(n, 2) = n(n - 1) / 2$                   | edges of $K_n$ , and links between $n$ buildings                                    | 6.2     |
| $\sum \deg^-(v) = \sum \deg^+(v) =  E $    | in a digraph both totals equal the number of arcs                                   | 6.4     |
| $a_{ij} = 1$ when $i$ and $j$ are adjacent | the adjacency matrix, symmetric for an undirected graph                             | 6.5     |
| every degree even                          | an <b>Euler circuit</b> exists, given the graph is connected                        | 6.6     |
| exactly two odd degrees                    | an <b>Euler path</b> exists, given the graph is connected                           | 6.6     |
| $v - e + f = 2$                            | <b>Euler's formula</b> for a connected planar graph: vertices, edges, faces         | 6.7     |
| $e \leq 3v - 6$                            | necessary for a simple planar graph with $v$ at least 3                             | 6.7     |
| $e \leq 2v - 4$                            | the same bound for a triangle-free planar graph                                     | 6.7     |
| $e = v - 1$                                | a tree on $v$ vertices has one fewer edge                                           | 7.1     |
| $n = mi + 1$                               | a full $m$ -ary tree: $n$ vertices in total, $i$ of them internal                   | 7.2     |
| $l = (m - 1)i + 1$                         | the same tree's leaf count                                                          | 7.2     |
| leaves at most $m^h$                       | an $m$ -ary tree of height $h$ , with equality when it is full and level            | 7.2     |

| FORMULA                                            | WHAT THE LETTERS MEAN                                                            | SECTION  |
|----------------------------------------------------|----------------------------------------------------------------------------------|----------|
| $m = 2k, \quad n = 2k + 1$                         | even and odd, for some integer $k$                                               | 9.1      |
| $p \rightarrow q \equiv \neg q \rightarrow \neg p$ | a conditional and its contrapositive agree, which licenses the method            | 1.2, 9.1 |
| $P(a)$ , and $P(k) \rightarrow P(k + 1)$           | <b>induction:</b> base case plus step gives $P(n)$ for every $n$ at or above $a$ | 9.2      |
| $a_n = a + (n - 1)d$                               | arithmetic: first term $a$ , common difference $d$ , position $n$                | 9.3      |
| $S_n = n(a + a_n) / 2$                             | the arithmetic series: first plus last, times half the count                     | 9.3      |
| $a_n = ar^{n-1}$                                   | geometric: first term $a$ , common ratio $r$                                     | 9.3      |
| $S_n = a(r^n - 1) / (r - 1)$                       | the geometric series, for $r$ not equal to 1                                     | 9.3      |
| $S = a / (1 - r)$                                  | the infinite geometric sum, when $r$ lies between $-1$ and $1$                   | 9.3      |
| $\Sigma k = n(n + 1) / 2$                          | the first $n$ whole numbers                                                      | 9.3      |
| $\Sigma k^2 = n(n + 1)(2n + 1) / 6$                | the first $n$ squares                                                            | 9.3      |
| $\Sigma k^3 = (n(n + 1) / 2)^2$                    | the first $n$ cubes, which is the square of the first sum                        | 9.3      |
| $x^k - c_1x^{k-1} - \dots - c_k = 0$               | the characteristic equation of a linear recurrence of order $k$                  | 9.4      |
| $a_n = Ax_1^n + Bx_2^n$                            | two distinct characteristic roots; $A$ and $B$ come from the initial conditions  | 9.4      |
| $a_n = (A + Bn)x_1^n$                              | one repeated root, which buys the extra factor of $n$                            | 9.4      |



| FORMULA                                   | WHAT THE LETTERS MEAN                                              | SECTION |
|-------------------------------------------|--------------------------------------------------------------------|---------|
| $\bar{x} = (\sum x_i) / n$                | the mean: the total over the count                                 | 10.1    |
| $\sigma^2 = (\sum (x_i - \bar{x})^2) / n$ | the variance: the mean of the squared deviations                   | 10.1    |
| $\sigma = \sqrt{(\sigma^2)}$              | the standard deviation, back in the data's own units               | 10.1    |
| $P(E) = n(E) / n(S)$                      | favourable over total, for <b>equally likely outcomes only</b>     | 10.2    |
| $P(A') = 1 - P(A)$                        | the complement rule, which is the "at least one" tool              | 10.2    |
| $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ | the addition rule; the last term is 0 when the events are disjoint | 10.2    |
| $P(A \cap B) = P(A)P(B)$                  | the multiplication rule, for <b>independent events only</b>        | 10.2    |
| $P(B   A) = P(A \cap B) / P(A)$           | conditional probability, valid when $P(A)$ is above 0              | 10.2    |
| $P(A   B) = P(B   A)P(A) / P(B)$          | <b>Bayes:</b> it reverses the bar                                  | 10.2    |

One table, because they are one set of laws. Learn a row once and you have it in Part One, Part Two and Part Eight.

| LAW              | LOGIC                                                       | SETS                                             | BOOLEAN                                   |
|------------------|-------------------------------------------------------------|--------------------------------------------------|-------------------------------------------|
| Identity         | $p \wedge T \equiv p$                                       | $A \cap U = A$                                   | $A \cdot 1 = A$                           |
| Domination       | $p \vee T \equiv T$                                         | $A \cup U = U$                                   | $A + 1 = 1$                               |
| Idempotent       | $p \vee p \equiv p$                                         | $A \cup A = A$                                   | $A + A = A$                               |
| Double negation  | $\neg(\neg p) \equiv p$                                     | $(A')' = A$                                      | $(A')' = A$                               |
| Commutative      | $p \wedge q \equiv q \wedge p$                              | $A \cap B = B \cap A$                            | $A \cdot B = B \cdot A$                   |
| Associative      | $(p \vee q) \vee r \equiv p \vee (q \vee r)$                | $(A \cup B) \cup C = A \cup (B \cup C)$          | $(A + B) + C = A + (B + C)$               |
| Distributive     | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ | $A \cdot (B + C) = A \cdot B + A \cdot C$ |
| <b>De Morgan</b> | $\neg(p \wedge q) \equiv \neg p \vee \neg q$                | $(A \cap B)' = A' \cup B'$                       | $(A \cdot B)' = A' + B'$                  |
| Absorption       | $p \vee (p \wedge q) \equiv p$                              | $A \cup (A \cap B) = A$                          | $A + A \cdot B = A$                       |
| Negation         | $p \vee \neg p \equiv T$                                    | $A \cup A' = U$                                  | $A + A' = 1$                              |

Each law has a dual, obtained by swapping  $\wedge$  with  $\vee$ ,  $\cap$  with  $\cup$ ,  $\cdot$  with  $+$ , and  $T$  with  $F$ ,  $U$  with  $\emptyset$ ,  $1$  with  $0$ . Learn one half of the pair and derive the other (section 8.1).

## Which section does a question belong to?

### TEACH

### SIGNAL WORDS, AND WHERE THEY SEND YOU

Read the question, find its phrase here, turn to that section. On a paper where you answer four of six, deciding **which** four is worth more than any single formula.

| IF THE QUESTION SAYS                                         | GO TO    |
|--------------------------------------------------------------|----------|
| converse, inverse, contrapositive, truth table, tautology    | 1.2, 1.3 |
| quantifier, "for all", "there exists", predicate             | 1.4      |
| valid argument, rule of inference, "therefore"               | 1.5      |
| types of set, set-builder, roster form                       | 2.1      |
| union, intersection, difference, Venn diagram                | 2.2      |
| bit string, Cartesian product, ordered pair                  | 2.3      |
| reflexive, symmetric, antisymmetric, transitive              | 3.2      |
| equivalence relation, equivalence class, partition           | 3.3      |
| partially ordered, totally ordered, poset                    | 3.4      |
| one-to-one, onto, bijection, correspondence types            | 4.2      |
| composition, $f$ of $g$ , inverse function                   | 4.3      |
| how many ways, passwords, PIN codes, labels                  | 5.1      |
| arrangements of a word, committees, "at least one"           | 5.2      |
| "both", "only", overlapping groups, pigeonhole               | 5.3      |
| types of graph, valency, size, order, incident               | 6.1, 6.2 |
| Handshaking, sum of degrees, "can each vertex have"          | 6.3      |
| in-degree, out-degree, digraph in tabular form               | 6.4      |
| adjacency matrix, path matrix, memory management             | 6.5      |
| isomorphic, Eulerian, Hamiltonian, the bridges               | 6.6      |
| planar, Kuratowski, $K_5$ , $K_{3,3}$                        | 6.7      |
| minimum distance, visit every city and return                | 6.8      |
| rooted tree terminology, root, leaf, level, height           | 7.1      |
| chain letter, convocation cards, $m$ -ary, balanced, playoff | 7.2      |
| preorder, inorder, postorder, prefix, postfix                | 7.3      |
| Boolean expression, logic gate, the door and the alarm       | 8.1      |
| prove, show that, disprove, counterexample                   | 9.1      |
| prove for all $n$ , base case, "by induction"                | 9.2      |
| sequence, progression, $n$ th term, common difference        | 9.3      |
| recurrence relation, initial conditions, closed form         | 9.4      |
| mean, median, mode, standard deviation                       | 10.1     |

probability, sample space, independent, conditional

10.2

write a C++ program, code snippet, function called

11.1, 11.2

REFERENCE · R.2

## Exam Day Strategy

*On a paper where you answer four questions out of six, the choosing is worth more than any single formula. This section is about the three hours: which four to take, how long each part may have, and what to write when a question is defective or you are stuck.*

*By the end of this section you will know the paper's shape and how it has changed across sessions, a defensible time budget per mark, a method for choosing four questions in ten minutes, what each command word is asking for, and what to do about the printing errors these papers actually contain.*

### The shape of the paper

#### ● MUST MEMORISE

#### THE FORMAT, AND HOW IT HAS MOVED

The course has been renamed twice and its rubric has changed with the credit weighting. All five papers are still valid practice, but only the last two are the shape you will sit.

| SESSION          | CODE ON THE PAPER | INSTRUCTION             | TIME           | MARKS EACH  |
|------------------|-------------------|-------------------------|----------------|-------------|
| 2020/2021        | CSC311            | attempt any THREE       | 2 hours        | about 23    |
| 2021/2022        | CSC416            | attempt any THREE       | 2 hours        | about 23    |
| 2023/2024        | CSC 242           | not on the part we hold | not stated     | 23.5        |
| 2024/2025        | CSC416            | attempt any FOUR        | 3 hours        | 17.5        |
| <b>2025/2026</b> | <b>CSC 242</b>    | <b>answer any FOUR</b>  | <b>3 hours</b> | <b>17.5</b> |

#### WHERE

**17.5** is the marks per question on the current format, so four questions give  $4 \times 17.5 = 70$  marks. **23.5** is the 2023/2024 figure, reconciled part by part from the scan we hold; three such questions would give 70.5. The total is the same; the split is not. **the code on the paper** is not a different course. CSC311, CSC416 and CSC242 are one subject re-named, and the questions repeat across all three names.

**What this means for practice.** Work the 2024/2025 and 2025/2026 papers under the real rubric: four questions, three hours. Work the older two for their content, which recurs almost unchanged, but do not use them to rehearse timing.

◆ **Memory hook: four of six, seventeen and a half each, three hours.**

Three hours is 180 minutes for 70 marks. Reserve time at both ends rather than discovering you have none.

| PHASE           | MINUTES | WHAT HAPPENS                                  |
|-----------------|---------|-----------------------------------------------|
| Read and choose | 10      | read all six questions, score them, pick four |
| Answer          | 160     | four questions at <b>40 minutes each</b>      |
| Check           | 10      | every number recomputed, every part numbered  |

#### WHERE

**40 minutes** is 160 divided by 4. **the rate** that follows is 40 minutes for 17.5 marks, which is a shade over **2 minutes per mark**. Multiply the printed marks by 2 and you have the minutes that part deserves. **10 minutes to choose** feels like a lot and is not. Answering the wrong four questions costs far more than ten minutes.

| A PART WORTH | GIVE IT ABOUT | WHICH IS ROUGHLY                                 |
|--------------|---------------|--------------------------------------------------|
| 1 mark       | 2 minutes     | one sentence, or one line of working             |
| 2.5 marks    | 5 minutes     | a short computation with its formula named       |
| 4.5 marks    | 9 minutes     | an expression, a table and a sentence of comment |
| 7.5 marks    | 15 minutes    | five terms defined, each with an example         |
| 10 marks     | 20 minutes    | a full worked answer with a diagram              |

**The rule that saves the most marks:** when a part's time is up, leave it and move on, even mid-sentence. The first half of the next part is always worth more than the last quarter of this one. Come back if the check time survives.

◆ **Memory hook: two minutes a mark, ten at each end.**

## Choosing the four

### TEACH

### A TEN-MINUTE METHOD

These questions are not single problems. Each is a bundle of five to eight small parts, often from unrelated topics, and the parts carry their own marks. That changes the choosing completely: you are not looking for the question you can solve, you are looking for the question whose **parts** you can bank.

- 1 Read all six. Do not start writing.** Six questions at ninety seconds each is nine minutes, and it is the highest-value nine minutes of the paper.
- 2 Beside each part, write a tick, a cross or a question mark.** Tick means "I can do this now". Question mark means "probably, with thought". Cross means "no".
- 3 Add up the ticked marks per question.** That number, not your feeling about the topic, is what to choose on.
- 4 Take the four highest.** A question with 12 ticked marks and 5.5 crossed beats one you like the look of with 8 ticked.
- 5 Start with your strongest.** It banks marks early and settles you. Leave the weakest until last, when you know exactly how much time is left for it.
- 6 Number every part exactly as the paper numbers it.** If the paper labels a part "b i.", so do you. A marker who cannot find a part cannot award it.

**Where the marks hide.** On the 2025/2026 paper the parts run 10, 3, 2, 2.5, 5, 2, 2, 4, 4.5, 7, 5, 5.5, and so on. A five-part question you can half-do is worth more than a two-part question you cannot start. Never abandon a question because one part defeats you.

## What the command words are asking for

### ● MUST MEMORISE

### COMMAND WORDS ON THESE PAPERS

Every one of these appears on a paper you hold. Answering the wrong verb is the commonest way to write a page and score half of it.

| WORD                           | WANTS                                     | THE TEST OF A COMPLETE ANSWER                                       |
|--------------------------------|-------------------------------------------|---------------------------------------------------------------------|
| <b>Define</b>                  | the standard wording                      | could someone who has never met the term use it from your sentence? |
| <b>State</b>                   | the claim itself, no proof                | is it a complete sentence that could be quoted?                     |
| <b>Explain</b>                 | the definition plus why it is so          | have you written a "because"?                                       |
| <b>Differentiate</b>           | the <b>contrast</b> , not two definitions | does a sentence contain "whereas" or "but"?                         |
| <b>Discuss</b>                 | several points, developed                 | have you given as many points as there are marks?                   |
| <b>Short notes on</b>          | a paragraph each, <b>with an example</b>  | does every item have its own example?                               |
| <b>Show / Demonstrate</b>      | the working that leads there              | can each line be checked without you present?                       |
| <b>Prove</b>                   | an argument covering every case           | have you named the method in the first line?                        |
| <b>Compute / Find</b>          | a number, and the formula it came from    | is the formula written before the substitution?                     |
| <b>Draw / Illustrate</b>       | a labelled diagram                        | is every vertex, edge or region labelled?                           |
| <b>Present in tabular form</b> | an actual table with headed columns       | did you draw the rules, and total the columns?                      |
| <b>Write a C++ program</b>     | a complete program, input included        | does it have includes, main, and a return?                          |
| <b>Justify</b>                 | the answer and the reason together        | would the answer alone score nothing? then say why                  |

#### WHERE

"**with an example each**" is not decoration. When a part says it, roughly half that part's marks are in the examples, and an answer with five perfect definitions and no examples loses them. "**differentiate**" is the one most often misread. Foundations F.2 covered it: two definitions side by side is not a difference until you write the sentence that contrasts them. "**in detail**" or "clearly" attached to any verb means the marks are large and the expected answer is long.

◆ **Memory hook: answer the verb, not the topic.**



## Writing an answer that collects everything it earned

### TEACH

### SIX HABITS, NONE OF WHICH NEEDS EXTRA KNOWLEDGE

- ▶ **State the formula before you substitute.** The formula is usually a mark on its own, and it survives an arithmetic slip further down. Every worked example in this book opens with Given, Find and Formula for exactly that reason.
- ▶ **Name the rule or law you are using.** "by De Morgan", "by the Handshaking theorem", "by the product rule". It costs three words and turns an assertion into an argument.
- ▶ **Draw the diagram even when it is not asked for.** A graph question with a sketch is easier to mark and easier to get right, and no marker has ever deducted for a correct extra diagram.
- ▶ **Write the units and the object.** Not "9" but "9 vertices of degree 4". A bare number can be right and unrecognisable.
- ▶ **Check the sanity of every answer.** A degree sum must be even. A probability must lie between 0 and 1. A count of arrangements must be a whole number. Each of those has caught a real error in this book's own working.
- ▶ **If you run out of time, write the method in ordered lines.** "1. Sum of degrees is  $2|E| = 60$ . 2. Known vertices account for 41. 3. Solve the pair." That is most of the reasoning marks with none of the arithmetic.

### ⚠ TRAP

### THESE PAPERS CONTAIN REAL PRINTING ERRORS

Every paper in this book has at least one defect, and this manual quotes them exactly as printed rather than tidying them, so you meet them here first rather than in the hall. Two have made a part unanswerable as written: 2025/2026 Q3(a) names the same language twice, and 2024/2025 Q4(b) asks which relations are reflexive in both part i and part v.

#### What to do, in one line each:

- 1 **Do not stop.** A defect is not permission to leave the part blank, and it is never in your favour to do so.
- 2 **Say what you think is meant, in one sentence.** "Taking the second figure to refer to Efik, since otherwise the two counts are the same group."
- 3 **Answer that reading in full.** The marks are for the method, and the method does not change.
- 4 **If a second reading is plausible and cheap, give it too.** One extra line covering the alternative cannot lose marks and may win them.

**The same applies to a figure you cannot read.** 2023/2024 Q5(c) has two arrowheads that do not resolve even at high magnification. State the reading you are tabulating, tabulate it correctly, and say that the arrowheads on your own copy should be checked. Section 6.4 does exactly that, and it is the honest answer, not a hedge.

Do these in order. They are cheap and they recover marks that are already earned.

- 1 Count your questions.** Four, not three, not five. A fifth answer normally earns nothing and cost you time on the four that counted.
- 2 Check every part is numbered as the paper numbers it,** and that no part has been silently skipped.
- 3 Re-add every arithmetic answer.** Not re-derive: re-add. Most errors are in the last step.
- 4 Check the sanity list:** degree sums even, probabilities between 0 and 1, counts whole, a tree's edges one fewer than its vertices.
- 5 Make sure every "and give an example" has an example.**
- 6 Read your first answer again.** It was written when you were least settled and is where a whole part is most often missing.

## RECALL AND CHECK

- 1. How many questions do you answer, out of how many, and for how many marks each?**
- 2. How many minutes does a part worth 4.5 marks deserve?**
- 3. What do you write beside each part while choosing?**
- 4. What does "differentiate" want that two definitions do not give?**
- 5. What should you do when a printed question is defective?**
- 6. Name three sanity checks worth running in the last ten minutes.**

### ANSWERS

1. Four out of six, 17.5 marks each, 70 in total, in three hours.
2. About 9 minutes, at a shade over two minutes per mark.
3. A tick, a cross or a question mark, then total the ticked marks per question and take the four highest.
4. The contrast itself: a sentence with "whereas" in it.
5. Say in one sentence what you take it to mean, then answer that reading in full, and give the alternative if it is cheap.
6. Degree sums are even; probabilities lie between 0 and 1; counts of arrangements are whole numbers; a tree has one fewer edge than vertices.

## REFERENCE • R.3

# Trap Index and Cram Sheet

*Every trap this book has flagged, in one table, with the correction beside it. Then the drills the examiners repeat year after year, and the twenty facts to hold in your head walking in.*

*By the end of this section you will have a single revision pass that covers the mistakes most likely to cost you marks, the questions most likely to appear, and the facts most likely to be needed.*

## The trap index

### TEACH

### EVERY TRAP IN THIS BOOK, AND ITS CORRECTION

Read the middle column and see whether the right-hand column is already in your head. Anywhere it is not, go back to that section.

| SECTION | THE TRAP                                         | THE CORRECTION                                                                                                 |
|---------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| F.1     | whether $\mathbb{N}$ starts at 0 or 1            | either is defensible; <b>state which you use</b> , and it cannot be marked wrong                               |
| F.1     | repeating an element, or re-ordering a set       | $\{1, 2, 2, 3\}$ and $\{3, 1, 2\}$ are both just $\{1, 2, 3\}$                                                 |
| F.2     | "whenever" and "only if"                         | "q whenever p" is $p \rightarrow q$ ; "p only if q" is also $p \rightarrow q$ , not the reverse                |
| F.2     | "differentiate" answered with two definitions    | write the contrasting sentence: the one with "whereas" in it                                                   |
| 1.1     | rejecting the rows where p is false              | $p \rightarrow q$ is <b>true</b> whenever p is false; the only false row is T then F                           |
| 1.2     | negating by opposites                            | the negation of "stay at home" is "not stay at home", not "go out"                                             |
| 1.3     | "nearly equivalent"                              | agreement in every row or none; one differing row settles it                                                   |
| 1.4     | $\forall$ with an "and"                          | $\forall$ takes an <b>arrow</b> , $\exists$ takes an <b>and</b>                                                |
| 1.4     | reordering nested quantifiers                    | $\forall x \exists y$ allows a different y each time; $\exists y \forall x$ demands one y for all              |
| 1.5     | affirming the consequent, denying the antecedent | both invalid; only modus ponens and modus tollens go those ways                                                |
| 2.1     | equal versus equivalent sets                     | equal means same elements; equivalent means same <b>number</b> of them                                         |
| 2.1     | $\emptyset$ against $\{\emptyset\}$              | $n(\emptyset) = 0$ but $n(\{\emptyset\}) = 1$ ; the second is a singleton                                      |
| 2.1     | $\in$ against $\subseteq$                        | an element on the left of $\in$ , a set on the left of $\subseteq$ ; never mix                                 |
| 2.3     | a power set missing its two free members         | $\emptyset$ and A itself are always in it; count $2^n$ first, then list                                        |
| 3.1     | reading (a, b) either way round                  | (a, b) means a relates to b; the pair is the direction of the arrow                                            |
| 3.2     | antisymmetric as the opposite of symmetric       | a relation can be <b>both</b> ; antisymmetry only forbids two <i>different</i> elements pointing at each other |
| 3.2     | reflexive on most elements                       | every element needs its loop; three out of four is a fail                                                      |
| 3.4     | antisymmetric as "not symmetric" again           | loops are always allowed and never break antisymmetry                                                          |
| 4.1     | calling any relation a function                  | every input used, and each used exactly once                                                                   |

|      |                                                           |                                                                                      |
|------|-----------------------------------------------------------|--------------------------------------------------------------------------------------|
| 4.3  | assuming $f \circ g = g \circ f$                          | they are usually different; the exam asks for both to see whether you noticed        |
| 5.1  | reading English "or" as the sum rule                      | the sum rule needs the groups to be <b>disjoint</b> and one of them chosen           |
| 5.2  | permutation where a combination is meant                  | ask: if I swap two chosen objects, is that a different answer? yes means permutation |
| 6.4  | presenting an unreadable figure as read                   | state the reading you tabulate, and say the arrowheads should be re-checked          |
| 6.6  | Euler against Hamilton                                    | Euler uses each <b>edge</b> once, Hamilton each <b>vertex</b> once                   |
| 6.8  | nearest-neighbour offered as the minimum                  | a minimum is a claim about every alternative; list all 12 tours                      |
| 7.3  | prefix as the expression reversed                         | both prefix and postfix are read off the <b>tree</b> , never by moving symbols       |
| 9.1  | one letter for two arbitrary integers                     | $m = 2a + 1$ and $n = 2b + 1$ ; the same letter proves only the case $m = n$         |
| 9.1  | copying the course text's proof-by-cases algebra          | $8k^3 - 2k = 2(4k^3 - k)$ , and $(2k + 1)^3$ has $12k^2$ , not $6k^2$                |
| 9.1  | a claim whose truth depends on a convention               | state the convention first; for $n^2 + 3n + 2$ it decides the answer outright        |
| 9.2  | circular induction, unused hypothesis, asserted base case | start from one side, cite $P(k)$ by name, and evaluate both sides of the base case   |
| 9.3  | mixing the two indexing conventions                       | say whether the first term is $a_0$ or $a_1$ , then count the list to be sure        |
| 10.1 | quoting the mean when there is an outlier                 | the median resists outliers, the mean does not; say which you quote and why          |
| 10.2 | the course text's "mutually exclusive or disjoint"        | that sentence defines <b>independence</b> ; for real events the two are opposites    |
| 10.2 | the course text's $P(A \cup B) = P(A) + P(B)$             | that is the <b>disjoint</b> case; in general subtract $P(A \cap B)$                  |
| 11.1 | writing code as though it will be run                     | skeleton first, comment the numbered steps, indent, and keep the paper's own names   |
| R.2  | stopping at a defective question                          | state the reading in one sentence, answer it in full, add the alternative if cheap   |

## The drills the examiners repeat

### ● MUST MEMORISE

### FOURTEEN QUESTIONS THAT KEEP COMING BACK

Compiled by walking all five papers and counting. If your revision time runs out, this is the order to spend it in.

| DRILL                                                             | APPEARED IN                    | SECTION  |
|-------------------------------------------------------------------|--------------------------------|----------|
| Types of undirected graph, with diagrams                          | 20/21, 21/22, 23/24            | 6.2      |
| The travelling missionary, minimum tour from Detroit              | 20/21, 21/22, 23/24            | 6.8      |
| Chain letter or convocation cards, m-ary counting                 | 20/21, 21/22, 23/24            | 7.2      |
| Handshaking: state it, prove it, apply it                         | 21/22, 23/24, 24/25            | 6.3      |
| R1 to R6 relation classification                                  | 20/21, 23/24, 24/25            | 3.2      |
| Rooted-tree terminology, traversals, prefix and postfix           | 20/21, 21/22, 24/25            | 7.1, 7.3 |
| Truth tables, converse, inverse, contrapositive                   | every year                     | 1.2, 1.3 |
| Counting: passwords, chairs, PIN codes, word arrangements         | most years                     | 5.1, 5.2 |
| In-degree and out-degree of a digraph, in tabular form            | 21/22, 23/24, 25/26            | 6.4      |
| The five function correspondence types, with a diagram            | 20/21, 21/22                   | 4.2      |
| Bit-string representation of a set                                | 23/24, 24/25                   | 2.3      |
| Inclusion and exclusion word problems                             | 21/22, 25/26                   | 5.3      |
| Balanced m-ary tree, leaves at height h                           | 20/21, 21/22, 24/25            | 7.2      |
| $f \circ g$ and $g \circ f$ for $f(x) = x^2 + 1$ , $g(x) = x + 2$ | 23/24 and 24/25, word for word | 4.3      |

#### WHERE

"every year" means it appears on all five papers we hold. "word for word" on the last row is literal: the 2023/2024 and 2024/2025 papers set the identical composition question, the later one adding "State if  $f \circ g = g \circ f$ ." the **section column** is where the drill is taught and worked; each of those sections carries the quoted question itself.

**What the table is really telling you.** Six of the fourteen are graph or tree questions, and the official course text teaches none of them. That is the whole shape of this course in one observation.

◆ **Memory hook: graphs, trees, relations and counting are most of the paper, and none of the first three is in the course text.**

## The cram sheet

### ● MUST MEMORISE

### TWENTY FACTS TO WALK IN WITH

| FACT                                                                                                               | SECTION  |
|--------------------------------------------------------------------------------------------------------------------|----------|
| 1 $p \rightarrow q$ is false in exactly one row: $p$ true, $q$ false                                               | 1.2      |
| 2 The contrapositive $\neg q \rightarrow \neg p$ is the only one of the three that agrees with $p \rightarrow q$   | 1.2      |
| 3 $\neg(p \wedge q) \equiv \neg p \vee \neg q$ , and the same law in sets and in Boolean                           | 1.3      |
| 4 $\forall$ takes an arrow, $\exists$ takes an and                                                                 | 1.4      |
| 5 A set of $n$ elements has $2^n$ subsets, and a truth table with $n$ letters has $2^n$ rows                       | 2.1      |
| 6 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$                                                                        | 2.2      |
| 7 Reflexive needs every element; antisymmetric allows loops; a relation may be symmetric and antisymmetric at once | 3.2      |
| 8 An equivalence relation is reflexive, symmetric and transitive, and it partitions the set                        | 3.3      |
| 9 $P(n, r) = n! / (n - r)!$ orders; $C(n, r) = n! / (r!(n - r)!)$ does not                                         | 5.2      |
| 10 "At least one" means total minus none                                                                           | 5.1, 5.2 |
| 11 $\sum \deg(v) = 2 E $ , so the degree sum is always even and the odd-degree vertices always even in number      | 6.3      |
| 12 $K_n$ has $C(n, 2) = n(n - 1) / 2$ edges                                                                        | 6.2      |
| 13 Euler circuit: every degree even. Euler path: exactly two odd                                                   | 6.6      |
| 14 $v - e + f = 2$ , and a simple planar graph obeys $e \leq 3v - 6$                                               | 6.7      |
| 15 A tree has $e = v - 1$ ; a full $m$ -ary tree has $n = mi + 1$ and $l = (m - 1)i + 1$                           | 7.1, 7.2 |
| 16 An $m$ -ary tree of height $h$ has at most $m^h$ leaves                                                         | 7.2      |
| 17 Preorder is root first, inorder root in the middle, postorder root last, and all three come off the tree        | 7.3      |
| 18 Induction is base case, hypothesis, step, and the step must cite the hypothesis                                 | 9.2      |
| 19 $a_n = a + (n - 1)d$ and $a_n = ar^{n-1}$ ; the $n - 1$ is the number of steps taken                            | 9.3      |
| 20 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ , and $P(B A) = P(A \cap B) / P(A)$                                   | 10.2     |

#### WHERE

$|E|$  and  $|V|$  are the numbers of edges and vertices.  $i$  is the count of **internal** vertices of a full  $m$ -ary tree and  $l$  its leaves.  $d$  is a common difference and  $r$  a common ratio.  $n$  means "how many" throughout, and its meaning is fixed by the sentence it sits in, not by the letter.

◆ **Memory hook for the graph block: degrees double the edges, complete graphs pick two, Euler counts odd vertices, and a tree is one edge short.**

## TEACH

## THE SANITY CHECKS, IN ONE PLACE

Run these on every answer before you leave it. Each one has caught a real error while this book was being written.

| IF YOU COMPUTED                    | IT MUST BE                           | BECAUSE                                                     |
|------------------------------------|--------------------------------------|-------------------------------------------------------------|
| a sum of degrees                   | even                                 | it equals $2 E $                                            |
| the number of odd-degree vertices  | even                                 | the same theorem                                            |
| a probability                      | between 0 and 1                      | the first axiom                                             |
| two complementary probabilities    | adding to 1                          | the complement rule                                         |
| a count of arrangements            | a whole number                       | you are counting objects                                    |
| $C(n, r)$                          | equal to $C(n, n - r)$               | choosing $r$ is choosing which $n - r$ to leave             |
| the edges of a tree                | one fewer than the vertices          | $e = v - 1$                                                 |
| the leaves of a full $m$ -ary tree | agreeing with both formulas          | $n = mi + 1$ and $l = (m - 1)i + 1$ must give the same tree |
| a set of deviations from a mean    | summing to zero                      | that is what the mean is                                    |
| a minimum tour                     | no longer than every tour you listed | a minimum is a claim about all of them                      |



## RECALL AND CHECK

1. Which of the converse, inverse and contrapositive agrees with the original conditional?
2. State the Handshaking theorem and one thing it immediately tells you.
3. When is a relation both symmetric and antisymmetric?
4. Give the two conditions for an Euler circuit and for an Euler path.
5. How many edges has a tree on 30 vertices?
6. What is the difference between the mean and the median in the presence of an outlier?
7. Name three drills that have appeared on at least three of the five papers.

---

### ANSWERS

1. Only the contrapositive.
2.  $\sum \deg(v) = 2|E|$ . It tells you at once that the degree sum is even, so no graph can have an odd degree sum, and that the number of odd-degree vertices is even.
3. When no two **different** elements are related in both directions, which includes any relation made only of loops.
4. Circuit: connected and every vertex of even degree. Path: connected and exactly two vertices of odd degree.
5. 29, since  $e = v - 1$ .
6. The median is unaffected by an extreme value; the mean is dragged towards it.
7. Any three of: types of undirected graph, the minimum tour, the chain letter, Handshaking, the R1 to R6 classification, rooted-tree terminology and traversals, the digraph tabulation, balanced m-ary trees.

## PAST PAPERS

# All Five Papers, Solved in Full

*Every question of every paper we hold, answered in paper order at exam length. The teaching parts quoted each question as printed and worked it slowly; this part shows what the answer looks like when the clock is running and you have forty minutes for seventeen and a half marks.*



## 2025/2026 Solved in Full

*The newest paper, and the template for the one you will sit. Six questions of 17.5 marks, answer any four, three hours. Every part is answered here in paper order; each question is quoted as printed in the section that teaches it, and the pointer is given so you can read the two side by side.*

*By the end of this section you will have seen a complete seventy-mark answer to this paper, and the reasoning behind choosing which four questions to take.*

| TEACH     |              |                 | THE PAPER AT A GLANCE |              |
|-----------|--------------|-----------------|-----------------------|--------------|
| SESSION   | CODE PRINTED | INSTRUCTION     | TIME                  | PER QUESTION |
| 2025/2026 | CSC 242      | answer any FOUR | 3 hours               | 17.5 marks   |

**Where the topics sit.** Q1 mixes definitions from Parts One, Two and Six. Q2 is sets and relations. Q3 is pure counting. Q4 is graph short notes plus logic plus one counting part. Q5 is Boolean algebra plus planarity. Q6 is matrices plus counting.

**The defects on this paper,** all quoted as printed and flagged in the sections that carry them: Q1(a) prints "Prepositional logic" and "Path Matrice", and runs "(2marks)" unspaced on item v only. Q2(a) is missing a word. **Q3(a) names the same language twice**, which makes seven marks unanswerable exactly as written; the answer below states the reading it adopts before computing anything.

Score each question by the marks you can bank, not by the topic you like. Worked here on the assumption that you have read this book:

| QUESTION | WHAT IT IS                                               | HARDEST PART                                             | TAKE IT?                                     |
|----------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------|
| Q1       | five differences, nine set types, one small computation  | the five differences at 2 marks each: it is all recall   | <b>yes</b> , highest density of recall marks |
| Q2       | five reasons, a product, three relation checks           | Q2(e), classifying the CGPA table with proof             | <b>yes</b>                                   |
| Q3       | inclusion and exclusion, combinations, prayer partners   | Q3(c), which needs three different readings              | <b>yes</b> if counting is solid              |
| Q4       | five graph short notes, logic in computing, SCHOOL       | nothing: it is five definitions and one formula          | <b>yes</b> , the easiest 17.5 on the paper   |
| Q5       | Boolean tables, Kuratowski, two planarity proofs         | Q5(b)(iii), 6 marks of counting edges against a bound    | reserve                                      |
| Q6       | adjacency matrix, memory, PIN codes, three-digit numbers | Q6(c)(ii), which asks about the question's own structure | reserve                                      |

**The recommendation: Q1, Q2, Q3, Q4.** They are 45 marks of recall and standard computation and 25 of judgement. Q5 and Q6 are not harder in themselves, but each has one part that can absorb fifteen minutes, and on a four-of-six paper that is the deciding consideration.

**All six are answered below anyway**, because the point of a past paper is not to rehearse a choice but to remove the need for one.

## Question One

### ANSWER

2025/2026 Q1 · 17.5 MARKS

#### A. FIVE DIFFERENCES, WITH TWO EXAMPLES EACH (10 MARKS)

*Quoted as printed in section 6.1.* Each pair needs a contrasting sentence, not two definitions: see F.2.

**i. Degree versus valency of a node.** They are two names for the same quantity, the number of edge ends meeting at a vertex, whereas "valency" is the older term borrowed from chemistry and "degree" is the one this syllabus uses elsewhere. Examples: in a triangle every vertex has degree 2 and equally valency 2; a pendant vertex has degree 1 and valency 1.

**ii. Predicate versus propositional logic.** A proposition is either true or false outright, whereas a predicate has a variable in it and only becomes true or false once the variable is given a value. Examples: "3 is prime" is a proposition; "x is prime" is a predicate  $P(x)$ , true at  $x = 3$  and false at  $x = 4$ .

**iii. Universal versus existential quantifier.**  $\forall x$  claims the predicate holds for **every** value in the domain, whereas  $\exists x$  claims it holds for **at least one**. Examples: over the integers  $\forall x(x^2 \geq 0)$  is true;  $\exists x(x^2 = 4)$  is true, and  $\forall x(x^2 = 4)$  is false.

**iv. Adjacency versus path matrix.** The adjacency matrix records whether two vertices are joined by a **single edge**, whereas the path matrix records whether one can be reached from the other by **any** sequence of edges. Examples: in the path  $a$  to  $b$  to  $c$ , the adjacency matrix has 0 in the  $(a, c)$  position but the path matrix has 1; in a complete graph the two matrices agree everywhere off the diagonal.

**v. Logical matrix versus relation matrix.** A logical matrix is any matrix whose entries are the truth values 0 and 1, whereas a relation matrix is a logical matrix used specifically to record a relation, with  $m_{ij} = 1$  exactly when  $(i, j)$  is in the relation.

Examples: an adjacency matrix is a logical matrix; the matrix of "divides" on  $\{1, 2, 3, 4\}$  is a relation matrix.

#### B. I. NINE TYPES OF SET (3 MARKS) II. TWO WAYS TO DESCRIBE A SET (2 MARKS)

*Quoted as printed in section 2.1.*

**i.** Empty  $\{\}$ , singleton  $\{5\}$ , finite  $\{1, 2, 3\}$ , infinite  $\mathbb{N}$ , equal ( $\{1, 2\}$  and  $\{2, 1\}$ ), equivalent ( $\{1, 2\}$  and  $\{a, b\}$ ), subset ( $\{1\}$  of  $\{1, 2\}$ ), universal ( $U$ , everything under discussion), and power set (all subsets of a set).

**ii. Roster form**, listing the members between braces, and **set-builder form**, stating the property they share:  $\{x \mid x \text{ is a positive prime below } 10\}$ .

#### C. $A = \{11, 12, 13, 14\}$ , $B = \{13, 14, 15, 16\}$ (2.5 MARKS)

*Quoted as printed in section 2.2.*

$$A \cap B = \{13, 14\}$$

$$A - B = \{11, 12\}$$

$$B - A = \{15, 16\}$$

Each is checked by reading  $A$  once and asking of each member whether it is in  $B$ .

## Question Two

### ANSWER

2025/2026 Q2 · 17.5 MARKS

#### A. FIVE JUSTIFICATIONS FOR THE SUBJECT (5 MARKS)

*Quoted as printed in Foundations F.1.* One mark each, so five distinct points with an application named in each.

**1. It is the mathematics of the computer itself**, which stores and processes separated values rather than a continuum. **2. Logic is the basis of circuit design and of program correctness**, as Part Eight shows for gates and Part One for conditions. **3. Graphs and trees are the standard data models** for networks, file systems, compilers and routing. **4. Counting gives complexity and capacity analysis**: how many passwords, how many states, how many operations. **5. Proof and induction justify algorithms**, which is how one shows a program is right for every input rather than for the ones tested.

#### B. $S \times C$ WHERE $S$ HAS 3 AND $C$ HAS 2 MEMBERS (2 MARKS)

*Quoted as printed in section 2.3.* The Cartesian product is every ordered pair with the first from  $S$  and the second from  $C$ , so there are  $3 \times 2 = 6$  of them:

$\{(\text{John}, \text{CSC701}), (\text{John}, \text{CSC702}), (\text{Mary}, \text{CSC701}), (\text{Mary}, \text{CSC702}), (\text{Peter}, \text{CSC701}), (\text{Peter}, \text{CSC702})\}$

#### C. IS $R$ REFLEXIVE? (2 MARKS)

*Quoted as printed in section 3.3.*  $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$  on  $\{1, 2, 3, 4\}$ .

**Yes.** Reflexivity requires  $(a, a)$  for **every**  $a$  in the set. All four loops  $(1,1), (2,2), (3,3), (4,4)$  are present, so  $R$  is reflexive. The extra pairs  $(1,2)$  and  $(2,3)$  do not affect reflexivity.

#### D. IS $R$ AN EQUIVALENCE RELATION? (4 MARKS)

*Quoted as printed in section 3.3.* Three properties, each checked explicitly, which is what "detailed explanation" is asking for.

**Reflexive:**  $(1,1), (2,2), (3,3), (4,4)$  are all present. Yes.

**Symmetric:** the non-loop pairs are  $(1,3)$  and  $(3,1)$ ,  $(2,4)$  and  $(4,2)$ . Each appears with its reverse. Yes.

**Transitive:**  $(1,3)$  with  $(3,1)$  needs  $(1,1)$ , present;  $(3,1)$  with  $(1,3)$  needs  $(3,3)$ , present;  $(2,4)$  with  $(4,2)$  needs  $(2,2)$ , present;  $(4,2)$  with  $(2,4)$  needs  $(4,4)$ , present. No other composable pair exists. Yes.

**So  $R$  is an equivalence relation**, and it partitions  $\{1, 2, 3, 4\}$  into the classes  **$\{1, 3\}$**  and  **$\{2, 4\}$** . Naming the classes is the line that shows you know what an equivalence relation is for.

#### E. PARTIALLY VERSUS TOTALLY ORDERED, AND THE CGPA TABLE (4.5 MARKS)

*Quoted as printed in section 3.4.*

**The difference.** In a partially ordered set some pairs may be incomparable, whereas in a totally ordered set **every** two elements are comparable. Both orders are reflexive, antisymmetric and transitive; total order adds comparability.

**The table.** A has 4.80, B has 4.50, C has 4.10. Order them by "has a CGPA at least as high as". Every pair can be compared, because any two real numbers can:  $4.80 \geq 4.50 \geq 4.10$ .

**Proof that it is a total order.** Reflexive, since each student's CGPA is at least as high as itself. Antisymmetric, since if two are each at least as high as the other their CGPAs are equal and, these three being distinct, the students are the same. Transitive, by the transitivity of  $\geq$  on the reals. Comparable, since  $\geq$  compares any two reals. Therefore **totally ordered**, and the chain is A, then B, then C.

## Question Three

### ANSWER

2025/2026 Q3 · 17.5 MARKS

#### A. 100 STUDENTS, TWO LANGUAGES (7 MARKS)

*Quoted as printed in section 5.3.*

**State the reading first.** The question as printed names Ibibio twice, for both the 72 and the 43, which would make the two figures describe one group and leave nothing to compute. The only reading that makes the question answerable, and the one taken here, is that **72 speak Ibibio and 43 speak Efik**. Say that in one line before computing anything.

**(i) Both.** Every student speaks at least one, so the union is 100:

$$n(I \cup E) = n(I) + n(E) - n(I \cap E)$$

$$100 = 72 + 43 - n(I \cap E)$$

$$n(I \cap E) = 115 - 100 = \mathbf{15}$$

**(ii) Ibibio only.**  $72 - 15 = \mathbf{57}$

**(iii) Efik only.**  $43 - 15 = \mathbf{28}$

**Check:**  $57 + 15 + 28 = 100$ . The three regions of the diagram account for every student, which is the one-line verification worth writing.

#### B. FOUR FROM SIX BOYS AND FOUR GIRLS, AT LEAST ONE BOY (5 MARKS)

*Quoted as printed in section 5.2.* "At least one" means total minus none.

$$\text{total committees} = C(10, 4) = \mathbf{210}$$

$$\text{with no boy, that is all four girls} = C(4, 4) = \mathbf{1}$$

$$\text{at least one boy} = 210 - 1 = \mathbf{209}$$

**Why not add up the cases.**  $C(6,1)C(4,3) + C(6,2)C(4,2) + C(6,3)C(4,1) + C(6,4) = 24 + 90 + 80 + 15 = 209$  as well, but it is four computations where the complement is one.

#### C. TEN MALE AND TEN FEMALE PRAYING IN PAIRS (5.5 MARKS)

*Quoted as printed in section 5.2.*

**(i) Sessions.** Each session pairs one male with one female and every person prays once per round, so ten pairs form at a time and there are **10** sessions if each male is to pray with each female.

**(ii) Prayer units.** Every male paired with every female is  $10 \times 10 = \mathbf{100}$  distinct units, by the product rule.

**(iii) Choice of prayer partners.** Fix the order of the males and assign the females to them; that is an arrangement of ten distinct people in ten positions:

$$10! = \mathbf{3628800}$$

**Say which reading you used.** The three parts are answered here as "one round of ten pairs at a time, ten rounds in all", and stating that protects every mark below it.



## Question Four

### ANSWER

2025/2026 Q4 · 17.5 MARKS

#### A. SHORT NOTES WITH EXAMPLES, FIVE ITEMS (10 MARKS)

*Quoted as printed in section 6.1.* Two marks each, so each item wants a definition and an example.

**i. Size of a graph.** The number of edges, written  $|E|$ . In the complete graph  $K_4$  the size is 6; in a tree on 10 vertices the size is 9.

**ii. Valency of a vertex.** The number of edge ends meeting at it, the same quantity as its degree. A pendant vertex has valency 1; every vertex of  $K_5$  has valency 4.

**iii. In-degree and out-degree.** In a directed graph the in-degree of  $v$  is how many arrows arrive at  $v$  and the out-degree how many leave. If  $a$  to  $b$  and  $c$  to  $b$  are the only arcs, then  $b$  has in-degree 2 and out-degree 0, while  $a$  has out-degree 1 and in-degree 0.

**iv. Graph isomorphism.** Two graphs are isomorphic when there is a one-to-one correspondence between their vertices that preserves adjacency: the same graph drawn twice. A square drawn as a square and the same four vertices drawn as a diamond are isomorphic;  $K_3$  and a path on three vertices are not, since they have different numbers of edges.

**v. Eulerian graph.** A graph containing a circuit that uses every edge exactly once and returns to its start, which happens exactly when the graph is connected and every vertex has even degree. A triangle is Eulerian; the Königsberg graph, with four odd-degree vertices, is not.

#### B. THE PLACE OF LOGIC IN COMPUTER SCIENCE (2 MARKS)

*Quoted as printed in section 1.5.* Logic supplies the rules that separate a valid argument from an invalid one, and computing rests on it at every level: circuit design builds directly from AND, OR and NOT (Part Eight); program conditions and loop tests are propositions; database queries are predicate logic; and proving a program correct is proving an implication.

#### C. THREE ACCEPTABLE LOGIC STATEMENTS LEADING TO A CONCLUSION (3 MARKS)

*Quoted as printed in section 1.5.* A valid argument, laid out as premises and conclusion:

If a program compiles, it has no syntax error.

If it has no syntax error, the parser accepted it.

This program compiles.

∴ The parser accepted it.

**Naming the rules earns the marks:** the first two premises chain by **hypothetical syllogism** to give "if it compiles, the parser accepted it", and with the third premise **modus ponens** gives the conclusion.

#### D. ARRANGEMENTS OF THE LETTERS OF SCHOOL (2.5 MARKS)

*Quoted as printed in section 5.2.* Six letters with O repeated twice:

$$6! / 2! = 720 / 2 = 360$$

**Why the division.** The two O's are identical, so each distinct arrangement has been counted  $2! = 2$  times by  $6!$ , once for each way of ordering them.

## Question Five

### ANSWER

2025/2026 Q5 · 17.5 MARKS

#### A. I. THE LABORATORY DOOR (4.5 MARKS)

Quoted as printed in section 8.1. Let  $A = 1$  when a valid card is presented,  $B = 1$  when the fingerprint is correct,  $D = 1$  when the door opens. The door opens only when both are presented, so

$$D = A \cdot B$$

which is 1 in exactly one of the four cases.

| A | B | $D = A \cdot B$ | DOOR  |
|---|---|-----------------|-------|
| 0 | 0 | 0               | shut  |
| 0 | 1 | 0               | shut  |
| 1 | 0 | 0               | shut  |
| 1 | 1 | 1               | opens |

#### A. II. THE ALARM (3 MARKS)

Let  $F$  and  $S$  be the two sensors and  $L$  the alarm. "Either or" here is the **inclusive** or, since the alarm must certainly sound if both fire:

$$L = F + S$$

With only the smoke sensor active,  $F = 0$  and  $S = 1$ , so  $L = 0 + 1 = 1$ : **the alarm sounds**. Had exclusive or been used the alarm would fall silent when both sensors detected danger, which is the sentence that secures the mark.

#### B. I. KURATOWSKI'S THEOREM (2 MARKS)

Quoted as printed in section 6.7. A graph is planar if and only if it contains no subgraph that is a subdivision of  $K_5$  or of  $K_{3,3}$ . In words: those two graphs are the only obstructions to planarity, and any non-planar graph has a copy of one of them hidden inside it.

#### B. II. WHY $K_5$ AND $K_{3,3}$ MATTER (2 MARKS)

Because by Kuratowski's theorem they are the **minimal** non-planar graphs: every non-planar graph contains a subdivision of one of them, so the whole question of planarity reduces to looking for these two. They are the reference points against which any drawing is tested.

#### B. III. SHOWING BOTH ARE NON-PLANAR (6 MARKS)

**$K_5$** . It has  $v = 5$  vertices and, being complete,  $e = C(5, 2) = 10$  edges. For a simple connected planar graph with  $v \geq 3$ , Euler's formula forces

$$e \leq 3v - 6 = 3 \times 5 - 6 = 9$$

but  $10 > 9$ , so the bound fails and  **$K_5$  is not planar**.

**$K_{3,3}$** . It has  $v = 6$  and  $e = 3 \times 3 = 9$ . It is bipartite, so it has no triangle, and the triangle-free bound applies:

$$e \leq 2v - 4 = 2 \times 6 - 4 = 8$$

but  $9 > 8$ , so  **$K_{3,3}$  is not planar**.

**Say why the second bound is different.** Using  $e \leq 3v - 6$  on  $K_{3,3}$  gives 12, which 9 satisfies, and the argument would prove nothing. The tighter bound is available precisely because a bipartite graph has no odd cycle and therefore no triangle. That sentence is worth a mark on its own.

## Question Six

### ANSWER

2025/2026 Q6 · 17.5 MARKS

#### A. I. THE ADJACENCY MATRIX, EXPLAINED (2.5 MARKS)

*Quoted as printed in section 6.5.* For a graph on  $n$  vertices labelled 1 to  $n$ , the adjacency matrix is the  $n$  by  $n$  array whose entry  $a_{ij}$  is 1 when vertices  $i$  and  $j$  are joined by an edge and 0 when they are not. For an undirected graph it is symmetric about the main diagonal, since an edge joining  $i$  to  $j$  equally joins  $j$  to  $i$ , and the diagonal is 0 unless the graph has loops.

#### A. II. MEMORY MANAGEMENT IN AN ADJACENCY MATRIX (4 MARKS)

**The cost.** The matrix always occupies  $n^2$  entries, whatever the number of edges, because every pair has a slot whether or not it is used.

**When that is wasteful.** A sparse graph, where the edges are far fewer than  $n^2$ , stores overwhelmingly zeros. A road network with a thousand junctions and three thousand roads still needs a million entries.

**Two ways it is managed.** Store only half of it, since symmetry makes the upper triangle sufficient for an undirected graph, roughly halving the space. Or store one **bit** per entry rather than one integer, since every entry is 0 or 1.

**The alternative.** An adjacency **list** keeps, for each vertex, only the vertices actually adjacent to it, so it uses space in proportion to the edges rather than to  $n^2$ . The trade is speed: the matrix answers "are  $i$  and  $j$  adjacent" in one lookup, the list has to search.

#### B. ATM PIN CODES OF 4 TO 6 DIGITS (5 MARKS)

*Quoted as printed in section 5.1.* Ten digits, repeats allowed, so a code of length  $k$  can be formed in  $10^k$  ways. The three lengths are alternatives, so add them:

$$10^4 + 10^5 + 10^6 = 10000 + 100000 + 1000000 = \mathbf{1110000}$$

**Both rules appear here.** The product rule gives each  $10^k$ , one factor of 10 per position; the sum rule joins the three lengths, because a code has exactly one length.

#### C. I. THREE-DIGIT NATURAL NUMBERS (1 MARK)

*Quoted as printed in section 5.1.* The first digit cannot be 0, so it has 9 choices; the other two have 10 each:

$$9 \times 10 \times 10 = \mathbf{900}$$

**Check:** the three-digit numbers run from 100 to 999, and  $999 - 100 + 1 = 900$ .

#### C. II. THE PARTS OF THE TASK, AND THEIR TYPE (4 MARKS)

The task of forming a three-digit number has **three** parts, choosing the hundreds digit, the tens digit and the units digit. They are **compulsory** and not alternative: every one must be performed to produce a number, which is exactly the condition the product rule requires. Had they been alternatives, only one would be carried out and the sum rule would apply, as it does across the three PIN lengths in part (b) above.

#### C. III. WHICH RULE APPLIES (1 MARK)

The **product rule**, since the three digit choices are performed one after another and every combination of them gives a different number.

| TEACH    |                           | MARK RECONCILIATION |
|----------|---------------------------|---------------------|
| QUESTION | PARTS                     | TOTAL               |
| Q1       | $10 + 3 + 2 + 2.5$        | 17.5                |
| Q2       | $5 + 2 + 2 + 4 + 4.5$     | 17.5                |
| Q3       | $7 + 5 + 5.5$             | 17.5                |
| Q4       | $10 + 2 + 3 + 2.5$        | 17.5                |
| Q5       | $4.5 + 3 + 2 + 2 + 6$     | 17.5                |
| Q6       | $2.5 + 4 + 5 + 1 + 4 + 1$ | 17.5                |

Every question reconciles to 17.5, so no part has been misread and none is missing from this answer. Doing this arithmetic in the first minute of the exam is worth the thirty seconds: a question whose printed parts do not add to 17.5 has a part you have not noticed.

#### PAST PAPER

## 2024/2025 Solved in Full

*The same format as the newest paper, four questions of 17.5 from six, three hours, and by some distance the hardest of the five. It sets two C++ programs, a 17.5-mark graph scenario built entirely around one trap, and a proof question. Every part is answered here.*

*By the end of this section you will have a complete answer to the hardest paper in the set, including its two programs, its Handshaking scenario and the proof in Q1(c)(ii).*

| TEACH     |              |                  |         | THE PAPER AT A GLANCE |
|-----------|--------------|------------------|---------|-----------------------|
| SESSION   | CODE PRINTED | INSTRUCTION      | TIME    | PER QUESTION          |
| 2024/2025 | CSC416       | attempt any FOUR | 3 hours | 17.5 marks            |

**This paper writes its marks without brackets:** "4.5 marks", not "(4.5 marks)". Small, but it tells you at a glance which paper you are holding.

**Its defects,** all quoted as printed in the sections that carry them. Q2 welds its part letter to its first numeral, printing "bi." then "ii.", "iii.". Q3(c) prints De Morgan with the overbar over B alone and a dash where an equals sign belongs. Q3(d) closes a set with two round brackets. **Q4(b) asks which relations are reflexive twice**, as part i and again as part v. Q6(b) numbers its parts i, iv, v, vi, vii, skipping ii and iii, and two superscripts in its expression cannot be read at any available resolution.

**The trap the paper is built around** is Q5, which states that the graph has five vertices of degree 4 and then asks how many vertices of degree 4 it has. The answer is not five.

## Question One

### ANSWER

2024/2025 Q1 · 17.5 MARKS

#### A. CONVERSE, CONTRAPOSITIVE AND INVERSE OF THREE CONDITIONALS (4.5 MARKS)

Quoted as printed in section 1.2. Identify p and q first; that is where the marks are lost, not in the three rewrites.

i. "If it snows tonight, then I will stay at home." p = it snows tonight, q = I will stay at home.

**Converse:** If I stay at home tonight, then it will snow.

**Contrapositive:** If I do not stay at home, then it will not snow.

**Inverse:** If it does not snow tonight, then I will not stay at home.

ii. "I go to the beach whenever it is a sunny summer day." The **whenever** clause is the hypothesis, so p = it is a sunny summer day, q = I go to the beach. Writing it the other way round loses every mark that follows.

**Converse:** If I go to the beach, then it is a sunny summer day.

**Contrapositive:** If I do not go to the beach, then it is not a sunny summer day.

**Inverse:** If it is not a sunny summer day, then I do not go to the beach.

iii. "When I stay up late, it is necessary that I sleep until noon." "It is necessary that q" makes q the **conclusion**, so p = I stay up late, q = I sleep until noon.

**Converse:** If I sleep until noon, then I stayed up late.

**Contrapositive:** If I do not sleep until noon, then I did not stay up late.

**Inverse:** If I do not stay up late, then I do not sleep until noon.

#### B. FOUR TRUTH TABLES (4 MARKS)

Quoted as printed in section 1.1.

i.  $p \leftrightarrow \neg p$

| P | $\neg P$ | $P \leftrightarrow \neg P$ |
|---|----------|----------------------------|
| T | F        | F                          |
| F | T        | F                          |

**False in every row: it is a contradiction**, which is worth saying rather than leaving the marker to notice.

ii.  $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

| P | Q | $\neg P$ | $Q \rightarrow \neg P$ | $P \leftrightarrow Q$ | THE WHOLE |
|---|---|----------|------------------------|-----------------------|-----------|
| T | T | F        | F                      | T                     | F         |
| T | F | F        | T                      | F                     | F         |
| F | T | T        | T                      | F                     | F         |
| F | F | T        | T                      | T                     | T         |

**True in one row only**, when both are false.

iii.  $(p \oplus q) \vee (p \oplus \neg q)$

| P | Q | $P \oplus Q$ | $P \oplus \neg Q$ | THE WHOLE |
|---|---|--------------|-------------------|-----------|
| T | T | F            | T                 | T         |
| T | F | T            | F                 | T         |
| F | T | T            | F                 | T         |
| F | F | F            | T                 | T         |

**True in every row: a tautology.** The reason is worth one line:  $p \oplus q$  and  $p \oplus \neg q$  are negations of each other, so exactly one of them is true in any row and their disjunction cannot fail.

#### iv. $((p \rightarrow q) \rightarrow r) \rightarrow s$

Four letters, so the table has  $2^4 = 16$  rows. Rather than fill sixteen rows blindly, work out where it is false and say so.

Write  $X$  for  $(p \rightarrow q) \rightarrow r$ . The whole expression  $X \rightarrow s$  is false **only** when  $X$  is true and  $s$  is false.  $X$  is true unless  $p \rightarrow q$  is true and  $r$  is false; so  $X$  is true when  $r$  is true (four ways to choose  $p$  and  $q$ ), or when  $r$  is false and  $p \rightarrow q$  is false, which is  $p$  true and  $q$  false. That is **five** combinations of  $p, q, r$ .

**So the expression is false in exactly 5 of the 16 rows**, namely those with  $s$  false and  $(p, q, r)$  equal to  $(T, T, T)$ ,  $(T, F, T)$ ,  $(F, T, T)$ ,  $(F, F, T)$  or  $(T, F, F)$ , and **true in the other 11**. Draw the full table if time allows; state this if it does not.

#### C. I. "THE HOME TEAM WINS WHENEVER IT IS RAINING" (3 MARKS)

Quoted as printed in section 1.2. With  $p$  = "It is raining" and  $q$  = "The home team wins", the statement is  $p \rightarrow q$ .

**Contrapositive:** If the home team does not win, then it is not raining.

**Converse:** If the home team wins, then it is raining.

**Inverse:** If it is not raining, then the home team does not win.

#### C. II. PROVE ONLY THE CONTRAPOSITIVE AGREES (6 MARKS)

**To prove:** of the converse, the inverse and the contrapositive, only the contrapositive has the same truth value as  $p \rightarrow q$  in every case.

**Proof, by exhaustive truth table.** There are only four assignments, so this covers every case and is a complete proof, not an illustration.

| P | Q | $P \rightarrow Q$ | $\neg Q \rightarrow \neg P$ | $Q \rightarrow P$ | $\neg P \rightarrow \neg Q$ |
|---|---|-------------------|-----------------------------|-------------------|-----------------------------|
| T | T | T                 | T                           | T                 | T                           |
| T | F | F                 | F                           | T                 | T                           |
| F | T | T                 | T                           | F                 | F                           |
| F | F | T                 | T                           | T                 | T                           |

**Read the columns.** The contrapositive column is identical to the original in all four rows, so  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ . The converse and the inverse each differ from the original in **two** rows, the second and the third, so neither is equivalent to it.  $\square$

**Using the statement from c(i), as the question asks.** Take the case where it is not raining and the home team wins, which is row 3. "The home team wins whenever it is raining" is **true**: nothing was promised about dry days. Its contrapositive, "if the home

team does not win it is not raining", is also true, vacuously. But its converse, "if the home team wins it is raining", is **false**, since the team won in the dry; and its inverse, "if it is not raining the home team does not win", is false for the same reason. One concrete case, and it exhibits the difference the table proves in general.

**A second line worth adding:** the converse and the inverse have identical columns, so they are equivalent to each other. They are contrapositives of one another.



## Question Two

### ANSWER

2024/2025 Q2 · 17.5 MARKS

#### A. FIVE QUANTIFIED TRANSLATIONS (7.5 MARKS)

Quoted as printed in section 1.4.  $C(x)$  = x has a cat,  $D(x)$  = x has a dog,  $F(x)$  = x has a ferret, domain = the students in the class. Because the domain is already the students, no  $S(x)$  is needed and no arrow appears.

- i.  $\exists x (C(x) \wedge D(x) \wedge F(x))$
- ii.  $\forall x (C(x) \vee D(x) \vee F(x))$
- iii.  $\exists x (C(x) \wedge F(x) \wedge \neg D(x))$
- iv.  $\neg \exists x (C(x) \wedge D(x) \wedge F(x))$ , equivalently  $\forall x \neg (C(x) \wedge D(x) \wedge F(x))$
- v.  $\exists x C(x) \wedge \exists y D(y) \wedge \exists z F(z)$

**Part v is the one that separates candidates.** "For each of the three animals there is a student who has it" allows a **different** student for each animal, so it needs three separate quantifiers. Writing  $\exists x (C(x) \wedge D(x) \wedge F(x))$  would say one student has all three, which is part i, not part v.

**Part iv is the negation of part i**, which is worth saying: it makes the pair easy to check and shows the structure was seen rather than guessed.

#### B. I. WHEN $F \circ G = G \circ F$ (2 MARKS)

Quoted as printed in section 4.3. With  $f(x) = ax + b$  and  $g(x) = cx + d$ :

$$(f \circ g)(x) = f(cx + d) = a(cx + d) + b = acx + ad + b$$

$$(g \circ f)(x) = g(ax + b) = c(ax + b) + d = acx + bc + d$$

The  $x$  terms already agree, so the two are equal for every  $x$  exactly when the constants agree:

$$ad + b = bc + d$$

That single equation is both necessary and sufficient, which is what the question asked for.

#### B. II. IF $f$ AND $f \circ g$ ARE ONE-TO-ONE, IS $g$ ? (2 MARKS)

**Yes**, and here is the proof.

**Proof.** Suppose  $g(x_1) = g(x_2)$ . Applying  $f$  to both sides,  $f(g(x_1)) = f(g(x_2))$ , that is  $(f \circ g)(x_1) = (f \circ g)(x_2)$ . Since  $f \circ g$  is one-to-one,  $x_1 = x_2$ . Therefore  $g$  is one-to-one.  $\square$

**Note what was not used:** the hypothesis that  $f$  is one-to-one never appears. The result holds from  $f \circ g$  alone, and saying so shows the argument was followed rather than recited.

#### B. III. $f \circ g$ AND $g \circ f$ FOR $x^2 + 1$ AND $x + 2$ (2 MARKS)

$$(f \circ g)(x) = f(x + 2) = (x + 2)^2 + 1 = x^2 + 4x + 5$$

$$(g \circ f)(x) = g(x^2 + 1) = (x^2 + 1) + 2 = x^2 + 3$$

**They are not equal.** One value settles it: at  $x = 1$  the first gives 10 and the second gives 4. This identical question is set again on the 2023/2024 paper.

#### C. "FEMALE AND A PARENT IMPLIES SOMEONE'S MOTHER" (4 MARKS)

Quoted as printed in section 1.4. Over a domain of all people, with  $F(x)$  =  $x$  is female,  $P(x)$  =  $x$  is a parent, and  $M(x, y)$  =  $x$  is the mother of  $y$ :

$$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$$

**Three things to point at.** The domain is all people, so the universal quantifier takes an **arrow**. "Someone's mother" is an existence claim about a second person, so it needs its own  $\exists y$  inside. And the  $\exists y$  sits **inside** the implication, because a different child may serve each mother.

## Question Three

### ANSWER

2024/2025 Q3 · 17.5 MARKS

#### A. WHY $(A \times B) \times (C \times D)$ AND $A \times (B \times C) \times D$ DIFFER (3.5 MARKS)

*Quoted as printed in section 2.3.*

**The point is the shape of the elements, not their number.** A typical element of the first is  $((a, b), (c, d))$ : an ordered pair whose two entries are themselves ordered pairs. A typical element of the second is  $(a, (b, c), d)$ : an ordered triple whose middle entry is a pair.

A pair of pairs is not a triple, so no element of one set is an element of the other, and the two sets are not equal. **The Cartesian product is not associative.**

**What is true.** The two sets have the same **size**,  $n(A)n(B)n(C)n(D)$  either way, and there is an obvious one-to-one correspondence between them. Equal size is not equality, and saying both halves is what a 3.5-mark answer looks like.

#### B. C++ PROGRAM FOR THE POWER SET (6 MARKS)

*Quoted as printed, and answered with a compiled and run listing, in section 11.2.* The method in one line: count a mask from 0 to  $2^n - 1$  and print element  $i$  whenever bit  $i$  of the mask is set. That enumeration produces the eight subsets in exactly the order the paper prints them, including the empty set as  $\{ \}$ .

#### C. THE FIRST DE MORGAN LAW BY SET-BUILDER NOTATION (3 MARKS)

*Quoted as printed in section 2.2.*

**Note on the printed line first.** The paper prints the law with the overbar sitting over  $B$  alone and with a dash where an equals sign belongs. The law being asked for is that **the complement of an intersection is the union of the complements**, and that is what is established here.

**To prove:**  $(A \cap B)' = A' \cup B'$ .

**Proof, by a chain of equivalences on an arbitrary element.**

$$\begin{aligned} x &\in (A \cap B)' \\ &\equiv \neg(x \in A \cap B) \quad [\text{definition of complement}] \\ &\equiv \neg(x \in A \wedge x \in B) \quad [\text{definition of intersection}] \\ &\equiv \neg(x \in A) \vee \neg(x \in B) \quad [\text{De Morgan for propositions, section 1.3}] \\ &\equiv x \in A' \vee x \in B' \quad [\text{definition of complement, twice}] \\ &\equiv x \in A' \cup B' \quad [\text{definition of union}] \end{aligned}$$

Since an arbitrary  $x$  belongs to one side exactly when it belongs to the other, the two sets are equal.  $\square$

**Name the step that does the work.** The whole proof is the propositional De Morgan law with set membership substituted for the propositions, and saying so is the point of the question.

#### D. I. BIT STRING OF THE COMPLEMENT (2.5 MARKS)

*Quoted as printed in section 2.3.* With  $U = \{1, \dots, 10\}$ , the set  $\{1, 3, 5, 7, 9\}$  is 1010101010. The complement flips every bit:

**0101010101**, which is  $\{2, 4, 6, 8, 10\}$

**The complement is a bitwise NOT**, which is why bit strings are used for sets in the first place.

#### D. II. UNION AND INTERSECTION BY BIT STRINGS (2.5 MARKS)

$\{1, 2, 3, 4, 5\}$  is 1111100000 and  $\{1, 3, 5, 7, 9\}$  is 1010101010.

**Union** is bitwise OR, 1 where either has a 1: **1111101010**, which is  $\{1, 2, 3, 4, 5, 7, 9\}$

**Intersection** is bitwise AND, 1 only where both have a 1: **1010100000**, which is  $\{1, 3, 5\}$

**Check by eye:** the intersection is the odd numbers up to 5, which is exactly what the two sets share.

## Question Four

### ANSWER

2024/2025 Q4 · 17.5 MARKS

#### A. TRUTH VALUES OVER THE REAL NUMBERS (4 MARKS)

Quoted as printed in section 1.4. Each answer needs a witness if it is true and a counterexample if it is false; the verdict alone is worth little.

i.  $\exists x(x^3 = -1)$ . **TRUE**. Witness  $x = -1$ , since  $(-1)^3 = -1$ .

ii.  $\exists x(x^4 < x^2)$ . **TRUE**. Witness  $x = 1/2$ :  $x^4 = 1/16$  and  $x^2 = 1/4$ , and  $1/16 < 1/4$ . In general it holds for every  $x$  strictly between  $-1$  and  $1$  other than  $0$ .

iii.  $\forall x((-x)^2 = x^2)$ . **TRUE**. For any real  $x$ ,  $(-x)^2 = (-1)^2 x^2 = x^2$ .

iv.  $\forall x(2x > x)$ . **FALSE**. Counterexample  $x = 0$ , which gives  $0 > 0$ , false. Also  $x = -1$ , which gives  $-2 > -1$ , false. **The claim is only true for positive  $x$** , and this is the part where the domain "all real numbers" is doing the work.

#### B. R1 TO R6 ON {1, 2, 3, 4} (7.5 MARKS)

Quoted as printed in section 3.2. One table answers all five parts and is far easier to mark than five sentences.

| RELATION | REFLEXIVE | SYMMETRIC | ANTISYMMETRIC | TRANSITIVE |
|----------|-----------|-----------|---------------|------------|
| R1       | no        | no        | no            | no         |
| R2       | no        | yes       | no            | no         |
| R3       | yes       | yes       | no            | no         |
| R4       | no        | no        | yes           | yes        |
| R5       | yes       | no        | yes           | yes        |
| R6       | no        | no        | yes           | yes        |

i and v. **Reflexive: R3 and R5**. Only these contain all four loops (1,1), (2,2), (3,3), (4,4).

ii. **Symmetric: R2 and R3**. In each, every pair appears with its reverse.

iii. **Antisymmetric: R4, R5 and R6**. None of them contains both (a, b) and (b, a) for two different elements.

iv. **Transitive: R4, R5 and R6**. Every composable pair of arrows has its shortcut present.

**On the repeated part.** The paper asks "Which of these relations are reflexive?" as part i and again as part v. Answer it twice, identically, and add one line: the intended part was almost certainly **irreflexive** or asymmetric, and if irreflexive was meant the answer is **R4 and R6**, the two containing no loop at all. That line costs nothing and covers both readings.

#### C. C++ ISPERFECTSQUARE BY BINARY SEARCH (6 MARKS)

Quoted as printed, and answered with a compiled and run listing, in section 11.2. The method in one line: search the range  $[1, \text{num}]$ , compare the square of the midpoint with  $\text{num}$ , keep the half that can still contain the answer, and return the midpoint on a hit or  $-1$  when the range empties. The function keeps the paper's own name and capitalisa-

tion, and its return type is `int` rather than `bool`, because the question asks for the square root to be returned.

## Question Five

### ANSWER

2024/2025 Q5 · 17.5 MARKS

*Quoted whole, as printed, in section 6.3.* The scenario: 24 vertices, 30 edges, five vertices of degree 4, seven pendant vertices, seven of degree 2, and every other vertex of degree 3 or 4.

#### A. HOW MANY VERTICES OF DEGREE 4? (4 MARKS)

**Not five.** The question says five vertices *are* of degree 4 and then says all other vertices have degree 3 or 4, so more of them are hiding in the remainder. Work it out.

degree sum =  $2|E| = 2 \times 30 = \mathbf{60}$  [Handshaking]

named vertices =  $5 + 7 + 7 = \mathbf{19}$ , leaving  $24 - 19 = \mathbf{5}$

their degree =  $4 \times 5 + 1 \times 7 + 2 \times 7 = 20 + 7 + 14 = \mathbf{41}$

degree left for the last five =  $60 - 41 = \mathbf{19}$

Let four of them have degree 4 and three of them degree 3, with four + three = 5 and  $4(\text{four}) + 3(\text{three}) = 19$ . Substituting,  $4(\text{four}) + 3(5 - \text{four}) = 19$ , so four = **4** and three = **1**.

**So the graph has  $5 + 4 = 9$  vertices of degree 4.**

**Check:** the full degree list is five 4s, seven 1s, seven 2s, four 4s and one 3, summing to  $20 + 7 + 14 + 16 + 3 = 60$ , and the vertices count  $5 + 7 + 7 + 4 + 1 = 24$ . Both agree.

#### B. DEGREE-4 VERTICES IN THE CORRESPONDING COMPLETE GRAPH (1 MARK)

The corresponding complete graph on the same 24 vertices is  $K_{24}$ , in which every vertex is joined to all 23 others, so every degree is 23.

**Therefore the number of vertices of degree 4 in it is 0.** State the reading in the same breath: "taking the corresponding complete graph to be  $K_{24}$ ". A one-mark part rewards the observation, not a calculation.

#### C. DOES THE HANDSHAKING THEOREM HOLD HERE? (3 MARKS)

**Statement.** In any undirected graph,  $\sum \deg(v) = 2|E|$ : the degrees sum to twice the number of edges, because each edge contributes exactly one to the degree of each of its two ends.

**Applied.** Here  $2|E| = 60$ , and the degree list found in part (a) sums to 60. The two agree, so the theorem holds for this graph.

**Proof of the theorem itself.** Count the pairs  $(v, e)$  where vertex  $v$  is an end of edge  $e$ . Counting by vertices gives  $\sum \deg(v)$ ; counting by edges gives  $2|E|$ , since every edge has exactly two ends. One quantity counted two ways must give the same total.  $\square$

**A consequence worth adding:** the degree sum is always even, so a graph with an odd degree sum cannot exist. That is the fact part (a) relied on.

#### D. C++ SNIPPET FOR THE REMAINING VERTICES (3 MARKS)

*Answered with a compiled and run listing in section 11.2.* The snippet computes  $24 - 19 = 5$  vertices left,  $60 - 41 = 19$  degree left to share, then solves the pair of equations in one line to print four of degree 4 and one of degree 3.

#### E. HANDSHAKING IN DATA COMMUNICATION (2 MARKS)

In a communication network model the vertices are devices and the edges are links. The theorem says the total number of **link ends**, that is the sum over devices of how many links each has, is exactly twice the number of links.

**Two uses.** Capacity planning: if a network is to have 50 links, the port counts across all devices must total 100, which sizes the hardware before anything is bought. And validation: if a device inventory reports an odd total number of ports, the inventory is wrong, because no network can have an odd total. That second use is a real error check and is the better half of the answer.

**F. N VERTICES, N EDGES, NO DEGREE 0 OR 1 (3 MARKS)**

**To prove:** if  $G$  has  $n$  vertices and  $n$  edges and no vertex has degree 0 or 1, then every vertex has degree exactly 2.

**Proof.** By Handshaking,  $\sum \deg(v) = 2|E| = 2n$ . No vertex has degree 0 or 1, so every one of the  $n$  vertices has degree at least 2, and therefore

$$\sum \deg(v) \geq 2n$$

But the sum **equals**  $2n$ . If any single vertex had degree 3 or more, the sum would exceed  $2n$ , since the other  $n - 1$  vertices already contribute at least 2 each. That is impossible, so no vertex exceeds degree 2. Combined with every degree being at least 2, every degree is exactly 2.  $\square$

**What such a graph looks like,** worth one extra line: every vertex of degree 2 and  $n$  edges means  $G$  is a disjoint union of cycles.

**G. WHAT IT MEANS FOR  $G$  AND  $H$  TO BE ISOMORPHIC (1.5 MARKS)**

$G$  and  $H$  are **isomorphic** when there is a one-to-one correspondence between their vertex sets that preserves adjacency: two vertices are joined in  $G$  exactly when their partners are joined in  $H$ . In plain words, they are the same graph drawn differently, with the vertices renamed.

**The consequence a marker looks for:** isomorphic graphs must agree on every structural property, so the same number of vertices, the same number of edges and the same list of degrees. Any disagreement in those proves they are not isomorphic.



## Question Six

### ANSWER

2024/2025 Q6 · 17.5 MARKS

#### A. THREE TRAVERSALS OF THE GIVEN TREE (6 MARKS)

Quoted as printed in section 7.3, where the tree is redrawn and each traversal is walked step by step. The tree has 17 vertices: a is the root with children b and c; b has d, e, f; e has j, k; k has n, o; c has g, h, i; h has l, m; m has p, q.

i. **Preorder** visits the root, then each subtree in turn:

a, b, d, e, j, k, n, o, f, c, g, h, l, m, p, q, i

ii. **Inorder** visits the leftmost subtree, then the root, then the remaining subtrees:

d, b, j, e, n, k, o, f, a, g, c, l, h, p, m, q, i

iii. **Postorder** visits every subtree first and the root last:

d, j, n, o, k, e, f, b, g, l, p, q, m, h, i, c, a

**Two checks worth the ten seconds.** All three lists have 17 entries. Preorder starts at the root and postorder ends at it.

#### B. I. THE ROOTED TREE OF THE EXPRESSION (2 MARKS)

Quoted as printed in section 7.3. Two superscript characters in the printed expression could not be read at any available resolution, so this answers the plain reading  $(3 - 4 \cdot 2) - (5 + (5 - 2))$  and says in one line what changes otherwise.

**The tree.** The root is the outer subtraction. Its left subtree is  $3 - 4 \cdot 2$ , whose root is a subtraction with 3 on the left and a multiplication of 4 and 2 on the right. Its right subtree is  $5 + (5 - 2)$ , whose root is an addition with 5 on the left and a subtraction of 5 and 2 on the right.

**Its value:**  $(3 - 8) - (5 + 3) = -5 - 8 = -13$ .

**If the two marks are exponents,** the tree gains two exponent nodes, one above each bracketed group, and the values change; the structure of the answer does not.

#### B. IV. PREFIX NOTATION (2 MARKS)

-- 3 · 4 2 + 5 - 5 2

#### B. V. POSTFIX NOTATION (2 MARKS)

3 4 2 · - 5 5 2 - + -

**Both are read off the tree,** prefix by visiting each node before its subtrees and postfix by visiting it after. Neither is the other reversed, and neither is produced by moving the symbols of the infix expression about by eye.

#### B. VI. LEAVES OF A QUINARY TREE OF HEIGHT 3 (2.5 MARKS)

**Say what is determined first.** A quinary tree of height 3 could have as few as 3 leaves; what m and h fix is the **maximum**. With  $m = 5$  and  $h = 3$ :

$$\text{leaves} = m^h = 5^3 = 125$$

**Cross-check by the other formula.** The full quinary tree of height 3 has internal vertices  $1 + 5 + 25 = 31$ , and  $l = (m - 1)i + 1 = 4 \times 31 + 1 = 125$ . The two agree.

#### B. VII. PROVE THE $M^H$ BOUND (3 MARKS)

**Proof, by induction on the height h.**

**Base case,  $h = 0$ .** A tree of height 0 is a single vertex, which is both root and leaf: 1 leaf, and  $m^0 = 1$ .

**Hypothesis.** Every  $m$ -ary tree of height  $k$  has at most  $m^k$  leaves.

**Step.** Let  $T$  have height  $k + 1$ . Removing the root leaves at most  $m$  subtrees, one per child, each of height at most  $k$ , so each has at most  $m^k$  leaves by the hypothesis. Every leaf of  $T$  lies in one of them, so  $T$  has at most  $m \times m^k = m^{k+1}$  leaves.  $\square$

**When the bound is attained:** exactly when the tree is full and every leaf sits at level  $h$ . Saying so shows the bound is sharp, not merely true.

## TEACH

## MARK RECONCILIATION

| QUESTION | PARTS                             | TOTAL |
|----------|-----------------------------------|-------|
| Q1       | $4.5 + 4 + 3 + 6$                 | 17.5  |
| Q2       | $7.5 + 2 + 2 + 2 + 4$             | 17.5  |
| Q3       | $3.5 + 6 + 3 + 2.5 + 2.5$         | 17.5  |
| Q4       | $4 + 7.5 + 6$                     | 17.5  |
| Q5       | $4 + 1 + 3 + 3 + 2 + 3 + 1.5$     | 17.5  |
| Q6       | $2 + 2 + 2 + 2 + 2 + 2 + 2.5 + 3$ | 17.5  |

All six reconcile, including Q6, whose printed numbering skips ii and iii. That is worth checking rather than assuming: if the parts had not added to 17.5, a part really would be missing rather than merely mislabelled.

## PAST PAPER

# 2023/2024 Solved in Full

*The paper we hold only in part, and the one whose marks are weighted differently: 23.5 per question rather than 17.5. Everything from Question Three onward is answered here, and the missing half is named rather than invented.*

*By the end of this section you will have complete answers to the three questions this scan carries, and a clear statement of what it does not carry.*

| SESSION   | CODE PRINTED | INSTRUCTION              | TIME       | PER QUESTION |
|-----------|--------------|--------------------------|------------|--------------|
| 2023/2024 | CSC 242      | not on the pages we hold | not stated | 23.5 marks   |

**Questions One and Two are not in our possession.** The scan opens at Question Three; page one carries only the title line. This manual does not invent them, and neither should you: if a past paper is quoted to you with questions that are not on the original, the questions are somebody's guess.

**What that costs you is small.** The three questions we do hold contain five of the fourteen repeat-offender drills of section R.3, including the minimum tour and the chain letter, both of which appear on two other papers as well.

**Its defects,** all quoted as printed. **Question Four has no number at all:** after Q3(d) the page runs straight into "(a) Explain in details ... types of undirected graph", and the third part of that question has no letter either. "(1 marks)" is printed three times and "(2 marks)" throughout, with a singular value and a plural noun. And two arrowheads in the Q5(c) digraph **cannot be resolved** at any available magnification, which the answer below states rather than hides.

**The mark weighting is the thing to notice.** At 23.5 a question, three answers give 70.5. Practising this paper against a 17.5-mark clock would leave you a third short on every question.

## Question Three

### ANSWER

2023/2024 Q3 · 23.5 MARKS

#### A. BIT STRINGS OVER $U = \{1, \dots, 10\}$ (3 MARKS)

Quoted as printed in section 2.3. The convention is fixed by the question: the  $i$ -th bit is 1 when  $i$  is in the set. Ten elements, so ten bits, and position  $i$  holds element  $i$ .

| SET                      | BIT STRING |
|--------------------------|------------|
| (i) {3, 4, 5}            | 0011100000 |
| (ii) {1, 3, 6, 10}       | 1010010001 |
| (iii) {2, 3, 4, 7, 8, 9} | 0111001110 |

**Check each by counting the 1s:** three, four and six, matching the sizes of the three sets. That is the fastest verification available and it catches a dropped or shifted bit at once.

#### B. C++ PROGRAM TAKING THE SETS DYNAMICALLY (7 MARKS)

Quoted as printed, and answered with a compiled and run listing, in section 11.2. The load-bearing word is **dynamically**: the sets must arrive through `cin` at run time, not be written into the source. The program reads how many sets, then for each one reads its size and its members, builds a string of ten '0' characters and sets position  $m - 1$  to '1' for each member  $m$ . Its output on these three sets is exactly the three strings above.

#### C. $f \circ g$ AND $g \circ f$ , AND ARE THEY EQUAL? (3.5 MARKS)

Quoted as printed in section 4.3. With  $f(x) = x^2 + 1$  and  $g(x) = x + 2$ , both from the reals to the reals:

$$(f \circ g)(x) = f(g(x)) = f(x + 2) = (x + 2)^2 + 1 = x^2 + 4x + 5$$

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = (x^2 + 1) + 2 = x^2 + 3$$

**They are not equal**, and the question explicitly asks you to say so. One value proves it: at  $x = 1$  the first gives 10 and the second gives 4.

**Composition is not commutative**, which is the general fact behind it. The 2024/2025 paper sets this identical pair of functions.

#### D. I. THE DIVIDES RELATION ON $\{1, 2, 3, 4\}$ (2 MARKS)

Quoted as printed in section 3.1.  $R = \{(a, b) \mid a \text{ divides } b\}$  means  $b$  is a multiple of  $a$ . Take each  $a$  in turn and list the multiples of  $a$  that lie in the set:

1 divides 1, 2, 3, 4

2 divides 2 and 4

3 divides 3

4 divides 4

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$ , eight pairs.

**Note it is not symmetric:** (1, 2) is in  $R$  but (2, 1) is not, since 2 does not divide 1. It is reflexive, antisymmetric and transitive, so it is a **partial order**, which is section 3.4.

#### D. II TO V. R1 TO R6 CLASSIFIED (8 MARKS)

Quoted as printed in section 3.2. One table answers all four parts.

| RELATION | REFLEXIVE | SYMMETRIC | ANTISYMMETRIC | TRANSITIVE |
|----------|-----------|-----------|---------------|------------|
| R1       | no        | no        | no            | no         |
| R2       | no        | yes       | no            | no         |
| R3       | yes       | yes       | no            | no         |
| R4       | no        | no        | yes           | yes        |
| R5       | yes       | no        | yes           | yes        |
| R6       | no        | no        | yes           | yes        |

**(ii) Reflexive:** R3 and R5

**(iii) Symmetric:** R2 and R3

**(iv) Antisymmetric:** R4, R5 and R6

**(v) Transitive:** R4, R5 and R6

**Give one reason per verdict where a relation fails,** since 2 marks a part is buying justification and not just a list. For example R1 is not reflexive because (3, 3) is absent, and not symmetric because (3, 4) has no (4, 3).

**Note that  $R6 = \{(3, 4)\}$  is antisymmetric and transitive,** both vacuously: there is no offending pair to break either, and a single arrow composes with nothing. Vacuous truth from section 9.1, doing real work.

## Question Four

### ANSWER

2023/2024 Q4 · 23.5 MARKS

*This question carries no number on the paper, and its third part carries no letter. Both are reproduced as printed in sections 6.2, 6.8 and 7.1.*

#### A. TYPES OF UNDIRECTED GRAPH, IN DETAIL WITH DIAGRAMS (9 MARKS)

*Quoted as printed in section 6.2.* Nine marks for a catalogue, so give a definition, a distinguishing feature and a sketch for each. This is the single most repeated question on the course, appearing on three of the five papers.

| TYPE                                           | DEFINITION                                             | HOW TO DRAW IT                                   |
|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|
| <b>Simple graph</b>                            | no loops, at most one edge between any two vertices    | a triangle                                       |
| <b>Multigraph</b>                              | two vertices may be joined by several edges            | two dots with three curves between them          |
| <b>Pseudograph</b>                             | a multigraph that also allows loops                    | the same, with a loop at one dot                 |
| <b>Null or empty graph</b>                     | vertices but no edges at all                           | four isolated dots                               |
| <b>Trivial graph</b>                           | a single vertex and no edge                            | one dot                                          |
| <b>Complete graph <math>K_n</math></b>         | every pair of vertices joined                          | $K_4$ : a square with both diagonals             |
| <b>Regular graph</b>                           | every vertex has the same degree                       | a square: every vertex of degree 2               |
| <b>Cycle graph <math>C_n</math></b>            | a single closed ring of $n$ vertices                   | a pentagon                                       |
| <b>Path graph</b>                              | a line of vertices, no closing edge                    | four dots in a row                               |
| <b>Wheel graph</b>                             | a cycle plus one hub joined to every vertex            | a pentagon with a centre dot joined to all five  |
| <b>Bipartite graph</b>                         | vertices split into two groups, edges only across      | two rows, edges only between rows                |
| <b>Complete bipartite <math>K_{m,n}</math></b> | bipartite with every cross pair joined                 | $K_{3,3}$ : three above, three below, nine edges |
| <b>Connected graph</b>                         | a path exists between every pair                       | any single piece                                 |
| <b>Weighted graph</b>                          | each edge carries a number                             | the Detroit map of part (b)                      |
| <b>Subgraph</b>                                | a graph made from some of another's vertices and edges | a triangle inside $K_4$                          |

**Draw every one you name.** The question says "with diagrammatic illustration", so the sketches are not optional and are probably half the nine marks. Small, labelled, five seconds each.

#### B. THE TRAVELLING MISSIONARY, MINIMUM DISTANCE (10 MARKS)

Quoted as printed in section 6.8, where all twelve tours are tabulated. Five cities, every pair joined, with the distances read off the printed map:

|              | GRAND RAPIDS | KALAMAZOO | SAGINAW | TOLEDO | DETROIT |
|--------------|--------------|-----------|---------|--------|---------|
| Grand Rapids | 0            | 56        | 113     | 167    | 147     |
| Kalamazoo    | 56           | 0         | 137     | 133    | 135     |
| Saginaw      | 113          | 137       | 0       | 142    | 98      |
| Toledo       | 167          | 133       | 142     | 0      | 58      |
| Detroit      | 147          | 135       | 98      | 58     | 0       |

**How many tours there are.** Starting and ending at Detroit, the other four cities can be visited in  $4! = 24$  orders, but each tour and its reverse cover the same route, so there are  $24 / 2 = 12$  distinct tours. Twelve is small enough to list, and the question says **minimum**, which is a claim about every alternative.

**The answer.** The shortest tour measures **458** miles:

**Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit**

$$98 + 113 + 56 + 133 + 58 = 458$$

**Say why a nearest-neighbour walk is not an answer.** Starting at Detroit the nearest city is Toledo at 58, then Kalamazoo at 133, then Grand Rapids at 56, then Saginaw at 113, then home at 98, which totals 458 as well. It agrees here, and that is luck: nearest-neighbour carries no guarantee, so on its own it establishes an upper bound and nothing more. The enumeration is what makes 458 a minimum.

### C. THREE APPLICATION DOMAINS OF TREE MODELS (4.5 MARKS)

Quoted as printed in section 7.1. This part has no letter on the paper. Three domains, each with an example, at 1.5 marks apiece.

**1. File systems and hierarchical storage.** A directory tree has one root, every file and folder has exactly one parent, and there are no cycles, which is precisely a rooted tree. Example: a path such as /home/user/notes.txt is the unique route from the root to a leaf, and the uniqueness of that path is a property of trees, not of graphs generally.

**2. Compilers and expression evaluation.** A parser turns source code into a syntax tree whose internal vertices are operators and whose leaves are operands, and the compiler then walks it. Example: section 7.3 builds the tree of  $(3 - 4 \cdot 2) - (5 + (5 - 2))$  and reads its postfix form straight off a postorder walk, which is exactly what a stack machine consumes.

**3. Searching and decision procedures.** A binary search tree keeps data ordered so that each comparison discards half the remaining possibilities, and a decision tree encodes a sequence of tests. Example: the binary search of section 11.2 halves its range each pass, so it reaches an answer in the number of halvings rather than in the number of items.

**Two more if the marks look generous:** organisational charts and family trees, where the hierarchy is the point; and prefix codes such as Huffman coding, where the tree structure is what makes the code unambiguous.

## Question Five

### ANSWER

2023/2024 Q5 · 23.5 MARKS

#### A. DEFINE THE HANDSHAKING THEOREM (3 MARKS)

*Quoted as printed in section 6.3.*

**Statement.** In any undirected graph  $G = (V, E)$ , the sum of the degrees of all the vertices equals twice the number of edges:

$$\sum \deg(v) = 2|E|$$

**Why.** Each edge has exactly two ends, and each end adds one to the degree of the vertex it meets, so counting edge ends by vertices and by edges must give the same total.

**Two consequences worth adding for three marks.** The degree sum is always **even**, so no graph can have an odd degree sum. And the number of vertices of odd degree is always even, since the odd degrees must pair up to leave an even total.

#### B. I. CAN THE PILOT VISIT EXACTLY THREE OF FIVE CITIES? (3 MARKS)

*Quoted as printed in section 6.3.* Model the five cities as vertices and each direct route as an edge. "Visit exactly three from each" means every vertex has degree exactly 3.

**Compute.** Five vertices, each of degree 3:

$$\sum \deg(v) = 5 \times 3 = 15$$

But Handshaking says this sum must equal  $2|E|$ , which is **even**, and 15 is odd.

**So no such flight plan exists.** It is impossible, and the impossibility is not about scheduling or geography: no graph on five vertices can have every degree 3.

**The general rule behind it,** worth one line: a graph in which every vertex has degree  $d$  and there are  $n$  vertices requires  $nd$  to be even, so  $n$  and  $d$  cannot both be odd.

#### B. II. SUPPORT THE ANSWER WITH A DIAGRAM (2 MARKS)

Draw the five cities as five dots and attempt it. Give four of them three edges each and the fifth is forced to have an even number, because the twelve edge ends already placed must pair off. Any attempt leaves one vertex at degree 2 or 4.

**Then draw what does work,** which shows you understood rather than merely failed: with **six** cities, every vertex of degree 3 is possible, since  $6 \times 3 = 18$  is even.  $K_{3,3}$  is exactly such a graph, and so is the triangular prism.

#### C. THE DIRECTED GRAPH, IN TABULAR FORM (6 MARKS)

*Quoted as printed in section 6.4, where the reading is set out in full.*

**State the reading first.** The three multi-arc bundles are clear. The two crossing diagonals in the middle are **not**: at the available resolution their arrowheads can be read either as two arcs  $a$  to  $c$  plus one arc  $b$  to  $d$ , or as one each way. The table below tabulates the first reading. **Re-read the arrowheads on your own copy**; the method, which is what the six marks are for, does not change.

**(i) Vertices.**  $a, b, c, d, e$ . **Total 5.** Note  $e$  has no arc at all and is still a vertex.

**(ii) Edges,** counted as bundles:  $b$  to  $a$  is 1 arc,  $a$  to  $b$  is 2,  $d$  to  $a$  is 2,  $b$  to  $c$  is 2,  $d$  to  $c$  is 3,  $a$  to  $c$  is 2,  $b$  to  $d$  is 1. **Total 13.**

**(iii) and (iv) In-degree and out-degree.**



| VERTEX       | OUT-DEGREE | WHERE TO               | IN-DEGREE | WHERE FROM                   |
|--------------|------------|------------------------|-----------|------------------------------|
| a            | 4          | 2 to b, 2 to c         | 3         | 1 from b, 2 from d           |
| b            | 4          | 1 to a, 2 to c, 1 to d | 2         | 2 from a                     |
| c            | 0          | none                   | 7         | 2 from b, 3 from d, 2 from a |
| d            | 5          | 2 to a, 3 to c         | 1         | 1 from b                     |
| e            | 0          | none                   | 0         | none                         |
| <b>Total</b> | <b>13</b>  |                        | <b>13</b> |                              |

**The check that makes the answer safe.** Both totals must equal the number of arcs, because every arc leaves exactly one vertex and arrives at exactly one. Here both are 13 and the edge count is 13, so the tabulation is internally consistent whatever the two uncertain arrowheads turn out to be.

#### D. THE CONVOCATION CHAIN LETTER (9.5 MARKS)

*Quoted as printed in section 7.2.* Model it as a rooted tree: the first person is the root, and each person who sends the cards on has four children. Those who received but did not send are the **leaves**, and there are 100 of them. Nobody receives more than one card, which is what makes it a tree rather than a general graph.

**The tree is full**, because everyone who sends sends to exactly four, so every internal vertex has exactly  $m = 4$  children.

**(i) How many have seen the card (6 marks).** For a full  $m$ -ary tree with  $i$  internal vertices, the leaves number  $l = (m - 1)i + 1$ , so

$$100 = 3i + 1$$

$$3i = 99, \text{ so } i = \mathbf{33}$$

and the total number of vertices is

$$n = mi + 1 = 4 \times 33 + 1 = \mathbf{133}$$

**So 133 people have seen the card, including the first person.** Check with the other identity:  $n = i + l = 33 + 100 = 133$ . The two agree, which is the verification worth writing down.

**(ii) How many sent it out (3.5 marks).** The senders are exactly the internal vertices, so **33** people sent the card on, and the first person is one of them.

**The trap in the wording:** "100 people who read it but did not send it out" describes the **leaves**, not the whole readership. A candidate who takes 100 as the answer to part (i) has answered a question the paper did not ask.

| TEACH    |                                         | MARK RECONCILIATION |
|----------|-----------------------------------------|---------------------|
| QUESTION | PARTS                                   | TOTAL               |
| Q3       | 1 + 1 + 1 + 7 + 3.5 + 2 + 2 + 2 + 2 + 2 | 23.5                |
| Q4       | 9 + 10 + 4.5                            | 23.5                |
| Q5       | 3 + 3 + 2 + 1 + 1 + 2 + 2 + 6 + 3.5     | 23.5                |

All three reconcile to **23.5**, not 17.5. That is how we know this paper used a different rubric from 2024/2025 and 2025/2026, and it is the kind of thing worth checking on any paper before you budget your time.

## PAST PAPER

# 2021/2022 Solved in Full

*Two credit units, two hours, attempt any three. A different rubric from the paper you will sit, but almost every question on it has reappeared since, so it is the best drill in the set even though it is the wrong shape for rehearsing timing.*

*By the end of this section you will have complete answers to all five questions, and a clear view of which of them recur on the current format.*

| TEACH     |              | THE PAPER AT A GLANCE |                   |         |
|-----------|--------------|-----------------------|-------------------|---------|
| SESSION   | CODE PRINTED | CREDIT                | INSTRUCTION       | TIME    |
| 2021/2022 | CSC416       | TWO                   | attempt any THREE | 2 hours |

**This paper closes up its mark text:** "5marks", "6½marks", "12marks", with no space before the word. Two parts break the pattern and print "(10 marks)" and "(5 marks)". Quoted as printed throughout.

**Its defects.** Q5(b) prints "There of 25 students" for "There are", and its numbers admit two readings; the answer below states which it takes. Q1(a), Q2(c) and Q4(a) punctuate commands as questions. Q5(d)(i) prints its connective as a **circled minus**, which can only mean exclusive or. And the digraph in Q1(b) is **not reliably readable** at the resolution we hold, which the answer says rather than hides.

**Marks per question:** Q1 23.5, Q2 23.5, Q3 23.5, Q5 23.5, but **Q4 totals 24** as printed (5.5 + 8.5 + 10). One question on this paper is worth half a mark more than the others. Not something you can act on, but exactly the sort of thing to notice in the first minute rather than assume away.

**What recurs.** Q1(c) is the types-of-graph question, set on three papers. Q3(a) is the chain letter, set on three. Q4(c) is the Detroit tour, set on three. Q2(c) is the function-correspondence question, set on two. Q5(d)(ii) is word for word a part of the 2024/2025 paper.

## Question One

### ANSWER

2021/2022 Q1 · 23.5 MARKS

#### A. LIST THE MEMBERS OF FIVE SETS (5 MARKS)

Quoted as printed in section 2.1. Set-builder into roster form, one mark each.

(i)  $\{x \mid x \text{ is a real number such that } x^2 = 1\}$ .  $x = 1$  or  $x = -1$ , and both are real:  $\{1, -1\}$ .

(ii)  $\{x \mid x \text{ is a positive prime number less than } 20\}$ . A prime has exactly two positive divisors, so 1 is not prime:  $\{2, 3, 5, 7, 11, 13, 17, 19\}$ , eight members.

(iii)  $\{x \mid x \text{ is a perfect square and } x < 100\}$ . **State the reading.** As printed the condition does not say  $x$  is a positive integer, so strictly it admits 0 as well. Taking  $x$  to be a whole number:  $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$ , ten members; omit the 0 if only positive integers are intended, giving nine. Say which you used and the mark is safe either way.

(iv)  $\{x \mid x \text{ is an integer such that } x^2 = 2\}$ . The only solutions are  $\sqrt{2}$  and  $-\sqrt{2}$ , and neither is an integer, as section 9.1 proves. So the set is  $\emptyset$ , the empty set, with 0 members.

(v)  $\{x \mid x \text{ is an integer such that } 2x = x\}$ . Subtracting  $x$  from both sides gives  $x = 0$ . So  $\{0\}$ , a singleton with one member.

**Parts iv and v are the discriminating ones.** One is empty and the other is a singleton containing zero; writing  $\emptyset$  for both, or  $\{0\}$  for both, is the error the pair is there to catch. Section 2.1 covers exactly that distinction.

#### B. THE DIRECTED GRAPH, IN TABULAR FORM (6½ MARKS)

Quoted as printed in section 6.4.

**State the reading first.** This figure is drawn small with soft arrowheads, and at the resolution we hold **the arc count and their directions are not reliably readable**. What follows is the method, which is what the marks are for; read the arrowheads off your own copy and fill the numbers in.

**The method, in four steps.**

- 1 List the vertices** and count them. Here a, b, c, d and the isolated e, so five. An isolated vertex is still a vertex, and it is usually why it was drawn.
- 2 List the arcs as bundles**, one row per ordered pair, with how many arcs run that way. Curved parallel arcs are separate arcs.
- 3 Tabulate** each vertex with its out-degree and its in-degree, naming where each goes and comes from.
- 4 Total both columns.** Each must equal the arc count, since every arc leaves exactly one vertex and arrives at exactly one. If they do not agree, an arrowhead has been misread.

**The layout to reproduce:**

| VERTEX       | ARCS OUT | OUT-DEGREE | ARCS IN | IN-DEGREE  |
|--------------|----------|------------|---------|------------|
| a            | ...      | ...        | ...     | ...        |
| b            | ...      | ...        | ...     | ...        |
| c            | ...      | ...        | ...     | ...        |
| d            | ...      | ...        | ...     | ...        |
| e            | none     | 0          | none    | 0          |
| <b>Total</b> |          | <b> E </b> |         | <b> E </b> |

**The equal totals are the whole point.** Section 6.4 works the same tabulation on the 2023/2024 digraph, where the reading is firmer, and there both totals come to 13.

### C. TYPES OF UNDIRECTED GRAPH, WITH DIAGRAMS (12 MARKS)

*Quoted as printed in section 6.2, where the full catalogue is tabled.* Twelve marks, the largest single part on the paper, and the most repeated question on the course.

**Name, define and draw each.** Simple, multigraph, pseudograph, null, trivial, complete  $K_n$ , regular, cycle, path, wheel, bipartite, complete bipartite  $K_{m,n}$ , connected, weighted, subgraph. Fifteen types is more than twelve marks needs; give ten with diagrams rather than fifteen without.

**The three distinctions the marks turn on.** A **simple** graph forbids loops and repeated edges; a **multigraph** permits repeated edges; a **pseudograph** permits loops as well. Getting those three apart, with a sketch each, is worth more than listing the other twelve names.

## Question Two

### ANSWER

2021/2022 Q2 · 23.5 MARKS

#### A. LABELLING THE CHAIRS OF AN AUDITORIUM (6 MARKS)

*Quoted as printed in section 5.1.* Each label is an uppercase English letter followed by a positive integer not exceeding 100. Both parts must be chosen, so this is the **product rule**.

letters: 26    integers 1 to 100: 100

labels =  $26 \times 100 = 2600$

**Two things to state.** "Not exceeding 100" and "positive" together give the integers 1 to 100, which is 100 values and not 99 or 101. And each label names one chair, so 2600 chairs can be labelled differently, which is what "the largest number" asks for.

#### B. SIMPLE COMMUNICATION LINKS BETWEEN FOUR BUILDINGS (7½ MARKS)

*Quoted as printed in section 6.2.* The question names four buildings. A simple link joins two of them, and a link from X to Y is the same link as from Y to X, so this is a **combination**, not a permutation:

links =  $C(4, 2) = 4 \times 3 / 2 = 6$

**The diagram** is  $K_4$ : four dots, each joined to the other three, drawn as a square with both diagonals. Label the four dots with the four buildings and count the lines: six.

**Watch the count of buildings.** The 2020/2021 paper sets the same question with **five** buildings, where the answer is  $C(5, 2) = 10$ . Read the list and count it rather than remembering the number.

#### C. FIVE TYPES OF FUNCTION CORRESPONDENCE, WITH A DIAGRAM (10 MARKS)

*Quoted as printed in section 4.2, where all five are drawn.* Two marks each, and each needs a two-column arrow diagram.

| TYPE                            | WHAT THE ARROWS DO                                       | HOW TO DRAW IT                                                   |
|---------------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| <b>i. One-to-one, not onto</b>  | no two inputs share an output, but some output is unused | 3 inputs, 4 outputs, three arrows to three different targets     |
| <b>ii. Onto, not one-to-one</b> | every output is used, but two inputs share one           | 4 inputs, 3 outputs, every target hit and one hit twice          |
| <b>iii. One-to-one and onto</b> | a perfect pairing, a bijection                           | 3 inputs, 3 outputs, one arrow each way                          |
| <b>iv. Neither</b>              | two inputs share an output and some output is unused     | 3 inputs, 3 outputs, two arrows to one target, one target unused |
| <b>v. Not a function</b>        | an input with no arrow, or an input with two             | 3 inputs, one with no arrow leaving it                           |

**The line that earns part v.** A function needs **every** input used and each used **exactly once**. Type v fails one of those, and saying which one your diagram breaks is the difference between a drawing and an answer.

## Question Three

### ANSWER

2021/2022 Q3 · 23.5 MARKS

#### A. THE CONVOCAION INVITATION CARDS (10 MARKS, SPLIT 5 AND 5)

*Quoted as printed in section 7.2.* Model it as a rooted tree: the first person is the root, each sender has exactly four children, and the 100 who read but did not send are the **leaves**. Nobody receives more than one card, which is what makes the structure a tree.

**(i) How many have seen the card.** For a full  $m$ -ary tree,  $l = (m - 1)i + 1$  with  $m = 4$  and  $l = 100$ :

$$100 = 3i + 1, \text{ so } 3i = 99 \text{ and } i = \mathbf{33}$$

$$n = mi + 1 = 4 \times 33 + 1 = \mathbf{133}$$

**133 people have seen the card**, including the first.

**(ii) How many sent it out.** The senders are the internal vertices: **33**.

**Check:**  $n = i + l = 33 + 100 = 133$ . The two routes agree.

**This is the same problem as 2020/2021 Q3(b) and 2023/2024 Q5(d)**, told three ways. Learn it once.

#### B. FOUR REAL-LIFE SCENARIOS FOR GRAPH MODELS, TWO EXAMPLES EACH (6 MARKS)

*Quoted as printed in section 6.1.* Say what the vertices are and what the edges are in each; that is the modelling and it is what the marks are for.

**1. Transport networks.** Vertices are places, edges are routes. Examples: a road map where edges carry distances, as in Q4(c) below; an airline route map where an edge means a direct flight.

**2. Social networks.** Vertices are people, edges are relationships. Examples: a friendship network, where the edge is undirected because friendship is mutual; a follower network, where it is directed because following is not.

**3. Utility and supply networks.** Vertices are junctions or substations, edges are pipes or cables. Examples: an electricity grid; a water distribution network, where edge weights are capacities.

**4. Scheduling and dependency.** Vertices are tasks, edges are "must come before". Examples: a project plan, where a cycle would mean an impossible schedule; a timetable conflict graph, where an edge means two exams cannot be at the same time.

#### C. KEY TERMINOLOGIES OF A ROOTED TREE, WITH A DIAGRAM (7½ MARKS)

*Quoted as printed in section 7.1, where one labelled diagram carries every term at once.*

**Open with the definition.** A rooted tree is a tree in which one vertex is designated the **root**, and every edge is then directed away from it, so every other vertex has exactly one path from the root.

**The terms, each in a line:** **root**, the designated vertex; **parent**, the vertex just above another; **child**, a vertex just below its parent; **siblings**, vertices sharing a parent; **ancestors**, everything on the path back to the root; **descendants**, everything below; **leaf** or terminal vertex, one with no children; **internal vertex**, one with at least one child; **subtree**, a vertex together with all its descendants; **level**, the distance from the root, which is 0 at the root itself; and **height**, the largest level in the tree.

**Then the diagram, labelled.** One tree with the root marked, one parent and child pair marked, a pair of siblings, a leaf, an internal vertex, and the levels numbered down the left edge. Marking the terms on a single tree is far stronger than eleven separate sketches.

**Close with the two counting facts,** which cost a line and show the definitions were understood: a tree on  $v$  vertices has exactly  $v - 1$  edges, and there is exactly one path between any two vertices.

## Question Four

### ANSWER

2021/2022 Q4 · 24 MARKS AS PRINTED

#### A. BRIEFLY EXPLAIN THE HANDSHAKING THEOREM (5½ MARKS)

*Quoted as printed in section 6.3.*

**Statement.** In any undirected graph, the sum of the degrees of all the vertices is twice the number of edges:  $\sum \deg(v) = 2|E|$ .

**Why it is true.** Count the pairs (vertex, edge-end-meeting-it). Counting by vertices gives  $\sum \deg(v)$ ; counting by edges gives  $2|E|$ , since every edge has exactly two ends. The same quantity counted two ways must agree.

**Why it is called that.** If everyone at a party shakes some hands, the total number of hands shaken across all people is twice the number of handshakes, because each handshake involves two people.

**Its two consequences,** which is where the extra marks are. The degree sum is always **even**, so no graph can have an odd degree sum. And the number of vertices of **odd** degree is always even.

**An example.** A triangle has three vertices of degree 2 and three edges:  $2 + 2 + 2 = 6 = 2 \times 3$ .

#### B. WHAT A BALANCED M-ARY TREE MEANS, WITH EXAMPLES (8½ MARKS)

*Quoted as printed in section 7.2.* Build the answer in three layers, because the question asks for one term whose definition needs two others.

**1. m-ary.** A rooted tree is **m-ary** when every internal vertex has **at most**  $m$  children. At  $m = 2$  it is called binary, at  $m = 3$  ternary, at  $m = 5$  quinary.

**2. Full.** An  $m$ -ary tree is **full** when every internal vertex has **exactly**  $m$  children, none fewer.

**3. Balanced.** An  $m$ -ary tree of height  $h$  is **balanced** when every leaf is at level  $h$  or level  $h - 1$ . In words: the leaves are all at the bottom or one step above it, so no branch is much longer than another.

**Examples, which the question demands.** A binary tree with root, two children and four grandchildren, all four leaves at level 2, is balanced and full. Add two great-grandchildren under one leaf and it is **no longer balanced**, because there are now leaves at level 2 and level 3 while the height is 3. A tree that is one long chain of five vertices is  $m$ -ary for any  $m$  but as unbalanced as it can be.

**Why balance matters,** worth one line: a balanced tree of height  $h$  holds close to  $m^h$  leaves, so searching it costs about  $h$  steps, whereas the chain costs as many steps as it has vertices. That is the whole reason the term exists.

#### C. THE TRAVELLING MISSIONARY FROM DETROIT (10 MARKS)

*Quoted as printed in section 6.8, where all twelve tours are tabulated.* Same five cities and same ten distances as the 2020/2021 and 2023/2024 papers.

**Count the tours first.** Fixing Detroit as start and end, the other four cities can be ordered in  $4! = 24$  ways, and each tour is identical to its own reverse, so there are  $24 / 2 = 12$  distinct tours.

**The minimum is 458 miles:**



**Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit**

$$98 + 113 + 56 + 133 + 58 = \mathbf{458}$$

**List all twelve to justify the word "minimum".** Ten marks is buying the enumeration, not the arithmetic: a minimum is a claim about every alternative, so exhibiting one short tour proves only that 458 is achievable, never that nothing beats it.

## Question Five

### ANSWER

2021/2022 Q5 · 23.5 MARKS

#### A. FOUR COMPUTING DOMAINS FOR GRAPH MODELS, ONE EXAMPLE EACH (6 MARKS)

*Quoted as printed in section 6.1.* Computing domains, not real-life scenarios; the two questions look alike and want different answers.

**1. Networking and routing.** Vertices are routers, edges are links. Example: a shortest-path algorithm choosing a route through the network.

**2. Compilers.** Vertices are program points, edges are possible jumps. Example: a control-flow graph, used to find unreachable code.

**3. Databases and the web.** Vertices are pages or records, edges are links or foreign keys. Example: ranking pages by how many others link to them, which is a question about in-degree.

**4. Operating systems.** Vertices are processes and resources, edges are "holds" and "waits for". Example: deadlock detection, which is the search for a cycle in that graph.

#### B. THE SIZE OF THE DISCRETE STRUCTURES CLASS (4 MARKS)

*Quoted as printed in section 5.3.*

**Note the printing first.** The paper prints "There of 25 students" where it means "There are". More importantly the sentence can be read two ways: 25 as the number majoring in Computer Science, or 25 as the class total. Only the first makes the question answerable, and that is the reading taken here. Say so in one line.

**The computation.** Every student majors in one or the other or both, so the class is the union:

$$\begin{aligned}n(C \cup M) &= n(C) + n(M) - n(C \cap M) \\&= 25 + 13 - 8 \\&= \mathbf{30} \text{ students}\end{aligned}$$

**Check with the three regions:** Computer Science only is  $25 - 8 = 17$ , Maths only is  $13 - 8 = 5$ , both is 8, and  $17 + 5 + 8 = 30$ . Drawing that as two overlapping circles is worth doing.

#### C. WHEN IS A VERTEX INCIDENT TO AN EDGE? (3½ MARKS)

*Quoted as printed in section 6.1.*

**Definition.** A vertex  $v$  is **incident** to an edge  $e$  when  $v$  is one of the two endpoints of  $e$ . Equivalently,  $e$  is incident to  $v$ . Incidence relates a vertex to an **edge**; adjacency relates a vertex to another **vertex**, and the two words are not interchangeable.

**Example.** In a triangle on vertices 1, 2, 3 with edges  $e_1 = \{1, 2\}$ ,  $e_2 = \{2, 3\}$ ,  $e_3 = \{1, 3\}$ , vertex 2 is incident to  $e_1$  and  $e_2$  and not to  $e_3$ . Vertices 1 and 2 are **adjacent**, because an edge joins them.

**The link to degree,** worth a line: the degree of a vertex is the number of edges incident to it, counting a loop twice because both its ends meet the same vertex.

#### D. TWO TRUTH TABLES (5 MARKS AND 5 MARKS)

*Quoted as printed in section 1.1.*

**(i)  $(p \leftrightarrow q) \ominus (p \leftrightarrow \neg q)$ .** The paper prints the connective as a **circled minus**. There is no such standard connective; in this position it can only mean **exclusive or**, normally written  $\oplus$ . Say that in one line, then answer.

| P | Q | $P \leftrightarrow Q$ | $\neg Q$ | $P \leftrightarrow \neg Q$ | THE WHOLE |
|---|---|-----------------------|----------|----------------------------|-----------|
| T | T | T                     | F        | F                          | T         |
| T | F | F                     | T        | T                          | T         |
| F | T | F                     | F        | T                          | T         |
| F | F | T                     | T        | F                          | T         |

**True in every row: a tautology.** And the reason is worth stating:  $p \leftrightarrow q$  and  $p \leftrightarrow \neg q$  are negations of one another, so exactly one of them holds in any row, which is precisely when an exclusive or is true.

**(ii)  $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$ .**

| P | Q | $\neg P$ | $Q \rightarrow \neg P$ | $P \leftrightarrow Q$ | THE WHOLE |
|---|---|----------|------------------------|-----------------------|-----------|
| T | T | F        | F                      | T                     | F         |
| T | F | F        | T                      | F                     | F         |
| F | T | T        | T                      | F                     | F         |
| F | F | T        | T                      | T                     | T         |

**True in one row only**, when  $p$  and  $q$  are both false. This is word for word the same proposition as 2024/2025 Q1(b)(ii), so one table answers a part of two papers.

## TEACH

## MARK RECONCILIATION

| QUESTION | PARTS               | TOTAL     |
|----------|---------------------|-----------|
| Q1       | 5 + 6.5 + 12        | 23.5      |
| Q2       | 6 + 7.5 + 10        | 23.5      |
| Q3       | 5 + 5 + 6 + 7.5     | 23.5      |
| Q4       | 5.5 + 8.5 + 10      | <b>24</b> |
| Q5       | 6 + 4 + 3.5 + 5 + 5 | 23.5      |

**Q4 is half a mark heavier than the rest**, as printed. That is a fact about the paper, not an error in this arithmetic, and it is the kind of thing worth checking rather than assuming: a question whose parts do not total what the others do either carries an extra part or is worth more.

## PAST PAPER

# 2020/2021 Solved in Full

*The oldest paper we hold, and the only one printed under the course code CSC311. Two credit units, two hours, attempt any three. Five of its fifteen parts reappear on later papers almost unchanged, so it repays working even though its rubric is not yours.*

*By the end of this section you will have complete answers to all five questions, including the play-off tree, the password count and the six relations on the integers.*

| SESSION   | CODE PRINTED  | CREDIT | INSTRUCTION       | TIME    |
|-----------|---------------|--------|-------------------|---------|
| 2020/2021 | <b>CSC311</b> | TWO    | attempt any THREE | 2 hours |

**The code fixes the lineage.** CSC311 here, CSC416 on 2021/2022 and 2024/2025, CSC 242 on 2023/2024 and 2025/2026: one course, renamed twice, with the same title, department and semester throughout. That is why all five papers are valid practice.

**Its defects.** Q5(a)(vi) prints "Contigency" for contingency. Q3(a) and Q1(c) punctuate commands as questions. Q4(a) is missing an article and asks for "a tree diagram for five games" although the playoff can finish in three or four. And the  $e_1$  to  $e_6$  figure in Q1(c) is **only partly readable**: the sixth edge cannot be settled as a loop or an ordinary edge at this resolution.

**Marks per question:** Q1 23.5, Q2 23.5, Q3 23.5, Q4 23.5, but **Q5 totals 22.5** as printed. One mark short of the others.

**What recurs.** Q1(a) is the types-of-graph question, on three papers. Q2(a) is the Detroit tour, on three. Q3(b) is the chain letter, on three. Q2(b) and Q2(c) reappear on 2021/2022, and Q4(b) is word for word 2021/2022 Q2(c).

## Question One

### ANSWER

2020/2021 Q1 · 23.5 MARKS

#### A. TYPES OF UNDIRECTED GRAPH, WITH EXAMPLES (12 MARKS)

Quoted as printed in section 6.2, where the full catalogue is tabled. The single most repeated question on this course, worth 12 marks here, 12 on 2021/2022 and 9 on 2023/2024.

**Name, define and illustrate each:** simple, multigraph, pseudograph, null, trivial, complete  $K_n$ , regular, cycle, path, wheel, bipartite, complete bipartite  $K_{m,n}$ , connected, weighted and subgraph. Note this paper says "with the aid of relevant examples" rather than "diagrammatical illustrations", so a named instance ( $K_4$ ,  $C_5$ ,  $K_{3,3}$ ) counts alongside a sketch.

**The distinction the marks turn on** is simple against multigraph against pseudograph: no repeats and no loops, repeats allowed, repeats and loops allowed. Ten types with examples beats fifteen names without.

#### B. COMMUNICATION LINKS BETWEEN FIVE BUILDINGS (6½ MARKS)

Quoted as printed in section 6.2. The question names **five** buildings. A simple link joins two of them, and the link from X to Y is the same link as from Y to X, so order does not matter:

$$\text{links} = C(5, 2) = 5 \times 4 / 2 = \mathbf{10}$$

**The diagram** is  $K_5$ : five dots arranged in a pentagon, each joined to the other four, which draws as a pentagon with all five diagonals. Ten lines, counted.

**Count the buildings, do not remember the number.** The 2021/2022 paper sets this with **four** buildings and its answer is  $C(4, 2) = 6$ .

**Worth one extra line:**  $K_5$  is exactly the graph that section 6.7 proves non-planar, so these ten links cannot be laid out on a flat plan without two of them crossing. That is a genuine engineering observation and it costs one sentence.

#### C. INCIDENCE, AND THE EDGES EACH VERTEX IS INCIDENT TO (5 MARKS)

Quoted as printed in section 6.1.

**Definition.** A vertex  $v$  is **incident** to an edge  $e$  when  $v$  is one of the two endpoints of  $e$ . Incidence relates a vertex to an **edge**; adjacency relates a vertex to another **vertex**.

**State the reading of the figure.** Vertices 1, 2 and 3 with edges  $e_1$  to  $e_5$  are clear, and  $e_1$ ,  $e_2$  are a parallel pair, as are  $e_4$ ,  $e_5$ . The sixth edge is drawn as a long curve on the right and **cannot be settled** at this resolution as a loop at vertex 2 or an edge from 2 to 3. The table below takes it as a **loop at vertex 2** and gives the other reading underneath.

| VERTEX | INCIDENT EDGES            | DEGREE |
|--------|---------------------------|--------|
| 1      | $e_1, e_2, e_3$           | 3      |
| 2      | $e_1, e_2, e_4, e_5, e_6$ | 6      |
| 3      | $e_3, e_4, e_5$           | 3      |
| 4      | none                      | 0      |

**Why vertex 2's degree is 6 and not 5.** A **loop** counts twice towards the degree, because both of its ends meet the same vertex.

**The Handshaking check, which settles that the reading is at least consistent.**

Six edges give a degree sum of  $2 \times 6 = 12$ , and  $3 + 6 + 3 + 0 = 12$ . Under the other reading, with  $e_6$  joining 2 and 3, the degrees are 3, 5, 4 and 0, which also sum to 12.

Both readings pass, so the check cannot decide between them, and saying that is more honest than pretending it can.

**Vertex 4 is the point of the question.** It is incident to no edge at all, which is why it was drawn. An isolated vertex has degree 0 and is still a vertex.

## Question Two

### ANSWER

2020/2021 Q2 · 23.5 MARKS

#### A. THE TRAVELLING MISSIONARY FROM DETROIT (10 MARKS)

*Quoted as printed in section 6.8, where all twelve tours are tabulated.* The same figure and the same ten distances as 2021/2022 and 2023/2024.

**Twelve tours**, since fixing Detroit leaves  $4! = 24$  orders for the other four cities and each tour equals its own reverse:  $24 / 2 = 12$ .

**The minimum is 458 miles:**

**Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit**

$$98 + 113 + 56 + 133 + 58 = 458$$

**The question says "minimum"**, so list all twelve. Exhibiting one short tour shows 458 is achievable; only the enumeration shows nothing beats it.

#### B. FOUR REAL-LIFE SCENARIOS FOR GRAPH MODELS, TWO EXAMPLES EACH (6 MARKS)

*Quoted as printed in section 6.1.* Say what the vertices and the edges are in each; that is the modelling, and it is what is being marked.

**1. Transport networks.** Vertices are places, edges are routes. Examples: a road map with distances on the edges, exactly as in part (a); an airline map where an edge means a direct flight.

**2. Social networks.** Vertices are people, edges are relationships. Examples: friendship, where the edge is undirected because it is mutual; following, where it is directed because it is not.

**3. Utility and supply networks.** Vertices are junctions, edges are cables or pipes. Examples: an electricity grid; a water network where the edge weights are capacities.

**4. Scheduling and dependency.** Vertices are tasks, edges are "must come before". Examples: a project plan, in which a cycle would mean an impossible schedule; an exam timetable, in which an edge means two papers clash and cannot share a slot.

#### C. KEY TERMINOLOGIES OF A ROOTED TREE, WITH A DIAGRAM (7½ MARKS)

*Quoted as printed in section 7.1.* Identical to 2021/2022 Q3(c).

**Definition first.** A rooted tree is a tree with one vertex designated the **root**; every edge is then read as directed away from it, and every other vertex has exactly one path back to it.

**The terms: root, parent, child, siblings, ancestors, descendants, leaf, internal vertex, subtree, level** (0 at the root) and **height** (the largest level).

**Then one labelled diagram** carrying every term at once, with the levels numbered down the left edge. Eleven labels on one tree beats eleven separate sketches, and it is faster to draw.

**Close with the counting facts:**  $v$  vertices means  $v - 1$  edges, and there is exactly one path between any two vertices.

## Question Three

### ANSWER

2020/2021 Q3 · 23.5 MARKS

#### A. WHAT A BALANCED M-ARY TREE MEANS, WITH EXAMPLES (7½ MARKS)

*Quoted as printed in section 7.2.* Three definitions stacked, and the question needs all three.

**m-ary:** every internal vertex has **at most**  $m$  children. **Full:** every internal vertex has **exactly**  $m$  children. **Balanced,** for a tree of height  $h$ : every leaf sits at level  $h$  or at level  $h - 1$ .

**Examples.** A binary tree with a root, two children and four grandchildren has all four leaves at level 2: full and balanced. Hang two more vertices under one of those leaves and it is **no longer balanced**, since the height becomes 3 while three leaves remain at level 2. A chain of five vertices is  $m$ -ary for every  $m$  and as unbalanced as a tree can be.

**Why it matters:** a balanced tree of height  $h$  holds close to  $m^h$  leaves, so reaching any one of them costs about  $h$  steps, while the chain costs one step per vertex. That is the reason the word exists.

#### B. THE CHAIN LETTER (8 MARKS)

*Quoted as printed in section 7.2.* The first person is the root, each sender has exactly four children, and the 100 who read but did not send are the leaves.

$$l = (m - 1)i + 1, \text{ so } 100 = 3i + 1 \text{ and } i = \mathbf{33}$$

$$n = mi + 1 = 4 \times 33 + 1 = \mathbf{133}$$

**133 people have seen the letter**, including the first, and **33 people sent it out**, those being the internal vertices.

**Check:**  $33 + 100 = 133$ .

**The same problem appears as convocation cards** on 2021/2022 Q3(a) and 2023/2024 Q5(d). One method, three costumes.

#### C. FOUR APPLICATION AREAS IN COMPUTING, ONE EXAMPLE EACH (8 MARKS)

*Quoted as printed in section 6.1.* Computing areas, not real-life scenarios: part (b) already took those.

**1. Networking and routing.** Vertices are routers, edges are links. Example: computing a shortest path across the network.

**2. Compilers and program analysis.** Vertices are program points, edges are possible jumps. Example: a control-flow graph, used to find code that can never run.

**3. Databases and the web.** Vertices are pages or records, edges are links or keys. Example: ranking a page by how many others point at it, which is a question about its in-degree.

**4. Operating systems.** Vertices are processes and resources, edges are "holds" and "waits for". Example: detecting deadlock, which is searching that graph for a cycle.



## Question Four

### ANSWER

2020/2021 Q4 · 23.5 MARKS

#### A. I. A TREE DIAGRAM FOR THE PLAYOFF (4 MARKS)

*Quoted as printed in section 7.2.* Two teams, at most five games, first to three wins takes the playoff.

**Read the question carefully.** It says "a tree diagram for five games", but the series **stops** the moment a team reaches three wins, so many branches end at three or four games. The tree must show every way the playoff can finish, not five games on every branch.

**How to draw it.** Start at the root. Each vertex has two children, one for each team winning the next game, labelled A and B. Stop a branch as soon as one letter has appeared three times along it. The result is a binary tree whose leaves are the completed series, the shortest with three games (AAA and BBB) and the longest with five.

#### A. II. IN HOW MANY WAYS CAN THE PLAYOFF OCCUR? (3½ MARKS)

Count the leaves, which is easier by cases than by drawing.

**A wins in three games:** AAA, and there is 1 way.

**A wins in four:** A's third win is the last game, so exactly one of the first three is a B, which is  $C(3, 1) = 3$  ways.

**A wins in five:** the last game is A's third win, so exactly two of the first four are B, which is  $C(4, 2) = 6$  ways.

A wins in  $1 + 3 + 6 = 10$  ways

**By symmetry B wins in 10 ways too,** so the playoff can occur in

$10 + 10 = 20$  different ways

**The trap here is  $2^5 = 32$ .** That would be the answer if five games were always played, but the series stops early, so the shorter sequences are single outcomes rather than groups of them. Saying that sentence shows you saw the difference.

#### B. FIVE TYPES OF FUNCTION CORRESPONDENCE, WITH A DIAGRAM (10 MARKS)

*Quoted as printed in section 4.2.* Word for word the same question as 2021/2022 Q2(c).

| TYPE                            | WHAT THE ARROWS DO                                       | HOW TO DRAW IT                                       |
|---------------------------------|----------------------------------------------------------|------------------------------------------------------|
| <b>i. One-to-one, not onto</b>  | no two inputs share an output, but some output is unused | 3 inputs, 4 outputs, three arrows to three targets   |
| <b>ii. Onto, not one-to-one</b> | every output used, but two inputs share one              | 4 inputs, 3 outputs, every target hit, one hit twice |
| <b>iii. One-to-one and onto</b> | a perfect pairing, a bijection                           | 3 inputs, 3 outputs, one arrow each                  |
| <b>iv. Neither</b>              | two inputs share an output and one output is unused      | 3 inputs, 3 outputs, two arrows to one target        |
| <b>v. Not a function</b>        | an input with no arrow, or an input with two             | 3 inputs, one with no arrow leaving it               |

**Say which rule type v breaks.** A function uses **every** input and uses each **exactly once**; naming the one your drawing violates is what turns a sketch into an answer.

**C. I.  $(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R)$  (3 MARKS)**

Quoted as printed in section 1.3. Eight rows, and the two columns must agree in all eight.

| P | Q | R | $P \rightarrow Q$ | $P \rightarrow R$ | LEFT SIDE | $Q \wedge R$ | RIGHT SIDE |
|---|---|---|-------------------|-------------------|-----------|--------------|------------|
| T | T | T | T                 | T                 | T         | T            | T          |
| T | T | F | T                 | F                 | F         | F            | F          |
| T | F | T | F                 | T                 | F         | F            | F          |
| T | F | F | F                 | F                 | F         | F            | F          |
| F | T | T | T                 | T                 | T         | T            | T          |
| F | T | F | T                 | T                 | T         | F            | T          |
| F | F | T | T                 | T                 | T         | F            | T          |
| F | F | F | T                 | T                 | T         | F            | T          |

**The two shaded columns agree in all eight rows, so the two are logically equivalent.**  $\square$

**The one-line reason,** worth adding: a conditional promises something only when p holds, and promising q and promising r separately is the same promise as promising both together.

**C. II.  $P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$  (3 MARKS)**

| P | Q | $P \leftrightarrow Q$ | $P \wedge Q$ | $\neg P \wedge \neg Q$ | RIGHT SIDE |
|---|---|-----------------------|--------------|------------------------|------------|
| T | T | T                     | T            | F                      | T          |
| T | F | F                     | F            | F                      | F          |
| F | T | F                     | F            | F                      | F          |
| F | F | T                     | F            | T                      | T          |

**Identical in all four rows, so the two are equivalent.**  $\square$

**In words:** the biconditional says p and q agree, and there are exactly two ways to agree, both true or both false. The right-hand side is that sentence written out.

## Question Five

### ANSWER

2020/2021 Q5 · 22.5 MARKS AS PRINTED

#### A. DEFINE AND DRAW THE TRUTH TABLE FOR SIX TERMS (6 MARKS)

*Quoted as printed in section 1.1.* One mark each, so a definition and a table apiece.

**i. Negation.**  $\neg p$  is true exactly when  $p$  is false. T gives F, F gives T.

**ii. Conjunction.**  $p \wedge q$  is true only when both are. TT gives T; the other three rows give F.

**iii. Exclusive or.**  $p \oplus q$  is true when exactly one is. TT gives F, TF gives T, FT gives T, FF gives F.

**iv. Implication.**  $p \rightarrow q$  is false only when  $p$  is true and  $q$  false. TT gives T, TF gives F, FT gives T, FF gives T.

**v. Tautology.** A compound proposition true in **every** row, whatever its letters are. Example:  $p \vee \neg p$ , whose column is T, T.

**vi. "Contingency", as printed, which is contingency.** A compound proposition that is neither a tautology nor a contradiction: true in some rows and false in others. Example:  $p \wedge q$ , whose column is T, F, F, F. Quote the misspelling, then define the real term.

**Name the third member of the family,** which is free marks: a **contradiction** is false in every row, such as  $p \wedge \neg p$ . Tautology, contradiction and contingency are the three possibilities, and every compound proposition is exactly one of them.

#### B. HOW MANY POSSIBLE PASSWORDS? (5 MARKS)

*Quoted as printed in section 5.1.* Six to eight characters, each an uppercase letter or a digit, and at least one digit.

**Count the alphabet.** 26 letters plus 10 digits = **36** symbols, of which 26 are not digits.

**Use the subtraction rule,** because "at least one digit" is the signal: count all passwords of a given length and remove those with no digit at all.

| LENGTH | ALL                    | NO DIGIT              | AT LEAST ONE DIGIT |
|--------|------------------------|-----------------------|--------------------|
| 6      | $36^6 = 2176782336$    | $26^6 = 308915776$    | 1867866560         |
| 7      | $36^7 = 78364164096$   | $26^7 = 8031810176$   | 70332353920        |
| 8      | $36^8 = 2821109907456$ | $26^8 = 208827064576$ | 2612282842880      |

**The three lengths are alternatives,** since a password has exactly one length, so add them by the sum rule:

$$1867866560 + 70332353920 + 2612282842880 = \mathbf{2684483063360}$$

**Two things worth writing.** Both counting rules appear here: the product rule inside each length, the sum rule across the three. And this number is far beyond what a 32-bit integer holds, which is why section 11.1 insists on `long long` for a count like this.

#### C. I. FIVE ORDERED PAIRS IN EACH RELATION (5 MARKS)

*Quoted as printed in section 3.1.* These are relations on the **integers**, so the pairs are not drawn from a small set and any five that satisfy the rule will do. Include a negative and a zero where the rule allows; it shows the rule was read rather than pattern-matched.

| RELATION | RULE                | FIVE PAIRS                             |
|----------|---------------------|----------------------------------------|
| R1       | $a \leq b$          | (1,1), (1,2), (0,5), (-3,0), (-7,-2)   |
| R2       | $a > b$             | (2,1), (5,0), (0,-1), (-2,-7), (10,9)  |
| R3       | $a = b$ or $a = -b$ | (3,3), (3,-3), (-4,4), (0,0), (7,7)    |
| R4       | $a = b$             | (1,1), (2,2), (0,0), (-5,-5), (9,9)    |
| R5       | $a = b + 1$         | (2,1), (1,0), (0,-1), (-4,-5), (11,10) |
| R6       | $a + b \leq 3$      | (1,1), (0,3), (3,0), (-2,5), (-10,0)   |

**C. II. WHICH ARE REFLEXIVE, SYMMETRIC, TRANSITIVE? (6½ MARKS)**

| RELATION           | REFLEXIVE | SYMMETRIC | TRANSITIVE |
|--------------------|-----------|-----------|------------|
| R1: $a \leq b$     | yes       | no        | yes        |
| R2: $a > b$        | no        | no        | yes        |
| R3: $a = \pm b$    | yes       | yes       | yes        |
| R4: $a = b$        | yes       | yes       | yes        |
| R5: $a = b + 1$    | no        | no        | no         |
| R6: $a + b \leq 3$ | no        | yes       | no         |

**Give the reason for each "no", since that is what 6.5 marks buys.**

**R1** is not symmetric:  $1 \leq 2$  but  $2$  is not  $\leq 1$ .

**R2** is not reflexive, since no  $a$  is greater than itself, and not symmetric for the same reason as R1. It is transitive:  $a > b$  and  $b > c$  give  $a > c$ .

**R3** is reflexive since  $a = a$ ; symmetric since  $a = -b$  gives  $b = -a$ ; and transitive, because  $a = \pm b$  and  $b = \pm c$  force  $a = \pm c$ .

**R4** is equality itself, which is reflexive, symmetric and transitive, and is therefore an **equivalence relation**.

**R5** fails all three:  $a$  is never  $a + 1$ ;  $a = b + 1$  does not give  $b = a + 1$ ; and  $a = b + 1$  with  $b = c + 1$  gives  $a = c + 2$ , not  $c + 1$ .

**R6** is not reflexive, since  $2 + 2 = 4$  is not  $\leq 3$ . It is symmetric, because  $a + b = b + a$ . It is not transitive: (3, 0) and (0, 3) are both in R6 but (3, 3) is not, since  $3 + 3 = 6$ .

**The counterexamples are the answer.** A single explicit pair settles each "no", and R6's failure of transitivity is the one worth writing out in full, because the relation is symmetric and looks well behaved until it is tested.

| QUESTION | PARTS                  | TOTAL       |
|----------|------------------------|-------------|
| Q1       | $12 + 6.5 + 5$         | 23.5        |
| Q2       | $10 + 6 + 7.5$         | 23.5        |
| Q3       | $7.5 + 8 + 8$          | 23.5        |
| Q4       | $4 + 3.5 + 10 + 3 + 3$ | 23.5        |
| Q5       | $6 + 5 + 5 + 6.5$      | <b>22.5</b> |

**Q5 is one mark short of the other four** as printed. Together with the 2021/2022 paper, where Q4 is half a mark heavy, that makes two of the five papers we hold with a question that does not match its neighbours. Check the arithmetic on your own paper in the first minute; it takes thirty seconds and it tells you whether a part has been missed.

## MOCK EXAMINATION ONE

# Sit This Paper Under Exam Conditions

*Three hours, no notes, no phone. Six questions of 17.5 marks; answer any four. This paper is built to the 2025/2026 template and spreads its six questions across the whole book, so that whichever four you choose you will have used most of what you have read. Full answers follow. Attempt everything before you turn to them.*



3 hours

70 marks

6 questions of 17.5

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|

choose any four

# Question Paper

Time: three hours. Answer **any FOUR** questions. All questions carry **17.5 marks**. Total 70 marks.

## △ READ THIS FIRST

## THESE ARE NOT THE EXAMINER'S WORDS

**Every question on this paper was written for this manual.** Not one of them is quoted from a real paper, and that is why no question here carries an **As printed** bar: that bar is reserved for the examiner's own words, and nothing on these pages is the examiner's. What this paper copies is the *shape* of the 2025/2026 paper, six questions of 17.5 marks with any four to be answered in three hours, and the mix of topics that all five papers we hold have settled into.

**When you want the real wording, go to the Past Papers part.** All five papers are quoted there exactly as printed and answered in full. Sit this mock first, then those.

**How to use it.** Time yourself. Spend the first two minutes adding up the printed marks of the six questions and choosing your four, by the method in R.2. Then give each question forty minutes and stop when the forty minutes are gone, even mid-sentence. Marking your own half-finished answer teaches you more about your pace than a leisurely complete one does.

## QUESTION ONE

17.5 MARKS

**(a)** Explain, with **two examples in each case**, the difference between:

- (i) a tautology and a contradiction
- (ii) a subset and a proper subset
- (iii) an injective function and a surjective function
- (iv) an Euler circuit and a Hamilton circuit
- (v) a permutation and a combination **(10 marks)**

**(b)** State De Morgan's laws for sets, and verify one of them using  $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ ,  $A = \{1, 2, 3, 4, 5\}$  and  $B = \{4, 5, 6, 7\}$ . **(2.5 marks)**

**(c)** For the same  $A$  and  $B$ , find  $A \cup B$ ,  $A \cap B$ ,  $A - B$ ,  $B - A$ , and the number of subsets of  $A \cap B$ . **(5 marks)**

## QUESTION TWO

17.5 MARKS

**(a)** In a class of 120 students, 60 offer Mathematics, 55 offer Physics and 50 offer Chemistry. 25 offer both Mathematics and Physics, 20 offer both Physics and Chemistry, 22 offer both Mathematics and Chemistry, and 10 offer all three. Draw the Venn diagram and find the number of students who offer (i) at least one of the three subjects, (ii) none of the three, (iii) exactly one of the three. **(7 marks)**

**(b)** A relation  $R$  is defined on the set  $\{1, 2, 3, 4\}$  by  $a R b$  if and only if  $a - b$  is even. Show that  $R$  is an equivalence relation and give the partition it induces. **(5.5 marks)**

**(c)** Let  $S = \{(1,1), (1,2), (2,3), (3,4)\}$  on the set  $\{1, 2, 3, 4\}$ . State, **with a reason in each case**, whether  $S$  is reflexive, symmetric, antisymmetric and transitive. **(5 marks)**

### QUESTION THREE

17.5 MARKS

- (a) How many distinct arrangements are there of the letters of the word **STATISTICS**? **(3 marks)**
- (b) A committee of 5 is to be formed from 7 men and 6 women. In how many ways can this be done if (i) there is no restriction, (ii) exactly 3 women must serve, (iii) at least 4 men must serve? **(6 marks)**
- (c) A password is 5 characters long, drawn from the 26 capital letters and the 10 digits with repetition allowed, and it must begin with a letter. How many such passwords are there? **(3.5 marks)**
- (d) Write a C++ program that reads a positive integer  $n$  and prints all the positive divisors of  $n$  on one line, and state what your program prints when  $n = 28$ . **(5 marks)**

### QUESTION FOUR

17.5 MARKS

- (a) Write short notes, **with one example each**, on:
- (i) a complete graph
  - (ii) a bipartite graph
  - (iii) a regular graph
  - (iv) a subgraph
  - (v) an Eulerian graph **(10 marks)**
- (b) A graph  $G$  on the vertices  $v_1$  to  $v_5$  has the adjacency matrix below. Draw  $G$ , write down its degree sequence, and verify the handshaking theorem for it. **(4 marks)**

|       | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ |
|-------|-------|-------|-------|-------|-------|
| $v_1$ | 0     | 1     | 1     | 0     | 0     |
| $v_2$ | 1     | 0     | 1     | 1     | 0     |
| $v_3$ | 1     | 1     | 0     | 1     | 1     |
| $v_4$ | 0     | 1     | 1     | 0     | 1     |
| $v_5$ | 0     | 0     | 1     | 1     | 0     |

- (c) A full binary tree has 8 leaves. How many internal vertices, how many vertices altogether, and how many edges does it have? State the formula you use. **(3.5 marks)**



### QUESTION FIVE

17.5 MARKS

- (a) Construct a truth table for  $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$  and say what the table establishes. **(5 marks)**
- (b) Simplify the Boolean expression  $F = A \cdot B + A \cdot B' + A' \cdot B$ , naming the law used at each step. **(4 marks)**
- (c) A machine starts only when at least two of its three safety switches are closed. Write the Boolean expression for the start signal, give its truth table, and draw the circuit. **(5 marks)**
- (d) Use a truth table to verify that  $\neg(p \wedge q)$  is logically equivalent to  $\neg p \vee \neg q$ , and name the law. **(3.5 marks)**

### QUESTION SIX

17.5 MARKS

- (a) Prove by mathematical induction that  

$$1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)(2n+1)/6$$
 for every positive integer  $n$ . **(6 marks)**
- (b) Prove by contraposition that if  $n^2$  is even then  $n$  is even, where  $n$  is an integer. **(3 marks)**
- (c) Solve the recurrence relation  $a_n = 6a_{n-1} - 8a_{n-2}$  given  $a_0 = 3$  and  $a_1 = 10$ . **(5 marks)**
- (d) A box holds 5 red and 7 blue balls. Two balls are drawn one after the other **without replacement**. Find the probability that (i) both are red, (ii) at least one is blue. **(3.5 marks)**

### TEACH

MARK RECONCILIATION, BEFORE YOU START WRITING

| QUESTION | PARTS           | TOTAL |
|----------|-----------------|-------|
| Q1       | 10 + 2.5 + 5    | 17.5  |
| Q2       | 7 + 5.5 + 5     | 17.5  |
| Q3       | 3 + 6 + 3.5 + 5 | 17.5  |
| Q4       | 10 + 4 + 3.5    | 17.5  |
| Q5       | 5 + 4 + 5 + 3.5 | 17.5  |
| Q6       | 6 + 3 + 5 + 3.5 | 17.5  |

Do this arithmetic on the real paper too. It takes thirty seconds and it catches the part you have not noticed.

## Answers

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*All six questions are answered, not just a recommended four, because the point of a mock is to remove the need for a choice rather than to rehearse one. Each answer is written at the length forty minutes allows. Mark yourself against it honestly: a right final number with no working scores a fraction of the marks, and the marks lost that way are the easiest in the whole paper to win back.*

**A. FIVE DIFFERENCES, TWO EXAMPLES EACH (10 MARKS)**

Two marks each, so each item wants a **contrasting sentence** and then two examples. Writing two definitions side by side and leaving the reader to spot the difference is the standard way to lose half of these marks: see F.2.

**i. Tautology and contradiction.** A tautology is a compound proposition that is **true in every row** of its truth table, whereas a contradiction is **false in every row**. Examples of tautologies:  $p \vee \neg p$ , and  $(p \wedge q) \rightarrow p$ . Examples of contradictions:  $p \wedge \neg p$ , and  $\neg(p \vee \neg p)$ .

**ii. Subset and proper subset.**  $A \subseteq B$  allows the two sets to be equal, whereas  $A \subset B$  requires  $A \subseteq B$  **and**  $A \neq B$ , so  $B$  must have at least one member  $A$  lacks. Examples:  $\{1, 2\} \subseteq \{1, 2\}$  is true while  $\{1, 2\} \subset \{1, 2\}$  is false;  $\{1\} \subset \{1, 2\}$  and  $\{1\} \subseteq \{1, 2\}$  are both true.

**iii. Injective and surjective.** An injective function sends **no two different inputs to the same output**, whereas a surjective function **reaches every element of the codomain**. The first is a condition on the domain side, the second on the codomain side, which is why a function can have one without the other. Examples: with  $A = \{1, 2\}$  and  $B = \{a, b, c\}$ , the map  $1 \rightarrow a, 2 \rightarrow b$  is injective but not surjective, since  $c$  is never reached; with  $A = \{1, 2, 3\}$  and  $B = \{a, b\}$ , the map  $1 \rightarrow a, 2 \rightarrow a, 3 \rightarrow b$  is surjective but not injective, since  $1$  and  $2$  share an image.

**iv. Euler circuit and Hamilton circuit.** An Euler circuit uses every **edge** exactly once and returns to its start, whereas a Hamilton circuit visits every **vertex** exactly once and returns to its start. One is about edges and the other about vertices, and neither implies the other. Examples: two triangles joined at a single vertex have an Euler circuit, since every degree is even, but no Hamilton circuit, because the shared vertex would have to be visited twice;  $K_4$  has a Hamilton circuit but no Euler circuit, since all four of its vertices have the odd degree 3.

**v. Permutation and combination.** A permutation counts **arrangements**, so order matters, whereas a combination counts **selections**, so order does not; the two differ by the  $r!$  orderings of each chosen set. Examples: choosing a president, a secretary and a treasurer from 5 people is  $P(5, 3) = \mathbf{60}$ ; choosing a committee of 3 from the same 5 people is  $C(5, 3) = \mathbf{10}$ , and  $60 / 10 = 3! = 6$ .

**B. DE MORGAN'S LAWS FOR SETS, ONE VERIFIED (2.5 MARKS)**

**The laws.** For any sets  $A$  and  $B$  inside a universal set  $U$ ,

$$(A \cup B)' = A' \cap B' \quad \text{and} \quad (A \cap B)' = A' \cup B'$$

In words: the complement of a union is the intersection of the complements, and the complement of an intersection is the union of the complements. The operation flips and the complement moves inward.

**Verifying the first.** With  $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ ,  $A = \{1, 2, 3, 4, 5\}$  and  $B = \{4, 5, 6, 7\}$ :

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}, \text{ so } (A \cup B)' = \mathbf{\{8\}}$$

$$A' = \{6, 7, 8\} \text{ and } B' = \{1, 2, 3, 8\}, \text{ so } A' \cap B' = \mathbf{\{8\}}$$

The two sides agree, which is what was to be shown. **Compute both sides separately and only then compare them.** Working from one side towards the other is how a verification turns into a circular argument.

**C. THE FIVE RESULTS (5 MARKS)**

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A \cap B = \{4, 5\}$$

$$A - B = \{1, 2, 3\}$$

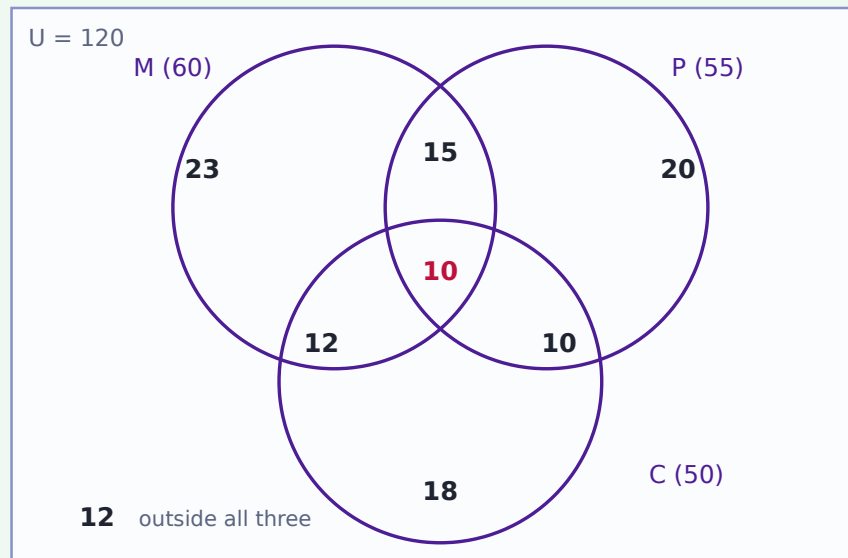
$$B - A = \{6, 7\}$$

$A \cap B$  has 2 members, so it has  $2^2 = 4$  subsets, namely  $\{\}$ ,  $\{4\}$ ,  $\{5\}$  and  $\{4, 5\}$

**Two checks worth the ink.**  $A - B$  and  $B - A$  are different, which is the whole reason the two are asked for separately; and  $|A \cup B| = |A| + |B| - |A \cap B| = 5 + 4 - 2 = 7$ , which is what the first line lists.

**A. THE THREE-SUBJECT CLASS OF 120 (7 MARKS)**

**Fill the diagram from the middle outward.** That single habit answers every question of this kind, and it is the reason the middle figure, 10, is given first.



*Every region carries a number, and the seven inside the circles add to 108 while the eighth accounts for the rest of the class.*

**Working the regions out.** Write M, P and C for the three sets. The centre is given as 10. Each two-subject region is the stated overlap less the centre:

$$\text{M and P only: } 25 - 10 = \mathbf{15}$$

$$\text{P and C only: } 20 - 10 = \mathbf{10}$$

$$\text{M and C only: } 22 - 10 = \mathbf{12}$$

Each single-subject region is the whole set less the three regions already placed inside it:

$$\text{M only: } 60 - 15 - 12 - 10 = \mathbf{23}$$

$$\text{P only: } 55 - 15 - 10 - 10 = \mathbf{20}$$

$$\text{C only: } 50 - 12 - 10 - 10 = \mathbf{18}$$

**(i) At least one.** By inclusion and exclusion for three sets,

$$n(M \cup P \cup C) = 60 + 55 + 50 - 25 - 20 - 22 + 10 = \mathbf{108}$$

**(ii) None.**  $120 - 108 = \mathbf{12}$

**(iii) Exactly one.**  $23 + 20 + 18 = \mathbf{61}$

**Check:** exactly two is  $15 + 10 + 12 = 37$ , exactly three is 10, and  $61 + 37 + 10 = 108$ , which is part (i) again by a different route. The diagram is worth marks in its own right here, so draw it even when the arithmetic alone would do.

**B. A R B WHEN A - B IS EVEN (5.5 MARKS)**

**Listing R first** makes the three checks a matter of looking. On  $\{1, 2, 3, 4\}$ ,  $a - b$  is even exactly when  $a$  and  $b$  are both odd or both even, so

$$R = \{(1,1), (1,3), (3,1), (3,3), (2,2), (2,4), (4,2), (4,4)\}$$

**Reflexive.** For every  $a$ ,  $a - a = 0$ , which is even, so  $(a, a)$  is in  $R$  for all four elements. Yes.

**Symmetric.** If  $a - b$  is even then  $b - a = -(a - b)$  is even too, so  $(b, a)$  is in  $R$  whenever  $(a, b)$  is. Yes.

**Transitive.** If  $a - b$  and  $b - c$  are both even, then  $a - c = (a - b) + (b - c)$  is a sum of two even numbers and therefore even. Yes.

**So  $R$  is an equivalence relation,** and the partition it induces is

**$\{1, 3\}$  and  $\{2, 4\}$ ,** the odd members and the even members

Naming the classes is the line that shows you know what an equivalence relation is *for*: it cuts the set into blocks that share a property, and every element lands in exactly one block.

### C. THE FOUR PROPERTIES OF $S$ , WITH REASONS (5 MARKS)

$S = \{(1,1), (1,2), (2,3), (3,4)\}$  on  $\{1, 2, 3, 4\}$ . A property fails the moment one witness fails, and **the witness is what earns the mark**: a bare "no" scores nothing.

| PROPERTY      | HOLDS?     | THE WITNESS                                                                                                    |
|---------------|------------|----------------------------------------------------------------------------------------------------------------|
| Reflexive     | <b>No</b>  | $(2, 2)$ is not in $S$ , and reflexivity needs $(a, a)$ for <i>every</i> $a$                                   |
| Symmetric     | <b>No</b>  | $(1, 2)$ is in $S$ but $(2, 1)$ is not                                                                         |
| Antisymmetric | <b>Yes</b> | the only pair appearing with its reverse is $(1, 1)$ , where the two entries are equal, so nothing violates it |
| Transitive    | <b>No</b>  | $(1, 2)$ and $(2, 3)$ are in $S$ but $(1, 3)$ is not                                                           |

**The trap in this part** is assuming that failing symmetry means passing antisymmetry, or the reverse. They are independent:  $S$  here is neither symmetric nor reflexive yet is antisymmetric, and a relation can be both symmetric and antisymmetric at once, which happens exactly when it contains nothing but loops.

**A. ARRANGEMENTS OF STATISTICS (3 MARKS)**

**Count the letters before anything else.** STATISTICS has 10 letters: S three times, T three times, I twice, and A and C once each. Check:  $3 + 3 + 2 + 1 + 1 = 10$ .

$$10! / (3! \times 3! \times 2!) = 3628800 / (6 \times 6 \times 2) = 3628800 / 72 = \mathbf{50400}$$

**Why the division.** The  $10!$  counts the three S's as though they were distinguishable, so each genuinely distinct arrangement has been counted  $3!$  times over for the S's,  $3!$  times over for the T's and  $2!$  times over for the I's. Dividing by the product removes exactly that overcount.

**B. A COMMITTEE OF 5 FROM 7 MEN AND 6 WOMEN (6 MARKS)**

Thirteen people in all. Order does not matter in a committee, so every part is a combination.

**(i) No restriction.**

$$C(13, 5) = 13! / (5! \times 8!) = \mathbf{1287}$$

**(ii) Exactly 3 women.** Three of the 6 women and, to fill the committee, two of the 7 men. The two choices are both compulsory, so multiply:

$$C(6, 3) \times C(7, 2) = 20 \times 21 = \mathbf{420}$$

**(iii) At least 4 men.** With only 5 seats, "at least 4 men" means exactly 4 or exactly 5. The two cases are alternatives, so add:

$$C(7, 4) \times C(6, 1) = 35 \times 6 = 210$$

$$C(7, 5) \times C(6, 0) = 21 \times 1 = 21$$

$$\text{total} = 210 + 21 = \mathbf{231}$$

**Both rules appear here,** and saying which is which is worth a mark: the product rule joins choices that are all made, the sum rule joins cases only one of which happens. Note also that the complement trick used in the 2025/2026 paper does *not* help in (iii), because "fewer than 4 men" is itself four cases.

**C. FIVE CHARACTERS, BEGINNING WITH A LETTER (3.5 MARKS)**

The alphabet has  $26 + 10 = 36$  characters. The first position is restricted to the 26 letters; the other four are unrestricted and repetition is allowed, so each has 36 choices. All five choices are made, so the product rule applies:

$$26 \times 36^4 = 26 \times 1679616 = \mathbf{43670016}$$

**Deal with the restricted position first.** Doing the free positions first and trying to correct afterwards is where this question goes wrong.

**D. THE DIVISORS OF N (5 MARKS)**

$d$  divides  $n$  exactly when  $n \% d$  is 0, so the whole program is one loop over the candidates from 1 to  $n$  with that one test inside it.

DIVISORS.CPP

```

// Mock One Question Three (d): every positive divisor of n, on one line.
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a positive integer: "; // i. prompt
    cin >> n;

    if (n < 1) { // ii. guard the precondition
        cout << "n must be positive." << endl;
        return 0;
    }

    cout << "Divisors of " << n << ":"; // iii. head the line
    for (int d = 1; d <= n; d++) { // iv. test every candidate
        if (n % d == 0) cout << " " << d; // v. d goes into n exactly
    }
    cout << endl; // vi. close the line
    return 0;
}

```

WHAT IT PRINTED FOR N = 28

```
Enter a positive integer: Divisors of 28: 1 2 4 7 14 28
```

**So the answer to state is 1, 2, 4, 7, 14, 28.** The prompt and the answer share a line because the 28 you type is not echoed to the output by the program; that is what the captured run above shows.

**What earns the marks on paper.** The loop bound  $d \leq n$  and the test  $n \% d == 0$  are the two things being examined. Writing  $d < n$  loses the divisor  $n$  itself, and starting at  $d = 0$  divides by zero. If you have time, add the line that  $28 = 1 + 2 + 4 + 7 + 14$ , so 28 is a perfect number: it costs one sentence and shows you read your own output.



**A. FIVE SHORT NOTES WITH AN EXAMPLE EACH (10 MARKS)**

Two marks each: one for the definition, one for the example. A note without an example is half an answer here.

**i. Complete graph.** A simple graph in which every pair of distinct vertices is joined by exactly one edge, written  $K_n$  on  $n$  vertices; it has  $C(n, 2) = n(n - 1)/2$  edges. Example:  $K_4$ , a square with both diagonals drawn, has  $4 \times 3 / 2 = 6$  edges.

**ii. Bipartite graph.** A graph whose vertices split into two sets such that every edge joins a vertex of one set to a vertex of the other, and no edge lies inside either set. Example:  $K_{3,3}$ , three houses each joined to three utilities, which has  $3 \times 3 = 9$  edges and no triangle at all.

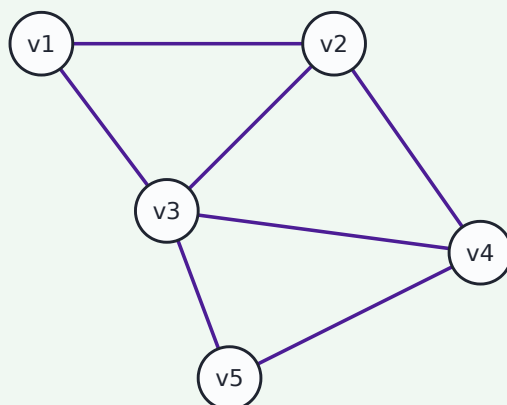
**iii. Regular graph.** A graph in which every vertex has the same degree; if that common degree is  $k$  it is called  $k$ -regular. Example: a cycle on five vertices is 2-regular, and  $K_4$  is 3-regular.

**iv. Subgraph.** A graph  $H$  whose vertices are among the vertices of  $G$  and whose edges are among the edges of  $G$ , with both endpoints of each kept edge also kept. Example: a single triangle inside  $K_4$  is a subgraph of it, obtained by deleting one vertex and the three edges at it.

**v. Eulerian graph.** A connected graph containing a circuit that uses every edge exactly once and returns to its start, which happens exactly when every vertex has even degree. Example: a triangle is Eulerian, each vertex having degree 2; the four-vertex graph of the Königsberg bridges is not, all four of its vertices having odd degree.

**B. DRAWING G FROM ITS ADJACENCY MATRIX (4 MARKS)**

**Read the matrix above the diagonal only.** The matrix is symmetric because the graph is undirected, so the upper triangle holds every edge once and reading the whole matrix double-counts them. The 1's above the diagonal sit at (1,2), (1,3), (2,3), (2,4), (3,4), (3,5) and (4,5), which is **7 edges**.



*G drawn from the matrix. Any drawing with the same seven edges is equally correct: the picture is not unique, the edge list is.*

**The degree sequence** is read off as the row sums:

$\deg(v_1) = 2$ ,  $\deg(v_2) = 3$ ,  $\deg(v_3) = 4$ ,  $\deg(v_4) = 3$ ,  $\deg(v_5) = 2$ , so the sequence written in decreasing order is **(4, 3, 3, 2, 2)**

**Verifying the handshaking theorem.** The sum of the degrees is  $4 + 3 + 3 + 2 + 2 = 14$ , and twice the number of edges is  $2 \times 7 = 14$ . They agree, as the theorem requires, because each edge contributes exactly 1 to each of its two endpoints. Note too that the graph has exactly **two** vertices of odd degree, and two is even, as the corollary to the theorem demands.

**C. THE FULL BINARY TREE WITH 8 LEAVES (3.5 MARKS)**

**The formula.** A full  $m$ -ary tree with  $i$  internal vertices has  $n = m \cdot i + 1$  vertices in all and  $l = (m - 1) \cdot i + 1$  leaves. Here  $m = 2$  and  $l = 8$ .

$$8 = (2 - 1) \cdot i + 1 = i + 1, \text{ so } i = 7 \text{ internal vertices}$$

$$n = 2 \times 7 + 1 = 15 \text{ vertices altogether}$$

$$\text{edges} = n - 1 = 14, \text{ since every tree on } n \text{ vertices has } n - 1 \text{ edges}$$

**Check by adding up:** 7 internal + 8 leaves = 15 vertices, which is the  $n$  the first formula gave. Two formulas agreeing is a real check; quoting one of them twice is not.

**A. THE TRUTH TABLE OF THE HYPOTHETICAL SYLLOGISM (5 MARKS)**

Three letters, so  $2^3 = 8$  rows. Build the columns in the order the connectives are applied, and never try to fill the last column directly.

| P | Q | R | $P \rightarrow Q$ | $Q \rightarrow R$ | $(P \rightarrow Q) \wedge (Q \rightarrow R)$ | $P \rightarrow R$ | THE WHOLE |
|---|---|---|-------------------|-------------------|----------------------------------------------|-------------------|-----------|
| T | T | T | T                 | T                 | T                                            | T                 | <b>T</b>  |
| T | T | F | T                 | F                 | F                                            | F                 | <b>T</b>  |
| T | F | T | F                 | T                 | F                                            | T                 | <b>T</b>  |
| T | F | F | F                 | T                 | F                                            | F                 | <b>T</b>  |
| F | T | T | T                 | T                 | T                                            | T                 | <b>T</b>  |
| F | T | F | T                 | F                 | F                                            | T                 | <b>T</b>  |
| F | F | T | T                 | T                 | T                                            | T                 | <b>T</b>  |
| F | F | F | T                 | T                 | T                                            | T                 | <b>T</b>  |

**What the table establishes.** The last column is T in all eight rows, so the proposition is a **tautology**. That is exactly what makes **hypothetical syllogism** a valid rule of inference: whenever both premises  $p \rightarrow q$  and  $q \rightarrow r$  hold,  $p \rightarrow r$  cannot fail. Say that sentence. A table with no conclusion under it answers only half of "say what the table establishes".

**Where the F's sit is the interesting part.** The conjunction is F in rows 2, 3, 4 and 6, and in every one of those rows the implication is vacuously T whatever  $p \rightarrow r$  does. Only the four rows where both premises hold are doing any work, and in all four  $p \rightarrow r$  is T.

**B. SIMPLIFYING  $A \cdot B + A \cdot B' + A' \cdot B$  (4 MARKS)**

Name a law on every line. The marks here are for the names, not for the final expression.

| STEP | EXPRESSION                            | LAW USED     |
|------|---------------------------------------|--------------|
| 0    | $A \cdot B + A \cdot B' + A' \cdot B$ | the given F  |
| 1    | $A \cdot (B + B') + A' \cdot B$       | distributive |
| 2    | $A \cdot 1 + A' \cdot B$              | complement   |
| 3    | $A + A' \cdot B$                      | identity     |
| 4    | $(A + A') \cdot (A + B)$              | distributive |
| 5    | $1 \cdot (A + B)$                     | complement   |
| 6    | <b><math>A + B</math></b>             | identity     |

**So  $F = A + B$ .** Check it against the truth table: F is 0 only when  $A = 0$  and  $B = 0$ , since each of the three products needs at least one of them to be 1, and  $A + B$  is 0 in exactly that one case.

**Steps 4 to 6 are the ones people miss.**  $A + A' \cdot B$  looks as though nothing can be done with it, because the usual distributive law of ordinary algebra does not run in that di-

rection. In Boolean algebra it does: addition distributes over multiplication as well as the other way round, and that second distributive law is what turns  $A + A'B$  into  $A + B$ .

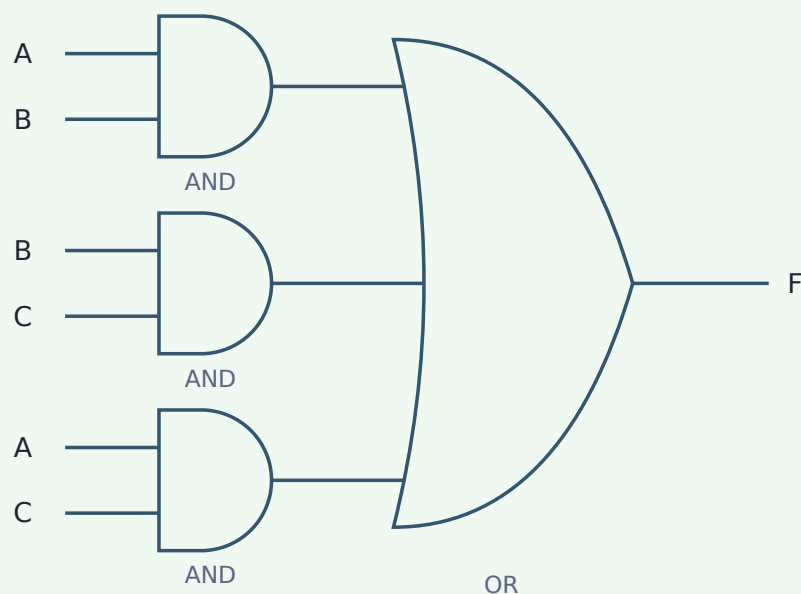
### C. AT LEAST TWO OF THREE SWITCHES (5 MARKS)

Let A, B and C be 1 when the corresponding switch is closed, and let F be 1 when the machine starts. "At least two" is the same as "some particular two", and the three particular pairs are alternatives, so they are joined by Boolean addition:

$$F = A \cdot B + B \cdot C + A \cdot C$$

| A | B | C | $A \cdot B$ | $B \cdot C$ | $A \cdot C$ | F | MACHINE |
|---|---|---|-------------|-------------|-------------|---|---------|
| 0 | 0 | 0 | 0           | 0           | 0           | 0 | stopped |
| 0 | 0 | 1 | 0           | 0           | 0           | 0 | stopped |
| 0 | 1 | 0 | 0           | 0           | 0           | 0 | stopped |
| 0 | 1 | 1 | 0           | 1           | 0           | 1 | starts  |
| 1 | 0 | 0 | 0           | 0           | 0           | 0 | stopped |
| 1 | 0 | 1 | 0           | 0           | 1           | 1 | starts  |
| 1 | 1 | 0 | 1           | 0           | 0           | 1 | starts  |
| 1 | 1 | 1 | 1           | 1           | 1           | 1 | starts  |

F is 1 in exactly the **4** rows that have two or three 1's among the inputs, which is the check that the expression says what the question says.



Three AND gates, one for each pair, feeding a single OR gate. Six input wires are drawn because each of A, B and C is used twice; on a real board one wire would fan out to two gates.

### D. VERIFYING DE MORGAN'S LAW FOR PROPOSITIONS (3.5 MARKS)

| P | Q | $P \wedge Q$ | $\neg(P \wedge Q)$ | $\neg P$ | $\neg Q$ | $\neg P \vee \neg Q$ |
|---|---|--------------|--------------------|----------|----------|----------------------|
| T | T | T            | <b>F</b>           | F        | F        | <b>F</b>             |
| T | F | F            | <b>T</b>           | F        | T        | <b>T</b>             |
| F | T | F            | <b>T</b>           | T        | F        | <b>T</b>             |
| F | F | F            | <b>T</b>           | T        | T        | <b>T</b>             |

**The two bold columns agree row for row**, so the two propositions are logically equivalent. This is **De Morgan's law**, and the sentence that earns the last half mark is what it means: to deny that both hold is to claim that at least one fails.

**Note it is the same law as in Q1(b)**, written in the other notation. Negation there was complement, conjunction was intersection and disjunction was union. One law, three costumes, as R.1 sets out side by side.

**A. INDUCTION FOR THE SUM OF SQUARES (6 MARKS)**

Write the claim down and name it before starting, so that the reader knows what  $P(n)$  refers to on every later line.

$$\mathbf{P(n):} \quad 1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$$

**Step 1, the basis.** At  $n = 1$  the left side is  $1^2 = 1$  and the right side is  $1 \times 2 \times 3 / 6 = 6 / 6 = 1$ . They agree, so  $P(1)$  is true.

**Step 2, the inductive hypothesis.** Assume  $P(k)$  holds for some particular positive integer  $k$ , that is

$$1^2 + 2^2 + \dots + k^2 = k(k+1)(2k+1)/6$$

**Step 3, the inductive step.** Add the next square to both sides and simplify the right:

$$\begin{aligned} 1^2 + \dots + k^2 + (k+1)^2 &= k(k+1)(2k+1)/6 + (k+1)^2 \\ &= (k+1)[k(2k+1) + 6(k+1)]/6 \quad (\text{taking out } (k+1)/6) \\ &= (k+1)(2k^2 + k + 6k + 6)/6 \\ &= (k+1)(2k^2 + 7k + 6)/6 \\ &= (k+1)(k+2)(2k+3)/6 \end{aligned}$$

and the claim  $P(k+1)$  reads

$$(k+1)(k+2)(2(k+1)+1)/6 = (k+1)(k+2)(2k+3)/6$$

which is the same expression. So  $P(k)$  implies  $P(k+1)$ .

**Conclusion.**  $P(1)$  is true and  $P(k)$  implies  $P(k+1)$  for every  $k$ , so by the principle of mathematical induction  $P(n)$  is true for every positive integer  $n$ .

**The step that is worth two marks on its own** is factorising  $2k^2 + 7k + 6$  as  $(k+2)(2k+3)$ . Multiply it back out to check:  $2k^2 + 3k + 4k + 6$ , which is  $2k^2 + 7k + 6$ . Without that factorisation the two sides look different and the proof appears to fail when it has in fact succeeded.

**A numerical sanity check** costs one line and is worth writing at the end: at  $n = 4$  the sum is  $1 + 4 + 9 + 16 = 30$ , and the formula gives  $4 \times 5 \times 9 / 6 = 180 / 6 = \mathbf{30}$ .

**B. IF  $N^2$  IS EVEN THEN  $N$  IS EVEN, BY CONTRAPOSITION (3 MARKS)**

**State the contrapositive first.** The claim "if  $n^2$  is even then  $n$  is even" has contrapositive

if  $n$  is **odd** then  $n^2$  is **odd**

and an implication and its contrapositive are logically equivalent, so proving the one proves the other.

**Proof.** Let  $n$  be odd. Then  $n = 2k + 1$  for some integer  $k$ , by the definition of odd. Squaring,

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

and  $2k^2 + 2k$  is an integer, so  $n^2$  has the form  $2m + 1$  and is odd. The contrapositive holds, and therefore so does the original claim.

**Why contraposition and not a direct proof.** A direct proof would start from " $n^2$  is even", that is  $n^2 = 2m$ , and would have to extract information about  $n$  from a fact about  $n^2$ , which needs a square root. The contrapositive starts from a statement about  $n$  it-

self, which can be written down in the form  $2k + 1$  at once. **Choose the direction whose hypothesis is the easier one to write in symbols.**

**C. SOLVING  $A_n = 6A_{n-1} - 8A_{n-2}$  (5 MARKS)**

This is linear, homogeneous, of order 2 and with constant coefficients, so the characteristic equation applies. Replace  $a_n$  by  $r^n$  and divide through by  $r^{n-2}$ :

$$r^2 = 6r - 8, \text{ that is } r^2 - 6r + 8 = 0$$

$$(r - 2)(r - 4) = 0, \text{ so } r = 2 \text{ or } r = 4$$

**Two distinct roots**, so the general solution is

$$a_n = A \cdot 2^n + B \cdot 4^n$$

**Fitting the two given terms.**

$$n = 0: A + B = 3$$

$$n = 1: 2A + 4B = 10, \text{ that is } A + 2B = 5$$

$$\text{subtracting, } B = 2, \text{ and then } A = 3 - 2 = 1$$

$$a_n = 2^n + 2 \cdot 4^n$$

**Check against the recurrence itself**, which is the check the examiner can see you doing. From the recurrence,  $a_2 = 6 \times 10 - 8 \times 3 = 60 - 24 = 36$ . From the formula,  $2^2 + 2 \times 4^2 = 4 + 32 = 36$ . They agree.

**D. TWO BALLS WITHOUT REPLACEMENT (3.5 MARKS)**

There are  $5 + 7 = 12$  balls. "Without replacement" is the signal that the second probability is **conditional** on the first draw, so the denominator falls from 12 to 11.

**(i) Both red.**

$$P(\text{both red}) = (5/12) \times (4/11) = 20 / 132 = 5 / 33$$

**(ii) At least one blue.** "At least one" is the signal for the complement, and the opposite of "at least one blue" is "no blue at all", which is exactly "both red":

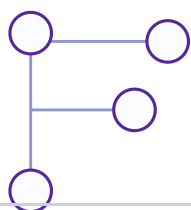
$$P(\text{at least one blue}) = 1 - 5 / 33 = 28 / 33$$

**Do not compute (ii) by cases.** One blue then one red, one red then one blue, and both blue is three fractions to get right where the complement is one subtraction. In decimals the answers are about **0.152** and **0.848**, and those two adding to 1 is the check.

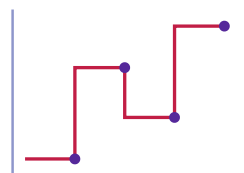
## MOCK EXAMINATION TWO

# Sit This One After the First

*Same shape, different ground. Where Mock One leaned on sets, counting and graphs, this paper leans on logic, functions, order, statistics and probability, and it asks for a tree traversal that no past paper we hold has asked for but that the syllabus plainly carries. Between the two mocks every part of this book is examined at least once.*



logic  
order and functions  
statistics and chance





# Question Paper

Time: three hours. Answer **any FOUR** questions. All questions carry **17.5 marks**. Total 70 marks.

## △ READ THIS FIRST

WRITTEN FOR THIS MANUAL, NOT PRINTED BY AN EXAMINER

**Nothing on this paper is quoted from a real examination.** As in Mock One, no question carries an **As printed** bar, because that bar belongs to the examiner's own words alone. The five genuine papers are quoted and answered in the Past Papers part.

**Sit Mock One first.** This one is deliberately weighted towards the parts of the syllabus that Mock One touched lightly, so taking them in order gives you full coverage; taking this one alone leaves gaps.

## QUESTION ONE

17.5 MARKS

(a) Define, **with one example each**, the following terms: a proposition, a tautology, a contradiction, a contingency, and logical equivalence. **(5 marks)**

(b) Construct the truth table for  $p \vee (q \wedge \neg r)$ . **(4 marks)**

(c) Write the converse, the inverse and the contrapositive of the statement "If a number is divisible by 6, then it is divisible by 3", and state which of the four statements are true, giving a counter-example where one is false. **(4.5 marks)**

(d) Express the following argument in symbols and test its validity, naming the rule of inference or the fallacy involved: "If it rains, the match is cancelled. The match was not cancelled. Therefore it did not rain." **(4 marks)**

## QUESTION TWO

17.5 MARKS

(a) Let  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ ,  $A = \{2, 4, 6, 8, 10\}$  and  $B = \{1, 2, 3, 4, 5\}$ . Find  $A \cup B$ ,  $A \cap B$ ,  $A'$ ,  $A - B$  and  $A \triangle B$ . **(5 marks)**

(b) Prove that  $(A \cup B)' = A' \cap B'$  by taking an arbitrary element of one side and showing it lies in the other. **(4 marks)**

(c) The function  $f$  from the real numbers to the real numbers is defined by  $f(x) = 3x - 5$ . Show that  $f$  is a bijection and find  $f^{-1}$ . **(5 marks)**

(d) With  $g(x) = x^2$  and  $h(x) = x + 3$  on the real numbers, find  $(g \circ h)(x)$  and  $(h \circ g)(x)$ , evaluate each at  $x = 2$ , and say whether  $g$  and  $h$  commute. **(3.5 marks)**

### QUESTION THREE

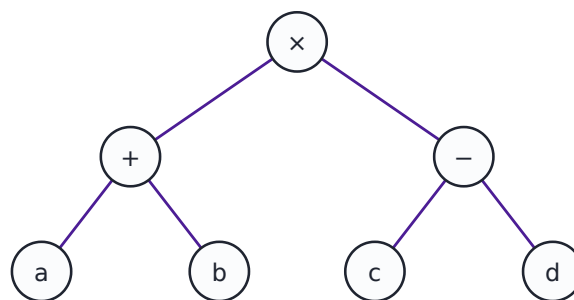
17.5 MARKS

- (a) Define, **with one example each**, the five properties a relation on a set may have. **(6 marks)**
- (b) Consider the relation "divides" on the set  $\{1, 2, 3, 4, 6, 12\}$ . List its ordered pairs, show that it is a partial order, and say with a reason why it is **not** a total order. **(6 marks)**
- (c) Write down the matrix of  $R = \{(1,2), (2,2), (2,3), (3,1)\}$  on  $\{1, 2, 3\}$  and use the **matrix alone** to decide whether  $R$  is reflexive and whether it is symmetric. **(3 marks)**
- (d) Give a relation on  $\{1, 2, 3\}$  that is **both** symmetric and antisymmetric, and explain why no contradiction is involved. **(2.5 marks)**

### QUESTION FOUR

17.5 MARKS

- (a) A simple graph has 8 vertices, every one of degree 3. How many edges has it? Does such a graph exist? Justify your answer. **(3 marks)**
- (b) A connected simple planar graph has 10 vertices and 15 edges. Use Euler's formula to find its number of faces, and check the graph against the bound  $e \leq 3v - 6$ . **(2.5 marks)**
- (c) For the binary tree drawn below, give the preorder, inorder and postorder traversals, and write down the arithmetic expression the tree represents. **(6 marks)**



- (d) State the handshaking theorem and use it to show that no graph can have five vertices every one of which has degree 3. **(3 marks)**
- (e) Write down the adjacency matrix of  $K_4$  and use it to find the number of walks of length 2 from  $v_1$  to  $v_2$ . **(3 marks)**

### QUESTION FIVE

17.5 MARKS

**(a)** The marks of ten students in a test were

45, 52, 38, 61, 45, 70, 55, 45, 62, 47

Find the mean, the median, the mode, the range, and both the population and the sample standard deviation. **(7 marks)**

**(b)** A fair die is rolled twice. Find the probability that (i) the sum of the two rolls is 8, (ii) at least one six appears, (iii) the sum is 8 given that the first roll is a 3. **(5 marks)**

**(c)** Three machines A, B and C produce 50%, 30% and 20% of a factory's output, and 3%, 4% and 5% of their output respectively is defective. An item picked at random from the day's production is found to be defective. What is the probability that it was made by machine A? **(5.5 marks)**

### QUESTION SIX

17.5 MARKS

**(a)** Prove by mathematical induction that

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

for every positive integer  $n$ . **(4 marks)**

**(b)** The 4th term of an arithmetic progression is 13 and the 10th term is 31. Find the first term, the common difference, and the sum of the first 20 terms. **(5 marks)**

**(c)** Solve the recurrence relation  $a_n = 4a_{n-1} - 4a_{n-2}$  given  $a_0 = 1$  and  $a_1 = 6$ . **(4.5 marks)**

**(d)** Write a C++ program that reads the first term  $a$ , the common difference  $d$  and a count  $n$ , and prints the sum of the first  $n$  terms of the arithmetic progression. State what it prints for the values you found in part (b). **(4 marks)**

### TEACH

MARK RECONCILIATION, BEFORE YOU START WRITING

| QUESTION | PARTS               | TOTAL |
|----------|---------------------|-------|
| Q1       | 5 + 4 + 4.5 + 4     | 17.5  |
| Q2       | 5 + 4 + 5 + 3.5     | 17.5  |
| Q3       | 6 + 6 + 3 + 2.5     | 17.5  |
| Q4       | 3 + 2.5 + 6 + 3 + 3 | 17.5  |
| Q5       | 7 + 5 + 5.5         | 17.5  |
| Q6       | 4 + 5 + 4.5 + 4     | 17.5  |

**Question Four has five parts** where every other question on this paper has three or four. That is deliberate: a five-part question is easier to run out of time on, and noticing it in the first minute is exactly what the reconciliation habit is for.

## Answers

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*All six answered again. Two habits are worth watching for as you mark: whether you stated the reading or the law before computing, and whether you wrote a check at the end. Those two lines are where the difference between a pass and a good grade tends to sit, and neither costs more than half a minute.*

**A. FIVE DEFINITIONS WITH AN EXAMPLE EACH (5 MARKS)**

One mark each, so one clean sentence and one example each. Do not write three sentences on the first and run out of time.

**Proposition.** A declarative sentence that is either true or false, but not both. Example: "7 is a prime number", which is true. Non-example, worth adding if you have room: "Close the door", which is a command and has no truth value.

**Tautology.** A compound proposition that is true in every row of its truth table, whatever its components. Example:  $p \vee \neg p$ .

**Contradiction.** A compound proposition that is false in every row. Example:  $p \wedge \neg p$ .

**Contingency.** A compound proposition that is neither a tautology nor a contradiction, so it is true in some rows and false in others. Example:  $p \wedge q$ , which is true only when both hold.

**Logical equivalence.** Two propositions are logically equivalent when they have the same truth value in every row, written  $P \equiv Q$ . Example:  $p \rightarrow q$  is equivalent to  $\neg p \vee q$ .

**B. THE TRUTH TABLE OF  $P \vee (Q \wedge \neg R)$  (4 MARKS)**

Three letters, so 8 rows, and one working column for each connective in the order it is applied.

| P | Q | R | $\neg R$ | $Q \wedge \neg R$ | $P \vee (Q \wedge \neg R)$ |
|---|---|---|----------|-------------------|----------------------------|
| T | T | T | F        | F                 | T                          |
| T | T | F | T        | T                 | T                          |
| T | F | T | F        | F                 | T                          |
| T | F | F | T        | F                 | T                          |
| F | T | T | F        | F                 | F                          |
| F | T | F | T        | T                 | T                          |
| F | F | T | F        | F                 | F                          |
| F | F | F | T        | F                 | F                          |

**Read the result back in words**, which is free and shows the table was understood: the compound is true in 5 of the 8 rows, namely whenever p is true, and otherwise only when q is true and r is false.

**C. CONVERSE, INVERSE, CONTRAPOSITIVE (4.5 MARKS)**

Let p be "the number is divisible by 6" and q be "the number is divisible by 3", so the statement is  $p \rightarrow q$ .

| NAME           | FORM                        | IN WORDS                                      | TRUE?        |
|----------------|-----------------------------|-----------------------------------------------|--------------|
| Original       | $p \rightarrow q$           | if divisible by 6 then divisible by 3         | <b>True</b>  |
| Converse       | $q \rightarrow p$           | if divisible by 3 then divisible by 6         | <b>False</b> |
| Inverse        | $\neg p \rightarrow \neg q$ | if not divisible by 6 then not divisible by 3 | <b>False</b> |
| Contrapositive | $\neg q \rightarrow \neg p$ | if not divisible by 3 then not divisible by 6 | <b>True</b>  |

**The counter-example, which is what the marks are for.** Take **9**. It is divisible by 3 but not by 6, so it makes the converse false; and it is not divisible by 6 yet is divisible by 3, so the same number makes the inverse false. **One number kills both**, and that is no accident: the converse and the inverse are contrapositives of each other, so they always stand or fall together.

**The original and the contrapositive are true** for the same reason, being equivalent:  $6 = 2 \times 3$ , so any multiple of 6 is a multiple of 3.

#### D. SYMBOLISING AND TESTING THE ARGUMENT (4 MARKS)

Let  $p$  be "it rains" and  $q$  be "the match is cancelled".

Premise 1:  $p \rightarrow q$

Premise 2:  $\neg q$

Conclusion:  $\therefore \neg p$

**The argument is valid**, and the rule is **modus tollens**. It is valid because there is no way for both premises to be true and the conclusion false: if  $p$  were true, the first premise would force  $q$ , which the second premise denies.

**The trap this question is set to catch** is the fallacy that looks almost the same. From  $p \rightarrow q$  and  $\neg p$  one may **not** conclude  $\neg q$ ; that is the fallacy of denying the hypothesis, and here it would say that if it did not rain the match went ahead, which the premises do not support. The match may have been cancelled for some other reason.

**A. FIVE SET OPERATIONS (5 MARKS)**

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$$

$$A \cap B = \{2, 4\}$$

$$A' = \{1, 3, 5, 7, 9\}$$

$$A - B = \{6, 8, 10\}$$

$$A \triangle B = (A - B) \cup (B - A) = \{6, 8, 10\} \cup \{1, 3, 5\} = \{1, 3, 5, 6, 8, 10\}$$

**Check the last one by counting.**  $|A \triangle B| = |A| + |B| - 2|A \cap B| = 5 + 5 - 4 = 6$ , and six members are listed. The 2 in that formula is the whole content of the symmetric difference: the shared members are removed once for each set.

**The complement needs the universal set**, which is why the question supplies  $U$ . A complement written without saying what it is taken inside is meaningless, and 7 and 9 appearing in  $A'$  while belonging to neither  $A$  nor  $B$  is the visible sign that  $U$  is doing work.

**B. PROVING  $(A \cup B)' = A' \cap B'$  (4 MARKS)**

**Two sets are equal when each is contained in the other**, so the proof has two halves. Take  $x$  arbitrary.

**First half.** Suppose  $x \in (A \cup B)'$ . Then  $x \notin A \cup B$ , so  $x$  is in neither  $A$  nor  $B$ . Hence  $x \in A'$  and  $x \in B'$ , so  $x \in A' \cap B'$ . This gives  $(A \cup B)' \subseteq A' \cap B'$ .

**Second half.** Suppose  $x \in A' \cap B'$ . Then  $x \notin A$  and  $x \notin B$ , so  $x$  is in neither, so  $x \notin A \cup B$ , so  $x \in (A \cup B)'$ . This gives  $A' \cap B' \subseteq (A \cup B)'$ .

**Both inclusions hold, so the sets are equal.** Write that closing line: the two halves are the work, and the conclusion is the mark.

**Why "arbitrary" matters.** The element  $x$  was never assumed to be anything in particular, so the argument covers every element at once. Choosing a specific  $x$ , or verifying the law on one example as Q1(b) of Mock One did, illustrates the law but does not prove it.

**C.  $f(x) = 3x - 5$  IS A BIJECTION (5 MARKS)**

**Injective.** Suppose  $f(x_1) = f(x_2)$ . Then

$$3x_1 - 5 = 3x_2 - 5, \text{ so } 3x_1 = 3x_2, \text{ so } x_1 = x_2$$

so no two different inputs share an output.

**Surjective.** Let  $y$  be any real number and put  $x = (y + 5)/3$ , which is a real number because 3 is not zero. Then

$$f(x) = 3 \cdot (y + 5)/3 - 5 = (y + 5) - 5 = y$$

so every  $y$  in the codomain is reached.

**Being both,  $f$  is a bijection**, and a bijection is exactly the condition under which an inverse function exists. Making  $x$  the subject of  $y = 3x - 5$  gives

$$f^{-1}(x) = (x + 5) / 3$$

**Check both ways round**, since an inverse must undo the function from either side:

$$f(f^{-1}(x)) = 3(x + 5)/3 - 5 = x, \text{ and } f^{-1}(f(x)) = ((3x - 5) + 5)/3 = x.$$

**D. COMPOSING  $g(x) = x^2$  AND  $h(x) = x + 3$  (3.5 MARKS)**

Read  $g \circ h$  as "g after h": the right-hand function is applied first. That is the single most common slip in this topic.

$$(g \circ h)(x) = g(h(x)) = g(x + 3) = (x + 3)^2$$

$$(h \circ g)(x) = h(g(x)) = h(x^2) = x^2 + 3$$

**At  $x = 2$ :**

$$(g \circ h)(2) = (2 + 3)^2 = 5^2 = 25$$

$$(h \circ g)(2) = 2^2 + 3 = 4 + 3 = 7$$

**So  $g$  and  $h$  do not commute:** 25 and 7 differ, and one counter-example is enough to settle a claim made for all  $x$ . Composition of functions is not commutative in general, and this is the standard demonstration of it.

**The extra line worth writing if there is time.** The two compositions agree exactly when  $(x + 3)^2 = x^2 + 3$ , that is  $x^2 + 6x + 9 = x^2 + 3$ , that is  $6x = -6$ , so they agree at the single point  $x = -1$  and nowhere else. Agreeing at a point is not commuting.



**A. THE FIVE PROPERTIES, WITH EXAMPLES (6 MARKS)**

Throughout,  $R$  is a relation on a set  $A$ . Each definition is worth about a mark and each example about a fifth of one, so state the definitions precisely and keep the examples tiny.

| PROPERTY      | DEFINITION                                                | EXAMPLE ON $\{1, 2\}$     |
|---------------|-----------------------------------------------------------|---------------------------|
| Reflexive     | $(a, a) \in R$ for every $a \in A$                        | $\{(1,1), (2,2), (1,2)\}$ |
| Irreflexive   | $(a, a) \notin R$ for every $a \in A$                     | $\{(1,2), (2,1)\}$        |
| Symmetric     | whenever $(a, b) \in R$ , also $(b, a) \in R$             | $\{(1,2), (2,1)\}$        |
| Antisymmetric | if $(a, b)$ and $(b, a)$ are both in $R$ then $a = b$     | $\{(1,1), (1,2)\}$        |
| Transitive    | whenever $(a, b)$ and $(b, c)$ are in $R$ , also $(a, c)$ | $\{(1,2), (2,2), (1,1)\}$ |

**Reflexive and irreflexive are not opposites.** A relation can be neither:  $\{(1,1)\}$  on  $\{1, 2\}$  has a loop at 1 and none at 2, so it fails both definitions. "Not reflexive" means **some** loop is missing; "irreflexive" means **every** loop is missing. Losing this mark is losing it to a quantifier, not to a relation.

**B. DIVISIBILITY ON  $\{1, 2, 3, 4, 6, 12\}$  (6 MARKS)**

**The pairs**, listed by first element so that none is missed. Read each row as "this divides these".

| FROM | PAIRS                                | HOW MANY |
|------|--------------------------------------|----------|
| 1    | (1,1) (1,2) (1,3) (1,4) (1,6) (1,12) | 6        |
| 2    | (2,2) (2,4) (2,6) (2,12)             | 4        |
| 3    | (3,3) (3,6) (3,12)                   | 3        |
| 4    | (4,4) (4,12)                         | 2        |
| 6    | (6,6) (6,12)                         | 2        |
| 12   | (12,12)                              | 1        |

That is **18** ordered pairs in all. Note that 1 relates to everything and 12 is related to by everything, which is the shape of a divisibility order on a set of divisors.

**Reflexive.** Every  $n$  divides itself, since  $n = 1 \times n$ , so all six loops are present. Yes.

**Antisymmetric.** Suppose  $a$  divides  $b$  and  $b$  divides  $a$ , with  $a$  and  $b$  positive. Then  $a \leq b$  and  $b \leq a$ , so  $a = b$ . Yes.

**Transitive.** If  $a$  divides  $b$ , say  $b = a \cdot s$ , and  $b$  divides  $c$ , say  $c = b \cdot t$ , then  $c = a \cdot (s \cdot t)$ , so  $a$  divides  $c$ . Yes.

**Reflexive, antisymmetric and transitive, so it is a partial order.**

**Why it is not a total order.** A total order requires every two elements to be **comparable**. Here **2 and 3** are not: 2 does not divide 3 and 3 does not divide 2, so neither  $(2, 3)$  nor  $(3, 2)$  is in the relation. One such pair is enough, though 4 and 6 are another, as are 3 and 4.

**C. READING A MATRIX (3 MARKS)**

Number the rows and columns 1, 2, 3, and put 1 in position  $(i, j)$  when  $(i, j)$  is in  $R$ :

|   | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0 | 1 | 0 |
| 2 | 0 | 1 | 1 |
| 3 | 1 | 0 | 0 |

**Reflexive?** The test in the matrix is that the **main diagonal is all 1's**. Here the diagonal reads 0, 1, 0, so  **$R$  is not reflexive**: the missing loops are at 1 and at 3.

**Symmetric?** The test is that the matrix **equals its own transpose**, that is, it is a mirror image about the main diagonal. Here the  $(1, 2)$  entry is 1 while the  $(2, 1)$  entry is 0, so  **$R$  is not symmetric**.

**This is why the matrix representation is worth learning.** Both questions are answered by looking rather than by searching a list of pairs, and the same two glances work for a relation with a hundred pairs.

#### D. SYMMETRIC AND ANTISYMMETRIC AT ONCE (2.5 MARKS)

**Take  $R = \{(1,1), (2,2)\}$**  on  $\{1, 2, 3\}$ , or indeed any relation made only of loops, including the empty relation.

**Symmetric**, because the only pairs in  $R$  are  $(1,1)$  and  $(2,2)$ , and each has its own reverse already in  $R$ .

**Antisymmetric**, because the definition only bites when  $(a, b)$  and  $(b, a)$  are both in  $R$ , and here that happens only with  $a = b$ , which is exactly what the definition then demands.

**Why there is no contradiction.** The two properties conflict only over a pair  $(a, b)$  with  $a \neq b$ : symmetry would insist that  $(b, a)$  be present, and antisymmetry would forbid it. A relation containing no such pair is answerable to neither demand, so it satisfies both definitions vacuously. The two properties are not opposites, and a relation is both exactly when it is a set of loops.

**A. EIGHT VERTICES, EACH OF DEGREE 3 (3 MARKS)**

By the handshaking theorem the degrees sum to twice the number of edges:

$$8 \times 3 = 2|E|, \text{ so } 24 = 2|E| \text{ and } |E| = 12$$

**Such a graph does exist.** The sum of the degrees, 24, is even, which is necessary; and a concrete example settles it beyond doubt: the eight corners of a cube, each joined by an edge to the three corners next to it, is a simple graph with 8 vertices, all of degree 3, and 12 edges, one for each edge of the cube.

**Name a graph rather than argue.** "The degree sum is even so it exists" is not a proof, because an even degree sum is necessary but not sufficient. Exhibiting one graph is a complete answer, and it is shorter.

**B. FACES BY EULER'S FORMULA (2.5 MARKS)**

For a connected planar simple graph drawn without crossings,  $v - e + f = 2$ . With  $v = 10$  and  $e = 15$ ,

$$f = 2 - v + e = 2 - 10 + 15 = 7$$

**The check against the bound.** For a simple connected planar graph with  $v \geq 3$ ,

$$e \leq 3v - 6 = 3 \times 10 - 6 = 24$$

and  $15 \leq 24$ , so nothing here contradicts planarity. **Say what the check does and does not prove.** Failing the bound would prove the graph is not planar; passing it proves nothing on its own, since  $K_{3,3}$  passes the same bound and is not planar. The question grants planarity, so Euler's formula may be used; the bound is only being confirmed as consistent.

**C. THE THREE TRAVERSALS OF THE TREE (6 MARKS)**

Each traversal visits the root, the left subtree and the right subtree in a fixed order, and the name says **where the root goes**: pre means first, in means between, post means last.

| TRAVERSAL | ORDER             | RESULT               |
|-----------|-------------------|----------------------|
| Preorder  | root, left, right | $\times + a b - c d$ |
| Inorder   | left, root, right | $a + b \times c - d$ |
| Postorder | left, right, root | $a b + c d - \times$ |

**Working the preorder through,** since the marks are for the method: visit the root  $\times$ ; then the left subtree, whose root is  $+$ , then its left child  $a$ , then its right child  $b$ ; then the right subtree, whose root is  $-$ , then  $c$ , then  $d$ . That gives  $\times + a b - c d$ .

**The expression the tree represents** is read off the inorder traversal, with the brackets that the tree shape implies and the flat string does not:

$$(a + b) \times (c - d)$$

**Say why the brackets are needed.** The inorder string  $a + b \times c - d$ , read with the usual precedence, would mean  $a + (b \times c) - d$ , which is a different calculation. Inorder alone loses the structure; preorder and postorder do not, which is why machines use them.

**D. THE HANDSHAKING THEOREM, APPLIED (3 MARKS)**

**The theorem.** In any undirected graph, the sum of the degrees of all the vertices equals twice the number of edges:

$$\sum \deg(v) = 2|E|$$

because each edge has two ends and so contributes exactly 1 to the degree of each of its two endpoints.

**The application.** Suppose a graph had five vertices, each of degree 3. Then

$$\sum \deg(v) = 5 \times 3 = 15$$

But  $2|E|$  is even for every graph, being twice a whole number, and 15 is odd. No number of edges can satisfy  $2|E| = 15$ , so **no such graph exists**.

**This is the corollary in disguise:** the number of odd-degree vertices in any graph is even, and five vertices of degree 3 would be five odd-degree vertices.

#### E. WALKS OF LENGTH 2 IN $K_4$ (3 MARKS)

$K_4$  joins every pair of its four vertices, so its adjacency matrix has 0 down the diagonal and 1 everywhere else:

|       | $v_1$ | $v_2$ | $v_3$ | $v_4$ |
|-------|-------|-------|-------|-------|
| $v_1$ | 0     | 1     | 1     | 1     |
| $v_2$ | 1     | 0     | 1     | 1     |
| $v_3$ | 1     | 1     | 0     | 1     |
| $v_4$ | 1     | 1     | 1     | 0     |

The number of walks of length 2 from  $v_1$  to  $v_2$  is the  $(1, 2)$  entry of  $A^2$ , which is row 1 of  $A$  multiplied into column 2 of  $A$ :

$$(0 \times 1) + (1 \times 0) + (1 \times 1) + (1 \times 1) = 0 + 0 + 1 + 1 = 2$$

**Naming them confirms it:** the two walks are  $v_1$  to  $v_3$  to  $v_2$ , and  $v_1$  to  $v_4$  to  $v_2$ . The first two products vanish precisely because a walk cannot step from  $v_1$  to itself or from  $v_2$  to itself: there are no loops in  $K_4$ .

**A. SIX STATISTICS FROM TEN MARKS (7 MARKS)**

**Sort first.** The median and the range both need the sorted list, and sorting takes one line:

38, 45, 45, 45, 47, 52, 55, 61, 62, 70

**Mean.** The ten marks total

$45 + 52 + 38 + 61 + 45 + 70 + 55 + 45 + 62 + 47 = 520$ , so the mean is  $520 / 10 = 52$

**Median.** Ten values is an even count, so the median is the mean of the 5th and 6th of the sorted list:

$(47 + 52) / 2 = 99 / 2 = 49.5$

**Mode.** 45 occurs three times and no other value occurs more than once, so the mode is 45.

**Range.**  $70 - 38 = 32$

**The two standard deviations.** Both start from the same column of squared deviations about the mean 52.

| X            | 45 | 52 | 38  | 61 | 45 | 70  | 55 | 45 | 62  | 47 |
|--------------|----|----|-----|----|----|-----|----|----|-----|----|
| $X - 52$     | -7 | 0  | -14 | 9  | -7 | 18  | 3  | -7 | 10  | -5 |
| $(X - 52)^2$ | 49 | 0  | 196 | 81 | 49 | 324 | 9  | 49 | 100 | 25 |

the squared deviations total **882**

population variance  $\sigma^2 = 882 / 10 = 88.2$ , so  $\sigma = \sqrt{88.2} = 9.39$  to two decimals

sample variance  $s^2 = 882 / 9 = 98$ , so  $s = \sqrt{98} = 9.90$  to two decimals

**The deviations must sum to zero**, and they do:  $-7 + 0 - 14 + 9 - 7 + 18 + 3 - 7 + 10 - 5 = 0$ . That one line catches an arithmetic slip in the mean before it poisons every figure below it.

**Which divisor, and say why.** Divide by  $n$  when the ten marks are the whole group you are describing, and by  $n - 1$  when they are a sample from a larger group whose spread you are estimating. The sample figure is always the larger of the two, here 9.90 against 9.39, because the divisor is smaller.

**B. TWO ROLLS OF A FAIR DIE (5 MARKS)**

The sample space is the  $6 \times 6 = 36$  ordered pairs, all equally likely. Ordered, because the first and second rolls are distinguishable, and (3, 5) is not the same outcome as (5, 3).

**(i) The sum is 8.** The favourable pairs are (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 of them:

$P(\text{sum} = 8) = 5 / 36$

**(ii) At least one six.** "At least one" calls for the complement. No six at all leaves 5 faces for each roll:

$P(\text{no six}) = 25 / 36$ , so  $P(\text{at least one six}) = 1 - 25/36 = 11 / 36$

**Counting them directly gives the same 11** and shows where  $6 + 6 = 12$  goes wrong: the six pairs with a six first and the six with a six second share the single pair

(6, 6), so the answer is  $6 + 6 - 1 = 11$ , which is inclusion and exclusion again.

**(iii) The sum is 8 given a first roll of 3.** Conditioning on the first roll shrinks the sample space to the 6 pairs beginning with 3, and exactly one of them, (3, 5), sums to 8:

$$P(\text{sum} = 8 \mid \text{first} = 3) = 1 / 6$$

**Note the two rolls are independent** and yet the conditional probability,  $1/6$ , differs from the unconditional  $5/36$ . There is no contradiction: independence is between the two **rolls**, whereas the event "the sum is 8" is not an event about one roll at all.

### C. WHICH MACHINE MADE THE DEFECTIVE ITEM (5.5 MARKS)

**Name the events before writing a single fraction.** Let A, B and C be the events that the item came from that machine, and let D be the event that it is defective. The question gives

$$P(A) = 0.50, P(B) = 0.30, P(C) = 0.20$$

$$P(D \mid A) = 0.03, P(D \mid B) = 0.04, P(D \mid C) = 0.05$$

**Step 1, the total probability of a defect.** The three machines cover the whole output and no item comes from two of them, so

$$\begin{aligned} P(D) &= P(A)P(D \mid A) + P(B)P(D \mid B) + P(C)P(D \mid C) \\ &= 0.50 \times 0.03 + 0.30 \times 0.04 + 0.20 \times 0.05 \\ &= 0.015 + 0.012 + 0.010 = \mathbf{0.037} \end{aligned}$$

**Step 2, Bayes' theorem** turns the conditioning round:

$$P(A \mid D) = P(A)P(D \mid A) / P(D) = 0.015 / 0.037 = \mathbf{15 / 37}, \text{ about } \mathbf{0.405}$$

**Read the answer back.** Machine A makes half the output but only about 40.5% of the defects, because it is the most reliable of the three. The three posteriors are  $15/37$ ,  $12/37$  and  $10/37$ , and they add to 1, which is the check to write.

**The mistake this question is set to catch** is answering 0.03. That is  $P(D \mid A)$ , the chance that an item from A is defective, and the question asks the reverse,  $P(A \mid D)$ . Bayes' theorem exists precisely because those two are different numbers.

**A. THE SUM OF THE FIRST N ODD NUMBERS (4 MARKS)**

$$\mathbf{P(n):} 1 + 3 + 5 + \dots + (2n - 1) = n^2$$

**Basis.** At  $n = 1$  the left side is the single term  $2 \times 1 - 1 = 1$  and the right side is  $1^2 = 1$ .  $P(1)$  holds.

**Hypothesis.** Assume  $P(k)$ , that  $1 + 3 + \dots + (2k - 1) = k^2$ .

**Step.** The next odd number is  $2(k + 1) - 1 = 2k + 1$ . Adding it to both sides,

$$1 + 3 + \dots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = \mathbf{(k + 1)^2}$$

which is exactly  $P(k + 1)$ . So  $P(k)$  implies  $P(k + 1)$ , and by the principle of mathematical induction  $P(n)$  holds for every positive integer  $n$ .

**The one line that is easy to drop** is the identification of the next term. Writing "+  $(2k + 1)$ " without saying it came from  $2(k + 1) - 1$  leaves the reader to check that the series really does continue that way, and it is a mark.

**B. THE ARITHMETIC PROGRESSION (5 MARKS)**

In an arithmetic progression the  $n$ th term is  $a_n = a + (n - 1)d$ . The two facts given become two equations:

$$a + 3d = 13 \quad (\text{the 4th term})$$

$$a + 9d = 31 \quad (\text{the 10th term})$$

**Subtracting the first from the second** removes  $a$  in one step:

$$6d = 18, \text{ so } d = \mathbf{3}, \text{ and then } a = 13 - 9 = \mathbf{4}$$

**The sum of the first 20 terms.**

$$S_n = n/2 \cdot [2a + (n - 1)d]$$

$$S_{20} = 20/2 \cdot [2 \times 4 + 19 \times 3] = 10 \times [8 + 57] = 10 \times 65 = \mathbf{650}$$

**Check the two given terms against  $a$  and  $d$ :** the 4th term is  $4 + 3 \times 3 = 13$  and the 10th is  $4 + 9 \times 3 = 31$ . Both match, so the sum rests on solid ground.

**Watch the index convention.** The 4th term uses  $(n - 1) = 3$ , not 4. Using  $a + 4d$  here is the single commonest error in this question, and it gives  $d = 3$  with  $a = 1$ , which then fails the check above.

**C. SOLVING  $A_n = 4A_{n-1} - 4A_{n-2}$  (4.5 MARKS)**

$$r^2 = 4r - 4, \text{ that is } r^2 - 4r + 4 = 0$$

$$(r - 2)^2 = 0, \text{ so } r = \mathbf{2}, \text{ a repeated root}$$

**A repeated root changes the shape of the solution.** With a single root  $r$  of multiplicity two the general solution is

$$a_n = (A + Bn) \cdot r^n = (A + Bn) \cdot 2^n$$

and not  $A \cdot 2^n + B \cdot 2^n$ , which collapses to a single constant times  $2^n$  and cannot meet two independent starting conditions.

**Fitting the two given terms.**

$$n = 0: (A + 0) \times 1 = 1, \text{ so } A = \mathbf{1}$$

$$n = 1: (1 + B) \times 2 = 6, \text{ so } 1 + B = 3 \text{ and } B = \mathbf{2}$$

$$\mathbf{a_n = (1 + 2n) \cdot 2^n}$$

**Check against the recurrence.** It gives  $a_2 = 4 \times 6 - 4 \times 1 = 20$ , and the formula gives  $(1 + 4) \times 4 = 20$ . They agree.

#### D. THE PROGRAM FOR THE SUM OF AN AP (4 MARKS)

The closed form of part (b) is one line of arithmetic. The loop beneath it is not required by the question, but it costs three lines and turns the program into its own check, which is worth writing when a program is being marked on paper.

APSUM.CPP

```
// Mock Two Question Six (d): the sum of the first n terms of an AP.
#include <iostream>
using namespace std;

int main() {
    int a, d, n;
    cout << "Enter a, d and n: ";           // i. prompt for all three
    cin >> a >> d >> n;

    int sum = n * (2 * a + (n - 1) * d) / 2; // ii. the closed form
    cout << "S_" << n << " = " << sum << endl;

    int running = 0;                        // iii. the same sum, term by term
    for (int k = 0; k < n; k++) {
        running += a + k * d;               // iv. the (k+1)th term
    }
    cout << "term by term: " << running << endl;
    return 0;
}
```

WHAT IT PRINTED FOR A = 4, D = 3, N = 20

```
Enter a, d and n: S_20 = 650
term by term: 650
```

**So the program prints 650**, which is the answer part (b) reached by hand. Two routes to one number is the strongest check a short program can carry.

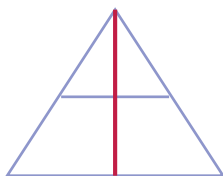
**The subtlety worth a sentence on paper.** In  $n * (2 * a + (n - 1) * d) / 2$  the multiplication happens before the division, so the product  $20 \times 65 = 1300$  is formed first and then halved. Writing  $n / 2 * (...)$  instead would divide 20 by 2 correctly here but would silently discard a half whenever n is odd, because integer division throws away the remainder. **Divide last.**



## MOCK EXAMINATION THREE

# The Hardest of the Three

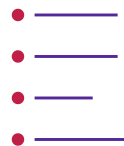
*Harder than anything the five papers have set, on purpose. Every question here asks you to **prove** something or to **rule something out**, rather than to compute a number and stop. If you can pass this paper in three hours you are over-prepared for the real one, which is the only comfortable place to be on the morning of an examination.*



prove it

or rule it out

no marks for a bare number



Q1 to Q6

# Question Paper

Time: three hours. Answer **any FOUR** questions. All questions carry **17.5 marks**. Total 70 marks.

## △ READ THIS FIRST

## DELIBERATELY HARDER THAN THE REAL THING

**Nothing here is quoted from a real examination**, and this paper is harder than any of the five we hold. That is deliberate, and it is the one respect in which this mock does **not** imitate the real paper. Do not read a poor score on it as a prediction.

**What it does imitate is the marking.** Every part asks for a justification, so a correct number with nothing under it is worth close to nothing, exactly as on the real paper when the phrase "with a detailed explanation" or "justify your answer" appears. Four of the five genuine papers use one of those phrases somewhere.

**If you are short of time, sit Questions One, Four and Six.** Those three carry the proof techniques, and proof is the part of this syllabus that the official course text covers worst.

## QUESTION ONE

17.5 MARKS

- (a) Show, **without using a truth table** and naming the law used at each step, that  $(p \wedge q) \vee (p \wedge \neg q)$  is logically equivalent to  $p$ . **(4 marks)**
- (b) Determine whether  $(p \rightarrow q) \wedge (\neg p \rightarrow r) \rightarrow (q \vee r)$  is a tautology. **(5 marks)**
- (c) Express in symbols, and then negate, the statement "Every student in the class has solved at least one problem". Say in plain words what the negation asserts. **(4 marks)**
- (d) Test the validity of the following argument, justifying your answer: "All computer scientists are logical. Some logical people are musicians. Therefore some computer scientists are musicians." **(4.5 marks)**

## QUESTION TWO

17.5 MARKS

- (a) A relation  $R$  is defined on the set of all integers by  $a R b$  if and only if  $a - b$  is divisible by 5. Prove that  $R$  is an equivalence relation and describe its equivalence classes. **(6 marks)**
- (b) The function  $f$  from the integers to the integers is defined by  $f(n) = 2n + 1$ . Determine whether  $f$  is injective and whether it is surjective. If  $f$  is not a bijection, restrict its codomain so that it becomes one. **(4 marks)**
- (c) Let  $R = \{(1,1), (1,3), (2,2), (3,1), (3,3), (4,4)\}$  on  $\{1, 2, 3, 4\}$ . Write down its matrix, read the five properties off the matrix, and give the partition induced if  $R$  is an equivalence relation. **(4 marks)**
- (d) Prove that if  $f$  and  $g$  are both injective then  $g \circ f$  is injective. **(3.5 marks)**

**QUESTION THREE****17.5 MARKS**

- (a) How many of the integers from 1 to 1000 inclusive are divisible by 3 or by 5 but **not** by 15? **(5 marks)**
- (b) How many distinct arrangements are there of the letters of the word **MISSISSIPPI**, and in how many of them do all four S's stand together? **(5 marks)**
- (c) A password is 8 characters long, drawn from the 26 small letters, the 26 capital letters and the 10 digits with repetition allowed, and it must contain **at least one digit**. How many such passwords are there? **(4 marks)**
- (d) Show that among any 13 people two must have been born in the same month, and find the least number of people that guarantees three share a birth month. **(3.5 marks)**

**QUESTION FOUR****17.5 MARKS**

- (a) Prove that in any undirected graph the number of vertices of odd degree is even. **(4 marks)**
- (b) A tree has two vertices of degree 4, three vertices of degree 3, and every one of its remaining vertices of degree 1. How many vertices has the tree altogether? **(5 marks)**
- (c) Show that no simple graph can have the degree sequence (5, 5, 4, 3, 2, 1). **(3.5 marks)**
- (d) Use Euler's formula to prove that every simple connected planar graph with at least three vertices has a vertex of degree at most 5. **(5 marks)**

**QUESTION FIVE****17.5 MARKS**

- (a) Simplify  $F = A'BC + AB'C + ABC' + ABC$  to a minimal sum of products, naming the law used at each step. **(5 marks)**
- (b) Draw the logic circuit for your simplified expression and say in words what function it computes. **(3 marks)**
- (c) A function of three inputs is 1 exactly when an **odd** number of its inputs are 1. Write down its sum of products form from the truth table, and show that **no two** of its four product terms can be combined, so that the expression is already as short as a sum of products can be. **(4.5 marks)**
- (d) Write a C++ program that reads three inputs, each 0 or 1, and prints the value of the majority function of part (a). State what it prints for the input 1 0 1. **(5 marks)**

## QUESTION SIX

17.5 MARKS

- (a) Prove by **strong** induction that every integer  $n \geq 2$  can be written as a product of prime numbers. **(5 marks)**
- (b) Solve the recurrence relation  $a_n = 2a_{n-1} + 3$  given  $a_0 = 1$ . **(4.5 marks)**
- (c) A bag holds 4 white, 5 black and 6 green balls. Three are drawn together without replacement. Find the probability that all three are of the same colour. **(4 marks)**
- (d) Prove by contradiction that there is no largest prime number. **(4 marks)**

## TEACH

MARK RECONCILIATION, BEFORE YOU START WRITING

| QUESTION | PARTS           | TOTAL |
|----------|-----------------|-------|
| Q1       | 4 + 5 + 4 + 4.5 | 17.5  |
| Q2       | 6 + 4 + 4 + 3.5 | 17.5  |
| Q3       | 5 + 5 + 4 + 3.5 | 17.5  |
| Q4       | 4 + 5 + 3.5 + 5 | 17.5  |
| Q5       | 5 + 3 + 4.5 + 5 | 17.5  |
| Q6       | 5 + 4.5 + 4 + 4 | 17.5  |

Six questions, four parts each, every part reconciling to 17.5. On this paper the reconciliation tells you something else as well: no part is worth fewer than 3 marks, so there is nothing here that can be dashed off in a line. Budget accordingly.

## MOCK EXAMINATION THREE

# Answers

*Mark these strictly. On this paper the number at the end of a part is worth almost nothing on its own: the marks are in the justification, and a proof that reaches the right conclusion by an argument that does not hold is worth less than an honest attempt that stops halfway. Where an answer below names a law or a definition, check that your own script named it too.*

**A.  $(P \wedge Q) \vee (P \wedge \neg Q) \equiv P$ , BY THE LAWS (4 MARKS)**

Three steps, and each one is a mark. The instruction "without using a truth table" means the table earns nothing here even if it is correct.

| STEP | EXPRESSION                            | LAW USED                                       |
|------|---------------------------------------|------------------------------------------------|
| 0    | $(p \wedge q) \vee (p \wedge \neg q)$ | the given proposition                          |
| 1    | $p \wedge (q \vee \neg q)$            | distributive                                   |
| 2    | $p \wedge T$                          | negation, since $q \vee \neg q$ is always true |
| 3    | <b>p</b>                              | identity                                       |

**Say what it means**, which costs a sentence and shows the manipulation was understood: whether  $q$  holds or not is irrelevant to the compound, because the two disjuncts between them cover both cases and each requires  $p$ . This is the propositional twin of the Boolean simplification  $AB + AB' = A$ .

**B. IS  $(P \rightarrow Q) \wedge (\neg P \rightarrow R) \rightarrow (Q \vee R)$  A TAUTOLOGY? (5 MARKS)**

**Yes.** The table has  $2^3 = 8$  rows and the last column is true in every one.

| P | Q | R | $P \rightarrow Q$ | $\neg P \rightarrow R$ | BOTH | $Q \vee R$ | THE WHOLE |
|---|---|---|-------------------|------------------------|------|------------|-----------|
| T | T | T | T                 | T                      | T    | T          | <b>T</b>  |
| T | T | F | T                 | T                      | T    | T          | <b>T</b>  |
| T | F | T | F                 | T                      | F    | T          | <b>T</b>  |
| T | F | F | F                 | T                      | F    | F          | <b>T</b>  |
| F | T | T | T                 | T                      | T    | T          | <b>T</b>  |
| F | T | F | T                 | F                      | F    | T          | <b>T</b>  |
| F | F | T | T                 | T                      | T    | T          | <b>T</b>  |
| F | F | F | T                 | F                      | F    | F          | <b>T</b>  |

**The argument in two lines, which is the better answer** if the question does not insist on a table. Suppose both premises hold. Either  $p$  is true, in which case  $p \rightarrow q$  gives  $q$ , so  $q \vee r$  holds; or  $p$  is false, in which case  $\neg p$  is true and  $\neg p \rightarrow r$  gives  $r$ , so  $q \vee r$  holds again. There is no third case, so the conclusion cannot fail while the premises hold. That is a **proof by cases**, and it explains *why* the table came out as it did.

**Rows 4 and 8 are the ones to look at.** In both,  $q \vee r$  is false, so if the whole thing is to be a tautology the premise must be false there, and it is. An implication with a false hypothesis is true whatever its conclusion does, which is exactly the vacuous case that makes so many people distrust the truth table for  $\rightarrow$ .

**C. SYMBOLISING AND NEGATING (4 MARKS)**

**Declare the domains and the predicate before writing a quantifier.** Let  $x$  range over the students in the class, let  $y$  range over the problems set, and let  $S(x, y)$  mean "student  $x$  has solved problem  $y$ ". Then

$$\forall x \exists y S(x, y)$$

**The negation**, pushed inward one quantifier at a time. Each step flips one quantifier and moves the negation past it:

$$\begin{aligned}\neg \forall x \exists y S(x, y) \\ \equiv \exists x \neg \exists y S(x, y) \\ \equiv \exists x \forall y \neg S(x, y)\end{aligned}$$

**In plain words:** there is a student in the class who has solved no problem at all.

**The two traps.** First, the order of the quantifiers matters:  $\exists y \forall x S(x, y)$  would say there is one single problem that every student has solved, which is a much stronger claim than the original. Second, the negation is **not** "no student has solved any problem"; that would be  $\forall x \forall y \neg S(x, y)$ . Denying that everyone succeeded requires only one failure.

#### D. THE VALIDITY OF THE SYLLOGISM (4.5 MARKS)

Let  $C(x)$  mean  $x$  is a computer scientist,  $L(x)$  that  $x$  is logical and  $M(x)$  that  $x$  is a musician.

Premise 1:  $\forall x (C(x) \rightarrow L(x))$   
Premise 2:  $\exists x (L(x) \wedge M(x))$   
Conclusion:  $\exists x (C(x) \wedge M(x))$

**The argument is invalid**, and the way to prove an argument invalid is to exhibit one situation in which every premise is true and the conclusion is false. Take a universe of just two individuals, 1 and 2, and set

$$C = \{1\}, \quad L = \{1, 2\}, \quad M = \{2\}$$

**Premise 1 holds:** the only computer scientist is 1, and 1 is logical.

**Premise 2 holds:** 2 is both logical and a musician.

**The conclusion fails:** the only computer scientist is 1, and 1 is not a musician, so no individual satisfies  $C(x) \wedge M(x)$ .

Premises true and conclusion false in the same situation, so the argument is **not valid**.

**Where the reasoning goes wrong.** Being logical is the middle term, and the two premises attach to *different parts* of it: the first says the computer scientists sit somewhere inside the logical people, the second says some logical people are musicians, and nothing forces those two regions of the logical people to overlap. Drawing three circles makes this visible at once, and a Venn diagram is a perfectly acceptable way to present this answer provided the counter-example is named.

**A. A – B DIVISIBLE BY 5, ON THE INTEGERS (6 MARKS)**

**Use the definition of "divides" explicitly**, because every one of the three checks is an application of it: 5 divides  $m$  means  $m = 5k$  for some integer  $k$ .

**Reflexive.** For any integer  $a$ ,  $a - a = 0 = 5 \times 0$ , and 0 is an integer, so 5 divides  $a - a$  and  $a R a$ . This holds for every  $a$ , so  $R$  is reflexive.

**Symmetric.** Suppose  $a R b$ , so  $a - b = 5k$  for some integer  $k$ . Then

$$b - a = -(a - b) = 5(-k)$$

and  $-k$  is an integer, so 5 divides  $b - a$  and  $b R a$ .

**Transitive.** Suppose  $a R b$  and  $b R c$ , so  $a - b = 5k$  and  $b - c = 5m$  for integers  $k$  and  $m$ . Adding,

$$a - c = (a - b) + (b - c) = 5k + 5m = 5(k + m)$$

and  $k + m$  is an integer, so  $a R c$ .

**$R$  is reflexive, symmetric and transitive, so it is an equivalence relation.**

**The classes.** Two integers are related exactly when they leave the **same remainder on division by 5**, since if  $a = 5q + r$  and  $b = 5q' + r$  then  $a - b = 5(q - q')$ , and conversely. There are five possible remainders, so there are exactly **five** classes:

$$[0] = \{\dots, -10, -5, 0, 5, 10, \dots\}$$

$$[1] = \{\dots, -9, -4, 1, 6, 11, \dots\}$$

$$[2] = \{\dots, -8, -3, 2, 7, 12, \dots\}$$

$$[3] = \{\dots, -7, -2, 3, 8, 13, \dots\}$$

$$[4] = \{\dots, -6, -1, 4, 9, 14, \dots\}$$

**Every integer lies in exactly one of them**, which is the partition the relation induces and the sentence that finishes the answer. Note that the classes are infinite while there are only finitely many of them: a partition does not have to cut a set into small pieces, only into pieces that do not overlap and leave nothing out.

**B.  $f(n) = 2n + 1$  FROM THE INTEGERS TO THE INTEGERS (4 MARKS)**

**Injective, yes.** Suppose  $f(n_1) = f(n_2)$ . Then  $2n_1 + 1 = 2n_2 + 1$ , so  $2n_1 = 2n_2$ , so  $n_1 = n_2$ .

**Surjective, no.** Every value of  $f$  is odd, because  $2n$  is even for every integer  $n$  and one more than an even number is odd. So no even integer is an image. A single witness settles it: if  $f(n) = 2$  then  $2n + 1 = 2$ , so  $n = 1/2$ , which is **not an integer**, and therefore 2 has no preimage.

**Making it a bijection.**  $f$  is injective but not surjective, so the codomain is too large. Shrink it to exactly the set of values  $f$  actually takes:

$f$  from the integers to  **$\{2k + 1 : k \text{ is an integer}\}$** , the set of odd integers

This is now surjective by construction, since any odd integer can be written  $2k + 1$  and is then  $f(k)$ , and it is still injective because shrinking the codomain cannot make two inputs collide. So it is a bijection.

**The general principle worth stating:** any injective function becomes a bijection onto its range. It is surjectivity that a codomain can be blamed for; injectivity is a property of the rule itself.

**C. READING THE FIVE PROPERTIES OFF A MATRIX (4 MARKS)**

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 1 | 0 | 1 | 0 |
| 4 | 0 | 0 | 0 | 1 |

| PROPERTY      | TEST IN THE MATRIX                                       | VERDICT                                                       |
|---------------|----------------------------------------------------------|---------------------------------------------------------------|
| Reflexive     | the main diagonal is all 1's                             | <b>Yes</b> , it reads 1, 1, 1, 1                              |
| Irreflexive   | the main diagonal is all 0's                             | <b>No</b> , and it cannot be, the relation being reflexive    |
| Symmetric     | the matrix equals its transpose                          | <b>Yes</b> , the only off-diagonal 1's are at (1,3) and (3,1) |
| Antisymmetric | no off-diagonal position has 1 in both (i, j) and (j, i) | <b>No</b> , (1,3) and (3,1) are both 1 with $1 \neq 3$        |
| Transitive    | every two-step path has a one-step edge                  | <b>Yes</b> , checked below                                    |

**Transitivity checked in full**, since it is the one property the matrix does not settle at a glance. The only composable pairs involving different elements are (1,3) with (3,1), needing (1,1), present; (3,1) with (1,3), needing (3,3), present; (1,3) with (3,3), needing (1,3), present; and (1,1) with (1,3), needing (1,3), present. Everything else is a loop composed with itself. So  $R$  is transitive.

**$R$  is reflexive, symmetric and transitive, so it is an equivalence relation**, and the partition it induces is

**{1, 3}, {2}, {4}**

**A class of one is still a class.** Elements 2 and 4 are related to nothing but themselves, and a partition must cover the whole set, so each forms its own block. Leaving them out is the usual way to lose the last mark here.

#### D. THE COMPOSITION OF TWO INJECTIONS (3.5 MARKS)

**Set it up first.** Let  $f$  be a function from  $A$  to  $B$  and  $g$  a function from  $B$  to  $C$ , both injective, so that  $g \circ f$  is a function from  $A$  to  $C$ .

**Proof.** Suppose  $(g \circ f)(x_1) = (g \circ f)(x_2)$  for some  $x_1$  and  $x_2$  in  $A$ . By the definition of composition this says

$$g(f(x_1)) = g(f(x_2))$$

Now  $f(x_1)$  and  $f(x_2)$  are elements of  $B$ , and  $g$  is injective, so equal images force equal inputs:

$$f(x_1) = f(x_2)$$

And  $f$  is injective, so by the same reasoning

$$x_1 = x_2$$

Since  $x_1$  and  $x_2$  were arbitrary,  $g \circ f$  sends no two different inputs to the same output, and is therefore injective. ■



**Notice the order in which the two hypotheses are used:**  $g$  first, then  $f$ , which is the reverse of the order in which the functions are applied. That is not a coincidence, and saying so is worth half a mark: the proof runs backwards along the chain  $x \rightarrow f(x) \rightarrow g(f(x))$ , peeling off the outermost function first.

**A. DIVISIBLE BY 3 OR 5 BUT NOT BY 15 (5 MARKS)**

**Count each family with a division.** The number of integers from 1 to 1000 that a given  $d$  divides is the whole-number part of  $1000 / d$ .

divisible by 3:  $1000 / 3 = 333.33\dots$ , so **333**

divisible by 5:  $1000 / 5 = \mathbf{200}$

divisible by 15:  $1000 / 15 = 66.66\dots$ , so **66**

**Why 15 and not something else.** An integer is divisible by both 3 and 5 exactly when it is divisible by 15, because 3 and 5 have no common factor above 1. This is the step to state: writing  $3 \times 5 = 15$  without saying why the two conditions combine that way is the part examiners probe.

**Divisible by 3 or by 5**, by inclusion and exclusion:

$$333 + 200 - 66 = \mathbf{467}$$

**Now remove the multiples of 15.** Every multiple of 15 is divisible by 3, so all 66 of them are inside that 467 and must come out:

$$467 - 66 = \mathbf{401}$$

**Check by regions.** Divisible by 3 only:  $333 - 66 = 267$ . Divisible by 5 only:  $200 - 66 = 134$ . Their sum is  $267 + 134 = \mathbf{401}$ , computed a different way. The phrase "or but not both" would have meant the same thing here, which is worth noticing: "divisible by 15" and "divisible by both" are the same set.

**B. MISSISSIPPI (5 MARKS)**

**Count the letters first.** MISSISSIPPI has 11 letters: M once, I four times, S four times, P twice. Check:  $1 + 4 + 4 + 2 = 11$ .

$$11! / (4! \times 4! \times 2!) = 39916800 / (24 \times 24 \times 2) = 39916800 / 1152 = \mathbf{34650}$$

**All four S's together.** Glue them into a single block. The block then behaves as one letter, so the objects to arrange are

[SSSS], M, I, I, I, I, P, P that is **8** objects, with the four I's alike and the two P's alike

$$8! / (4! \times 2!) = 40320 / 48 = \mathbf{840}$$

**Why the block is not itself rearranged.** Ordinarily a block of  $k$  items would be permuted internally in  $k!$  ways, but the four S's are identical, so their  $4!$  internal orders all give the same word. Multiplying by  $4!$  here is the standard error and gives 20160.

**A sanity check on the size:** 840 out of 34650 is about one arrangement in 41, which is plausible, since insisting that four particular letters be adjacent is a severe restriction. An answer larger than 34650 would be impossible on its face.

**C. EIGHT CHARACTERS WITH AT LEAST ONE DIGIT (4 MARKS)**

The alphabet has  $26 + 26 + 10 = \mathbf{62}$  characters. "At least one" is the signal for the **complement**: count all passwords and remove those with no digit at all.

$$\text{all passwords: } 62^8 = \mathbf{218340105584896}$$

$$\text{with no digit, only the 52 letters: } 52^8 = \mathbf{53459728531456}$$

$$\text{at least one digit} = 62^8 - 52^8 = \mathbf{164880377053440}$$

**Leaving the answer as  $62^8 - 52^8$  is acceptable** and is what most of the marks are for. If you do evaluate it, note that the answer is about 75.5% of all passwords, which is

the kind of remark that shows the number was understood rather than copied off a calculator.

**Why not count directly.** "Exactly one digit, exactly two digits, and so on up to eight" is eight terms, each a product of a binomial coefficient and two powers. It gives the same answer and takes ten times as long.

#### **D. TWO PIGEONHOLE ARGUMENTS (3.5 MARKS)**

**The first.** Name the objects and the boxes, which is where the marks are. The **objects** are the 13 people; the **boxes** are the 12 months of the year; each person goes into the box of their birth month, and every person has exactly one birth month, so every object goes into some box. Since  $13 > 12$ , the pigeonhole principle says some box holds at least two people, that is, **two people share a birth month**.

**The second** needs the generalised form:  $N$  objects in  $k$  boxes force some box to hold at least  $\lceil N / k \rceil$ . Three in one box is guaranteed as soon as

$\lceil N / 12 \rceil \geq 3$ , that is  $N > 24$ , so the least such  $N$  is **25**

**Why 24 is not enough**, and this sentence is worth a mark of its own: with 24 people it is possible for exactly two to be born in each of the twelve months, and then no month holds three. The principle guarantees a crowded box only from the twenty-fifth person onward, and exhibiting the arrangement that defeats 24 is what proves the answer is least.

**A. THE NUMBER OF ODD-DEGREE VERTICES IS EVEN (4 MARKS)**

**Split the sum, which is the whole idea.** Let  $V$  be the vertex set and divide it into the vertices of even degree and the vertices of odd degree. By the handshaking theorem

$$\sum_{\text{even}} \deg(v) + \sum_{\text{odd}} \deg(v) = \sum_V \deg(v) = 2|E|$$

**Two of those three quantities are visibly even.** The total  $2|E|$  is even, being twice a whole number. The first sum is even, being a sum of even numbers. Therefore

$$\sum_{\text{odd}} \deg(v) = 2|E| - \sum_{\text{even}} \deg(v) = \text{even} - \text{even} = \text{even}$$

**Finish with the parity argument.** That last sum consists entirely of odd numbers. A sum of odd numbers is odd when there is an odd count of them and even when there is an even count. Since the sum is even, the count must be even. So the number of vertices of odd degree is even. ■

**The consequence worth naming:** no graph has exactly one odd-degree vertex, or three, or five. This is the corollary that answers "can such a graph exist" questions in one line, and part (b) of Mock Two Question Four used it.

**B. THE TREE WITH DEGREES 4, 4, 3, 3, 3 AND THE REST 1 (5 MARKS)**

**Two facts about the tree, then equate them.** Let  $n$  be the total number of vertices. A tree on  $n$  vertices has exactly  $n - 1$  edges, so by the handshaking theorem

$$\sum \deg(v) = 2(n - 1) = 2n - 2$$

**Now add the degrees up from the description.** There are two vertices of degree 4, three of degree 3, and the remaining  $n - 5$  vertices have degree 1:

$$\sum \deg(v) = 2 \times 4 + 3 \times 3 + (n - 5) \times 1 = 8 + 9 + n - 5 = n + 12$$

**Equating the two expressions** gives one equation in one unknown:

$$n + 12 = 2n - 2, \text{ so } n = 14$$

**Check every figure.** The tree has 14 vertices and 13 edges. Its leaves number  $14 - 5 = 9$ , and the degrees total  $8 + 9 + 9 = 26$ , which is  $2 \times 13$  as the theorem requires. A tree with 9 leaves and 5 branch vertices is entirely believable, so nothing here is out of proportion.

**The step that is easy to miss** is that "the rest" have degree 1 and there are  $n - 5$  of them, not some fixed number. Writing the unknown into the degree sum is what makes the equation solvable.

**C. NO SIMPLE GRAPH HAS DEGREE SEQUENCE (5, 5, 4, 3, 2, 1) (3.5 MARKS)**

**The parity test does not settle it,** and saying so first shows you tried the cheap test: the degrees sum to  $5 + 5 + 4 + 3 + 2 + 1 = 20$ , which is even, so the handshaking theorem is satisfied with 10 edges. Something else must do the work.

**The argument.** The sequence has six terms, so the graph has six vertices, and it contains **two vertices of degree 5**. In a simple graph with six vertices, a vertex of degree 5 is joined to *all five* of the others, since it has no loops and no repeated edges available to it. So call the two of them  $u$  and  $w$ .

Every other vertex is joined to  $u$ , and every other vertex is joined to  $w$ . Hence **every vertex apart from  $u$  and  $w$  has degree at least 2**. But the sequence contains a vertex of degree **1**, which would have to be joined to both  $u$  and  $w$  and so have degree at least 2. That is a contradiction, so no such simple graph exists. ■

**Where "simple" is doing the work.** Without it a vertex could reach degree 5 through loops or parallel edges and would not need to touch all five other vertices, and the argument collapses. Whenever a degree-sequence question is set, look first at whether the largest degree forces adjacency to everything.

**D. EVERY SIMPLE CONNECTED PLANAR GRAPH HAS A VERTEX OF DEGREE AT MOST 5 (5 MARKS)**

**By contradiction.** Suppose  $G$  is simple, connected and planar with  $v \geq 3$  vertices, and suppose **every** vertex has degree at least 6.

**The lower bound from the assumption.** By the handshaking theorem,

$$2e = \sum \deg(v) \geq 6v, \text{ so } e \geq 3v$$

**The upper bound from planarity.** Euler's formula  $v - e + f = 2$ , together with the fact that every face of a simple graph is bounded by at least three edges and every edge borders at most two faces, gives the standard bound

$$e \leq 3v - 6$$

**The two bounds cannot both hold.** They would give

$$3v \leq e \leq 3v - 6, \text{ so } 3v \leq 3v - 6, \text{ so } 0 \leq -6$$

which is false. The assumption is therefore untenable, and **some vertex has degree at most 5**. ■

**Read the result back.** This is the reason five colours are always enough for a planar map by an easy argument, and it is also why  $K_5$ , in which every vertex has degree 4, is not ruled out by this test and needs the edge-count argument instead. A theorem that excludes a shape is only as useful as the shapes it fails to exclude.

**A. SIMPLIFYING  $A'BC + AB'C + ABC' + ABC$  (5 MARKS)**

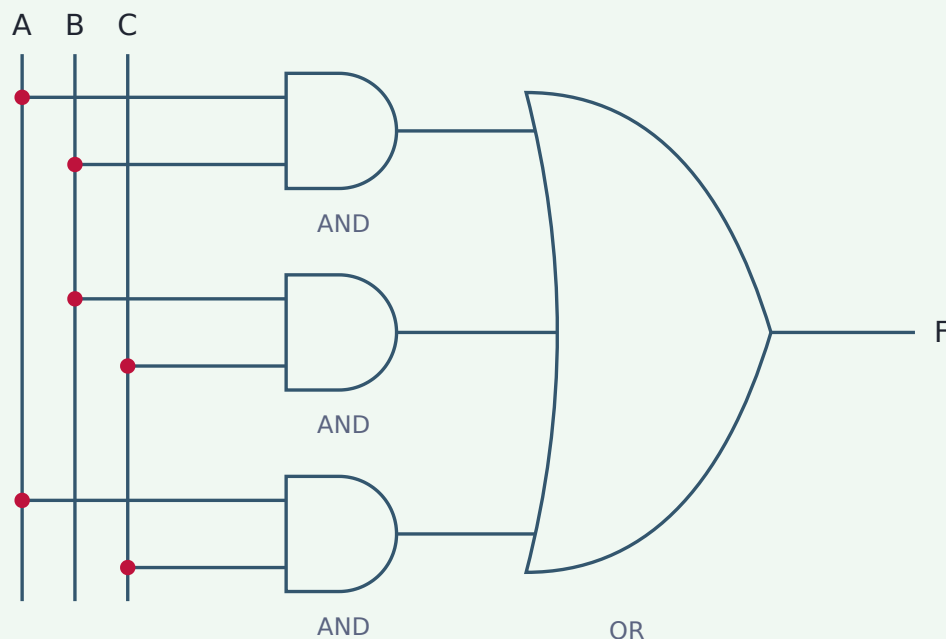
**The trick is to use ABC three times.** In Boolean algebra  $X + X = X$ , so a term may be duplicated as often as it is useful without changing the expression. That is the idempotent law, and it is the step that makes this question tractable.

| STEP | EXPRESSION                                   | LAW USED                    |
|------|----------------------------------------------|-----------------------------|
| 0    | $A'BC + AB'C + ABC' + ABC$                   | the given F                 |
| 1    | $A'BC + AB'C + ABC' + ABC + ABC + ABC$       | idempotent, $X = X + X + X$ |
| 2    | $(A'BC + ABC) + (AB'C + ABC) + (ABC' + ABC)$ | commutative and associative |
| 3    | $BC(A' + A) + AC(B' + B) + AB(C' + C)$       | distributive                |
| 4    | $BC \cdot 1 + AC \cdot 1 + AB \cdot 1$       | complement                  |
| 5    | <b><math>AB + BC + AC</math></b>             | identity                    |

**Check by counting the 1's.** The original F is 1 in exactly four rows: 011, 101, 110 and 111. The simplified  $AB + BC + AC$  is 1 exactly when some two of the inputs are 1, which is those same four rows. Four product terms have become three, and each has shrunk from three literals to two.

**B. THE CIRCUIT, AND WHAT IT COMPUTES (3 MARKS)**

**It is the majority function of three inputs:** the output is 1 exactly when at least two of A, B and C are 1. It is the circuit behind any two-out-of-three voting arrangement, such as three sensors on a machine or three copies of a computation compared against one another.



Three vertical rails carry A, B and C down the diagram, and a dot marks each place a gate taps one of them. A line crossing a rail with no dot on it is not connected to that rail, which is the standard convention and the reason the dots are drawn at all.

**Six taps, three rails.** Each input is used by exactly two of the three AND gates, which the drawing shows directly; the version of this circuit that draws six separate input wires is not wrong, but it hides the fact that there are only three signals.

### C. THE ODD-PARITY FUNCTION, AND WHY IT WILL NOT SIMPLIFY (4.5 MARKS)

**Build the table and take the rows where F is 1.** Each such row contributes one product term, with a letter written plain where the input is 1 and complemented where it is 0.

| A | B | C | NUMBER OF 1'S | F | PRODUCT TERM |
|---|---|---|---------------|---|--------------|
| 0 | 0 | 0 | 0             | 0 |              |
| 0 | 0 | 1 | 1             | 1 | $A'B'C$      |
| 0 | 1 | 0 | 1             | 1 | $A'BC'$      |
| 0 | 1 | 1 | 2             | 0 |              |
| 1 | 0 | 0 | 1             | 1 | $AB'C'$      |
| 1 | 0 | 1 | 2             | 0 |              |
| 1 | 1 | 0 | 2             | 0 |              |
| 1 | 1 | 1 | 3             | 1 | $ABC$        |

$$F = A'B'C + A'BC' + AB'C' + ABC$$

**Why no two terms combine.** The only way to shorten a sum of products is the rule  $XY + XY' = X$ , which needs two terms that agree in every letter but one. Writing each term as the row it came from, the four rows are 001, 010, 100 and 111, and every pair of them differs in **two** positions:

| PAIR OF ROWS | POSITIONS WHERE THEY DIFFER | HOW MANY |
|--------------|-----------------------------|----------|
| 001 and 010  | B and C                     | 2        |
| 001 and 100  | A and C                     | 2        |
| 001 and 111  | A and B                     | 2        |
| 010 and 100  | A and B                     | 2        |
| 010 and 111  | A and C                     | 2        |
| 100 and 111  | B and C                     | 2        |

All six pairs differ in two positions, none in exactly one, so the combining rule never applies and the four-term expression above is already as short as a sum of products can be. ■

**Why this happens, in one sentence:** changing a single input flips the count of 1's from odd to even, so it always flips F, and two rows one step apart can never both be rows where F is 1. The function is the parity of the three inputs, and parity is the standard example of a function that a sum of products handles badly.

### D. THE MAJORITY FUNCTION AS A PROGRAM (5 MARKS)

The Boolean product and sum of part (a) are written with the bitwise operators & and |, which act on 0 and 1 exactly as  $\cdot$  and  $+$  do in Boolean algebra. That correspondence is

the point of the program, so make it visible in a comment.

MAJORITY.CPP

```
// Mock Three Question Five (d): the majority of three 0/1 inputs.
#include <iostream>
using namespace std;

// F = AB + BC + AC, with & for the Boolean product and | for the sum.
int majority(int a, int b, int c) {
    return (a & b) | (b & c) | (a & c);
}

int main() {
    int a, b, c;
    cout << "Enter A, B and C as 0 or 1: "; // i. prompt
    cin >> a >> b >> c;

    if (a > 1 || b > 1 || c > 1) { // ii. reject anything else
        cout << "Each input must be 0 or 1." << endl;
        return 0;
    }

    cout << "F = " << majority(a, b, c) << endl; // iii. report
    return 0;
}
```

WHAT IT PRINTED FOR THE INPUT 1 0 1

```
Enter A, B and C as 0 or 1: F = 1
```

**So the program prints F = 1**, which agrees with the truth table: with A = 1, B = 0 and C = 1, the term AC is 1 and that is enough.

**Two remarks worth a mark each on paper.** First, & and | here are the **bitwise** operators, not the logical && and ||; on inputs restricted to 0 and 1 the two pairs agree, but && and || also stop evaluating as soon as the answer is settled, which is a difference that matters as soon as the operands do anything besides hold a value. Second, the guard rejecting inputs above 1 is not decoration: with a = 2 the expression a & b would compute a bit pattern rather than a truth value, and the program would print something that is not 0 or 1.



**A. EVERY INTEGER  $n \geq 2$  IS A PRODUCT OF PRIMES (5 MARKS)**

**Say at the outset that a single prime counts as a product of one prime.** Without that convention the basis case fails and the whole proof stalls on a technicality.

**Basis.**  $n = 2$  is prime, so it is a product of primes, namely of the one prime 2.

**Strong inductive hypothesis.** Let  $k \geq 2$  and assume that **every** integer  $m$  with  $2 \leq m < k$  is a product of primes.

**Inductive step.** Consider  $k$ . There are two cases and they are exhaustive.

**Case 1,  $k$  is prime.** Then  $k$  is a product of one prime and there is nothing to do.

**Case 2,  $k$  is composite.** Then by the definition of composite,  $k = a \times b$  for some integers  $a$  and  $b$  with  $2 \leq a < k$  and  $2 \leq b < k$ . Both  $a$  and  $b$  fall in the range covered by the hypothesis, so each is a product of primes. Multiplying those two products together expresses  $k$  as a product of primes.

In both cases  $k$  is a product of primes, so the hypothesis for all values below  $k$  yields the claim at  $k$ . By the principle of strong induction, every integer  $n \geq 2$  is a product of primes. ■

**Why ordinary induction will not do here,** and this is what the word "strong" in the question is testing. Ordinary induction would let you assume the claim only for  $k - 1$ , and knowing that  $k - 1$  factorises tells you nothing whatever about  $k$ . What the proof actually needs is the claim at  $a$  and at  $b$ , two numbers that may lie anywhere below  $k$ . Strong induction is exactly the form that makes the whole range below  $k$  available.

**B. SOLVING  $A_n = 2A_{n-1} + 3$  (4.5 MARKS)**

This is not homogeneous, because of the constant 3, so the characteristic equation does not apply directly. The standard move is to absorb the constant by shifting the sequence.

**The shift to use.** For  $a_n = c \cdot a_{n-1} + b$  with  $c \neq 1$ , add  $b / (c - 1)$  to both sides. Here  $c = 2$  and  $b = 3$ , so the amount to add is  $3 / (2 - 1) = 3$ . Put  $b_n = a_n + 3$ . Then

$$b_n = a_n + 3 = 2a_{n-1} + 3 + 3 = 2a_{n-1} + 6 = 2(a_{n-1} + 3) = 2b_{n-1}$$

**which is a plain geometric progression.** Its first term is  $b_0 = a_0 + 3 = 1 + 3 = 4$ , so

$$b_n = 4 \cdot 2^n = 2^{n+2}$$

**Undo the shift** to get back to the sequence that was asked for:

$$a_n = 2^{n+2} - 3 = 4 \cdot 2^n - 3$$

**Check the first three terms against the recurrence**, which is the only check that tests both the formula and the starting value:

| N | FROM THE RECURRENCE    | FROM THE FORMULA      |
|---|------------------------|-----------------------|
| 0 | given as 1             | $4 \times 1 - 3 = 1$  |
| 1 | $2 \times 1 + 3 = 5$   | $4 \times 2 - 3 = 5$  |
| 2 | $2 \times 5 + 3 = 13$  | $4 \times 4 - 3 = 13$ |
| 3 | $2 \times 13 + 3 = 29$ | $4 \times 8 - 3 = 29$ |

**Iterating instead is a legitimate route** and gets the same answer:  $a_n = 2^n a_0 + 3(2^{n-1} + \dots + 2 + 1) = 2^n + 3(2^n - 1) = 4 \cdot 2^n - 3$ , using the sum of a geometric progression. Whichever route you take, **state which one you are taking** in the first line.

### C. THREE BALLS OF THE SAME COLOUR (4 MARKS)

The bag holds  $4 + 5 + 6 = 15$  balls, and three are taken together, so order does not matter and every part of this is a combination.

all selections of three:  $C(15, 3) = (15 \times 14 \times 13) / (3 \times 2 \times 1) = 2730 / 6 = 455$

**The favourable selections, one colour at a time.** The three colours are alternatives, only one of which can happen, so they are added rather than multiplied:

all three white:  $C(4, 3) = 4$

all three black:  $C(5, 3) = 10$

all three green:  $C(6, 3) = 20$

favourable in all:  $4 + 10 + 20 = 34$

$P(\text{all the same colour}) = 34 / 455$ , about **0.075**

**The fraction will not reduce:**  $455 = 5 \times 7 \times 13$  and  $34 = 2 \times 17$ , so they share no factor. Saying so saves you from hunting for a cancellation that is not there.

**A check on plausibility.** About one selection in thirteen is monochrome, which is believable: the largest colour class has only 6 of the 15 balls, so three matching is a genuine coincidence rather than a near-certainty. Any answer above about 0.2 should have prompted a recount.

### D. THERE IS NO LARGEST PRIME (4 MARKS)

**Proof by contradiction, so begin by assuming the opposite**, and say that you are doing so.

**Suppose** there is a largest prime. Then the primes can be listed completely and finitely:

$p_1 = 2, p_2 = 3, p_3 = 5, \dots, p_k$ , the largest

**Construct a number the list cannot account for.** Let

$$N = (p_1 \times p_2 \times \dots \times p_k) + 1$$

Then  $N > 1$ , so by part (a) of this question  $N$  has at least one prime divisor; call it  $q$ .

**Now trap  $q$ .** Suppose  $q$  were one of the primes on the list. Then  $q$  divides the product  $p_1 \times \dots \times p_k$ , and  $q$  divides  $N$  by choice, so  $q$  divides their difference:

$$N - (p_1 \times \dots \times p_k) = 1$$

But no prime divides 1, since every prime is at least 2. So  $q$  is a prime that is **not** on the list, contradicting the assumption that the list contained every prime.

**The assumption is therefore false, and there is no largest prime. ■**

**The point that is worth stating and is usually got wrong:**  $N$  itself need not be prime. The argument does not claim it is. It claims only that *some* prime divides  $N$  and that no prime on the list can. For instance  $2 \times 3 \times 5 \times 7 \times 11 \times 13 + 1 = 30031$ , which is  $59 \times 509$  and so not prime at all; but 59 is a prime that was missing from the list, which is exactly what the proof promised.

**Notice this answer used part (a).** The step "N has a prime divisor" is precisely the theorem proved by strong induction earlier in this question, and pointing that out is free.